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Phased Array Antennas

SECOND EDITION

Robert C. Hansen

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Phased Array Antennas

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Phased Array Antennas

Second Edition

R. C. HANSEN
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Second Edition

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Authorized India Edition

This book is dedicated to those who made Microwave Scanning Antennas possible:

Nicolas A. Begovich
Robert W. Bickmore
Jesse L. Butler
Lorne K. De Size
Robert S. Elliott
Richard C. Johnson
H. C. Ko
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Preface to the First Edition

Although array antennas have many decades of history, the last two decades have experienced a maturation, both in the understanding and design of arrays, and in the use of large sophisticated arrays. Radars utilizing electronic scanning arrays are in common use, from airport surveillance to missile detection and tracking; names of U.S. military systems, such as Aegis, Patriot, and Pave Paws, are well known. This book is a comprehensive treatment of all aspects of phased arrays; much has changed since the only other such work, *Microwave Scanning Antennas*, appeared in 1966. Most noteworthy has been the parallel development of inexpensive computer power and the theoretical understanding of nearly all aspects of phased array design. Design algorithms suitable for computers are emphasized here, with numerical tips and short algorithms sprinkled throughout the chapters. The work is prepared from the dual viewpoint of a design engineer and an antenna array analyst.

Chapter 2, on basic array characteristics, which covers grating lobes, quantization lobes, bandwidth, and directivity follows an introductory chapter. Highly efficient linear aperture and array synthesis techniques, including sum and difference patterns, are covered in Chapter 3. Chapter 4 treats synthesis of planar arrays. Array elements are covered in Chapter 5 and include not only the classic dipoles and slots, but TEM horns and patches. In Chapter 6, feeds for linear and planar arrays, both fixed beam and scanning, are examined; photonic time delay and feeders are included. Array performance is strongly affected by mutual impedance. Chapter 7 investigates ways of calculating this for various arrays elements, including an extensive treatment of ways of calculating array performance with mutual effects included. Among these are unit cell, spectral moment method, finite impedance matrix, and scattering techniques. Finite arrays are examined in Chapter 8, including the recently developed Gibbsian models. Next, Chapter 9 is an extensive view of superdirective arrays; the implications of high-temperature superconductors for antennas is an important feature. Multiple-beam arrays, as opposed to multiple-beam reflector feeds, are treated in Chapter 10.

Included are one- and two-dimensional Butler and Rotman lenses, and the practical meaning of beam orthogonality. Conformal arrays, ranging from ring arrays to arrays on cones, are covered next; much previously unpublished material is included in this chapter. Finally, Chapter 12 discusses array diagnostics, waveguide simulators in depth, and array tolerances. Extensive references to the archival literature are used in each chapter to offer additional sources of data.

ROBERT C. HANSEN

Tarzana, CA

Preface to the Second Edition

Several specialized types of phased arrays have attracted attention since the first edition. Connected dipole arrays offer wide bandwidth compared to a conventional array; these are discussed in detail in Chapter 12. (The old Chapter 12 is now Chapter 15). Reflectarrays provide reduced fabrication costs compared to a phased array. And retrodirective arrays offer interesting capabilities for data links. Both of these technologies are the subject of Chapter 13. The combination of reflectors and arrays is addressed in Chapter 14, both for focal plane arrays, including coma correction, and near-field Cassegrainian and Gregorian antennas.

Updates and additions have been made to existing sections: time delay deployment options for corporate fed arrays; fundamental limitations on Artificial Magnetic Conductors; Substrate Integrated Waveguide to replace rectangular waveguide; antennas for 60 GHz and beyond; impedances matching capabilities and limitations including Bode criterion limitations; elaboration of Scan Impedance and Scan Element Pattern calculations and measurements; and finally comments on completely overlapped sub-arrays.

CHAPTER ONE

Introduction

1.1 ARRAY BACKGROUND

Discovery of the first works on array antennas is a task best left to historians, but the two decades before 1940 contained much activity on array theory and experimentation. Some of the researchers were G. H. Brown, E. Bruce, P. S. Carter, C. W. Hansell, A. W. Ladner, N. E. Lindenblad, A. A. Pistolkors, S. A. Schelkunoff, G. C. Southworth, E. J. Sterba, and T. Walmsley. Primary journals were *Proc. IRE*, *Proc. IEE*, *BSTJ*, *RCA Review*, and *Marconi Review*. During World War II, much array work was performed in the United States and Britain. Interest in arrays returned in the early 1960s, with research projects at Lincoln Laboratories, General Electric, RCA, Hughes and others. Some of the array conferences are mentioned in the annotated reference list in Section 1.3.

A salient event was the publication by Academic Press of the three-volume book *Microwave Scanning Antennas* (*MSA*), with volume 1 appearing in 1964, and volumes 2 and 3 in 1966. This work was the first extensive coverage of phased arrays, with emphasis on mutual coupling theory, which is the basis of all array characteristics. After 30 years, *MSA* is still in print, through Peninsula Publishing.

It is the purpose of this book to present a thorough and extensive treatment of phased arrays, adding to and updating the array portions of *MSA*. The scope of the book is all types of arrays except adaptive, for which several excellent books exist; see references at the end of the chapter. Multiple-beam arrays are included. Because most arrays operate at frequencies that allow spacing above ground to be sufficiently large to preclude the ground affecting the array internal parameters, all arrays are presumed to be in free space. Active arrays, that is, those containing active devices, are not treated, nor are array-related circuit components, except for phasers, which are discussed briefly. It is also assumed that all array elements are identical, although the impedance matching may vary with the element position. A semantic difficulty

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arises with the phrase "phased array". For some people, this implies beam steering or scanning. But for others all arrays are phased; fixed beam broadside arrays are also phased. There are more important questions of terminology; these are addressed next.

1.2 SYSTEMS FACTORS

Important array factors for the systems designer are broadside pattern, gain versus angles, element input impedance, and efficiency. For all regular arrays, the pattern is given by the product of the element pattern and the pattern of the isotropic array, where the array elements are replaced by isotropes. However, the element excitations must be those of the real array; as discussed later, these are found by solving equations associated with a self-impedance and mutual-impedance or admittance matrix. In general, each element of an array will have a different input impedance. For a fixed beam array these are called "embedded impedances"; the obsolete and misleading term "active impedance" is deprecated. A scanning array not only has different element impedances, but each of them varies with scan angle. These element input impedances are called *scan impedances*.

The pattern of array gain versus angles is called *scan element pattern*; this term replaces *active element pattern*. The *scan element pattern* (SEP) is an extremely useful design factor. The element pattern and mutual coupling effects are subsumed into the *scan element pattern*; the overall radiated pattern is the product of the *scan element pattern* and the pattern of an isotropic array of elements scanned to the proper angle. The isotropic array factor incorporates the effects of array size and lattice, while the *scan element pattern*, as mentioned, incorporates element pattern, backscreen if used, and mutual coupling. Since the *scan element pattern* is an envelope of array gain versus scan angles, it tells the communications system or radar designer exactly how the array performs with scan, whether blind angles exist, and whether matching at a particular scan angle is advantageous. *Scan element pattern* is used for antenna gain in the conventional range equations. For an infinite array, the SEP is the same for all elements, but for a finite array each element sees a different environment, so that the SEP is an overall array factor. Use of infinite array *scan element patterns* allows array performance to be separated into this SEP and edge effects. Formulas for both finite array and infinite array *scan element pattern* are derived later; edge effects are also discussed later.

A similar parameter, appropriate for backscattering from antenna arrays, is the *scattering scan element pattern* (SSEP). This parameter gives the backscattered field intensity from an array element, when the array is excited by an incident plane wave. This then is different from the SEP, which relates radiated field intensity to total radiated power. The radar cross section (RCS) relates reradiated field intensity to incident field intensity, with a $4\pi R^2$ factor. The SSEP is this ratio of reradiated to incident intensity; a convenient normalization is to the broadside value. Just as in the case of a radiating array, the scattering array finite size and edge effects have been separated, so that the SSEP relates the effects of element design and array lattice. It can then be used to make design trades for type of element and lattice; the features

due to the array size are included simply by multiplying by the isotropic array factor. Of course, SSEP is related to the RCS pattern. It can be considered as the RCS pattern of one unit cell of the array.

System factors also arise in arrays used for wideband baseband (no carrier) applications. The one-way (communications) range equation, written without explicit wavelength dependence, is

$$P_r = \frac{P_t G A_e}{4\pi R^2} \tag{1.1}$$

where as usual P_r and P_t are received and transmitted powers, R is the range, and G and A_e are the gain of one antenna and the effective area of the other. Both gain and effective area include an impedance mismatch factor:[1] $(1-|\Gamma|^2)$. It is assumed that P_t is fixed, independent of frequency. If the GA_e product is relatively constant over the frequency band of interest, then the signal is transferred without significant dispersion, providing that the antenna and matching unit phase are well behaved also (Hansen and Libelo, 1995). Otherwise significant dispersion can occur.

From a casual look at array antennas, one might assume a planar array to be a constant effective area antenna. However, for a regularly spaced array of low-gain elements, as the frequency increases from nominal half-wave spacing, the gain increases until the first grating lobe appears, with the gain then dropping back to the original level. Further increases in frequency produce additional rises in gain followed by drops as grating lobes appear. The net result is that over a wide bandwidth the gain of an array is at best roughly constant and equal to the half-wave spaced value (Hansen, 1972). This does not include effects of embedded element impedance mismatch with frequency, a phenomenon that further greatly reduces gain. Thus the regularly spaced array is not a candidate for compensation of dispersion. An array with pseudorandom spacing does not experience the appearance of regular grating lobes as frequency is increased. The fraction of power in the sidelobes is roughly constant in a well-designed nonuniformly spaced array, and thus the gain is roughly constant with frequency. Of more importance, however, is the fact that very large numbers of elements are needed to achieve even moderately low sidelobe levels. Thus these types of arrays are not suitable for dispersion compensation either. Arrays of higher gain elements experience, in addition, the dispersion introduced by the elements themselves and are even less suitable.

1.3 ANNOTATED REFERENCE SOURCES

Many textbooks discuss arrays, but the books and digests listed here provide in-depth resources on phased arrays.

[1]Note that "effective length", which is defined as open circuit voltage divided by incident electric field, does not include impedance mismatch, and is therefore useless by itself.

Microwave Scanning Antennas, R. C. Hansen, Ed., 3 vols., Academic Press, 1964, 1966 [Peninsula Publishing, 1985, 442 pp., 400 pp., 422 pp. (Peninsula combined volumes)].

This, the first extensive work on phased arrays, is still quite useful. Volume 1 has a chapter on aperture distributions. Volume 2 includes array theory, and infinite and finite array analysis; probably the first development of the spectral domain analysis technique for arrays. Feeds, frequency scanning, and multiple beams are covered in vol. 3; multiple beams by Butler of matrix fame.

Proceedings of the 1964 RADC Symposium on Electronically Scanned Array Techniques and Applications, report RADC-TDR-64-225, AD-448 481.

Contained here are early papers on phase quantization errors, ferrite and semiconductor phasers, and beam forming matrices.

The Theory and Design of Circular Antenna Arrays, James D. Tillman, University of Tennessee Engineering Experiment Station, 1966, 235 pp.

This treatise on ring arrays and concentric ring arrays applies sequence theory of azimuthal modes, called symmetrical components in electric power work, to the analysis of impedance and pattern. Array scanning is also discussed.

Proceedings of the 1970 NELC Conformal Array Conference, TD-95, Naval Electronics Lab. Center, AD-875 378.

Both ring arrays and cylindrical arrays are treated in papers, both theoretically and for applications.

Phased Array Antennas, A. A. Oliner and G. H. Knittel, Artech, 1972, 381 pp.

This book is a record of the 1970 Phased Array Antenna Symposium held at Polytechnic Institute of Brooklyn. Included are many papers on impedance calculations, blind angles, and so on, and also on practical aspects, such as scan compensation and feeding and phasing.

Theory and Analysis of Phased Array Antennas, N. Amitay, V. Galindo, and C. P. Wu, Wiley–Interscience, 1972, 443 pp.

Arrays of waveguide radiators are the subject here. The spectral domain method is used extensively. Small finite arrays are solved via equations over the modes and elements. This work is one of the first using multimode spectral analysis.

Proceedings of the 1972 NELC Array Antenna Conference, TD-155, 2 Parts, Naval Electronics Lab. Center, AD-744 629, AD-744 630.

Many papers cover array techniques and components; adaptive arrays, and conformal arrays.

Theory and Application of Antenna Arrays, M. T. Ma, Wiley–Interscience, 1974, 413 pp.

Transform analysis and synthesis of fixed beam arrays is covered, along with many general array examples. The effect of ground on arrays represents a significant part of this book.

Conformal Antenna Array Design Handbook, R. C. Hansen, Ed., Naval Air Systems Command, 1982, AD-A110 091.

This report summarizes a decade of Navair-supported work on cylindrical and conical slot arrays, including mutual impedance algorithms.

Antenna Theory and Design, R. S. Elliott, Prentice-Hall, 1981, 594 pp.

This text is an excellent source for waveguide slot array analysis and synthesis. Sidelobe envelope shaping is treated in detail.

The Handbook of Antenna Design, A. W. Rudge, K. Milne, A. D. Olver, and P. Knight, Eds., IEE/Peregrinus, 1983, vol. 2, 945 pp.

This handbook contains chapters on linear arrays, planar arrays, conformal arrays, ring arrays, and array signal processing. Extensive data are included on array analysis and synthesis, including mutual coupling effects.

Proceedings of the 1985 RADC Phased Array Symposium, H. Steyskal, Ed., report RADC-TR-85-171, AD-A169 316.

This symposium record contains papers on microstrip arrays, adaptive arrays, and scan impedance, among others. A second volume has restricted distribution.

Antenna Handbook, Y. T. Lo and S. W. Lee, Van Nostrand Reinhold, 1988.

This handbook contains chapters on array theory, slot arrays, periodic and aperiodic arrays, practical aspects, and multiple-beam arrays.

Antenna Engineering Handbook, R. C. Johnson and H. Jasik, McGraw-Hill, 1993.

This updated edition of an old classic contains chapters on array theory, slot arrays, frequency scan and phased arrays, and conformal arrays.

Phased Array Antenna Handbook, R. J. Mailloux, Artech, 1994, 534 pp.

This specialized handbook covers most array topics, with emphasis on analysis and synthesis. A chapter covers limited scan arrays and time delayed arrays.

Phased Array-Based Systems and Applications, N. Fourikis, Wiley–Interscience, 1997.

This book emphasizes systems aspects of arrays.

1.3.1 Adaptive Antenna Reference Books

Compton, R. T., Jr., *Adaptive Antennas*, Prentice-Hall, 1988.

Hudson, J. E., *Adaptive Array Principles*, IEE/Peregrinus, 1981.

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CHAPTER TWO

Basic Array Characteristics

This chapter is concerned with basic characteristics of linear and planar arrays, primarily with uniform excitation. The theory of, and procedures for, the design of array distributions to produce narrow-beam, low-sidelobe patterns, or shaped beams, are covered in detail in Chapter 3. Impedance effects due to mutual coupling are treated in Chapter 7. Covered here are such parameters as pattern, beamwidth, bandwidth, sidelobes, grating lobes, quantization lobes, and directivity.

2.1 UNIFORMLY EXCITED LINEAR ARRAYS

2.1.1 Patterns

In general, the excitation of an array consists of an amplitude and a phase at each element. This discrete distribution is often called an aperture distribution, where the discrete array is the aperture. The far-field radiation pattern is just the discreet Fourier transform of the array excitation. The array pattern is the product of the isolated element pattern and the isotropic array factor; this is the "forced excitation" problem. To achieve this, the element drives are individually adjusted so that the excitation of each element is exactly as desired. More common is the "free excitation" situation, where the element drives are all fixed, and the element excitations are those allowed by the *scan impedance*. The latter is discussed in detail in Chapter 7. Here the concern will be only with the forced excitation array, where the excitations are constant in amplitude, but may have a scan phase.

A common notation in the antenna literature is used here, where λ is wavelength, d is element spacing, $k = 2\pi/\lambda$, and the angular variable is u. The latter is $u = (\sin\theta - \sin\theta_0)$ where θ_0 is the scan angle. Uniform (equal spacing) is assumed in this chapter; unequally spaced arrays are discussed in Chapter 3. Although it is simpler

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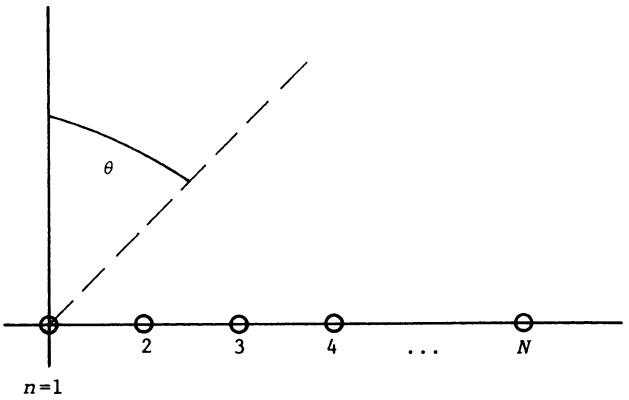


FIGURE 2.1 Linear array geometry.

to have a coordinate system axis in the center of a linear array, complications ensue for even and odd numbers of elements. A more general case starts the coordinate system at one end of the array, as shown in Figure 2.1. The pattern, sometimes called a space factor, is

$$F(u) = \sum A_n \exp[jkd(n-1)u] \tag{2.1}$$

A_n is the complex excitation, which for much of this section will be assumed constant.

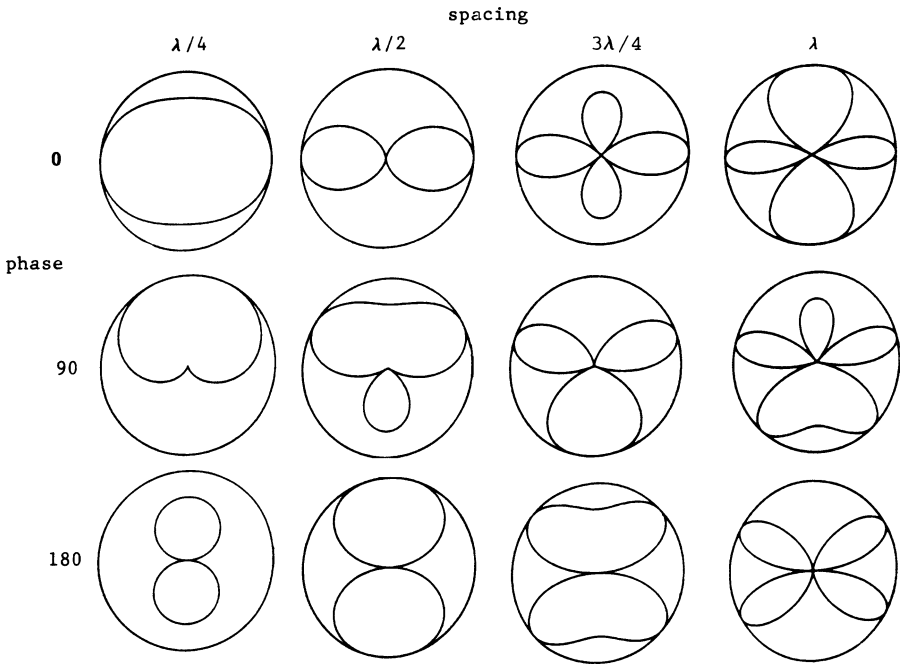


FIGURE 2.2 Two-element array patterns.

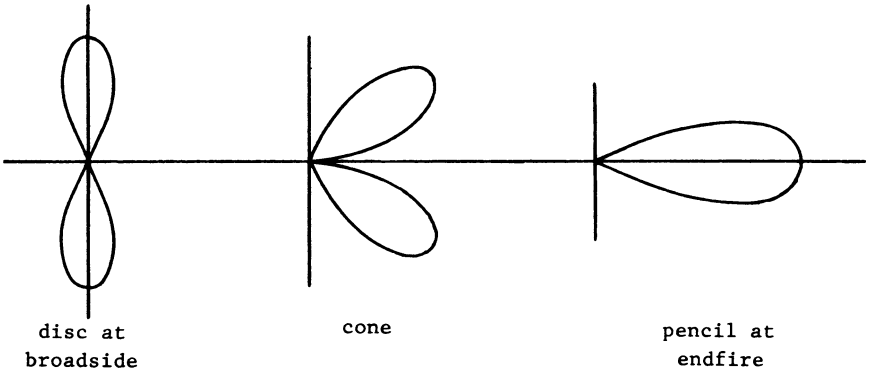


FIGURE 2.3 Linear array beams.

For uniform excitation, the array pattern becomes a simple result, where the exponential in Eq. (2.2) can be discarded, leaving a real pattern times an exponential:

$$F(u) = \exp[j\pi(N-1)u]\frac{\sin\frac{1}{2}Nk\,du}{N\sin\frac{1}{2}k\,du} \tag{2.2}$$

This interelement phase shift is $kd\sin\theta_0$. By varying this phase shift, the beam position can be scanned. Figure 2.2 shows patterns produced by the various spacings and phases (Southworth, 1930).

Many linear arrays are designed to produce a narrow beam. Figure 2.3 depicts how the beam changes with scan. With no scan the narrow beam is omnidirectional around the array axis. As the beam is scanned, this "disk" beam forms into a conical beam as shown in the center sketch. When the 3-dB point gets to 90°, a singular situation occurs. Beyond this scan angle the beam has two peaks, and the "beamwidth" will double as the outside 3-dB points are used. Finally, at endfire a pencil beam results; thus a linear array at broadside yields directivity in one dimension while at endfire it yields directivity in two dimensions. It might be expected as a result that the endfire beamwidth is broader; this will be shown next.

2.1.2 Beamwidth

The half-power points on a uniform array pattern are found by putting $\sin\frac{1}{2}Nk\,du_3/(N\sin\frac{1}{2}k\,du_3) = \sqrt{0.5}$. Figure 2.4 gives the solution of this as a function of the number of elements in the array. For $N \geq 7$, the variation in normalized beamwidth Nu_3 is $<1\%$, and the error is only 5% for $N = 3$. Thus for large arrays, the half-power points are given simply by $\frac{1}{2}Nk\,du_3 = \pm 0.4429$. For a beam scanned at angle θ_0, this gives the 3-dB-beamwidth θ_3 as

$$\theta_3 = \arcsin\left(\sin\theta_0 + 0.4429\frac{\lambda}{Nd}\right) - \arcsin\left(\sin\theta_0 - 0.4429\frac{\lambda}{Nd}\right) \tag{2.3}$$

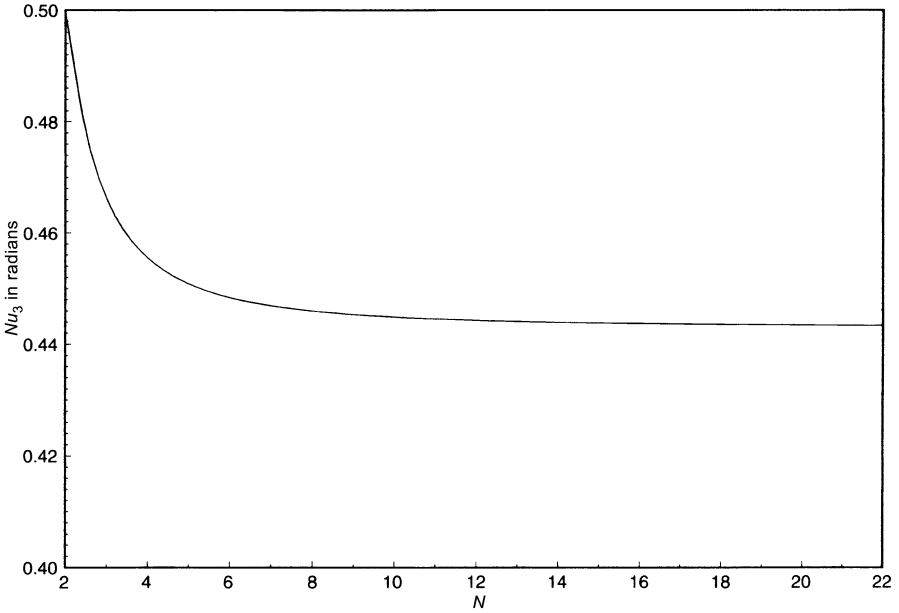


FIGURE 2.4 Normalized beamwidth versus number of elements.

For large N, this reduces to

$$\theta_3 \simeq \frac{0.8858\lambda}{Nd\cos\theta_0} \tag{2.4}$$

The beam collapse near endfire, where the 3-dB point is at 90°, occurs for a scan angle of

$$\theta_0 = \arcsin\left(1 - 0.4429\frac{\lambda}{Nd}\right) \tag{2.5}$$

The beamwidth broadening near endfire is shown in Figure 2.5 for several arrays. For large N, the endfire beamwidth is

$$\theta_3 \simeq 2\sqrt{\frac{0.8858\lambda}{Nd}} \tag{2.6}$$

The accuracy of this is better than 1% for $Nd/\lambda > 4$. The endfire beamwidth is larger than the broadside value by $2.14\sqrt{Nd/\lambda}$. Thus the endfire pencil beam is broader than the broadside pancake beam.

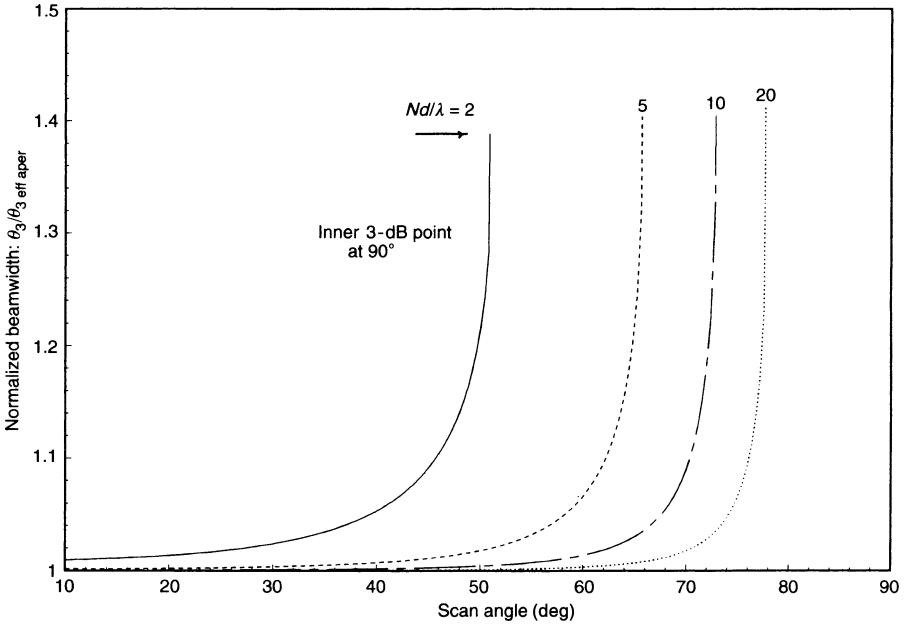


FIGURE 2.5 Beamwidth broadening versus scan angle.

2.1.3 Sidelobes

Uniform array nulls and sidelobes are well behaved and equally spaced. The nulls occur at $u = n/N$, with $n = 1$ to $N - 1$. The peaks of $F(u)$ occur for u that are solutions of $N \tan \pi u = \tan N\pi u$. For large N, this reduces to $\tan N\pi u = N\pi u$; the first solution is $Nu = 1.4303$. A convenient term is sidelobe ratio, which is the ratio of the main beam amplitude to that of the first sidelobe. For large arrays, the sidelobe ratio (SLR) is the same as that for uniform line sources, and is independent of the main beam angle. It is 13.26 dB. For smaller arrays the value of u for the first sidelobe is shown in Figure 2.6. Figure 2.7 gives SLR versus number of elements. Arrays of <8 elements are shown to experience a significant sidelobe ratio degradation. The uniform linear array has a sidelobe envelope that decays as $1/\pi u$, and, as will be discussed in Chapter 3, this decay allows a low-Q and tolerance-insensitive array design.

2.1.4 Grating Lobes

The array pattern equation [Eq. (2.2)] allows the inference that a maximum pattern value of unity occurs whenever $u = N$. If d/λ and θ_0 are chosen properly, only one main beam will exist in "visible" space, which is for $-90 \leq \theta \leq 90°$. Large spacings will produce additional main beams called grating lobes (GL); this is because the larger spacing allows the waves from each element to add in phase at the GL

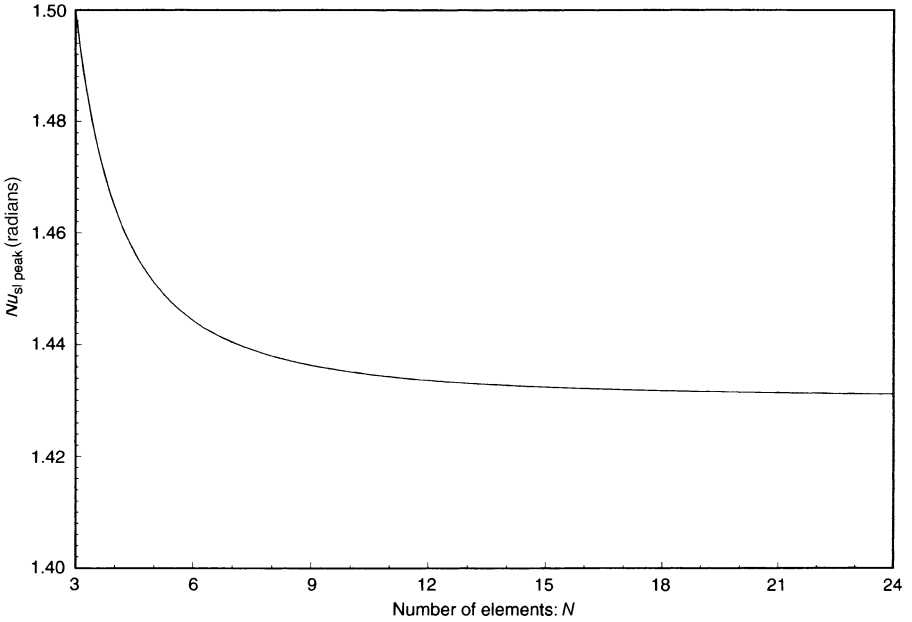


FIGURE 2.6 First sidelobe position versus number of elements.

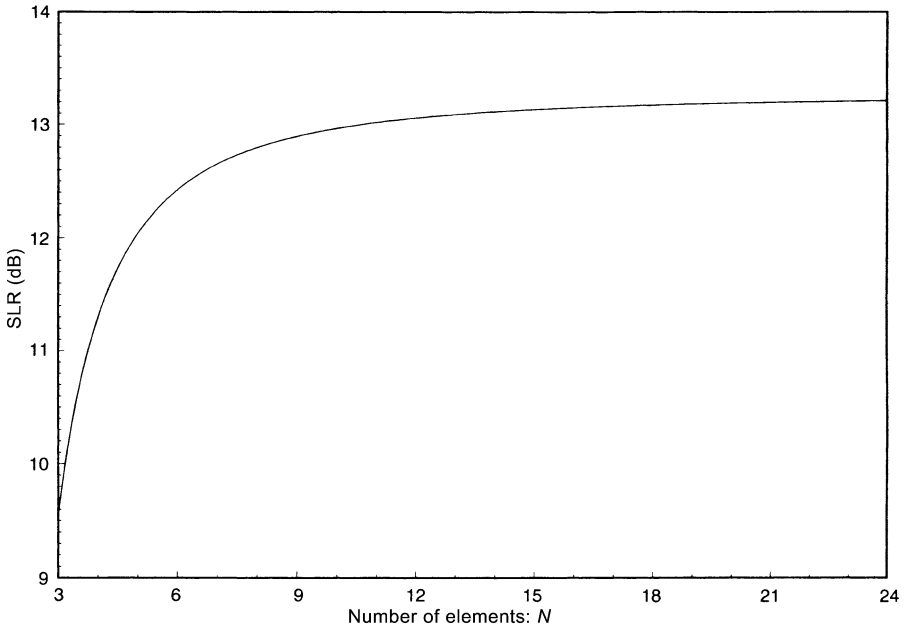


FIGURE 2.7 Sidelobe ratio versus number of elements.

angle as well as at the main beam angle. The equation for grating lobes is easily determined

$$\frac{d}{\lambda} = \frac{n}{\sin\theta_0 - \sin\theta_{GL}} \tag{2.7}$$

For half-wave spacing, a grating lobe appears at -90° for a beam scanned to $+90^\circ$. A one-wavelength spacing allows grating lobes at $\pm 90^\circ$ when the main beam is broadside. Figure 2.8 shows a grating lobe at -45° when the beam is scanned to $\pm 45^\circ$ for a spacing of 0.707λ. The onset of grating lobes versus scan angle is shown in Figure 2.9. The common rule that half-wave spacing precludes grating lobes is not quite accurate, as part of the grating lobe may be visible for extreme scan angles.

For any scan angle it is desirable to keep all of the grating lobes out of visible space. In principle one could adjust the spacing so that the grating lobe amplitude at the edge of visible space is just equal to the sidelobe level. However, the sides of the grating lobe are steep, which means that tight tolerances would be required to avoid an excessive amount of grating lobe. A better scheme puts the pattern null adjacent to the grating lobe at -90°; this comfortably excludes the entire grating lobe. Figure 2.10 depicts part of a pattern, where the main beam is at θ_0 and the grating lobe peak at θ_{GL}. When the $\theta_{GL} - \theta_1$ null is placed at 90°, the grating

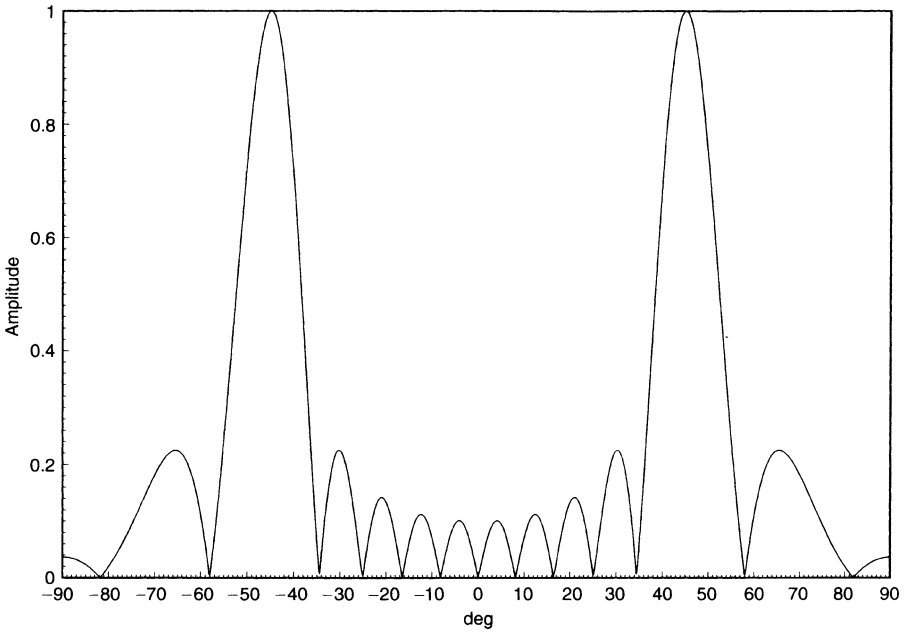


FIGURE 2.8 Array pattern scanned to 45°, $d/\lambda = 0.7071$, $N = 10$.

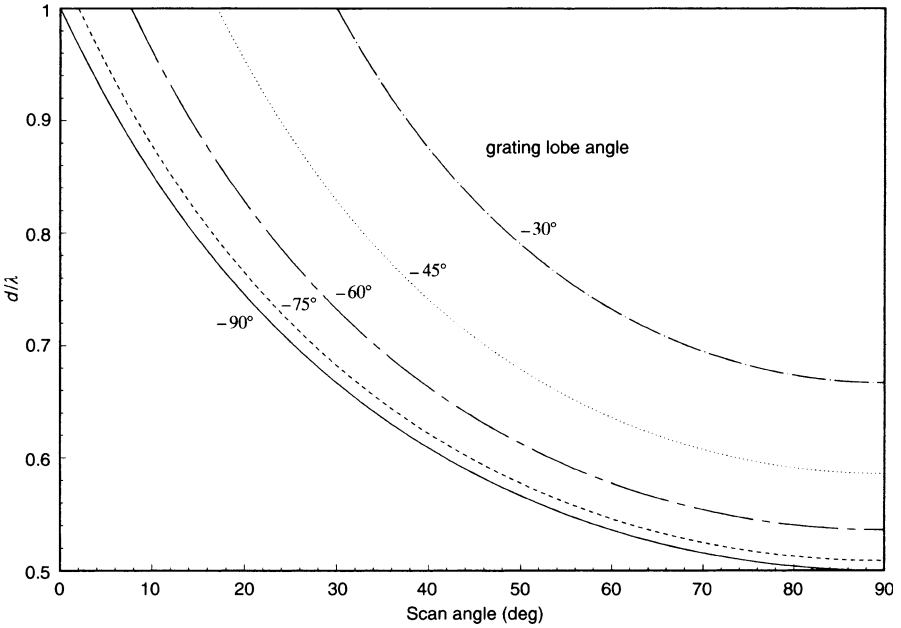


FIGURE 2.9 Grating lobe appearance versus element spacing and scan.

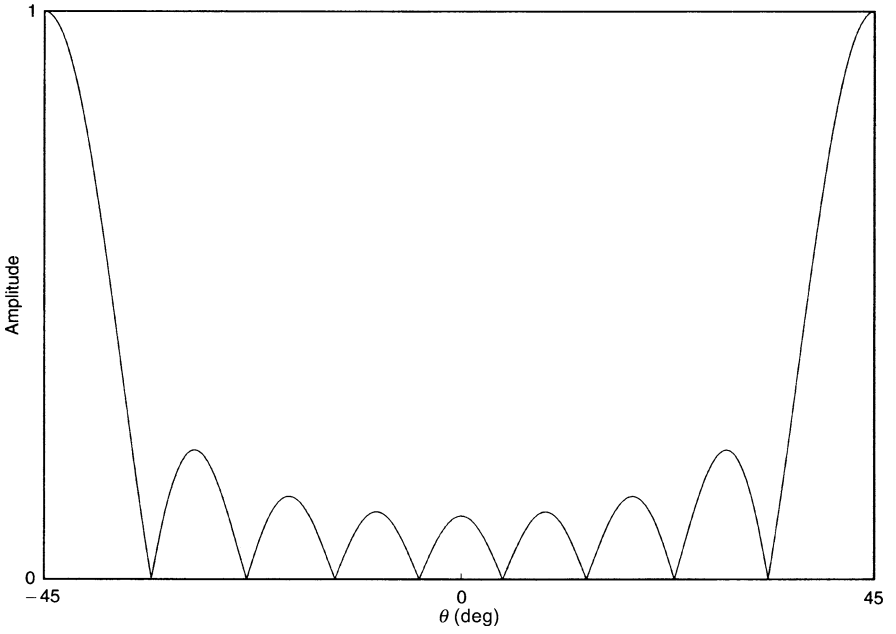


FIGURE 2.10 Array pattern with grating lobe.

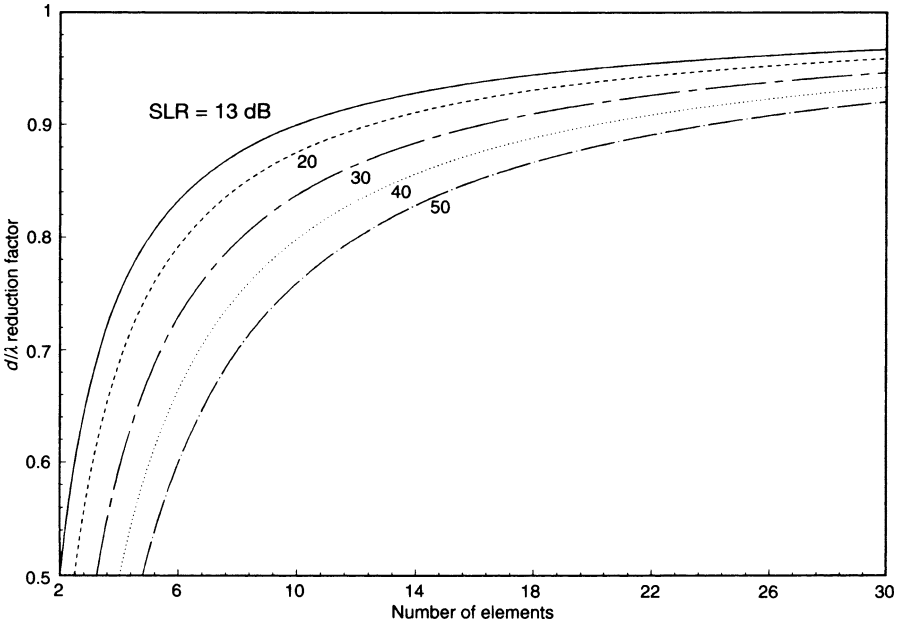


FIGURE 2.11 Element spacing reduction for grating lobe null at 90°.

lobe is in invisible space. The array spacing reduction required to accomplish this is given by[1]

$$\frac{d}{\lambda} = \frac{N - \sqrt{1 + B^2}}{N(1 + \sin \theta_0)} \tag{2.8}$$

This general formula applies to amplitude tapered distributions as well. The parameter B is a taper constant that is discussed in Chapter 3. This spacing reduction for grating lobe null placement is shown in Figure 2.11, where SLR is a parameter. Recall that for the uniform array the SLR = 13.26-dB curve applies.

2.1.5 Bandwidth

Bandwidth (BW) of an array is affected by many factors, including change of element input impedances with frequency, change of array spacing in wavelengths that may allow grating lobes, change in element beamwidth, and so on. When an array is scanned with fixed units of phase shift, provided by phasers, there is also a bandwidth limitation as the position of the main beam will change with frequency. When the array is scanned with time delay, the beam position is independent of frequency to first order. But with fixed phase shift, the beam movement is easily calculated. Beam

[1]This analysis is due to David Munger and Richard Phelan.

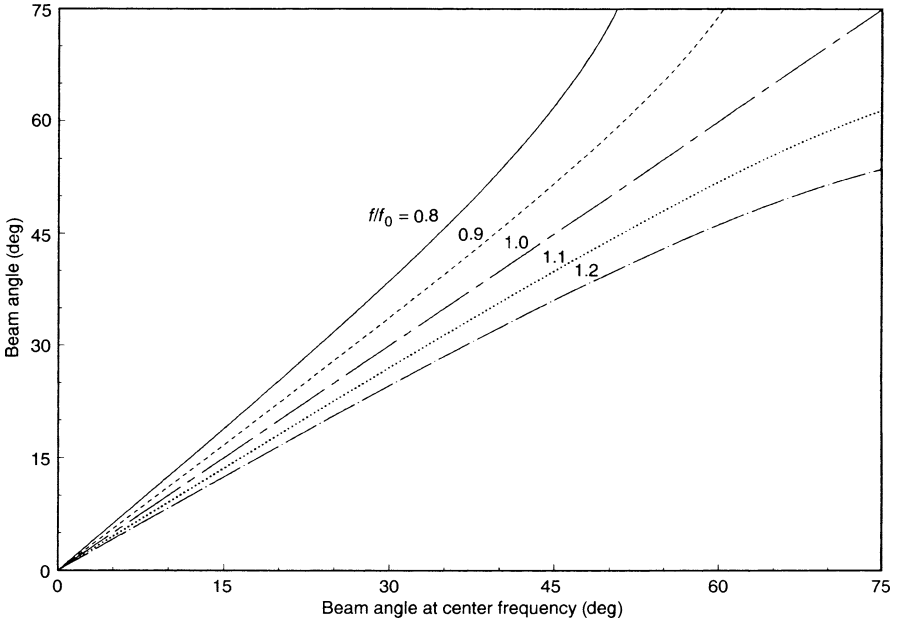


FIGURE 2.12 Beam angle shift with frequency.

angle θ is simply related to scan angle θ_0 by $\sin\theta = (f_0/f)\sin\theta_0$; Figure 2.12 shows this behavior. To calculate steering bandwidth, assume that the main beam has moved from scan angle θ_0 to the 3-dB points for frequencies above and below nominal. Let subscripts 1 and 2 represent the lower and upper frequencies, respectively. Fractional bandwidth is then given by

$$\text{BW} = \frac{f_2 - f_1}{f_0} = \frac{(\sin\theta_2 - \sin\theta_1)\sin\theta_0}{\sin\theta_1 \sin\theta_2} \tag{2.9}$$

For large arrays,

$$\text{BW} \simeq \frac{\theta_3}{\sin\theta_0} \tag{2.10}$$

The BW is then given by

$$\text{BW} \simeq \frac{0.866\lambda}{L\sin\theta_0} \tag{2.11}$$

for a uniform array, and

$$\text{BW} \simeq \frac{\lambda}{L\sin\theta_0} \tag{2.12}$$

for tapered arrays. When the beam angle is 30°, the commonly used formula for fractional BW results:

$$\mathrm{BW} \simeq \frac{2\lambda}{L} \tag{2.13}$$

As a result, long arrays have smaller bandwidth in terms of beam shift at the band edges. See also Section 2.3.2.

2.2 PLANAR ARRAYS

2.2.1 Array Coordinates

In this section, the characteristics of linear arrays are extended to planar arrays. Spherical coordinates with the polar axis normal to the plane of the array are convenient (see Fig. 2.13). This figure shows an array of elements on a rectangular lattice with even number of elements along the x-axis and also along the y-axis. For most pencil beam applications, the array excitations are symmetric, so that the pattern is given by summing over only one-half of the elements along each axis:

$$F(u,v) = \sum_{n=1}^{N/2}\sum_{m=1}^{M/2} A_{nm} \cos\left[\left(n-\frac{1}{2}\right)k\,d_x u\right] \cos\left[\left(m-\frac{1}{2}\right)k\,d_y v\right] \tag{2.14}$$

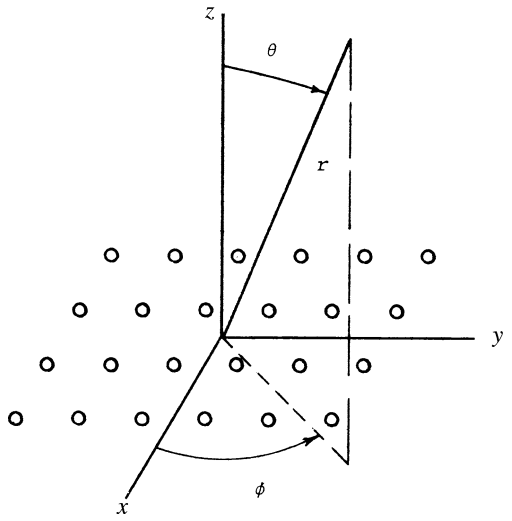


FIGURE 2.13 Spherical coordinate system and rectangular lattice.

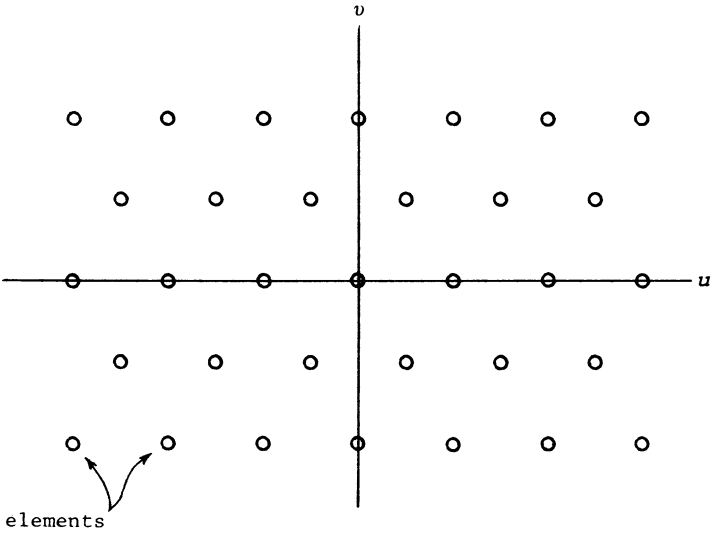


FIGURE 2.14 Triangular lattice.

The direction cosine plane variables are

$$\begin{aligned} u &= \sin\theta\cos\phi - \sin\theta_0\cos\phi_0 \\ v &= \sin\theta\sin\phi - \sin\theta_0\sin\phi_0 \end{aligned} \tag{2.15}$$

with θ_0, ϕ_0 the beam pointing angles. The element spacings are d_x and d_y. The interelement phase shifts needed to scan the beam are

$$\begin{aligned} \Phi_u &= k\,d_x u_0 = k\,d_x \sin\theta_0\cos\phi_0 \\ \Phi_v &= k\,d_y v_0 = k\,d_y \sin\theta_0\sin\phi_0 \end{aligned} \tag{2.16}$$

A triangular lattice, as shown in Figure 2.14, is often used, as it allows slightly larger element spacing without appearance of grating lobes.

2.2.2 Beamwidth

For scanning in a principal plane, the beamwidth is just that given before in Eq. (2.3). Cross-plane beamwidth is more complex, as both the array length in the principal plane, L, and the array width, W, affect it. Taking principal plane scan, with $\phi_0 = 0$ and $\theta = \theta_0$, the direction cosine variables are

$$u = \frac{L}{\lambda}\sin\theta_0(1 - \cos\phi) \quad v = \frac{W}{\lambda}\sin\theta_0\sin\phi \tag{2.17}$$

The half-power values u_3 and v_3 are solutions of

$$\operatorname{sinc} \pi u_3 \operatorname{sinc} \pi v_3 = \frac{\pi^2 u_3 v_3}{\sqrt{2}} \tag{2.18}$$

The cross-plane beamwidth is found from

$$\phi_{3y} = 2 \arcsin(\sin \theta_0 \sin \phi_3) \tag{2.19}$$

where the $\sin \theta_0$ factor represents a projected aperture. For large arrays, an excellent approximation is

$$\phi_{3y} \simeq 2 \sin \theta_0 \sin \phi_3 \tag{2.20}$$

The array aspect ratio W/L is a convenient parameter, as the root ϕ_{3y} can be determined as a function of $(W/\lambda) \sin \theta_0$. The cross-plane beamwidth can also be written in terms of v_3:

$$\frac{W}{\lambda} \phi_{3y} \simeq 2v_3 \tag{2.21}$$

Figure 2.15 shows beamwidth in the scan plane as a function of array length and scan angle; Figure 2.16 gives cross-plane beamwidth as a function of array aspect ratio and normalized scan parameter. It can be seen that for a normalized scan parameter of 1 or greater, the cross-plane beamwidth is close to the nominal value.

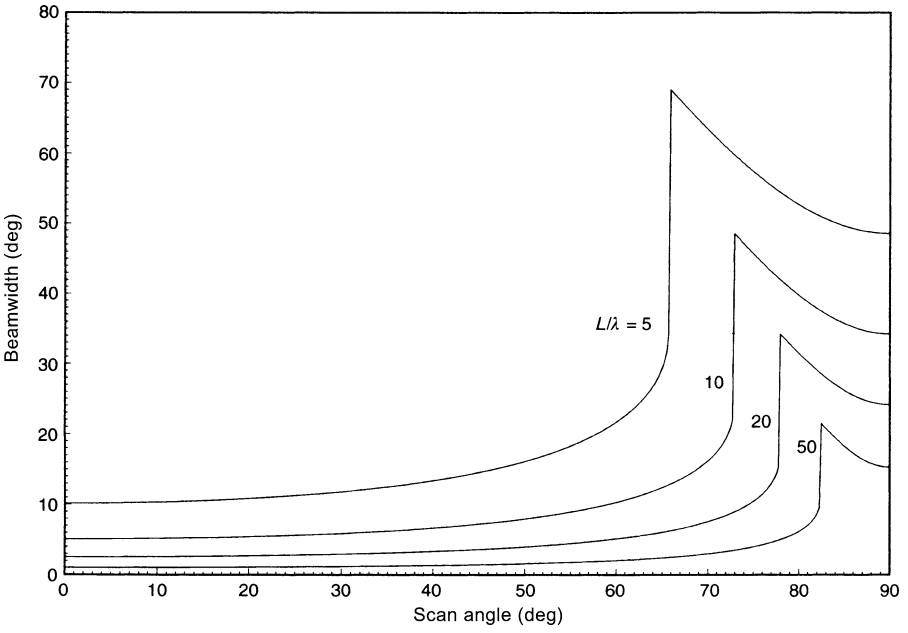


FIGURE 2.15 Uniform square aperture beamwidth in scan plane.

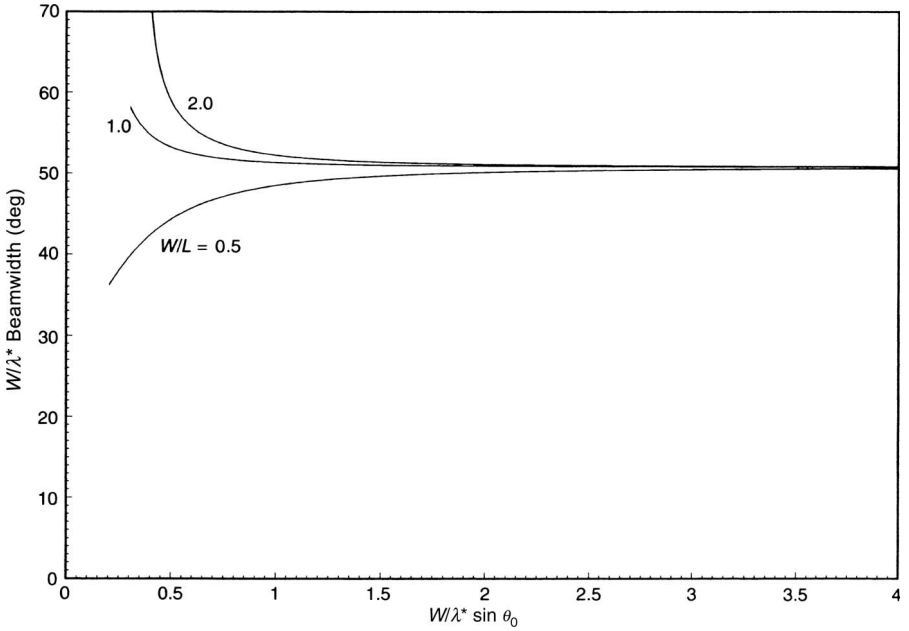


FIGURE 2.16 Uniform rectangular aperture beamwidth in cross-plane.

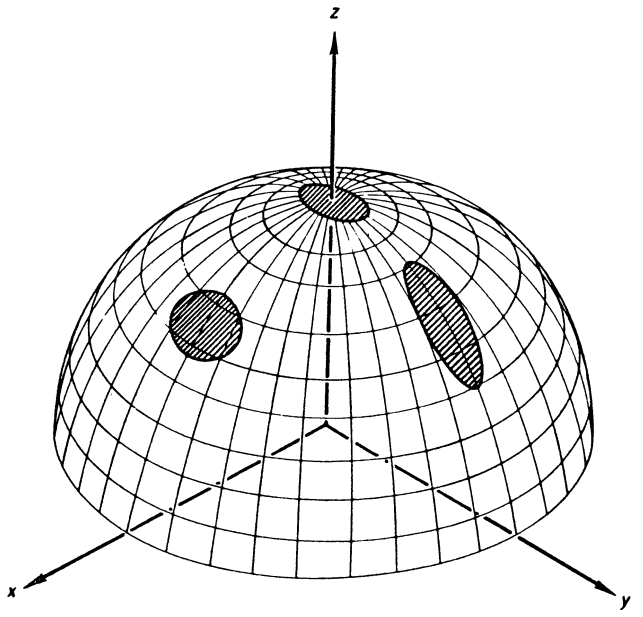


FIGURE 2.17 Beam shape versus scan position for a pencil beam.

As the beam is scanned, its shape changes as the projected aperture of the array changes. For rectangular arrays, the 3-dB beam contour is approximately elliptical. However, for scans not in the principal planes, the combination of projected aperture width and length and the scan angles results in an elliptical beam whose major diameter is generally not oriented along the scan plane or principal planes (Elliott, 1966). Since the area of an ellipse is proportional to the product of major and minor diameters, the 3-dB beam area, called areal beamwidth, is proportional to the product of the major- and minor-axis beamwidths. The areal beamwidth is to first order independent of azimuth angle ϕ, although the beam shape and orientation of major axis may change with θ. Figure 2.17 shows several beams as they change with scan.

2.2.3 Grating Lobes: Rectangular Lattice

A rectangular lattice with scanning in either principal plane behaves exactly like a linear array, as described in Section 2.2. For other scan angles the situation is less simple. The u,v-plane, sometimes called the direction cosine plane, was developed by Von Aulock (1960) and is extremely useful for understanding grating lobe behavior. The grating lobe positions can be plotted in the u,v-plane; they occur at the points of an inverse lattice, that is, the lattice spacing is λ/d_x and λ/d_y (see Fig. 2.18). All real angles, representing visible space, are inside or on the unit circle. The latter represents $\theta = 90^\circ$. Angles outside the unit circle are "imaginary", or in invisible space. When

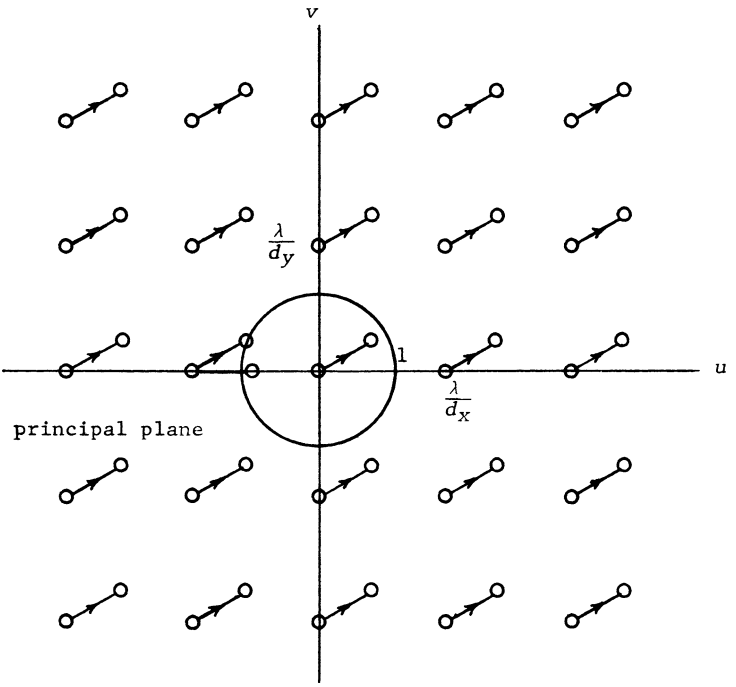


FIGURE 2.18 Direction cosine plane.

the main beam is scanned, the origin of the u,v plot moves to a new value, and all grating lobes move correspondingly. However, the unit circle remains fixed. Thus, for scan in the u-plane ($\phi = 0$) the main beam point moves toward $+1$ for $\theta > 0$, and all GL points move the same amount. When the GL just outside the unit circle moves enough to intersect the unit circle, that GL becomes visible. Thus the distance between the broadside GL and the unit circle must be no larger than unity, or the GL will not intersect the unit circle before the main beam stops at the right side of the unit circle. The result is

$$u_{\mathrm{GL}} \le \frac{\lambda}{d_x} - 1 \qquad \text{or} \qquad \frac{d_x}{\lambda} \ge \frac{1}{1 + \sin\theta} \tag{2.22}$$

The inequality is such that exceeding the equality allows a grating lobe to appear. Scan along the v-axis ($\phi = 90^\circ$) is analogous:

$$\frac{d_y}{\lambda} \ge \frac{1}{1 + \sin\theta} \tag{2.23}$$

If the array spacings are sufficiently large, scan to any pair of angles may produce one or more grating lobes. The limiting cases are of interest, and these are of two types: diagonal GL, and tangential GL. The diagonal plane scan sketched in Figure 2.19 produces a GL when the diagonal point intersects the unit circle normally. The u,v values that allow this are

$$\begin{aligned} u_{\mathrm{GL}} &= \frac{SQ - 1}{SQd_x/\lambda} & v_{\mathrm{GL}} &= \frac{SQ - 1}{SQd_y/\lambda} \\ \sin\theta_{\mathrm{GL}} &= SQ - 1 & \tan\phi_{\mathrm{GL}} &= \frac{d_x}{d_y} \end{aligned} \tag{2.24}$$

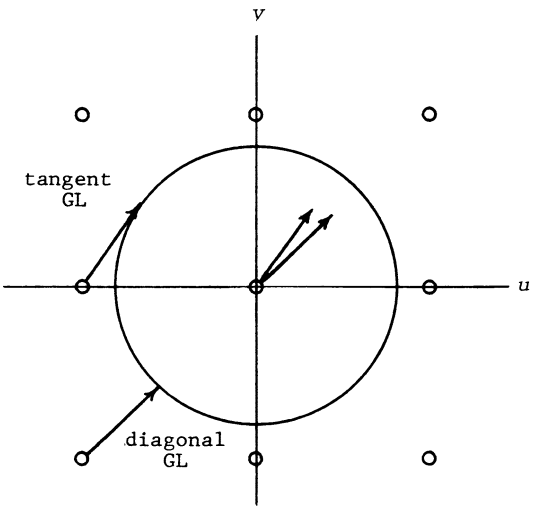


FIGURE 2.19 Grating lobe limiting cases.

where $SQ = \sqrt{dx^2/\lambda^2 + dy^2/\lambda^2}$. This diagonal plane GL occurs only for $SQ \geq 2$. In general, a grating lobe appears whenever the propagation constant is real; at the transition it is zero:

$$\frac{\beta}{k} = \sqrt{1 - (u - n\lambda/d_x)^2 - (v - m\lambda/d_y)^2} \tag{2.25}$$

Scan in principal planes corresponds to $n = 0$ or $m = 0$. The diagonal case is for $n = 1$, $m = 1$. Sufficiently large spacings allow several GL to appear, with the result that the propagation constant is real for larger n and/or larger m. The tangential limiting case occurs when the GL is located on the u-axis just left of the unit circle. A diagonal scan can place the GL just tangent to the circle (see Fig. 2.19). Only d_x is involved here, and the limiting values are

$$\begin{aligned} u_{\mathrm{GL}} &= \frac{1 - d_x^2/\lambda^2}{d_x/\lambda} & v_{\mathrm{GL}} &= \sqrt{1 - d_x^2/\lambda^2} \\ \sin\theta_{\mathrm{GL}} &= \frac{\sqrt{1 - d_x^2/\lambda^2}}{d_x/\lambda} & \sin\phi_{\mathrm{GL}} &= \frac{d_x}{\lambda} \end{aligned} \tag{2.26}$$

The minimum value of d_x/λ is $1/\sqrt{2}$. The v tangent case is analogous, as might be expected, with u and v interchanged and d_x and d_y interchanged. For unrestricted scan angles, the principal plane GL appears first, controlled by the smaller of λ/d_x and λ/d_y. The u,v-plane not only gives an excellent physical picture of GL occurrence, but also allows the formulas to be derived easily. In summary, if $d_x/\lambda < 1/(1 + \sin\theta)$ and $d_y/\lambda < 1/(1 + \sin\theta)$, no grating lobes will exist for the rectangular lattice.

2.2.4 Grating Lobes: Hexagonal Lattice

Isosceles triangular lattices are sometimes used; see the brick waveguide array in Chapter 7. A commonly used special case is the equilateral triangular lattice, or regular hexagonal lattice. Let the elements be all equally spaced by $2d$, and let the x-axis go through a row of elements spaced this distance apart. Then $d_x = d$ and $d_y = \sqrt{3}d/2$, where d_x and d_y are half the distance to the next element along the x- or y-axis. The hexagonal array is simply analyzed by breaking it into two interlaced rectangular lattice arrays. The pattern is then the sum of the two array patterns. The grating lobe locations for no beam scan are simply

$$u_{\mathrm{GL}} = \pm\frac{n\lambda}{2d} \qquad v_{\mathrm{GL}} = \pm\frac{m\pi}{\sqrt{3}d} = \pm\frac{m\lambda}{2d_y} \tag{2.27}$$

The inverse lattice in the u,v-plane is shown in Figure 2.20; the GL points are equidistant by λ/d. Scan in the u-plane gives the same results as for the rectangular lattice.

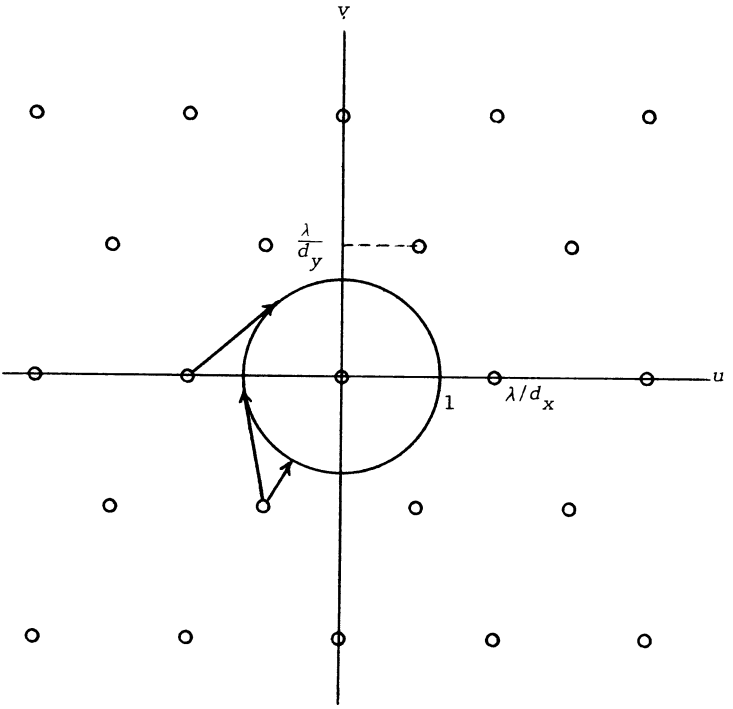


FIGURE 2.20 Direction cosine plane.

Diagonal plane scan, approaching the unit circle normally, yields a GL for

$$\begin{aligned} u_{\mathrm{GL}} &= \frac{1 - d/\lambda}{2d/\lambda} \qquad v_{\mathrm{GL}} = \frac{\sqrt{3}(1 - d/\lambda)}{2d/\lambda} \\ \sin\theta_{\mathrm{GL}} &= \frac{1 - d/\lambda}{d/\lambda} \qquad \phi_{\mathrm{GL}} = \frac{\pi}{3} \end{aligned} \tag{2.28}$$

The minimum value of d/λ for this lobe to appear is $1/\sqrt{2}$. Now there are two tangent cases, where the GL normally on the u-axis becomes tangent to the unit circle, and where the normally diagonal GL becomes tangent to the unit circle. The first of these two cases occurs for

$$\begin{aligned} u_{\mathrm{GL}} &= \frac{1 - d^2\lambda^2}{d/\lambda} \qquad v_{\mathrm{GL}} = \sqrt{1 - d^2/\lambda^2} \\ \sin\theta_{\mathrm{GL}} &= \frac{\sqrt{1 - d^2/\lambda^2}}{d/\lambda} \qquad \sin\phi_{\mathrm{GL}} = \frac{d}{\lambda} \end{aligned} \tag{2.29}$$

The minimum value of d/λ for this GL to appear is $1/\sqrt{2}$.

Tangency from the diagonal point is somewhat more complicated, and occurs for

$$u_{GL} = \frac{\sqrt{1 - d^2/\lambda^2}}{2}\left(\frac{\sqrt{1 - d^2/\lambda^2}}{d/\lambda} + \sqrt{3}\right)$$
$$v_{GL} = \frac{\sqrt{1 - d^2/\lambda^2}}{2}\left(\frac{\sqrt{3(1 - d^2/\lambda^2)}}{d/\lambda} - 1\right) \tag{2.30}$$
$$\sin\theta_{GL} = \sqrt{\frac{1 - d^2/\lambda^2}{d/\lambda}} \qquad \phi_{GL} = \frac{\pi}{3} - \arcsin\frac{d}{\lambda}$$

Comparing the square lattice and the hexagonal lattice, both have grating lobe appearance when $\sin\theta = \lambda/d - 1$. The square lattice element area is d^2, while that for the hexagonal lattice is $2d^2/\sqrt{3}$. Thus the hexagonal lattice requires only 0.866 as many elements to give the same grating lobe free area. This results in a saving of 15% in number of elements (Sharp, 1961; Lo and Lee, 1965).

2.3 BEAM STEERING AND QUANTIZATION LOBES

Interelement phase shift is necessary to provide beam scan, the devices that produce this phase shift are called *phasers*. Row and column phasing is the simplest, even for circular arrays. Thus each phaser is driven by a command to produce a specified x- and a specified y-axis phase. The steering bits affect the precision of the beam steering, control bandwidth, and produce phase quantization lobes. Each of these will be discussed.

2.3.1 Steering Increment

The smallest steering increment is related to the smallest phase bit (Hansen, 1983).

$$\frac{\theta_{err}}{\theta_3} = \frac{1}{2^M 1.029} \tag{2.31}$$

The factor 1.029 is the circular distribution normalized beamwidth of Table 4.1, and M is the number of bits. Table 2.1 gives the steering increment divided by the half-power beamwidth θ_3. Note that in this case the largest bit is π. Thus 4 bits gives a steering least count of 0.061 beamwidth, or roughly $1/16$ of a beamwidth.

TABLE 2.1 Steering Precision Phase Bits

Phaser Bits	θ_{err}/θ_3
3	$0.121 \sim 1/8$
4	$0.0607 \sim 1/16$
5	$0.0304 \sim 1/32$

Adding a bit, of course, decreases the steering increment by a factor of 2. Adequate steering precision will often be provided by 4 bits. This phaser would be located at each element; the phases would be 180, 90, 45, and 22.5°.

2.3.2 Steering Bandwidth

For an electronic scanning array, bandwidth is usually defined such that at the band edges the main beam has moved through one-half of the beamwidth, that is, the gain has dropped 3 dB. For an array scanned by phase alone, the bandwidth is given approximately by $\mathrm{BW} = \theta_3/\sin\theta_0$. Using the rough value of $\theta_3 \simeq \lambda/L$, for a 30° scan the bandwidth is given by $\mathrm{BW} \simeq 2\lambda/L$, an oft-quoted, but imprecise, result.

Use of time delay for some phaser bits will increase the bandwidth; the amount of the increase will be determined here. Let the center frequency be represented by f_0. Let the upper and lower band edges be represented by f_2 and f_1. Similarly, let the sine of the center frequency main beam angle at the scan limit be S_0 and at the band edges be S_2 and S_1. Further, let N be the number of time delay bits. The frequency excursions that define the bandwidth are now given by

$$\begin{aligned}(2^N - 1)f_1S_0 + f_0S_0 &= 2^N f_1 S_1 \\ (2^N - 1)f_2S_0 + f_0S_0 &= 2^N f_2 S_2\end{aligned} \tag{2.32}$$

For small bandwidths, the band edges may be approximated by $f_2 = f_0 + \Delta_2$ and $f_1 = f_0 - \Delta_1$. The normalized bandwidth is then $(\Delta_1 + \Delta_2)$. For large scan angles, the expressions for S_2 and S_1 can also be simplified. However, for the small scan angle used here, this simplification becomes inaccurate as A approaches unity. Thus the exact equations were solved.

It is convenient to normalize the bandwidth by a factor $\sin\theta_0/\theta_3$. When all bits are phase, the normalized bandwidth is 1. Changing each bit from phase to time delay roughly doubles the bandwidth factor, and the bandwidth. Note that the bandwidth increases faster than 2^M, as it must since when all bits are time delay the steering bandwidth is infinite (see Table 2.2). However, the increase is less for larger arrays.

TABLE 2.2 Time Delay Bits versus Bandwidth, $\theta_0 = 30°$, $\theta_3 = 1°$

Normalized	
Bits of Time Delay	Bandwidth Factor
0	1
1	2.002
2	4.016
3	8.124
4	17.016

2.3.3 Time Delay Deployment

Many phased arrays utilize a corporate feed, and for wideband arrays, the question arises on where to best put the necessary time delays. First, the corporate feed without sub-arrays is considered. The gold standard uses at each element a multibit phaser plus a delayer, with the delays staircased to correspond with the scan angle. See Figure 2.21. This approach, although it provides wideband scanning, is seldom used as it requires at each element a delay switchable from λ to $kL \sin \theta$. Note that the time delay is neither true nor false. Conversion from delay length to time is trivial:

$$L/\lambda = t_{\text{nanosec}}\, f_{\text{gigahertz}} \tag{2.33}$$

A deployment sometimes suggested places the smallest delay bit (either 0 or λ) at each of the first level of combiners, just after the elements. The next delay bit (either 0 or 2λ) is placed at each of the second level of combiners, and so on. However a careful evaluation of this topology shows that delay bit values of 0 or $n\lambda$ will not provide frequency independent beam steering (Hansen, 2002).

A variant on this second topology averages the delays at each combiner over the scan range; then the phaser bits are optimized for that scan range. Beam steering is excellent, but some high (-6 dB) quantization lobes occur.

Still another deployment that provides wideband scanning utilizes odd-numbered elements with zero delay and even-numbered elements with phase of $kd \sin \theta$, where d is the element spacing, provided by a phaser at the element. At the first combiner level even arms have zero delay while odd arms have a delay that includes the previous phaser bits and all the smaller delay bits. Figure 2.22 shows an example for a 16 element array with a scan of 45°. This topology uses only half of the delayers used in the gold standard; the number of phasers is the same.

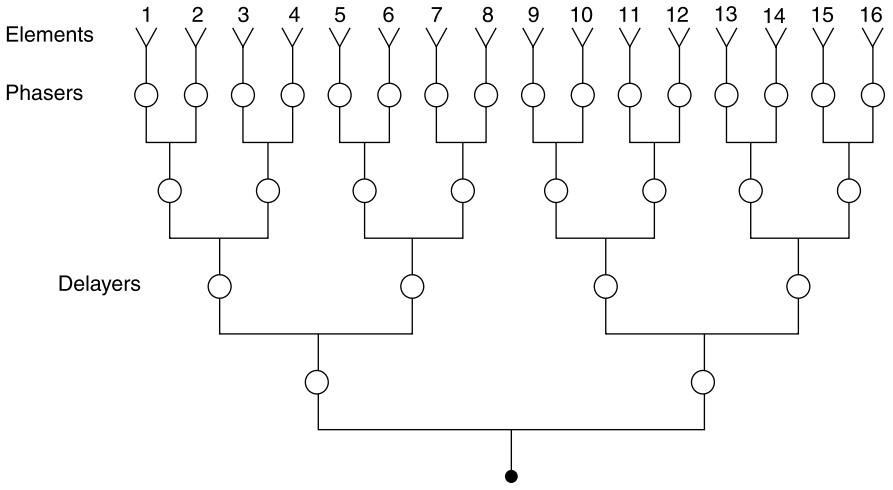


FIGURE 2.21 A 16-element corporate feed.

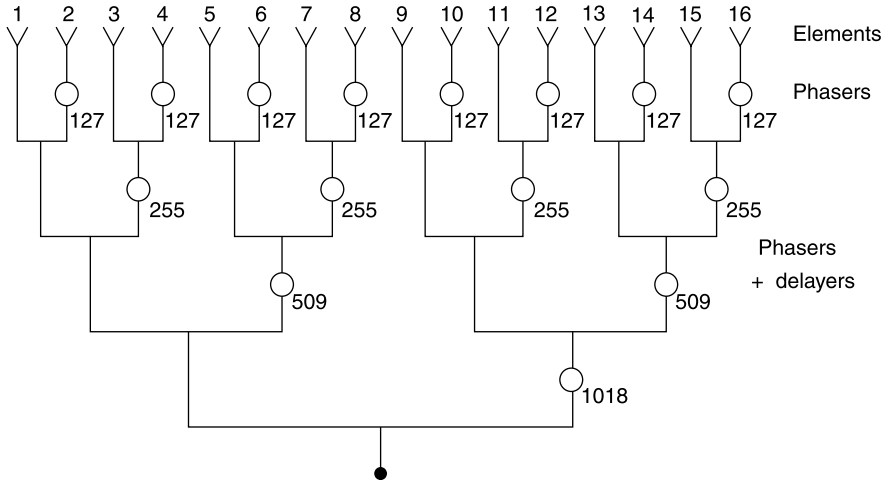


FIGURE 2.22 The phasers and delayers for a 45° scan.

Turning now to an array of sub-arrays, as sketched in Figure 2.23, each element has a phaser; delayers are located in the corporate feed. The delay times are staircased, and the perfect steering bandwidth obtained is the phaser bandwidth multiplied by the number of sub-arrays. For example, phaser bandwidth is approximately $\theta_3/\sin\theta_0$. An array with 16 sub-arrays would have 16 times that steering bandwidth. A caution is that for sub-arrays of more than three elements quantization lobes may appear, and pseudo-randomization may not decollimate all of the QL (Hansen and Charlton, 1999).

2.3.4 Phaser Quantization Lobes

Most phasers are now digitally controlled, whether the intrinsic phase shift is analog or digital. Such phasers have a least phase, corresponding to one bit. An M-bit phaser has phase bits of $2\pi/2^M$, $2\pi/2^{M-1}, \ldots, \pi$. The ideal linear phase curve for electronic scanning is approximated by a stair-step phase, producing a sawtooth error curve;

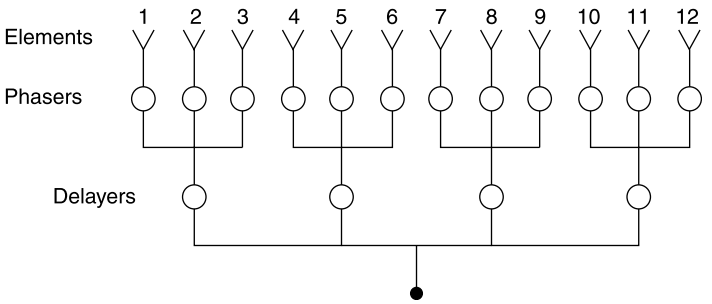


FIGURE 2.23 The average delays at the third combiner level for a 45° scan.

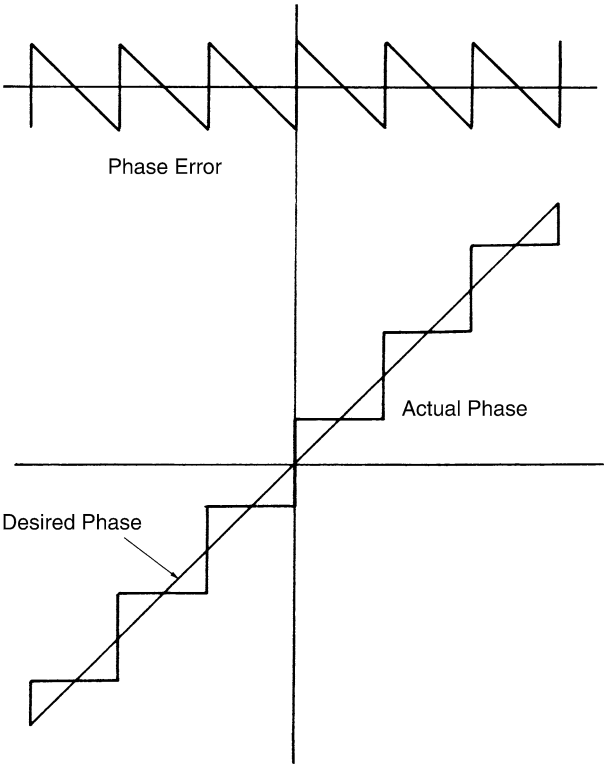


FIGURE 2.24 Phase quantization errors.

see Figure 2.24. Since the array is itself discrete, the positions of the elements on the sawtooth are important. There are two well defined cases. The first case is when the number of elements is less than the number of steps. In this case the phase errors assume a random nature. This can be evaluated by approximating the phase variance by $1/3$ of the peak sawtooth error of $\pi/2^M$. The variance is

$$\sigma^2 = \frac{\pi^2}{3 \cdot 4^M} \tag{2.34}$$

and the RMS sidelobe level is σ^2/G. For uniform excitation of the array, the RMS sidelobe level is σ^2/N. Figure 2.25 gives RMS sidelobe level due to random phaser quantization error, for broadside radiation. At scan angle θ_0 the main beam is reduced by approximately $\cos^2 \theta_0$, so that the RMS sidelobe level with respect to the main beam is reduced. A modest gain decrease accompanies the increased sidelobes:

$$\frac{G}{G_0} \simeq 1 - \sigma^2 \tag{2.35}$$

This random gain decrease is given in Table 2.3.

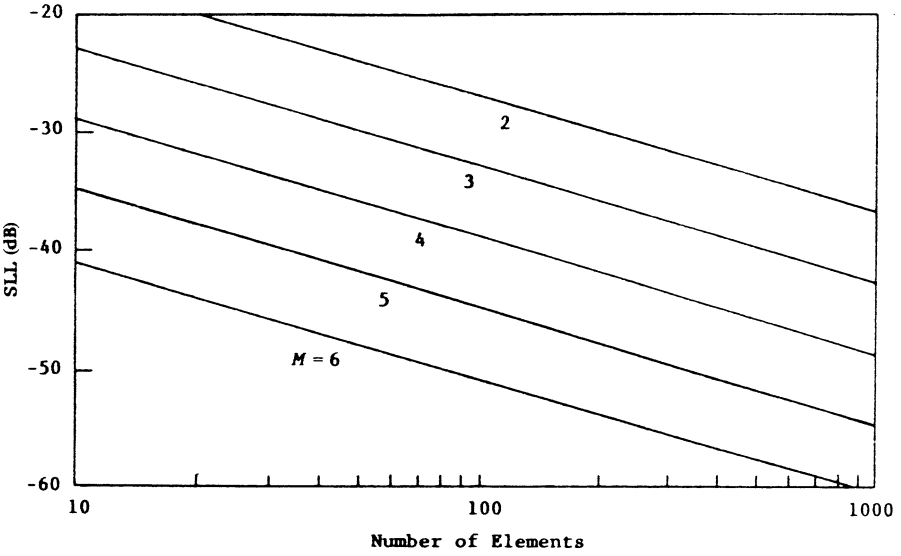


FIGURE 2.25 The RMS sidelobe level from phaser quantization.

TABLE 2.3 Phaser Gain Decrease for Small Arrays

M	Gain Decrease (dB)
2	1.000
3	0.229
4	0.056
5	0.014

The second case has two or more elements per phase step, and the discrete (array) case is approximated by a continuous case. This quantization produces a set of lobes called quantization lobes (QL), that have predictable amplitudes and positions. An N-element array has an end-to-end phase of $(N-1)kd \sin \theta_0$. The number of elements per phase step, with M bits of phase, is J:

$$J = \frac{N}{(N-1)2^M(d/\lambda)\sin \theta_0} \tag{2.36}$$

This allows the largest scan angle for which $J \geq 2$ to be found. For large arrays, N approximately cancels, leaving $\sin \theta_0 \simeq 1/[(d/\lambda)2^{M+1}]$. For half-wave element spacing, the maximum angles are given in Table 2.4. Larger angles produce pattern errors that are more complex. For J between 1 and 2, there is a transition region between the random sidelobes regime ($J < 1$) discussed and the QL regime.

The QL angle θ_q is governed by the step width W:

$$\frac{W}{\lambda} = \frac{1}{2^M \sin \theta_0} \tag{2.37}$$

TABLE 2.4 Maximum Scan Angle for Well Defined QL

M bits	$\theta_{\max}$ (deg)
1	30
2	14.48
3	7.18
4	3.58
5	1.79

This gives a peak sawtooth error of $\beta = \pi/2^M$, as mentioned above. The QL amplitudes are given by $\text{sinc}(\beta \pm i\pi)$, where $i = 0$ gives the main beam. The first two QL and the main beam are given by Miller (1964):

$$\begin{aligned} &\text{Main Beam} && \text{sinc}\,\beta \\ &\text{First QL} && \frac{\sin\beta}{\pi \pm \beta} \\ &\text{Second QL} && \frac{\sin\beta}{2\pi \pm \beta} \end{aligned} \tag{2.38}$$

When $\beta = \pi/2$, the main beam and first QL are equal, clearly a bad case. But this only occurs for one-bit phasers, an extreme case. Table 2.5 gives lobe amplitudes versus number of phaser bits.

Gain decrease is given approximately by the main beam decrease (Allen et al., 1960):

$$\frac{G}{G_0} \simeq \text{sinc}^2\,\beta \tag{2.39}$$

These data allow the array designer to make an intelligent trade on phaser bits.

For rectangular lattices, the grating lobe appearances derived for linear arrays are still valid along the u- and along the v-axes. These intersect to form bands, where the grating lobes tend to appear, as shown in Figure 2.26. The discrete spacing of the grating lobes depends, of course, on the scan angles. As always, only grating lobes on bands within the unit circle represent visible QL.

TABLE 2.5 Phaser Quantization Lobe Amplitudes

M	Main Lobe (dB)	QL_1 (dB)	QL_2 (dB)
1	−3.92	−3.92	−13.46
2	−0.912	−10.45	−14.89
3	−0.224	−17.13	−19.31
4	−0.056	−23.58	−24.67
5	−0.014	−29.84	−30.38
6	0	−35.99	−36.26

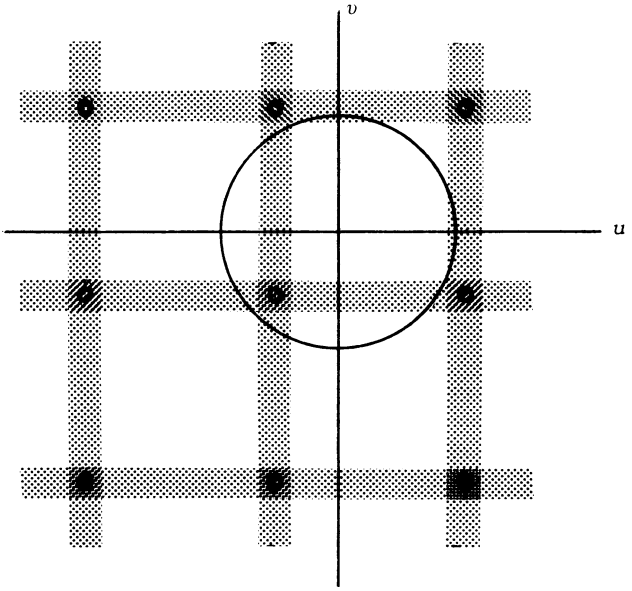


FIGURE 2.26 Quantization lobe strips for square lattice.

When a hexagonal lattice is used with a hexagonal phaser network, which is a two-dimensional structure, the QL bands follow the principal diagonals. In most cases, only two bands would cross the unit circle. However, hexagonal networks require many circuit board crossovers, and hence are seldom used. A more common scheme uses row and column phasers: these produce QL bands parallel to the u- and v-axes (Nelson, 1969). Figure 2.27 shows a hexagonal lattice with row and column phasers. Additional QL bands can occur in this situation. In summary, the hexagonal lattice offers grating lobe advantages, but no quantization lobe advantages when row-column (u,v) phasers are used.

2.3.5 Sub-array Quantization Lobes

An array composed of contiguous sub-arrays, when scanned, has a phase consisting of stair-steps, with one step over each sub-array. This, just as in the case of digital phasers, produces quantization lobes, except that a sub-array has at least two elements. So the QL are always deterministic here. Call the sawtooth width (sub-array width) W, and the peak sawtooth error β, as before. For large arrays, the discrete sub-array may be replaced by a continuous aperture, but with the sawtooth phase. Consider N equal sub-arrays, and with uniform excitation. The pattern is

$$F(u) = \frac{1}{2}\sum_{n=1}^{N} \exp\left\{ j\left[\left(\frac{1}{2}k\,du - N\beta\right)p + \beta(2n - 1 - N)\right]\right\} dp \qquad (2.40)$$

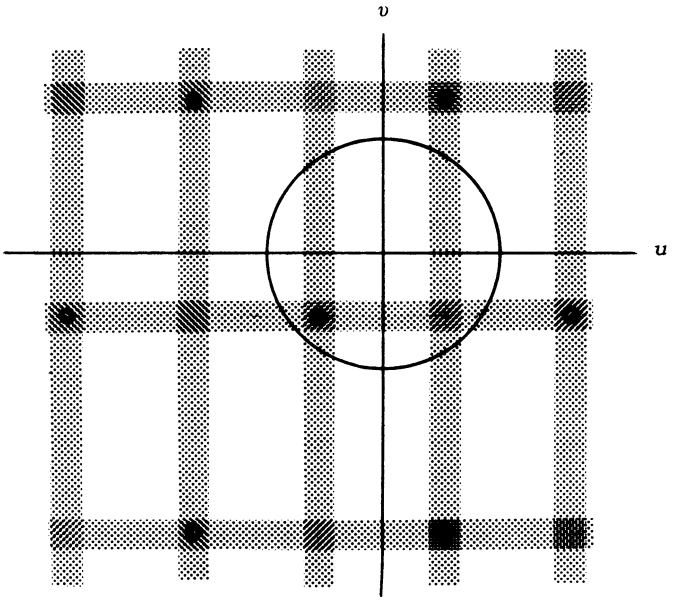


FIGURE 2.27 Quantization lobe strips for hexagonal lattice.

where the integration is over each step, and

$$u = (\sin\theta_0 - \sin\theta) \qquad N_1 = \frac{2n - N}{N} \qquad \text{and} \qquad N_2 = \frac{2n - 2 - N}{N}$$

Convenient variables are

$$\begin{aligned} w &= \frac{W}{\lambda}(\sin\theta_0 - \sin\theta) \quad \beta = \pi v \\ v &= \frac{W}{\lambda}\sin\theta \quad v_0 = \frac{W}{\lambda}\sin\theta_0 \end{aligned} \tag{2.41}$$

The integration and summation can both be performed in closed form, with the result

$$F(u) = \frac{\sin N\pi w}{N\sin\pi w}\operatorname{sinc}\pi v \tag{2.42}$$

This is immediately recognized as the pattern of a uniform array of N isotropic elements spaced W wavelengths apart, times the pattern of a uniform line source of length W wavelengths, with main beam at $\theta = v = 0$. For W larger than one wavelength, grating lobes will exist from the first factor. These GL have unit amplitudes and positions for $w = 0, 1, 2, \ldots$. Of these the first is the main beam. These lobes are weighted by the sinc beam. With the GL spaced by unity in w, and with the line

beamwidth of 2 (in v), the first GL is always at the edge of, or in, the sinc main beam. For example, for no scan the first GL is at the sinc factor first null. As the array beam scans, it moves away from the sinc peak, thus decreasing. Concomitantly the GL moves from the sinc null toward the peak. Thus the level of the first quantization is closely that of the sinc factor at the first GL position: $v = v_0 - 1$. The next QL will occur at the second GL: $v = v_0 - 2$, with the sinc pattern sidelobes there given the QL amplitude. An exact determination of QL peaks would require a differentiation of Eq. (2.31), with the result set to zero. As long as N is large, the GL beamwidth will be small compared with the sinc beamwidth, and the results given here will be accurate. When the N-element array pattern beam occurs at a null in the sub-array sinc pattern, the QL bifurcates, with a small lobe each side of the null (Mailloux, 1994). These are of little interest owing to their low amplitude. The QL amplitudes, and that of the main beam, are formed from Eq. (2.38), using the sub-array β of course. The QL locations are approximately at the GL angles:

$$\sin \theta_{\mathrm{GL}} = \sin \theta_0 \pm \frac{m\lambda}{W} \qquad m = 1, 2, 3, \ldots . \tag{2.43}$$

Approximate quantization lobe amplitudes are given by sinc πv_q; these versus $v_0 = (W/\lambda)\sin \theta_0$ are given in Table 2.6 for various levels. The table allows various combinations of sub-array size and scan angle to be traded. For example, QL no higher than uniform sidelobe level obtains for $W/\lambda = 2$ at 5.37° scan, or for $W/\lambda = 1$ at 10.79° scan. These amplitudes are relative to the main beam at broadside. Figure 2.28 shows QL for an array with half-wave spacing and 75 elements. Each sub-array contains five elements; scan angle is 3°, with $\beta/\pi = 0.1308$. The closest QL are −16.69 dB at −20.34°, and −18.98 dB at 26.89°.

A more precise determination of QL behavior would use a discrete sub-array instead of a continuous sub-array. However, the accuracy of the model used here is adequate, and it is simpler.

These results point up the reasons why these lobes are called quantization lobes rather than grating lobes. Although they occur at the grating lobe angles, their appearance and amplitude are functions of scan angle. While grating lobes appear even for zero scan, phaser and sub-array quantization lobes do not exist for zero scan. Their amplitude increases with scan. Further grating lobes tend to have high amplitude,

TABLE 2.6 Sub-array Quantization Lobes

β/π	Main Lobe (dB)	QL_1 (dB)	QL_2 (dB)
0.025	−0.01	−31.83	−32.26
0.05	−0.04	−25.61	−26.48
0.1	−0.14	−19.23	−20.97
0.15	−0.32	−15.39	−18.02
0.2	−0.58	−12.62	−16.14
0.3	−1.33	−8.69	−14.06
0.4	−2.42	−5.94	−13.30
0.5	−3.92	−3.92	−13.46

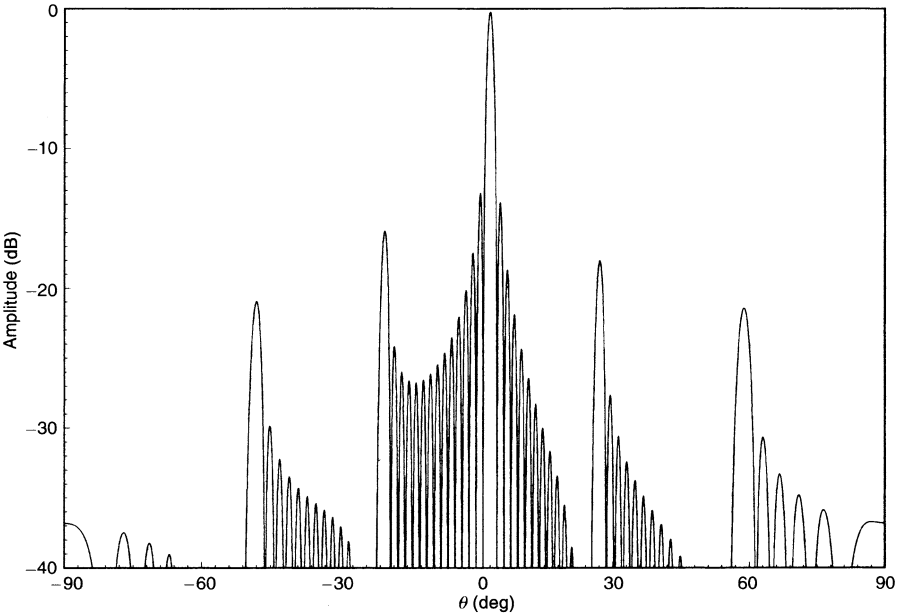


FIGURE 2.28 The 15 sub-arrays, each 5 elements, $\lambda/2$ spacing, scan 3°.

diminished from main beam amplitude only by a slowly varying element pattern. Quantization lobes, on the other hand, are typically much lower than the main beam, and each additional QL is significantly lower than the previous one.

2.3.6 QL Decollimation: Overlapped Sub-arrays

There are several schemes for decollimation of phaser quantization lobes, of which the best is the phase added technique (Smith and Guo, 1983; Hansen and Charlton, 1999). In this technique, the element phases are calculated to a number of bits that would provide sufficiently low quantization lobes if used. Call this number of bits N. The phasers are driven by a smaller number of bits, M. A set of random numbers, one for each phaser, with each number N bits long, is generated and stored. The random number table is then truncated to M bits and stored separately. When the set of phaser drive bits is calculated for a given scan direction, the N-bit random numbers are added, one to each phaser drive word. The phaser drive words are then truncated to M bits, and the M-bit random numbers are then subtracted, one for each phaser drive word. The resulting words are then used to drive the phasers. The result of this is a decollimation of the quantization lobe, and a small rise in the average sidelobe envelope. It is important that the results of this method, unlike those of some others, have zero mean value. The RMS sidelobe level is given by

$$\mathrm{SLL}_{\mathrm{RMS}} = \frac{\alpha^2}{(1-\alpha^2)N_{\mathrm{elem}}} \tag{2.44}$$

TABLE 2.7 QL Decollimation

Phaser Bits	α	RMS SLL (dB)
3	0.2267	−44.1
4	0.1134	−50.1
5	0.0567	−56.1

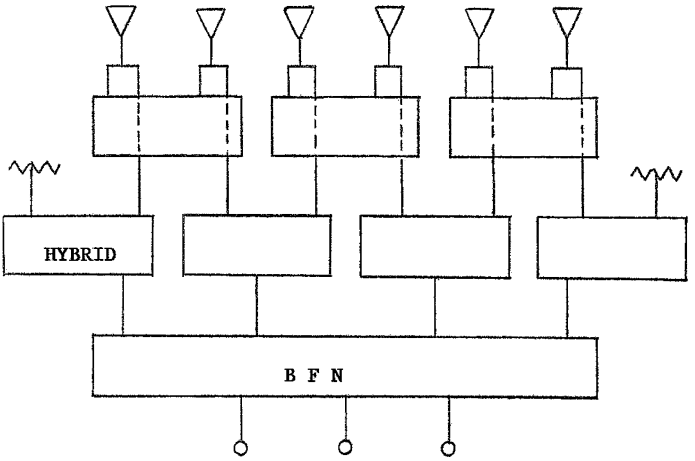


FIGURE 2.29 Overlapped sub-arrays.

where the parameter $\alpha = \pi/(\sqrt{3}2^M)$, and the array has N_{elem} elements. The number of phaser bits is M, where again the largest is π (see Table 2.7).

How decollimation works on large elements or sub-arrays needs to be investigated, but the gain drop cannot be recovered. The only scheme for significantly reducing these effects, and obviating the gain loss, uses partially overlapped sub-arrays (Mailloux, 1994). As shown in Figure 2.29, the output of each element is divided into two or more outputs. This allows two or more configurations of sub-arrays to be connected to the same element. These sub-array outputs can then be combined to produce a virtual amplitude taper. The result is reduced QL and restoration of gain. Unfortunately, the feed structure is very complex even for a linear array. The element pattern significantly reduces the effect of these quantization lobes, as it has the lowest value at the extreme scan angle, for which the QL are the largest.

2.4 DIRECTIVITY

2.4.1 Linear Array Directivity

The directivity of a linear array, as usual, is the integrated power radiation pattern over a sphere divided by the power density at the angle of interest. With θ the angle from the

normal to the array, the directivity becomes

$$G = \frac{4\pi|F|^2}{\int_0^{2\pi}\int_{-\pi/2}^{\pi/2} |F(\theta,\,\phi)|^2 \cos\theta\, d\theta\, d\phi} \tag{2.45}$$

where G is used here for directivity, although it is also often used for gain. Since conduction losses of these arrays are usually much less than radiation resistances, the gain and directivity are essentially equal, except for impedance mismatch effects. For isotropic elements, the array factor is rotationally symmetric, leading immediately to a simplification:

$$G = \frac{2|F|^2}{\int_{-\pi/2}^{\pi/2} |F(\theta)|^2 \cos\theta\, d\theta} \tag{2.46}$$

For a uniform broadside array, the closed form pattern expression is inserted to yield

$$G = \frac{N^2}{\int_0^{\pi/2} \dfrac{\sin^2 \frac{1}{2}Nk\,du}{\sin^2 \frac{1}{2}k\,du} \cos\theta\, d\theta} \tag{2.47}$$

A change of variable to u gives

$$G = \frac{N^2 d/\lambda}{\int_0^{d/\lambda} \dfrac{\sin^2 \frac{1}{2}Nk\,du}{\sin^2 \frac{1}{2}k\,du}\, du} \tag{2.48}$$

This can be integrated with the help of an expansion (Whittaker and Watson, 1952):

$$\begin{aligned} \frac{N^2 d/\lambda}{G} &= \int_0^{d/\lambda} \left[N + 2\sum_{n=1}^{N-1} (N-n)\cos nk\,du \right] du \\ &= N\frac{d}{\lambda} + \frac{2d}{\lambda}\sum_{n=1}^{N-1} (N-n)\,\mathrm{sinc}\, nkd \end{aligned} \tag{2.49}$$

The resulting directivity, where $\mathrm{sinc}\, x$ is $\sin(x)/x$, is

$$G = \frac{N^2}{N + 2\sum_{n=1}^{N-1} (N-n)\,\mathrm{sinc}\, nkd} \tag{2.50}$$

Figure 2.30 shows array directivity versus spacing for various arrays from 2 to 24 elements. Note that the directivity drops abruptly at the appearance of the first grating lobe, to about the value obtained at half-wave spacing. Although not shown

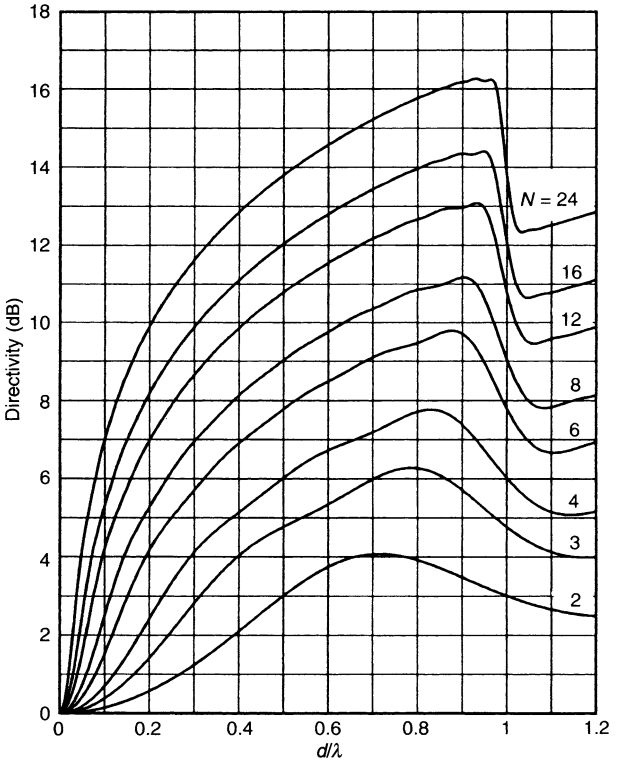


FIGURE 2.30 Array of isotropic elements.

on this graph, as the spacing increases further the directivity rises until a second grating lobe appears, at which point there is another drop to roughly the same half-wave spaced value (Hansen, 1972a). In fact, the directivity is equal to N whenever the spacing is a multiple of 0.5λ. The minor oscillations below the directivity peak depend on whether the boundary of real space coincides with a peak sidelobe or with a sidelobe null. This behavior can also be interpreted as the effect of mutual coupling between elements varying as the spacing changes. The summation in this equation can be recognized as a weighted sum of the mutual resistances, since the virtual mutual resistance between two isotropic radiators spaced by s is simply 120 sinc ks.

For scan angles away from broadside, the directivity becomes

$$G = \frac{N^2}{N + 2\sum_{n=1}^{N-1}(N-n)\operatorname{sinc} nkd\ \cos(nkd \sin\theta_0)} \tag{2.51}$$

This result will produce directivity changes near grating lobe incidence similar to those of Figure 2.30. A special case is endfire, which gives directivity of

$$G = \frac{N^2}{N + 2\sum_{n=1}^{N-1}(N-n)\operatorname{sinc} 2nkd} \tag{2.52}$$

This replicates the results of a broadside array, but with spacing halved. Thus directivity equals N for spacing that is a multiple of quarter wavelength.

For half-wave spacing, the sinc argument in Eq. (2.50) is $n\pi$, which makes the directivity independent of scan angle, and equal to N. In contrast, a uniformly excited continuous line source (Hansen, 1964) has an endfire directivity of $4L/\lambda$, but a broadside directivity of $2L/\lambda$.

For large N, maximum directivity for a large array occurs at a spacing just under λ for broadside operation and just under $\lambda/2$ for endfire operation; the directivity approaches $2N$. When expressed in terms of aperture length $L = (N - 1)d$, it is evident that these maximum directivities approach the same values as those of the continuous line source, namely, $2L/\lambda$ at broadside and $4L/\lambda$ at endfire.

An array of isotropic elements with an amplitude taper has a space factor of

$$F(u) = \sum_{n=1}^{N} A_n \exp j(nk\,du) \tag{2.53}$$

When this is inserted into Eq. (2.46) and integrated, the resulting directivity is

$$G = \frac{\left|\sum_{n=1}^{N} A_n\right|^2}{\sum_{n=1}^{N}\sum_{m=1}^{N} A_n A_m^* \operatorname{sinc}(n-m)kd} \tag{2.54}$$

Even with complex excitation coefficients, the numerator and denominator are real. For the uniform array, all $A_n = 1$, this reduces to Eq. (2.50). Symmetric arrays allow the directivity sums to be expressed over half of the array. For example, for N even, the directivity becomes

$$G = \frac{2\left|\sum_{n=1}^{N/2} A_n\right|^2}{\sum_{n=1}^{N/2}\sum_{m=1}^{N/2} A_n A_m^* [\operatorname{sinc}(n+m-1)kd + \operatorname{sinc}(n-m)kd]} \tag{2.55}$$

2.4.2 Directivity of Arrays of Short Dipoles

Arrays of short dipoles are of interest primarily because the directivity expressions can be integrated in closed form. Parallel short dipoles provide directivity in the ϕ dimension; the appropriate element pattern to use in the directivity equation is

$$f(\theta, \phi) = \sqrt{1 - \cos^2\theta \sin^2\phi} \tag{2.56}$$

Collinear short dipoles provide directivity in the plane of the array; an appropriate element pattern is $\cos\theta$. The resulting directivity for a uniform broadside linear

array of parallel short dipoles is

$$G = \frac{3N^2/2}{N + 3\sum_{n=1}^{N-1}(N-n)\left[\dfrac{\sin nkd}{nkd} + \dfrac{\cos nkd}{(nkd)^2} - \dfrac{\sin nkd}{(nkd)^3}\right]} \tag{2.57}$$

For the special case of half-wave spacing, the result becomes

$$G = \frac{3N^2/2}{N + 3\sum_{n=1}^{N-1}\dfrac{(-1)^n(N-n)}{\pi^2 n^2}} \tag{2.58}$$

Collinear short dipoles or slots yield a similar directivity:

$$G = \frac{3N^2/2}{N - 6\sum_{n=1}^{N-1}(N-n)\left[\dfrac{\cos nkd}{(nkd)^2} - \dfrac{\sin nkd}{(nkd)^3}\right]} \tag{2.59}$$

Again the special case of half-wave spacing yields

$$G = \frac{3N^2/2}{N - 6\sum_{n=1}^{N-1}\dfrac{(N-n)(-1)^n}{\pi^2 n^2}} \tag{2.60}$$

For both dipole orientations, the mutual resistance is given by the bracketed trig terms times a constant. Although these directivity expressions are simple, short elements are seldom used in arrays because of their high reactance (susceptance), low-radiation resistance, and narrow bandwidth. Half-wave or resonant elements are almost always preferred.

One might infer that directivity of general dipole or slot or patch arrays could be calculated using mutual resistances, a concept pioneered by Bloch et al. (1953). This finding proves to be true, and will be discussed next.

2.4.3 Directivity of Arrays of Resonant Elements

Directivity will be derived for a uniform broadside linear array of half-wave dipoles in any orientation. As before, directivity is the ratio of peak power density to input power:

$$G = \frac{4\pi r^2 E_0 H_0^*}{P} \tag{2.61}$$

For a linear array of half-wave dipoles, the peak fields E_0 are

$$E_0 = \frac{60}{r}\sum I_n \qquad H_0 = \frac{E_0}{120\pi} \tag{2.62}$$

Here the dipole currents are I_n. For uniform excitation, all $I_n = I_0$, and the numerator is

$$120\left[\sum I_n\right]^2 = 120I_0^2N^2 \tag{2.63}$$

The input power is

$$P = \sum \mathrm{Re}(I_n V_n^*) = \sum I_n \sum I_m^* R_m = I_0^2 \sum \sum R_{nm} \tag{2.64}$$

where the mutual resistance between elements is R_{nm}. Finally, the directivity becomes

$$G = \frac{120N^2}{\sum_{n=1}^{N}\sum_{m=1}^{N} R_{nm}} = \frac{120N^2}{R_0N + 2\sum_{n=1}^{N-1}(N-n)R_n} \tag{2.65}$$

The self-resistance is R_0, and R_n is the mutual resistance between dipoles nd apart. Directivity for parallel and collinear linear arrays of half-wave dipoles has been calculated using compact and efficient computer algorithms for mutual impedances (Hansen, 1972b; Hansen and Brunner, 1979). Results for parallel dipole arrays are shown in Figure 2.31 while directivity for collinear dipole arrays are shown in

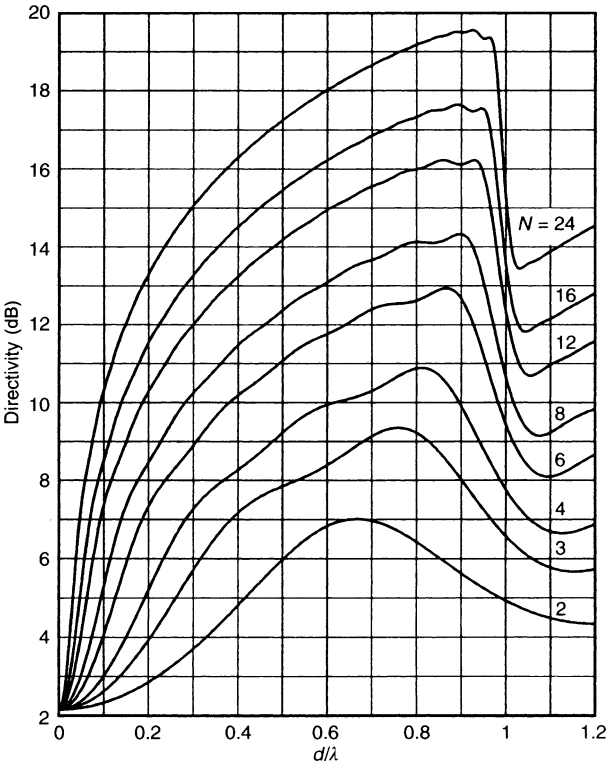


FIGURE 2.31 Parallel array of half-wave dipoles.

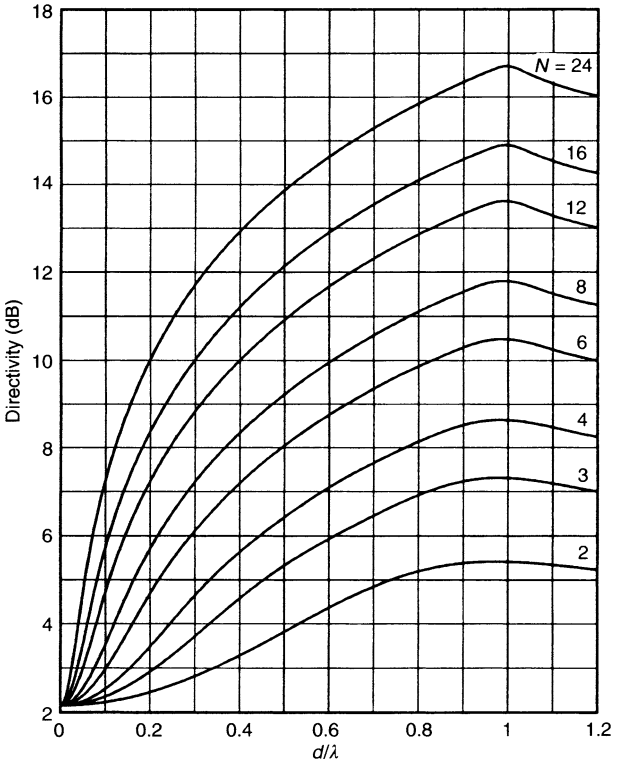


FIGURE 2.32 Collinear array of half-wave dipoles.

Figure 2.32. Notice that the parallel dipole array exhibits grating lobe directivity drops similar to those with isotropic elements. Both experience strong mutual coupling. However, for collinear arrays the mutual coupling is less and thus there is less grating lobe effect for these arrays. At the same time, the parallel array gives roughly 3 dB more directivity at the peak, due to narrowing the beam in the transverse plane.

For a fixed element spacing and number of elements, maximum directivity does not occur at broadside or at endfire. This is discussed in Chapter 9 on Superdirective Arrays.

2.4.4 Planar Array Directivity

The expression for planar array directivity is

$$G = \frac{4\pi F_0^2}{\int_0^{2\pi}\int_0^{\pi} F^2(\theta)\sin\theta\, d\theta\, d\phi} \tag{2.66}$$

For an array of isotropic elements located on a rectangular lattice, with a broadside beam, the numerator of the directivity expression becomes

$$4\pi\left[\sum_{n=1}^{N/2}\sum_{m=1}^{M/2}A_{nm}\right]^2 \tag{2.67}$$

In the denominator, F^2 is a fourfold sum:

$$\begin{aligned}F^2 = &\sum_n\sum_m\sum_p\sum_q A_{nm}A_{pq}\cos(2n-1)\alpha\cos(2m-1)\beta\cos(2p-1)\alpha\\ &\times\cos(2q-1)\beta\end{aligned} \tag{2.68}$$

with $\alpha = (\pi d_x/\lambda)u$ and $\beta = (\pi d_y/\lambda)v$. Liberal and frequent use of trigonometric identities allows the four-cosine product to become a sum of eight terms, where the integer factors A and B occur in all possible combinations. The general term is

$$\frac{1}{8}\cos(A\alpha \pm B\beta) \tag{2.69}$$

and the values of A and B are

$$A = \begin{cases} n+p-1 \\ n-p \end{cases} \qquad B = \begin{cases} m+q-1 \\ m-q \end{cases} \tag{2.70}$$

These can be recast into the form

$$\cos(a\cos\phi + b\sin\phi) = \cos\left[\sqrt{a^2+b^2}\sin\left(\phi + \arctan\frac{a}{b}\right)\right] \tag{2.71}$$

This equation can be integrated in ϕ:

$$\int_0^{2\pi}\cos\left[\sqrt{a^2+b^2}\,\sin\left(\phi+\arctan\frac{a}{b}\right)\right]d\phi = 2\pi J_0\left(\sqrt{a^2+b^2}\right) \tag{2.72}$$

and this result allows the θ integration to be performed

$$2\pi\int_0^{\pi}J_0\left(\sqrt{a^2+b^2}\right)\sin\theta\,d\theta = 4\pi\ \text{sinc}\ C \tag{2.73}$$

where $\sqrt{A^2 + B^2} = C \sin\theta$. Finally, the directivity can be written in terms of a fourfold sum:

$$D = \frac{4\left(\sum_{n=1}^{N/2}\sum_{m=1}^{M/2} A_{nm}\right)^2}{\sum_{n=1}^{N/2}\sum_{m=1}^{M/2}\sum_{p=1}^{N/2}\sum_{q=1}^{M/2} A_{nm}A_{pq}[S_1 + S_2 + S_3 + S_4]} \tag{2.74}$$

where

$$\begin{aligned}
S_1 &= \operatorname{sinc} 2\pi\sqrt{(n+p-1)^2 d_x^2/\lambda^2 + (m+q-1)^2 d_y^2/\lambda^2} \\
S_2 &= \operatorname{sinc} 2\pi\sqrt{(n+p-1)^2 d_x^2/\lambda^2 + (m-q)^2 d_y^2/\lambda^2} \\
S_3 &= \operatorname{sinc} 2\pi\sqrt{(n-p)^2 d_x^2/\lambda^2 + (m+q-1)^2 d_y^2/\lambda^2} \\
S_4 &= \operatorname{sinc} 2\pi\sqrt{(n-p)^2 d_x^2/\lambda^2 + (m-q)^2 d_y^2/\lambda^2}
\end{aligned} \tag{2.75}$$

Unfortunately, this result does not simplify appreciably for either uniform excitation or half-wave spacing. The terms S_i are only zero for special sets of integers, because the cross-coupling impedance terms are not generally zero.

Although this directivity expression is for a rectangular array with rectangular lattice, it can readily be modified for other geometries. For example, a flat plane array, where the waveguide linear array sticks are of different lengths, is handled by adjusting the summation limits. Other lattices, such as hexagonal, can also be accommodated.

In general, it is necessary to use approximations for planar array directivity. For example, the directivity is approximately

$$G \simeq 2G_x G_y \tag{2.76}$$

where G_x and G_y are the linear array directivities along x and along y. The factor of 2 gives a result that compares well with exact square array calculations (Hansen, 1972a); it also accounts for a pencil beam radiating on one side only.

The previous derivation was for directivity of a planar array of isotropic elements. Directivity of an array of actual elements can be computed whenever the mutual impedance between any two elements can be calculated. Following the method of Section 2.4.3 the general directivity formula is

$$D = \frac{120\left[\sum_n \sum_m A_{nm}\right]^2}{\sum_n \sum_m \sum_p \sum_q A_{nm}A_{pq}R_{nmpq}} \tag{2.77}$$

Here the sums go over all the nm elements, and R_{nmpq} is a mutual resistance between the nmth element and the pqth element.

An approximate understanding of planar array effective aperture is provided by considering the directivity of a uniformly excited planar aperture. Let the aperture be in the y,z-plane of a spherical coordinate system, but a coordinate system where $\sin\theta$ is replaced by $\cos\theta$. The aperture normal is then $\theta_0 = \pi/2$, $\phi_0 = 0$. The space factor is $F = \text{sinc}\,\pi u\ \text{sinc}\,\pi v$ where

$$u = \frac{L}{\lambda}(\cos\theta - \cos\theta_0), \qquad v = \frac{W}{\lambda}\sin\theta\sin\phi \tag{2.78}$$

Scan in the $\phi_0 = 0$ plane is assumed, which gives for directivity

$$G = \frac{4\pi}{\int_0^{2\pi}\int_{-\pi/2}^{\pi/2} F^2 \sin\theta\, d\theta\, d\phi} \tag{2.79}$$

The ϕ integral can be simplfied by changing the variable to v. The necessary cosine is inserted since it is essentially unity over the main beam for large apertures. The error is larger at angles where the sidelobe energy is low. This gives the ϕ integral approximately as

$$\int_{-\pi/2}^{\pi/2} F^2\, d\phi \simeq \frac{\lambda}{W\sin\theta} \tag{2.80}$$

The directivity is now approximately

$$\frac{4\pi}{G} \simeq \frac{\lambda}{W}\int_0^{\pi} \frac{\sin^2\pi u}{\pi^2 u^2}\, d\theta \tag{2.81}$$

A similar change of u variable produces a result (King and Thomas, 1960) which is valid as long as the beam is narrow in both planes:

$$G \simeq \frac{4\pi LW\sin\theta_0}{\lambda^2} \tag{2.82}$$

When the angle θ is changed back to that measured from the array normal, the effective aperture varies as $\cos\theta_0$. However, as the beam broadens with scan, the approximations become disabling. For angles near endfire, the same ϕ approximation is used, but the θ integration is approximated differently. Let

$$u = \frac{L}{2\lambda}(\theta + \theta_0)(\theta - \theta_0) \tag{2.83}$$

which gives near endfire

$$\frac{4\pi}{G} \simeq \frac{\lambda^2}{LW}\int_0^1 \frac{\sin^2\pi u}{\pi^2 u^2\sqrt{(2\lambda u/L)^2 - \theta_0^2}}\, du \tag{2.84}$$

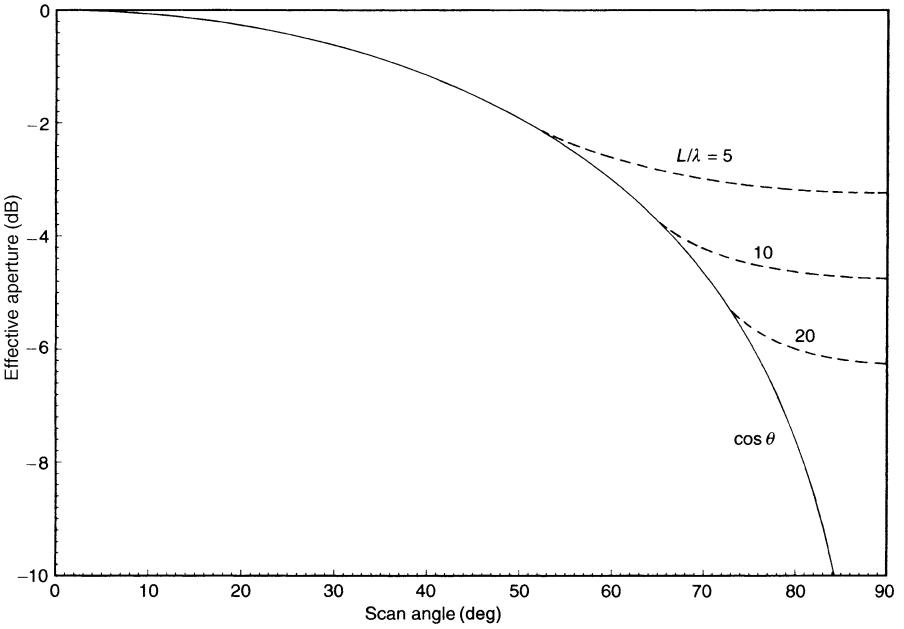


FIGURE 2.33 Effective aperture versus scan angle.

At endfire the result is

$$G \simeq 3\sqrt{2}\pi \frac{W}{\lambda}\sqrt{\frac{L}{\lambda}} \tag{2.85}$$

Figure 2.33 shows how the effective aperture varies with principal plane scan, and from this the range of usefulness of the $\cos\theta_0$ effective aperture may be observed. The $\cos\theta_0$ value decays to the endfire value for

$$\cos\theta_0 = \frac{3}{2\sqrt{2L/\lambda}} \tag{2.86}$$

It should be used only for scan angles well above this critical value. Note that none of these $\cos\theta$ effective aperture results includes *scan impedance* effects (see Chapter 7).

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CHAPTER THREE

Linear Array Pattern Synthesis

3.1 INTRODUCTION

Linear arrays of radiating antenna elements have been developed over many decades. This chapter brings together the important developments that provide a detailed understanding of array design and synthesis. Included are the design of narrow-beam, low-sidelobe arrays, shaped beam arrays, arrays designed by constrained optimization, and aperiodic arrays. Many older data have been recalculated using the excellent available computing power, as well as appropriate new supplementary data.

3.1.1 Pattern Formulations

The fields radiated from a linear array are a superposition (sum) of the fields radiated by each element in the presence of the other elements. Each element has an excitation parameter, which is current for a dipole, voltage for a slot, and mode voltage for a multiple-mode element. This chapter is concerned primarily with the "forced excitation" problem, where the drive of each element is individually adjusted so that the excitation is as desired. This adjustment of the drive accommodates the mutual coupling among feeds that changes the element input impedance (admittance). A more common situation is that of "free excitation": The feed network is designed to produce the desired excitations, but as the scan angle or frequency changes, the element "scan impedance" change will result in a change of excitation. These matters are discussed more fully in Chapter 7.

The excitation of each element will be complex, with amplitude and phase. This discrete distribution is often called an aperture distribution, where the array is the aperture. The far-field radiation pattern is the discrete Fourier transform (FT) of the array excitation. The pattern is sometimes called the space factor. A commonly used array notation is appropriate for this chapter: The angle from broadside is θ, the element

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spacing is d, and the array variable u is

$$u = \frac{d}{\lambda}(\sin\theta - \sin\theta_0) \tag{3.1}$$

Although the far-field pattern can be written as a simple sum of amplitude and exponential phase path length terms, it is convenient to provide a real pattern in a formulation valid for both even and odd numbers of elements. This formulation refers the phase to the array center, which results in the real pattern, and is

$$F(u) = \sum_{n=1}^{N} A_n \exp\left[j2\pi(n-1)u\right] \tag{3.2}$$

Excitation coefficients are A_n, the array has N elements, and the beam peak is at θ_0. The array is represented in Figure 3.1.

To understand how the element contributions combine, the unit circle approach of Schelkunoff (1943) replaces the exponential factor by a new variable:

$$w = \exp(j2\pi u) \tag{3.3}$$

This gives the pattern as a polynomial in w:

$$F(u) = \sum_{n=1}^{N} A_n w^{n-1} \tag{3.4}$$

Physical space, commonly called real space, corresponds to part of the unit circle in the complex w plane. The polar angle is $\psi = 2\pi u$; the part of the unit circle covered is for

$$-kd(1 + \sin\theta_0) \le \psi \le kd(1 - \sin\theta_0) \tag{3.5}$$

With half-wave spacing, w traverses the unit circle once. Wavelength spacing produces two traverses; and so on. The polynomial in w has $N - 1$ roots that may or may not lie

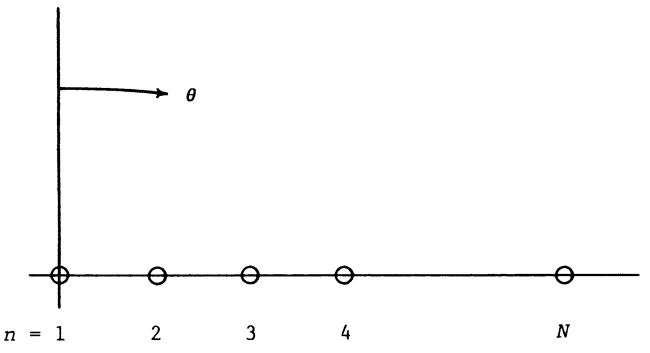


FIGURE 3.1 Linear array geometry.

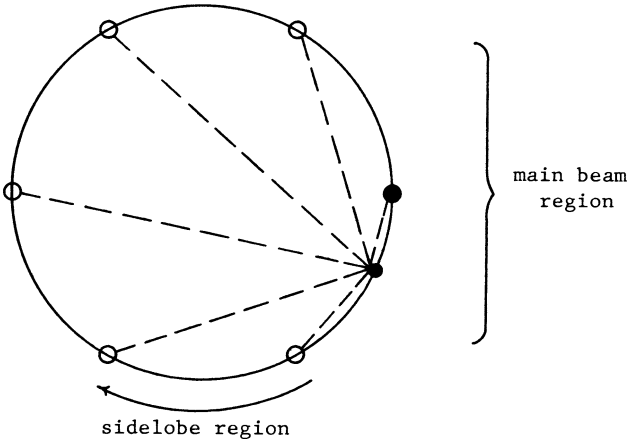


FIGURE 3.2 Unit circle for seven-element array.

on the unit circle. The pattern is given by the product of the distances from the observation point in w (on the circle) to each of the zeros (roots) (see Fig. 3.2). As w moves around the circle, lobes form, then decay. When zeros are located on the unit circle, pattern nulls are produced; zeros off the unit circle may give pattern minima. A simple case is the uniform array:

$$F(u) = \frac{\sin N\pi u}{N \sin \pi u} \exp\left[j\pi(N-1)u\right] = \frac{w^N - 1}{w - 1} \tag{3.6}$$

Roots are equally spaced on the circle by $2\pi/N$, and the exponential factor gives the phase at the center of the array. This control of pattern by movement of w-plane zeros will be utilized later, in shaped beam synthesis. Readers may recognize that the unit circle technique developed by Schelkunoff is exactly the Z transform used many years later in circuit analysis.

3.1.2 Physics versus Mathematics

For uniform excitation, calculation of array performance is relatively easy; tapered distributions used to reduce sidelobes are less easy. In the days before computers (BC), aperture distributions were chosen for their easy integrability to closed-form solutions. For example, a cosine to the nth power, on a pedestal, allows the sidelobe level to be adjusted, and is readily calculated. There are, however, two disadvantages to these distributions. In the example there are two parameters: exponent and pedestal height. For given values of these, all important characteristics (beamwidth, aperture efficiency, sidelobe level, beam efficiency, etc.) can be calculated. But there is no easy way to choose the two parameters to optimize efficiency for a given sidelobe level, for example. More fundamental is the second disadvantage: these distributions, depending on the values of parameters selected, may be quite inefficient in terms of beamwidth, aperture efficiency, and so on. Another example is the popular Gaussian

with a pedestal (a truncated Gaussian). It is multiple parameter distribution, and, what is worse, the Fourier transform is complex. Thus in modern antenna work easy mathematics has yielded to good physics: Aperture distributions should be designed by proper placement of pattern-function zeros, preferably using a single parameter. This assures good physics, which means highly efficient, low-Q apertures. Design can be handled via the computer. Taylor was a pioneer in this approach to narrow-beamwidth low-sidelobe distributions; his design principles are discussed next.

3.1.3 Taylor Narrow-Beam Design Principles

A common pattern requirement is for high directivity and low sidelobes, useful for radar, communications, mapping, and so on. *Sidelobe level* (SLL) is the amplitude of the highest sidelobe, usually that closest to the main beam, normalized to the main beam peak. It is convenient also to use *sidelobe ratio* (SLR), which is the inverse of SLL. The main beam may be fixed at broadside or at some other angle, or it may be electronically scanned to any desired angle. The terms "electronic scanning array" and "phased array" are synonymous. For maximum directivity without superdirectivity, uniform excitation is used. When lower sidelobes than those provided by uniform excitation are required, the aperture amplitude is tapered (apodized, shaded) from the center to the ends. There are some general rules, derived by Taylor and others.

Taylor (1953) and his colleagues developed some important general rules for low-sidelobe patterns. These are listed here.

1. Symmetric amplitude distributions give lower sidelobes.
2. $F(u)$ should be an entire function of u.
3. A distribution with a pedestal produces a far-out sidelobe envelope of $1/u$.
4. A distribution going linearly to zero at the ends produces a far-out sidelobe envelope of $1/u^2$.
5. A distribution that is nonzero at the ends (pedestal) is more efficient.
6. Zeros should be real (located on the w unit circle).
7. Far-out zeros should be separated by unity (in u).

In the sections that follow, these principles will be employed to produce highly efficient and robust (low-Q) excitations and the corresponding patterns.

Few methods exist for the design of efficient array distributions for prescribed patterns; most syntheses utilize continuous apertures where the aperture distribution is sampled to obtain the array excitation values. As long as the array spacing is at or near half-wave, and the number of elements is not small, this procedure works well. When the number of elements is too small, a discrete $\bar{n}$ synthesis of the type presented in Section 3.6 can be used. The cost is that a new synthesis is required for each different number of elements. A more sophisticated method matches the zeros of the desired continuous pattern to those of the discrete array pattern (see Section 3.9.2). Of course, the Dolph–Chebyshev synthesis of Section 3.2 is discrete from the start, but these designs are seldom used because of the high farther-out sidelobes.

3.2 DOLPH–CHEBYSHEV ARRAYS

3.2.1 Half-Wave Spacing

A half-wave spaced array yields maximum directivity for a given sidelobe ratio when all sidelobes are of equal height. Dolph (1946) recognized that Chebyshev polynomials were ideally suited for this purpose. In the range ± 1, the polynomial oscillates with unit amplitude, while outside this range it becomes monotonically large. See, for example, Figure 3.3 for the graph of a ninth-order Chebyshev polynomial. The polynomial can be simply expressed in terms of trig type functions as

$$T_n(x) = \begin{cases} (-1)^n \cosh(n \operatorname{arccosh} |x|) & x < -1 \\ \cos(n \arccos x) & |x| \leq 1 \\ \cosh(n \operatorname{arccosh} x) & x \geq 1 \end{cases} \tag{3.7}$$

It can also be readily calculated through recursion. A direct correspondence between the array polynomial and the Chebyshev polynomial is not feasible, because the array main beam must be symmetric and have zero slope at the center. Since an N-element array has $N-1$ zeros, an $(N-1)$-degree Chebyshev, which has the same number of zeros, is appropriate. The oscillatory part of the Chebyshev polynomial is mapped once onto the sidelobes on one side of the main beam; the main beam up to the peak is mapped onto the $x > 1$ region of the Chebyshev polynomial. The transformation that performs this is $x = x_0 \cos(u/2)$. This transforms the Chebyshev polynomial

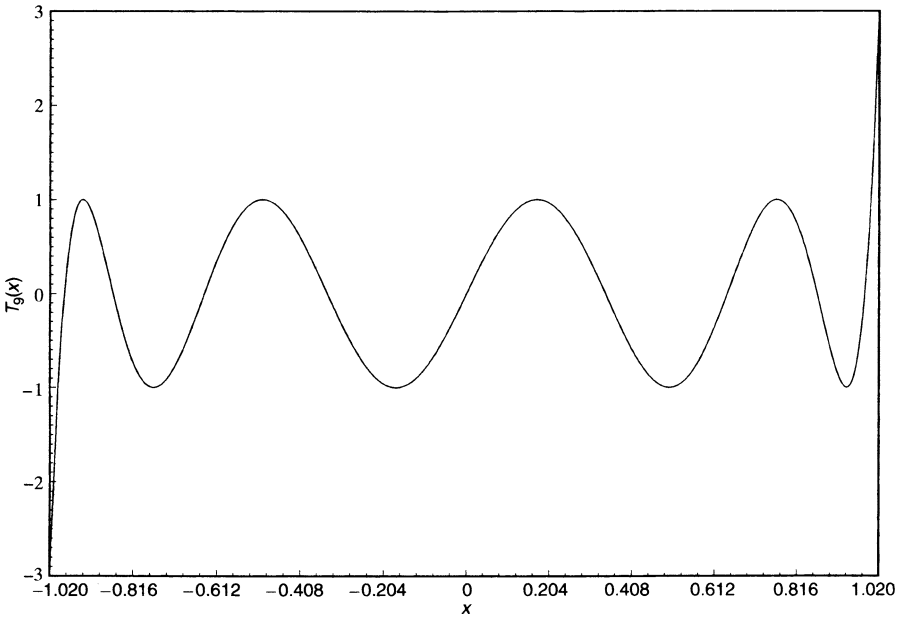


FIGURE 3.3 Chebyshev polynomial.

$T_{N-1}(x)$ to the pattern function $F(u)$. The pattern variable is given by $u = kd \sin \theta$. Thus the pattern factor is simply

$$F(u) = T_{N-1}\left(x_0 \cos\frac{u}{2}\right) \tag{3.8}$$

As θ varies from $-\pi/2$ to $\pi/2$, x varies from $(x_0 \cos \pi d/\lambda)$ up to $\theta = 0$, and then back to $(x_0 \cos \pi d/\lambda)$. For half-wavelength spacing, the minimum value of x is zero. The maximum allowable spacing is determined by the need to prevent x falling below -1. This gives the maximum element spacing for a broadside array as

$$\frac{d}{\lambda} = \frac{1}{\pi}\arccos\left(-\frac{1}{x_0}\right) \tag{3.9}$$

which approaches one-wavelength spacing for large arrays.

The voltage sidelobe ratio is given by

$$\text{SLR} = T_{N-1}(x_0) \qquad \text{SLR} > 1 \tag{3.10}$$

Given the SLR, the mapping constant x_0 is given by

$$\begin{aligned} x_0 &= \cosh\frac{\operatorname{arccosh} \text{SLR}}{N-1} \\ &= \frac{1}{2}\left[\text{SLR} + \sqrt{\text{SLR}^2 - 1}\,\right]^{1/N} + \frac{1}{2}\left[\text{SLR} - \sqrt{\text{SLR}^2 - 1}\right]^{1/N} \end{aligned} \tag{3.11}$$

The second form is given because some computers do not have inverse hyperbolic function subroutines. The Dolph formulation is valid only for array spacings of 0.5λ and larger. A typical array pattern is given in Figure 3.4. Using the array center as phase reference, the space factor may be written as a series:

$$F(u) = \sum_{n=1}^{N} A_n \exp\left\{j\left[(2n - N - 1)\frac{u}{2}\right]\right\} \tag{3.12}$$

This finite series is inverted to produce the coefficients:

$$A_n = \frac{1}{N}\sum F\left(\frac{2\pi m}{N}\right) \exp\left\{-j\left[(2n - N - 1)\frac{\pi m}{N}\right]\right\} \tag{3.13}$$

When there is substitution in the Chebyshev polynomial, the coefficients are given as a sum of Chebyshev polynomials:

$$A_n = \frac{1}{N}\sum_{m} T_{N-1}\left(x_0 \cos\frac{\pi m}{N}\right) \exp\left\{-j\left[(2n - N - 1)\frac{\pi m}{N}\right]\right\} \tag{3.14}$$

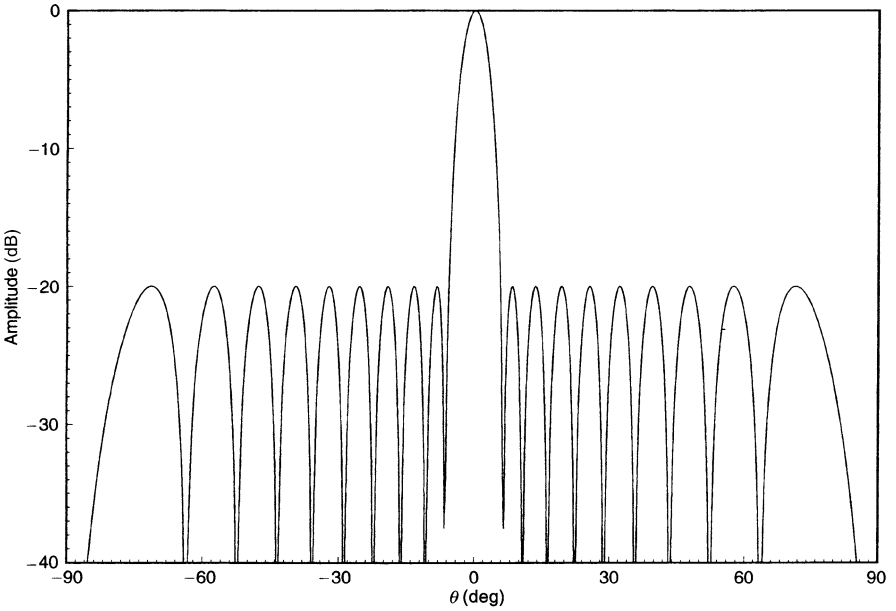


FIGURE 3.4 The 20-element Dolph–Chebyshev array pattern, $d = 0.5\lambda$.

Because the pattern is symmetric, additional simplifications can be made. The formulas developed by Stegen (1953) are convenient for calculating the excitation coefficients. In these formulas, the elements are numbered from the array center, and are different for odd and even N. For even $N = 2M$:

$$A_n = \frac{2}{N}\left[\mathrm{SLR} + 2\sum_{m=1}^{M-1} T_{N-1}\left(x_0 \cos\frac{\pi m}{N}\right)\cos\left((2n-1)\frac{\pi m}{N}\right)\right] \quad 1 \le n \le M \quad (3.15)$$

For odd $N = 2M + 1$:

$$A_n = \frac{2}{N}\left[\mathrm{SLR} + 2\sum_{m=1}^{M} T_{N-1}\left(x_0 \cos\frac{m\pi}{N}\right)\cos\frac{2nm\pi}{N}\right] \quad (3.16)$$

These results are valid for spacings $\geq \lambda/2$.

These formulas are awkward for large arrays; simplified approximate versions have been developed by Van der Maas (1954) and Barbiere (1952). However, with current computer capability, such approximations are unnecessary. Stegen's formulas can be implemented directly. An extensive table of coefficients, directivity, and beamwidth has been prepared by Brown and Scharp (1958), giving $N = 3(1)40$ and SLR = 10(1)40 dB for $d = \lambda/2$. However, there are significant errors near the upper values

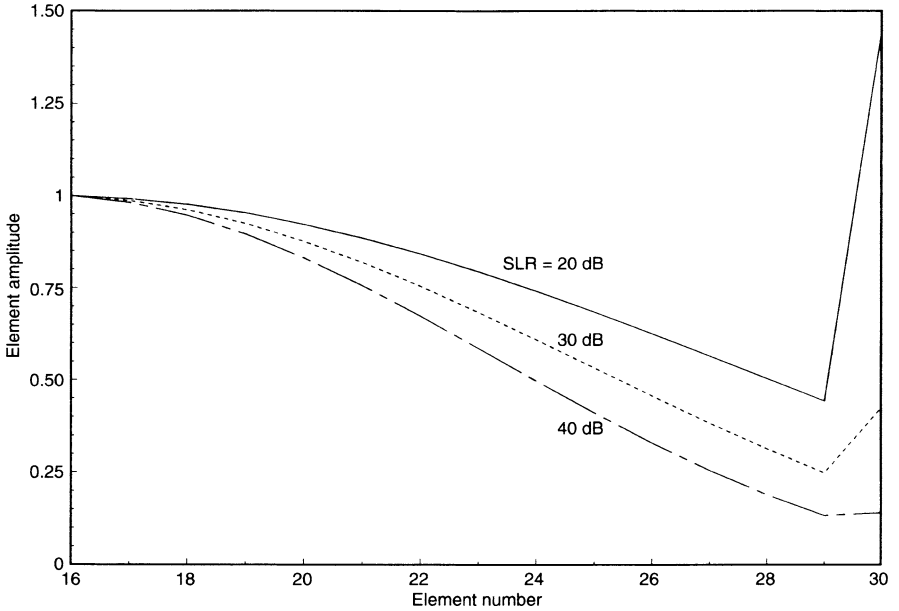


FIGURE 3.5 Dolph–Chebyshev 30-element array coefficients.

of N and SLR, presumably due to roundoff error. Users should compute their own values using the formulas above, or those that follow.

The array coefficients for a typical 30-element array are shown in Figure 3.5. It may be observed that none of the three distributions is monotonic, and that the 20 dB SLR values are larger at the ends than at the center. In general, the high edge values increase with larger arrays, and with lower SLR. This curve points up a major difficulty in the utilization of Dolph–Chebyshev arrays: The nonmonotonic distributions and end spikes are difficult to achieve. A further disadvantage is that the equal level far-out sidelobes tend to pick up undesired interference and clutter. For these reasons, Dolph–Chebyshev arrays are seldom used. Figure 3.6 gives values of the maximum N for a given sidelobe ratio that produces a monotonic distribution, and also the maximum N versus sidelobe ratio that produces an end spike no larger than the center excitation value.

Beamwidth is simply calculated from the pattern Eq. (3.12); however, a root finder needs to be used to find the beamwidth values from the equation

$$\sqrt{2}T_{N-1}\left(x_0 \cos\frac{u_3}{2}\right) = T_{N-1}(x_0) \tag{3.17}$$

Figure 3.7 gives normalized beamwidth in degrees for a half-wave spaced array with number of elements from 10 to 100 versus sidelobe ratio. It can be seen that the increase in beamwidth with SLR is roughly linear as expected.

Directivity of a Dolph–Chebyshev array is calculated from the directivity formulas of Chapter 2 using the excitation coefficients given above. Rather than show

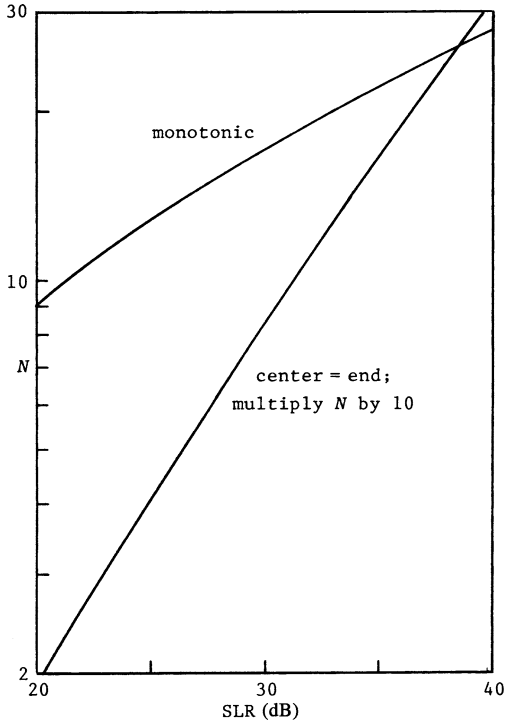


FIGURE 3.6 Maximum N for a given SLR.

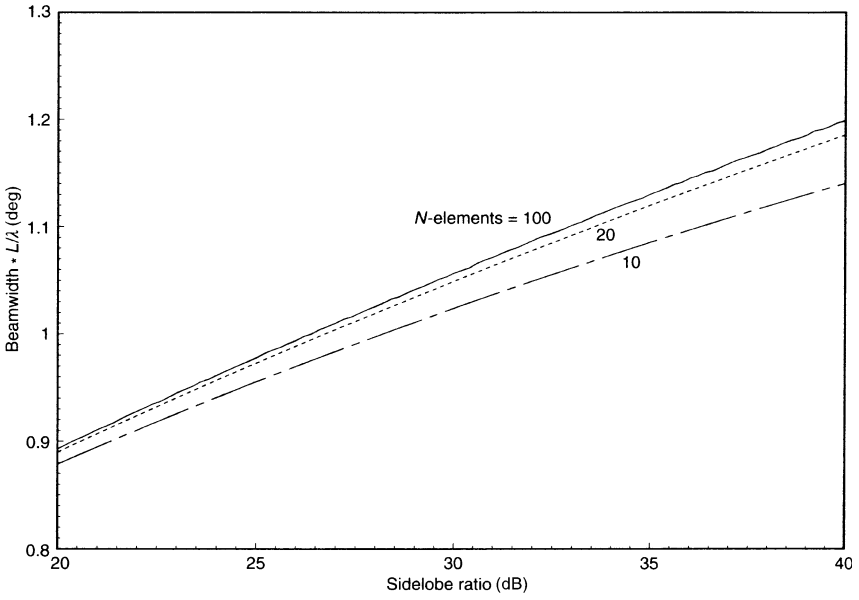


FIGURE 3.7 Dolph–Chebyshev distribution beamwidth.

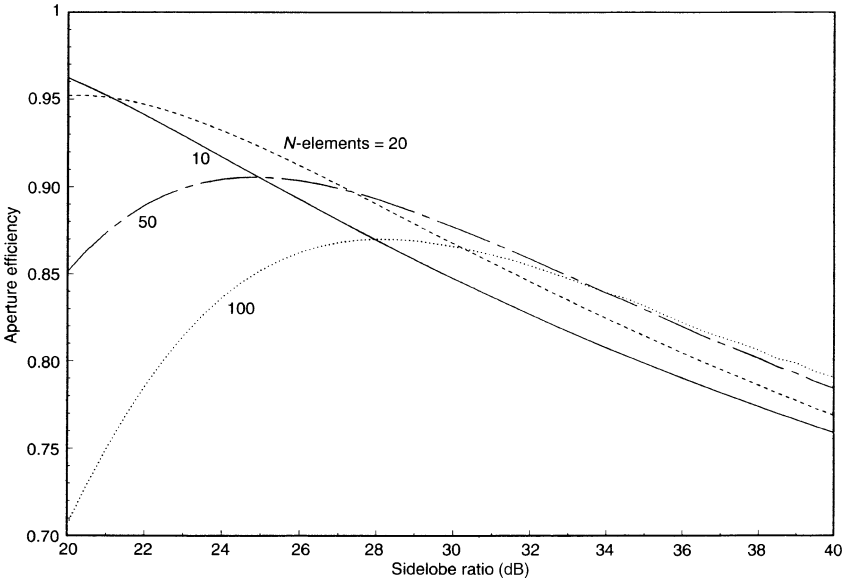


FIGURE 3.8 Dolph–Chebyshev array efficiency, $d = 0.5\lambda$.

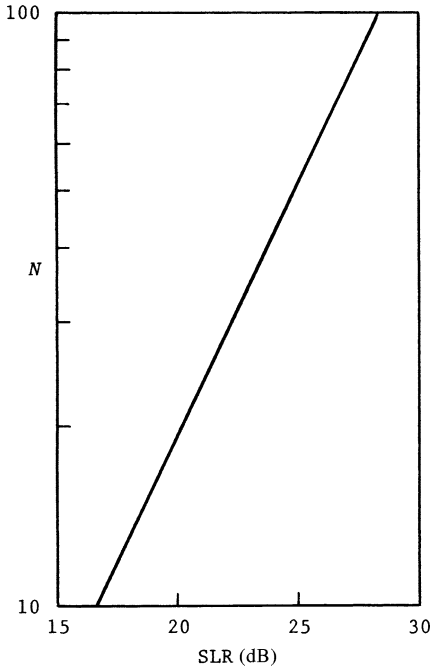


FIGURE 3.9 Dolph–Chebyshev array with maximum efficiency.

directivity, it is more useful to show array efficiency, which is the ratio of array directivity to that of a uniformly excited array of the same number of elements and spacing. Figure 3.8 shows array efficiency for half-wave spaced arrays. It is interesting to note that the efficiency peaks for a particular sidelobe ratio for each number of array elements. The reason is because the sidelobe energy is a function primarily of the sidelobe ratio, while the mainbeam energy primarily depends on the number of elements. Thus as the sidelobe energy becomes a larger fraction of the total, the efficiency decreases. The values of number of elements and sidelobe ratio that provide maximum efficiency are shown in Figure 3.9. This curve is fitted approximately by $\log N \simeq 0.08547\ \mathrm{SLR}_{\mathrm{dB}} - 0.4188$.

3.2.2 Spacing Less Than Half-Wave

The derivations in Section 3.2.1 are limited to half-wave spacing or greater, because smaller spacings only utilize a smaller part of the oscillatory region of the polynomial. For N odd, Riblet (1947) showed that this restriction could be removed; the pattern factor is formed by starting near the end of the Chebyshev $+1$ region, tracing the oscillatory region to the other end, and then retracing back to the start end and up the monotonic portion to form the main beam half. The exact starting point depends on Nkd. Since the M-order Chebyshev has $M - 1$ oscillations, which are traced twice, and since the trace from 0 to 1 and back forms the sidelobes, the pattern factor always has an odd number of sidelobes on each side, and thus an even number of zeros. As a result, only an odd number of elements can be formed into a Chebyshev array for spacing less than half-wave using the Riblet synthesis. The pattern function is given by

$$\begin{aligned} F(u) &= T_M(a\cos(u) + b) \\ a &= \frac{z_0 + 1}{1 - \cos(kd)} \qquad b = \frac{z_0\cos(kd) + 1}{\cos(kd) - 1} \end{aligned} \tag{3.18}$$

As before, $u = kd\sin\theta$ and $M = (N - 1)/2$. The value of z_0 is

$$z_0 = \cosh\frac{\operatorname{arccosh}(\mathrm{SLR})}{M} \tag{3.19}$$

Formulas have been developed by many, including DuHamel (1953), Brown (1957, 1962), Drane (1963, 1964), and Salzer (1975). The formulas of Drane are suitable for computer calculation of superdirective arrays (see Chapter 9) and thus are used here. Again the elements are numbered starting with zero at the center; the array excitation coefficients are

$$A_n = \frac{\varepsilon_n}{4M}\sum \varepsilon_m\varepsilon_{M_{2-m}}T_n(x_n)T_M(ax_n + b) + (-1)^nT_M(b - ax_n) \tag{3.20}$$

where $\varepsilon_i = 1$ for $i = 0$ and $\varepsilon_i = 2$ for $i > 0$. The variable $x_n = \cos n\pi/M$. The integers M_1 and M_2 are the integer parts of $M/2$ and $(M + 1)/2$, respectively. Small spacings may require double-precision arithmetic due to the subtraction of terms.

For half-wave spacing, the a and b above reduce to

$$a = \frac{1}{2}(z_0 + 1) \qquad b = \frac{1}{2}(z_0 - 1) \tag{3.21}$$

For half-wave spacing, the two approaches give identical results. For this case, the pattern factor of Eq. (3.8) is equal to the pattern factor of Eq. (3.18):

$$T_{N-1}\left(x_0^{N-1} \cos\frac{u}{2}\right) \equiv T_M\left(\frac{z_0^M(\cos(u) + 1) + \cos(u) - 1}{2}\right) \tag{3.22}$$

DuHamel (1953) extended the Dolph–Chebyshev design to endfire arrays, but only for spacing less than half-wave. This restriction is not serious, as spacing is customarily made quarter-wave at endfire to avoid a large back lobe. For any scan angle, u is modified as usual for scanned arrays:

$$u = ka(\sin\theta - \sin\theta_0) \tag{3.23}$$

where θ_0 is the main beam scan angle. The interelement phase shift is $kd \sin\theta_0$, and the coefficients a and b become

$$a = \frac{3 + z_0 + 2\sqrt{2(z_0 + 1)}\cos kd}{2\sin^2 kd} \tag{3.24}$$

$$b = \frac{\left(\sqrt{z_0 + 1} + \sqrt{2}\cos kd\right)^2}{2\sin^2 kd} \tag{3.25}$$

Drane (1968) has shown that a large Dolph–Chebyshev array has a directivity independent of scan angle for any spacings between half- and full-wave, as long as the spacing admits no grating lobes.

McNamara (2006) has shown that Zolotarev polynomials (see also, Section 3.7.4) can be used to synthesize a Chebyshev pattern for an even number of elements, where again the element spacing is less than half-wave.

3.3 TAYLOR ONE-PARAMETER DISTRIBUTION

3.3.1 One-Parameter Design

Taylor (1955) [summarized here] recognized that to produce a linear aperture distribution with a sidelobe envelope approximating a $1/u$ falloff, the uniform amplitude $\sin(x)/x$ pattern could be used as a starting point. He understood that the height of

each sidelobe is controlled by the spacing between the aperture pattern factor zeros on each side of the sidelobe. Since the sinc pattern has a $1/u$ sidelobe envelope, it was necessary only to modify the close-in zeros to reduce the close-in sidelobes. The far-out zeros were left at the integers. The shifting was accomplished by setting zeros equal to

$$u = \sqrt{n^2 + B^2} \tag{3.26}$$

where B is a positive real parameter. The pattern function is obtained by writing the sinc function as the ratio of two infinite products. Then the zeros are shifted as mentioned. In this form the pattern function, where C is a constant, is

$$F(u) = \frac{C \prod_{n=1}^{\infty} \left[1 - \dfrac{u^2 - B^2}{n^2}\right]}{\prod_{n=1}^{\infty} \left[1 + \dfrac{B^2}{n^2}\right]} \tag{3.27}$$

This is more simply written as

$$\begin{aligned} F(u) &= \frac{\sinh \pi\sqrt{B^2 - u^2}}{\pi\sqrt{B^2 - u^2}} \qquad u \le B \\ F(u) &= \frac{\sin \pi\sqrt{u^2 - B^2}}{\pi\sqrt{u^2 - B^2}} \qquad u \ge B \end{aligned} \tag{3.28}$$

Now the constant $C = \sinh(\pi B)/\pi B$. This pattern function then is a modified sinc pattern, with the single parameter B controlling all characteristics: sidelobe level, beamwidth, directivity efficiency, beam efficiency, and so on. A transition from the sinc to the hyperbolic form occurs on the side of the main beam at $u = B$. For smaller u the hyperbolic form provides the central part of the main beam, with a peak value of $\sinh(\pi B)/\pi B$. The trig form provides the lower part of the main beam and the sidelobes. The sidelobe ratio is immediately that of the sinc function multiplied by the beam peak value. In dB, this is

$$\text{SLR} = 20 \log \frac{\sinh \pi B}{\pi B} + 13.2614 \text{ dB} \tag{3.29}$$

A typical Taylor one-parameter pattern, plotted versus u, is shown in Figure 3.10.

The beamwidth is found by applying a root finder to the equation

$$\sqrt{2} F(u_3) = \frac{\sinh \pi B}{\pi B} \tag{3.30}$$

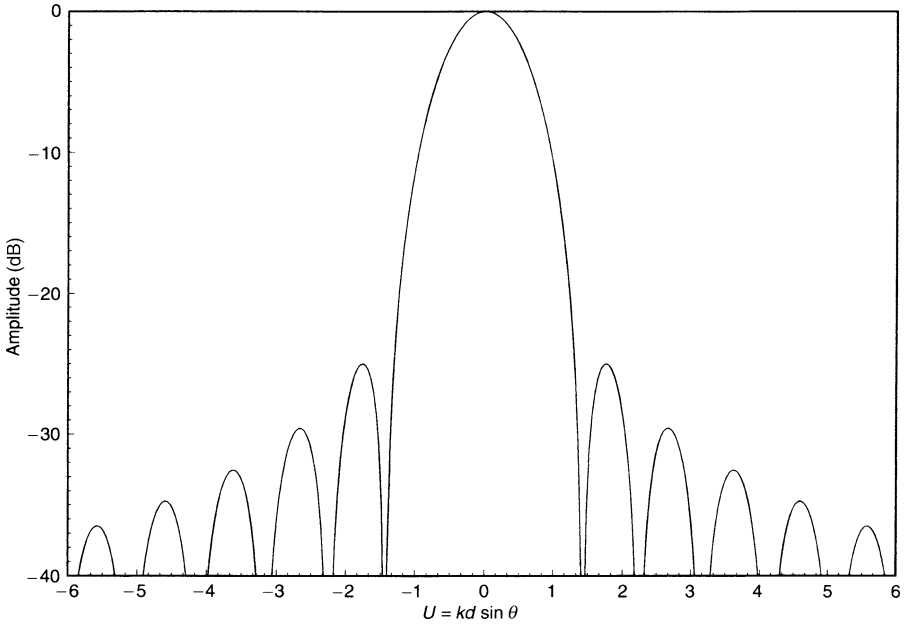


FIGURE 3.10 Taylor one-parameter pattern, SLR = 25 dB.

These values are given in Table 3.1 for sidelobe ratios up to 50 dB. The u_3 beamwidth is converted to physical angle θ_3 using

$$\frac{\theta_3}{\lambda/L} = \frac{2L}{\lambda} \arcsin \frac{u_3}{L/\lambda} \tag{3.31}$$

For long linear apertures, the normalized beamwidth is independent of L/λ and is

$$\frac{\theta_3}{\lambda/L} \simeq 2u_3 \tag{3.32}$$

TABLE 3.1 Taylor One-Parameter Characteristics

SLR (dB)	B	u_3 (rad)	η_t	$\eta_t u_3$	Edge Taper (dB)
13.26	0	0.4429	1	0.443	0
15	0.3558	0.4615	0.993	0.457	2.5
20	0.7386	0.5119	0.933	0.478	9.2
25	1.0229	0.5580	0.863	0.481	15.3
30	1.2762	0.6002	0.801	0.481	21.1
35	1.5136	0.6391	0.751	0.480	26.8
40	1.7415	0.6752	0.709	0.479	32.4
45	1.9628	0.7091	0.674	0.478	37.9
50	2.1793	0.7411	0.645	0.478	43.3

Figure 3.11 shows normalized beamwidth for apertures of length 5, 10, and 50 wavelengths. It can be seen that there is only a small difference between 5 and 10 wavelength apertures for the larger sidelobe apertures. Then, for most cases the approximate formula is satisfactory.

For some mapping and radiometer applications, the fraction of energy radiated in the main beam null-to-null is important, which is called beam efficiency η_b. The beam efficiency is calculated by integrating the square of the pattern function over the main beam, and dividing by the integral of the square of the pattern function over the entire pattern. This would normally require numerical integration over all of the pattern sidelobes, an expensive and relatively inaccurate task. However, it will be seen later that the taper efficiency η_t also involves this integration over all the pattern sidelobes. This efficiency allows the beam efficiency to be written in terms of the taper efficiency and an integral only over the main beam, a relatively simple and fast integration. This result is

$$\eta_b = \frac{2L\eta_t}{F_0^2\lambda} \int_0^{\theta_1} |F|^2 \cos\theta \, d\theta \tag{3.33}$$

Figure 3.12 shows beam efficiency versus sidelobe ratio. The curve is independent of L/λ for L/λ as small as 5.

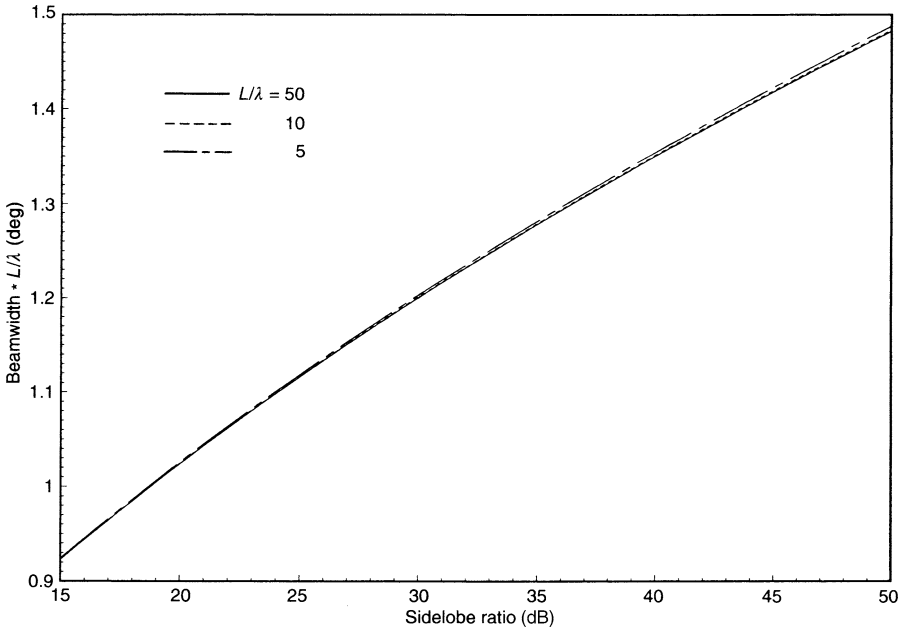


FIGURE 3.11 Taylor one-parameter distribution beamwidth.

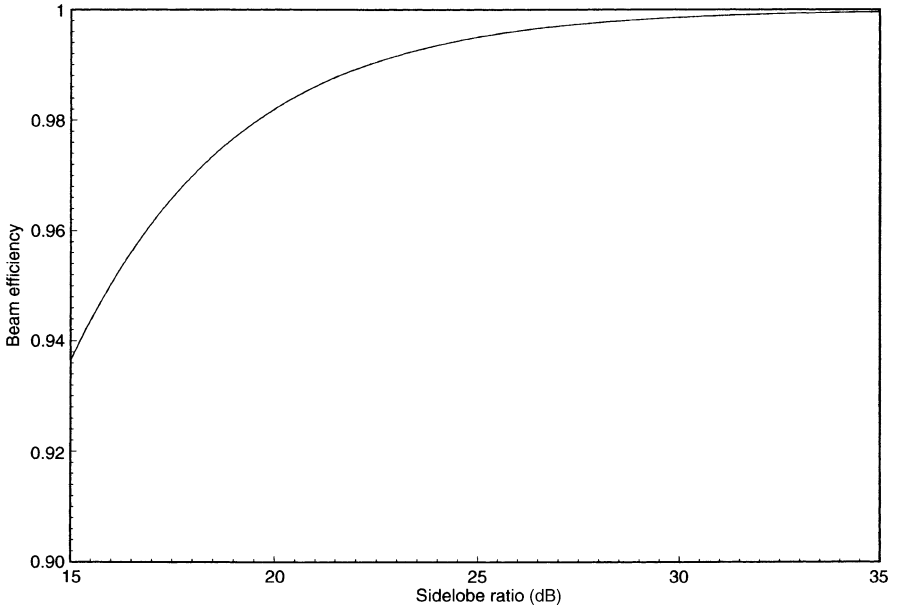


FIGURE 3.12 Taylor one-parameter beam efficiency.

The aperture distribution is the inverse transform of Eq. (3.28):

$$g(p) = I_0\left(\pi B\sqrt{1-p^2}\right) \tag{3.34}$$

where p is the distance from aperture center to aperture end, with $p = 1$ at the end. The parameter I_0 is the zero-order modified Bessel function. The pedestal height is $1/I_0(\pi B)$. Aperture amplitude distributions from center to end are given for several SLRs in Figure 3.13. Aperture (taper) efficiency is given by

$$\eta_t = \frac{|F(0)|^2}{\displaystyle\int_{-L/2\lambda}^{L/2\lambda} |F(u)|^2 du} \tag{3.35}$$

Because of the rapid sidelobe decay with u, the integral limits may be approximated by infinity even for modest apertures. The integration can then be reduced to a tabulated integral (Rothman, 1949):

$$\eta_t = \frac{2\sinh^2 \pi B}{\pi B \bar{I}_0(2\pi B)} \tag{3.36}$$

where $\bar{I}_0$ is the integral of I_0 from zero to the argument. Values of aperture efficiency are shown in Table 3.2.

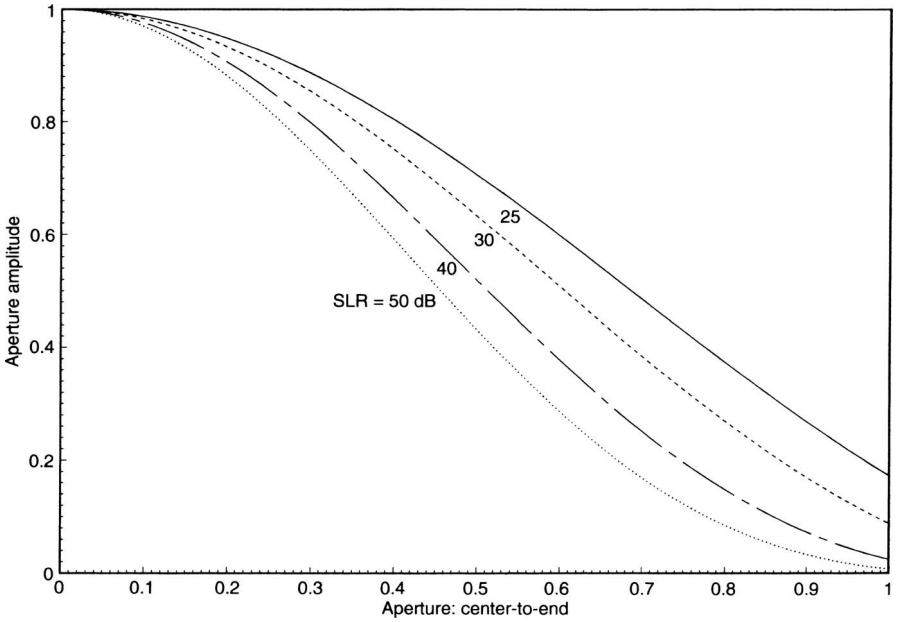


FIGURE 3.13 Taylor one-parameter aperture distribution.

TABLE 3.2 Taylor One-Parameter Efficiencies

SLR (dB)	B	η_t	η_b
13.26	0	1	0.9028
15	0.3558	0.9931	0.9364
20	0.7386	0.9329	0.9820
25	1.0229	0.8626	0.9950
30	1.2762	0.8014	0.9986
35	1.5136	0.7509	0.9996
40	1.7415	0.7090	0.9999
45	1.9628	0.6740	1.0000
50	2.1793	0.6451	1.0020

3.3.2 Bickmore–Spellmire Two-Parameter Distribution

Occasionally, it is desirable to provide an envelope taper more rapid than $1/u$. The two-parameter family of distributions developed by Bickmore and Spellmire (1956) allows this. The two-parameter pattern factor is

$$F(u) = \frac{J_v \pi\sqrt{u^2 - C^2}}{(\pi\sqrt{u^2 - C^2})^v} \tag{3.37}$$

A value of $v = 1/2$ yields the one-parameter pattern while $v > 1/2$ gives envelope tapers more rapid than $1/u$. When $v = -1/2$, the "ideal" pattern of Eq. (3.28) results. The parameter C controls initial sidelobe level. Note that these two parameters are independent, unlike the obsolete multiple-parameter distributions previously discussed. The aperture distribution is

$$g(p) = \frac{J_{v-1/2}\left(j\pi C\sqrt{1-p^2}\right)}{\left(j\pi C\sqrt{1-p^2}\right)^{v-1/2}} \tag{3.38}$$

When $v < 1/2$, singularities occur at the aperture ends. A value of $v = 3/2$, for example, gives a $1/u^2$ sidelobe envelope taper. These more rapid tapers are seldom employed because of their low-aperture efficiency.

3.4 TAYLOR *N*-BAR APERTURE DISTRIBUTION

The Taylor $\bar{n}$ *distribution* was developed as a compromise between the Dolph–Chebyshev or "ideal" aperture with its constant-level sidelobes, and the $1/u$ sidelobe envelope falloff of the $\sin(x)/x$ pattern. The goal of this $\bar{n}$ distribution is to obtain higher efficiency while retaining most of the advantages of a tapered distribution. To accomplish this, Taylor started from a different point from that used for the one-parameter distribution. Here his starting point is the "ideal" linear aperture, which has a pattern of equal-level sidelobes like the Dolph–Chebyshev array distribution. Thus for any linear phase uniform aperture, it provides the highest directivity for a given sidelobe ratio. The equal-level sidelobes are provided by a cosine function, with the main beam provided by a hyperbolic cosine:

$$\begin{aligned} F(u) &= \cosh \pi\sqrt{A^2 - u^2} \qquad u \le A \\ F(u) &= \cos \pi\sqrt{u^2 - A^2} \qquad u \ge A \end{aligned} \tag{3.39}$$

The single parameter A controls the distribution and pattern characteristics. Pattern zeros $z_n = \pm\sqrt{A^2 + (n - 1/2)^2}$. The sidelobe ratio is the value at $u = 0$:

$$\text{SLR} = \cosh A \tag{3.40}$$

and the parameter in terms of SLR is

$$A = \frac{1}{\pi} \ln\left(\text{SLR} + \sqrt{\text{SLR}^2 - 1}\right) \tag{3.41}$$

The 3-dB beamwidth is given by

$$u_3 = \frac{1}{\pi}\left[\ln^2\left(\mathrm{SLR} + \sqrt{\mathrm{SLR}^2 - 1}\right) - \ln^2\left(\mathrm{SLR}/\sqrt{2} + \sqrt{\mathrm{SLR}^2/2 - 1}\right)\right]^{1/2} \tag{3.42}$$

There is a transition on the side of the main beam at $u = A$ between the trig and hyperbolic forms. This canonical distribution is never used, as it not only has high far-out sidelobes, but the distribution has an infinite spike at the ends. Of course, this is the limiting value of the Dolph–Chebyshev distribution as the number of elements increases. However the "ideal" distribution led to a widely used distribution, the Taylor $\bar{\mathrm{n}}$ (Taylor, 1953).

Taylor utilized the first few equal sidelobes of the "ideal" distribution to provide increased efficiency, and shifted zeros of the far-out sidelobes to produce a $1/u$ envelope to reduce interference and clutter, and to provide a low-Q distribution. To do this, the u scale is stretched slightly by a factor σ, with σ slightly greater than unity. Thus the close-in zero locations are not changed significantly. At some point, a zero will fall at an integer; from this transition point on, the zeros occur at $\pm\mathrm{n}$. The pattern then has $\bar{\mathrm{n}}$ roughly equal sidelobes; beyond $u = \bar{\mathrm{n}}$ the sidelobe envelope decays as $1/u$. The first $\bar{\mathrm{n}}$ lobes are not of precisely equal heights, as the transition region allows some decay of the sidelobes near the transition point. The envelope beyond the transition point is slightly different from $1/u$. The zeros are

$$\begin{aligned} z_n &= \pm\sigma\sqrt{A^2 + (n - \tfrac{1}{2})^2} \qquad 1 \le n \le \bar{\mathrm{n}} \\ z_n &= \pm n \qquad\qquad\qquad\qquad\quad \bar{\mathrm{n}} \le n \end{aligned} \tag{3.43}$$

The dilation parameter is

$$\sigma = \frac{\bar{\mathrm{n}}}{\sqrt{A^2 + (\bar{\mathrm{n}} - \frac{1}{2})^2}} \tag{3.44}$$

The pattern function incorporating these zeros was first written as an infinite product on the zeros, starting with the expansion of the "ideal" aperture. For both conceptual and computational reasons, it is advantageous to incorporate the regularly spaced zeros into a trig function, with the $\bar{\mathrm{n}} - 1$ modified zeros in an explicit finite product:

$$F(u) = \frac{\sin \pi u}{\pi u}\prod_{n=1}^{\bar{\mathrm{n}}-1}\frac{1 - u^2/z_n^2}{1 - u^2/n^2} \tag{3.45}$$

This pattern function can also be written as a superposition of sinc beams each side of the central beam spaced at unit intervals in u:

$$F(u) = \sum_{n=-(\bar{\mathrm{n}}-1)}^{\bar{\mathrm{n}}-1} F(n, A, \bar{\mathrm{n}})\ \mathrm{sinc}[\pi(u + n)] \tag{3.46}$$

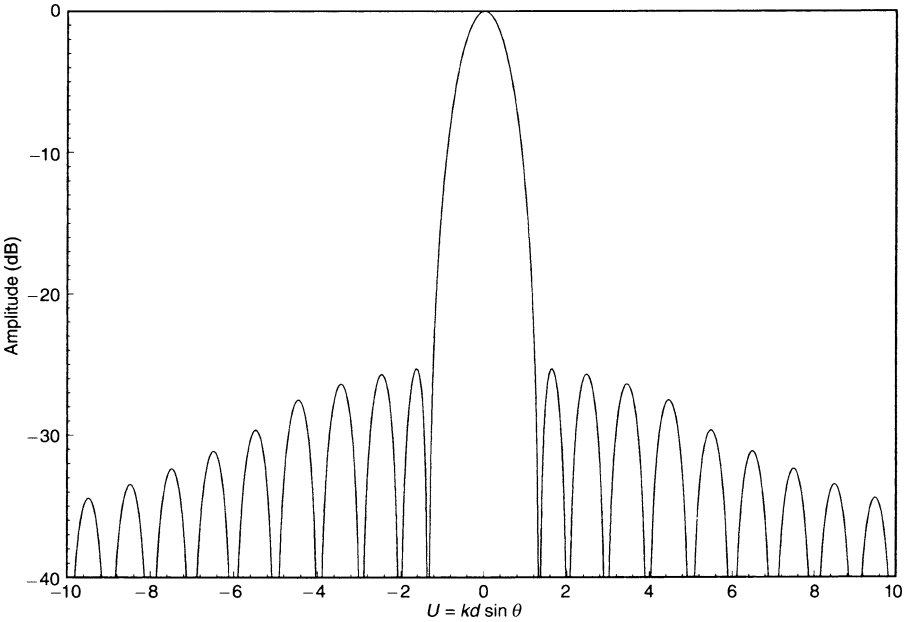


FIGURE 3.14 Taylor $\bar{n}$ pattern, SLR = 25 dB, $\bar{n} = 5$.

Figure 3.14 shows a typical Taylor $\bar{n}$ pattern in u. It may be observed that the first four sidelobes have an envelope closer to constant level than to $1/u$; the $\bar{n}$ transition forces a gradual change in sidelobe height. For large $\bar{n}$, the nearly constant sidelobes are more pronounced.

The beamwidth is accurately given by σu_3^i, where u_3^i is the "ideal" beamwidth as given in Eq. (3.42). Table 3.3 gives the parameter A versus sidelobe ratio, along with the beamwidth and dilation parameter. Normalized beamwidth, BW $\times L/\lambda$, is shown in Figure 3.15. A continuous curve is not used as the $\bar{n}$ value for each SLR is different. Also shown is the beamwidth curve for the Taylor *one-parameter distribution*. As expected, the $\bar{n}$ distribution is significantly better, especially for high SLR.

TABLE 3.3 Taylor $\bar{n}$ Characteristics

SLR			σ				
(dB)	A	u_3	$\bar{n} = 2$	$\bar{n} = 4$	$\bar{n} = 6$	$\bar{n} = 8$	$\bar{n} = 10$
20	0.9528	0.4465	1.1255	1.1027	1.0749	1.0582	1.0474
25	1.1366	0.4890		1.0870	1.0683	1.0546	1.0452
30	1.3200	0.5284		1.0693	1.0608	1.0505	1.0426
35	1.5032	0.5653			1.0523	1.0459	1.0397
40	1.6865	0.6000			1.0430	1.0407	1.0364
45	1.8697	0.6328				1.0350	1.0328
50	2.0530	0.6639					1.0289

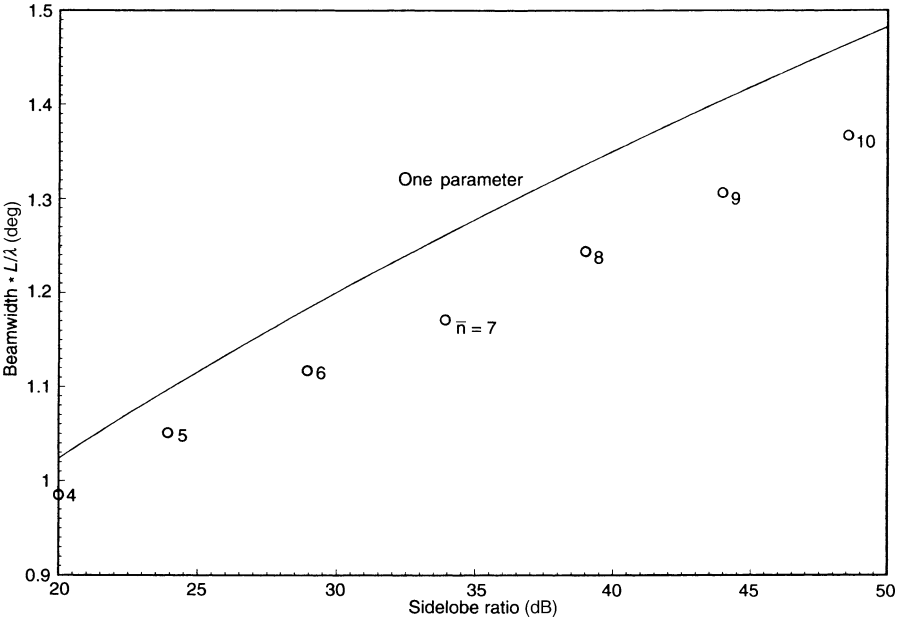


FIGURE 3.15 Taylor distribution beamwidth.

The aperture distribution is best expressed as a finite Fourier series:

$$g(p) = 1 + 2\sum_{n=1}^{\bar{n}-1} F(n, A, \bar{n}) \cos n\pi p \tag{3.47}$$

Again the aperture variable p is zero at the center and unity at the ends. The coefficient used in this formula, and in the previous pattern formula, is

$$F(n, A, \bar{n}) = \frac{[(\bar{n}-1)!]^2}{(\bar{n}-1+n)!(\bar{n}-1-n)!} \prod_{m=1}^{\bar{n}-1} \left[1 - \frac{n^2}{z_m^2}\right] \tag{3.48}$$

where $F(0, A, \bar{n}) = 1$. The amplitude distribution for one-half of the $\bar{n}$ aperture is given in Figure 3.16. In comparison with Figure 3.13 for the *one-parameter distribution*, the $\bar{n}$ "tails" are much higher. That is, the distribution pedestal is higher, leading to better efficiency in accordance with Taylor's rules. Tables of aperture distribution and coefficients are given for SLR = 20(5)40 dB and $\bar{n} = 3(1)10$ by Hansen (1964). More extensive tables are available in Brown and Scharp (1958).

Directivity and aperture efficiency are easily calculated. It is easier to integrate the aperture distribution, since it is expressed as a Fourier series of orthogonal terms,

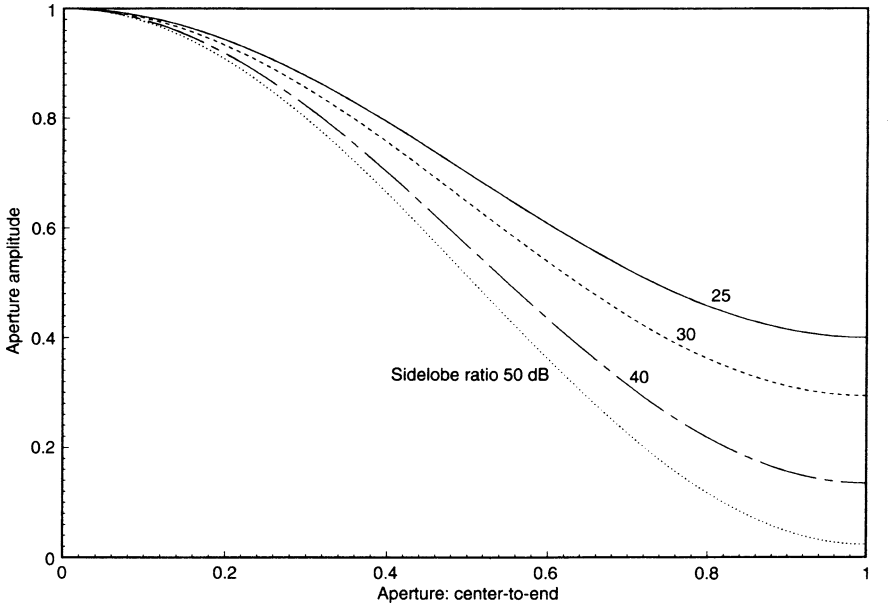


FIGURE 3.16 Taylor $\bar{n}$ aperture distribution.

than to integrate the pattern function; this expression is accurate for apertures large in wavelengths (Friis and Lewis, 1947).

$$\eta = \frac{\left| \int_0^1 g(p)\, dp \right|^2}{\int_0^1 g^2(p)\, dp} \tag{3.49}$$

By using Eq. (3.47) for the distribution function, the numerator of the efficiency expression is unity since the cosine terms integrate to zero. In the denominator, the g^2 product gives unity plus two single series plus a double series. Each term of the single series integrates to zero. The double series integral is zero for n not equal to m, and gives one-half for $n = m$. Thus the aperture efficiency is

$$\frac{1}{\eta} = 1 + 2 \sum_{n=1}^{\bar{n}-1} F^2(n, A, \bar{n}) \tag{3.50}$$

Calculation is easy even for large values on $\bar{n}$. For each sidelobe ratio there is a value of $\bar{n}$ that gives maximum efficiency because, as $\bar{n}$ increases from a small value, the main beam energy decreases faster than the sidelobe energy increases. For large $\bar{n}$, however, the main beam energy changes more slowly. Typical $\bar{n}$ aperture efficiency values are shown in Figure 3.17. Since each SLR value has a different $\bar{n}$ value, a continuous curve is not used. For comparison, the efficiency of the *one-parameter distribution* is also shown. Table 3.4 gives the values of $\bar{n}$ that yield maximum efficiency. Also

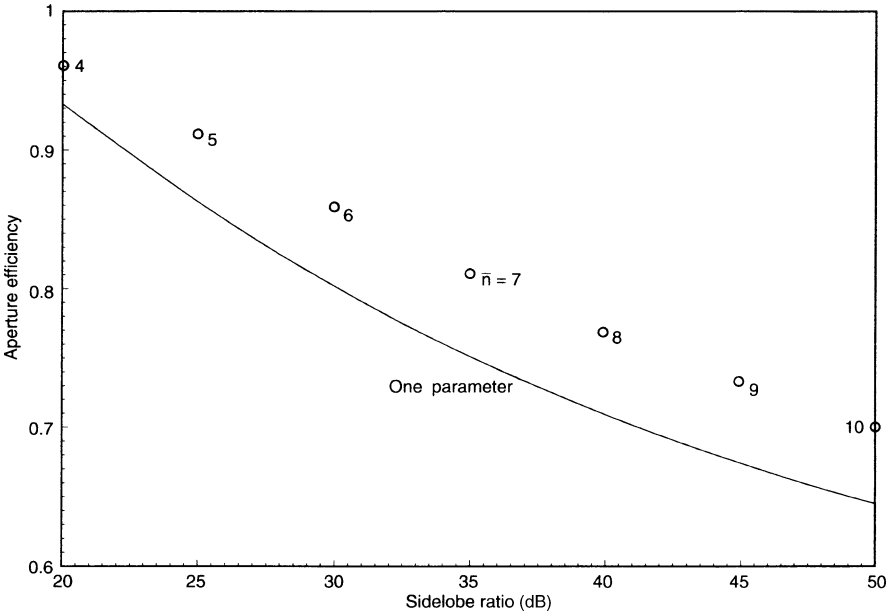


FIGURE 3.17 Taylor distribution aperture efficiency.

TABLE 3.4 Maximum Efficiency and Monotonic n̄ Values

	Max η_t Values		Monotonic	
SLR	n̄	η_t	n̄	η_t
20	6	0.9667	3	0.9535
25	12	0.9252	5	0.9105
30	23	0.8787	7	0.8619
35	44	0.8326	9	0.8151
40	81	0.7899	11	0.7729
45				
50				

shown are the largest values of n̄ that allow a monotonic distribution, along with the corresponding aperture efficiencies. The efficiency penalty for choosing a monotonic n̄ over the maximum efficiency n̄ is modest, as may be observed.

Beam efficiency is calculated using Eq. (3.33), with the aperture efficiency obtained above. Table 3.5 gives values for SLR from 20 to 35 dB with typical n̄ values for each. These results may be compared with those of Figure 3.12 for the *one-parameter distribution*.

It is important to discuss the proper range of values of n̄. Too large a value of n̄ for a particular sidelobe ratio will, as shown, give a nonmonotonic aperture distribution; a large peak may even be produced at the ends of the aperture. Too large a value of n̄ will also not allow the transition zone zeros to behave properly. Figure 3.18 shows the

TABLE 3.5 Taylor $\bar{n}$ Beam Efficiency

SLR (dB)	$\bar{n}$	η_b
20	4	0.9658
25	5	0.9861
30	6	0.9947
35	7	0.9980

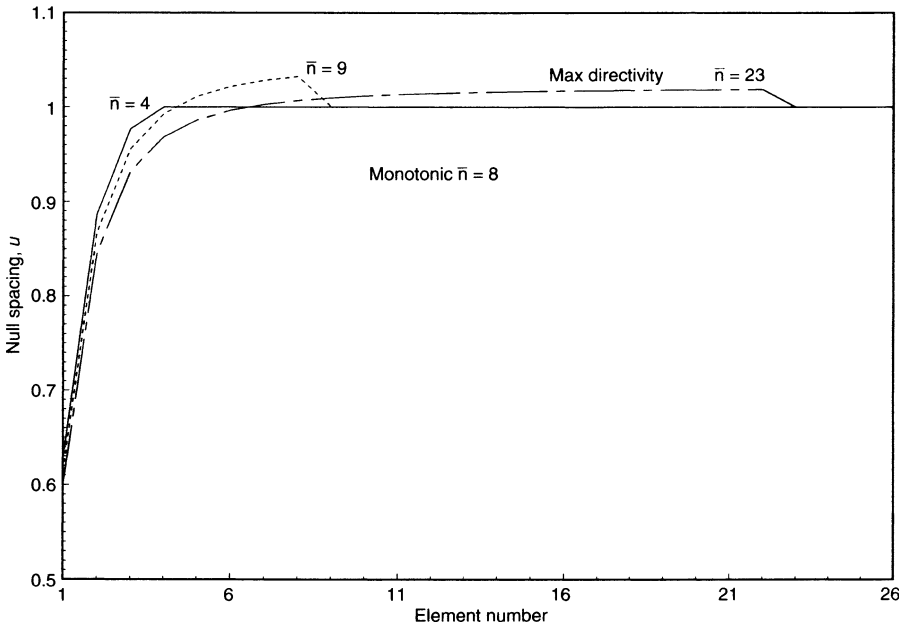


FIGURE 3.18 Taylor $\bar{n}$ distribution null spacing, SLR = 30 dB.

behavior of zero spacings on $\bar{n}$ for an SLR of 30 dB. The largest overshoot occurs for $\bar{n} = 9$, just above the largest monotonic value of $\bar{n} = 8$. Overshoots occur out to (and beyond) the maximum efficiency $\bar{n} = 23$. However, an $\bar{n} = 4$ shows no overshoot; $\bar{n} = 5$ and 6 give small overshoots. These characteristics occur also for other values of SLR, but for different $\bar{n}$ value.

In summary, the Taylor $\bar{n}$ *distribution* is widely used because it gives slightly better efficiency and beamwidth than the Taylor *one-parameter distribution*, for the same sidelobe level.

3.5 LOW-SIDELOBE DISTRIBUTIONS

3.5.1 Comparison of Distributions

Low sidelobes, which might be roughly defined as -30 to -60 dB, are of interest for several reasons: reduction of radar and communications intercept probability,

reduction of radar clutter and jammer vulnerability, and increasing spectrum congestion in satellite transmissions. From the previous sections, it is clear that a low sidelobe distribution should be heavily tapered in amplitude, and from Taylor's rules have a reasonable pedestal height. The $\bar{n}$ space factor appears to be ideal for these applications.

The design of a high-performance low-sidelobe pattern should again emphasize the pattern zeros. The close-in zeros are adjusted to obtain a few nearly equal sidelobes at the design sidelobe level, while farther-out zeros are placed to give a $1/u$ envelope. This scientifically designed pattern and distribution is in stark contrast to the classical World War II era distributions, which were all chosen for easy integrability. For example, the Hamming distribution is a cosine of a doubled argument, on a pedestal (Blackman and Tukey, 1958). This distribution is given by

$$g(p) = 0.54 + 0.46 \cos \pi p = a + b \cos \pi p \tag{3.51}$$

giving a value of unity in the center and an end value of 0.08. The excitation efficiency is

$$\eta_t = \frac{a^2}{a^2 + 2b^2} = 0.7338 \tag{3.52}$$

and the half-power beamwidth is $u_3 = 0.6515$. The pattern function has zeros at $u = 2, 3, 4, \ldots$, with another zero at $u = a/(a - b) = 2.5981$. The pattern function is

$$F(u) = \frac{[(a - b)u^2 - a] \sin \pi u}{\pi a u(u^2 - 1)} \tag{3.53}$$

The close spacing of the first two zeros produces an irregular sidelobe envelope; the highest sidelobe is the fourth at -42.7 dB. The first and second sidelobes are -44.0 and -56.0 dB.

A comparison of the Taylor one-parameter, Taylor $\bar{n}$, and Hamming distributions is illuminating. Since the Hamming SLR = 42.7 dB, this is used for the others as well. Table 3.6 gives the first 10 pattern zeros for these distributions, with values of $\bar{n} = 6$ and 10 used. All of the Taylor distributions exhibit a relatively smooth progression of zeros, but the Hamming has a singular discontinuity at the second zero. Table 3.7 gives the spacing between zeros corresponding to Table 3.6. Here the smooth progression of all the Taylor distributions is again evident, although there is a slight oscillation at the transition $\bar{n}$ for the $\bar{n} = 10$ case. Note that the Hamming null spacings change from roughly 0.6 to 0.4 to 1, clearly an inefficient way to design an aperture distribution. Finally, Table 3.8 gives the beamwidth in u and the aperture efficiency. The Hamming is roughly 3% poorer in both indices. Although no data are given, the Taylor distributions have an intrinsically lower Q. That is, errors in aperture excitation will affect the Hamming distribution more, owing to its higher Q.

Because phase errors affect low-sidelobe patterns strongly, it may be expected that measurement distance will change such patterns. The conventional far-field distance

TABLE 3.6 Zeros for SLR = 42.7 dB

	Taylor	Taylor		
n	One-Parameter	$\bar{n} = 6$	$\bar{n} = 10$	Hamming
1	2.112	1.894	1.897	2
2	2.732	2.398	2.396	2.598
3	3.550	3.173	3.166	3
4	4.412	4.069	4.056	4
5	5.335	5.020	5.002	5
6	6.282	6	5.978	6
7	7.243	7	6.970	7
8	8.214	8	7.974	8
9	9.190	9	8.984	9
10	10.172	10	10	10

TABLE 3.7 Null Spacings for SLR = 42.7 dB

	Taylor	Taylor		
n	One-Parameter	$\bar{n} = 6$	$\bar{n} = 10$	Hamming
1	0.619	0.503	0.499	0.598
2	0.799	0.776	0.770	0.402
3	0.881	0.895	0.890	1
4	0.923	0.951	0.946	1
5	0.947	0.980	0.976	1
6	0.961	1.0	0.993	1
7	0.970	1.0	1.003	1
8	0.977	1.0	1.011	1
9	0.981	1.0	1.016	1
10	0.985	1.0	1.0	1

TABLE 3.8 Comparison of Distributions for SLR = 42.7 dB

	Taylor	Taylor		
	One-Parameter	$\bar{n} = 6$	$\bar{n} = 10$	Hamming
u_3	0.694	0.637	0.635	0.651
η_t	0.690	0.754	0.755	0.734
$u_3\eta_t$	0.478	0.480	0.480	0.478

of $2L^2/\lambda$, where L is array length, for measurement and operation, may not be sufficient for low-sidelobe designs. As the observation distance moves in from infinity, the first sidelobe rises and the null starts filling. Then the sidelobe becomes a shoulder on the now wider main beam, and the second null rises. This process continues as the distance decreases. To first order, the results are dependent only on design sidelobe level. Details are given in Chapter 15.

3.5.2 Average Sidelobe Level

It is sometimes desirable to be able to estimate the average sidelobe level for an antenna. Because the average level and the pattern envelope are approximately related to the antenna directivity, this can be done; the Taylor one-parameter distribution will be used as an example. Since most antenna patterns have a sidelobe envelope with $1/u$ asymptotic decay, this example is widely applicable. The average power sidelobe level is defined here as the integral from the first null to $\pi/2$:

$$\begin{aligned} \mathrm{SLL}_{\mathrm{ave}} &= \frac{\lambda}{\pi^2 F_0^2 L} \int_{u_0}^{L/\lambda} \frac{du}{u^2 - B^2} \\ &= \frac{\lambda}{2\pi^2 F_0^2 LB} \ln \frac{(L/\lambda - B)(u_0 + B)}{(L/\lambda + B)(u_0 - B)} \end{aligned} \tag{3.54}$$

The first null is at $u_0 = \sqrt{1 + B^2}$. For long apertures, $L/\lambda \gg B$, which gives

$$\mathrm{SLL}_{\mathrm{ave}} \simeq \frac{\lambda}{\pi^2 F_0^2 LB} \ln \frac{u_0 + B}{u_0 - B} \tag{3.55}$$

The actual sidelobe power is less because of the sidelobe shape. A typical shape, independent of the envelope, is $\sin^2 \pi u$, giving an integral of $1/2$. Thus the sidelobe

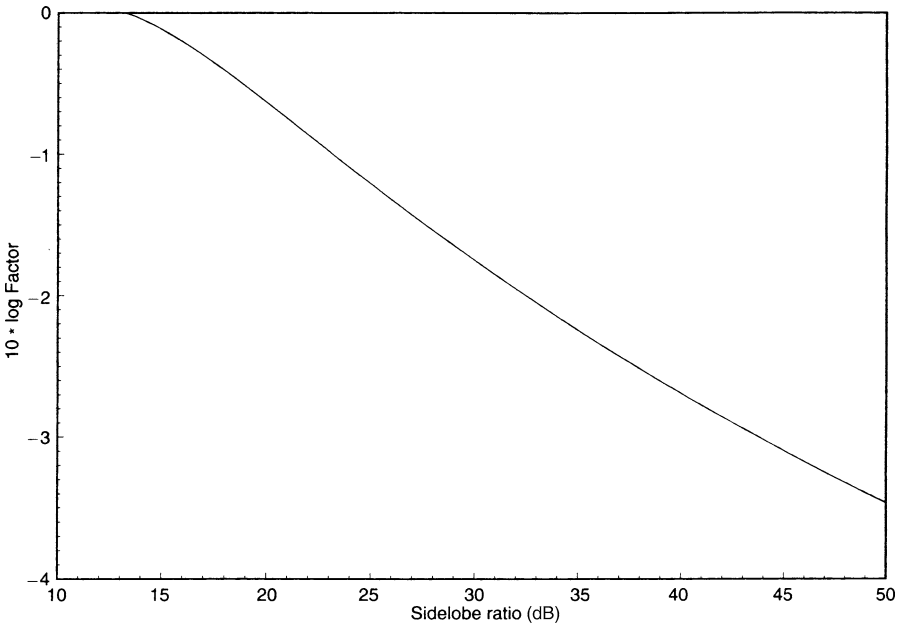


FIGURE 3.19 Average sidelobe level factor.

power is 3 dB below the envelope. The directivity is $G = 2\eta_t L/\lambda$, with η_t the aperture efficiency, and average sidelobe level may be written in terms of directivity:

$$\mathrm{SLL}_{\mathrm{ave}} \simeq \frac{\eta_t}{2\pi^2 F_0^2 GB} \ln\left(\frac{u_0 + B}{u_0 - B}\right) \tag{3.56}$$

Given an SLR, B is fixed, and the right-hand factor can be calculated; it is shown in Figure 3.19 and represents a correction of <3 dB for high SLR. In dB terms, the average sidelobe level becomes

$$\mathrm{SLR}_{\mathrm{ave}} \simeq G_{\mathrm{dB}} + (\mathrm{SLR} - 13.26\,\mathrm{dB}) - 10\log\left[\frac{\eta_t}{2B} \ln\frac{u_0 + B}{u_0 - B}\right] + 9.95\,\mathrm{dB} \tag{3.57}$$

As expected, higher directivity yields lower average sidelobe level, as does SLR.

3.6 VILLENEUVE *N*-BAR ARRAY DISTRIBUTION

For most linear array designs, the number of elements is sufficiently large so that the continuous Taylor distributions previously discussed can be sampled. Villeneuve (1984) developed an elegant method for the design of Taylor $\bar{\mathrm{n}}$ arrays for a small number of elements. This was accomplished by adjusting $\bar{\mathrm{n}} - 1$ close-in zeros of the Dolph–Chebyshev array polynomial (instead of the close-in zeros of the continuous aperture function). The Chebyshev polynomial is written in product form so that the close-in zeros can be shifted. The Taylor dilation factor σ is also used. The result is the pattern function

$$\begin{aligned} F_m(w) \equiv E(\psi) = \exp jN\psi/2 \frac{\sin\left(\dfrac{2N+1}{2}\psi\right)}{\sin\left(\frac{1}{2}\psi\right)} \\ \times \frac{\displaystyle\prod_{m=1}^{\bar{\mathrm{n}}-1} \sin\left(\frac{\psi - \psi'_m}{2}\right) \sin\left(\frac{\psi + \psi'_m}{2}\right)}{\displaystyle\prod_{m=1}^{\bar{\mathrm{n}}-1} \sin\left(\frac{\psi - \dfrac{m2\pi}{2N+1}}{2}\right) \sin\left(\frac{\psi + \dfrac{m2\pi}{2N+1}}{2}\right)} \end{aligned} \tag{3.58}$$

where $\psi = kd\sin\theta$, $u = u_0 \cos\psi/2$, and $\psi' = \sigma\psi$.

This pattern is the discrete array equivalent of the Taylor $\bar{\mathrm{n}}$ for a continuous aperture. The dilation factor is

$$\sigma = \frac{2\pi\bar{\mathrm{n}}}{(2N+1)\psi_{\bar{\mathrm{n}}}} \tag{3.59}$$

The amplitude excitation coefficients are found by applying a finite transform to the pattern function. The coefficients are slightly different for even and odd numbers of elements. For an even number of elements $N = 2M$, the following relations hold:

$$\begin{aligned} u_0 &= \cosh\left[\frac{1}{2N-1}\ln\left(\eta+\sqrt{\eta^2-1}\right)\right] \\ \psi_p &= 2\arccos\left[\frac{1}{u_0}\cos(2p-1)\frac{\pi}{2(2N-1)}\right] \qquad p=1,\ldots,2N-1 \\ \sigma &= \frac{\bar{n}2\pi}{2N\psi_{\bar{n}}} = \frac{\bar{n}\pi}{2N\arccos\left[\frac{1}{u_0}\cos\left((2\bar{n}-1)\frac{\pi}{2(2N-1)}\right)\right]} \end{aligned} \tag{3.60}$$

The coefficients are now given by

$$\begin{aligned} a_p &= \frac{1}{2N}\sum_{m=-(\bar{n}-1)}^{\bar{n}-1} E\left(\frac{m2\pi}{2N}\right)\exp\{-j[(p-1/2)m2\pi/(2N)]\}, \quad -(N-1)<p<N \\ &= \frac{1}{2N}\left\{E(0)+2\sum_{m=1}^{\bar{n}-1}E\left(\frac{m2\pi}{2N}\right)\cos\left[(p-1/2)\frac{m\pi}{N}\right]\right\} \end{aligned} \tag{3.61}$$

where

$$\begin{aligned} E\left(\frac{m2\pi}{2N}\right) &= \frac{2N(-1)^m\prod_{q=1}^{\bar{n}-1}\sin\left(\frac{m\pi}{2N}-\frac{\psi_q'}{2}\right)\sin\left(\frac{m\pi}{2N}+\frac{\psi_q'}{2}\right)}{\sin\left(\frac{m\pi}{2N}\right)\sin\left(\frac{2m\pi}{2N}\right)\prod_{\substack{q=1\\q\neq m}}^{\bar{n}-1}\sin\left(\frac{(m-q)\pi}{2N}\right)\sin\left(\frac{(m+q)\pi}{2N}\right)} \\ E(0) &= \frac{2N\prod_{q=1}^{\bar{n}-1}\sin^2\left(\frac{\psi_q'}{2}\right)}{\prod_{q=1}^{\bar{n}-1}\sin^2\left(\frac{q\pi}{2N}\right)} \end{aligned} \tag{3.62}$$

For an odd number of elements, where $N = 2M + 1$, the relations are

$$\begin{aligned} \sigma &= \frac{\bar{n}\pi}{(2N+1)\arccos\left[\frac{1}{u_0}\cos\left((2\bar{n}-1)\frac{\pi}{4N}\right)\right]} \\ u_0 &= \cosh\left[\frac{1}{2N}\ln\left(\eta+\sqrt{\eta^2-1}\right)\right] \\ \psi_p &= 2\arccos\left[\frac{1}{u_0}\cos(2p-1)\frac{\pi}{4N}\right] \qquad p=1,\ldots,2N \end{aligned} \tag{3.63}$$

The coefficients become

$$a_p = \frac{1}{2N+1}\sum_{m=-(\bar{n}-1)}^{\bar{n}-1} E\left(\frac{m2\pi}{2N+1}\right)\exp\{-j[\,pm2\pi/(2N+1)]\} \quad -N \le p \le N$$

$$= \frac{1}{2N+1}\left[E(0) + 2\sum_{m=1}^{\bar{n}-1} E\left(\frac{m2\pi}{2N+1}\right)\cos\left(\frac{pm2\pi}{2N+1}\right)\right] \quad -N \le p \le N \qquad (3.64)$$

where

$$E\left(\frac{m2\pi}{2N+1}\right) = \frac{(2N+1)(-1)^m \prod_{q=1}^{\bar{n}-1} \sin\left(\frac{m\pi}{2N+1} - \frac{\psi_q'}{2}\right)\sin\left(\frac{m\pi}{2N+1} + \frac{\psi_q'}{2}\right)}{\sin\left(\frac{m\pi}{2N+1}\right)\sin\left(\frac{2m\pi}{2N+1}\right)\prod_{\substack{q=1\\ q\ne m}}^{\bar{n}-1}\sin\left(\frac{[m-q]\pi}{2N+1}\right)\sin\left(\frac{[m+q]\pi}{2N+1}\right)}$$

$$E(0) = \frac{(2N+1)\prod_{q=1}^{\bar{n}-1}\sin^2\left(\frac{\psi_q'}{2}\right)}{\prod_{q=1}^{\bar{n}-1}\sin^2\left(\frac{q\pi}{2N+1}\right)} \qquad (3.65)$$

These expressions appear complex, but are readily programmed for the computer.

TABLE 3.9 Villeneuve n̄ Array Efficiency

	SLR (dB)							
	25		30		35		40	
N	$\bar{n}$	η_t	$\bar{n}$	η_t	$\bar{n}$	η_t	$\bar{n}$	η_t
41	5	0.9026	7	0.8551	9	0.8098	12	0.7701
31	5	0.9000	7	0.8528	9	0.8078	12	0.7693
21	5	0.8947	7	0.8479	11	0.8060	11	0.7630
15	6	0.8917	7	0.8409	8	0.7949	8	0.7540
11	6	0.8807	6	0.8276	6	0.7825	6	0.7447
9	5	0.8674	5	0.8158	5	0.7732	5	0.7384
7	4	0.8468	4	0.7987	4	0.7608	4	0.7310
5	3	0.8182	3	0.7725	3	0.7448		

TABLE 3.10 Taylor $\bar{n}$ Array Efficiency

SLR (dB)	$\bar{n}$	η_t
25	5	0.9105
30	7	0.8619
35	9	0.8151
40	11	0.7729

The aperture efficiency of Villeneuve arrays has been computed by summing virtual mutual resistances (Hansen, 1985), using the formula

$$G = \frac{\left(\sum_{n=1}^{N} A_n\right)^2}{\sum_{n=1}^{N} \sum_{m=1}^{N} A_n A_m \operatorname{sinc}[(n-m)kd]} \tag{3.66}$$

Table 3.9 gives aperture efficiencies for Villeneuve odd arrays from 5 to 41 elements, with sidelobe ratios of 25–40 dB. For each case, $\bar{n}$ is chosen to be the largest that allows a monotonic distribution. These values can be compared with the corresponding Taylor $\bar{n}$ efficiencies in Table 3.10. These results show that there is only a modest penalty in arrays with as few as 10 elements. The tables allow the designer to determine when the size of the array requires use of a Villeneuve distribution.

3.7 DIFFERENCE PATTERNS

3.7.1 Canonical Patterns

A difference pattern typically has a null at broadside; these patterns are used for accurate determination of arrival angle. The widely used monopulse radar antenna employs a difference pattern. The aperture distributions considered here are those with a biphasal distribution, where one-half of the aperture is inphase, while the other half is 180° out of phase. The uniformly excited linear array, with the two halves of the array fed through a 180° hybrid, is a simple linear difference antenna. The hybrid provides a sum pattern that is sinc πu, and a difference pattern which is

$$F(u) = \frac{\sqrt{2}(1 - \cos \pi u)}{\pi u} \tag{3.67}$$

The difference distribution is normalized to make the integral of the aperture distribution squared equal to unity. For this simple difference pattern, the peak value is 1.0248, and the slope in the center is 0.7071. The first sidelobe is −10.57 dB below the peak. These high sidelobes make this uniform difference pattern unattractive.

The excitation efficiency is obtained from the formula for excitation efficiency for equiphase apertures (Friis and Lewis, 1947), modified to include the pattern peak location u_0 (again accurate for large apertures):

$$\eta_t = \frac{\left(\int g(p) \exp(j\pi p u_0)\, dp\right)^2}{2\int g^2(p)\, dp} \tag{3.68}$$

The result for the uniform difference aperture is $\eta_t = 0.7420$.

The sidelobes of the maximum slope difference pattern (Powers, 1967; Kirkpatrick, 1953) are even higher. This distribution is triangular, zero at the center and maximum at each edge. Using the same normalization, the peak (edge) value is 1.2247. The slope at the center is 0.8165 and the first sidelobe is -8.28 dB below the main beam. The pattern function is given by

$$F(u) = \frac{\sqrt{6}(\operatorname{sinc} \pi u - \cos \pi u)}{\pi u} \tag{3.69}$$

The excitation efficiency is $\eta_t = 0.5708$.

A third distribution is that providing maximum directivity. This maximum directivity pattern function is

$$F(u) = \operatorname{sinc}[\pi(u - u_m)] - \operatorname{sinc}[\pi(u + u_m)] \tag{3.70}$$

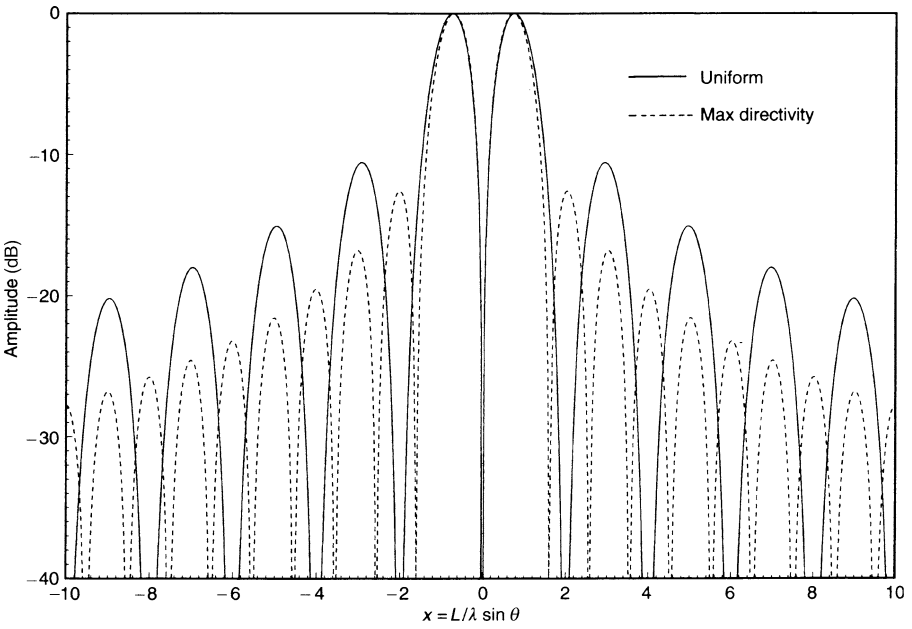


FIGURE 3.20 Uniform and maximum directivity difference patterns.

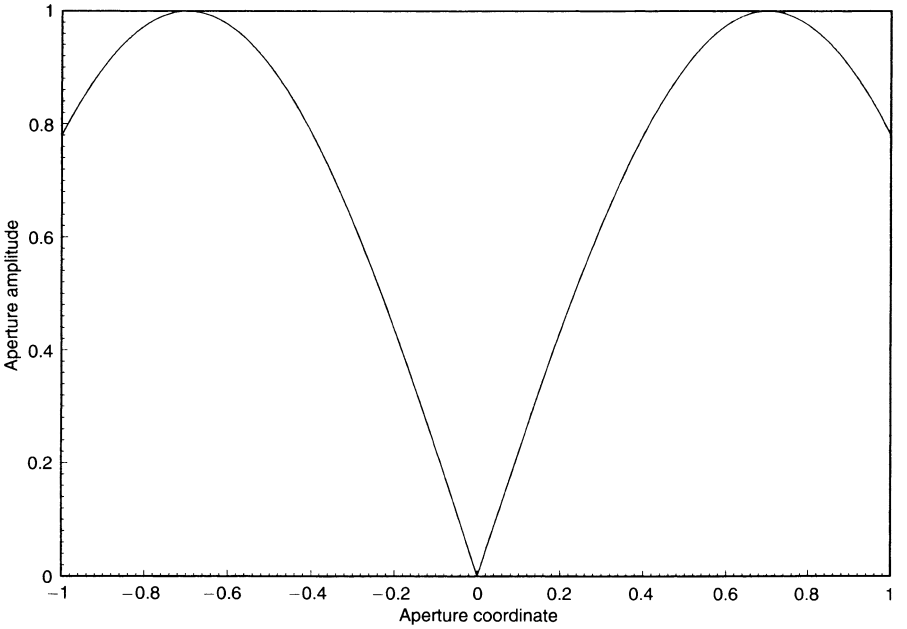


FIGURE 3.21 Maximum directivity difference pattern distribution.

where the constant $u_m = 0.715148$. The aperture distribution that corresponds is given by $g = \sin u_m \pi y$. The first sidelobe is -12.59 dB below the peak. The aperture efficiency is 0.6086. Figure 3.20 gives uniform and maximum directivity difference patterns. As might be expected, the latter has slightly narrower main difference beams and lower sidelobes. Of course the price is a complex aperture distribution, as shown in Figure 3.21. Each half of the distribution is a portion of a sine function. Here the peak of the aperture distribution has been normalized to unity for convenience in plotting.

A clever technique that provides a difference pattern (monopulse) from a planar array of waveguide edge slots has been demonstrated by Kinsey (1999). Alternate waveguide "sticks" are coupled to hybrid junctions to provide the monopulse tracking.

3.7.2 Bayliss Patterns

The high sidelobes of the preceding distributions can be reduced by applying the principles of Taylor to the difference pattern. This was done by Bayliss (1968), who produced a difference pattern analogous to the Taylor $\bar{n}$ pattern. Bayliss started with the Taylor "ideal" pattern function (see Section 3.4), then differentiated this to obtain a difference pattern. It was perhaps surprising that the resulting sidelobes were of irregular levels, with a nonmonotonic sidelobe envelope. An iterative procedure was used to adjust these zeros to yield equal-level sidelobes. Only four needed adjustment, but these four zeros depend on the SLR. Bayliss gives the four zeros, the Taylor

TABLE 3.11 Bayliss Roots and Parameters

	SLR (dB)					
	15	20	25	30	35	40
A	1.00790	1.22472	1.43546	1.64126	1.84308	2.04154
z_1	1.51240	1.69626	1.88266	2.07086	2.26025	2.45039
z_2	2.25610	2.36980	2.49432	2.62754	2.76748	2.91234
z_3	3.16932	3.24729	3.33506	3.43144	3.53521	3.64518
z_4	4.12639	4.18544	4.25273	4.32738	4.40934	4.49734
u_0	0.66291	0.71194	0.75693	0.79884	0.83847	0.87649

parameter A, and the difference peak location u_0 in terms of fourth-order polynomials in SLR (in dB). Table 3.11 gives the corrected Bayliss parameters, using the polynomial coefficients in Table 3.12. The other zeros are the same as for the linear Taylor $\bar{n}$ case:

$$z_n = \pm\sqrt{A^2 + n^2} \qquad n > 4 \tag{3.71}$$

The dilation factor is given by

$$\sigma = \frac{\bar{n} + \frac{1}{2}}{z_{\bar{n}}} \tag{3.72}$$

The pattern is provided by replacing the first $\bar{n}$ zeros of $\cos \pi u$ by the modified set. This results in a ratio of finite products for the pattern function:

$$F(u) = u \cos \pi u \frac{\prod_{n=1}^{\bar{n}-1} \left(1 - \dfrac{u^2}{\sigma^2 z_n^2}\right)}{\prod_{n=0}^{\bar{n}-1} \left(1 - \dfrac{u^2}{\left(n + \frac{1}{2}\right)^2}\right)} \tag{3.73}$$

TABLE 3.12 Bayliss Polynomial Coefficients

	C_1	$10C_2$	10^3C_3	10^5C_4	10^7C_5
A	0.30387530	−0.5042922	−0.27989	−0.343	−0.2
u_1	0.98583020	−0.3338850	0.14064	0.190	0.1
u_2	2.00337487	−0.1141548	0.41590	0.373	0.1
u_3	3.00636321	−0.0683394	0.29281	0.161	0.0
u_4	4.00518423	−0.0501795	0.21735	0.088	0.0
u_0	0.47972120	−0.1456692	−0.18739	−0.218	−0.1

The pattern function can also be written as a finite sum over $\bar{n}$ zeros:

$$F(u) = u \cos \pi u \sum_{n=0}^{\bar{n}-1} \frac{(-1)^n B_n}{\left(n + \frac{1}{2}\right)^2 - u^2} \tag{3.74}$$

Figure 3.22 shows a typical Bayliss pattern for SLR = 25 dB and $\bar{n} = 5$. As in the case of the Taylor $\bar{n}$ pattern, there is a smooth transition between the nearly equal-level sidelobes and the $1/u$ envelope sidelobes. The aperture distribution is also given by a finite $\bar{n}$ sum:

$$g(p) = \sum_{n=0}^{\bar{n}-1} B_n \sin\left(\frac{\pi p}{n + \frac{1}{2}}\right) \tag{3.75}$$

where the coefficients are

$$B_m = -(-1)^m \left(m - \tfrac{1}{2}\right)^2 \frac{\prod_{n=1}^{\bar{n}-1} \left(1 - \dfrac{\left(m + \frac{1}{2}\right)^2}{\sigma^2 z_n^2}\right)}{\prod_{\substack{n=0 \\ n \neq m}}^{\bar{n}-1} \left(1 - \dfrac{\left(m + \frac{1}{2}\right)^2}{\left(n + \frac{1}{2}\right)^2}\right)} \tag{3.76}$$

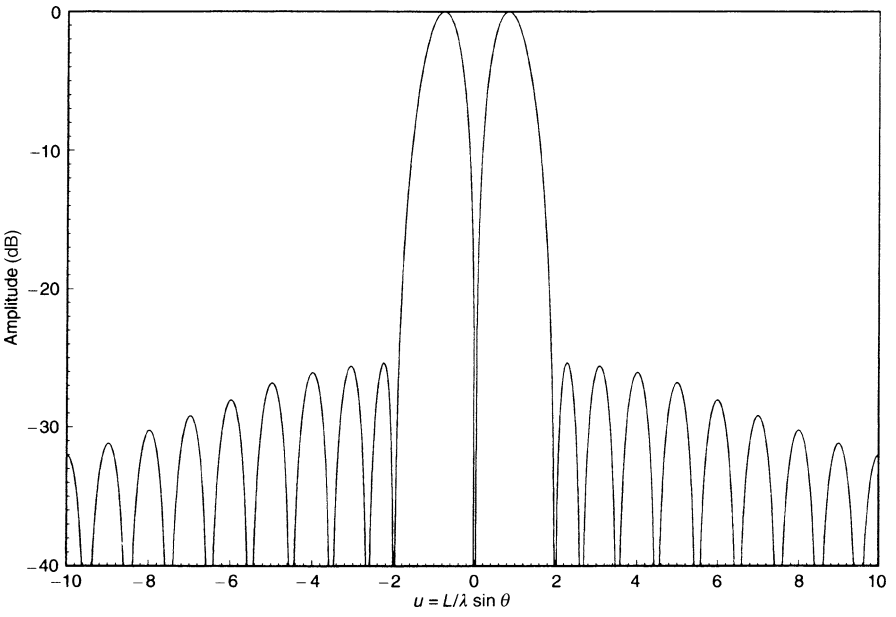


FIGURE 3.22 Bayliss difference pattern, $\bar{n} = 5$.

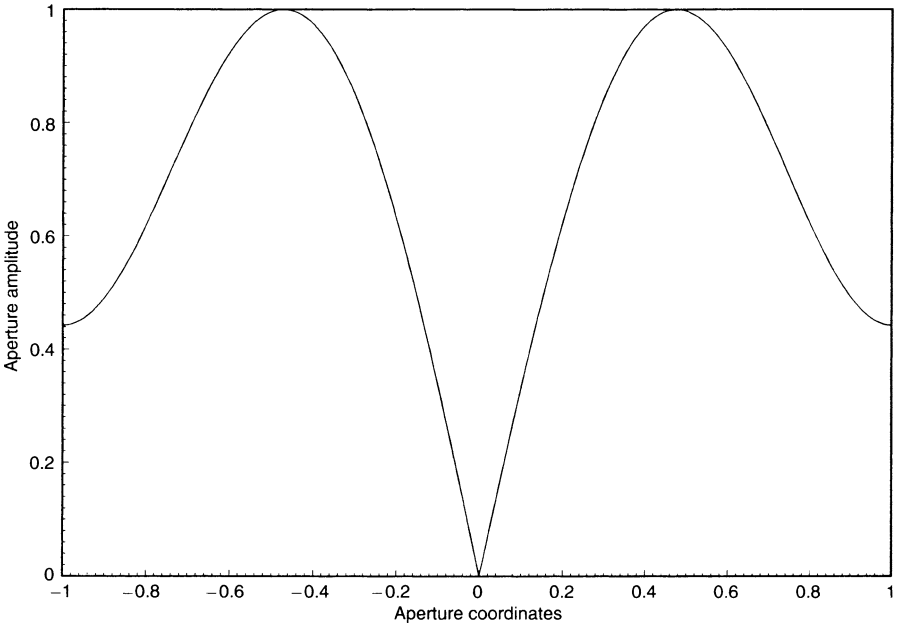


FIGURE 3.23 Bayliss difference pattern distribution.

Figure 3.23 shows the aperture distribution corresponding to the pattern of Figure 3.22. As expected from the Taylor principles, there is an appreciable pedestal at the aperture edges. The excitation efficiency for the Bayliss aperture is

$$\eta_t = 2u_0^2 \cos^2 \pi u_0 \frac{\sum_{n=0}^{\bar{n}-1} \dfrac{(-1)^n B_n}{\left[u_0^2 - \left(n+\frac{1}{2}\right)^2\right]^2}}{\sum_{n=0}^{\bar{n}-1} B_n^2} \tag{3.77}$$

The slope at the center is also given in terms of the coefficients:

$$S = \frac{2}{\pi} \sum_{n=0}^{\bar{n}-1} \frac{(-1)^n B_n}{\left(n+\frac{1}{2}\right)^2} \tag{3.78}$$

Table 3.13 gives Bayliss aperture efficiency and slope.

Bayliss difference patterns possess the same type of extended far-field distances as the low-sidelobe designs of Section 3.5. Again the first sidelobe height and null depth deteriorate as the distance decreases from $2L^2/\lambda$. The progressive raising and shouldering of sidelobes continues as R decreases. Details are given in Chapter 15.

TABLE 3.13 Bayliss Efficiency and Pattern Slope

SLR (dB)	$\bar{n}$	Aperture Efficiency	Normalized Slope
15	4	0.5959	0.9567
20	4	0.5846	0.8974
25	5	0.5633	0.8427
30	6	0.5393	0.7912
35	7	0.5162	0.7448
40	8	0.4951	0.7037

3.7.3 Sum and Difference Optimization

From the preceding discussions, it is clear that the array and aperture distributions that produce good sum pattern sidelobes are different from those that produce good difference pattern sidelobes. For example, the Taylor $\bar{n}$ and Bayliss $\bar{n}$ distributions are quite different. Ways have been developed to allow the good features of both sum and difference patterns to be realized. The two most important of these methods are the tandem feed, and the use of sub-arrays.

The tandem feed was invented by Kinsey (1970) and typically consists of a ladder network with a directional coupler at the end of each rung. The inboard arms of the couplers are connected along the rungs, while the outboard arms on one side are connected to the array elements, with the outboard arms on the other side terminated in loads. At the center of each of the ladder rails is a hybrid junction, whose outputs give the sum and difference patterns. The rail of couplers adjacent to the array provides the sum excitation, while both rails of couplers provide the difference excitation. See Figure 3.24 for a sketch of the configuration. This arrangement gives excellent control of both sum and difference patterns (Lopez, 1968; Jones and DuFort, 1971), although it is complex.

The sub-array configuration utilizes a linear array divided into sub-arrays, which need not be of equal length, although the array is symmetric about its center.

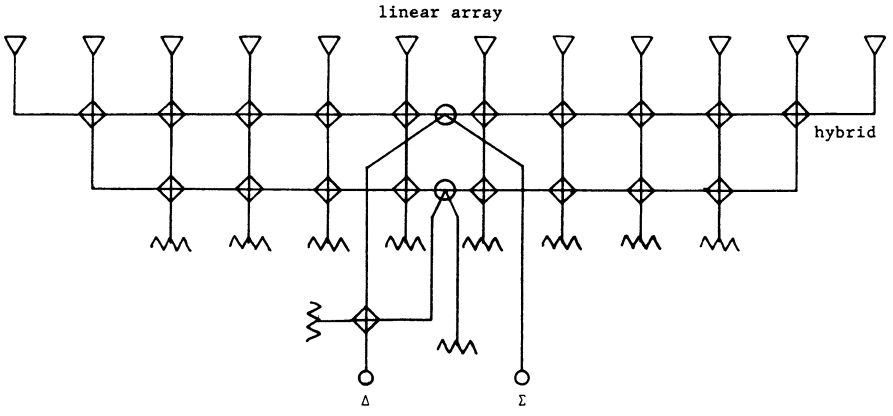


FIGURE 3.24 Ladder sum and difference pattern optimization.

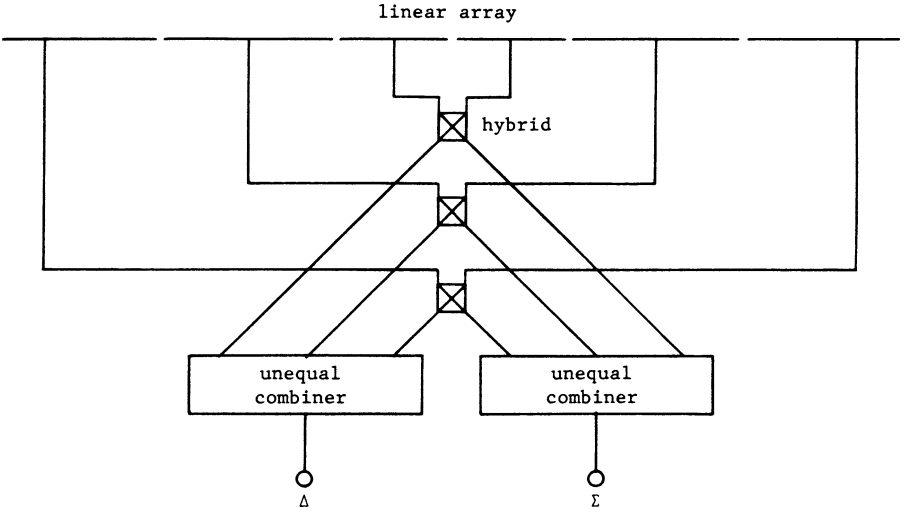


FIGURE 3.25 Sub-array sum and difference pattern optimization.

Symmetric pairs of sub-arrays are connected to a hybrid junction; see Figure 3.25. The sum ports of the hybrids are connected to a power combiner (divider) to provide the sum pattern, and similarly the difference ports of the hybrids are connected to a combiner to provide the difference pattern. Thus a stair-step approximation is provided to the desired sum pattern distribution, and a different stair-step approximation is provided to the desired difference pattern distribution (Kinsey, 1974). As more sub-arrays of smaller size are used, the stair-step approximations become better, and the resulting patterns are closer to the desired patterns. The same procedure has

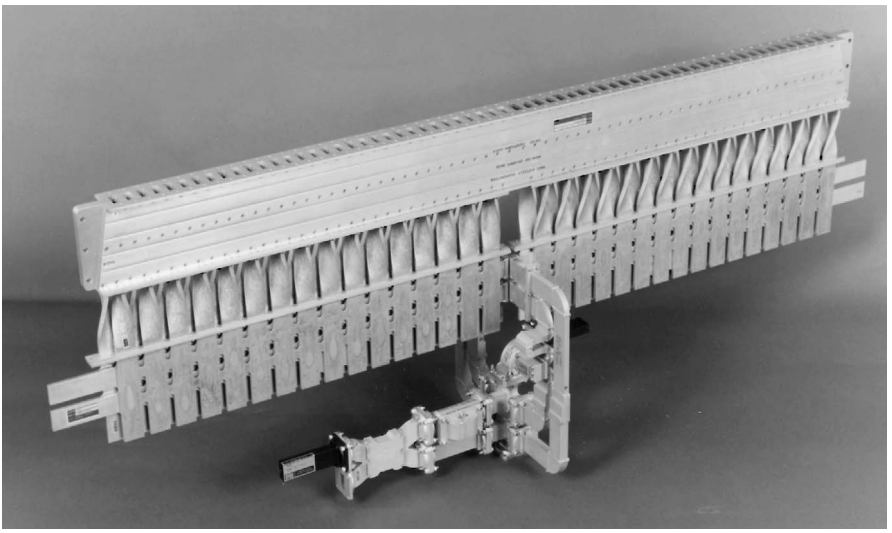


FIGURE 3.26 Lopez monopulse beamformer. (Courtesy Naval Research Lab.)

been applied to planar disk arrays, and is discussed in Chapter 4. Figure 3.26 shows a Lopez feed; the two waveguide rails are visible along the bottom (Rao et al., 1996). Rungs are not terminated with loads, thereby increasing efficiency.

3.7.4 Discrete Zolotarev Distributions

Zolotarev, a student of Chebyshev, developed a class of odd polynomials that gave an equi-ripple approximation in a stated interval, just as the even Chebyshev polynomials do. These were applied to microwave filter design by Levy (1970, 1971), who also gives historical and derivational details. It was recognized by McNamara (1985, 1993) that these polynomials could also be used to synthesize a difference pattern with equal-level sidelobes. The polynomial is defined as a hyperbolic cosine of an argument that is N times a function, giving a polynomial of order $2N + 1$. Unfortunately, the argument function contains elliptic integrals and Jacobi η and ζ functions. The synthesis procedure will be summarized briefly; for details consult McNamara (1993). Given an SLR, and an array of $2N$ elements, the order $2N - 1$ polynomial at the peak is set equal to SLR. The solution of this gives the Jacobi modulus, and the polynomial is completely determined. For half-wave spacing, the polynomial is expanded into a conventional polynomial; from these coefficients the array excitation coefficients may be found. Specific computational details for the special functions are given in the reference above. Figure 3.27 shows a ninth-order Zolotarev polynomial, where x_1 and x_3 are the starts of the equal sidelobe regimes. A typical difference pattern is shown in Figure 3.28, for a 20-element array with 30 dB SLR; the amplitudes over one-half of the array are shown in Figure 3.29.

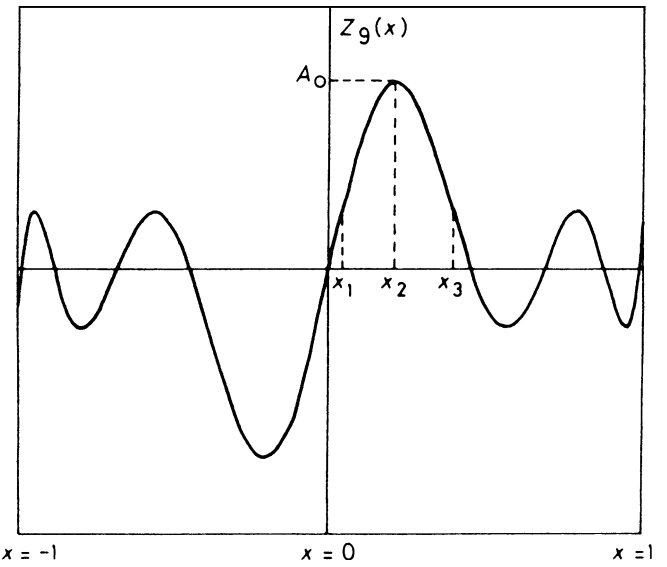


FIGURE 3.27 The Z_g Zolotarev polynomial. (Courtesy McNamara, D. A., "Direct Synthesis of Optimum Difference Patterns for Discrete Linear Arrays Using Zolotarev Distributions", *Proc. IEE*, Vol. MAP-140, Pt. H, Dec. 1993, pp. 495–500.)

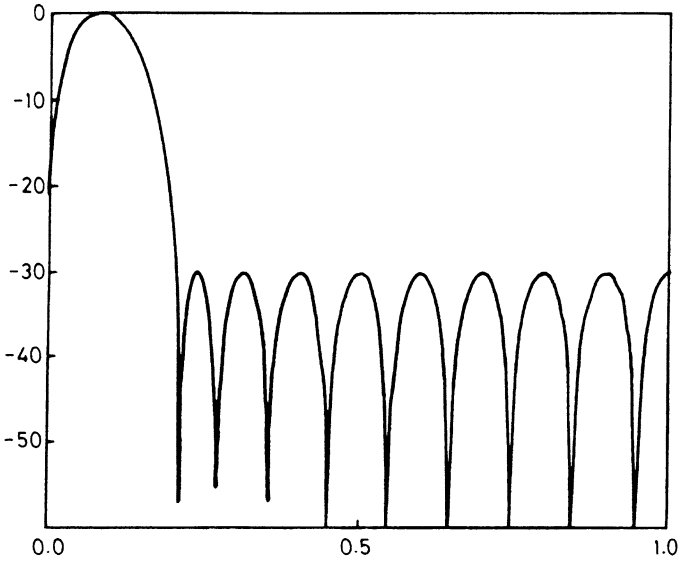


FIGURE 3.28 20-element Zolotarev pattern. (Courtesy McNamara, D. A., "Direct Synthesis of Optimum Difference Patterns for Discrete Linear Arrays Using Zolotarev Distributions", *Proc. IEE*, Vol. MAP-140, Pt. H, Dec. 1993, pp. 495–500.)

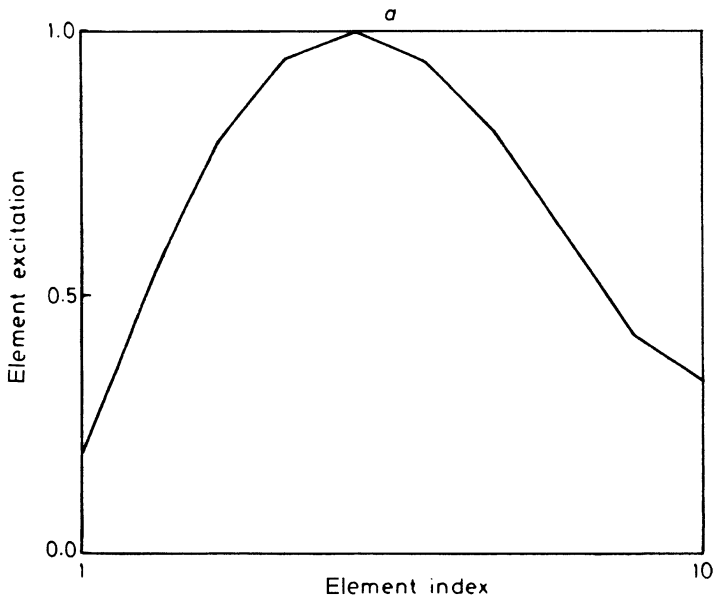


FIGURE 3.29 Array excitations. (Courtesy McNamara, D. A., "Direct Synthesis of Optimum Difference Patterns for Discrete Linear Arrays Using Zolotarev Distributions", *Proc. IEE*, Vol. MAP-140, Pt. H, Dec. 1993, pp. 495–500.)

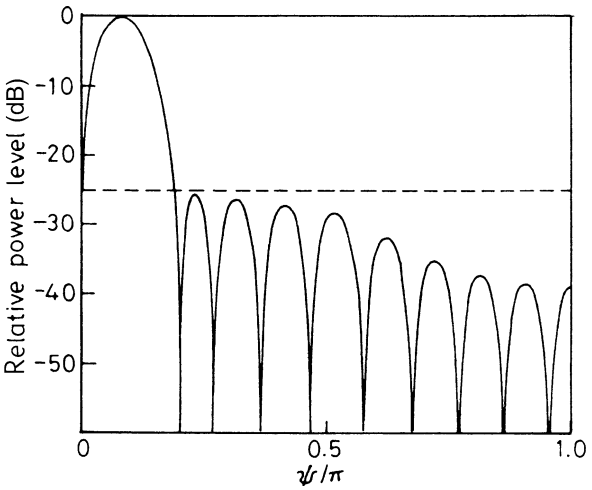


FIGURE 3.30 The $\bar{n}$ Zolotarev pattern. (Courtesy McNamara, D. A., "Performance of Zolotarev and Modified-Zolotarev Difference Pattern Array Distributions", *Proc. IEE*, Vol. MAP-141, Feb. 1994, pp. 37–44.)

A discrete $\bar{n}$ difference pattern, analogous to the discrete Villeneuve $\bar{n}$ sum pattern, has also been developed (McNamara, 1994). The starting point, as expected from the analog, is the equal-ripple Zolotarev polynomial. The first $\bar{n}$ zeros are kept, while farther-out zeros are replaced by those of a generic pattern, chosen to allow a proper correspondence between the two patterns. A maximum-slope difference pattern is used as the generic pattern. A transition region between the $\bar{n}$ Zolotarev zeros and the generic zeros is used as before, with a dilation factor. The procedure is somewhat tedious, but the results are excellent. Figure 3.30 shows the 20-element array pattern, with nominal 25-dB SLR, and $\bar{n} = 4$. The dilation factor is $\sigma = 1.01051$. The abscissa is $(2d/\lambda)\sin\theta$.

3.8 SIDELOBE ENVELOPE SHAPING

The sidelobe envelope of a pencil beam array may be controlled through adjustment of individual sidelobe heights, using techniques developed by Elliott (1976). The principle is based on the fact that the spacing between pattern function zeros controls the heights of the sidelobes. Moving two adjacent zeros closer together reduces that sidelobe. However, as each array has only a fixed number of zeros, moving zeros closer together means that some other zeros will be farther apart. In this technique, a canonical pattern whose beamwidth and average sidelobe level are roughly the same as those of the desired pattern is selected. This canonical pattern must have the same number of pattern zeros as the desired pattern; the pattern is then written as a product of zeros. For example, a Taylor one-parameter pattern or a Taylor $\bar{n}$ pattern might be used to start.

The shaping procedure begins by writing each zero of the desired pattern as that of the corresponding zero of the canonical pattern plus a small shift. With higher order effects discarded, the desired pattern can be expressed as the canonical pattern times a factor, which is 1 plus a sum over the pattern zeros. This sum involves the canonical pattern zeros and the unknown shifts in zeros. For narrow sidelobes, the sidelobe peak position is approximately halfway between the adjacent zeros. After inserting the sidelobe positions into the equation, a result among the desired and canonical levels of each sidelobe, and the various zeros, is obtained. This is a set of simultaneous equations, with the differences in sidelobe heights specified, and with unknown zero shifts. These equations are solved iteratively, with each resulting pattern used as a canonical pattern for the next step. Usually fewer than 10 iterations are needed. A gradient scheme, such as Newton–Raphson (Stark, 1970) provides a rapid way of finding the null shifts.

The resulting pattern is a product on the zeros that allows easy computation. The array excitation coefficients are less easily obtained. The product pattern function is multiplied out to obtain a polynomial. The coefficients are now the array excitation values. An example is a Taylor $\bar{n} = 8$ pattern with SLR = 20 dB, except for the three closest sidelobes on one side that should be at -30 dB. Figure 3.31 shows the resulting pattern (Elliott, 1977) that was obtained in three iterations; Figure 3.32

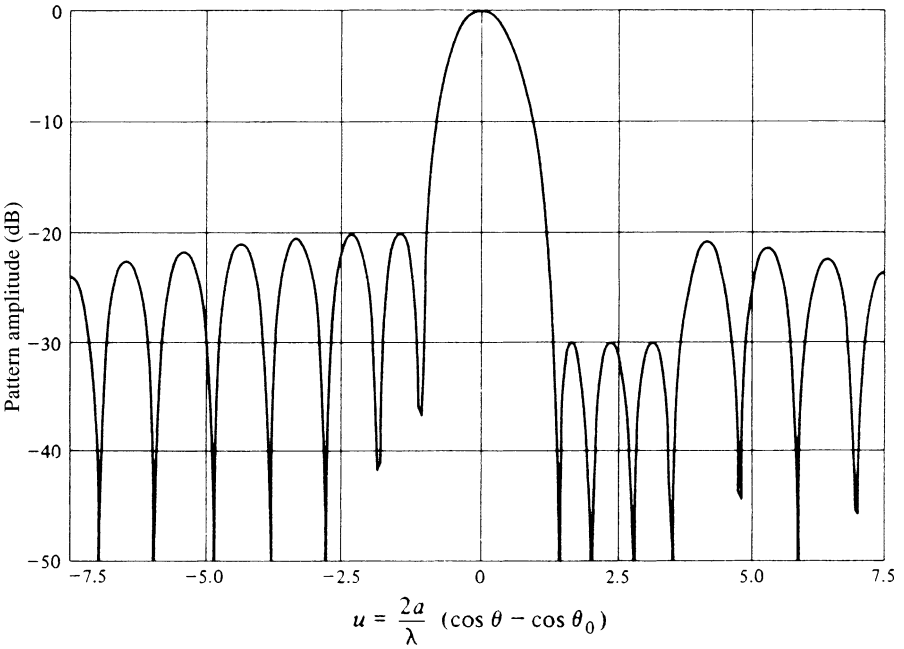


FIGURE 3.31 Shaped sidelobe pattern. (Courtesy Elliott, R. S., "Design of Line-Source Antennas for Sum Patterns with Sidelobes of Individually Arbitrary Height", *Trans. IEEE*, Vol. AP-24, 1976, pp. 76–83.)

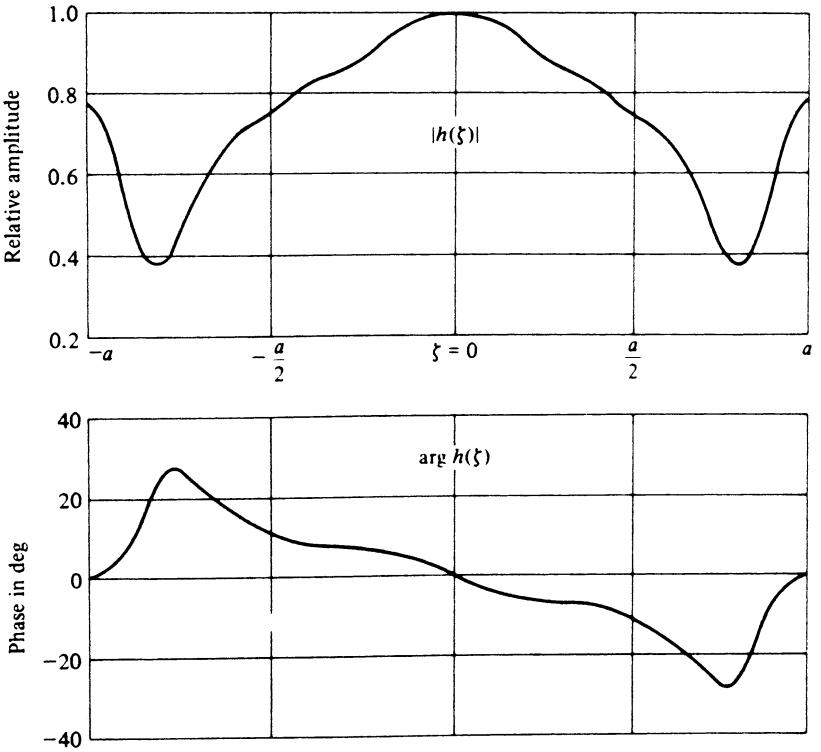


FIGURE 3.32 Shaped sidelobe pattern distribution. (Courtesy Elliott, R. S., "Design of Line-Source Antennas for Sum Patterns with Sidelobes of Individually Arbitrary Heights", *Trans. IEEE*, Vol. AP-24, 1976, pp. 76–83.)

shows the amplitude and phase of the resulting aperture distribution. The latter is of course complex, as the pattern is no longer symmetric. This pattern shaping by zero adjustment has proved to be a powerful and useful technique.

Another way of looking at this procedure is to write the array pattern in Z-transform form, where the pattern zeros are located on the unit circle (see Chapter 2). Moving two zeros closer together lowers the sidelobe that they control, but raises adjacent sidelobes. Maintaining the same beamwidth with lower overall sidelobes requires more zeros, hence a long array. Sidelobe nulls may be partially or completely filled by moving zeros off the unit circle, either inside or outside. In general, this allows multiple zero arrangements (and array excitations) that produce the same sidelobes. However, some excitations may be more robust than others, so all cases should be computed and compared.

When the array distribution is required to be real, as for resonant arrays or for some microstrip arrays, the same general synthesis technique can be used, except that additional complex roots are introduced (Ares et al., 1994).

3.9 SHAPED BEAM SYNTHESIS

3.9.1 Woodward–Lawson Synthesis

Earlier parts of this chapter concerned pencil beam (high-gain) pattern synthesis. Often, broad, shaped beams are desired. For example, the cosecant radar pattern, or a flat-top earth overage pattern. The prototype shaped beam synthesis method was developed by Woodward (1947) and Woodward and Lawson (1948); it uses an assemblage of virtual arrays that produce orthogonal sinc beams. Thus each sinc beam has its amplitude adjusted such that the beam peak fits a point on the pattern, with the fitting points equally spaced. Each constituent beam is formed from a virtual linear array with constant amplitude and linear phase. The nth beam is given by $\sin(N\pi u)/(N \sin \pi u)$, where the array has N elements with spacing d; $u = kd(\sin \theta - \sin \theta_n)$. The beams are spaced (in $\sin \theta$) apart by $n\pi/Nd$, so all beams except the nth have zeros at the peak of that beam. The desired pattern is then sampled at approximately $N\lambda/2d$ points, with the array excitations equal to the sample amplitude. The total array distribution is simply the sum of the constituent array excitations. Given a desired pattern $F(u)$ the synthesized pattern is

$$WL(u) = \sum_{n=1}^{N} F(u_n) \frac{\sin N\pi u}{N \sin \pi u} \tag{3.79}$$

The amount of pattern ripple is controlled by the element spacing. Figure 3.33 shows an example of a cosecant pattern synthesized from 21 beams, using a 21-element array with half-wave spacing. One zero is placed at $\theta = 0$. By shifting the zeros to allow the pattern dropoff to cross the axis, smaller ripples are obtained, as seen in Figure 3.34. Larger spacing than half-wave will allow grating lobes, while smaller spacing tends toward superdirectivity. Thus, half-wave spacing is desirable.

The Woodward–Lawson synthesis uses constituent beams derived from a uniform distribution, giving a -13.3-dB sidelobe level for long arrays. It is possible to start with the zeros of a low-sidelobe distribution, then to move the zeros off the unit circle (in or out) to shape the beam. However, the orthogonal beam spacing here is larger; the cost of achieving lower sidelobes (and higher directivity) on the opposite side of the cosecant pattern, for example, is larger pattern ripple or a larger aperture.

The Woodward–Lawson synthesis is important as it is applicable to multiple-beam antennas (see Chapter 10). It is also a good starting point for an iterative adjustment procedure of the type discussed in Section 3.8. The array excitations can be adjusted to minimize ripple, and to control the sidelobe envelope outside the shaped region. Intuitively one might expect the optimum cosecant pattern to have nulls in the low part of the pattern just adequate to give the prescribed sidelobe level. When the entire pattern can be specified, for example, in broadcasting antennas, a least-squares error function is then minimized by an iterative scheme: Newton–Raphson (see Section 3.8) or gradient (Perini and Idselis, 1972). Of gradient algorithms, the Fletcher–Powell (Fletcher and Powell, 1963) and Fletcher–Reeves (Fletcher and Reeves, 1964) are widely available; significant improvements have been made by

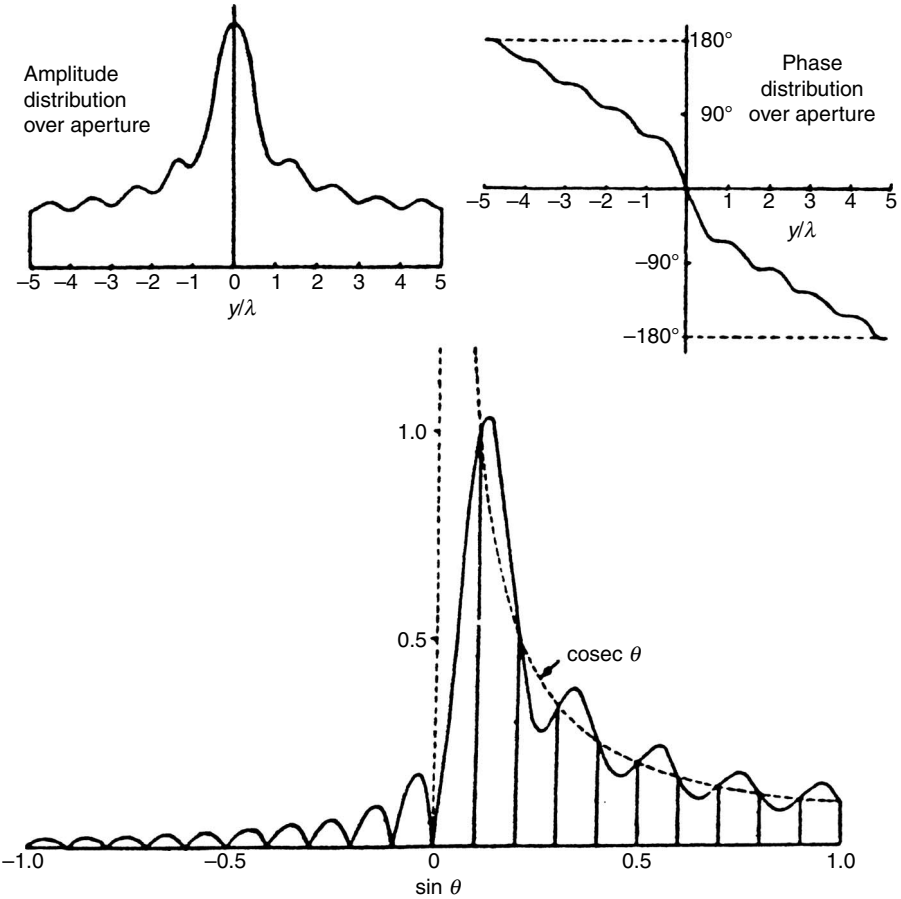


FIGURE 3.33 Shaped beam pattern. (Courtesy Woodward, P. M. and Lawson, J. D., "The Theoretical Precision with Which an Arbitrary Radiation Pattern May Be Obtained from a Source of Finite Size", *J. IEE*, Vol. 95, Part III, 1948, pp. 363–370.)

Fletcher (1970) and Gill and Murray (1972). Least-squares synthesis tends to produce unequal ripples; an approach to a true minimax synthesis results from a least-pth synthesis (Temes and Zai, 1969; Bandler and Charalambous, 1972). These techniques are not useful when sidelobe regions are involved because the number and position of sidelobes is usually not important; only the sidelobe envelope is important. Thus, the individual sidelobes cannot be effectively specified.

A major disadvantage of the Woodward–Lawson synthesis techniques is that in unit circle form, with the pattern written as a product of zeros, the roots in the shaped beam region are moved off the circle in pairs. Thus the available degrees of freedom are halved; these affect the number of ripples in the shaped region, hence the fidelity of the pattern match to the desired pattern. Next, a technique that utilizes all available degrees of freedom will be discussed.

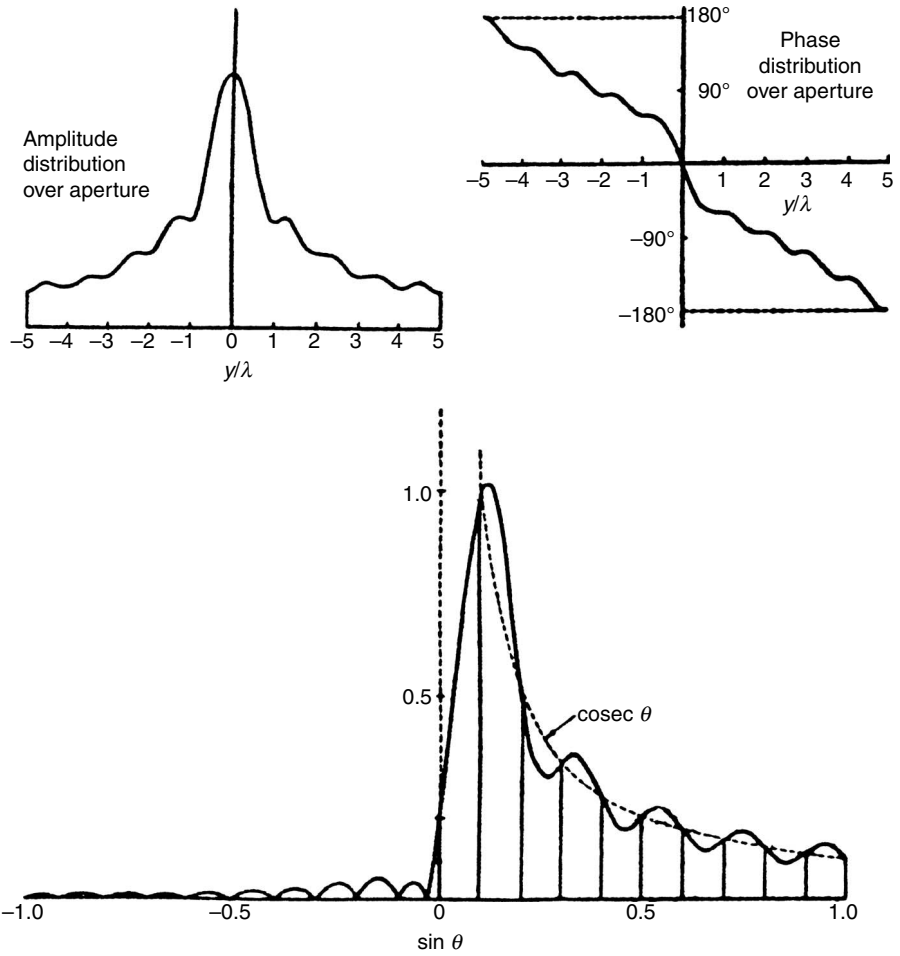


FIGURE 3.34 Improved shaped beam. (Courtesy Woodward, P. M. and Lawson, J. D., "The Theoretical Precision with Which an Arbitrary Radiation Pattern May Be Obtained from a Source of Finite Size", *J. IEE*, Vol. 95, Part III, 1948, pp. 363–370.)

3.9.2 Elliott Synthesis

Efficient synthesis techniques for shaped beams have been developed by Elliott (1982), where all zeros of the array polynomial (in unit circle terminology) are adjusted to optimize the pattern. Since factorization of complex polynomials is difficult, it is better to start the iterative synthesis process with a suitable canonical pattern that can be expressed in product form. A uniform linear array is a good choice; the roots are equally spaced on the unit circle (for half-wave spacing). The first step in the synthesis procedure is to iteratively perturb the roots on the unit circle, such that the sidelobes match the desired envelope in the sidelobe region, and such that the

sidelobes are raised to the desired shaped envelope in the shaped beam region. This iterative process assumes that the peak of each lobe occurs midway between the adjacent zeros. Each zero is perturbed by a delta increment, and the resulting pattern is calculated at each lobe peak. The differences between these lobe peaks and the desired values give differential derivatives; a set of simultaneous equations relates the deltas and the derivatives. This set is best solved by the iterative Newton–Raphson method (Stark, 1970). This process typically needs on the order of 10 iterations or fewer. This first-stage result is shown in an example: A 16-element array with half-wave spacing, designed for a shaped beam of $\text{cosec}^2\,\theta \cos\theta$ over a 40° sector, and for a sidelobe region consisting of four close sidelobes at −30 dB, with the rest at −20 dB. Figure 3.35 shows the resulting pattern (Elliott and Stern, 1984).

In the second step, the unit circle roots in the shaped region are moved radially off the circle; this fills in the nulls, producing a shaped beam with ripples. Again an iterative procedure is used, with the ripple peaks and dips assumed to occur at the unit circle points. However, when the roots are displaced from the unit circle the results show that the ripples are not symmetric, and that the sidelobes are affected (see Fig. 3.36). The iterative process must be rerun, to readjust the sidelobes, and balance the ripples. A final result is shown in Figure 3.37, which shows four symmetric ripple oscillations. Table 3.14 gives the root positions; note that two roots are outside and two roots

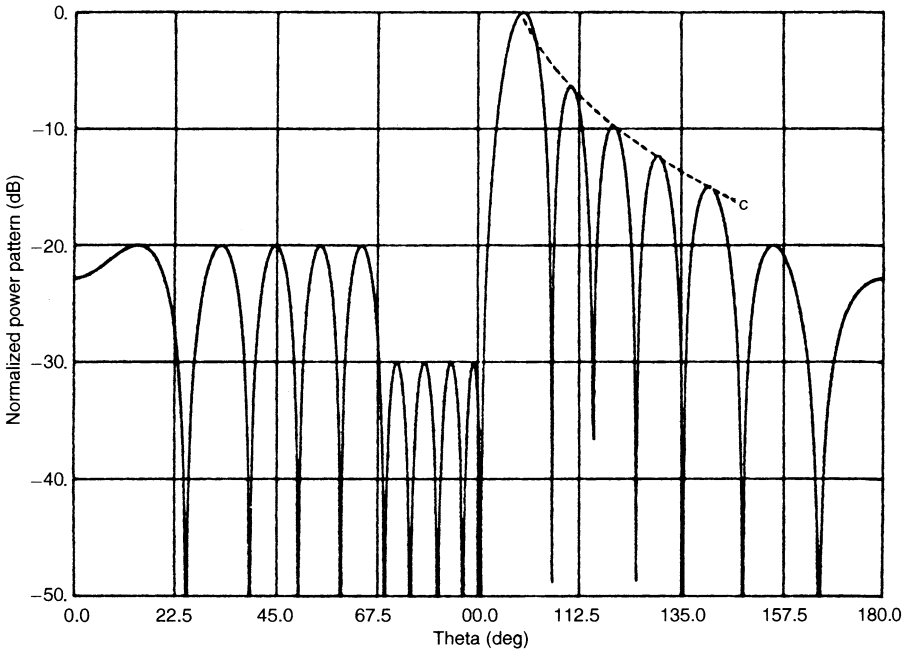


FIGURE 3.35 Shaped beam with prescribed sidelobes. (Courtesy Elliott, R. A. and Stern, G. J., "A New Technique for Shaped Beam Synthesis of Equispaced Arrays", *Trans. IEEE*, Vol. AP-32, Oct. 1984, pp. 1129–1133.)

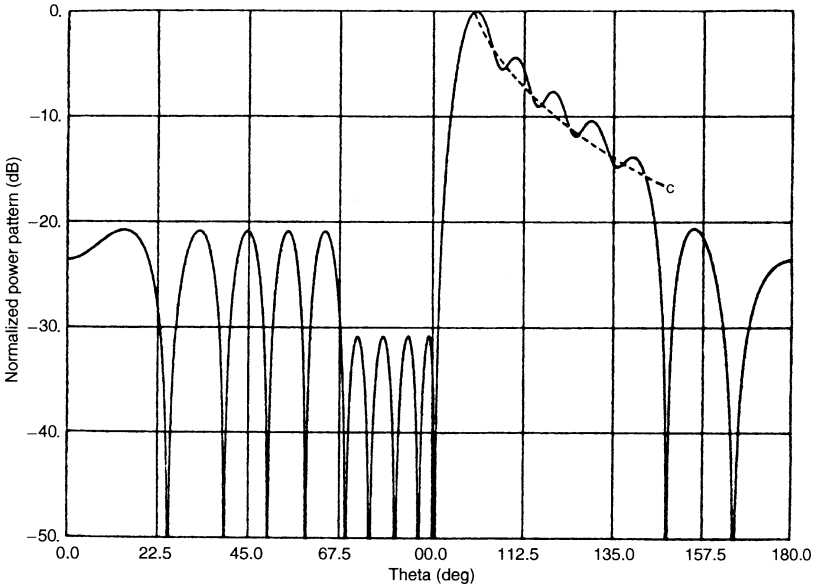


FIGURE 3.36 Shaped beam with null filling. (Courtesy Elliott, R. S. and Stern, G. J., "A New Technique for Shaped Beam Synthesis of Equispaced Arrays", *Trans. IEEE*, Vol. AP-32, Oct. 1984, pp. 1129–1133.)

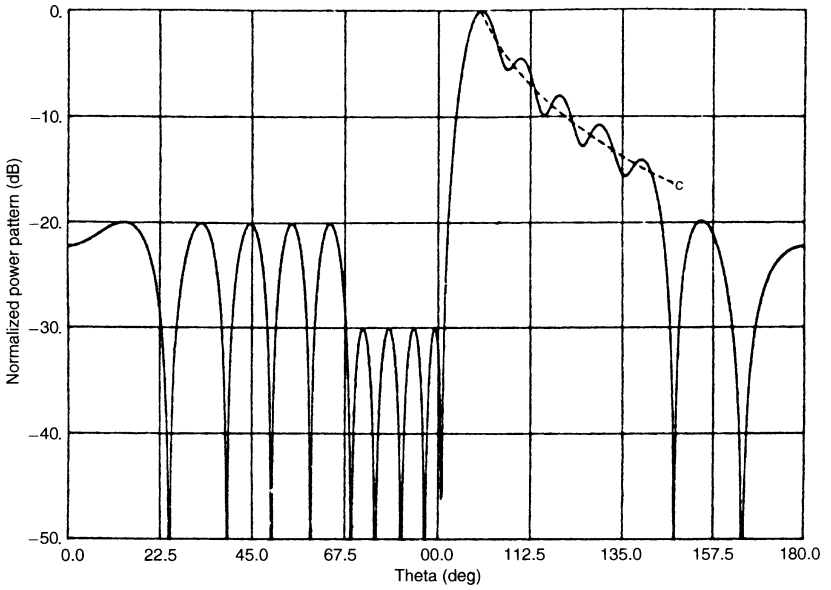


FIGURE 3.37 Shaped beam with readjusted sidelobes. (Courtesy Elliott, R. S. and Stern, G. J., "A New Technique for Shaped Beam Synthesis of Equispaced Arrays", *Trans. IEEE*, Vol. AP-32, Oct. 1984, pp. 1129–1133.)

TABLE 3.14 Unit Circle Root Locations

Radial Distance	Angular Distance (deg)
1.0000	−8.76
1.3219	−64.12
0.7565	−108.15
1.3588	−108.15
0.7360	−143.76
1.0000	−166.26
1.0000	171.24
1.0000	148.74
1.0000	126.24
1.0000	103.74
1.0000	81.24
1.0000	58.74
1.0000	36.24
1.0000	13.74

are inside the unit circle. Array excitations are found by multiplying the factors to get the array polynomial; the coefficients are the excitations. These are given for the example in Table 3.15. An interesting feature of this synthesis is that each shaped region root can be moved in or out. With M such roots, there are 2^M designs that produce the same pattern. The amplitude and phase of the element excitations will be different, and some values may be difficult to realize. For example, waveguide slots have practical limits on admittance values. The ratio of largest to smallest coupling should not be too large in order to have a robust design. Mutual coupling will also

TABLE 3.15 Array Element Excitation

Element	Amplitude	Phase (deg)
1	0.300	231.71
2	0.242	211.46
3	0.307	198.35
4	0.378	161.79
5	0.337	132.12
6	0.431	123.71
7	0.753	85.44
8	1.000	29.68
9	1.000	−29.68
10	0.753	−85.44
11	0.431	−123.71
12	0.337	−132.12
13	0.378	−161.79
14	0.307	−198.35
15	0.242	−211.46
16	0.300	−231.71

be different owing to the different excitations. It is desirable to compute most, or all, of the 2^M cases to determine the best choice in terms of element embedded impedance.

When the phase of the pattern is not important, a power pattern may be synthesized; this may allow additional degrees of freedom. Now the iterative procedure operates on the magnitude squared of the unit circle product (Orchard et al., 1985). Another option groups roots in pairs to produce a real array distribution (Ares et al., 1994, 1996).

When additional constraints are placed on the pattern, such as null positions, the array equations are overdetermined. These can be solved in a least mean squares sense by the singular value decomposition method (Ares et al., 1997).

3.10 THINNED ARRAYS

3.10.1 Probabilistic Design

Large arrays with half-wave spacing contain many elements, and may be unacceptable. Also, owing to the associated cost, weight, and so on, mutual coupling may seriously degrade scanning performance. Since half-power beamwidth is approximately proportional to λ/L, array length L cannot be reduced without increasing beamwidth. In some cases, array size is dictated by beamwidth, but the directivity of a filled array is not needed. Thus a thinned array can offer essentially the same beamwidth with less directivity and fewer elements. Directivity is approximately equal to the number of elements, N. Mutual coupling effects are also significantly reduced (Agrawal and Lo, 1972). The average sidelobe level (power) for a linear array is $1/N$. Regular thinning produces grating lobes, but these can be partially suppressed by randomizing[1] the element spacings. There is then the question of how to select the "random" or nonuniform spacings. Most of the analytical work started with a desired pattern expressed as a Fourier series, then devised various ways of fitting the array sum to the Fourier series. Sandler (1960) expands in terms of a set of uniformly spaced arrays; Harrington (1961) considers an array with small departures from uniform; Unz (1962) expands each array term in a Bessel function series; Ishimaru (1962) expands in a delta-function series. All of these approaches suffer from a common difficulty: In order to expand the desired pattern into a Fourier or any other series, the pattern must be known in detail. If only amplitude is known, the problem is much more difficult; ideally pattern amplitude and phase are given. Rarely, however, is phase specified, and for narrow-beam, low-sidelobe arrays, the amplitude cannot even be specified. Rather the sidelobe envelope is usually specified, or the envelope below that all sidelobes must lie is specified. No Fourier series can be fitted to a sidelobe envelope as the series represents the actual lobes. Thus, analytical efforts have been found to be of little use.

Computer selection of element spacings has also been pursued. However, the problem is enormously difficult as a simple calculation will show. Take an array

[1]If the array spacings are selected by a deterministic algorithm, the array is aperiodic instead of random.

100 wavelengths long, and allow possible element positions every eighth wavelength. Now take 40 elements, a 20% filled situation. There are

$$\binom{800}{40} \simeq 10^{70} \tag{3.80}$$

possible patterns to be evaluated if a direct search is used. Dynamic programming was tried as a search speedup, but is not applicable since the optimum position of an element depends on the positions of elements on both sides of the element. Other search speedups have also proved unsatisfactory. Optimization algorithms using conjugate gradient methods, such as Fletcher–Powell or Fletcher–Reeves (in most computer scientific subroutine libraries), will yield an optimum thinned array, say minimum sidelobe envelope for a given array length, but are exceedingly slow. Most of the random-array computer studies have been on small arrays (King et al., 1960; Andreasen, 1962; Redlich, 1973; Steinberg, 1976). Use of such arrays is limited by the type of sidelobe structure that usually appears. High sidelobes occur at random angles, with a wide range of sidelobe heights in between these high lobes. Thus, the computer design usually attempts to keep all sidelobes below a fixed value. Algorithmic comparisons have been made by Steinberg (1976).

A major advance in understanding the behavior of random thinned arrays was made by Lo (1964). He assumed that isotropic elements with unit excitation were placed randomly, giving a probability density function. From this the probability of a narrow-beam pattern with all sidelobes below a given level was calculated by approximating the autocorrelation function by a rectangle of width $2\lambda/L$. However, the accuracy is poor for probability values that are small, or for small arrays. A different method was used by Agrawal and Lo (1972) wherein the probability of the pattern function crossing the specified sidelobe level was examined. The resulting integral equation was solved numerically by dividing the interval into many short segments, with the probability of a crossing assumed constant over a segment. The resulting product in the limit gives a closed-form probability result that compares very well with Monte Carlo simulations of arrays as short as 5 wavelengths, for low as well as high probabilities. The Agrawal and Lo result is in terms of the array length L/λ, and a combined parameter α that is the number of elements N times power sidelobe level:

$$\alpha = \frac{N}{\text{SLR}} \tag{3.81}$$

Here the sidelobe ratio SLR is in power. The result for probability of sidelobes below $1/\text{SLR}$ is

$$P = [1 - \exp(-\alpha)] \exp\left(-\frac{2L}{\lambda}\sqrt{\frac{\pi\alpha}{3}}\exp(-\alpha)\right) \tag{3.82}$$

Thus the sidelobe level and the number of elements can be directly traded. Figure 3.38 shows P versus α for $L/\lambda = 10, 30, 100, 300, 1000$. For example, an SLR of 20 dB is

achieved with an 80% probability for $L = 1000\lambda$ with $N = 1028$; or an average element spacing of one wavelength. For large N, the formula may be simplified (Lo, 1964):

$$P = [1 - \exp(-\alpha)]^{6L/\lambda} \tag{3.83}$$

This gives a reasonably close fit for $P > 0.5$ and $L/\lambda > 100$. Figure 3.39 shows the approximate relationship for $P = 0.8$, 0.09, and 0.99 versus L/λ. The quantity plotted is 10 log N − SLR and again these results are only valid for large arrays. All of the results are for an unsymmetrical array. If the array is symmetric about its center, some degrees of freedom are lost, and the sidelobes are higher.

In using the curves of Figure 3.39, care must be taken that the resulting N is not greater than that of a filled array, $2L/\lambda$; for example, a filled array with $L = 100\lambda$ had $N = 200$ (half-wave spacing), which gives $\alpha = 0.44$ for SLR = 13.26 dB. From the graph the probability is near unity, which means that a 100λ aperture can achieve an SLR to 13.26 dB or better. However, for $L = 30\lambda$, $N = 60$, the value of α for the SLR is 2.83, which from the curve gives a probability near zero. This means that a thinned 30λ aperture will probably have sidelobes worse than −13 dB. Thus, for each aperture length there is a sidelobe ratio that is difficult to

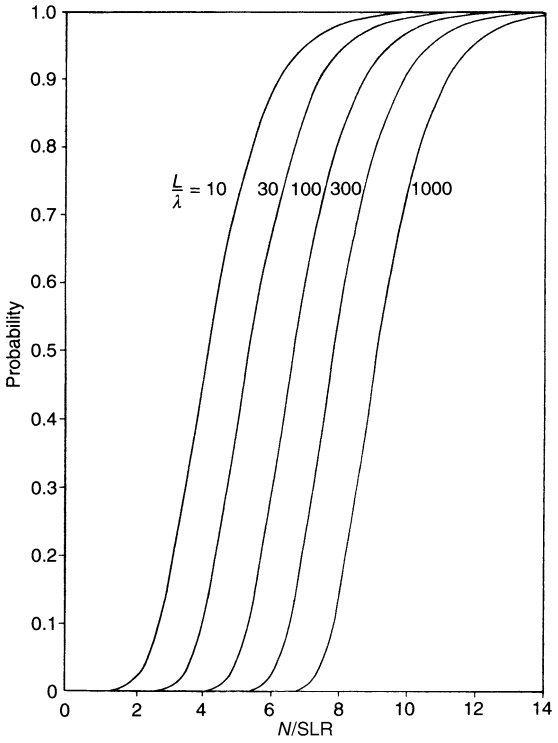


FIGURE 3.38 Probability of sidelobe ratio less than SLR (power).

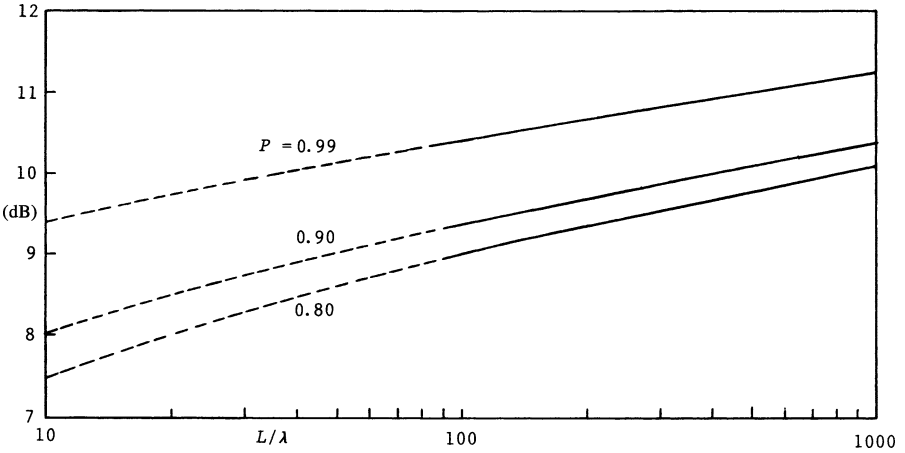


FIGURE 3.39 The 10 log N − SLR for thinned arrays.

exceed, with longer apertures yielding better SLR capability. For example, for 80% probability, Table 3.16 gives these values. Thus, for reasonably low sidelobes, very large arrays are required to achieve a significant thinning. But because these very large arrays have very many sidelobes, only a small fraction of these sidelobes need to be at the envelope level to greatly reduce directivity and beam efficiency. This is expected, since directivity is roughly proportional to the number of elements.

In evaluating a specific thinned array design, patterns must be calculated over the frequency band and over the two-dimensional scan range. Beam positioning increments, both in frequency and in angle, should be roughly one-half of the half-power beamwidth. This will require extensive calculations, but it will insure that the design has no high sidelobe "popups".

Beam efficiency is important for radiometer and power-transfer situations. A tapered sidelobe envelope will reduce the sidelobe power and increase beam efficiency. This result is especially important for planar arrays since there are so many more sidelobes at large angles from broadside.

The analysis and results above are for thinned arrays of isotropic elements. Half-wave elements will tend to reduce far-out sidelobes in one plane owing to the element

TABLE 3.16 Upper Limit of Sidelobe Ratio

L/λ	SLR (dB)
10	5.7
30	9.6
100	14.1
300	18.2
1000	22.9
3000	27.2

TABLE 3.17 Thinned Array with Directive Elements $L = 1000\lambda$, $P = 0.8$, SLR = 15 dB

W/λ	P	Filling Factor
Isotropic	0.8	0.163
0.5	0.87	0.163
2	0.97	0.651

pattern. Often it is advantageous to use elements larger than half-wave, and these further reduce far-out sidelobes. Steinberg (1976) has shown that the effect of element directivity is to reduce the effective array length used in Eq. (3.82). His result is

$$\frac{L_{\text{eff}}}{\lambda} = \frac{\mathrm{L}/\lambda}{\sqrt{\alpha W/\lambda}} \tag{3.84}$$

This is only valid for $W/\lambda > 1/2$, owing to approximations used in evaluating an integral. An example will show the effect of element directivity. Take $L/\lambda = 1000$, $P = 80\%$, and SLR = 15 dB. Then a thinned array of isotropic elements needs 325 elements. Keeping the SLR and the number of elements fixed, consider half-wave dipole-type elements ($W/\lambda = 1/2$) and 2λ sub-arrays. Table 3.17 gives the probabilities of a sidelobe ratio of 15 dB or greater. Then, as expected, directive elements reduce the sidelobes.

An interesting concept in thinned arrays uses fractal theory to randomize element placement (Kim and Jaggard, 1986). A three-element sub-array with regular 0.8λ spacing is the "generator", while the fractal random process (the "initiator") locates the sub-arrays at pseudo-random locations. An example of a 360λ array with 180 elements shows a highest sidelobe of -12.5 dB, but the sidelobe envelope is roughly constant at -20 dB. In contrast, if the elements were uniformly spaced, the first sidelobe would be -13.2 dB, and the envelope would taper down with angle off broadside to a value below -60 dB at 90°, but the beamwidth would be four times larger due to the much shorter array.

3.10.2 Space Tapering

A special kind of thinned array uses variable element spacing to produce an equivalent amplitude taper. The goal is to produce a sidelobe envelope that tapers down (Willey, 1962). As in all thinned arrays, the number of array elements must be large. It is particularly attractive in a distributed amplifier array where each element has its own transmitter–receiver module. Since it is much easier to make all modules alike, space tapering allows lower sidelobes to be realized, with all elements equally excited. Experience has shown, however, that the sidelobes do not fall off quite as fast as predicted by theory. Also, an occasional high sidelobe is observed. It may be necessary to make adjustments to the spacing or to the filling ratio.

Some calculations show the degree of thinning versus sidelobe level. The Taylor one-parameter distribution is used (see Section 3.3.1). The number of elements

TABLE 3.18 Spaced Tapered Linear Array

SLR (dB)	N/N_0	θ_3/θ_{30}
13.26	1	1
15	0.9133	1.042
20	0.7562	1.156
25	0.6647	1.260
30	0.6030	1.355
35	0.5577	1.443
40	0.5223	1.524

relative to the $\lambda/2$ filled array number N_0 is

$$N = \frac{N_0}{I_0(\pi B)} \int_0^1 I_0\left(\pi B\sqrt{1-p^2}\right) dp \tag{3.85}$$

This equation can be integrated (using formulas 11.4.10 and 10.2.13 of Abramowitz and Stegen, 1970) to give

$$N = \frac{N_0 \sinh \pi B}{\pi B\, I_0(\pi B)} \tag{3.86}$$

Table 3.18 gives the values and also shows the relative beamwidths. Even for SLR = 40 dB, the thinning is less than one-half. So, although space tapering does allow simulation of tapered amplitude distributions, the percentage of elements saved is modest. For rectangular arrays, the values for each coordinate are combined.

3.10.3 Minimum Redundancy Arrays

For some applications, such as aperture synthesis in radio astronomy, the spatial frequency spectrum is of use, and it is important to consider the simplest arrays that provide all spatial frequencies up to an upper limit. These arrays are called minimum redundancy arrays. The redundancy R is defined (Moffet, 1968) as

$$R = \frac{N(N-1)}{2\sum \text{spacings}} \tag{3.87}$$

where the array has N elements, and all multiples of unit spacing $< N_s$ are present. For very large arrays the lower limit on redundance is $4/3$ (Bedrosian, 1986). An example for a linear array, from Moffet, is sketched in Figure 3.40. In the figure are seven elements, with spacings 1,3,6,2,3,2. The spatial frequency coverage is up to 17, and the redundancy is 1.24. Finding such arrays for large N is an interesting problem in number theory.

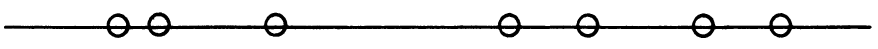


FIGURE 3.40 Seven-element minimum redundancy array.

ACKNOWLEDGMENT

Photograph courtesy of Dr. J. B. L. Rao.

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CHAPTER FOUR

Planar and Circular Array Pattern Synthesis

4.1 CIRCULAR PLANAR ARRAYS

4.1.1 Flat Plane Slot Arrays

Many aircraft radars are now equipped with flat plane slot arrays instead of dish antennas, where the outline is approximately circular. These are usually fixed narrow-beam antennas with controlled sidelobes. Pattern synthesis typically uses the product of two linear syntheses; one along the slot "sticks", and a second across the sticks. These are often designed as Taylor $\bar{n}$ patterns, as described in Chapter 3. Since the sticks near the edge are shorter, with fewer slots, than those at the center, the overall pattern will deviate somewhat from the product of the two orthogonal patterns. In particular, some adjustments may be necessary to produce the desired sidelobe envelopes in the principal planes. As these arrays are often divided into quadrants for azimuth and elevation monopulse, a compromise may be advisable between the sum pattern sidelobes, and the difference pattern sidelobes and slope (see Chapter 3). An example of the design, using the zero adjustment methods described in Chapter 3, of a planar array of inclined slots, providing a cosecant type pattern, with a Taylor $\bar{n}$ pattern in the cross-plane, is given by Erlinger and Orlow (1984).

Design of the array elements, to include mutual coupling, is more complex in that the entire array must be encompassed in the synthesis process. That synthesis is described in Chapter 6, and it can be extended to the flat plane array by using double sums for mutual coupling to include all slots. The iterative technique again converges rapidly. A typical flat plane slot array is shown in Figure 4.1 (see also Figs. 6.30 and 6.31).

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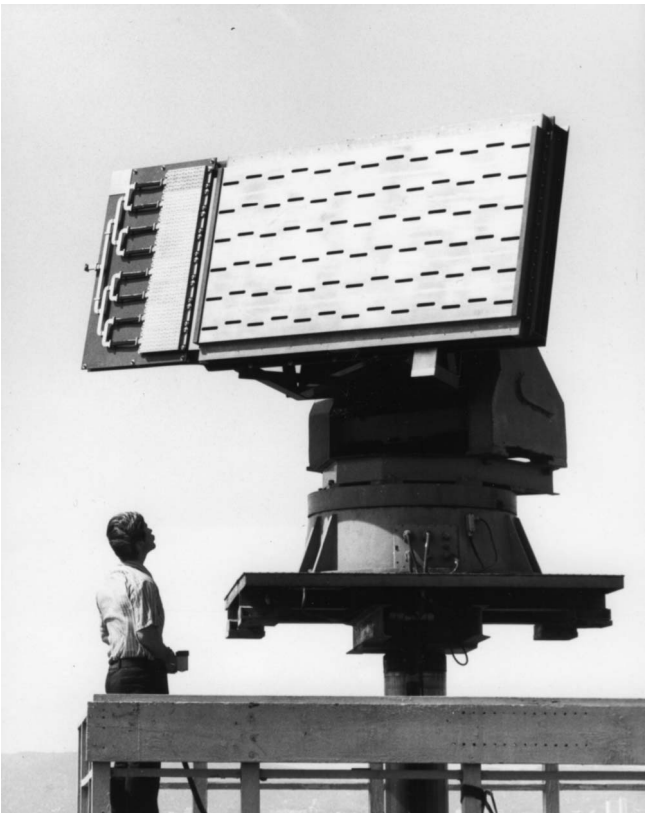


FIGURE 4.1 Flat plane array. (Courtesy Hughes Aircraft Co.)

Use of principal plane canonical patterns simplifies the synthesis process, but lower sidelobes can be obtained through synthesis of a rotationally symmetric pattern, as discussed in Section 4.1.2.

4.1.2 Hansen One-Parameter Pattern

A symmetric circular one-parameter distribution and pattern analogous to the Taylor one-parameter distribution was developed by Hansen (1976). Starting as before with the uniform amplitude pattern, which is $2J_1(\pi u)/\pi u$, the close-in zeros are shifted to suppress the close-in sidelobes. The single parameter is H, and the space factor (pattern) is

$$\begin{aligned} F(u) &= \frac{2J_1\left(\pi\sqrt{u^2 - H^2}\right)}{\pi\sqrt{u^2 - H^2}} \qquad u \geq H \\ F(u) &= \frac{2I_1\left(\pi\sqrt{H^2 - u^2}\right)}{\pi\sqrt{H^2 - u^2}} \qquad u \leq H \end{aligned} \tag{4.1}$$

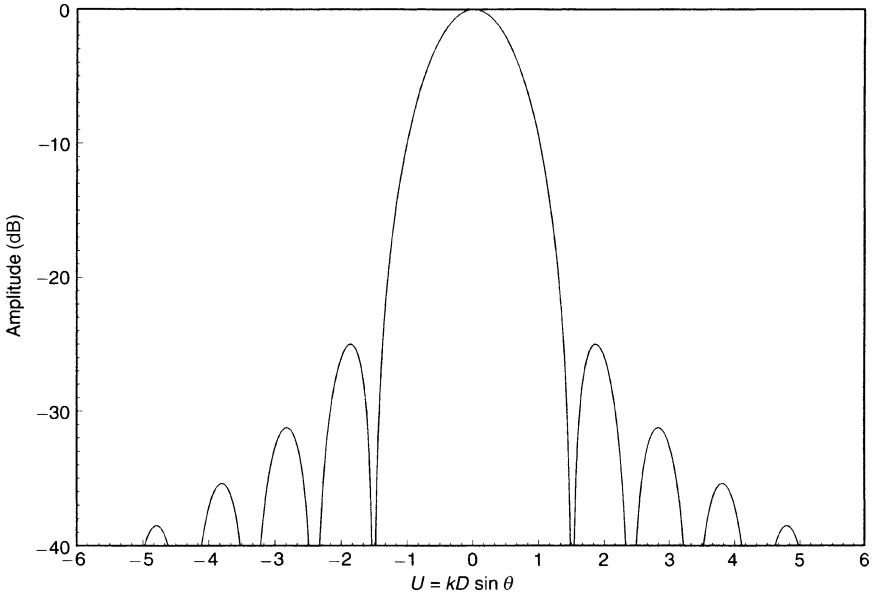


FIGURE 4.2 Hansen one-parameter pattern, SLR = 25 dB.

J_1 and I_1 are the Bessel function and modified Bessel function of the first kind and order one. In exactly the same manner as for the Taylor pattern, the expression for $u \geq H$ forms the sidelobe region, and part of the main beam, while the expression for $u \leq H$ forms the remainder of the main beam. The transition occurs at $u = H$. At this point the formula is 17.57 dB above the first sidelobe. From this point to the beam peak is $2I_1(\pi H)/\pi H$, so the sidelobe ratio is

$$\text{SLR} = 17.57\,\text{dB} + 20\log\frac{2I_1(\pi H)}{\pi H} \tag{4.2}$$

The one-parameter pattern is easy to use, as both J_1 and I_1 are represented by absolutely convergent series and are quickly generated by common computer subroutines. Figure 4.2 shows a typical one-parameter pattern, for 25-dB SLR. Table 4.1 gives H

TABLE 4.1 Characteristics of Hansen One-Parameter Distribution

SLR (dB)	H	u_3 (rad)	η_t	ηu_3	Edge Taper (dB)
17.57	0	0.5145	1	0.5145	0
20	0.4872	0.5393	0.9786	0.5278	4.5
25	0.8899	0.5869	0.8711	0.5113	12.4
30	1.1977	0.6304	0.7595	0.4788	19.3
35	1.4708	0.6701	0.6683	0.4478	25.8
40	1.7254	0.7070	0.5964	0.4216	32.0
45	1.9681	0.7413	0.5390	0.3996	38.0
50	2.2026	0.7737	0.4923	0.3809	43.9

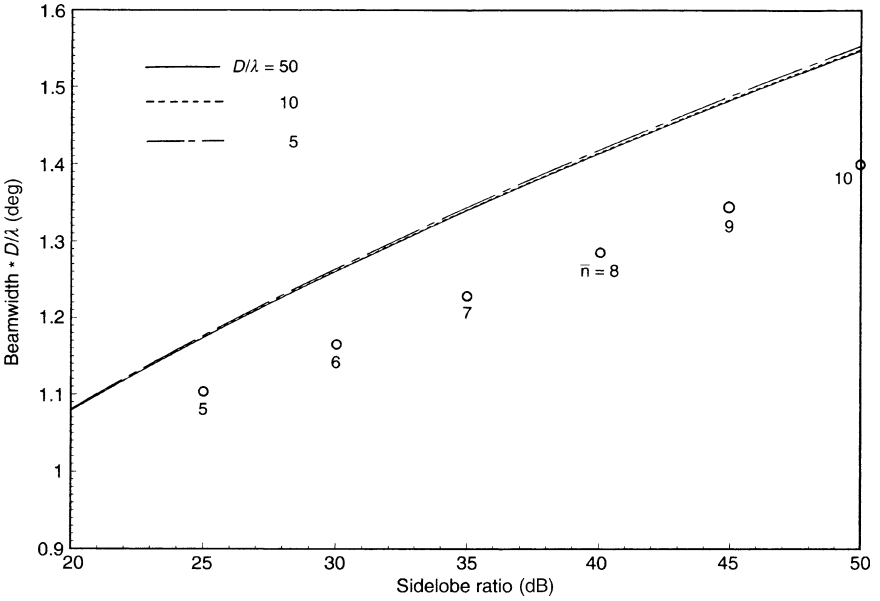


FIGURE 4.3 Hansen one-parameter distribution beamwidth.

and the normalized half-beamwidth u_3. Values up to 50 dB SLR are given, as the antenna art allows this performance. The 3-dB beamwidth u_3 is a solution of Eq. (4.3):

$$\frac{I_1(\pi H)}{2\pi H} = \frac{J_1\left(\pi\sqrt{u_3^2 - H^2}\right)}{\pi\sqrt{u_3^2 - H^2}} \tag{4.3}$$

The aperture distribution is

$$g(p) = I_0\left(\pi H\sqrt{1 - p^2}\right) \tag{4.4}$$

where p is the radial coordinate, zero at the center, and unity at the edge. I_0 is a zero-order modified Bessel function. Figure 4.3 gives the beamwidth times aperture diameter D divided by wavelength, and it may be seen that the beamwidth is nearly independent of D/λ, depending primarily on SLR. The dots will be discussed in the next section. Figure 4.4 shows the (half-) aperture amplitude distribution for several values of SLR. Edge values are given in Table 4.1. The aperture excitation efficiency η_t is found by integrating the distribution

$$\begin{aligned}\eta_t &= \frac{2\left(\int_0^{\pi} g(p)p\ dp\right)^2}{\pi^2\int_0^{\pi} g^2(p)p\,dp} \\ &= \frac{4I_1^2(\pi H)}{\pi^2 H^2[I_0^2(\pi H) - I_1^2(\pi H)]}\end{aligned} \tag{4.5}$$

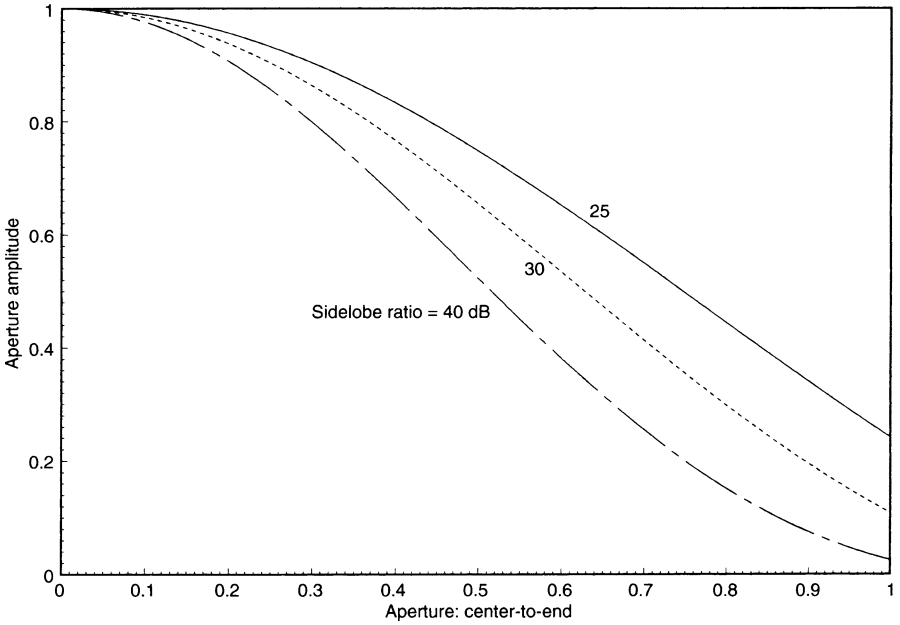


FIGURE 4.4 Hansen one-parameter aperture distribution.

This result for efficiency is readily calculated, and is shown in Table 4.1. This table also gives the product of efficiency and normalized beamwidth. In comparison with the circular Taylor $\bar{n}$ distribution, the Hansen distribution is slightly less efficient, but it is smoother and more robust.

Beam efficiency η_b, the fraction of power in the main beam, might require integration over the sidelobes, a tedious task. However, it can be written in terms of aperture efficiency η_t, which can be easily calculated from the aperture distribution. The aperture efficiency is

$$\frac{1}{\eta_t} = \frac{\pi^2 D^2}{2\lambda^2} \int_0^{\pi/2} \frac{E^2}{E_0^2} \sin\theta \, d\theta \tag{4.6}$$

while beam efficiency is

$$\eta_b = \frac{\displaystyle\int_0^{\theta_N} \frac{E^2}{E_0^2} \sin\theta \, d\theta}{\displaystyle\int_0^{\pi/2} \frac{E^2}{E_0^2} \sin\theta \, d\theta} \tag{4.7}$$

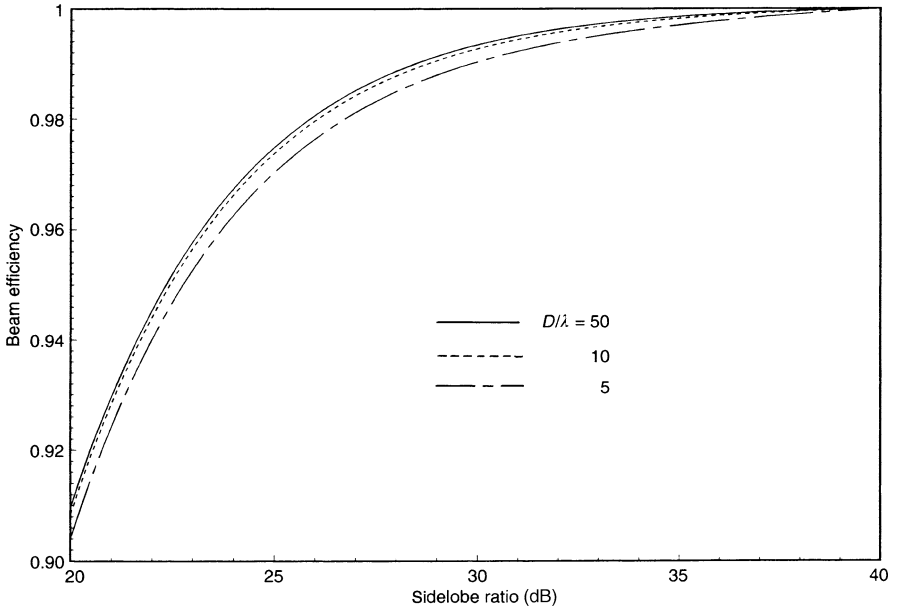


FIGURE 4.5 Hansen one-parameter distribution beam efficiency.

Here θ_N is the main beam null angle. Now the beam efficiency can be expressed by a simple integral over the main beam:

$$\eta_b = \frac{\pi^2 D^2 \eta_t}{2\lambda^2} \int_0^{\theta_N} \frac{E^2}{E_0^2} \sin\theta \, d\theta \tag{4.8}$$

Results are shown in Figure 4.5. For diameters of 10 wavelengths or more, η_b is nearly independent of D/λ. Because of the low sidelobes at all azimuth angles, the beam efficiency is much higher than that for a linear distribution.

4.1.3 Taylor Circular $\bar{n}$ Pattern

The circular Taylor $\bar{n}$ distribution offers a slightly higher efficiency, through a slightly slower decay of the first $\bar{n}$ sidelobes, exactly analogously to the linear Taylor $\bar{n}$ distribution. Again the aperture distribution is symmetric. The uniform $2J_1(\pi u)/\pi u$ pattern is the starting point; $\bar{n}$ zeros each side of the main beam are adjusted to provide roughly equal levels at the specified SLR. Farther-out zeros are those of the uniform pattern. As before, a dilation factor σ provides a transition between the $\bar{n}$ zeros, and those adjacent. The sidelobe ratio is related to the general parameter A:

$$\text{SLR} = \cosh A \tag{4.9}$$

The Taylor circular $\bar{n}$ space factor (pattern) is given by (Taylor, 1960)

$$F(u) = \frac{2J_1(\pi u)}{\pi u} \prod_{n=1}^{\bar{n}-1} \frac{1 - \dfrac{u^2}{u_n^2}}{1 - \dfrac{u^2}{\mu_n^2}} \tag{4.10}$$

The new zeros are

$$u_n = \pm\sigma \sqrt{A^2 + \left(n - \frac{1}{2}\right)^2} \tag{4.11}$$

and the dilation factor is

$$\sigma = \frac{\mu_{\bar{n}}}{\sqrt{A^2 + \left(\bar{n} - \dfrac{1}{2}\right)^2}} \tag{4.12}$$

The zeros of $J_1(\pi u)$ are μ_n. A typical $\bar{n}$ pattern is shown in Figure 4.6. Beamwidth is the ideal line source beamwidth u_3 times σ; Table 4.2 gives the parameter A, half-beamwidth u_3, and σ for various SLR and $\bar{n}$. Hansen (1960) gives tables of the aperture distribution.

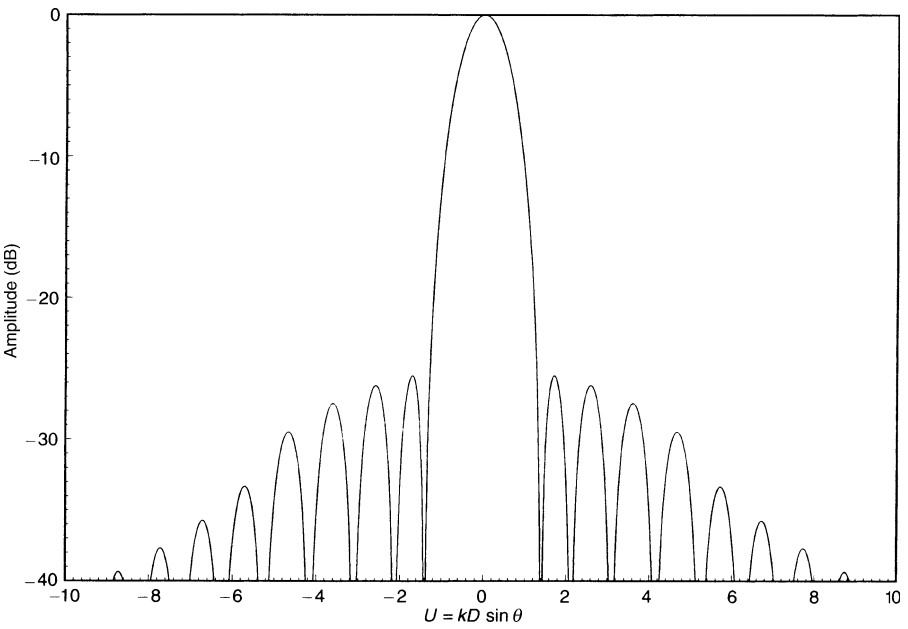


FIGURE 4.6 Taylor circular $\bar{n}$ pattern, SLR = 25 dB, $\bar{n} = 5$.

TABLE 4.2 Taylor Circular $\bar{n}$ Pattern Characteristics

SLR			σ							
(dB)	A	u_3	$\bar{n}=3$	$\bar{n}=4$	$\bar{n}=5$	$\bar{n}=6$	$\bar{n}=7$	$\bar{n}=8$	$\bar{n}=9$	$\bar{n}=10$
20	0.9528	0.4465	1.2104	1.1692	1.1398	1.1186	1.1028	1.0906	1.0810	1.0732
25	1.1366	0.4890	1.1792	1.1525	1.1296	1.1118	1.0979	1.0870	1.0782	1.0708
30	1.3200	0.5284	1.1455	1.1338	1.1180	1.1039	1.0923	1.0827	1.0749	1.0683
35	1.5032	0.5653		1.1134	1.1050	1.0951	1.0859	1.0779	1.0711	1.0653
40	1.6865	0.6000		1.0916	1.0910	1.0854	1.0789	1.0726	1.0670	1.0620
45	1.8697	0.6327			1.0759	1.0748	1.0711	1.0667	1.0624	1.0583
50	2.0530	0.6639				1.0636	1.0628	1.0604	1.0573	1.0542

The aperture distribution becomes (Taylor, 1960)

$$g(p) = \frac{2}{\pi^2} \sum_{n=0}^{\bar{n}-1} \frac{F_n J_0(\pi p \mu_n)}{J_0^2(\pi \mu_n)} \tag{4.13}$$

Note that the radial variable for circular Taylor $\bar{n}$ is $p = 2\pi\rho/D$. The coefficients F_m are

$$F_m = -J_0(\pi\mu_m) \frac{\prod_{n=1}^{\bar{n}-1} \left(1 - \dfrac{\mu_m^2}{u_n^2}\right)}{\prod_{\substack{n=1 \\ n \neq m}}^{\bar{n}-1} \left(1 - \dfrac{\mu_m^2}{\mu_n^2}\right)} \qquad m > 0$$
$$F_0 = 1 \tag{4.14}$$

Representative aperture distributions (half only) are shown in Figures 4.7 and 4.8. Too high a value of $\bar{n}$ causes instability in the null spacings, reflected in a nonmonotonic distribution. As larger values of $\bar{n}$ give higher efficiency, accurate for large apertures and constant phase, for each SLR there is an optimum trade on $\bar{n}$.

Aperture efficiency is obtained from

$$\eta_t = \frac{2\left(\int_0^{\pi} g(p) p \, dp\right)^2}{\int_0^{\pi} g^2(p) p \, dp} \tag{4.15}$$

Using the above $g(p)$, the terms in the numerator for $m > 0$ contain the factor $J_0(\pi\mu_m)$, which is zero. Only the $m = 0$ term remains. In the denominator, the integration of the product of the two series produces an integral of two orthogonal Bessel functions; only a single series remains:

$$\frac{1}{\eta_t} = 1 + \sum_{n=1}^{\bar{n}-1} \frac{F_n^2}{J_0^2(\pi\mu_n)} \tag{4.16}$$

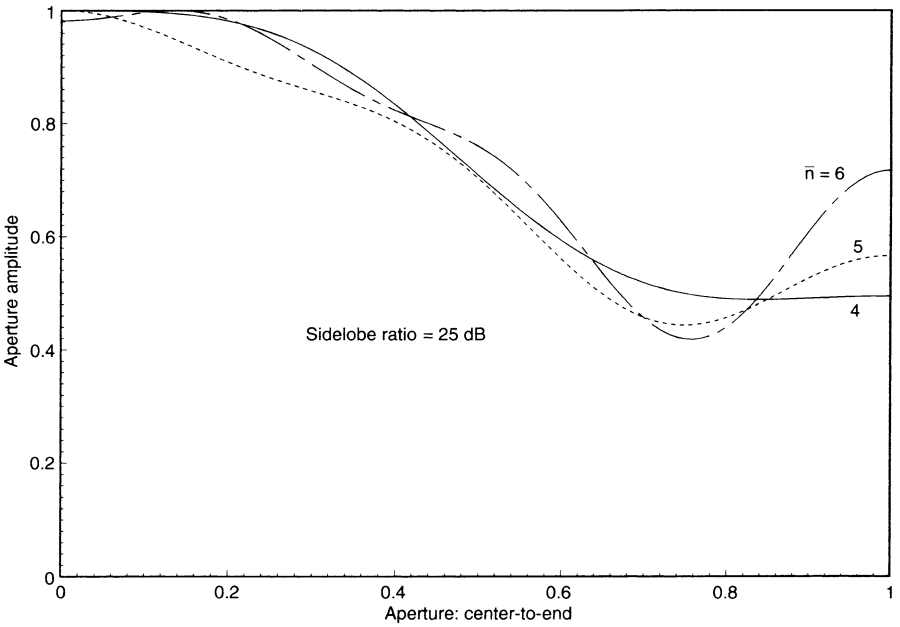


FIGURE 4.7 Taylor circular n̄ distribution (SLR = 25 dB).

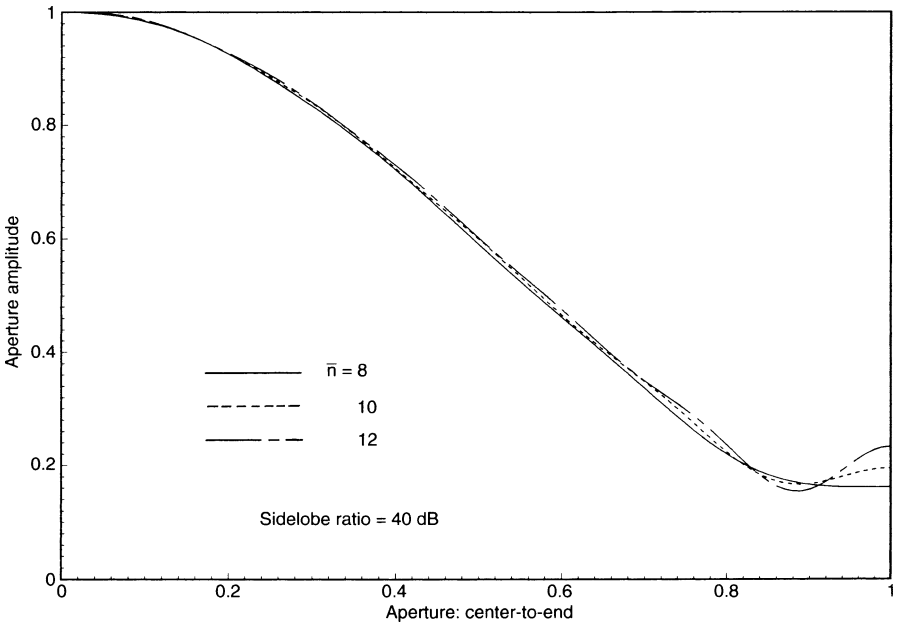


FIGURE 4.8 Taylor circular n̄ distribution (SLR = 40 dB).

TABLE 4.3 Taylor Circular $\bar{n}$ Efficiency

SLR (dB)	$\bar{n} = 4$	$\bar{n} = 5$	$\bar{n} = 6$	$\bar{n} = 8$	$\bar{n} = 10$
20	0.9723	0.9356	0.8808	0.7506	0.6238
25	0.9324	0.9404	0.9379	0.9064	0.8526
30	0.8482	0.8623	0.8735	0.8838	0.8804
35	0.7708	0.7779	0.7880	0.8048	0.8153
40	0.7056	0.7063	0.7119	0.7252	0.7365
45		0.6484	0.6494	0.6575	0.6663
50			0.5986	0.6016	0.6076

Source: Courtesy Rudduck, R. C. et al., "Directive Gain of Circular Taylor Patterns", *Radio Sci.*, Vol. 6, Dec. 1971, pp. 1117–1121.

TABLE 4.4 Taylor Circular $\bar{n}$ Beam Efficiency

SLR (dB)	$\bar{n}$	η_b
20	4	0.8271
25	5	0.9142
30	6	0.9621
35	7	0.9844
40	8	0.9938
45	9	0.9976
50	10	0.9991

Excitation efficiencies for several SLR and $\bar{n}$ are given in Table 4.3, from Rudduck et al. (1971). Beam efficiency again is from an integral over the main beam. Table 4.4 gives this versus SLR.

4.1.4 Circular Bayliss Difference Pattern

Before discussing the Bayliss $\bar{n}$ difference pattern, it is useful to consider a uniform circular aperture of radius a with the two halves out of phase by π. The ϕ integral for the pattern function reduces to

$$\cos\phi H_0(k\rho\sin\theta\cos\phi) \tag{4.17}$$

where H_0 is the zero-order Struve function (Hansen, 1995). The difference pattern in the $\phi = 0$ plane is then given by

$$F(\theta,\phi) = \cos\phi\int_0^a H_0(k\rho\sin\theta\cos\phi)\rho\,d\rho \tag{4.18}$$

The Struve function closely resembles the J_1 Bessel function, as seen in Figure 4.9, although there appears to be no similarity for higher orders. Figure 4.10 gives a pattern produced by numerical integration, with the amplitude such that the corresponding sum pattern peak is unity. The difference pattern peaks occur at $u_0 = \pm 2.653$, and their level is -3.19 dB. First sidelobes are at $u = \pm 9.775$, with level -14.90 dB, or 11.72 dB below the difference peak.

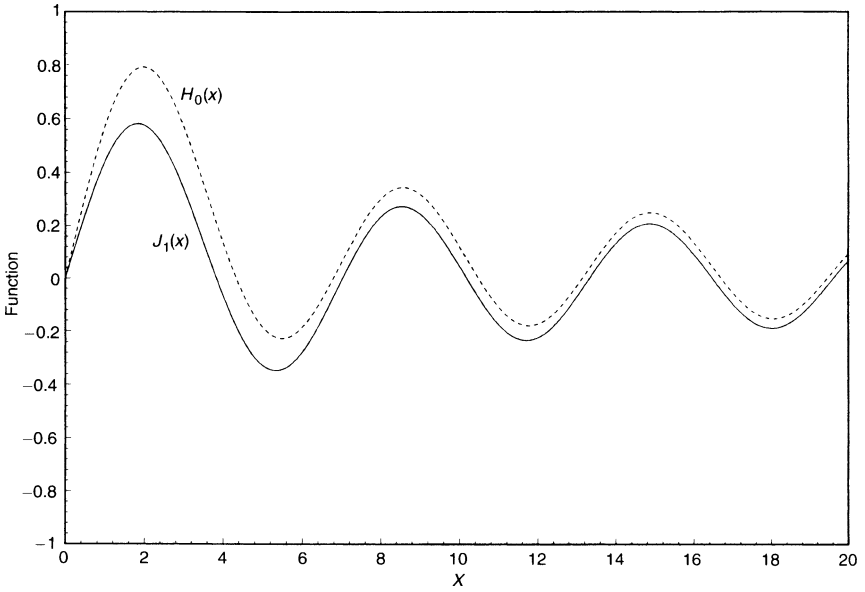


FIGURE 4.9 The parameter J_1 Bessel and H_0 Struve functions.

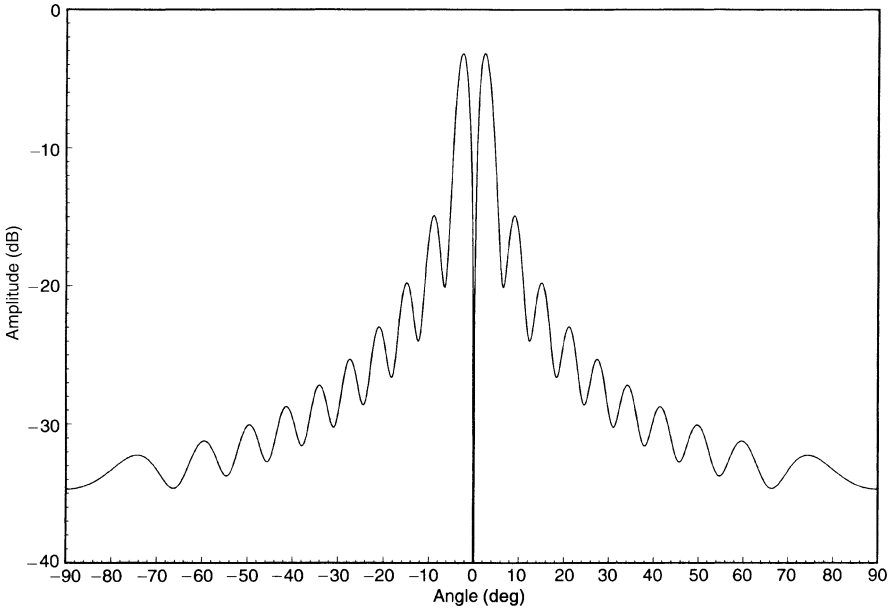


FIGURE 4.10 Uniform circular aperture difference pattern, $D = 20\lambda$.

The Bayliss difference pattern for a line source is discussed in Section 3.7.2. Bayliss (1968) developed a similar $\bar{n}$ difference pattern for a circular aperture. The starting pattern is the derivative of the "ideal" Taylor sum pattern $\cos(\pi\sqrt{u^2 - A^2})$, as the uniform circular aperture difference pattern does not exist in usable form. The derivative, with coefficients deleted, is

$$f_{\text{ideal}}(u) = u \sin\left(\pi\sqrt{u^2 - A^2}\right) \tag{4.19}$$

The $\bar{n}$ difference pattern is achieved by replacing $\bar{n}$ of the zeros of the ideal pattern by shifted zeros z_n:

$$f(u) = f_{\text{ideal}}(u) \prod_{n=1}^{\bar{n}} \frac{z_n^2 - u^2}{\mu_n^2 - u^2} \tag{4.20}$$

The new zeros are z_n; these replace the ideal pattern zeros μ_n. The space factor can be written in product form on the zeros:

$$f(u) = u \cos\phi \prod_{n=1}^{\bar{n}-1} \left(1 - \frac{u^2}{\sigma^2 z_n^2}\right) \prod_{n=\bar{n}}^{\infty} \left(1 - \frac{u^2}{\mu_n^2}\right) \tag{4.21}$$

As the zeros of the second product are those of $J_1'(\pi\mu_n)$, that product can be augmented with the missing zeros, and replaced by $J_1'(\pi u)$, with the result

$$f(u) = u \cos\phi J_1'(\pi u) \frac{\prod_{n=1}^{\bar{n}-1} \left(1 - \frac{u^2}{\sigma^2 z_n^2}\right)}{\prod_{n=0}^{\bar{n}-1} \left(1 - \frac{u^2}{\mu_n^2}\right)} \tag{4.22}$$

An alternate version sums over $\bar{n}$ terms:

$$f(u) = u \cos\phi J_1'(\pi u) \sum_{n=0}^{\bar{n}-1} \frac{B_n J_1(\pi\mu_n)}{\mu_n^2 - u^2} \tag{4.23}$$

In this difference pattern result, the zeros for $n > \bar{n}$ are given by $z_n = \pm\sqrt{A^2 + n^2}$. Because the ideal difference pattern had close-in sidelobes of irregular levels, Bayliss adjusted these $\bar{n}$ zeros to obtain a monotonic sidelobe envelope. This was done numerically, with fourth-order polynomials fitted to the results. The parameter A and the position of the difference pattern peak were also fitted. These parameters are the same for the linear Bayliss pattern, and are given in Table 3.11. The polynomial coefficients are given in Table 3.12. Figure 4.11 shows a circular Bayliss pattern for SLR = 25 dB, and $\bar{n} = 5$.

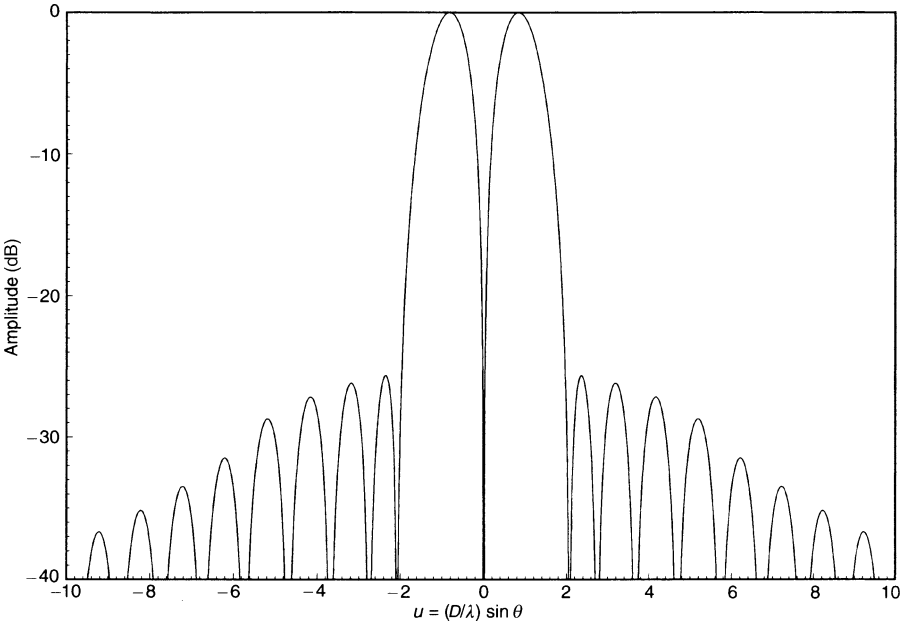


FIGURE 4.11 Circular Bayliss pattern, SLR = 25 dB, $\bar{n} = 5$.

The aperture distribution comes from the Bessel series:

$$g(p) = \cos\phi \sum_{n=0}^{\bar{n}-1} B_n J_0(\mu_n p) \tag{4.24}$$

where $p = \pi\rho$, with ρ the aperture radius from zero to one. The B_n coefficients are

$$B_m = \frac{2\mu_m^2 \prod_{n=1}^{\bar{n}-1}\left(1 - \dfrac{\mu_m^2}{\sigma^2 z_n^2}\right)}{J_1(\pi\mu_m) \prod\limits_{\substack{n=0 \\ n\neq m}}^{\bar{n}-1}\left(1 - \dfrac{\mu_m^2}{\mu_n^2}\right)} \qquad m = 0, 1, 2, \ldots, \bar{n} = 1 \tag{4.25}$$

$$B_0 = 0 \qquad m \geq \bar{n}$$

The dilation factor σ is given by

$$\sigma = \frac{\mu_{\bar{n}}}{\sqrt{A^2 + \bar{n}^2}} \tag{4.26}$$

Figure 4.12 shows the aperture distribution for the case of Figure 4.10. As expected, larger values of $\bar{n}$ for a given SLR produce a rise at the aperture edge, which can be higher than the center value. Figure 4.13 from a Bayliss report shows a quadrant of the two-dimensional (2D) distribution.

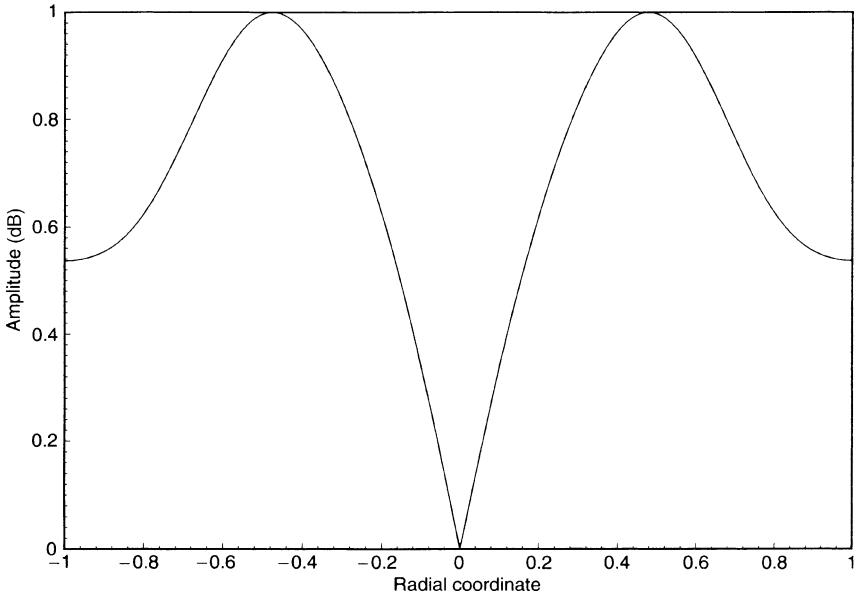


FIGURE 4.12 Circular Bayliss aperture distribution, SLR = 25 dB, $\bar{n} = 5$.

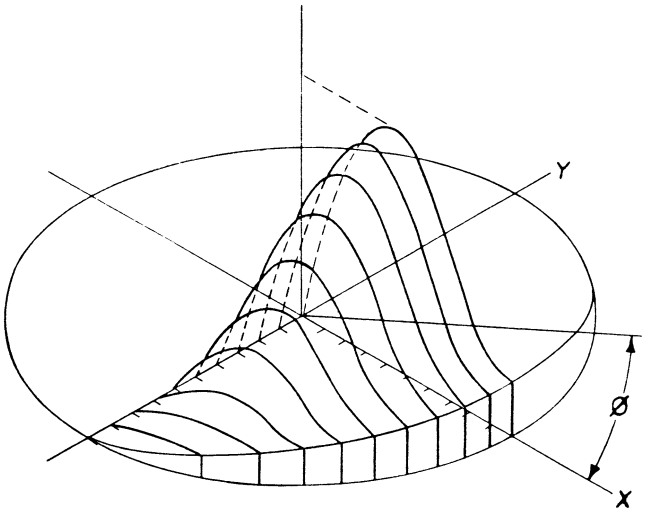


FIGURE 4.13 Circular Bayliss distribution, SLR = 35 dB, $\bar{n} = 10$. (Courtesy Bayliss, E. T., "Monopulse Difference Patterns with Low Sidelobes and Large Angle Sensitivity", Tech. Memo. 66-4131-3, Bell Labs., Dec. 1966.)

TABLE 4.5 Circular Bayliss Efficiency and Slope

SLR (dB)	$\bar{n}$	η_t	Normalized Slope
25	5	0.4931	0.8754
30	6	0.4593	0.8156
35	8	0.4285	0.7638
40	10	0.3988	0.7144
45	12	0.3717	0.6691
50	14	0.3475	0.6286

Aperture efficiency is defined as the directivity at the difference pattern peak, divided by the uniform aperture sum pattern directivity. Owing to the orthogonality of the aperture series, the directivity integral is readily evaluated, giving

$$\frac{1}{\eta_t} = \frac{\pi^4}{8} \sum_{n=0}^{\bar{n}-1} |B_n|^2 \left(1 - \frac{1}{\pi^2 \mu_n^2}\right) J_1^2(\pi \mu_n) \tag{4.27}$$

A second important parameter is slope, here normalized to that of the maximum slope aperture. This reference pattern is $J_2(\pi u)/\pi u$, and its slope is $(0.5/\pi)\sqrt{G_0}$, where G_0 is the directivity. For the Bayliss pattern, the normalized slope becomes

$$S = \sqrt{\eta_t} \sum_{n=0}^{\bar{n}-1} \frac{B_n J_1(\pi \mu_n)}{\mu_n^2} \tag{4.28}$$

Table 4.5 gives efficiency and slope for typical combinations of SLR and $\bar{n}$.

4.1.5 Difference Pattern Optimization

The use of sub-arrays, where each can be fed with a different amplitude (and phase) for sum and difference patterns, was discussed in Chapter 3 for linear arrays. The same principle was applied to circular planar arrays by Josefsson et al. (1977). The array is divided into sub-arrays, with quadrantal symmetry. The shape of each sub-array and the number of elements will generally be different; see Figure 4.14. Symmetric pairs of sub-arrays are then connected to hybrid junctions; these hybrids are connected to the sum and to the two difference combiner networks (see Fig. 4.15). As in the linear array case, a stair-step approximation is provided to the three desired pattern distributions. Excellent results have been obtained with only a small number of sub-arrays per quadrant. This configuration is especially attractive with printed circuit antennas, as the implementation is natural for realization in stripline.

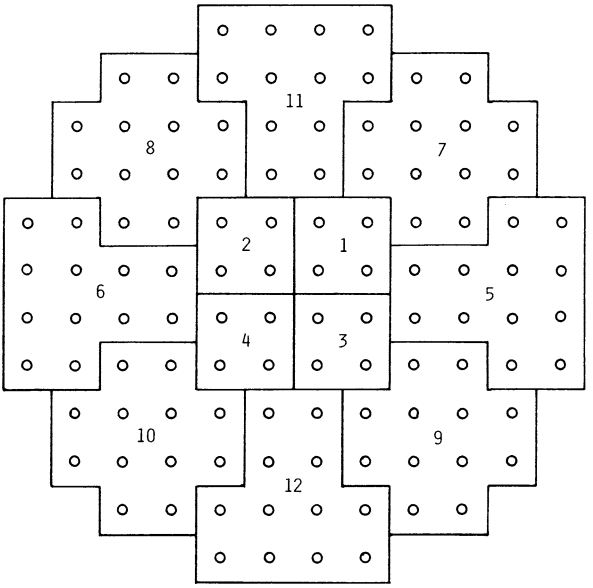


FIGURE 4.14 Planar array sub-arrays.

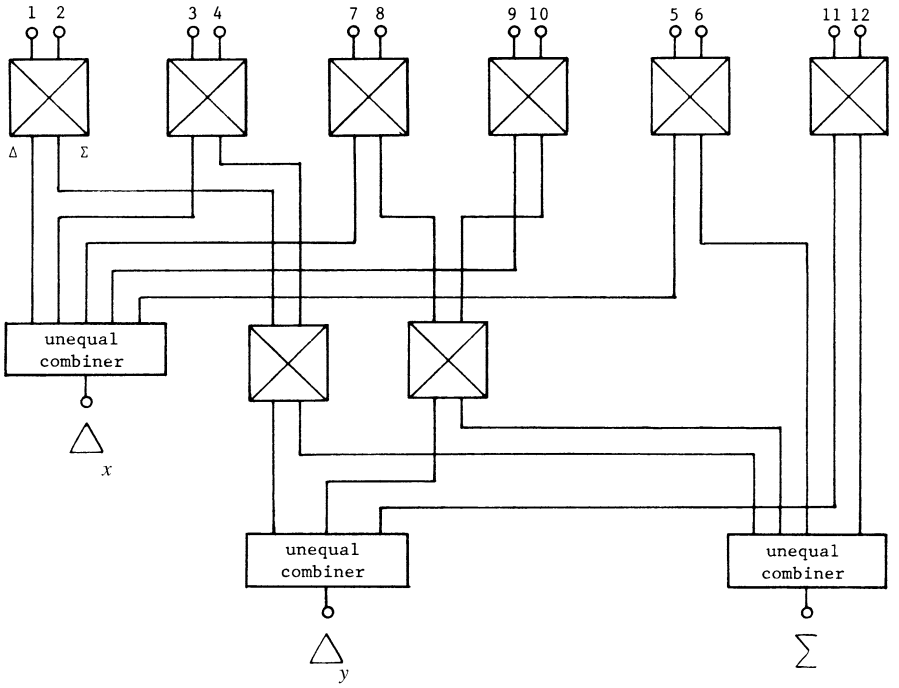


FIGURE 4.15 Feed network for sum and difference optimization.

4.2 NONCIRCULAR APERTURES

4.2.1 Two-Dimensional Optimization

When the aperture (array) is not circular, or when x and y distributions are not separable, recourse must be had to numerical techniques, such as constrained optimization. Unfortunately, a relatively simple technique (dynamic programming) is not applicable, as it requires the function ahead of the change to be independent of the change. But with all arrays, changes to one zero or to one excitation coefficient affect the entire pattern. Optimization programs, such as conjugate gradient are suitable provided the objective function can be well specified, but slow. Since sidelobe positions are not predictable, maximizing directivity subject to the requirement that all sidelobes be below a given envelope is difficult. Either a sidelobe location routine that searches for each and every sidelobe peak at each function (array pattern) evaluation is used, or each pattern must be digitized sufficiently finely to accurately locate all sidelobe peaks. Sidelobe searching algorithms only work well on canonical patterns, with regularly spaced, well-behaved sidelobes. Numerically optimized patterns, and measured patterns, typically have a nonmonotonic sidelobe envelope. Further, many "split" sidelobes occur that are much narrower. The typical fine structure of sidelobes defeats most algorithms, even for one-dimensional (1D) patterns. Fine sampling with filtering to find the envelope is feasible, but for 2D patterns it can be extraordinarily slow. Since an optimization may require hundreds of function evaluations, the computational cost may well be excessive.

For patterns where the array is small or modest in size, and the pattern features are coarse, optimization of directivity, slope, and so on, with constraints on sidelobe envelope, bandwidth, tolerances, and so on, may be feasible (see Perini and Idselis, 1972, for example). Several classes of optimization algorithms exist. First are gradient schemes. Simple gradient and conjugate gradient schemes tend to be fast, but they often stop at a local maximum, or hang up at a saddle point. More sophisticated schemes avoid both these problems and will usually find the global maximum (Fletcher, 1986). Some of these are those of Fletcher and Powell (1963), Fletcher and Reeves (1964), and Gill and Murray (1972). An algorithm due to Hersey et al. (1972) allows bounds to be placed over regions, such as sidelobe regions; a fine discretization provides a REMEZ exchange minimax optimization (Mucci et al., 1975). A review of unconstrained optimization methods is given by Nocedal (1991). Search techniques are generally inefficient, except for simplex methods. Simplex compares function values, forming a sequence of simplexes; each simplex is a set of points with a vertex where the function value is maximum. This vertex is reflected in the centroid of the other vertices, forming a new simplex. The value at this new vertex is evaluated, and the process is repeated. Special rules restart the process during stagnation. In the author's modest experience, an excellent algorithm is the Fletcher variable metric (Fletcher, 1970), available in Fortran from Harwell, both in analytical and numerical derivative form.

Because the sidelobe positions change as the excitation changes, an iterative procedure is usually used. Here the sidelobe positions for an appropriate canonical pattern

are used in the first iteration. At each iteration, the pattern zeros are determined with sidelobe peaks assumed midway between.

Synthesis with sidelobe constraints, or synthesis to minimize the oscillations in a shaped beam pattern, often minimize the sum of squares of the differences. However, use of least squares in the optimized function gives fits that are less good at the interval edges. A better scheme uses a least pth optimization, where the pth root of the sum of differences, each to the pth power, is minimized (Bandler and Charalambous, 1972; Temes and Zai, 1969). A value of $p = 10$ gives superior fits, while very large p values approach minimax results. However, the latter require careful floating-point number handling.

4.2.2 Ring Sidelobe Synthesis

For rectangular apertures, a sidelobe envelope that decays with θ is often desirable, analogous to the Hansen circular one-parameter pattern, or to the Taylor circular $\bar{n}$ pattern. Such a synthesis can be produced by the brute-force computer techniques of the previous section, or by the methods of Section 3.9.2, where the zeros of a (2D) polynomial are iteratively adjusted. This latter approach is far better.

A more analytical approach utilizes the Baklanov (1966) transformation, which converts the 2D problem to a 1D problem. Tseng and Cheng (1968) applied this transformation to a square array providing Chebyshev-type ring sidelobes. Kim and Elliott (1988) extended this to an adjustable topology of ring sidelobes. For a square array of $N \times N$ elements, let the pattern be

$$F(u, v) = 4 \sum_{n=1}^{N/2} \sum_{m=1}^{N/2} A_{nm} \cos[(2n-1)u] \cos[(2m-1)v] \tag{4.29}$$

The Baklanov transformation is

$$w = w_0 \cos u \cos v \tag{4.30}$$

Now assume the desired ring sidelobe pattern is given by a polynomial of order $N-1$ in the new variable w:

$$f_{N-1}(w) = \sum_{l=1}^{N/2} B_{2l-1} w^{2l-1} \tag{4.31}$$

Through use of trig expansions the array excitation coefficients A_{nm} are found from the polynomial coefficients B_l using

$$A_{nm} = \sum^{N/2} \frac{B_{2l-1}}{2^{2l-1}} \binom{2l-1}{2l-n} \binom{2l-1}{2l-m} \left[\frac{w_0}{2}\right]^{2l-1} \tag{4.32}$$

The sum starts at the larger of n and m. The parameter w_0 is related to the sidelobe envelope, and is determined iteratively. The synthesis procedure, to determine the B_l coefficients, can utilize either the zero adjusting process or the Orchard process of Section 3.9.2. In any case, the polynomial zeros must be determined so that sidelobe

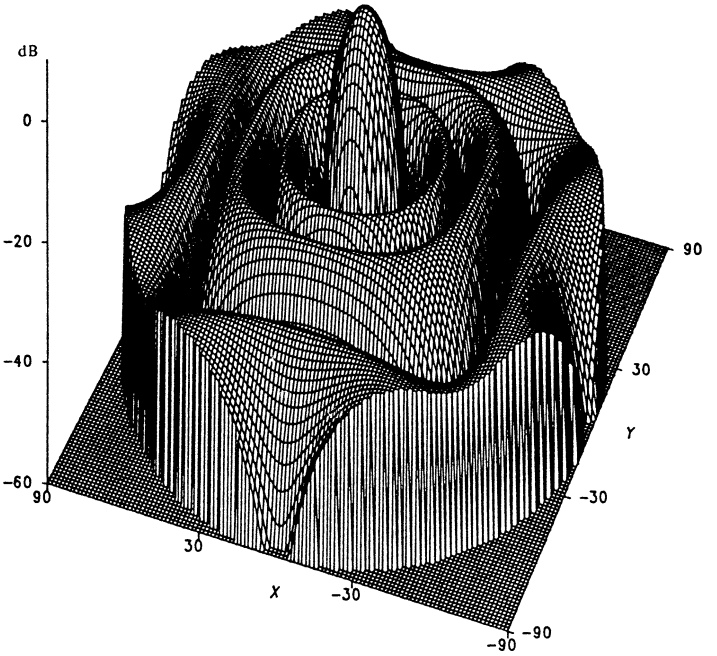


FIGURE 4.16 Baklanov ring sidelobe pattern. (Courtesy Kim, Y. U. and Elliott, R. S., "Extensions of the Tseng-Cheng Pattern Synthesis Technique", *J. Electromagnetic Waves Appl*, Vol. 2, No. 3/4, 1988, pp. 255–268.)

(ring) locations can be calculated. This procedure is repeated each iteration. Fortunately, <10 iterations are usually required. Further use of a projection algorithm is described by Kim (1988). An example is shown in Figure 4.16, which is a contour pattern for a 10×10 square array with first sidelobe at -30 dB (Kim, 1990). The sidelobe rings are not circular, but fit the aperture geometry. Such a synthesis provides higher gain for the same sidelobe level than a design based on the product of two linear distributions.

ACKNOWLEDGMENT

Photograph courtesy of Dr. Winifred Kummer.

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CHAPTER FIVE

Array Elements

Fixed-beam broadside arrays may employ low- or moderate-gain elements, but most arrays employ low-gain elements owing to the effects of grating and quantization lobes (see Chapter 3). This chapter is concerned with these low-gain elements. Moderate-gain elements, such as the spiral, helix, log-periodic, Yagi–Uda, horn, and backfire, generally do not have unique array properties, and are covered in numerous other books.

5.1 DIPOLES

5.1.1 Thin Dipoles

The dipole, with length L approximately a half-wavelength, is widely used because of its simple construction and good performance. Dipoles are made of two collinear and contiguous metallic rods or tubes with the feed between; or of conical conductors, typically hollow; or of strips or triangles printed on a thin dielectric support. Microstrip dipoles are discussed later. A strip dipole, where strip thickness is small compared to strip width, is equivalent to a cylindrical dipole of radius a equal to one-fourth the strip width w. (Lo, 1953). A half-wave dipole has a pattern symmetric about the dipole axis; with θ measured from the axis it is

$$f(\theta) = \frac{\cos(\pi/2 \cos\theta)}{\sin\theta} \tag{5.1}$$

The half-power beamwidth is 78.1°, and the directivity is $1.64 = 2.15$ dB. As the dipole length shortens, the pattern approaches that of a short dipole, $\sin\theta$, with a half-power beamwidth of 90° and directivity of $1.5 = 1.76$ dB. For lengths longer

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than a half-wavelength, the pattern sharpens, then breaks up. At $L = 0.625\lambda$, the main lobe is broadside, and the two sidelobes are small. But at $L = 0.75\lambda$, the sidelobes (at 45°) are larger than the main beam. Finally, the full-wave dipole with sinusoidal current distribution has a null at broadside. Pattern bandwidth then goes from short dipole length to length of roughly 0.6λ. The impedance bandwidth is usually limiting. It was shown by Abraham in 1898 that a vanishingly thin dipole has a sinusoidal current distribution. The Carter zero-order impedance theory discussed in Chapter 7 is adequate for mutual-impedance calculations, and for self-impedance of very thin ($L/a > 100$) dipoles. Full-wave dipoles, according to this theory, have zero feed current, so a higher order theory is needed for them. However, owing to their high input impedance and multilobe pattern, full-wave dipoles are rarely used. For antennas in the vicinity of half-wave in length, the King second-order theory (King, 1956) agrees well with experiments. Unfortunately, no satisfactory simple function fit to the variation of impedance versus frequency has been possible.

Practical dipole bandwidths can range up to roughly 30–40%; the open sleeve dipole discussed below provides an octave of impedance bandwidth (73%). Figures 5.1 and 5.2 give thin dipole input resistance and reactance versus dipole length, for three values of L/a. Capacitance between the dipole arms produces resonance below half-wave length, with fatter resonant dipoles being shorter than thin resonant dipoles. Figure 5.3 shows dipole shortening versus dipole radius. Moment method calculations also agree well with experiments (Hansen, 1990), and can give good results for very fat dipoles. The designer should start with the simple Carter results, then refine

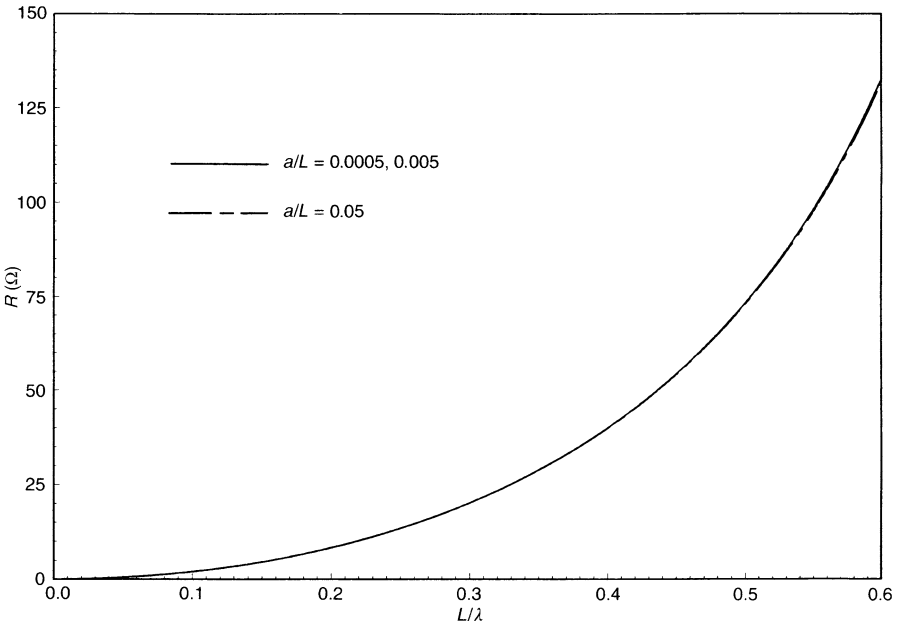


FIGURE 5.1 Resistance of wire dipole.

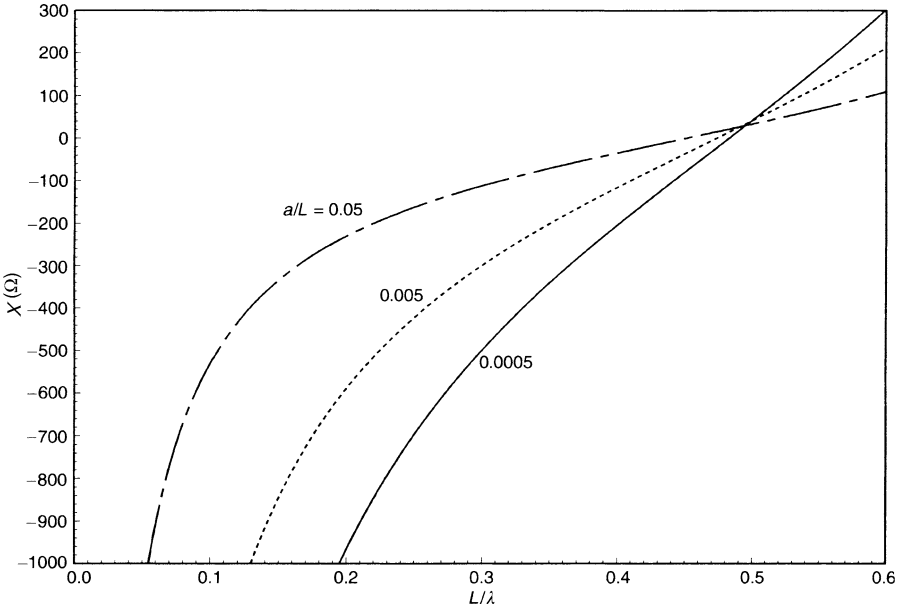


FIGURE 5.2 Reactance of wire dipole.

as needed with moment method, and of course, validate with measurements. The advent of network analyzers has greatly simplified the task of antenna impedance measurement.

Arrays with dipoles parallel to a ground or back screen are commonly used for beam angles around broadside. Such dipoles are fed by baluns, to convert the balanced two-wire dipole feed to a coax feed. Baluns then are normal to the screen, and utilize the roughly $\lambda/4$ spacing between dipole and screen. A simple but often used balun is

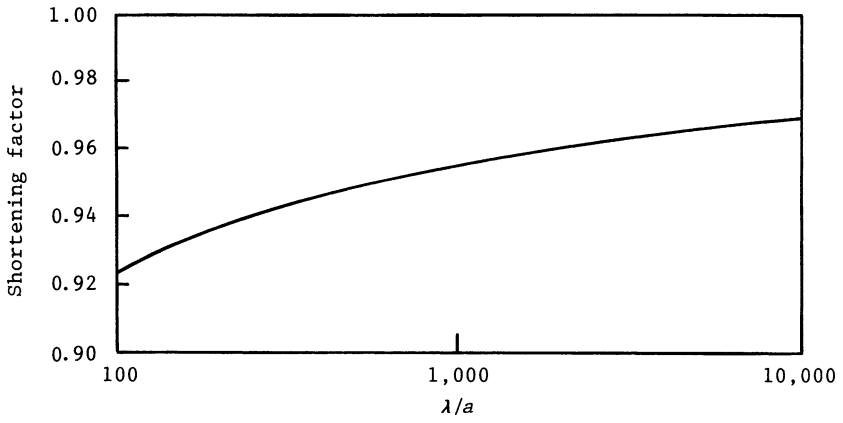


FIGURE 5.3 Resonant dipole shortening versus radius.

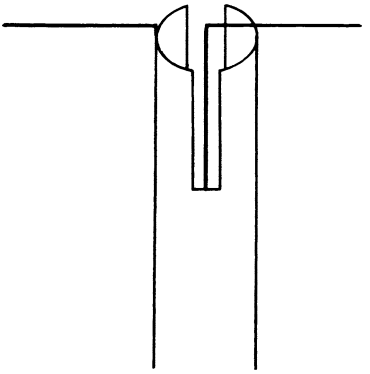


FIGURE 5.4 Split tube balun.

the split tube balun of Figure 5.4, where the split is $\lambda/4$ long. Wider bandwidths are provided by coaxial type I or III baluns,[1] where a coax feed line is enclosed by a cylinder connected at the screen. In the type III balun, a dummy outer conductor (to the coax) is added to form a two-wire line inside the outer cylinder (see sketch in Fig. 5.5). Circular or adjustable polarization is provided by crossed dipoles connected to two sets of balun posts at right angles to each other. A simple way of producing

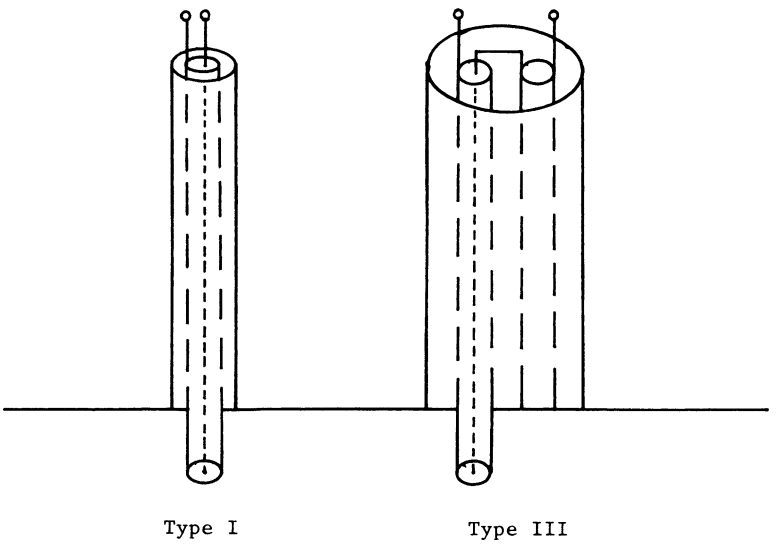


FIGURE 5.5 Coaxial baluns. (Courtesy Nelson, J. A. and Stavis, G., "Impedance Matching, Transformers and Baluns", in *Very High-Frequency Techniques*, Vol. 1, H. J. Reich, Ed., Radio Research Lab., McGraw-Hill, 1947.)

[1]Defined circa World War II by Nelson and Stavis (1947).

circular polarization uses crossed dipoles connected to a single balun. One dipole is larger and thinner, the other shorter and fatter, so that the respective phases are $\pm 45^\circ$. Bandwidth is significantly reduced, as is expected.

A dipole a distance h above a ground plane is equivalent to the dipole and its image (at a distance twice that of the dipole above the plane): Thus this "two-element array" provides directivity. The pattern, but not including the dipole pattern, is $\sin(kh \cos \theta)$. Input impedance, as shown in Figure 5.6, oscillates with h, but for small h becomes small. As the spacing decreases the pattern approaches $\cos \theta$. Directivity increases slowly as h decreases, as observed by Brown (1937). However, since the image current is reversed, as h decreases the heat loss increases. Thus the gain actually peaks. Figure 5.7 shows directivity versus h/λ, and also gain for several loss factors. The loss resistance shown is the equivalent resistance at the terminals. This finding is related to the surface resistance R_s by multiplying the latter by the metallic path length and dividing by the path width. For a half-wave dipole the path length is $1/2$, while the width is the dipole circumference:

$$R_L = \frac{R_s L}{4\pi a} = \frac{R_s L}{\pi w} \tag{5.2}$$

For example, a printed circuit dipole at 5000 MHz made of copper, and with a length/width ratio of 50, has a terminal resistance of 0.3 Ω, a small but nontrivial value. There is then a spacing between dipole and ground screen that maximizes the gain.

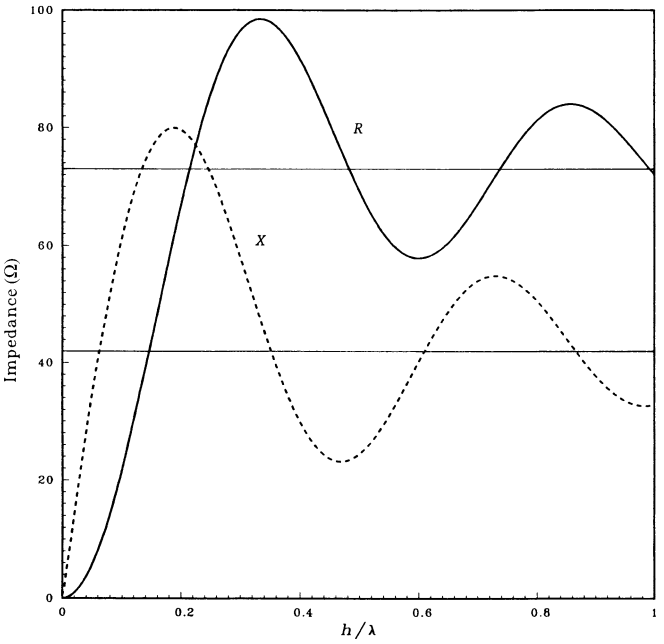


FIGURE 5.6 Impedance of half-wave dipole parallel over screen.

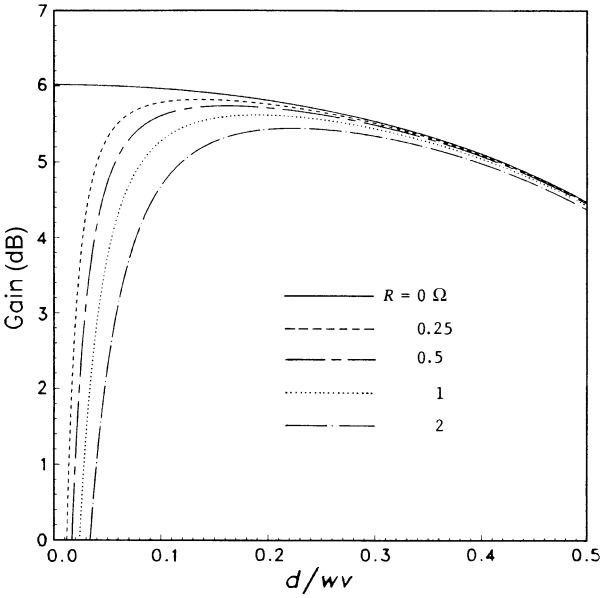


FIGURE 5.7 Gain of out of phase dipole pair with loss.

Direct (metallic) coupling of a dipole to a transmission line is not necessary; electromagnetic coupling may be used (Forbes, 1960). Here the shorted dipole is placed above a two-wire feed line at an angle to the line.

Dipoles with cross sections other than circular can, if the length/diameter is roughly ≥ 10, be equated to a cylindrical dipole. The most important is the thin strip dipole. Lo (1953) found the equivalent radius of polygons: for a strip dipole it is one-fourth the width; for a square conductor it is 0.59 times the side. An L shaped conductor has equivalent radius of 0.2 times the sum of the arm widths (Wolff, 1967). An elliptical conductor has equivalent radius half the sum of the major and minor radii (Balanis, 1982); also given there is a curve for a rectangular conductor.

Folded dipoles are used mostly in VHF–UHF arrays, as their primary advantage is the ability to match typical two-wire transmission lines. They are no more broadband than a fat dipole occupying the same volume. The impedance over a wide range of lengths (as seen in Fig. 5.8) has resonances at dipole length roughly $\lambda/3$ and $2\lambda/3$. In between, at length $\lambda/2$, is the normal operating point. The impedance is given by

$$Z = \frac{2(1+a)^2 Z_T Z_D}{(1+a^2)Z_D + 2Z_T} \tag{5.3}$$

where Z_D is the dipole impedance and Z_T is the shorted stub line impedance: $Z_T = jZ_0 \tan(\pi L/\lambda)$, with L the dipole length. The impedance can be adjusted by changing the ratio of the radius of the fed arms ρ_1 to the radius of the coupled arm ρ_2, and the center–center spacing d. The impedance transformation ratio is $(1+a)^2$, which is shown in Figure 5.9 versus spacing for several radius ratios. Figure 5.10 gives radius ratio versus spacing for several transformation ratios (Hansen, 1982).

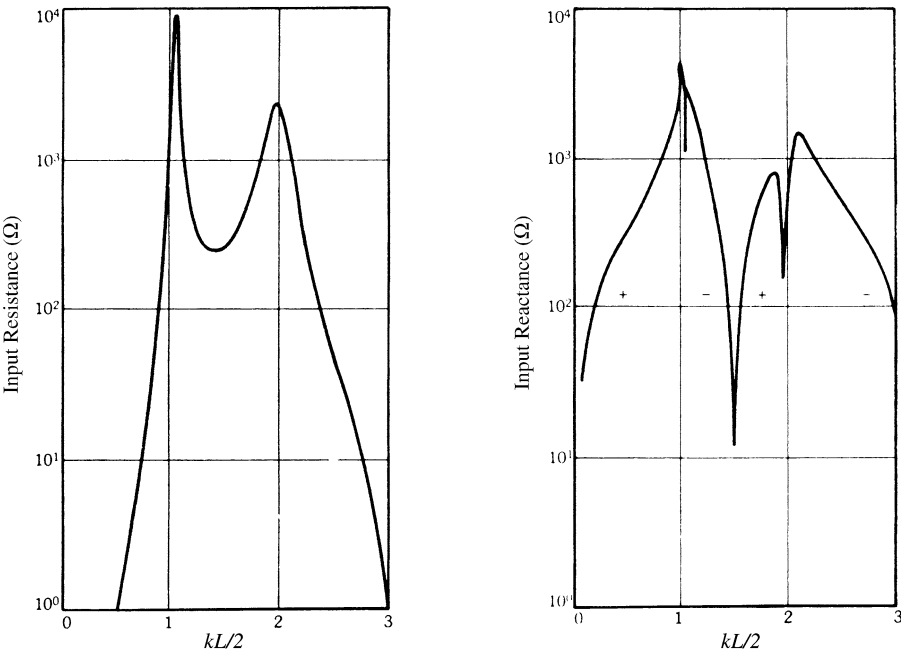


FIGURE 5.8 Folded dipole impedance, $d = 12.5a$, $a = 0.0005\lambda$. (Courtesy Rispin, L. W. and Chang, D. C., "Wire and Loop Antennas", in *Antenna Handbook—Theory, Applications, and Design*, Y. T. Lo and S. W. Lee, Eds., Van Nostrand Reinhold, 1988, Chapter 7.)

The effects of locating a dipole close to the ground plane can be mitigated by use of an Artificial Magnetic Conductor (AMC). This AMC layer presents a high impedence that prevents the ground plane from "shorting out" the dipole. Typically, an AMC consists of a closely packed array of conducting circles located a short distance above the ground plane. The center of each circle is connected to the ground plane by a post. This "mushroom" AMC was the first (Sievenpiper et al., 1999). Here the circle capacitance and the postinductance form a high impedance at the resonant frequency. A limitation on all AMC is the bandwidth. Away from the resonance frequency, all dipole parameters are affected: pattern, gain, and impedence. Hansen (2002) has shown that the range of usefulness is only for a phase change of $\pm 45^\circ$. Also, there is a fundamental limitation on AMC bandwidth (Brewitt-Taylor, 2007):

$$\frac{\Delta f}{f} \leq \pi \frac{h}{\lambda} \Delta\phi$$

Here, $\Delta f/f$ is the useable bandwidth, the dipole-screen spacing is h, and $\Delta\phi$ is the allowable phase excursion, $\pi/4$. This reduces to $\Delta f/f \leq h/4\lambda$. It appears that the only way to overcome this limit is to use a low-loss permeable layer. Walser[2] has shown that such a material would increase the bandwidth by μ. Prototype materials are "Metaferrites", a proprietary product of Metamaterials LLC.

[2]Private communication.

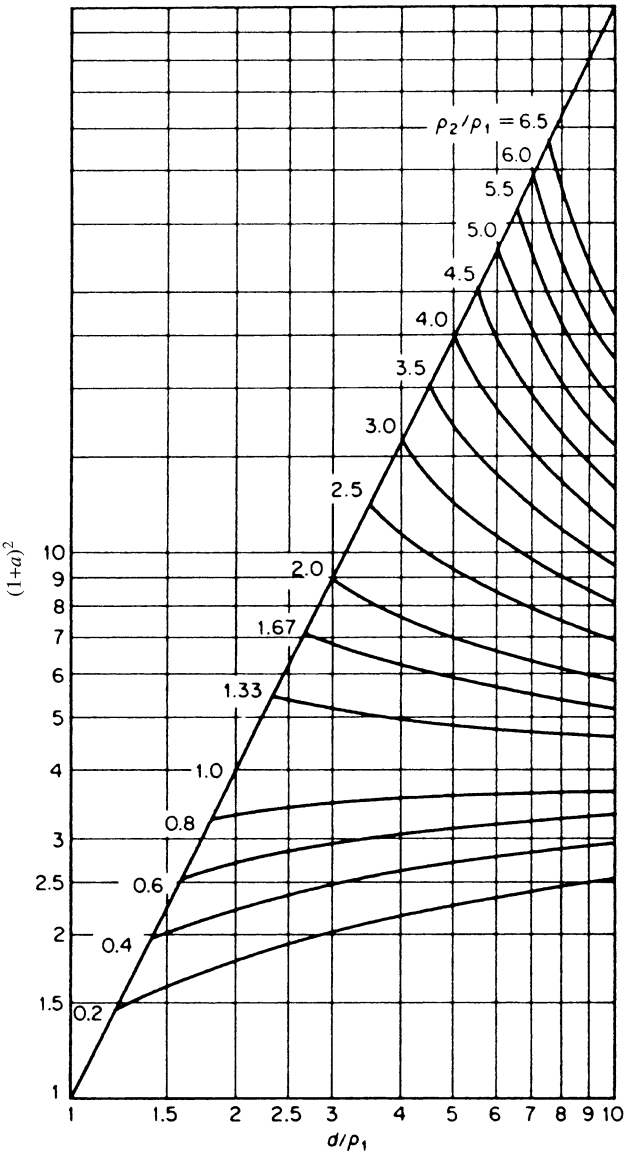


FIGURE 5.9 Impedance transformation ratio.

5.1.2 Bow-Tie and Open-Sleeve Dipoles

For very fat configurations, a bow-tie shape has demonstrated excellent bandwidth. Only numerical methods, such as the moment method, have proved useful in analyzing such fat dipoles. Figures 5.11 and 5.12 give impedance design data for bow-tie monopoles versus electrical half-length in degrees as calculated by Butler et al.

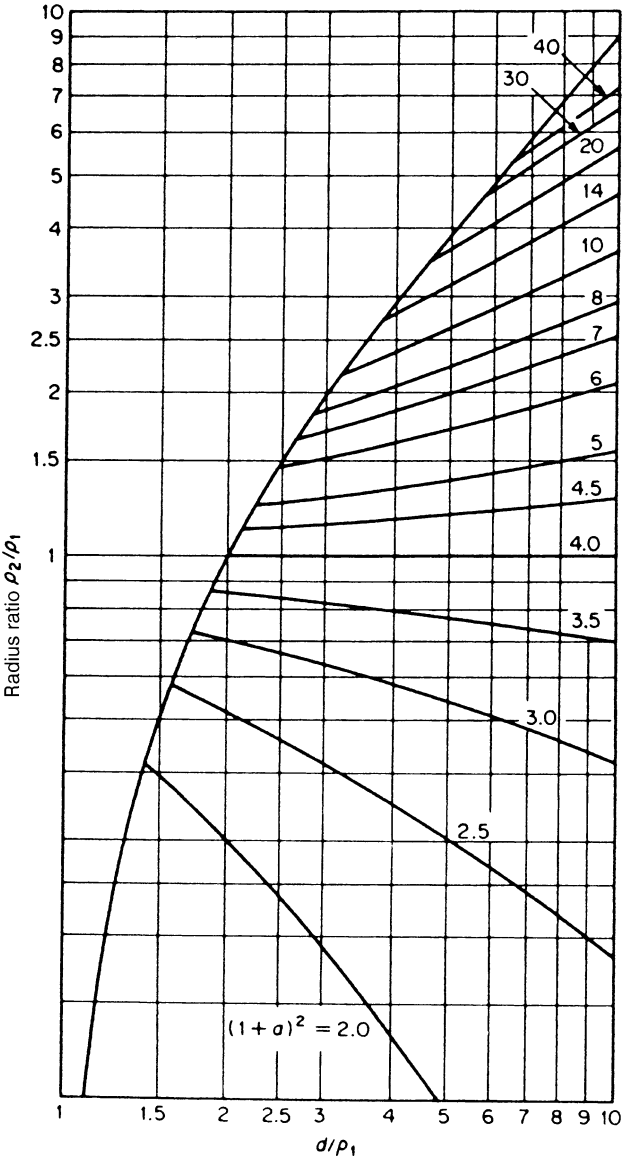


FIGURE 5.10 Impedance transformation.

(1979). The conical monopole, or biconical dipole, has slightly better bandwidth than its flat counterparts. An FDTD (finite difference time domain) analysis has been given by Maloney et al. (1990).

Sleeve dipoles (monopoles), wherein a cylindrical sleeve surrounds the central portion of the dipole, were developed during World War II at the Harvard Radio Research Laboratory (Bock et al., 1947). A significant improvement was made by

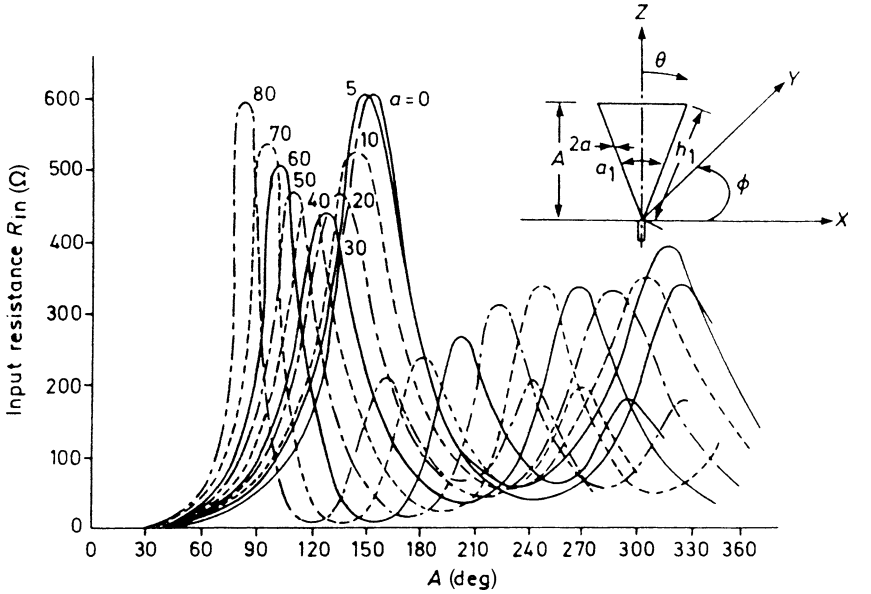


FIGURE 5.11 Bowtie monopole resistance. (Courtesy Butler, C. M. et al., "Characteristics of a Wire Biconical Antenna", *Microwave J.*, Vol. 22, Sept. 1979, pp. 37–40.)

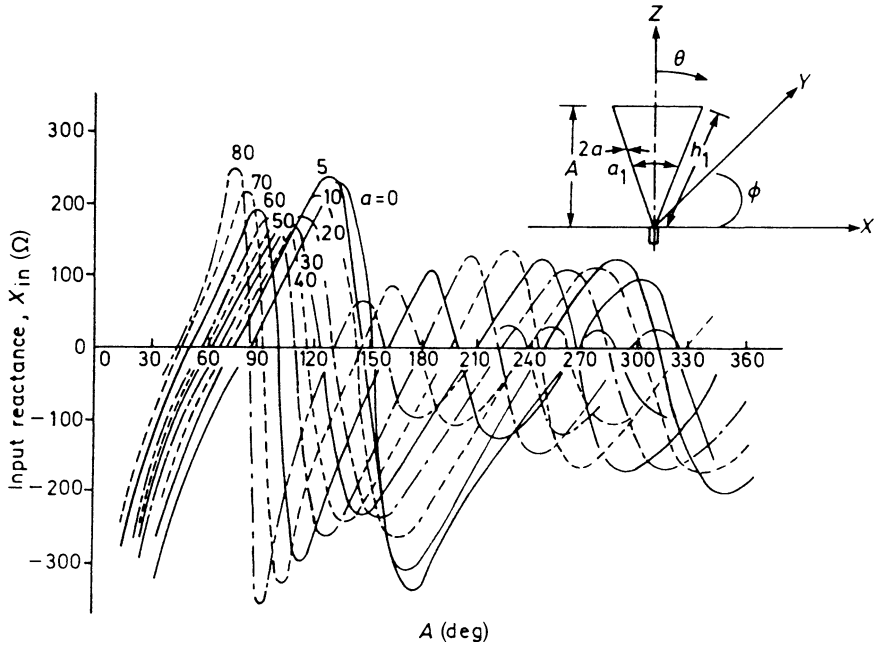


FIGURE 5.12 Bowtie monopole reactance. (Courtesy Butler, C. M. et al., "Characteristics of a Wire Biconical Antenna", *Microwave J.*, Vol. 22, Sept. 1979, pp. 37–40.)

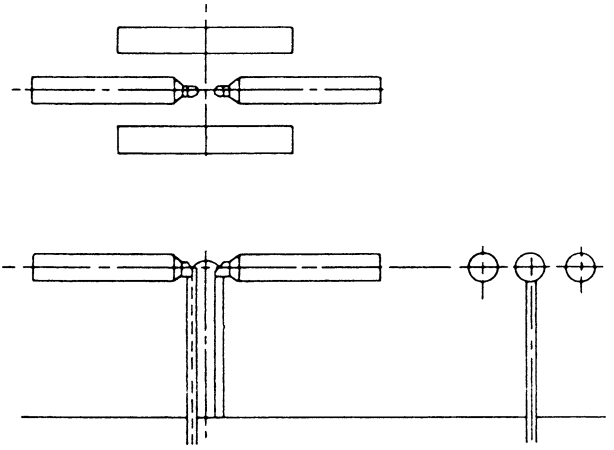


FIGURE 5.13 Open sleeve dipole. (Courtesy King, H. E. and Wong, J. L., "An Experimental Study of a Balun-Fed Open-Sleeve Dipole in Front of a Metallic Reflector", *Trans. IEEE*, Vol. AP-20, 1972, pp. 201–204.)

Bolljahn (1950) in opening the sleeve, allowing a more compact and versatile structure. Reports and theses were produced by H. B. Barkley, John Taylor, A. W. Walters, and others; King and Wong (1972) and Wong and King (1973) optimize impedance over an octave, while Wunsch (1988) presents a numerical analysis. The sleeve is open as it does not surround the dipole but consists only of two tubes or plates on opposite sides of the dipole. Figure 5.13 sketches a linearly polarized open sleeve dipole. Crossed dipoles with crossed sleeves for circular polarization (CP) can also be constructed. The crossed sleeves on each side can be replaced by a metallic circular or square disk. Bandwidth, for VSWR < 2 can be an octave. These dipoles have been arrayed over a ground plane with excellent results. The open sleeve dipole appears to be the low-gain element with the widest bandwidth, excluding, of course, resistively loaded elements where bandwidth is traded for efficiency. Again, moment methods—particularly patch versions—are the appropriate design tool.

5.2 WAVEGUIDE SLOTS

Waveguide slots were invented by Watson (1946, 1947) in 1943 at McGill University (Montreal, Canada), with related work by Stevenson, Cullen, and others in Britain. After World War II, slot work shifted to the Hughes Aircraft Microwave Laboratory at Culver City, California, under the leadership of Les Van Atta. Contributors there included Jim Ajioka, Al Clavin, Bob Elliott, Frank Goebbels, Les Gustafson, Ken Kelly, Lou Kurtz, Bernie Maxum, Joe Spradley, Lou Stark, Bob Stegen, George Stern, Ray Tang, and Nick Yaru.

A waveguide slot is simply a narrow slot cut into the broad wall or edge of a rectangular waveguide. Slots can be accurately milled, especially with numerically

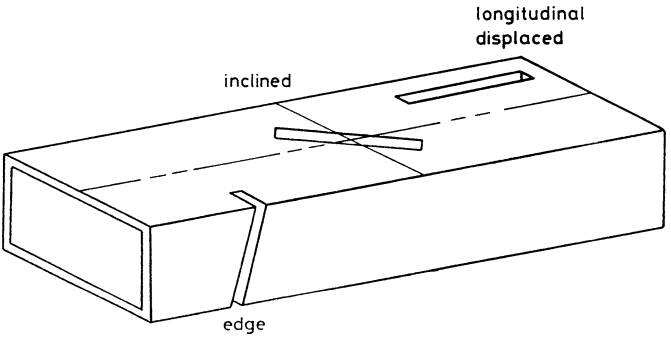


FIGURE 5.14 Waveguide slot types.

controlled machines, and the waveguide provides a linear feed that is low loss. The precise control of aperture distribution afforded by slot arrays has led to their replacing reflector antennas in many missile and aircraft radar systems, and their wide use in many other applications. A narrow slot along the center line of the waveguide broad wall does not radiate; to produce radiation, the slot must be displaced toward the edge or rotated about the centerline. Similarly, a slot in the narrow wall does not radiate if it is normal to the edge; rotating the slot couples it to the waveguide mode. Since an edge slot must usually be "wrapped around" to get resonant length, displacing such a slot toward the center-line is impractical. Figure 5.14 shows the three most important types. The displaced broad wall slot is often called a shunt slot as its equivalent circuit is a shunt admittance across the feed line. The rotated series slot, centered on the broad wall, is a series slot, as its equivalent circuit has a series impedance. And the edge slot is a shunt-type slot. The edge slot and displaced broadwall slot are most often used, with polarization often the deciding factor. In an edge slot, the E polarization is along the guide axis, while for displaced slots it is across the axis. Pattern behavior of waveguide slots is close to that of slots in an infinite ground plane, except at angles near grazing, where edge effects are important.

Resonant arrays (Chapter 6) require in-phase elements at half-guide-wavelength spacing. To cancel the guide phase advance, every other slot is placed on the opposite side of the center line (for longitudinal slots), or the rotation angle is reversed (for edge slots). Now the array pattern is composed of an array factor with double spacing, and an element consisting of a pair of slots. For longitudinal slots, the sub-array (dual slot element) lobes can be controlled through element spacing. With edge slots, however, cross-polarized lobes are produced at certain angles. These lobes can be reduced by replacing each slot by a closely spaced pair of slots with smaller inclination angle.

5.2.1 Broad Wall Longitudinal Slots

Admittance (or impedance) of a waveguide slot consists of an external contribution that can be computed from the equivalent slot in ground plane or Babinet equivalent

dipole, and an internal contribution due to energy storage in evanescent modes in the guide around the slot. Since the internal contribution is reactive, the conductance of a resonant slot can be found approximately from the Babinet dipole and from the coupling to the TE_{01} mode. Stevenson (1948) developed the formula

$$\frac{G}{Y} = \frac{480(a/b)}{\pi R_0(\beta/k)} \sin^2\left(\frac{\pi x}{a}\right) \cos^2\left(\frac{\pi\beta}{2k}\right) \tag{5.4}$$

where R_0 is the dipole resistance, x is the slot offset, a and b are the guide width and height, and β is the guide wavenumber. This formula is satisfactory for slot conductance, but there is no simple formulation for slot susceptance. A variational formulation of the susceptance problem was made by Oliner (1957a,b). He obtained closed-form simplifications that are of use, but not sufficiently accurate for array design. With the availability of powerful computers, a more complete evaluation of variational forms can be made. Yee (1974) developed such a formulation, and results from it are used below.

A different type of solution was developed by Khac and Carson (1973) and Khac (1974), who wrote coupled integral equations representing external and internal electric fields. The coupled integral equations are then solved by the moment method, using pulse expansion functions and delta testing functions. In both, wall thickness is included via Oliner's method of coupling external and internal fields by a waveguide transmission line, where the waveguide cross section is the slot and the waveguide length is the wall thickness. Satisfactory agreement with measured data has been realized with both approaches. Further improvement has been made by Elliott and his students, using piecewise sinusoidal expansion and testing functions, a Galerkin stationary formulation (Park et al., 1983).

Using the variational method, admittance of slots in WR-90 waveguide was calculated: resonant slot length versus slot offset and resonant conductance. Offset x is conveniently normalized to guide width a, while resonant length l_r is normalized to free-space wavelength λ_0. Figure 5.15 shows curves for $a = 0.9$, wall thickness $=$ 0.05, and slot width $= 1/16$, all in inches. Frequencies of 9.375 and 10 GHz are shown, with $b = 0.4$ and 0.2 in. Standard WR-90 has 0.4 height, but the half-height guide is of considerable interest for receiving arrays. Resonant conductance is shown in Figure 5.16 for one case; it matches well the Stevenson formula given earlier.

Next, the calculated admittance is normalized by resonant conductance G_r, with slot length normalized by resonant length. These universal curves were developed by Kaminow and Stegen (1954) on the basis of careful measurements. Figure 5.17 gives these data for WR-90 at 9375 MHz, again for two slot offsets. A plotting error in the original curves has been corrected. The curves are universal in the following sense. To first order, the variation of Y/G_r with l/l_r and with x/a is independent of β/k, that is, of frequency. Thus measurement or calculation of data for a given waveguide size at one frequency is adequate. For small offsets (0.05 and below), the normalized admittance variation is also independent of a/b. However, for larger offsets, data should be obtained for the exact a/b. "Universal" curves

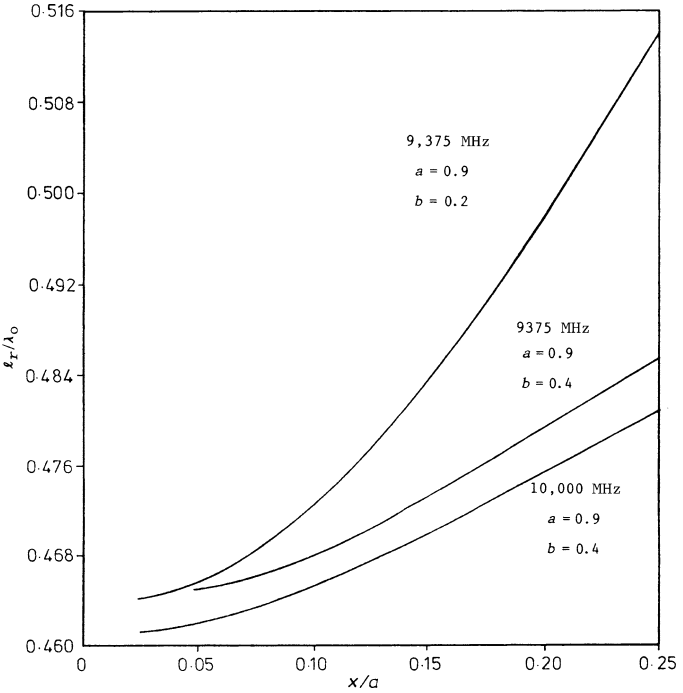


FIGURE 5.15 Longitudinal slot resonant length.

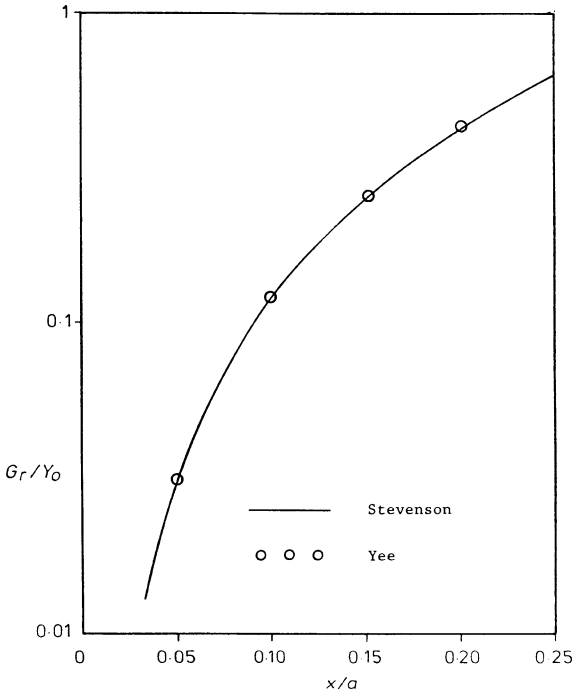


FIGURE 5.16 Longitudinal slot resonant conductance.

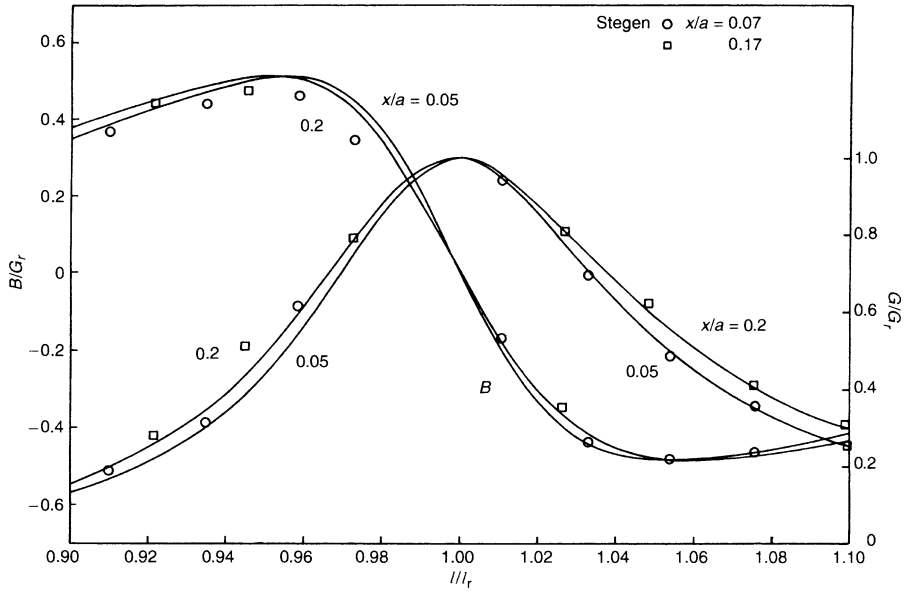


FIGURE 5.17 Longitudinal slot admittance, 9375 MHz, $a = 0.9$, $b = 0.4$. (Courtesy Stegen, R. J., "Longitudinal Shunt Slot Characteristics", Tech. Memo. 261, Hughes Aircraft Co., Nov. 1951.)

are shown in Figure 5.18 for half-height WR-90 guide at 9375 MHz; two slot offsets are shown.

The variational or moment method calculation of admittance is much too slow to use with iterative methods, such as those described in Chapter 6; polynomial fits are more suitable. However, the universal curves are difficult to fit with one expression. Dividing up the curve allows a more accurate fit, but care must be taken to avoid slope discontinuities at the joins as these affect the gradient methods used for array design. An alternative method uses impedance rather than admittance. When slot impedance is normalized to resonant resistance, the curves are as shown in Figures 5.19 and 5.20. Resistance is a straight line (to first order), with the slope independent of a/h. There is a slope change with β/k as shown. The reactance curve is a slightly curved line, which again is independent of β/k, and for small offsets is independent of a/b also. For larger offsets, the reactance curves depend on both x/a and a/b. Thus the impedance curves are just as "universal" as the admittance curves, and are much easier to fit with polynomials. If data are not available for a particular size and wall thickness of waveguide to be used in an array, it is necessary either to calculate or to measure sufficient data to plot universal curves. The impedance curves are preferable as they are slowly varying.

For most waveguide slots, an assumed sinusoidal field distribution is satisfactory. However, dielectric filled guides induce a slot that is better approximated by a half-cosinusoidal distribution (Elliott, 1983). Unfortunately, this means that mutual admittance requires numerical integration over both slots, which greatly impedes

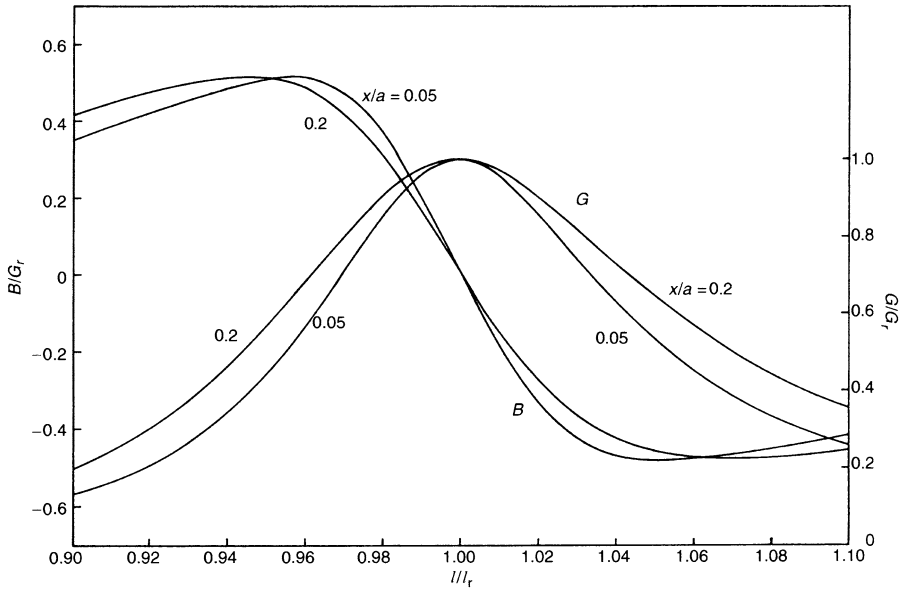


FIGURE 5.18 Longitudinal slot admittance, 9375 MHz, $a = 0.9$, $b = 0.2$.

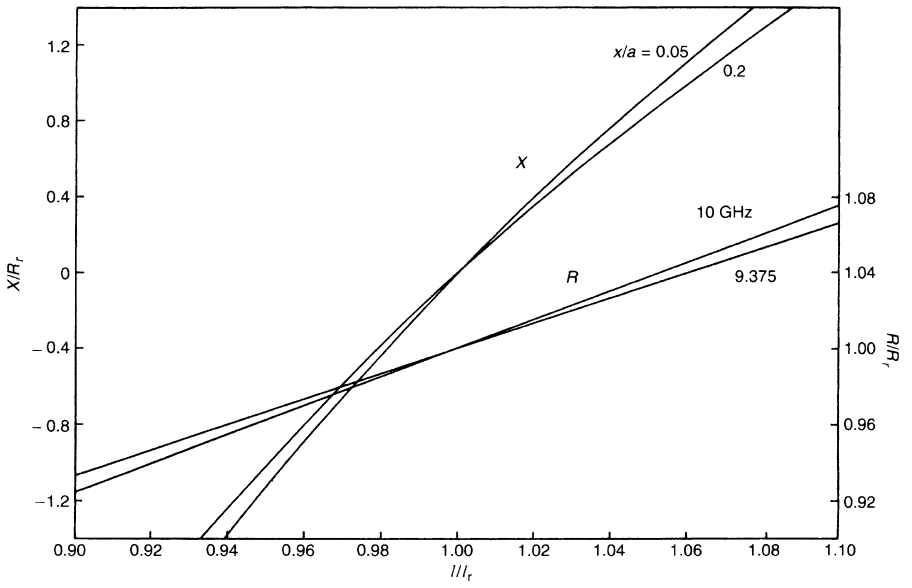


FIGURE 5.19 Longitudinal slot impedance, $a = 0.9$, $b = 0.4$.

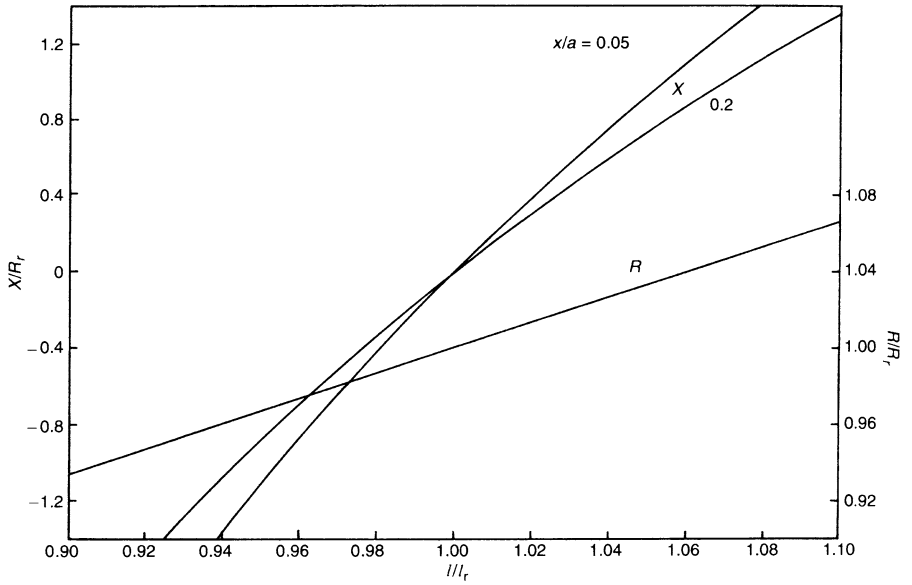


FIGURE 5.20 Longitudinal slot impedance, $a = 0.9$, $b = 0.2$.

the synthesis of slot array patterns. A simple relationship between the two mutual admittances has been given by Malherbe and Davidson (1984). Dielectric filling that occupies only part of the waveguide width can help suppress grating lobes; a moment method analysis of a transverse slot is given by Joubert (1995).

It is convenient to make waveguide slots with round ends, to accommodate milling machines. To first order, such a slot is equivalent to a rectangular slot of the same width and same area (Oliner, 1957a,b). For more accurate results, a moment method analysis can be used (Sangster and McCormick, 1987).

Longitudinal displaced broadwall slots are commonly used owing to the relative simplicity of design and manufacture. A more general slot is the compound slot, which is both displaced and rotated. The compound slot offers a wider range of amplitude and phase couplings. Design data are given by Rengarajan (1989).

5.2.2 Edge Slots

Edge slots are difficult to analyze owing to their wrap-around nature. For the same reason, the wall thickness has a significant effect on admittance. For a reduced-height guide, where the wrap-around is severe, even the pattern is difficult to calculate. At present, no theories for edge slots have appeared, using either variational or moment methods. Array design is based on measurements, and, even here, edge slots are difficult. Because of the strong mutual coupling between edge slots (as compared with displaced broad-wall slots) incremental conductance is usually measured. That is, a series of resonant slots is measured; then one slot is taped up and the

remaining slots are measured again. The resulting incremental conductance (Watson, 1946) is that of a resonant slot in the presence of mutual coupling. Trial and error is necessary to find the resonant length for a set of slots for a given angle, however. For inclination angles below 15°, the resonant conductance developed by Stevenson (Watson, 1946) varies as $\sin^2 \theta$:

$$\frac{G}{Y_0} = \frac{480a/b}{\pi R_0 (2a/\lambda)^4 \beta/k} \sin^2 \theta \tag{5.5}$$

This and the incremental values are shown in Figure 5.21 for WR-90 at 9375 MHz, for a 1/16-in. wide slot. Both appropriately follow a $\sin^2 \theta$ behavior, especially for small angles, with the incremental value larger than the single slot conductance. Resonant lengths will be different in an array, of course; so extensive measurements are necessary to develop sufficient data for design. Although many industrial organizations have done this, the available edge slot database is meager. A finite element analysis, where the slot region is filled with many cylinders of length equal to the wall thickness and of triangular cross section, has been given by Jan et al. (1996). This paper includes limited measured data.

The wrap-around nature of edge slots can be avoided by using an H slot (Chignell and Roberts, 1978), where a slot normal to the guide edge is augmented by slots at each end that are parallel to the edges. Thus the slot looks like an H. By making the outside arms asymmetric, the coupling can be varied; in fact circular polarization can be achieved (Hill, 1980). A moment method analysis of H slots is given by Yee and Stellitano (1992).

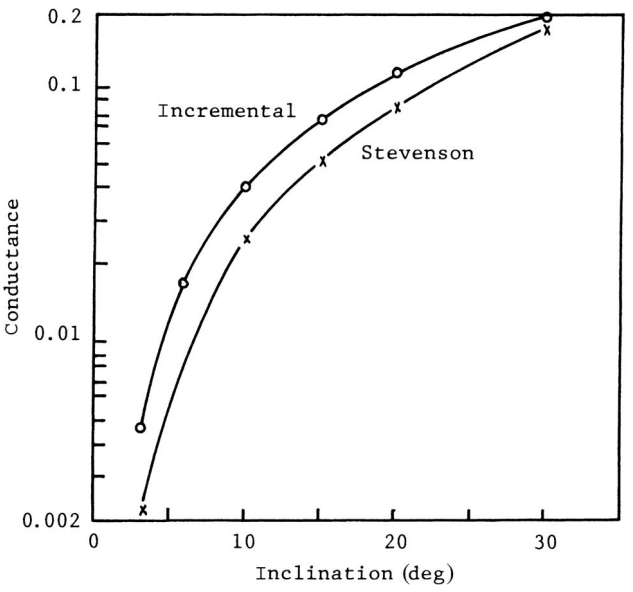


FIGURE 5.21 Edge slot conductance, 9375 MHz, $a = 0.0$, $b = 0.4$.

Other types of slots, such as probe coupled, iris excited, and crossed, are discussed by Oliner and Malech (1966).

5.2.3 Stripline Slots

Slots in stripline are typically narrow rectangular resonant slots cut into the outer conducting plate of the stripline, with the strip center conductor oriented normal to the slot and offset from the center, and with slot length adjusted for resonance. Conductance can then be calculated (Breithaupt, 1968). For example, a metal-clad dielectric sheet would be etched to produce the slot radiators, then a double-clad dielectric sheet would be etched on one side to produce the feed network. These two would be fastened together to make the stripline slot array. Although the external (radiation) behavior of a stripline slot is similar to that of a slot in an infinite ground plane, the internal admittance is strongly affected by the closely spaced ground plane underneath. The net result is to severely restrict the impedance bandwidth.

Another stripline slot configuration uses boxed stripline (stripline with shorting edges parallel to the strip conductor). Each array stick looks like a waveguide (see Fig. 5.22), except that the slots are all on the centerline (Park and Elliott, 1981). The strip conductor is symmetrically angled below each slot, with the coupling proportional to the strip inclination angle. Snaking the strip conductor with the slots along the centerline eliminates unbalance and loading problems experienced in the earlier work of Strumwasser ($\sim$1952),[3] where the strip was straight and the slots angled. The slots here are one wavelength apart (in the stripline medium), so no phase reversals are needed. Mutual coupling is accommodated in the design by techniques described in Chapter 7. Transverse displaced slots have also been used (Sangster and Smith, 1994).

5.2.4 Open-End Waveguides

Waveguide radiators are different from slots and dipoles in that evanescent modes are important. Indeed, these modes can be controlled to minimize scan effects. Because of this these elements are discussed in Chapter 7 on Mutual Coupling.

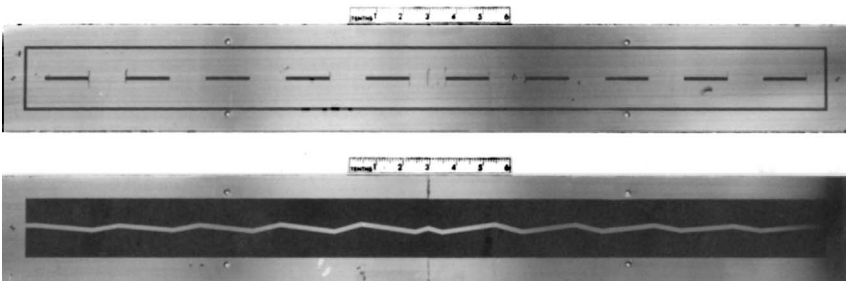


FIGURE 5.22 Boxed stripline array. (Courtesy Hughes Aircraft Co.)

[3]Unpublished Hughes report.

5.2.5 Substrate Integrated Waveguide

A way of reducing the cost and fabrication time of waveguide slot arrays uses SIW: A waveguide formed by rows of conducting pins in a parallel plate region. This waveguide was used early as a feed structure for a parallel plate radiator (Hirokawa and Ando, 1998). The next step formed virtual rectangular waveguide in a parallel plate region (Deslandes and Wu, 2001). Figure 5.23 is a sketch of an SIW with the top conducting plate removed. The parallel plate region can have a dielectric filling, or air. Equivalence between a standard rectangular waveguide and SIW has been given by Che et al. (2008):

$$a' = \frac{2a}{\pi} \operatorname{arc\,cot}\left[\frac{\pi w}{4a} \ln \frac{w}{4R}\right] \tag{5.6}$$

In this formula, a' is the SIW width; w is the distance between pins, center-to-center; R is the pin radius; a is the equivalent rectangular waveguide width. Equation (5.6) implies that $R < w/4$, $a > a'$. When $R > w/4$, then $a > a'$. With these formulas the standard waveguide tables can be used. Detailed SIW propagation calculations have been given by Wang et al. (2008). The SIS with slots include a longitudinal slot (Stephens et al., 2005), a longitudinal slot array (Yan et al., 2004), and a dual polarization inclined slot array (Park et al., 2005). Finally an array of SIS guides has been used to feed a reflector (Cheng et al., 2008a).

SIW have been used as phasers, where additional pins provide phase shift (Cheng et al., 2008b; Sellal et al., 2008).

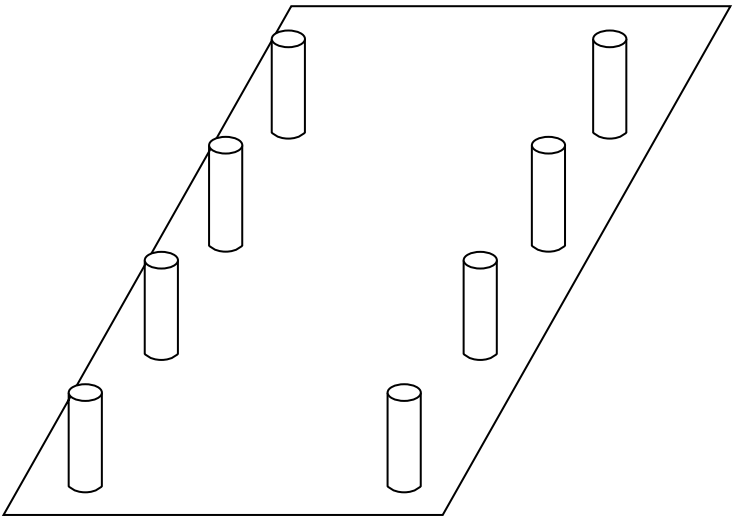


FIGURE 5.23 SIS; top plate removed.

5.3 TEM HORNS

5.3.1 Development of TEM Horns

Two conductors that flare out, either continuously or in steps, with a TEM mode between them constitute a TEM horn. Unlike a waveguide horn, where a feed mode cutoff exists, the TEM horn can exceed an octave in bandwidth. Low-frequency impedance performance is controlled primarily by the horn mouth. All of the various types of TEM horns discussed below are derived from the biconical antenna, as sketched by Schelkunoff (1943) in Figure 5.24. World War II work curved the vee dipole, and rotated it, as seen in Figure 5.25 (Alford, 1947), but details have not been found. A 2D tapered TEM horn was built of metal strips by Sengupta and Ferris (1971) (see Fig. 5.26). Three-dimensional (3D) TEM horns flare both in separation and in arm width, from throat to mouth, providing better impedance performance over a wide bandwidth (Evans and Kong, 1983). A sketch of a 3D TEM horn is shown in Figure 5.27. Kerr (1973) used a heavily ridged waveguide horn, where the ridge

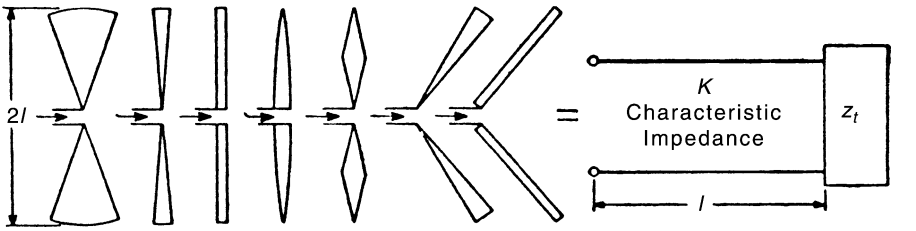


FIGURE 5.24 Transition from biconical to transmission line. (Courtesy Schelkunoff, S. A., *Electromagnetic Waves*, Van Nostrand, 1943, Section 11.6.)

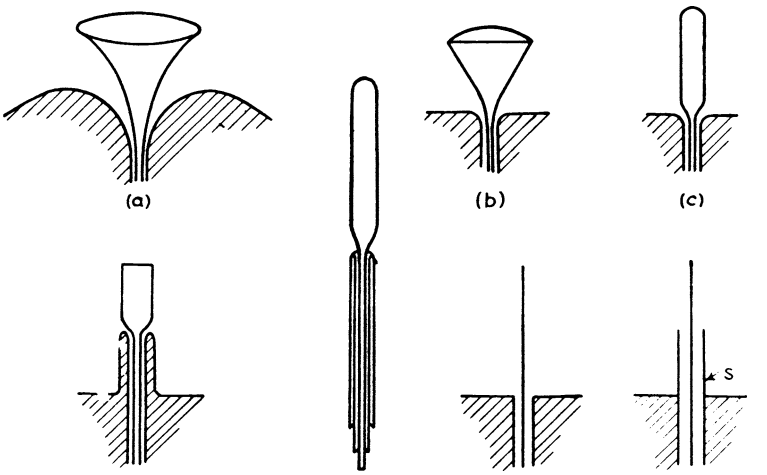


FIGURE 5.25 Sleeve dipole prototypes. (Courtesy Alford, A., "Broad-Band Antennas", in *Very High-Frequency Techniques*, Vol. 1, H. J. Reich, Ed., McGraw-Hill, 1947, Chapter 1.)

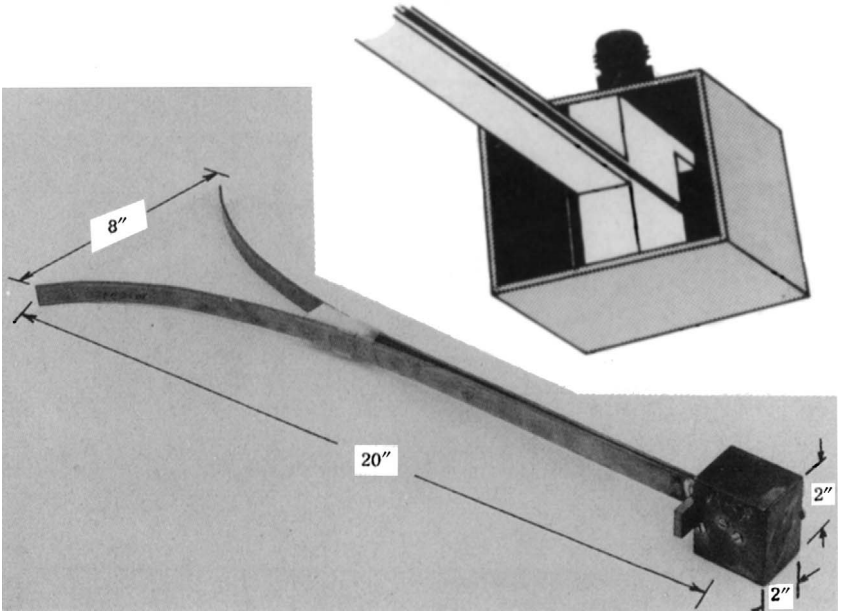


FIGURE 5.26 Rudimentary TEM horn. (Courtesy Sengupta, D. L. and Ferris, J. E., "Rudimentary Horn Antenna", *Trans. IEEE*, Vol. AP-19, Jan. 1971, pp. 124–126.)

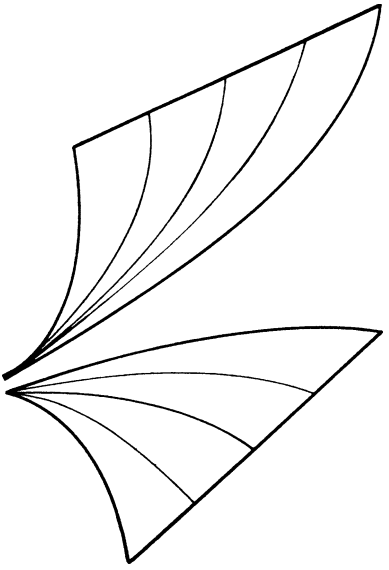


FIGURE 5.27 Sketch of TEM horn.

opening formed a roughly exponential taper from throat to mouth. He then found that the side walls could be removed, making a TEM horn. Top and bottom walls are also not important. The ridges alone constitute a good horn; the next step was to extend the conductor on each side, and to fabricate the horn by etching away a horn shaped opening in a plated dielectric sheet. These configurations were disclosed at the 1979 European Microwave Conference by Prasad and Mahapatra (1983) and by Gibson (1979). The latter carried the whimsical name "Vivaldi Antenna", but this term is deprecated as it has no content.

Several variants exist of these printed TEM horns. The metal comprising the sides of the horn may extend a short distance, or may connect to that of adjacent horns. The horn conductors may be thin, or even petal shaped (Gazit, 1988; Fourikis et al., 1993). The horn opening may be curved and roughly exponential; it may contain a fixed horn angle (like a vee dipole), or it may be stepped, as in a stepped waveguide horn. The constant angle and the stepped versions have been called "linear tapered", and "constant width" slot antennas (Yngvesson et al., 1985, 1989; Janaswamy et al., 1986; Schaubert, 1993). This nomenclature is poor, as "slot antenna" implies radiation through the slot, while these antennas are all endfire TEM horn antennas. With all of these printed antennas it is often convenient to use stripline, microstrip, slot line, coplanar guide, and so on, for feeding the horn. In all cases, there is much art in devising broadband feeds that transition from the transmission line to the horn. There are many types of printed circuit baluns; a microstrip balun is shown in Figure 5.28 (Simons et al., 1995).

In arraying printed TEM horns, it is important to distinguish between horns with finite width conductors and those where the conductors are part of a conducting sheet. The latter case allows currents produced by excitation of one horn to couple directly to adjacent horns. This may or may not be advantageous in impedance matching of large arrays.

5.3.2 Analysis and Design of Horns

Single TEM horns can be analyzed by surface patch moment methods (Janaswamy, 1989), in which the dielectric sheet is incorporated through perturbation theory. Gridded space methods are also applicable; Thiele and Taflove (1994) apply FDTD to isolated TEM horns, and to small arrays thereof. Results are good, but the computational process is not cost effective. Less powerful approaches approximate the field inside the horn by replacing the horn contour by many small steps, using slotline equations for each step, and finally equating power flow between steps. The resulting aperture distribution is used with a half-plane Green's function formulation to get patterns (Eleftheriades et al., 1991; Janaswamy and Schaubert, 1987). Results are not as good as those from mode matching at small steps in waveguide horns, but are better for horn mouths several wavelengths or more. Because of the many variables—mouth width, horn length, horn conductor geometry, dielectric sheet (if used) dielectric constant and thickness, and type of feed—it is not feasible to give design data, especially for the relatively small horns that are useful in arrays. Moment method analysis and design is recommended, starting with no dielectric and an ideal feed, then adding

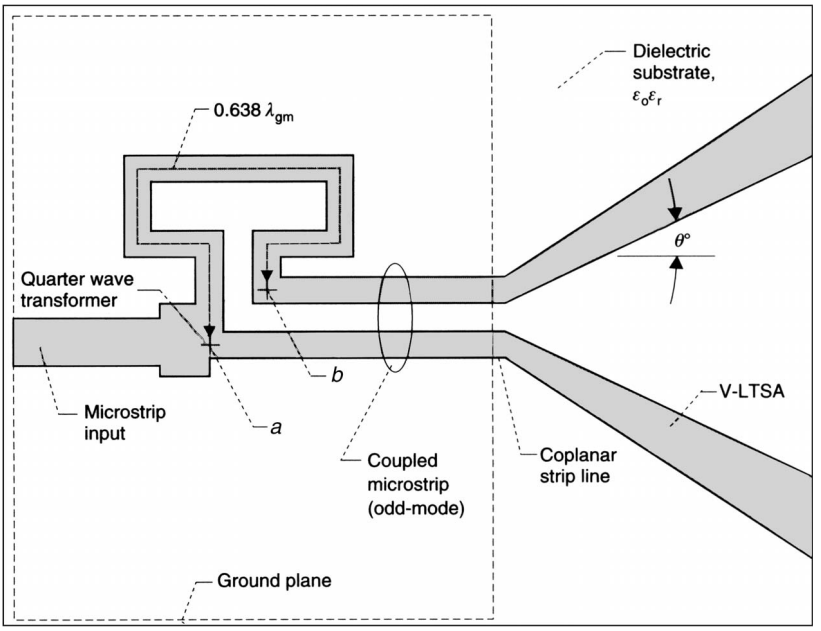


FIGURE 5.28 Microstrip balun. (Courtesy Simons, R. N. et al., "Integrated Uniplanar Transition for Linearly Tapered Slot Antenna", *Trans. IEEE*, Vol. AP-43, Sept. 1995, pp. 998–1002.)

these items. Users should be cautioned that 2D horns may have high cross polarization in the diagonal plane (Kim and Yngvesson, 1990). Polarization agility is obtained simply with two horns at right angles, or, in a unique arrangement, with four horns with a circular cavity underneath. Here the horns form a four-pointed star, with the horn mouths facing inward (Povinelli, 1987).

5.3.3 TEM Horn Arrays

The behavior of an array of TEM horns depends on whether the individual horns are connected only at the feeds, or whether direct current paths exist from each horn to the next. When the horn openings are etched out of a conducting layer, current paths exist from throat to mouth of each horn and to throat to mouth of adjacent horns. As a result, the embedded impedance versus frequency for a fixed beam array, or the *scan impedance* for a scanning array, will be different. For all but small arrays, moment methods are the best approach. Of course much can be learned from the behavior and design trades of infinite arrays, and in the spectral (Floquet) domain the number of equations to be solved is modest. Cooley et al. (1991) gave a moment method analysis of an infinite array of linearly tapered TEM horns; Schaubert et al. (1994) extend this to curved TEM horns. These analyses are able to predict important phenomena, such as blind angles in *E*-plane scan. Both linear polarizations can be provided by "egg-crating" two printed circuit boards together (Hunt and Ventresca, 1977). Another configuration

uses printed horns for one polarization, and a waveguide mode between the printed planes for the other polarization (Monser, 1981, 1984). In all configurations, the horn axes are in the main beam direction, as the horn lengths are greater than half-wave, thus not allowing the axes to be in the array plane. An extensive discussion of TEM horn arrays is given in Chapter 7.

5.3.4 Millimeter Wave Antennas

In recent years, much effort has been expended on combining solid-state circuit devices with antennas, primarily vee strip dipoles and TEM horns (Kotthaus and Vowinkel, 1989). At millimeter wavelengths, the power source or preamplifier–mixer can be produced on the same substrate as the antenna. Rutledge et al. (1978, 1983) give reviews of these. Many of the papers are concerned primarily with the solid-state devices rather than with the antenna, and are not discussed here. A variant on the printed circuit flat TEM horn is the square linear taper horn with dipole feed at the mouth opening; the horn walls are etched and metallized on a thick substrate, as in the feed (Guo et al., 1991; Rebeiz et al., 1990; Rebeiz and Rutledge, 1992).

Low-temperature cofired ceramic (LTCC) offers a low-loss substrate that is compatible with MMIC chips. A stacked patch on LTCC at 30 GHz (Panther et al., 2005), and a patch array at 30 GHz are typical (Alhalabi and Rebeiz, 2008). The bent arms in the last reference are not to reduce asymmetric feed modes (see Section 7.4.3), but rather to broaden the *E*-plane beamwidth at 20 GHz. A monopulse array at 32 GHz uses longitudinal slots (Liu et al., 2009). The TEM horn antennas are also good candidates for millimeter wave antennas (Yngvesson et al., 1989). "Tapered Slot Antenna" is poor nomenclature. They are not traveling wave antennas as often stated, but are true horns. More recent TEM horn work has used LTCC (Muldavin and Rebeiz, 1999).

Short-range communications at 60 GHz are currently in vogue, as this band offers a bandwidth with a minimum of interference due to the heavy atmospheric attenuation, of roughly 15 dB/km. Antennas for 60 GHz are mostly the same as for lower frequencies: patches and TEM horns. Murakami et al. (1998) used patches on each face of a four-sided pyramid, for indoor wireless LAN systems. Seki et al. (2005) used a patch with parasitic patches on LTCC. Arrays of patches on LTCC have been developed also for the unlicensed 57–64-GHz band (Lamminem et al., 2008). A TEM horn antenna on LTCC with an air cavity was used by Kim et al. (2006). A switchable antenna system uses pyramidal waveguide horns to allow optimization of system S/N (Celik et al., 2008). A monopulse slot array was developed by Liu et al. (2009).

Above 60 GHz, in the terahertz region, TEM horns are good choices. Kotthaus and Vowinkel (1989) used a double taper TEM horn at 80 GHz with a low ε substrate. A TEM horn on silicon substrate at 90 GHz was investigated by Rizk and Rebeiz (2002). The highest frequency horn appears to be that of Acharya et al. (1993) at 802 GHz. The horn was fabricated on a thin membrane of silicon oxide–silicon nitride.

Quasioptical terahertz antennas, including superconducting SIS antennas (Zmuidzinas and Richards, 2004) are outside the scope of this book.

Cheng et al. (2009) built a 79 GHz array of longitudinal slots. Zigzag monopoles were used by Zhang et al. (2006) at 110 GHz. Finally a Series Fed dipole array was developed for 28.3 GHz (IR) (Middlebrook et al., 2008).

5.4 MICROSTRIP PATCHES AND DIPOLES

A microstrip patch or dipole is a resonant metallic shape in the top microstrip conductor. Patches were invented by Deschamps (see Deschamps and Sichak, 1953), probably based on the earlier partial sleeve (shorted quarter-wave patch) work at the University of Illinois Antenna Laboratory. Further development was by Munson (1974) and others. Because of the extensive literature on microstrip antennas, only a cursory coverage is given here. Books include *Microstrip Antennas* (Bahl and Bhartia, 1980), *Microstrip Antenna Theory and Design* (James et al., 1981), *Handbook of Antenna Design* (2 volumes) (James and Hall, 1989), *Millimeter-Wave Microstrip and Printed Circuit Antennas* (Bhartia et al., 1991), *Broadband Patch Antennas* (Zurcher and Gardiol, 1995) and *Advances in Microstrip and Printed Antennas* (Lee and Chen, 1997). A useful review of microstrip design and fabrication techniques has been given by Gardiol (1988).

Patches can be of many shapes, but most are rectangular (nearly square) or circular. A patch may be fed by an in-plane microstrip feed line, as sketched in Figure 5.29 or by a coaxial cable connected to the underside of the ground plane, with the center conductor extending up to the patch. A typical inset microstrip line feed for matching is shown in Figure 5.30. Patches are typically square or circular, and may have electromagnetically coupled portions; the last are discussed later. Square and circular annular slots in stripline are also used. Polarization and pattern for patches are similar to those of half-wave dipoles or slots. The zero-order pattern for a square patch is given by

$$E_x = \cos\theta \,\operatorname{sinc}\left(\frac{kLv}{2}\right)\cos\left(\frac{kWu}{2}\right) \tag{5.7}$$

where $u = \sin\theta\cos\phi$ and $v = \sin\theta\sin\phi$. For a circular patch of radius a, with $\beta = ka\sin\theta$, the zero-order pattern is

$$E_x = \cos\theta[J_0(\beta) - \cos(2\phi)J_2(\beta)] \tag{5.8}$$

A typical measured pattern is shown in Figure 5.31 (Kerr, 1979). Cross polarization (XP) occurs in diagonal planes, with maximum XP levels as shown in Table 5.1 (Hansen, 1987). Note that the formulas in the reference are correct except that the square patch XP coefficient should be $2uw$ instead of $4uw$. However, the tables there are incorrect; the corrected table is:

	Circular Patch Cross Polarization	
ε	$\theta_{\max}$ (deg)	XPOL (dB)
1	51.956	−17.37
1.5	52.934	−20.36
2	53.404	−22.59
3	53.861	−25.84
4	54.085	−28.20
5	54.217	−30.36
7	54.367	−32.89
10	54.479	−35.91
15	54.565	−39.38

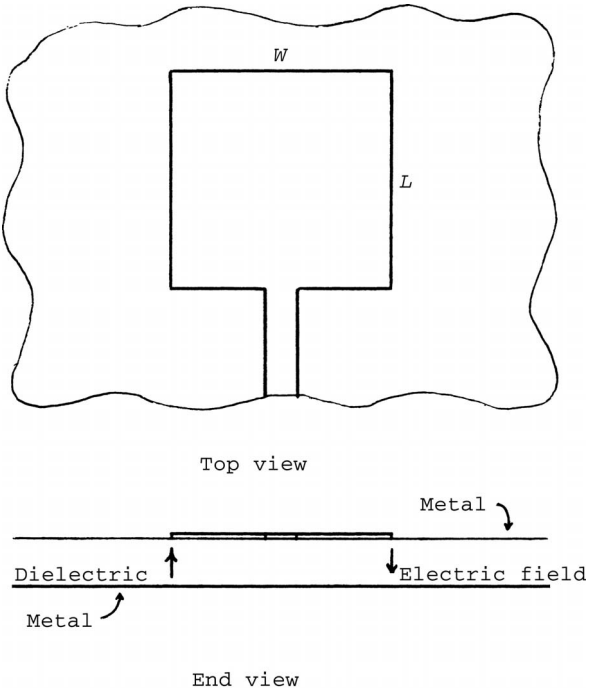


FIGURE 5.29 Microstrip patch geometry.

Circular polarization (CP) is achieved simply by using two feeds at 90°, with a 90° hybrid connected to the feed lines. If the hybrid is relatively wideband, the CP bandwidth will be the same as that for linear polarization. A simpler, but slightly narrower-band version, replaces the hybrid by a power divider, with an extra $\lambda/4$ line length in one feed. Single feeds can also be used to produce CP: the principle is to modify the patch so that the x-directed resonance is below and the y-directed resonance is above the operating band. When the phases differ by 90°, CP results. However, the bandwidth over which CP occurs is significantly reduced from the intrinsic patch

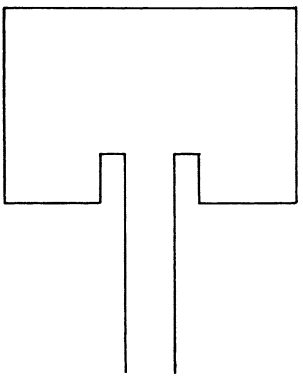


FIGURE 5.30 Inset feed impedance matching.

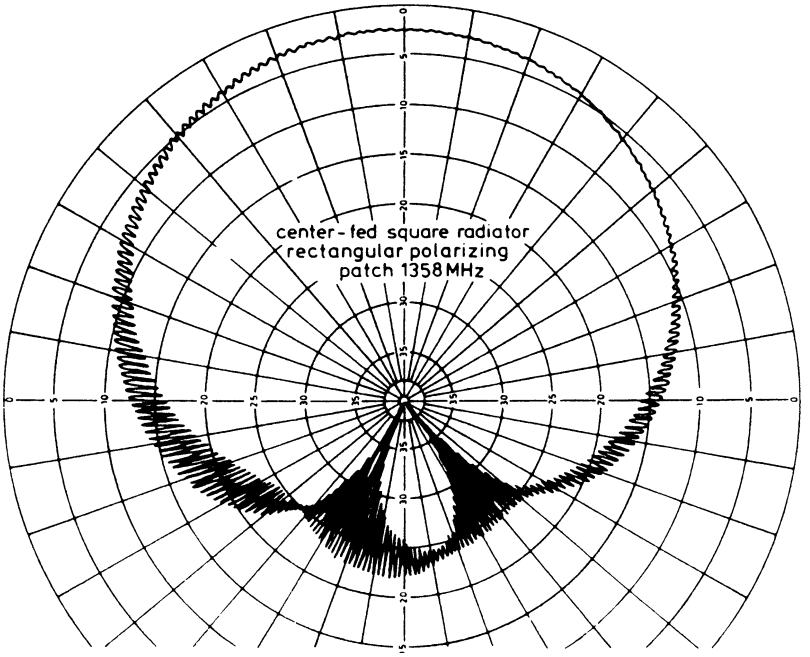


FIGURE 5.31 Typical patch axial ratio pattern. (Courtesy Kerr, J. L., "Microstrip Antenna Developments", *Proceedings Printed Circuit Antenna Technology Workshop*, New Mexico State University, Las Cruces, NM, Oct. 1979.)

bandwidth. For example, the patch is rectangular with one side shorter and one side longer than a square patch. A circular patch can be modified to an elliptical shape. Simple formulas relating dimensions to bandwidth were derived by Lo and Richards (1981). Other schemes for exciting CP are sketched in Figure 5.32.

Microstrip dipoles, where the patch width is small, giving a strip dipole, have been investigated extensively by Alexopoulos and colleagues (Uzunoglu et al., 1979). However, for thin substrates the bandwidth tends to be much smaller than that of a patch on the same substrate (Pozar, 1983).

TABLE 5.1 Maximum Cross-Polarization Levels

	Square Patch		Circular Patch	
	$\theta_{max,XP}$ (deg)	Max. XP (dB)	$\theta_{max,XP}$ (deg)	Max. XP (dB)
1	51.51	−17.93	51.96	−17.37
2	54.74	−23.54	53.40	−22.59
3	54.74	−26.82	53.86	−25.84
5	54.74	−31.07	54.22	−30.36
10	54.74	−36.95	54.48	−35.91

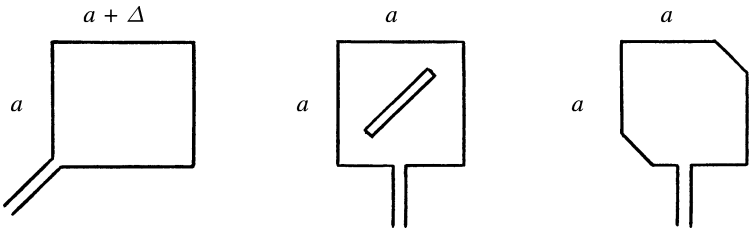


FIGURE 5.32 Circular polarization patches. (Courtesy Kerr, J. L., "Microstrip Antenna Developments". *Proceedings Printed Circuit Antenna Technology Workshop*, New Mexico State University, Las Cruces, NM, Oct. 1979.)

5.4.1 Transmission Line Model

The square patch acts as a half-wave resonant transmission line, with the end gaps producing the radiated field, as sketched in Figure 5.33. Simplest of the analytical models is the transmission line model (Sengupta, 1984), where an effective dielectric constant is used, and where the ends are terminated with a radiation conductance and susceptance. The effective dielectric constant varies with the ratio of patch thickness to width (Hammerstad, 1975):

$$\varepsilon_e = \frac{1}{2}(\varepsilon_r + 1) + \frac{\varepsilon_r - 1}{2\sqrt{1 + 12t/W}} \tag{5.9}$$

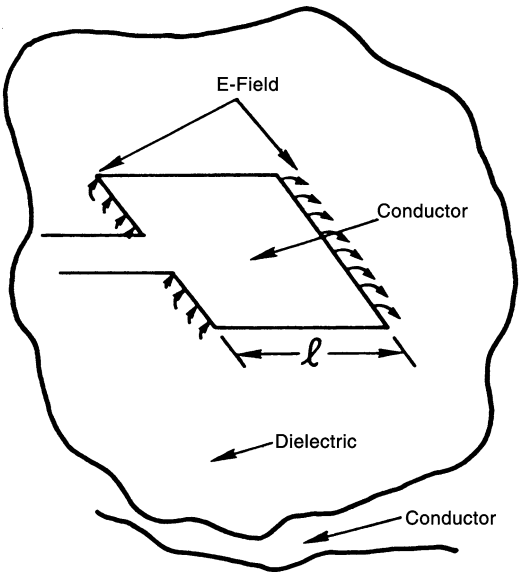


FIGURE 5.33 Radiating edge slots of patch.

A more complex formulation has been developed by Hammerstad and Jensen (1980), but the simpler form may be adequate. The radiation admittance implies a radiation Q, which is (Vandesande et al., 1979)

$$Q_{\text{rad}} = \frac{\sqrt{\varepsilon_e}\lambda_0}{4t} - \frac{\varepsilon_e \Delta L}{t} \tag{5.10}$$

where ΔL is the change in effective length produced by field fringing (Hammerstad, 1975):

$$\frac{\Delta L}{t} = 0.412 \frac{(\varepsilon_e + 0.300)(W/t + 0.262)}{(\varepsilon_e - 0.258)(W/t + 0.813)} \tag{5.11}$$

A more accurate, and again more complex, formula is given by Kirschning et al. (1981). An advantage of the transmission line approach is that the feed point impedance can easily be computed. Since the bandwidth for a matched antenna of small bandwidth is $1/(\sqrt{2}Q)$ for VSWR ≤ 2, the fractional bandwidth is approximately

$$\text{BW} \simeq \frac{4t}{\sqrt{2\varepsilon_r}\lambda_0} \tag{5.12}$$

That is, it is proportional to thickness in free space wavelengths, and inversely to square root of dielectric constant. Figure 5.34 shows bandwidth for VSWR < 2 for dielectric constants of 2.5, 5.0, and 10.0. Note that this approximate bandwidth does not include conduction and dielectric losses, which increase bandwidth.

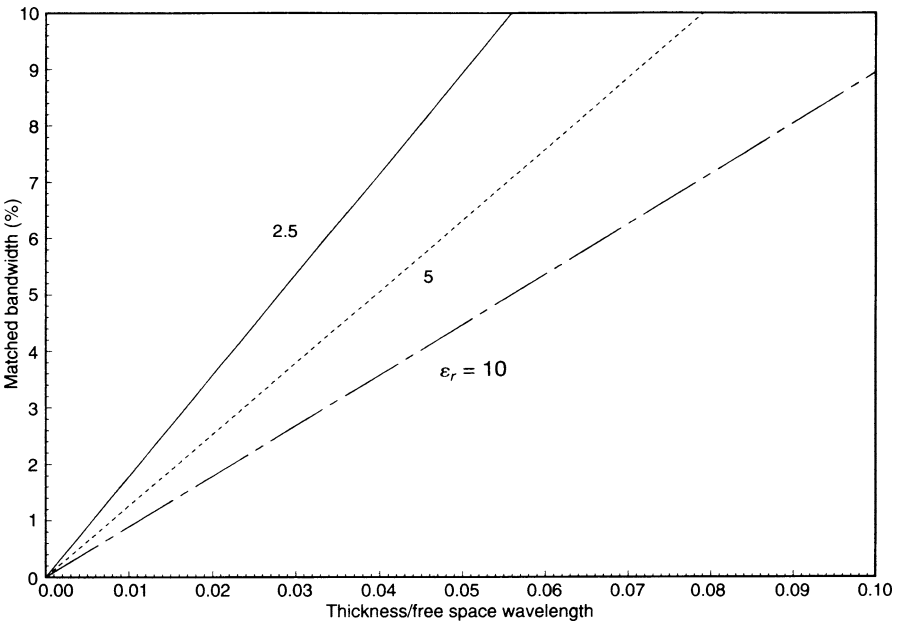


FIGURE 5.34 Patch bandwidth for VSWR = 2.

Surface wave effects are also not included, but these are usually important only for large arrays. Bandwidth for any other VSWR is given by

$$\mathrm{BW} = \mathrm{BW}_2 \frac{\sqrt{2}(\mathrm{VSWR} - 1)}{\sqrt{\mathrm{VSWR}}} \tag{5.13}$$

For half-power, VSWR = 5.828, and the bandwidth is $2\sqrt{2}$ times larger than the VSWR = 2 bandwidth.

Use of a low loss magnetic substrate can significantly increase bandwidth (Hansen and Burke, 2000).

5.4.2 Cavity and Other Models

The cavity model, developed by Lo and colleagues (Lo et al., 1979) at the University of Illinois-Urbana, and by Carver and colleagues (Carver and Mink, 1981) at New Mexico State University, represents the field between the patch and the ground plane as a series of rectangular (or circular) cavity modes. These modes are then matched to the gap admittance at the patch edges. Almost all of the radiated energy is from the dominant TM_{01} mode; from this a radiation Q_{rad} can be calculated. All modes contribute to the stored energy. Using the dominant mode current and field, the conduction and dielectric losses and Q values are obtained. Finally, the overall Q is simply $1/Q = 1/Q_{\mathrm{rad}} + 1/Q_{\mathrm{cond}} + 1/Q_{\mathrm{diel}}$. Bandwidth is found from Q as before, and the input impedance is calculated from the Q and the stored and radiated power. Although the cavity model is more accurate than the transmission line model in predicting resonant frequency, a small error or shift usually appears.

Moment methods have also been used to determine the currents on the conducting surfaces. Newman and Tulyathan (1981) used quadrilateral expansion and testing functions in the spatial domain. This avoids the difficulties associated with calculation of the spectral domain Green's function; the spectral domain is more appropriate for infinite array analysis via Floquet's theorem. Since that first paper, many have appeared; other types of subsectional expansion and test functions have been used, as have entire expansion and test functions. The latter are usually the cavity modes.

FDTD methods have also been used to analyze patch elements (Wu et al., 1992; Luebbers and Langdon, 1996). However, the cavity and moment method approaches are usually satisfactory, and are simpler.

5.4.3 Parasitic Patch Antennas

Bandwidth can be increased by adding one or more resonant circuits. Matthaei, Young, and Jones (1964) in their treatise gave solutions to the simultaneous equations developed by Fano:

$$\frac{\tanh na}{\cosh a} = \frac{\tanh nb}{\cosh b} \tag{5.14}$$

$$\cosh nb = \Gamma \cosh na \tag{5.15}$$

$$\sinh b = \sinh a - 2\delta \sin(\pi/2n) \tag{5.16}$$

Here there are n matching sections (including the antenna as one), δ is decrement ($1/Q$ at band edges), and a and b are parameters to be determined. The parameter Γ is the reflection coefficient. These simultaneous equations were solved by Matthaei et al. with results presented in graphical form. Subsequently Newton–Raphson (Stark, 1970) was used by this author to obtain precise values of a and b. Tables 5.2 and 5.3 give values of a, b, and δ for VSWR values of 2 and 5.828 (half-power). Note that $n = 1$ is for the antenna alone. The Bandwidth Improvement Factor is just δ for $n = 1$ divided by δ for $n > 1$. Table 5.4 gives these BWIF for several values of VSWR. Actual matching devices will have loss, and this will increase the bandwidth

TABLE 5.2 VSWR $\leq$ 2

n	a	b	δ
1	1.81845	0.32745	1.33333
2	1.03172	0.39768	0.57735
3	0.76474	0.36693	0.46627
4	0.62112	0.33112	0.42416
5	0.52868	0.30027	0.40264
∞	0	0	0.34970

TABLE 5.3 VSWR $\leq$ 5.828

n	a	b	δ
1	1.14622	0.65848	0.35355
2	0.76429	0.56419	0.17416
3	0.59982	0.47449	0.14394
4	0.50164	0.41026	0.13207
5	0.43483	0.36284	0.12589
∞	0	0	0.11032

TABLE 5.4 Maximum Bandwidth Improvement Factors

Number of Additional Matching Circuits	VSWR			
	1.5	2	3	5.828
1	2.6833	2.3094	2.1213	2.0301
2	3.4202	2.8596	2.5864	2.4563
3	3.8001	3.1435	2.8271	2.6772
4	4.0242	3.3115	2.9699	2.8083
00	4.6848	3.8128	3.3993	3.2049

at the expense of efficiency. Note that for the commonly used VSWR = 2, an infinite number of matching circuits will increase bandwidth 3.8 times.

A word of caution is needed on the use of the Bode (1949) criterion for BWIF.

$$\text{BWIF} = \frac{\pi}{\ln(1/\Gamma)} \tag{5.17}$$

Bode assumed constant Γ over the band of interest; this clearly invalidates his result. An exact correction was provided by Hansen (2008):

$$\text{BWIF} = \frac{1-\Gamma^2}{2\Gamma} \cdot \frac{\pi}{\ln(1/\Gamma)} \tag{5.18}$$

This equation gives results that are exactly the same as the Fano results (Table 5.4 for an infinite number of sections).

A modified matching scheme was suggested by Pues and Van De Capelle (1989). Here the load resistance Z_0 is not made equal to the element resistance R_0, but a factor T is introduced: $T = Z_0/R_0$. Normally the Q bandwidth product, QB, equals:

$$\text{QB} = \frac{\text{VSWR} - 1}{\sqrt{\text{VSWR}}} \tag{5.19}$$

This is maximized for $T = 0.5 * (\text{VSWR} + 1/\text{VSWR})$. Table 5.5 shows the optimum value of T and the improved QB. It is apparent that improvement is minor for low values of VSWR.

There are several clever ways of incorporating matching circuits into patch antennas. Perhaps the simplest is the parasitic patch, where a parasitic patch is supported above the driven patch by a (usually) low-dielectric constant material, such as foam or hexcel (see sketch, Fig. 5.35). The parasite length and width, and the spacing and dielectric constant add four more design variables. As a result these antennas were difficult to design until moment methods were applied. Kastner et al. (1988) use spectral domain sampling and FFT. Barlatey et al. (1990) use entire domain expansion and test functions (cavity modes) in the spatial domain. Tulintseff et al. (1991) use entire domain expansion and test functions plus a polynomial for edge currents, in the

TABLE 5.5 Bandwidth Improvement Due To Mismatched *L*

VSWR	T_{apt}	$\text{QB}_{\text{opt } T}$	QB $T = 1$
1	1	0	0
1.5	1.0833	0.417	0.408
2	1.2500	0.750	0.707
3	1.6667	1.333	1.155
4	2.1250	1.875	1.500
5.828	3.0000	2.828	2.000

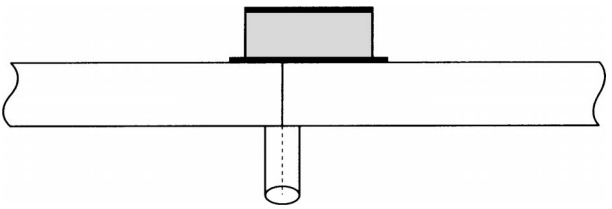


FIGURE 5.35 Probe fed patch with parasite.

spectral domain. These powerful numerical simulation techniques have allowed parasitic patch antennas to be designed, with roughly double the bandwidth of the driven patch alone. Many references on parasitic patches are listed.

Another way of incorporating a matching section is through electromagnetic coupling. Oltman (1977) reported a strip dipole electromagnetically coupled to an open ended microstrip line; an electromagnetically coupled patch was analyzed by Zhang et al. (1985). A modification utilized coupling through an aperture in either stripline or microstrip ground plane (Pozar, 1985). Sullivan and Schaubert (1986), using moment method, analyzed an open ended microstrip line electromagnetically coupled to a narrow slot (across the line), which in turn is electromagnetically coupled to the patch; see sketch in Figure 5.36. Reciprocity can be used to separate the analysis into two coupled problems: one for the antenna, and a second for the microstrip line (Pozar, 1986). Careful analyses have been done by Ittipiboon et al. (1991) and Yang and Shafai (1995). Again, with careful design the bandwidth can be at least doubled.

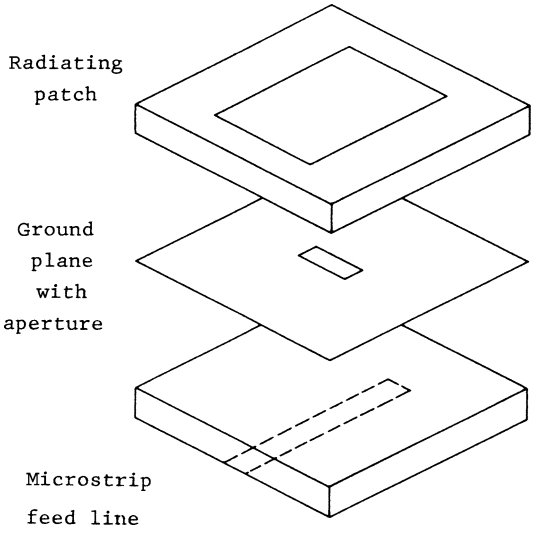


FIGURE 5.36 Aperture coupled patch.

5.4.4 Balanced-Fed Patches

Many patches are fed by balanced probes, driven 180° out of phase, and fed by a reactive power divider. Schuss and Hanfling (1987) have shown that in an E-plane scan the radiator becomes unbalanced with the resulting scattering matrix becoming unbalanced and nonreciprocal. Scan blindness can result due to this parasitic mode. Fortunately, this can be averted by using a well-matched resistive balun, or by altering the line lengths of the balun.

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CHAPTER SIX

Array Feeds

Fixed-beam arrays are usually either linear arrays or assemblies of linear arrays to make a planar array. Thus the linear array is a basic building block. When the array elements are in series along a transmission line, the array is termed "series". Similarly, when the elements are in parallel with a feed line or network, the array is termed "shunt". A further division of series feeds is into resonant or standing wave feeds, and traveling wave feeds. Shunt feeds are generally of the corporate type, or distributed. Each of these types is discussed in this chapter.

6.1 SERIES FEEDS

Waveguide slots lend themselves to series feeding, as they are spaced along the transmission feed line (waveguide). This is independent of whether the slot equivalent circuit is series or shunt (see Chapter 5). Dipoles are usually more suited to shunt feeds. Microstrip patches can be fed in series by microstrip lines connecting adjacent patches, but these arrays offer much less control over the array excitation. Since most series-fed arrays use waveguide slots, these are emphasized, both for resonant and for traveling wave arrays. Figure 6.1 shows an experimental single waveguide array with longitudinal slots.

6.1.1 Resonant Arrays

6.1.1.1 Impedance and Bandwidth. The most common resonant array is a waveguide with slots at half-guide wavelength spacings, with the guide ends shorted, and with power coupled in (or out) along the guide. The end shorts allow standing waves to exist, and the $\lambda_g/2$ spacing allows all slot radiators to be in phase. Such a waveguide array is called a "stick"; planar arrays are made up of closely packed

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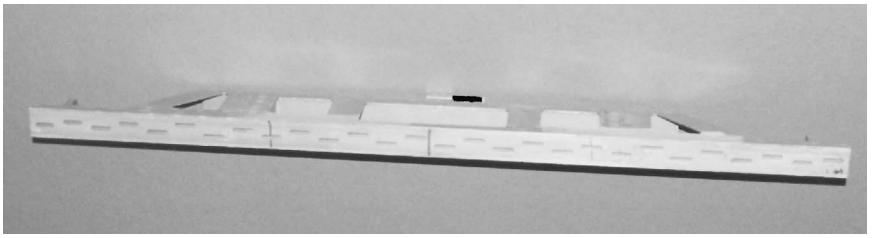


FIGURE 6.1 Longitudinal slot array. (Courtesy California State University Northridge.)

sticks. Ideally, the sum of slot conductances should equal unity, and the sum of slot susceptances should equal zero, for shunt slots. The obvious limitation of resonant feeds is that they are resonant at only one frequency! Over a bandwidth the slot excitations will change in amplitude and phase, thereby introducing pattern distortion and impedance mismatch. Three approaches to quantifying resonant array performance will be given. First, a simple end element equation; second, an analysis by Watson that is useful for uniformly excited (constant amplitude) arrays; and third, an exact analysis.

For a symmetric N-element array with element spacing d and propagation constant β, the last element is $\lambda/2$ out of phase when $N\pi d = N\pi + \pi$, or $f/f_0 = 1 + 1/N$. This gives a product of a number of elements times percent bandwidth $NB = 114$. Unfortunately, this simple merit factor calculation is much too optimistic.

Watson (1947) set up a product of factors, where each factor represents a length of transmission line and a shunt load admittance, Y_n. When multiple reflections between loads are neglected, the finite product can be approximated by a sum. His result, which is often quoted, is

$$\Gamma = \frac{2 - \sum_{n-1}^{N} Y_n - \exp(j2N\beta d) \sum_{n=1}^{N} Y_n \exp(-j2n\beta d)}{\left(2 + \sum_{n=1}^{N} Y_n\right) \exp(j2N\beta d) + \sum_{n=1}^{N} Y_n \exp(j2n\beta d)} \tag{6.1}$$

At center frequency f_0, $\beta d = \pi$, and the formula reduces to a simple result:

$$\Gamma = \frac{1 - \sum_{n=1}^{N} Y_n}{1 + \sum_{n=1}^{N} Y_n} \tag{6.2}$$

This is the genesis of the simple rule mentioned above, that the conductance sum be unity and that the susceptance sum be zero. For uniform excitation, $Y_n = 1/N$ and the series in Eq. (6.1) can be summed exactly to give

$$\Gamma = \frac{1 - \exp[j(N-1)\beta d] \dfrac{\sin N\beta d}{N \sin \beta d}}{3 \exp(j2N\beta d) + \exp[j(N+1)\beta d] \dfrac{\sin N\beta d}{N \sin \beta d}} \tag{6.3}$$

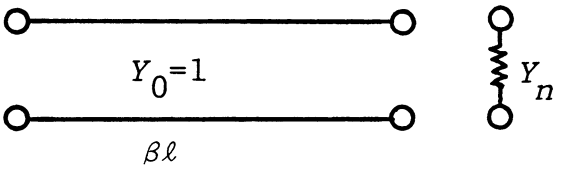


FIGURE 6.2 Prototype section.

The $(\sin N\beta d)/(N \sin \beta d)$ factor is the pattern of an N-element array, and it produces the oscillations in VSWR and Γ as βd changes. These oscillations were shown by Fry and Goward (1950) who plotted Eq. (6.3). The VSWR rises with increasing frequency, crossing a value of 2 at a normalized frequency below $1 + 1/2N$, and then oscillating about VSWR $= 2$. The merit factor is $NB = 89$ for $N = 20$. This value is more conservative than the previous simplistic merit factor of $NB = 114$.

A more accurate approach, which also gives results for tapered excitation, uses a cascade of A–B–C–D matrices (Altman, 1964). Each section of waveguide with a shunt admittance load (slot) is represented by a prototype section, as shown in Figure 6.2. For brevity, call $C = \cos \beta d$ and $S = \sin \beta d$. Then, the section matrix (Altman, 1964) is

$$\begin{pmatrix} C & jS \\ jS & C \end{pmatrix} \begin{pmatrix} 1 & 0 \\ Y_n & 1 \end{pmatrix} = \begin{pmatrix} C + jY_nS & jS \\ Y_nC + jS & C \end{pmatrix} \tag{6.4}$$

The matrix of the resonant array is then the product of the N section matrices, starting at one end, with a shorted quarter-wave section as the last matrix. A large Y_s simulates the short, and the overall matrix is

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \left[\prod_{n=1}^{N} \begin{pmatrix} C + jY_nS & jS \\ Y_nC + jS & C \end{pmatrix} \right] \begin{pmatrix} C_2 + jY_2S_2 & jS_2 \\ Y_2C_2 + jS_2 & C_2 \end{pmatrix} \tag{6.5}$$

The parameter C is used both as a trig term and as part of A–B–C–D, but the usage should be obvious. Subscript 2 is for the shorted section. Input impedance and reflection coefficient are readily obtained

$$Z_{\text{in}} = \frac{A + B}{C + D} \qquad \Gamma = \frac{2}{A + B + C + D} \tag{6.6}$$

Figure 6.3 shows VSWR versus normalized frequency for a uniform 20 element array. The curve is symmetric about f, so only half is shown. A 25-dB Taylor distribution array produces a similar curve with a damped oscillation. The parts of these curves below the first peak are plotted versus NB in Figure 6.4. From these data, the merit factor for uniform excitation is $NB = 66$ for VSWR $= 2$, and NB $= 50$ for VSWR $= 2$ for the 25-dB Taylor pattern.

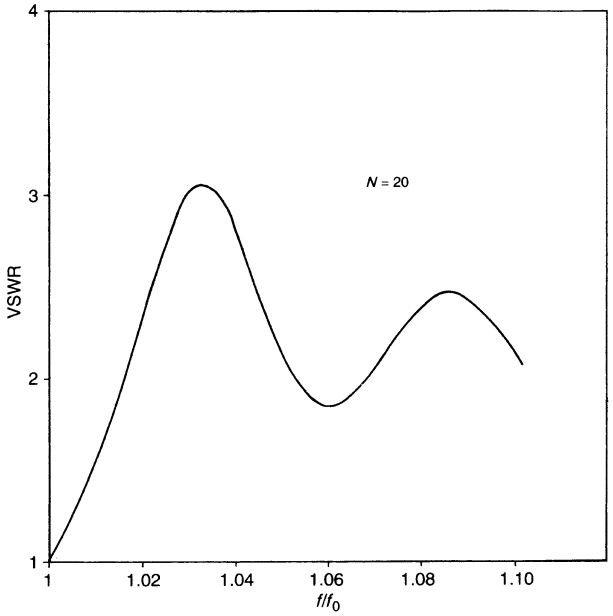


FIGURE 6.3 Uniform resonant array VSWR.

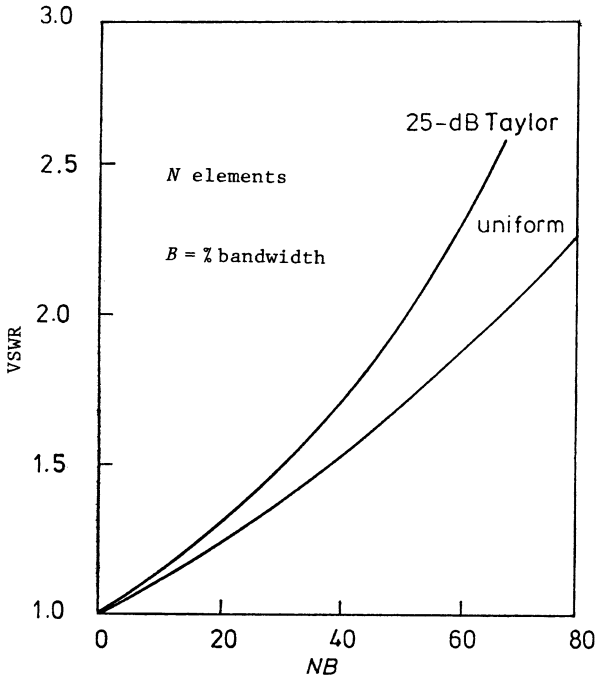


FIGURE 6.4 Resonant array bandwidth.

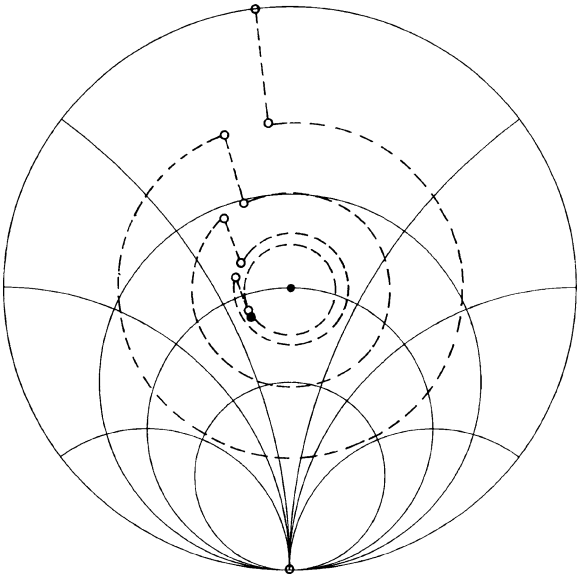


FIGURE 6.5 Resonant array impedance transformation, four elements, uniform, $f/f_0 = 0.96$.

An understanding of bandwidth limits is obtained from Smith chart plots. For uniform excitation, all B movements are equal, but the first few are larger and nearly radial on the Smith chart, as shown in Figure 6.5. These move the impedance toward match. With a tapered excitation (Fig. 6.6), the first few are smaller owing to the taper, and are smaller on the chart. The last few are larger, and tend to bypass the match point.

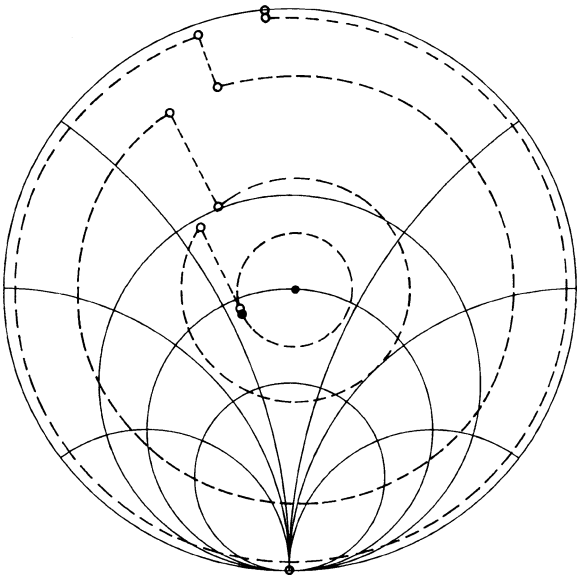


FIGURE 6.6 Resonant array impedance transformation, four elements, 25 dB, $f/f_0 = 0.96$.

In both cases, large numbers of elements give a nearly completed rotation around the chart for each step. Location of the feed near strongly coupled elements exacerbates the reduced bandwidth.

In practice, the bandwidth is smaller, owing to the change of slot admittance with frequency. For all types of waveguide slots the admittance change can be included, using moment methods, but the slot modeling is simpler and more accurate for broadwall slots than for edge slots. A further complication is beam splitting, which can occur when the two traveling waves that constitute the standing wave produce beams to the left and right of center (Kummer, 1966). This occurs at roughly twice the VSWR = 2 value of NB. Keeping VSWR ≤ 2 obviates these split beams.

6.1.1.2 Resonant Slot Array Design. Because waveguide slot arrays provide high efficiency with excellent pattern shaping and control, they deserve an exposition on their design. With the half-guide wavelength spacing requirement, it is necessary to change the coupling sign for every other slot. With longitudinal slots the N slots alternate on each side of the broadwall centerline. For inclined broadwall slots and for inclined edge slots, the inclination angle is simple reversed at the next slot. The result is broadside radiation in phase from all slots.

The difficult task is to obtain the slot lengths and offsets, or lengths and angles, that will provide the desired array distribution, while satisfying the resonant array conditions. Were it not for mutual impedance this would be a simple task. However, mutual coupling is both ubiquitous and necessary, and must be included. Thus the array design task must encompass all slots simultaneously. Two relationships allow a solution to exist. First, the sum of slot admittances (at resonance) must equal the specified input admittance, which is $1 + j0$ for a single stick, but will vary for sticks in a planar array. Second, the mode voltages of the slots must all be equal. In what follows, broadwall longitudinal slots are used as an example. Mutual impedance can be found by replacing the slots by Babinet equivalent strip dipoles, and these in turn can be replaced by cylindrical dipoles of diameter half the strip width (Tai and Long, 1993). It has been assumed here that the waveguide face is a flat ground plane. Carter's 1932 mutual impedance formulas can be used; a convenient FORTRAN formulation of these has been given by Hansen and Brunner (1979). For a dielectric-filled guide, a more accurate mutual impedance based on a cosinusoidal distribution along the slot can be used (Elliott, 1983); however, the mutual impedance calculation now requires numerical integration. Slot self-admittance includes the effects of energy storage inside the guide, and must be calculated more carefully or measured. These calculations are either of variational type (Yee, 1974) or of moment method type. The latter, initiated by Khac (1974), has been successfully exploited by Elliott and colleagues (Elliott, 1979; Elliott and Kurtz, 1978). See Chapter 5 for details.

Using the Stevenson slot result from Chapter 5, and the combined slot impedance Z_n, the slot admittance is

$$\frac{Y_n}{G_0} = \frac{480(a/b)C_n^2}{\pi(\beta/k)Z_n} \tag{6.7}$$

The coefficient C_n is given in Chapter 5, and Z_n is the inverse of the sum of slot self-admittance and slot mutual admittance. Using network equations, the combined slot impedance is given by

$$Z_n = \frac{480(a/b)C_n^2}{\pi(\beta/k)Y_n^0/G_0} + \sum_{m=1}^{N} \frac{V_m Z_{nm}}{V_n} \tag{6.8}$$

The first term includes slot self-admittance Y_n^0, and the sum covers all the mutual terms, where Z_{nm} is the mutual impedance between the n and the m dipoles. Slot voltage is V_n. Given the slot lengths and offsets, Eqs. (6.7) and (6.8) can be used to obtain embedded slot admittances. Since the sum is constrained,

$$\sum_{n=1}^{N} \frac{Y_n}{G_0} = 1 + j0 \tag{6.9}$$

This complex equation gives two real equations. Now the mode voltages are equated. From Chapter 5, these are

$$v_n = j\frac{2\sqrt{2a/b}\,C_n V_n}{\pi(\beta/k)Y_n/G_0} \tag{6.10}$$

The result is a set of $2N - 2$ equations that, together with Eq. (6.9), gives $2N$ equations for the $2N$ unknowns (slot length and offset). These nonlinear and transcendental equations can be readily solved via the Newton–Raphson method (Stark, 1970), where a gradient matrix is formed of derivatives of the equations, for each variable. A starting set of lengths and offsets must be provided. Through iteration, a solution is found that satisfies the input admittance and mode voltage equations. This technique generally converges in 6–10 iterations, and it has been used to design many linear and planar slot arrays, including those with shaped sidelobe envelopes (see Chapter 3). A sketch of a waveguide longitudinal slot array is shown in Figure 6.7.

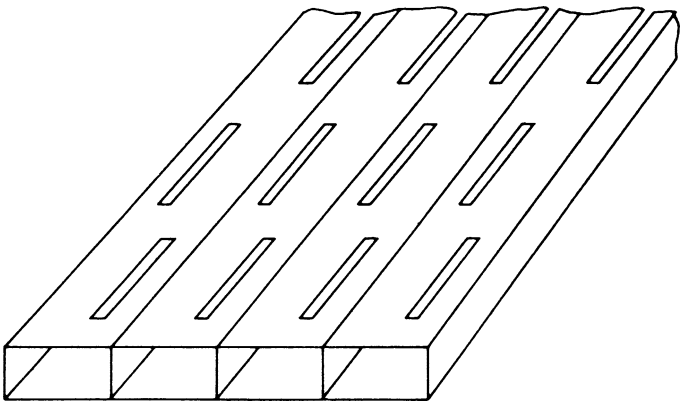


FIGURE 6.7 Longitudinal shunt slots.

6.1.2 Traveling Wave Arrays

A travelling wave (TW) array, like a resonant array, has radiating elements dispersed along a transmission line. However, this array is fed at one end, with a load at the other end. As the excitation wave travels along, the array part of it is radiated, leaving only a small part at the end to go to the load. Thus elements near the feed must couple lightly to the transmission lines, while elements near the load must couple heavily. To avoid reflections in phase, the elements must not be spaced at half-wave intervals, so that the TW array is nonresonant. Because of the nonresonant spacing, the pattern main beam will move slightly with frequency. Element spacing can be less than, or greater than, half-wave; this is discussed further below. Element coupled susceptance should be zero or smaller, so that the pseudo-random addition of these results is a small value. Given a desired array excitation it is necessary to solve a transmission line problem to obtain element coupling values.

6.1.2.1 Frequency Squint and Single-Beam Condition. Both TEM and dispersive feed lines are used. They may have phase velocity above (waveguide), equal to, or below (dielectric loaded coax) that of free space. For what range of spacings will only one beam exist? Two examples are representative. First is a TEM line, where the element couplings are π out-of-phase. The need for out-of-phase coupling will be explained later. With θ measured from broadside and d element spacing as usual, the interelement phase shift needed for a main beam at θ_0 is $kd \sin \theta_0$. Phase along the feed line is $kd + \pi$, where the π is the out-of-phase term. Thus the phase equation is

$$kd \sin \theta_0 + 2n\pi = kd - \pi \tag{6.11}$$

Each $n \geq 0$ represents a real beam, with the nth beam starting at $d/\lambda = (n + \frac{1}{2})/2$. Figure 6.8 shows beam angles versus d/λ for the first three beams. No beam with complete addition exists for $d/\lambda < 0.25$, and for $d/\lambda = 0.75$ two beams exist. So the useful range is $0.25 < d/\lambda < 0.75$. Either forward squint ($d/\lambda > 0.5$) or backward squint ($d/\lambda < 0.5$) is possible. Beam-squint change with frequency is indicated by the slope of the $n = 0$ line. Spacing of 0.5λ represents a special case of broadside radiation. However, this spacing allows all element admittances to add together, with the resulting narrowband performance of the standing-wave array of Section 6.1.1.

Now it will be evident why TW arrays with in-phase coupling are undesirable. The phase equation in this case is

$$kd \sin \theta_0 + 2n\pi = kd \tag{6.12}$$

This equation allows a forward endfire main beam for $n = 0$ ($\sin \theta_0 = 1$) at all spacings. Although beams for $n = 1$, 2, and so on, can be positioned at a desired angle, the endfire beam is always present.

The second example is that of a waveguide array with out-of-phase coupling. Again, the in-phase coupling allows a beam for all spacings, at $\sin \theta_0 = \beta/k$. For the out-of-phase case, the phase equation is

$$kd \sin \theta_0 + 2n\pi = \beta d - \pi \tag{6.13}$$

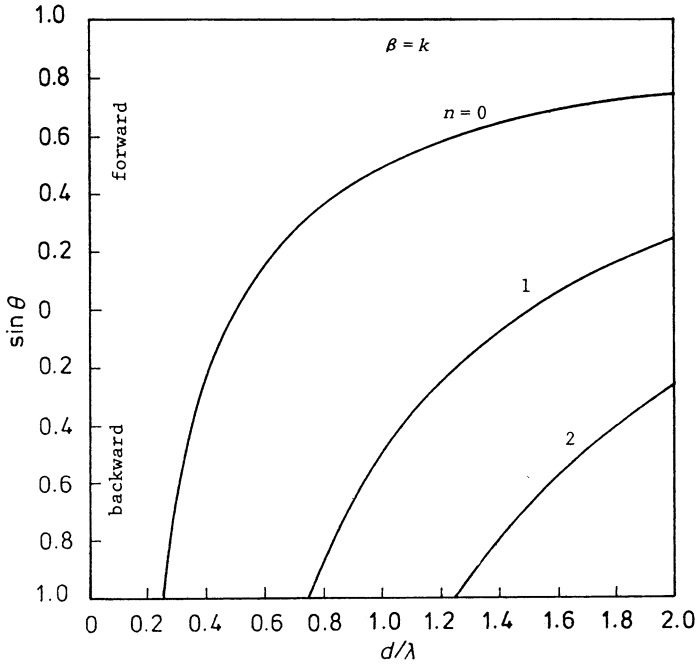


FIGURE 6.8 The TW array beam angle versus spacing.

Now both plus and minus values of n give real beams. Figure 6.9 shows beam position versus d/λ for the first three beams. The graph is plotted for $\beta/k = 0.6$. For $\beta/k = 0.5$, both $n = 1$ modes start at $d/\lambda = 1$. Since waveguides are usually operated with higher β/k, the range of operations is

$$\frac{0.5}{1 + (\beta/k)} < \frac{d}{\lambda} < \frac{1.5}{1 + (\beta/k)} \qquad \frac{\beta}{k} > 0.5 \tag{6.14}$$

For the example chosen, the $n = -1$ beam starts at $d/\lambda = 1.25$, where the $n = 1$ beam starts at $d/\lambda = 0.9375$. Not shown is the $n = 2$ beam starting at $d/\lambda = 1.563$. The figure gives a picture of the mode appearance, but is not useful for design because it is for $\beta/k = 0.6$ only. Figure 6.10 plots beam angle versus spacing for various values of β/k: $\beta/k = 0.5(0.1)0.9$. Using this graph, the trade of β/k and d/λ to give a fixed-beam angle may be made.

Change of beam angle with frequency is important, as can be seen from the slope of the curves. An explicit formula for slope is useful. Taking the derivative of sin θ_0 with respect to frequency, and multiplying to normalize the slope gives

$$f\frac{d \sin \theta_0}{df} = \frac{\lambda^2/\lambda_c^2}{\beta/k} + \frac{1}{2d/\lambda} \tag{6.15}$$

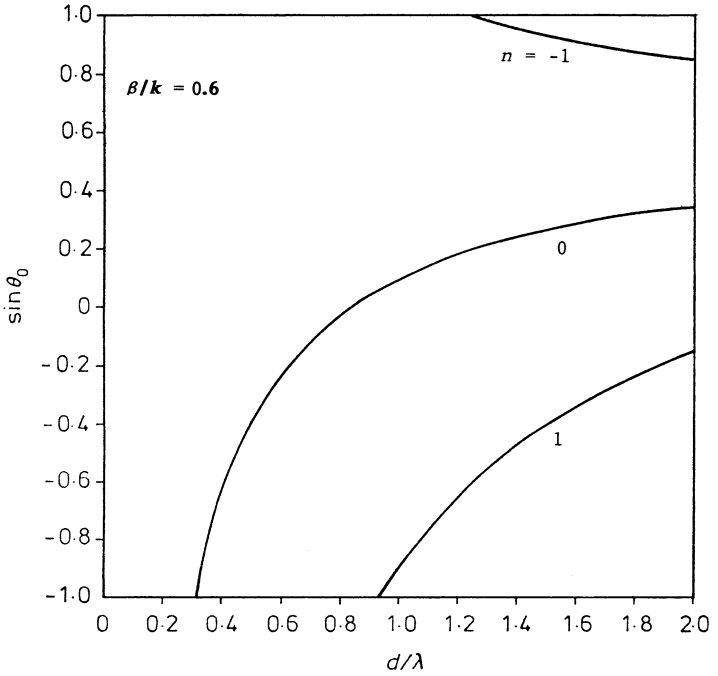


FIGURE 6.9 The TW array beam angle versus spacing.

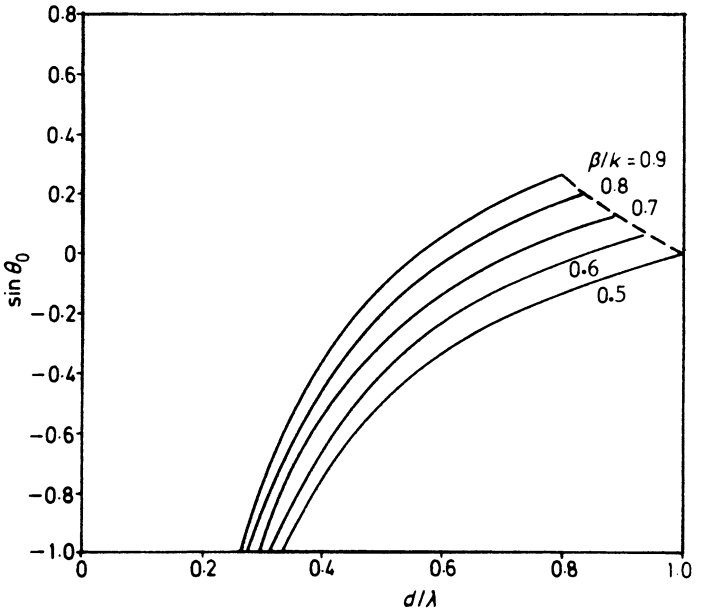


FIGURE 6.10 Single beam: angle versus spacing.

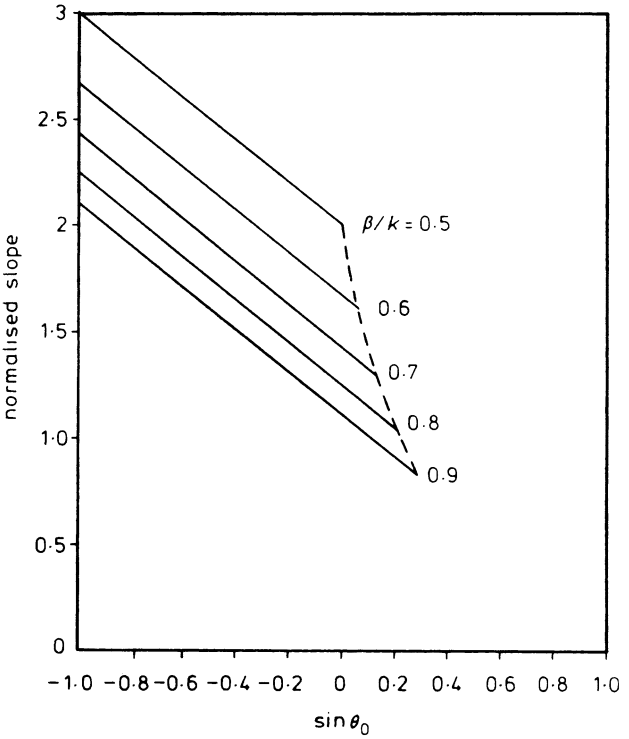


FIGURE 6.11 The TW array beam angle change versus frequency.

This contains three interrelated variables, but fortunately it simplifies to

$$f\frac{d\sin\theta_0}{df} = \frac{1}{\beta/k} - \sin\theta_0 \tag{6.16}$$

Normalized slope is plotted versus sin θ_0 in Figure 6.11. As expected, β/k nearer to unity gives lower slopes, as do larger values of sin θ_0. The curves stop at the angle where a second beam emerges; thus these curves indicate the practical design range. The beam angle θ_0, waveguide β/k, element spacing, and normalized slope can all be traded for an optimum design, given a particular application.

6.1.2.2 Calculation of Element Conductance. An exact calculation of the coupling in a traveling wave array would use $A-B-C-D$ matrices to combine each segment of waveguide (between slots) and one slot admittance with other segments, finding the overall input admittance, as is done for resonant arrays. However, since TW arrays usually have many elements, each coupling is then small, and since the element admittances add in a pseudo-random manner owing to the irrational spacing, it is usually sufficient to assume that the array is matched at any point. At each element

the power divides, with part producing the element excitation and the remainder going on to the next element after some waveguide attenuation. Let the desired array excitation power distribution be proportional to F_n, where $n = 1$ to N. Call the fraction of input power that is dissipated in the end load L,

$$L = \frac{P_l}{P_{\text{in}}} \tag{6.17}$$

and call the conductance distribution along the waveguide G_n. If waveguide loss is included, the factor s representing power loss between slots is appropriate:

$$s = \exp(-2\alpha d) \tag{6.18}$$

Interelement spacing is d and the attentuation coefficient α can be found in standard texts on electromagnetic theory. By using the sketch of Figure 6.12, the last coupling is written first, then the next, and so on.

$$G_N = \frac{F_N}{P_N} = \frac{F_N}{P_l} \qquad G_{N-1} = \frac{F_{N-1}}{P_{N-1}} = \frac{F_{N-1}}{F_N/s + P_l/s} \tag{6.19}$$

$$\begin{aligned} G_n &= \frac{F_n}{\sum_{m=n+1}^{N} F_m s^{n-m} + P_l s^{n-N}} \\ G_1 &= \frac{F_1}{P_{\text{in}} - F_1} \end{aligned} \tag{6.20}$$

The input power is the load power plus the sum of coupled powers, with each multiplied by the appropriate number of segment loss factor $1/s$:

$$P_{\text{in}} = \sum_{m=1}^{N} F_m s^{1-m} + P_l s^{1-N} \tag{6.21}$$

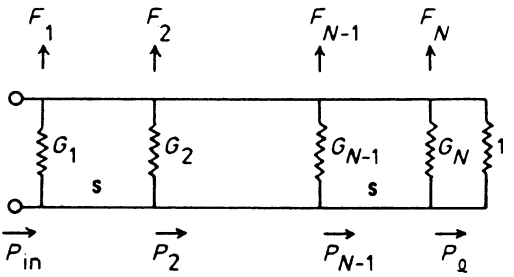


FIGURE 6.12 The TW array circuit.

Eqs. (6.20) and (6.21) can be used to eliminate P_{in}, giving P_l:

$$P_l = \frac{L}{1 - Ls^{1-N}} \sum_{n=1}^{N} \frac{F_n}{s^n} \tag{6.22}$$

Inserting this in the expression for G_n yields the conductance equation, after some rearrangement:

$$G_n = \frac{F_n / s^n}{\dfrac{1}{1 - Ls^{1-N}} \sum_{m=1}^{N} \dfrac{F_m}{s^m} - \sum_{m=1}^{n} \dfrac{F_m}{s^m}} \tag{6.23}$$

For negligible loss, of course, $s = 1$. Loss generally needs to be included only when it is large, or when the loss between elements is comparable to the smallest conductance. Usually the F_n excitation coefficients are obtained from a distribution, such as Taylor one-parameter or Taylor $\bar{n}$, and the scale factor of the F_n will make $P_{\text{in}} = 1$. Note that at each slot the fraction of power coupled out is $G/(1 + G)$. The continuous version of Eq. (9.169) is often seen (Kummer, 1966).

$$G(x) = \frac{F(x)}{\dfrac{1}{1 - L} \displaystyle\int_{-L/2}^{L/2} [F(y) + R(y)]dy - \int_{-L/2}^{x} [F(y) + R(y)]dy} \tag{6.24}$$

Here $R(y)$ is the loss per unit length of guide. However, as might be expected, sampling the continuous results to get slot coupling is less accurate than using the sum formula (Eq. 6.23). For small arrays, or for low-sidelobe arrays, it is important to use the more exact equation.

Figure 6.13 shows the conductance values for a 29-element uniformly excited array, for load power percentages of 5, 10, and 25%. It may be seen that the 5 and 10% curves go off-scale, while the 25% curve peaks just >0.1. It is always necessary to keep all conductances small; 0.1 is a good maximum. Not only are larger values often difficult to realize, but the impedance mismatch effect may no longer be negligible. Larger arrays, of course, will have lower conductances per slot. The curves for $N = 29$ may be converted to general-purpose curves (approximately) by multiplying values by $N/2 = 14.5$. Conductance values for different N are then obtained by dividing the new values by $N/2$. However, for precise results, a new calculation should be made after the number of slots is established based on the approximate scaling just described.

Figures 6.14 and 6.15 give conductances for the Taylor one-parameter distribution and for the Taylor $\bar{n}$ distribution ($\bar{n} = 5$). For both figures, $N = 29$ and SLR = 25 dB. Note that the heavy taper of the one-parameter distribution gives curves with a single peak, whereas the $\bar{n}$ distribution, which has a higher pedestal (see Chapter 3), tends to

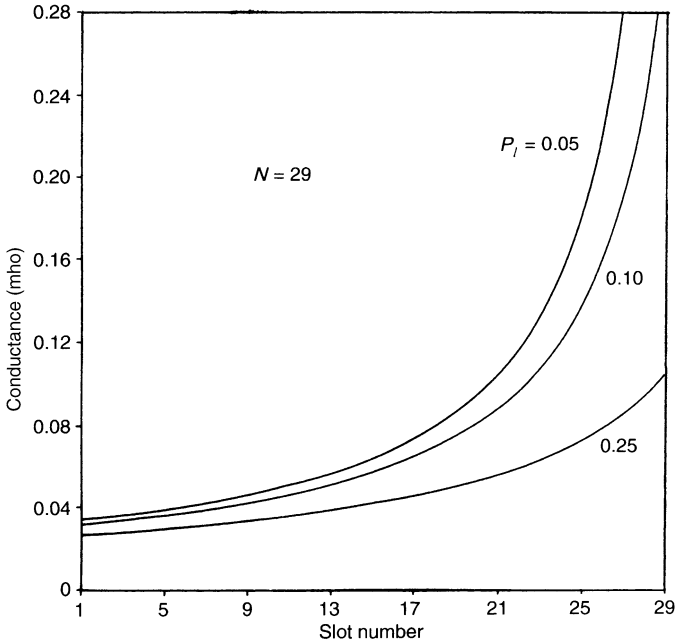


FIGURE 6.13 Uniform slot conductance.

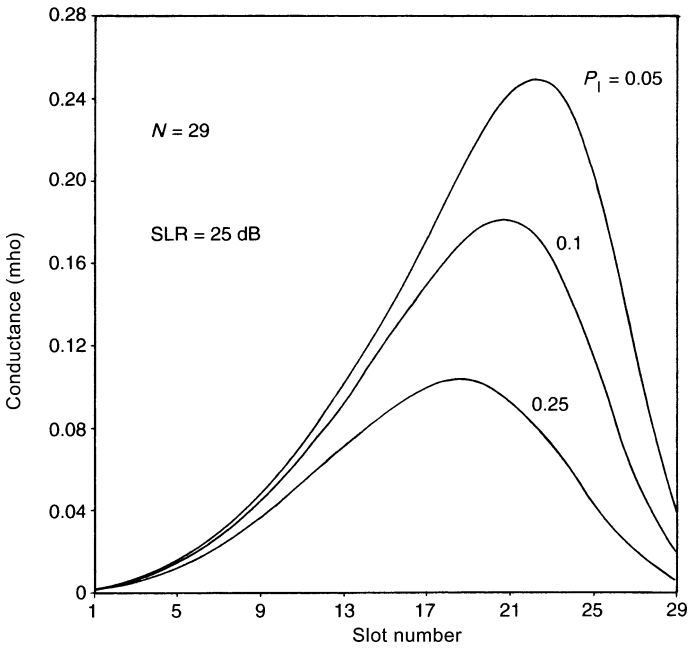


FIGURE 6.14 Taylor one-parameter slot conductance.

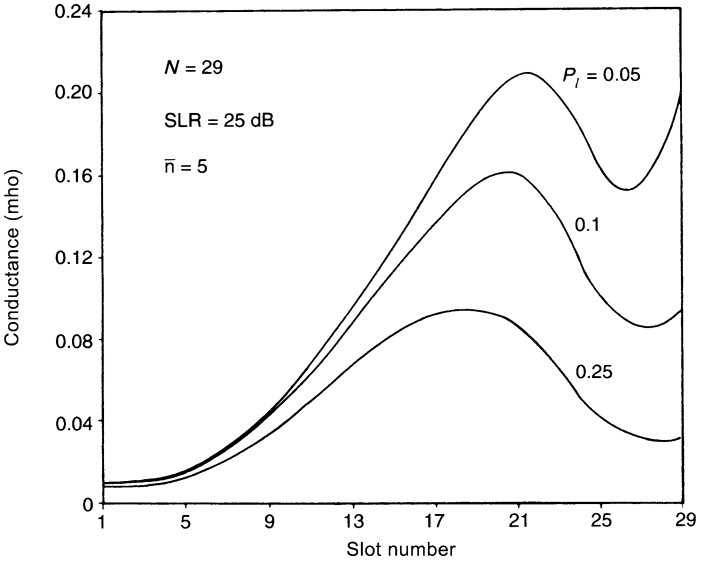


FIGURE 6.15 Taylor n̄ slot conductance.

rise again near the load, especially for small load power. Since all the curves for 5 and 10% load power have peak >0.1, it is in practice necessary either to increase load power, thereby reducing gain, or to increase the number of slots. Again, the curves can be scaled approximately using $N/2$.

6.1.2.3 TW Slot Array Design. Without the effects of mutual coupling, the design of a traveling-wave slotted waveguide array is simple: A table of conductances is obtained (see previous section) and the universal slot curves of Chapter 5 are used to find length and position of each slot, to give the proper resonant conductances. With mutual coupling, the slots interact strongly so that it is necessary to consider the entire array at once. This has been successfully done for broadwall shunt (displaced) slots by Elliott (1979), and the procedure can be applied equally well for any array in which mutual coupling between elements can be computed accurately. A salient exception is the inclined-edge slot waveguide array, as the replacement of slots by Babinet equivalent dipoles for the computation of mutual impedance is not sufficiently accurate owing to the wrap-around of the slots.

The design procedure for traveling-wave arrays is similar to, but more complicated than, that for resonant arrays. As is the case there, the mode voltage at each slot is proportional to the slot voltage divided by the *scan impedance*:

$$v_n = \frac{2\sqrt{a/b}\; C_n V_n}{\pi(\beta/k) Y_a^n / G_0} \tag{6.25}$$

Unlike the resonant array, the mode voltages are unequal, since the slots are spaced an irrational number of guide wavelengths apart. Notation is that of Section 6.1.1, except

the mode voltages are subscripted here. In fact, the nth mode voltage is related to the $(n-1)$st mode voltage and $(n-1)$st admittance, where elements are numbered starting at the load. This relation is

$$\frac{v_n}{v_{n-1}} = \cos\beta d + j\frac{Y_{n-1}}{G_0}\sin\beta d \tag{6.26}$$

The parameter Y_n is defined below, and d is the slot spacing along the guide. Combining with Eq. (6.25), an equation is obtained for the ratio of two successive slot voltages:

$$\frac{V_n}{V_{n-1}} = \frac{C_{n-1}Y_{n-1}^A}{C_nY_n^A}\left(\cos\beta d + j\frac{Y_{n-1}}{G_0}\sin\beta d\right) \tag{6.27}$$

Since the slot voltage is specified, if the fractional power into the load L, and all the slot lengths and offsets, are known, Eq. (6.27) represents $2N-2$ equations that must be satisfied. That is, using the correct slot parameters gives the desired slot voltage ratios. A perturbation technique is then used with starting values of slot lengths and offsets. The N-element array has $2N$ unknowns; the remaining two equations come from the input admittance. Like the resonant array it should be $1+j0$. However, input admittance here is not just a simple sum of slot active admittances; rather, a transmission-line concatenation must be used. By calling Y_n the guide admittance seen at the nth slot (including the guide admittance beyond the slot and the slot active admittance), it becomes:

$$\frac{Y_n}{G_0} = \frac{Y_n^A}{G_0} + \frac{Y_{n-1}\cos\beta d + jG_0\sin\beta d}{G_0\cos\beta d + jY_{n-1}\sin\beta d} \tag{6.28}$$

Starting at the slot nearest to the load where $Y_1 = Y_1^A + G_0$, Eq. (6.28) is applied successively until the input admittance is obtained. This should be $1+j0$ if the correct slot lengths and offsets were chosen. These two real equations, with the $2N-2$ mode voltage equations, give the $2N$ real equations necessary to allow a definitive problem. These also can be solved iteratively using a Newton–Raphson solution. Figure 6.16 depicts a traveling wave patch array; note the change in patch sizes to accommodate coupling resistance of the form of Figure 6.14. Figure 6.17 shows the AWACS array, in which the horizontal waveguide sticks are traveling wave arrays of edge slots. This is a very low sidelobe level array (−50 dB).

The TW arrays just discussed have a beam angle that changes with frequency, owing to the series nature of the feed. This beam scan can be removed by use of an equal-path-length feed, as sketched in Figure 6.18 (Cheston and Frank, 1970).

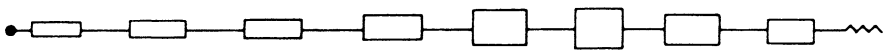


FIGURE 6.16 The TW patch array.

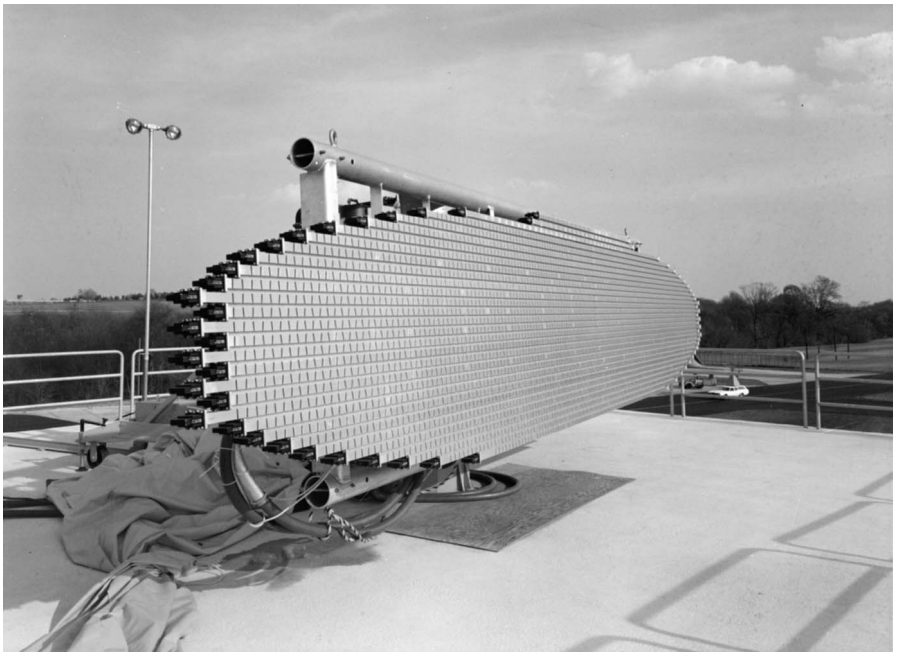


FIGURE 6.17 Low-sidelobe edge slot array. (Courtesy Westinghouse Electric Corp.)

Unfortunately, much of the low-cost simplicity of the series TW feed is lost. The series feed slots in this scheme do not radiate directly, but each feeds a waveguide. Radiating elements are arranged at the waveguide end. The geometry is arranged so that the waveguide path length from the input port to each radiating element is the same; thus no change in beam position with frequency will occur. Another advantage is that the radiators experience the same amount of waveguide dispersion. The radiating waveguides should be along the squint angle so that the aperture is broadside. This specifies the angle between feed and radiating guides.

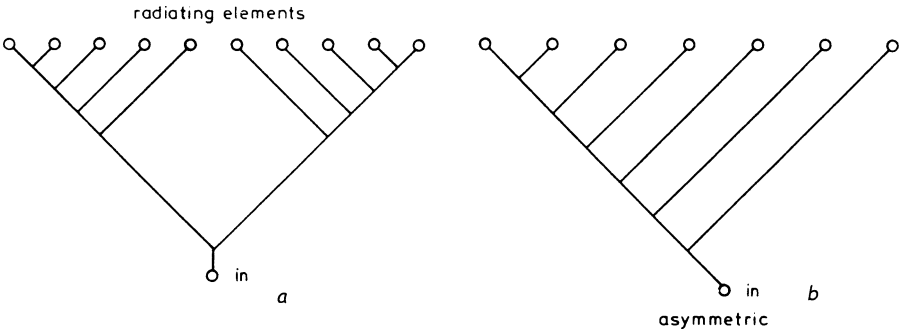


FIGURE 6.18 Equal-path-length feeds.

6.1.3 Frequency Scanning

Advantage can be taken of the beam squint with frequency by increasing the interelement path length such that a small frequency change scans the main beam. Such arrays are traveling wave arrays, but are important enough to warrant a separate section. Assume that the radiating elements spaced d apart are connected by a serpentine (snake) feed of length s and wave number β. The wavefront is defined by

$$kd \sin \theta_m = \beta s - 2m\pi \tag{6.29}$$

(See sketch in Figure 6.19.) To avoid an endfire beam, $kd \geq \beta s - 2m\pi$. More than one main beam can be avoided (see Section 2.1) by keeping

$$\frac{d}{\lambda} < \frac{1}{1 + \sin \theta_m} \tag{6.30}$$

The phase equation is often written as

$$\sin \theta_m = \frac{s\lambda}{d\lambda_g} - \frac{m\lambda}{d} \tag{6.31}$$

Clearly, larger s/λ, and correspondingly larger m, gives a faster change of beam angle with frequency. If the frequency scan passes through broadside there will be a significant change in impedance owing to the addition of all the element conductances. Element conductances, or couplings, are determined just as for a TW array. If the waveguide bends needed to make a snake are not well matched over the frequency band, it may be necessary to calculate the input admittance using the techniques developed for resonant arrays. The coupler reflections tend to produce a sidelobe at the conjugate beam direction; the level of this reflection sidelobe depends on the coupling conductances, and is higher when the scan is closer to broadside.

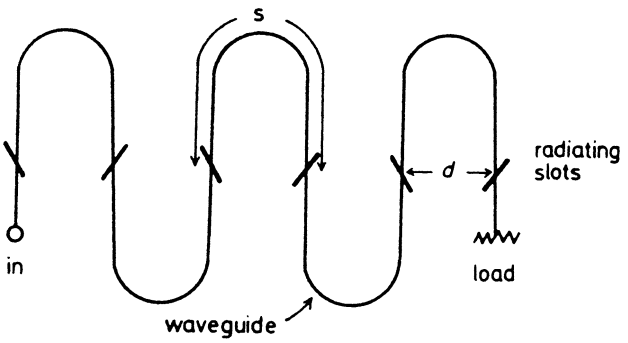


FIGURE 6.19 Serpentine feed geometry.

In a straight TW feed, losses are usually not important, but the extra length of the sinuous feed often introduces significant loss. Maximum gain occurs when $\alpha s = 1$, where s is the total snake feed length (Begovich, 1966). The corresponding feed loss is 4.3 dB. Maximum radar target tracking accuracy, a combination of gain and beamwidth, occurs for $\alpha s = 2$, with 8.6-dB feed loss (Begovich, 1966).

Dispersion in the snake feed will distort short pulses if the pulse length is less than or comparable to the feed travel time. Detailed calculations have been made by Bailin (1956) and Tseng and Cheng (1964).

As an example, assume each serpentine loop to be five free- space wavelengths ($s = 5\lambda_0$), a waveguide wavenumber ratio of $\beta/k = 0.6$ at center frequency, and a slot spacing of half free-space wavelength. The beam angle is given by

$$\sin\theta_0 = 10\left[\sqrt{1-\left(\frac{0.8f_0}{f}\right)^2} - 0.6\frac{f_0}{f}\right] \tag{6.32}$$

This is shown in Figure 6.20. A 10% bandwidth covers a 117° scan. A typical sinuous feed using sidewall couplers is shown in Figure 6.21. Sinuous feeds have also been used to feed cylindrical reflectors. Another way of packaging the snake is in a helix. Here a helical waveguide with the outside wall open is machined from a solid block; a cover is later fastened around the open top (Croney and Foster, 1964). Higher dispersion, and hence a larger slope of degrees per megahertz, can be obtained from a dispersive, periodically loaded waveguide, analogous to the waveguide delay lines used in traveling-wave tubes (Hockham and Wolfson, 1978).

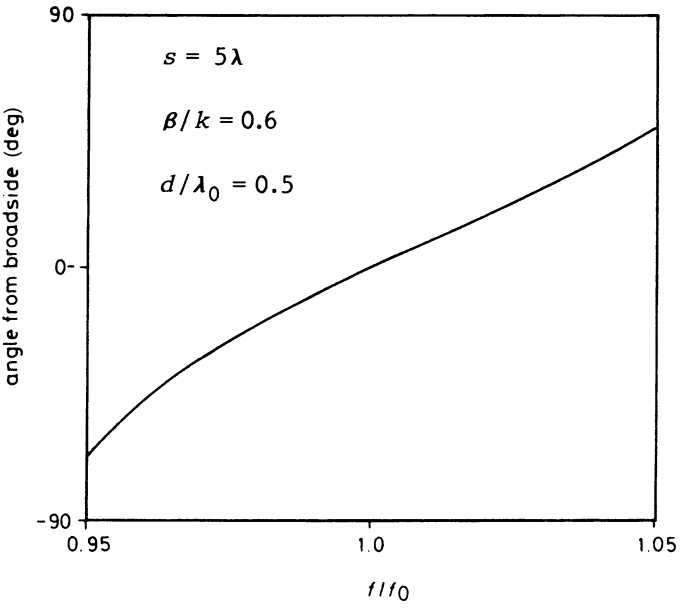


FIGURE 6.20 Frequency scan example.

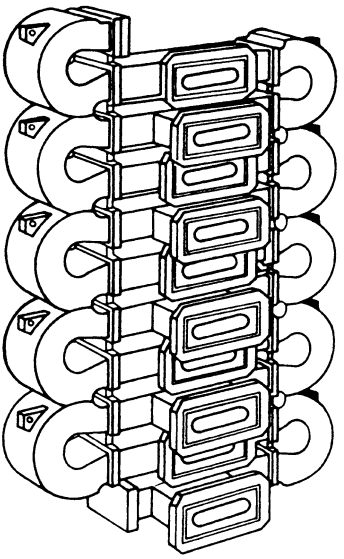


FIGURE 6.21 Serpentine feed. (Courtesy ITT Gilfillan.)

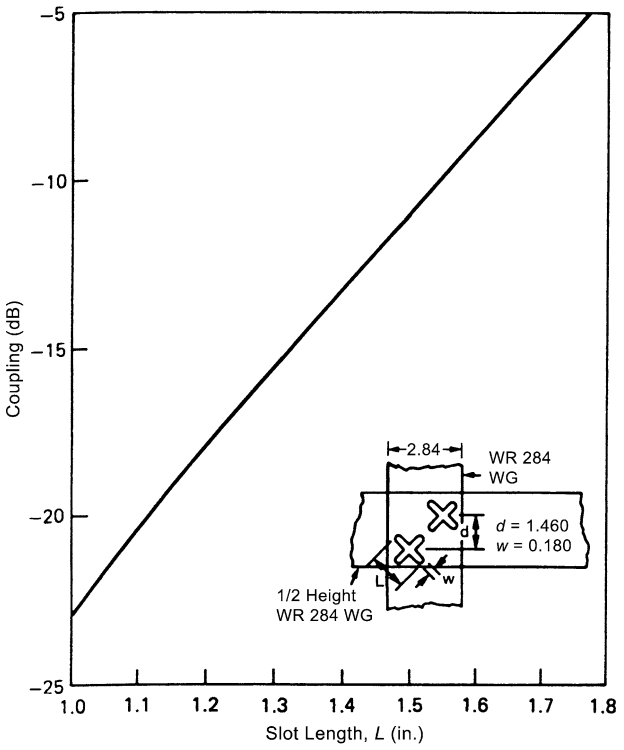


FIGURE 6.22 Cross-slot coupling. (Courtesy Ajioka, J. S., "Frequency-Scan Antennas", in *Antenna Engineering Handbook*, R. C. Johnson, Ed., McGraw-Hill, 1993, Chapter 19.)

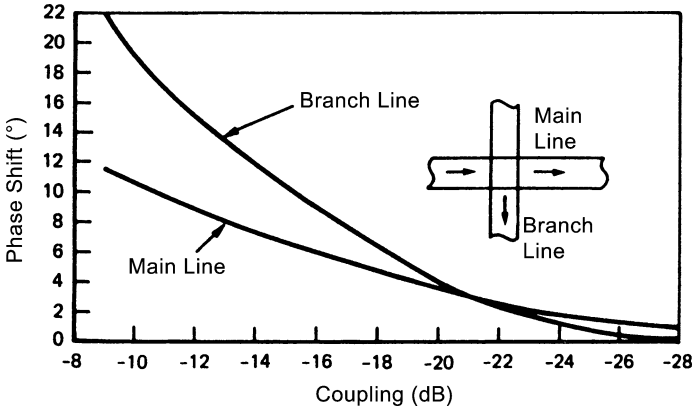


FIGURE 6.23 Cross-slot coupler phase shift. (Courtesy Ajioka, J. S., "Frequency-Scan Antennas", in *Antenna Engineering Handbook*, R. C. Johnson, Ed., McGraw-Hill, 1993, Chapter 19.)

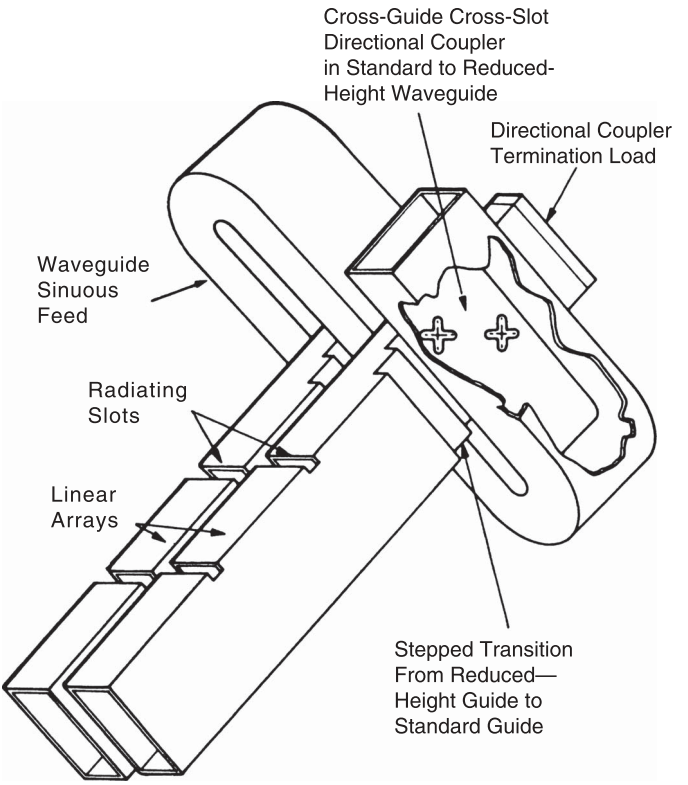


FIGURE 6.24 Serpentine feed on edge slot array. (Courtesy Ajioka, J. S., "Frequency-Scan Antennas", in *Antenna Engineering Handbook*, R. C. Johnson, Ed., McGraw-Hill, 1993, Chapter 19.)

When the operating frequency is too near the resonant array frequency, the reflections from the sinuous feed bends and couplers tend to add to give a significant reflection. Kurtz and Gustafson (Gustafson, 1957) developed impedance matching nondirective couplers as seen from the feed. This allows scanning through broadside with small degradation. The coupler used a rectangular guide normal to the narrow wall of the feed guide, with a centered shunt slot in that wall. A resonant inductive iris was placed in the feed guide, opposite the coupling slot.

For low-sidelobe designs, an improved 4-port coupler uses a cross-guide coupler with two pairs of crossed-coupling slots. A load for this directional coupler is placed at the end of the radiating waveguide stick (Ajioka, 1993). Figure 6.22 shows the coupling region for an S-band coupler. Phase shifts versus coupling are shown in Figure 6.23 for that coupler, and a typical implementation is shown in Figure 6.24 (Ajioka, 1993). An L-shaped coupling slot has been used by Park (1995). Figure 6.25 shows an airport surveillance radar antenna. Use of wideband directional couplers made out of aluminum billets by numerical controlled machining allows low sidelobes to be obtained.

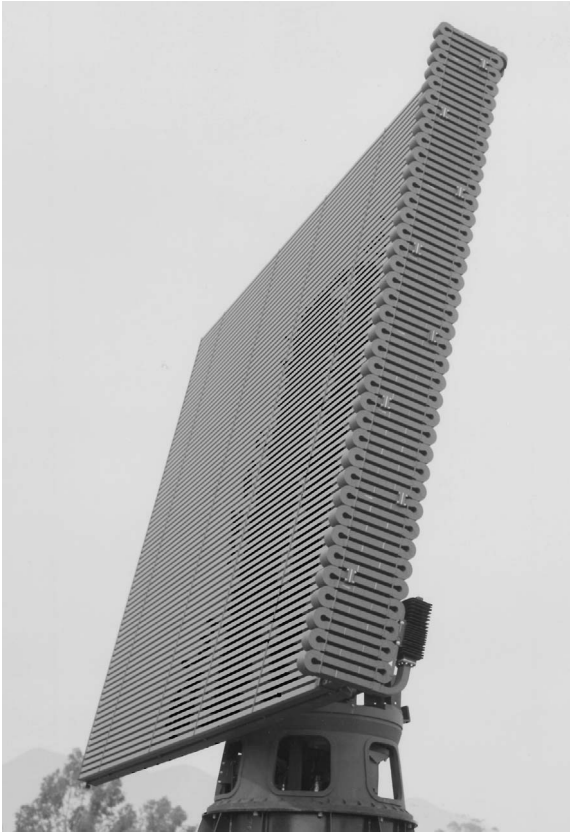


FIGURE 6.25 Airport surveillance antenna. (Courtesy ITT Gilfillan.)

6.1.4 Phaser Scanning

Linear series feeds may be scanned by use of phasers either in the feed line or adjacent to the elements; these are series phasers and parallel phasers, respectively. Figure 6.26 sketches these arrangements. With waveguide slot array sticks there is no room for phaser devices. Many other configurations do allow phasers to be incorporated, for example, the nonsquint feed of Figure 6.18. An advantage of series phasers is that each phaser handles only $1/N$ of the transmitted power, and each element path has the phaser loss. A disadvantage is that the basic interelement phase shift $\phi = kd \sin \theta_0$ must be multiplied by the number of the element. As almost all phasers are now digitally controlled, this is a minor disadvantage. Modern digital controls are inexpensive. For parallel phasers, the advantage is that all phasers have the same setting equal to ϕ. Disadvantages are that the first phaser must handle almost the entire power, and that the phaser losses are in series. In both topologies, there are systematic errors due to impedance mismatch; these are treated later.

A word about phasers is appropriate, although a full discussion is outside the scope of this book. Phasers, often called phase shifters, are primarily of two basic types: switched line and ferrite. Switched line phasers use semiconductor switches to switch in or out lengths of transmission line, where the latter is usually coax, microstrip, or stripline. Discrete components may be used, but more often the switches, loads, and controls are in integrated circuit packages that are integrated into microstrip or stripline. Monolithic microwave integrated circuits (MMIC) configurations are becoming more common, especially >10 GHz. Of course, these phasers utilize time delay, but because the largest bit is usually 180°, they are classed as phasers. Digital phasers are binary: A 5-bit phaser has phases of 180, 90, 46, 22.5, and 11.25°. The phaser design must keep the impedances matched as the bits switch, and in extreme applications must also match the losses. If amplifiers are incorporated to offset losses, the phasers become nonreciprocal.

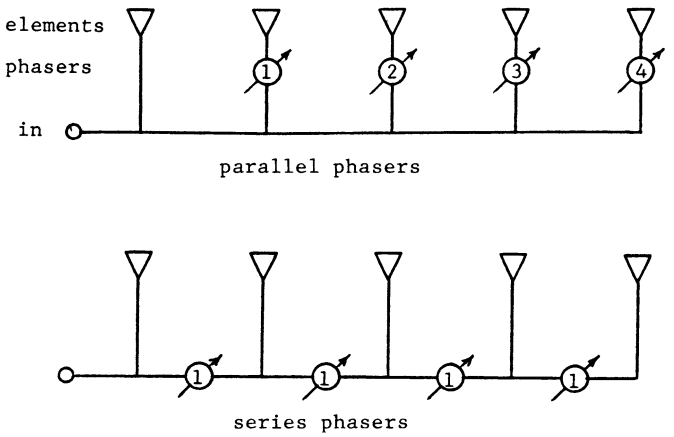


FIGURE 6.26 Linear series feeds.

Ferrite phasers are also of two types, and although these are intrinsically analog phasers, current practice has digitized the phase control. They are almost always embodied in waveguide. Toroidal ferrite phasers use one or several ferrite toroids of rectangular shape placed in a waveguide such that the axis of the toroid is along the guide axis. A drive wire threaded through the toroid provides either a positive or negative current pulse. The pulse drives the toroid to saturation, and the toroid is latched at a positive or negative remanent-induction point. The difference between the electrical lengths of the two states provides the phase shift. Typically, several toroids are placed serially, with lengths chosen to provide the N bits of phase shift. Advantages and disadvantages are obvious: these phasers are nonreciprocal, and must be reset after each pulse for radar operation; drive power is used only when switching, so that power and heat dissipation are minimized.

The Faraday rotation ferrite phaser uses a ferrite rod along the axis of the waveguide, with quarter-wave plates at each end to convert linear to circular polarization, and vice versa. In practice, the ferrite rod is plated with silver or gold to form a small ferrite waveguide. This ferrite guide is then placed inside the regular guide, with nonplated tapered transition sections on each end to match between the two guides. An external solenoidal coil around the waveguide provides the magnetization that controls the phase shift. Again, typical drive circuits are digitally controlled. These phasers are reciprocal, and are often used in high-power applications, such as radars. Sophisticated drive compensation circuits have been developed to adjust the drive as the temperature of the ferrite increases; the drive power must be applied at all times.

Useful references on phasers include a special issue of *Transactions of the IEEE* on array control devices (Whicker, 1974), a chapter on phasers and delayers (Ince and Temme, 1969), and a two-volume monograph (Koul and Bhat, 1991).

6.2 SHUNT (PARALLEL) FEEDS

6.2.1 Corporate Feeds

Corporate feeds are common in arrays of dipoles, open-end guides, and patches, and are named after the structure of organization charts, where the feed divides into two or more paths, then each path divides, and so on. Such feeds are commonly binary, but sometimes the divider tree includes three-way, or even five-way dividers, depending on the number of array elements. Figure 6.27 shows a simple binary parallel feed with phasers. As in the case of main line series feeds, the phase shift needed is progressively larger. Now, however, each path length has only one phaser. For wideband applications, where modulo 2π phase shift is inadequate, time delay units (delayers) can be used. In principle, a unit of delay is needed at each power division level. For wide-angle scanning, delayers of $\frac{1}{2}Nkd \sin \theta_0$ would be placed after the first power division; four delayers with half that at the next level; and finally eight phasers at the element level, where all phase delays are 2π. A more practical compromise for

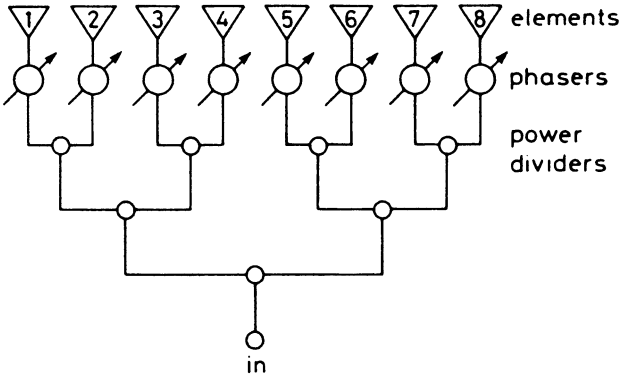


FIGURE 6.27 Parallel (corporate) feed (numbers are phase shift units).

large arrays is to have one or two levels of delayers in addition to the phasers at each element. Figure 6.28 shows such an example.

The critical component in the corporate feed is the power divider (combiner); bifurcated T waveguide or coaxial-line T junctions, or hybrid junctions, can be used. The former are simpler, but hybrid junctions reduce the effects of element impedance mismatch. Typical hybrids are the short-slot hybrid (Riblet, 1952; Levy, 1966) that use two guides with a contiguous narrow wall, the top-wall hybrid (Hadge, 1953), the magic tee (Young, 1947), and the ring hybrid (Tyrrell, 1947). Power dividers and hybrids can also be implemented in stripline or microstrip (Jones and Shimizu, 1958; Levy, 1966; Young, 1972). A broadband TEM line coupler was developed by Schiffman (1958). The popular Wilkinson divider (Wilkinson, 1960) incorporates resistive loads to improve the match; it is commonly a stripline divider with the load across the separated lines. Another stripline divider is the Lange coupler (Lange, 1969), which is a 90° hybrid utilizing interdigitated coupled lines. However, many microstrip patch arrays use the simple split line reactive power divider.

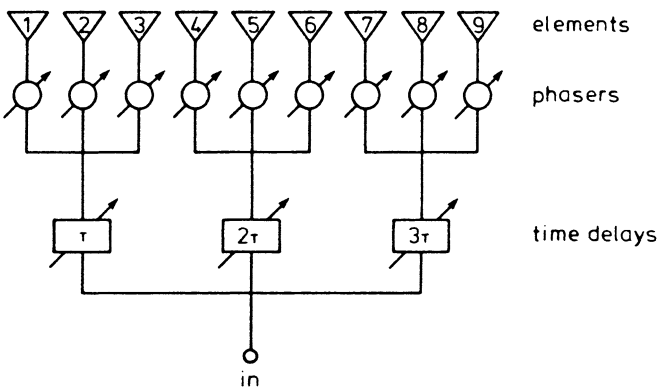


FIGURE 6.28 Phase-delay corporate feed.

6.2.2 Distributed Arrays

In a distributed array, each element is connected to its own receiver–transmitter module. Such modules often contain duplexers, circulators, filters, preamps, power amplifiers, phasers, and control components. Sometimes the element is part of the module. Several advantages accrue. Many low-power semiconductor sources may be used instead of a single high-power tube source. Feed network loss, which reduces signal to noise (S/N), is minimized. Graceful degradation allows system operation with slight gain and sidelobe changes as modules fail. Mean time before failure (MTBF) is greatly increased (Hansen, 1961). Against these advantages is the higher cost. The nominal few hundred dollar module has been a long-sought goal, but non-recurring engineering costs are high, especially for MMIC, so that a large number of production units is needed to make the modules cost-effective. Solid-state and tube modules are discussed by Ostroff et al. (1985); the current art is reviewed by Cohen (1996).

A module may be utilized for each element, or for a line array in an array scanning in only one plane. Figure 6.29 shows a module that feeds six printed dipoles. Modules for planar arrays are discussed in Section 6.3.

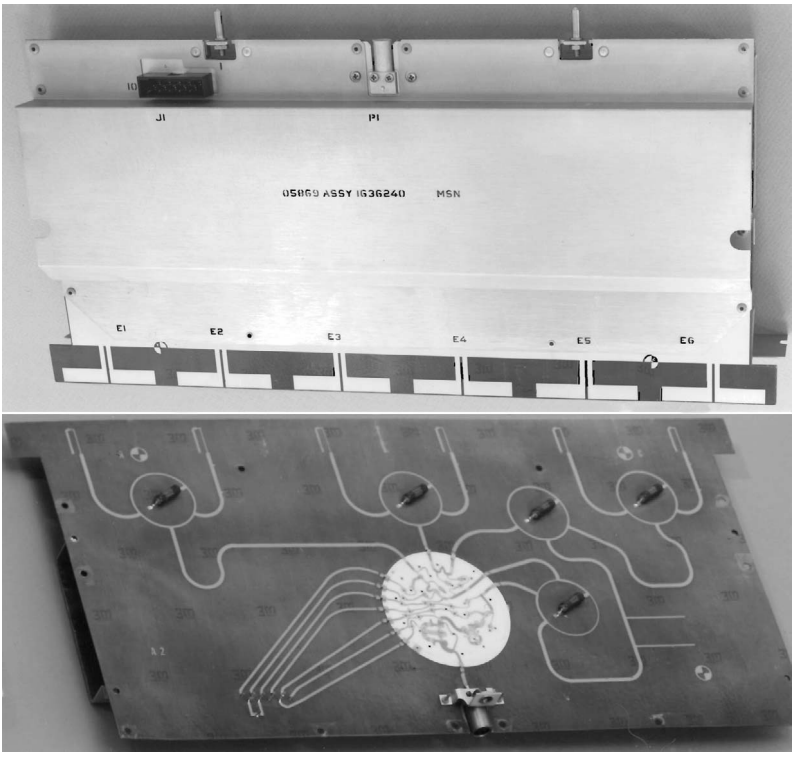


FIGURE 6.29 Sub-array module. (Courtesy Hughes Aircraft Co.)

Calculation of G/T for an active array is straightforward provided no resistances are used to provide amplitude taper; of course losses of all components, and active device gain and noise figure, must be known. Third-order intercept and the spur-free dynamic range (see Section 6.4.1) are also often important. When resistive tapers are used, care must be taken with G/T calculations (Lee, 1993). A comparison of active weighting (for amplitude taper) utilizing the gain of the MMIC preamplifiers, and passive weighting utilizing unequal combiners, has been given by Holzman and Agrawal (1996).

6.3 TWO-DIMENSIONAL FEEDS

6.3.1 Fixed-Beam Arrays

Planar arrays may be composed of linear arrays, or may be an assemblage of individual elements, such as open waveguides. When the main beam direction is fixed, the constituent branch linear arrays and the main feed line that feeds the linear arrays can be traveling wave or resonant, or corporate. Thus, the flat plane array (FPA), which has replaced parabolic reflector antennas in many aircraft and missile radars, is comprised of branch "sticks", which are resonant arrays of longitudinal shunt slots in waveguide, with the waveguide sticks fed by a resonant main feed. Figure 6.30



FIGURE 6.30 Flat plane slot array. (Courtesy Hughes Aircraft Co.)

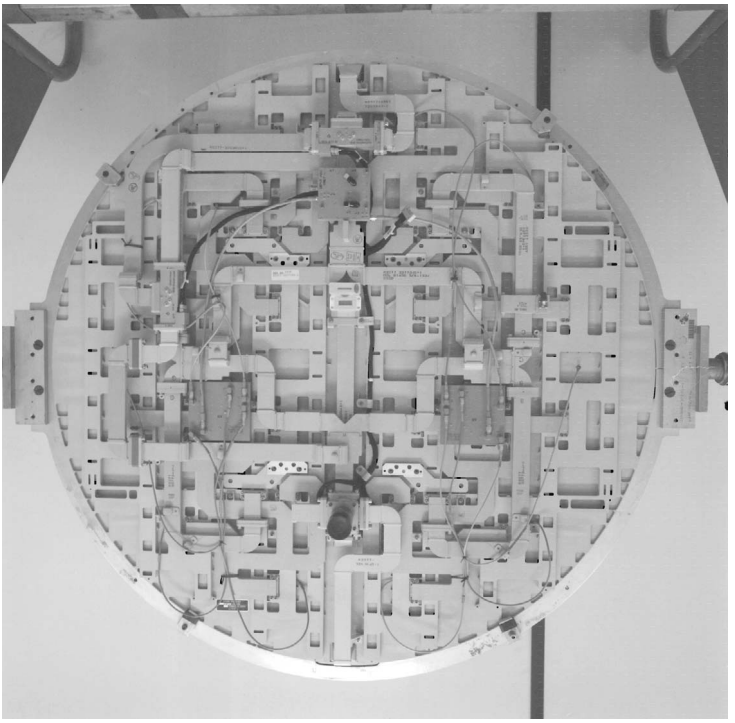


FIGURE 6.31 Optimized sum and difference feed network. (Courtesy Hughes Aircraft Co.)

shows the FPA, where each stick array is divided into sub-arrays, with five comprising the center stick. This is done to allow increased bandwidth since each resonant array now has fewer than 21 slots. The complete array has 1368 slots. Figure 6.31 shows the back, where the cross-guide feeds can be seen. A waveguide corporate feed excites the cross-guide feeds. A hybrid monopulse feed network is used to provide optimized sum and azimuth and elevation difference patterns. Figure 6.17 shows a very low (−50 dB) sidelobe edge slot waveguide array shaped to fit into a radome. Here the stick arrays are traveling wave arrays; since the array rotates in azimuth, the

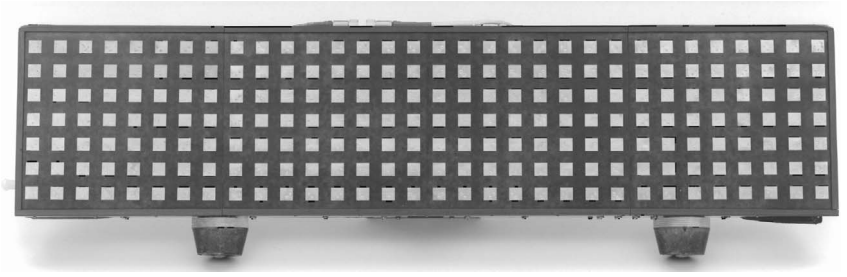


FIGURE 6.32 Dual polarization C-band SAR patch array. (Courtesy Technical University of Denmark.)

beam squint is easily accommodated. Achievement of such low sidelobe levels requires meticulous accommodation of edge effects and mutual coupling. Precise fabrication is essential for low-sidelobe antennas. Microstrip patch antennas with in-plane corporate feeds suffer from feed radiation and feed mutual coupling (Hall and Hall, 1988). A large C-band patch array using resonant sub-array feeds is shown in Figure 6.32 (Granholm et al., 1994).

6.3.2 Sequential Excitation Arrays

A planar array of linearly polarized elements may be sequentially oriented to produce circular polarization (Huang, 1986, 1995; Hall and Hall, 1988). Figure 6.33 sketches a circular patch array where the patches are oriented and phased for circular polarization (CP). The basic feature is a 2×2 sub-array with proper patch rotation and phasing. Advantages are better control of *E*- and *H*-plane beamwidths, and reduced mutual coupling (Huang, 1986). A major disadvantage is the large grating lobes in the diagonal planes. These are eliminated by using CP elements with sequential orientation and excitation. The same technique can be used with imperfect CP elements or feed errors to reduce grating lobes (Hall et al., 1989; Hall and Smith, 1994). Large sub-arrays, such as 4×4, may be used with new sequential rotation configurations to reduce sidelobes (Hall and Smith, 1994). Finally, linear polarized elements with sequential excitation can be arranged in rings (Iwasaki et al., 1995). A variation on linear polarization elements utilizes gridded patches, where each patch consists of a set of narrow parallel strips, with the strip directions rotated around the 2×2 sub-array as before. Use of gridded patches reduces the mutual coupling between polarizations (Brachat and Baracco, 1995).

6.3.3 Electronic Scan in One Plane

Phase shift due to interelement line length and frequency, or produced by phasers, can provide scan in one plane. Most electronically scanned arrays are of this type, owing

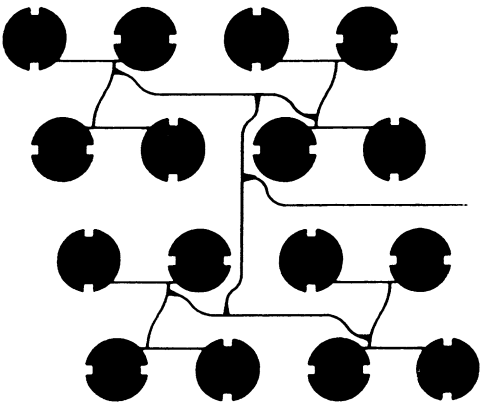


FIGURE 6.33 Sequentially rotated patch array.

to lower losses and cost. Scan in a second plane is often provided by mechanical rotation. The phasers may be in series with the feed line, or in parallel at the elements; see Section 6.3.4 for details. Figure 6.34 shows part of a planar array of L-band patches; horizontal sub-arrays of 18 elements are fed as a resonant array, driven by a phaser module. Phasers provide elevation scanning, with the modules fed by a

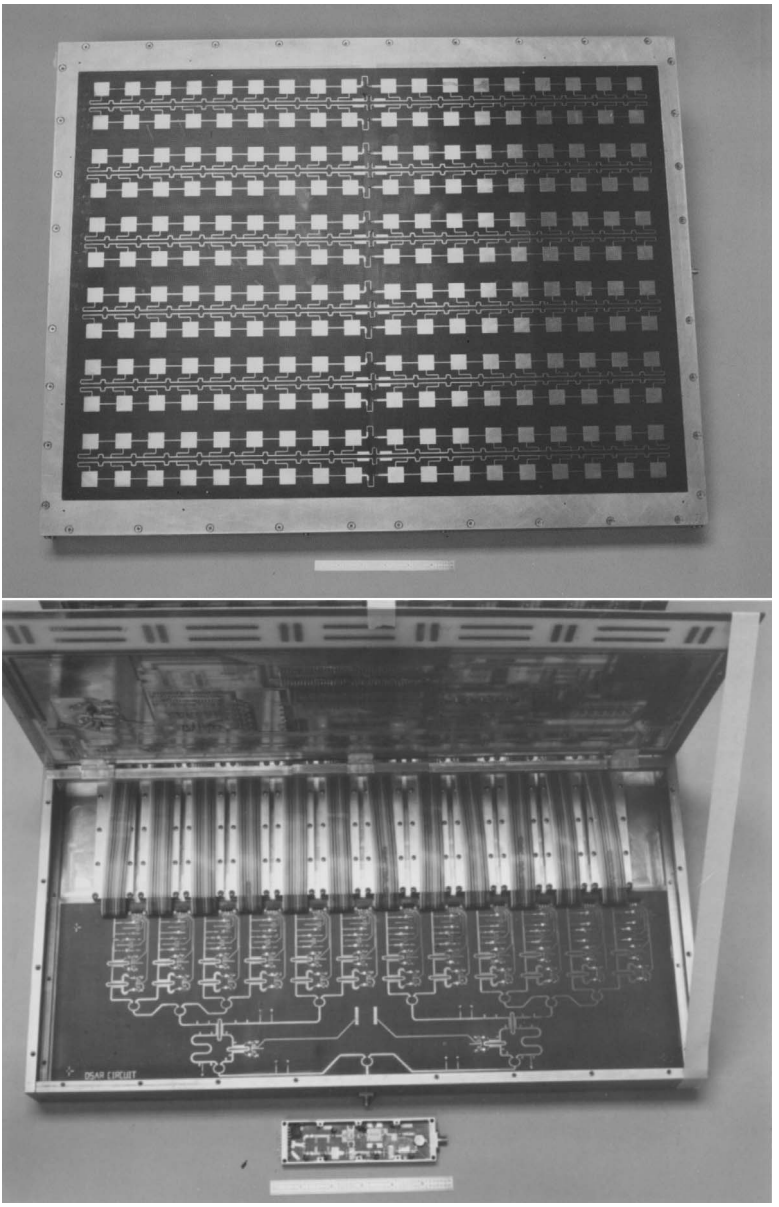


FIGURE 6.34 The SIR-C sub-array module. (Courtesy Ball Aerospace and Communications.)

corporate feed (Elachi, 1988). Figure 6.25 is a photograph of an airport surveillance radar utilizing frequency scan (see Section 6.3.4) in elevation and rotation in azimuth. Edge slots are used.

6.3.4 Electronic Scan in Two Planes

Most 2D arrays have radiating elements arranged on regular rows and columns, and the feeds tend to follow the same arrangement. In the case of fixed-beam planar waveguide slot arrays, the elements are integral with the feed. As discussed in Sections 6.1 and 6.2, the main and branch (secondary) feed components may be resonant, traveling wave, or corporate. Here a new criterion is used to demarcate feed topologies: the phaser locations in the feed network. Branch line topologies will be examined first.

Phasers can be located in series or in parallel with respect to the branch lines, as shown in Figure 6.26. The branch feed can be resonant, traveling wave, or corporate. Of course, with a TW feed the phaser settings must accommodate the varying phase along a TW feed. The main feed, which feeds all the branch feeds, can also be series or parallel. Thus for 3D arrays many feed combinations are possible; however, some lend themselves to fabrication better than others.

Figure 6.35 shows a series–series feed arrangment, with both branch line and main line feeds of TW type. An advantage is that all branch phasers have the same setting $\theta = kd \sin \theta_0$, and all main phasers have the same setting $\Phi = kd \sin \phi_0$. Both phases are corrected for the TW feed line phase shifts. With all 2D scan configurations it is possible to simplify the feed to 1D scan. For example, the branch lines can be fixed beam, and the main line can be a TW frequency scan feed. Series branch lines can be resonant; these are often used when there is no branch line scan.

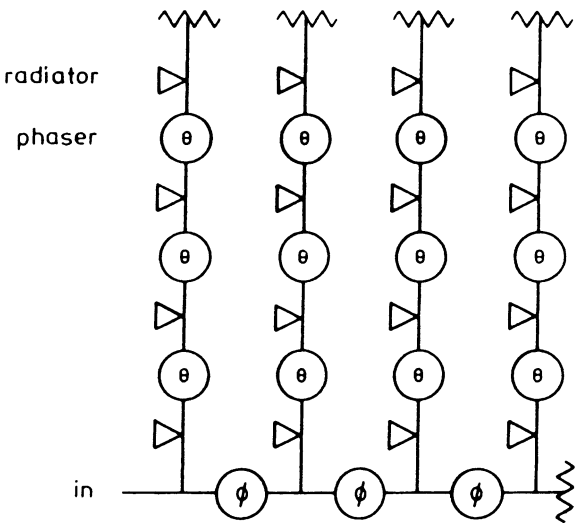


FIGURE 6.35 Series–series TW feed.

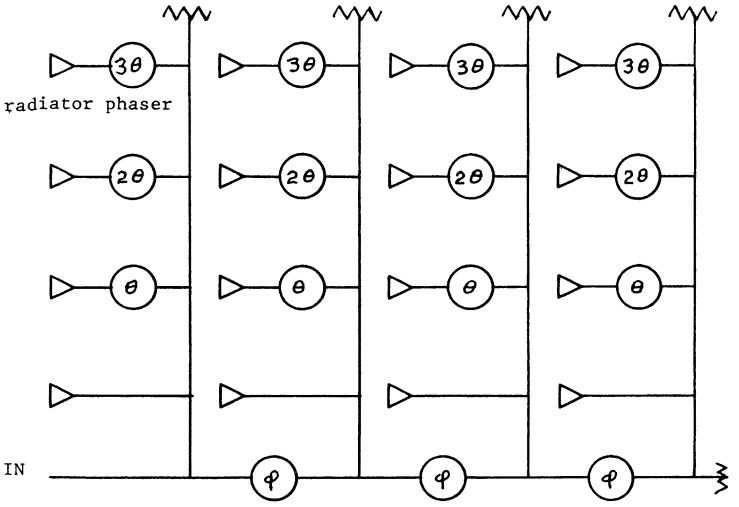


FIGURE 6.36 Parallel–series TW feed.

A parallel–series feed has a set of branch feed lines with parallel phasers, and a series phaser main feed line (see Fig. 6.36). Now the phase progression along the phasers of each branch feed are multiples of Φ. However, each parallel phaser handles equal power, and each element sees equal loss (for the branch feed). Again the branch lines and main line can be resonant or traveling wave.

A corporate–series feed has a corporate feed for each row of elements, with the corporate feeds fed from a series main line. Figure 6.37 sketches this topology. Again the main feed line can be of several types. Again the phaser values in a corporate feed are multiples; power is divided among the phasers.

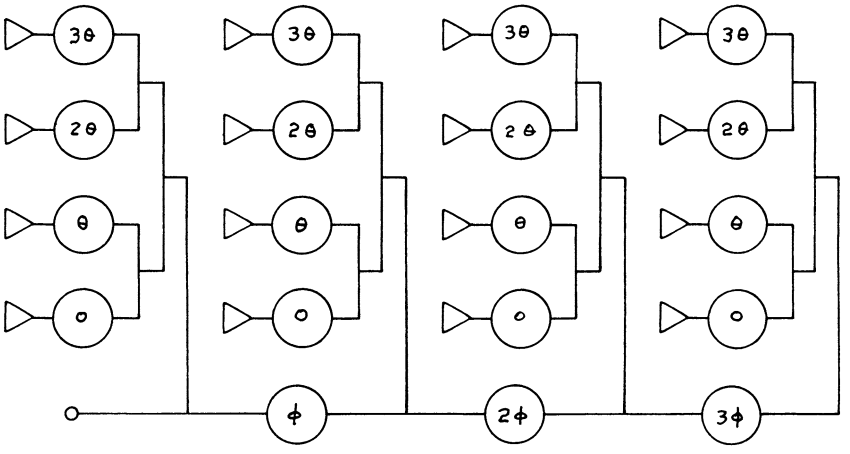


FIGURE 6.37 Corporate–series feed.

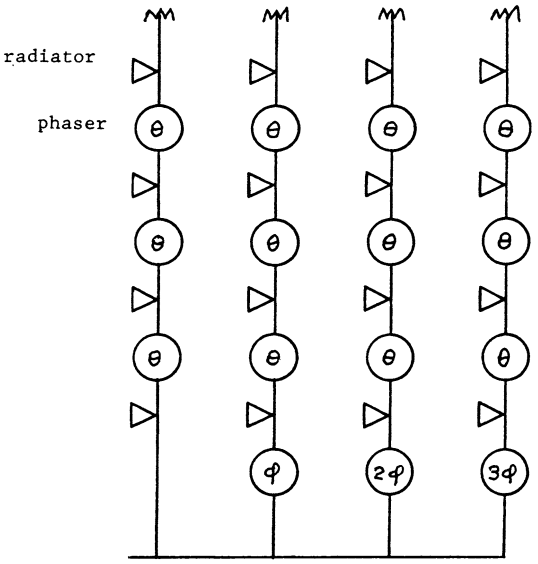


FIGURE 6.38 Series–parallel feed.

All of these branch feed schemes can also be used with parallel or corporate main feeds. Figure 6.38 shows a series–parallel topology, with series phaser TW branch lines, and a resonant main feed with parallel main phasers. Main phasers again progress in phase, and all main phasers conduct the same power (for uniform array excitation). A variation on this feed topology is the 20-GHz array shown in Figure 6.39

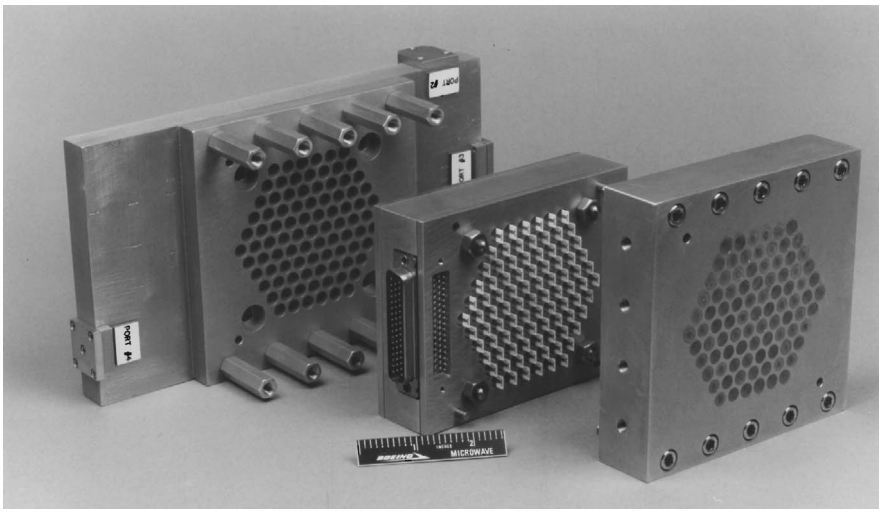


FIGURE 6.39 Hexagonal lattice array with modules. (Courtesy Boeing Defense and Space Group.)

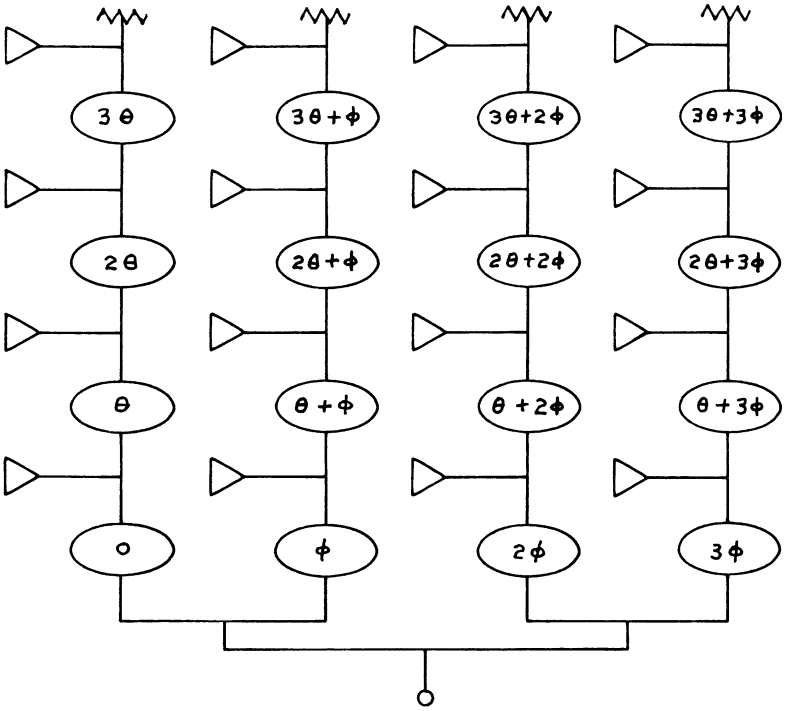


FIGURE 6.40 Series–corporate feed.

(Fitzsimmons et al., 1994). The rear block contains traveling wave waveguides with edge couplers to the circular guides. The center block contains phaser modules and dipole radiators; the right block is dielectric loaded guides (below air cutoff) and polarizers.

A series–corporate feed uses a corporate feed to excite the branch lines, as shown in Figure 6.40. The principal advantage of the corporate feed is intrinsically wider

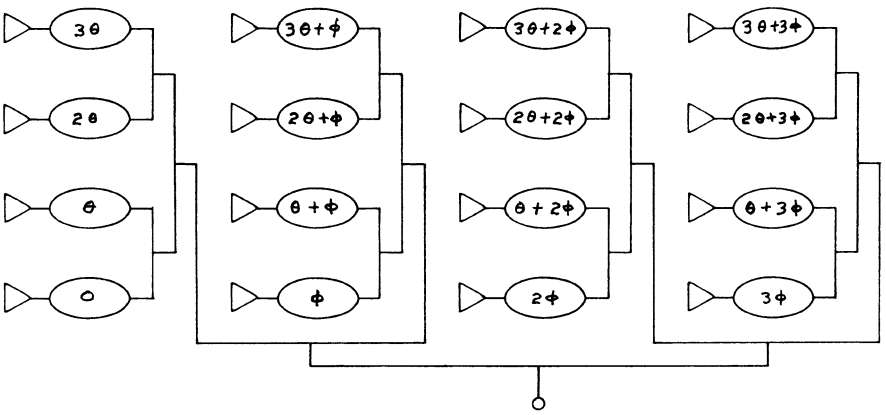


FIGURE 6.41 Full corporate feed.

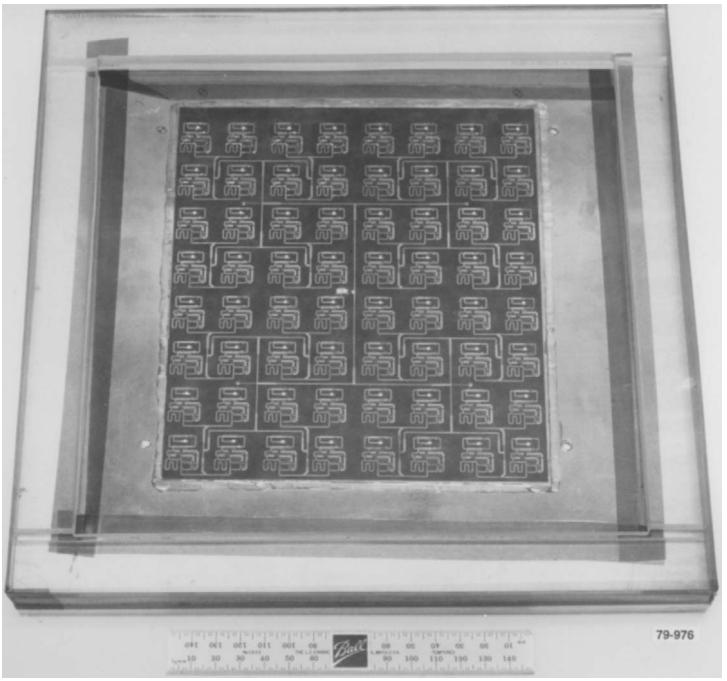


FIGURE 6.42 Corporate fed patch array. (Courtesy Ball Aerospace and Communications.)

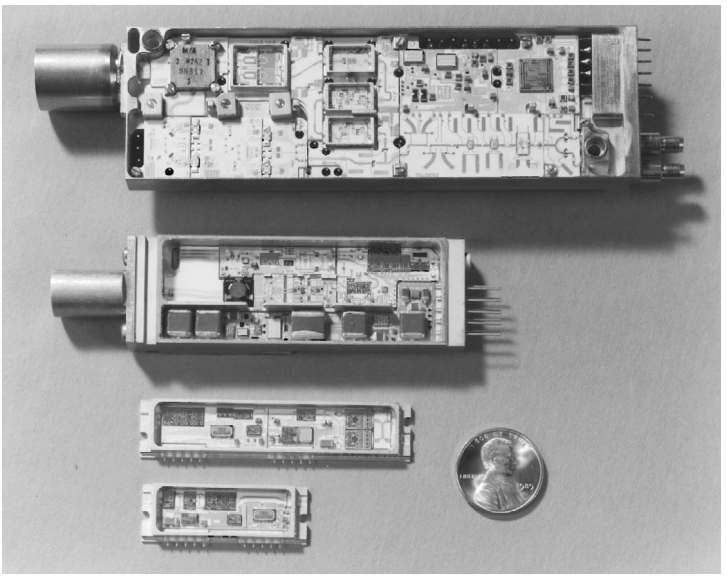


FIGURE 6.43 Four generations of modules. (Courtesy Texas Instruments.)

bandwidth; there is no interbranch or interelement phase that varies with frequency. All corporate feed lines are of equal length. To utilize this wideband potential, the phasers would be replaced by time delayers to maintain proper path lengths at all frequencies of concern.

Finally, a full corporate feed encompasses both branch and main feeds (see Fig. 6.41). Now each phaser requires a different phase value, but the power is

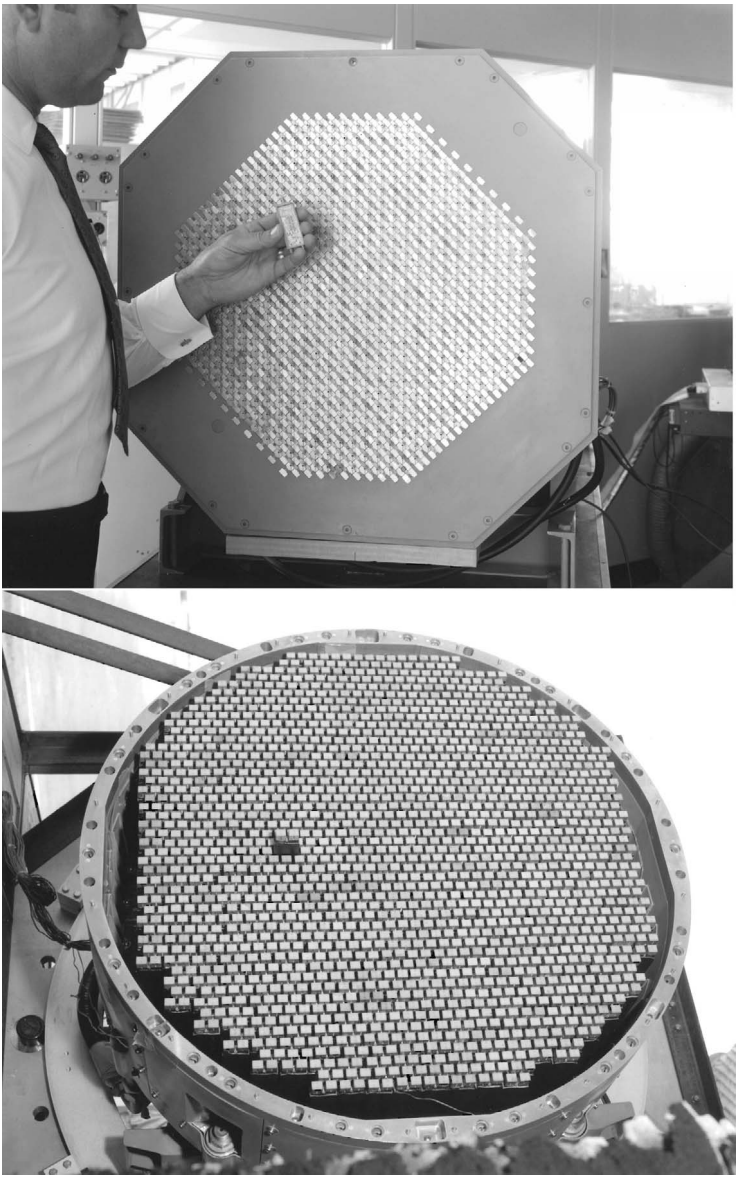


FIGURE 6.44 Array with TR modules. (Courtesy Texas Instruments.)

evenly divided. This type of feed is often used with microstrip patch arrays (see Fig. 6.42). As mentioned in the previous section, the improvement and size reduction of TR modules is continuing; Figure 6.43 shows four generations of X-band modules that feed circular waveguide and TEM horn radiators. Arrays using TR modules are shown in Figure 6.44; the top picture shows the MERA array, ~1965, while the bottom picture shows the RASSR array, ~1974.

6.4 PHOTONIC FEED SYSTEMS

The potential advantages of optical hardware for the care and feeding of a phased array are twofold: to provide time delay without bulky, heavy, and lossy transmission lines; and to reduce weight, volume, and cost. This rapidly developing field is briefly examined here. Two books on the subject are worthwhile: those by Zmuda and Toughlian (1994), and Kumar (1996). Also, ARPA has sponsored an annual series of Symposia on Photonic Systems for Antenna Applications, available to U.S. Department of Defense contractors. Proceedings of these are listed below:

PSAA-1	Dec. 1990	AD-B174 896L & 850L
2	Dec. 1991	AD-B174 895L
3	Jan. 1993	AD-B177 110L
4	Jan. 1994	AD-B193 121L
5	Jan. 1995	AD-B205 913
6	Jan. 1996	AD-B218 390
7	Jan. 1997	AD-B223 415L
8	Jan. 1998	AD-B233 444
9	Jan. 1999	AD-B242 534
10	Jan. 2000	AD-B255 657

First, delay line options are explored, including switched delay, and dispersive delay. Then less conventional approaches are considered.

6.4.1 Fiber Optic Delay Feeds

6.4.1.1 Binary Delay Lines. Photonic delay lines are used much as microwave delay lines are, with switches to select various delay segments. Of course, optical modulators and detectors are needed; these are discussed below. In the binary delay chain, semiconductor optical switches are used to switch in or out delay segments that are binary multiples of a minimum delay, say 1λ. An n-bit delay chain then provides 1λ, 2λ, $4\lambda, \ldots, 2^{n-1}\lambda$ delays. If phasers are not used for delays $<1\lambda$, the photonic delay chain can include these also. Figure 6.45 shows a block diagram of a delay chain; these are sometimes called BIFODEL (binary fiber optic delay) (Goutzoulis and Davies, 1990; Kumar, 1996). An efficient use of photonic delay is for sub-arrays only, where each antenna element is connected to a conventional phaser, with largest phase bit of 180°; each sub-array is then connected to a photonic delay chain (Ng et al., 1991; Lee et al., 1995).

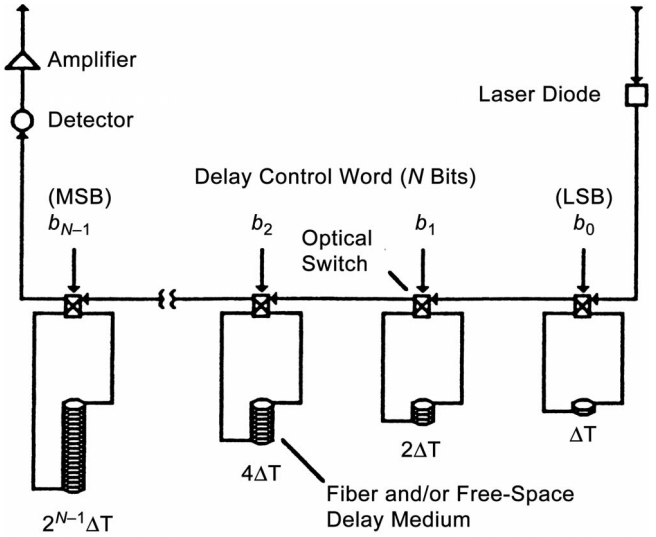


FIGURE 6.45 Switched binary fiber delay. (Courtesy Goutzoulis, A. P. and Davies, D. K., "Hardware-Compressive 2-D Fiber Optic Delay Line Architecture for Time Steering of Phased-Array Antennas", *Appl. Opt.*, Vol. 29, Dec. 20, 1990, pp. 5353–5359.)

Problems encountered in the utilization of photonic delay are connected with uniformity, stability and drift, and IP3 (see below). Unfortunately, a 1-dB optical nonuniformity produces a 2-dB microwave error. The IP3 is the third-order intercept point, and it applies to the amplifiers used to offset link loss. The amplifier noise floor and the IP3 determine the spur-free dynamic range, an important parameter for many applications. If an array uses 5-bit phasers, the smallest phase bit is 11.25°, or $\lambda/32$. At the other extreme, the longest time delay needed to scan an array of length L to θ_0 is $L \sin \theta_0$. For example, a 100λ array scanned to 60° needs a longest delay of 87λ. With binary steps there would be 7 bits of time delay. Maintaining delays of 64λ, 32λ, and 16λ, for example, to a small part of the least phaser bit $\lambda/32$, is a difficult and challenging task. If the allowable error is 2°, the time delay must be maintained to better than 1 part in 10,000. Photonic links are lossy, and the loss must be controlled and compensated so that the array amplitude taper is not unacceptably distorted.

Another type of delay chain uses two sets of switchable segments in series. This is called square-root delay (Soref, 1984; Goutzoulis and Davies, 1990). It provides a set of delays, with each being larger by a fixed delay: delay $= N\tau$. The number of delay lines is $2\sqrt{N}$, while the number of switches is $2N - 1$. This sequence of equal step delays would be suitable to excite an array at a fixed scan angle, but the square-root delay produces only one delay per switch setting. All the switch permutations would be used to produce the set of delays for the N array elements. Since there appears to be no way to latch (store) the delays, this type of delay chain is not useful for arrays.

The third-order intercept is obtained by sending two passband frequencies f_1 and f_2 through an amplifier. The third-order product is $2f_2 - f_1$, and IP3 is the power level

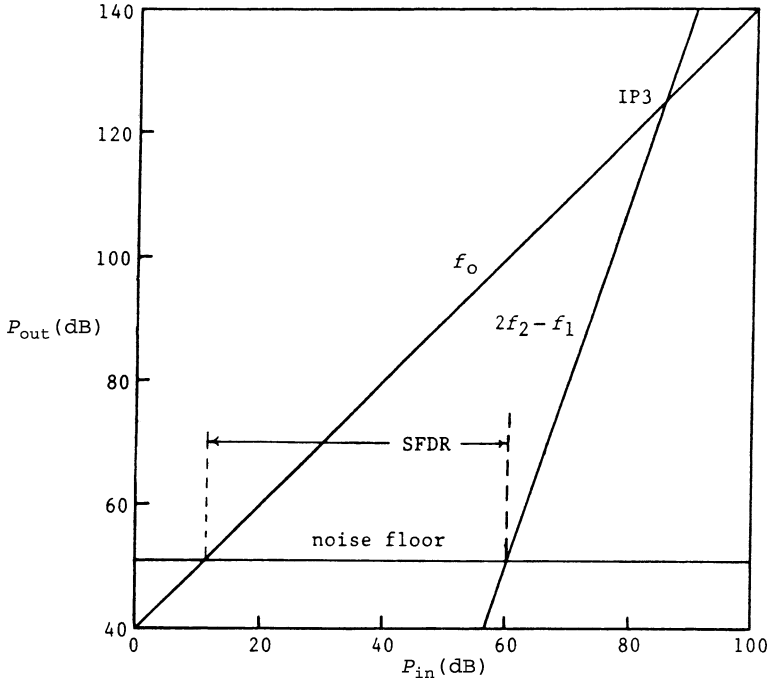


FIGURE 6.46 The IP3 and spur-free dynamic range.

where the linear (extrapolated) gain curve intersects the third-order product gain curve. The spur-free dynamic range is along the noise floor between the f_0 and $2f_2$ lines (see Fig. 6.46). IP3 of a cascade of amplifiers is simply calculated:

$$\frac{1}{\text{IP3}} = \frac{1}{\text{IP3}_1} + \frac{G_1}{\text{IP3}_2} \tag{6.33}$$

where the two amplifiers have IP3_1 and IP3_2, with gains G_1 and G_2. This IP3 is referred to the input. Referred to the output, the IP3 is (Lee et al., 1995)

$$\frac{1}{\text{IP3}} = \frac{1}{\text{IP3}_2} + \frac{1}{G_2 \cdot \text{IP3}_1} \tag{6.34}$$

Units of IP3 are $\text{dBm}/\text{Hz}^{2/3}$.

6.4.1.2 Acousto-Optical Switched Delay. An alternative to the use of switched series delays is an N-way "switch" that selects one of N delays. The optical switch is an acousto-optical Bragg cell, in which an applied microwave signal launches a traveling acoustic wave in the acousto-optic crystal. An incident optical beam is modulated by the phonon–phonon interactions, and is deflected by an angle proportional to the

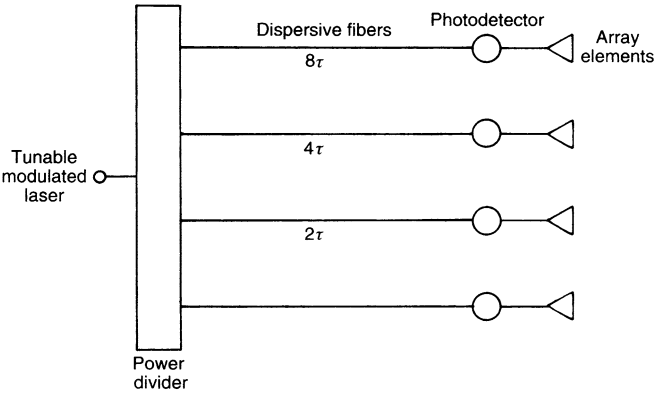


FIGURE 6.47 Beam scanning with dispersive delay.

microwave drive frequency. The optical beam then impinges on one of a cluster of optical fibers, each of which has a binary delay value. Outputs of all the fibers are collected in an optical power combiner, which in turn feeds a photo detector (Jemison and Herczfeld, 1993; Jemison et al., 1996). Figure 6.47 sketches such a configuration.

6.4.1.3 Modulators and Photodetectors. Laser light modulators are either direct or external. In direct modulation, the RF signal is impressed upon the DC current source for the laser. The resulting optical intensity is proportional to the bias current. Distributed feedback (DFB) semiconductor diode lasers are usually used for direct modulation as their insertion loss can be low, with good dynamic range, and low noise. Wavelengths are usually 1310 or 1515 nm. The laser module may contain, in addition to the DFB laser, a thermistor and thermoelectric cooler for monitoring and controlling the laser temperature, a photodiode for monitoring power, an optical isolator to reduce reflected light from the fiber, and impedance matching components. DFB laser bandwidth appears to be limited to 20–30 GHz; however, quantum well lasers may offer much wider bandwidths.

External modulators need a higher power (for good linearity) and low-noise light source; most use a diode-pumped solid-state NdYAG device. The laser output is coupled to a fiber, with the external modulator placed on the fiber. Modulators can be electro-optic or acousto-optic. Most common is the Mach–Zehnder (M–Z) electro-optic device fabricated in $LiNbO_3$ or in GaAs, or possibly in an electro-optical polymer. The M–Z is an interferometer in which the fiber is split into two paths, with the electro-optic material phase modulating one path due to RF bias; the recombined paths contain the RF-modulated optical carrier.

Acousto-optic modulators (Bragg cell) utilize an RF acoustic traveling wave along the crystal, which may be indium phosphide. An optical beam receives a single-sideband Doppler shift due to phonon–phonon interactions. Thus the optical carrier is modulated by the RF signal.

The PIN photodiodes are primarily used as broadside-coupled (instead of edge-coupled) detectors for fiber optics, and there is a trade between quantum efficiency

and modulation frequency; higher frequencies reduce efficiency. The frequency response is limited by the transit time of carriers out of the depletion region, and by the time constant of the junction region capacitance plus stray capacitance. Thinner depletion layers for higher frequencies absorb less light and hence are less efficient. At 10 GHz, efficiencies may be high, but at 60 GHz the efficiency may be as low as 30% (Wentworth et al., 1992). For such high microwave frequencies, waveguide structures with orthogonal carrier transport and absorption obviate the need for thin depletion layer. The photodiode collection aperture must match the fiber spot size to maximize collection efficiency. Antireflection coatings can be used to reduce reflected light. Finally, the photodiode circuit must be impedance matched to the appropriate system characteristic impedance. See Tamir et al. (1995) for papers on components.

6.4.2 Wavelength Division Fiber Delay

Delay can be selected by choosing the proper light wavelength, through use of dispersive fibers, or through use of frequency selective gratings.

6.4.2.1 Dispersive Fiber Delay. The RF modulated optical beam is sent through a length of highly dispersive fiber; the incremental delay is proportional to the wavelength shift of the laser carrier (Soref, 1992). A representative chromatic dispersion is 50 ps/mm of wavelength shift per kilometer of fiber. This dispersion is approximately constant over the laser tuning range. A variable delay can be obtained simply by tuning the laser wavelength, with the delay produced by a length of dispersive fiber. A simple (but expensive) array feed would have a laser oscillator for each array element, with each laser output going through a dispersive fiber then to a photodetector to the element. Dispersive fiber lengths are of the order of 1–10 km. As

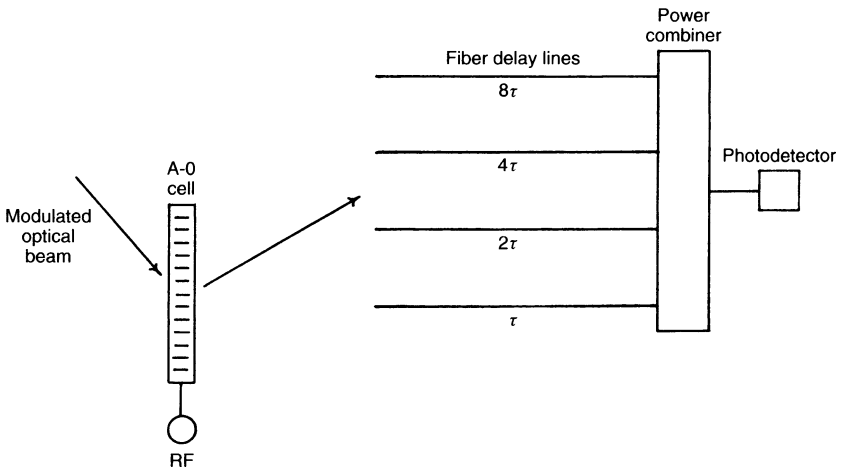


FIGURE 6.48 Acousto-optic (A-O) cell delay switching.

mentioned elsewhere, the delay must be stable and accurate to a small part of the smallest phaser bit in the array.

A more elegant architecture utilizes one tunable laser (per plane of scan) with the modulated output fanned out to a cluster of dispersive fibers. These would be designed for delays of τ, 2τ, 4τ, $(2N-1)\tau$, and each would drive one array element. Tuning the laser would change τ and thus scan the beam (see Fig. 6.48). For large arrays the fan-out loss can be appreciable. For a 2D scan, two tunable lasers would be used, and the azimuth and elevation delays would be combined at the element photodetectors (Johns et al., 1993; Frankel et al., 1996).

6.4.2.2 Bragg Fiber Grating Delay. A grating can be made in a length of fiber by focusing ultraviolet energy onto a periodic sequence of spots along the fiber, thereby producing a grating of different index spots of refraction. Such a grating will reflect an optical wave at the resonance frequency of the grating. A series of Bragg gratings can be spaced along a fiber, with each reflecting at a different optical wavelength. An optical signal enters the fiber with gratings through an optical circulator; the reflected signal path length (delay) depends on which grating is selected by the optical frequency. The circulator output then goes to an element, through a photodiode.

A more elaborate arrangement uses a fiber grating prism, where the prism consists of a set of Bragg grating lines, one fiber line per array element. Each line contains as many gratings as beam scan positions, so that each laser wavelength will give delays appropriate for that scan angle. To obtain the proper delays across the array, each fiber line has its gratings spaced farther apart than those of the previous line. Each line needs an optical circulator, and the tunable laser feeds into an optical divider. Again, fan-out loss must be considered (Soref, 1996; Tong and Wu, 1996; Zmuda et al., 1997; Chang et al., 1997) (see Fig. 6.49).

6.4.2.3 Traveling Wave Fiber Delay. Time delay can be achieved over a wide microwave frequency band through use of a traveling wave fiber that contains a

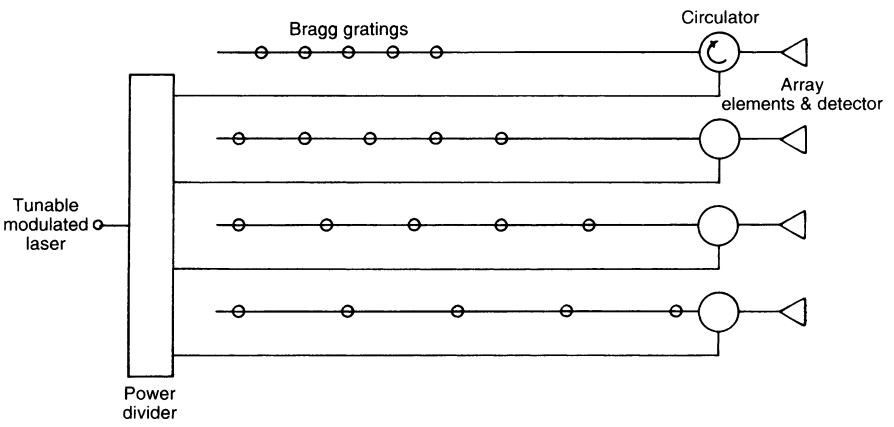


FIGURE 6.49 Bragg grating delay prism.

bidirectional coupler for each array element (Lee et al., 1997). Into one end of the TW fiber is fed the optical carrier modulated by f_1, while one modulated by f_2 is fed into the other end. At each coupler the outputs are mixed, with $f_2 - f_1$ providing the desired microwave signal. The beam scan is related to $f_2 + f_1$ so that, by changing f_2 and f_1, the signal can have the proper frequency and delay.

6.4.3 Optical Delay

Time delay can of course be produced by paths in media other than fibers, such as optical resonators. However, the relatively short path lengths, even with many bounces, are more suitable for phase rather than time delay. A spatial light modulator (SLM), which might be a Bragg cell or a liquid crystal used in the birefringent mode, can be used to switch the optical beam out of the path toward a prism. The prism reflects the beam back to another series SLM, thus incurring a delay of twice the distance to the prism (Dolfi et al., 1992, 1994). Here again the path lengths are of necessity relatively short.

6.4.4 Optical Fourier Transform

Since the array amplitude and phase distribution is just the discrete Fourier transform (FT) of the desired pattern, the element excitations can be found directly by sampling the optical FT. For a fixed-beam array, the modulated laser light beam illuminates a nonreflecting diaphragm containing a pinhole; the pinhole represents a narrow antenna beam. The diaphragm and pinhole can be replaced by a partially transparent screen with transmission that represents, for example, a Taylor main beam at the center and the Taylor sidelobe envelope around the main beam. The pinhole or screen illuminates an FT lens, with a 2D microlens–fiber array in the output plane. Each fiber then drives an array element.

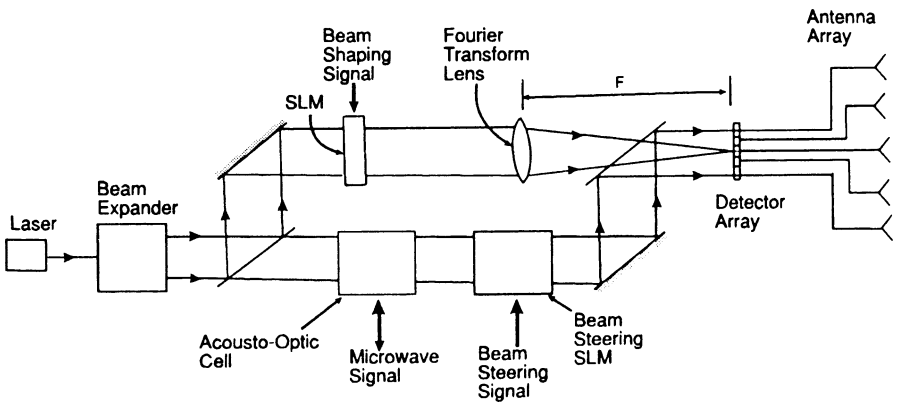


FIGURE 6.50 Fourier transform beam scanning. (Courtesy Zmuda, H. and Toughlian, E. N., Eds., *Photonic Aspects of Modern Radar*, Artech House, 1994.)

To obtain a scanning beam, a second laser source is introduced with a convenient IF difference frequency. The second source beam is deflected, either by an electro-optic spatial light modulator (Koepf, 1984) or by an acousto-optic cell SLM (Zmuda and Toughlian, 1994), and formed into a plane wave, which is then combined with the output of the FT lens (see Fig. 6.50). The beam deflection produces beam scan. However, the optical structure must be large to provide the time delay paths needed for a large array.

6.5 SYSTEMATIC ERRORS

A narrow band, fixed-beam array feed may be designed to be well matched. However, when the array is scanned in one or both planes, the mutual impedance contributions change owing to the changing relative phase between elements. These impedance changes will produce reflections internal to the feed, and may produce second-order beams caused by these systematic errors.

6.5.1 Parallel Phasers

When parallel phasers are located in the branch or element feed lines, the scan impedance changes will produce a small reflection at each element. This reflected energy passes through the phasers, and at the feed line is rereflected, owing to the light coupling inherent in an array of many elements. The rereflected energy now traverses the phaser again, as sketched in Figure 6.51, and is radiated as a second-order beam, with amplitude reduced by the scan impedance reflection coefficient Γ. Since there are two extra phaser trips, this beam will appear at an angle corresponding to an interelement phase three times that of the main beam. Not all this re-reflected energy is radiated; some is reflected back toward the phaser. Thus a set of second-order beams is produced, with phase delays of 3ϕ, 5ϕ, 7ϕ, and so on, and amplitudes of $|\Gamma|$, $|\Gamma^2|$, $|\Gamma^3|$, and so on. Beam positions are given by

$$\sin \theta_m = m \sin \theta_0 \pm \frac{p}{d/\lambda} \tag{6.35}$$

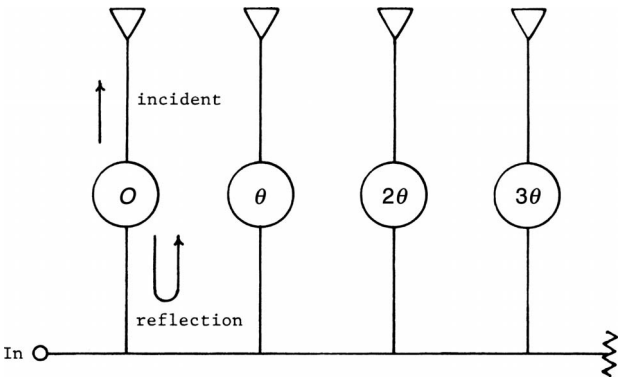


FIGURE 6.51 Second-order beam generation.

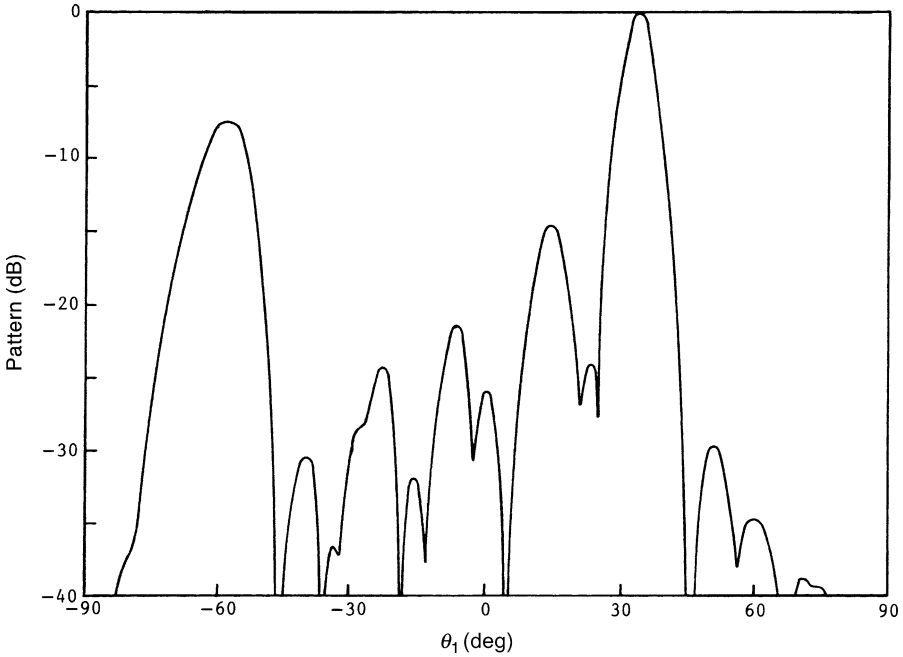


FIGURE 6.52 Second-order beams: parallel phasers.

TABLE 6.1 Systematic Error Second-Order Beams

Beam Angle (deg)	Beam Level (dB)	m	p
33.75	0	0	0
−60.81	−7.7	1	−1
−13.78	−15.4	3	−1
−4.55	−23.1	5	−2

Here the main beam is at angle θ_0, and the last term is a modulo 2π contribution. A typical example, measured on a 16-element array, but with all impedance terminations alike, is shown in Figure 6.52. The main beam is at 33.75°, with second-order beams at −60.80, −13.75, −4.55°. With a reflection coefficient of 0.17, these beams are down by $|\Gamma^m|$; the element spacing is 0.7λ. Table 6.1 gives the calculated values, which compare well with the measurement results of Spradley and Odlum (1956).

6.5.2 Series Phasers

When series phasers are used in the feed line, systematic errors can exist, but they tend to be smaller than those just discussed. Again multiple reflections occur, but now those in the feed line are of interest. Owing to multiple reflections in the feed line, the input

impedance and branch port voltages are given by an $A-B-C-D$ matrix cascade. When all branch impedances are equal, and the main feed is matched at each junction, the impedance can be written as a double sum involving *scan impedance*, characteristic impedances, intercoupler phase shifts, and so on (Kummer, 1966). The result is that the systematic errors produce a second-order beam of amplitude Γ/N, where there are N elements. In general, these systematic errors are not critical.

6.5.3 Systematic Error Compensation

Systematic errors may be reduced by special array designs that minimize *scan impedance* changes; these are discussed in Chapter 7. Compensation in the feed network is also possible; this allows the element face to be unchanged. Coupling circuits can be designed into the feed network to compensate for the systematic phase errors. These would connect between adjacent branch line feeds (Hannan et al., 1965). A simpler technique uses a coupling hole between adjacent waveguide feed lines (Cook and Pecina, 1963). Both of these techniques are relatively narrowband, as it is difficult to compensate for *scan impedance* and frequency changes together.

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CHAPTER SEVEN

Mutual Coupling

7.1 INTRODUCTION

It will be shown below that mutual coupling is responsible for all the unique characteristics of phased arrays. The task, then, is to understand it, and to make advantageous use of it.

This chapter is concerned with infinite arrays for two reasons. First, the essential characteristics of all scanning arrays exist in infinite arrays, and are most easily calculated there. Second, most array design starts with an infinite array, with finite array (edge) effects included near the end of the design process. These edge effects are the subject of Chapter 8.

Mutual coupling fundamentals are discussed first. There follow discussions of the basic analysis methods: spatial domain (element-by-element), spectral domain (periodic cell), scattering matrix; all may include moment method techniques. Finally, methods for compensation of *scan impedance* are covered.

7.2 FUNDAMENTALS OF SCANNING ARRAYS

7.2.1 Current Sheet Model

The simplest concept of a phased array is an infinite flat current sheet carrying a uniform current flow parallel to one of the coordinate axes, and is due to Wheeler (1948, 1965). This current is phased to radiate at an angle away from broadside. Such a current sheet can be used as a *gedanken* to derive key scanning properties of phased arrays. The sheet may be either receiving or transmitting, with the two situations giving related behavior. First, consider an incident plane wave (receiving) case. Figure 7.1 shows the current sheet in side view, both for a wave incident in the

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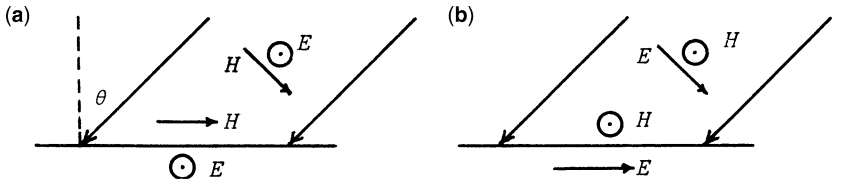


FIGURE 7.1 Electric current sheet: (a) *H*-plane; (b) *E*-plane.

E- and *H*-plane. If the current sheet is matched for normal incidence ($\theta = 0$) to $\eta\ \Omega$/square, then at other scan angles there is a mismatch. For *H*-plane incidence the incoming wave "sees" a section of current sheet which is wider than the section of wavefront; thus the apparent resistance is lower by $\cos\theta$. The reflection coefficient is now

$$\Gamma = \frac{\cos\theta - 1}{\cos\theta + 1} = -\tan^2\frac{\theta}{2} \tag{7.1}$$

The *E*-plane incidence gives the opposite result; the incoming wave sees a section of current sheet, which is longer than the section of wavefront, and thus the apparent resistance is higher by $1/\cos\theta$. The reflection coefficient is

$$\Gamma = \frac{\sec\theta - 1}{\sec\theta + 1} = \tan^2\frac{\theta}{2} \tag{7.2}$$

When the current sheet transmits instead of receives, the reflection coefficient signs will be reversed. Figure 7.2 shows the variation of current-sheet resistance for

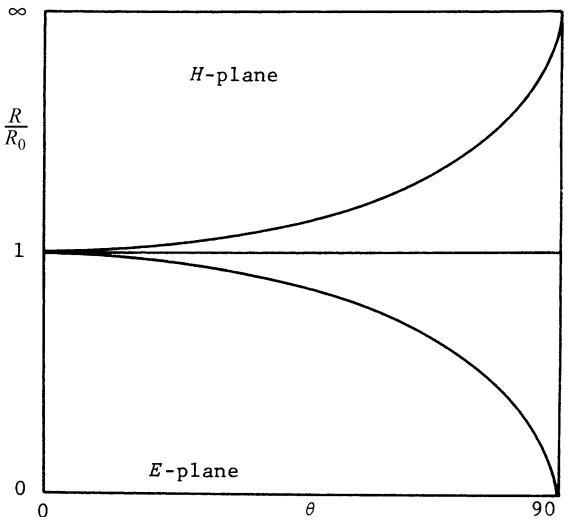


FIGURE 7.2 Current sheet scan resistance. (Courtesy Wheeler, H. A., "Simple Relations Derived from a Phased-Array Antenna Made of an Infinite Current Sheet", *Trans. IEEE*, Vol. AP-13, July 1965, pp. 506–514.)

E- and H-plane scans, for transmitting, with $R_0 = \eta$. Of course, this simple current sheet model gives no information about reactance. The electric current sheet requires an open-circuit reflector behind the sheet, where open circuit means $\varepsilon = 0$, $\eta = \infty$. Magnetic current sheets may also be considered, with E and H interchanged. Now the reflector behind the sheet is a short circuit $\sigma = \infty$. Thus the magnetic current sheet, unlike the electric current sheet, may be approximated physically. An array of short slots in a metal sheet, for example, provides such an approximation. This simple concept indicates the basic behavior of phased arrays: *scan resistance* increases in one scan plane, but decreases in the other. This trend will be observed in later sections where precise *scan impedances* or admittances are calculated.

Further insight can be obtained from another concept due to Wheeler (1965), namely, that of an ideal element pattern. Again, consider a transmitting current sheet, but let the current sheet be composed of short dipoles with patterns

$$\begin{aligned} &H\text{-plane} \qquad F(\theta) = 1 \\ &E\text{-plane} \qquad F(\theta) = \cos\theta \end{aligned} \tag{7.3}$$

The patterns are affected by the effective (projected) aperture, which varies as $\cos\theta$, where θ is measured from the normal to the array. Since the effective aperture broadens the beam, the radiation resistance is proportional to $\sec\theta$, so that the net effect is

$$\frac{R}{R_0} = \frac{F^2}{\cos\theta} \tag{7.4}$$

If the element power pattern were $\cos\theta$, the resistive part of the array impedance would be matched. Thus the "ideal" element pattern is $F(\theta) = \cos^{1/2}\theta$. This conical pattern is symmetric about the axis. The ideal element pattern has been approximated by a Huygens source, which is a crossed electric dipole and magnetic dipole, sometimes realized by a dipole and loop. However, the pattern of the Huygens source is

$$F(\theta) = \frac{1}{2}(1 + \cos\theta) = \cos^2\frac{\theta}{2} \tag{7.5}$$

and this is only a fair approximation to $\cos^{1/2}\theta$. In the section on scan compensation, it will appear that both electric and magnetic modes are needed. The Huygens source is then a crude approximation, giving a single electric and a single magnetic mode.

7.2.2 Free and Forced Excitations

Arrays may be analyzed by either of two viewpoints: free or forced excitation. In the forced excitation model, a constant driving voltage (current) is applied to each element, with the element phases adjusted to provide the desired scan angle. Each element has a *scan impedance* (admittance), which is the impedance of an element in an infinite array at the scan angle, and an associated *scan reflection coefficient.*

The array element currents (voltages) are the solution of an impedance (admittance) matrix equation

$$[V] = [Z][I] \tag{7.6}$$

This impedance matrix contains all the interelement mutual impedances Z_{ij}; the matrix is symmetric so $Z_{ij} = Z_{ji}$. The mutual impedances, in principle, are calculated between two elements, with all other elements open circuited. In practice, however, these impedances are almost always calculated with only the two elements present. Results are usually very good; take half-wave dipole elements that become quarter-wave wires when open circuited. These have very small scattering (coupling) cross sections. The overall array pattern is given by the sum over the array elements with the currents as coefficients, all multiplied by the isolated element pattern. From this the *scan element pattern* can be obtained by factoring out the array factor. This type of analysis is relatively easy to carry out. However, implementation of such an array is difficult as each element must be fed by a constant-voltage (current) source. Simple feed networks, in contrast, are of the constant-available-power type where an element impedance mismatch reduces the applied voltage.

Free excitation assumes that each area of the feed network is equivalent to a voltage (current) source in series with R_0. These are thus constant-available-power sources. Such sources, with constant incident power, are suited to a scattering analysis:

$$[V_r] = [S][V_i] \tag{7.7}$$

Here V_i and V_r are the incident and reflected voltage (current) vectors and S is the scattering matrix of coupling coefficients. The *scan reflection coefficient* is given by

$$\Gamma_s = \sum_p \sum_q S_{00,pq} \frac{A_{pq}}{A_{00}} \tag{7.8}$$

where S is the coupling coefficient between the "00" element and the "pq" element. The excitation coefficients differ by the progressive scan phase between the two elements, and may also differ in amplitude if a finite array with a tapered excitation is used. Although the scattering approach accurately represents most arrays, and is conceptually simple, there is no direct way of calculating the coupling coefficients. They may be obtained from the impedance (admittance) matrix:

$$[S] = \frac{[Z] - Z_0}{[Z] + Z_0} \tag{7.9}$$

where the Z_0 applies only to diagonal terms. Measurement of the scattering matrix is performed with all elements present and terminated in matched loads. For arrays of moderate gain elements (e.g., horns in a reflector feed array), this measurement is generally satisfactory. Resonant size elements (dipoles) have coupling in the H-plane that

decreases as $1/r^2$, while coupling in the *E*-plane decreases as $1/r$ (Wheeler, 1959). In an array, however, the coupling decay becomes asymptotic to $1/r^2$ as shown by Hannan (1966), Galindo and Wu (1968), and Steyskal (1974). This finding is borne out by measurements on large arrays (Amitay et al., 1964; Debski and Hannan, 1965). Phase measurements show that the coupled energy has the phase velocity of free space provided there is no external loading.

In some arrays, there is a beam position (other than $\theta = 90^\circ$) where the *scan element pattern* has a zero, or in different terms where the *scan reflection coefficient* has a magnitude of unity. Such an angle is called a "blind spot". It will be shown later that the appearance of these can be precluded by proper choice of array parameters.

7.2.3 Scan Impedance and Scan Element Pattern

A most important and useful parameter is *scan impedance*;[1] it is the impedance of an element as a function of scan angles, with all elements excited by the proper amplitude and phase. From this, the *scan reflection coefficient* is immediately obtained. Array performance is then obtained by multiplying the isolated element power pattern (normalized to 0 dB max) times the isotropic array factor (power) times the impedance mismatch factor $(1 - |\Gamma|^2)$. The isolated element pattern is measured with all other elements open-circuited. This result is not quite the same as with all other elements absent, except for canonical minimum scattering antennas (see Section 7.2.3.1). Here it is assumed that the array is sufficiently large that edge effects are negligible and that *scan impedance* is that for an infinite array. This simple performance expression allows the contributions of array lattice and element spacing, element type, and mutual coupling to be discerned.

An equivalent array performance expression combines the isolated element pattern and the impedance mismatch factor into a new parameter, the *scan element pattern* (SEP) (formerly active element pattern). Now the overall array performance is the product of the isotropic array factor and the *scan element pattern*. The former shows the array beamwidth and sidelobe structure for the scan angle of interest; the latter, like the isolated element pattern, is slowly varying, and shows array gain versus scan angle. It is important to note that *scan element pattern* provides the radar or communications system designer array gain, at the peak of the scanned beam, versus scan angles. The *scan element pattern* can be calculated from the *scan impedance* and the isolated element pattern as follows: At the peak of the scanned beam the array gain is

$$G(\theta) = Ng_s(\theta) = \frac{4\pi|E(\theta)|^2}{\eta P_{\text{avail}}} \tag{7.10}$$

where g_s is the scan element (power) pattern, and P_{avail} is the available power. The electric field can be written in terms of the isolated element pattern, normalized by

[1]The obsolete term "active impedance" is confusing, and is deprecated.

the scan element current $I_s(\theta)$:

$$E(\theta) = \frac{NI_s(\theta)E_{\text{iso}}(\theta)}{I_{\text{iso}}} \tag{7.11}$$

The isotropic element gain is expressed as the ratio of field squared to power:

$$g_{\text{iso}}(\theta) = \frac{4\pi|E_{\text{iso}}(\theta)|^2}{\eta|I_{\text{iso}}|^2 R_{\text{iso}}} \tag{7.12}$$

Power available to the array is written in terms of scan current and impedances:

$$P_{\text{avail}} = \frac{NV^2}{4R_g} = \frac{N|I_s(\theta)|^2|Z_s(\theta) + Z_g|^2}{4R_g} \tag{7.13}$$

Combining these results yields the *scan element pattern* expressed in terms of isolated element gain and impedances (Allen et al., 1962):

$$g_s(\theta) = \frac{4R_g R_{\text{iso}} g_{\text{iso}}(\theta)}{|Z_s(\theta) + Z_g|^2} \tag{7.14}$$

When the *scan element pattern* is normalized to unity at $\theta = 0$, the SEP can be expressed in terms of only the isolated element power pattern and the *scan impedance*:

$$g_s(\theta) = \frac{4R_s(0)g_{\text{iso}}(\theta)}{|Z_s(\theta) + Z_g|^2 g_{\text{iso}}(0)} \tag{7.15}$$

For the general case where the generator reactance is not zero, it is appropriate to use the conjugate reflection coefficient, defined as

$$\Gamma_* = \frac{Z_s^*(\theta) - Z_g}{Z_s(\theta) + Z_g} \tag{7.16}$$

This gives

$$1 - |\Gamma_*|^2 = \frac{4R_g R_s(\theta)}{|Z_s(\theta) + Z_g|^2} \tag{7.17}$$

which allows the *scan element pattern* to be expressed in terms of the *scan reflection coefficient*, the isolated element pattern, and the *scan resistance*:

$$g_s(\theta) = \frac{R_{\text{iso}} g_{\text{iso}}(\theta)}{R_s(\theta)}\left(1 - |\Gamma_*|^2\right) \tag{7.18}$$

Now a generator (source) conjugate matched to the *scan resistance* at $\theta = 0$ allows the final form of SEP to be realized.

$$g_s(\theta) = \frac{R_s(0) g_{\text{iso}}(\theta)}{R_s(\theta) g_{\text{iso}}(0)} \left[1 - |\Gamma_*(\theta)|^2\right] \tag{7.19}$$

Note that only for the special case where the generator impedance is real the SEP can be written as:

$$g_s(\theta) = \frac{R_{\text{iso}} g_{\text{iso}}(\theta)}{R_g} |1 - \Gamma(\theta)|^2 \tag{7.20}$$

The previous form is preferred as it explicates the roles of *scan resistance*, isolated element pattern, and *scan reflection coefficient*.

An often quoted approximate result for SEP needs to be examined. This purports to show a $\cos\theta$ variation (Hannan, 1964):

$$g_s(\theta) \simeq \frac{4\pi A_{\text{elem}}}{\lambda^2} \cos\theta \left[1 - |\Gamma(\theta)|^2\right] \tag{7.21}$$

where A_{elem} is the unit cell area of the element. In Section 7.4, it is shown that the *scan element pattern* behavior is closer to $\cos^{1.5}\theta$, the extra power contributed by the impedance mismatch. So it is better to use the generally applicable formula for SEP (Eq. 7.19) to get accurate and useful results. Extensive *scan impedance* data for dipole arrays are given in Section 7.4.1.

Alternate equations appropriate for computer analysis are expressed in terms of currents:

$$E(\theta, \phi) = \frac{60 F(\theta, \phi)}{r} \sum^{N} I_n \tag{7.22}$$

Here the sum is over the elements of the array, and the pattern function F is,

$$F(\theta, \phi) = \frac{\cos kh - \cos(kh\cos\theta)}{\sin kh \sqrt{1 - \cos^2\theta \cos^2\phi}} \tag{7.23}$$

The dipoles are along the x-axis and in the x–y plane. Thus the array axis is for $\phi = 0$. Dipole half-length is h and $k = 2\pi/\lambda$.

Power into the array, without matching or source impedance, is simply,

$$P = \sum^{N} I_n I_n^* R_{sn} \tag{7.24}$$

where R_{sn} is the *scan resistance* of the nth element. The gain equation now takes the form

$$G(\theta,\ \phi) = \frac{120F^2(\theta,\ \phi)\sum^N I_n \sum^N I_n^*}{\sum^N I_n I_n^* R_{sn}} \tag{7.25}$$

Since the SEP is gain per element, one gets

$$\text{SEP} = \frac{120F^2(\theta,\ \phi)\sum^N I_n \sum^N I_n^*}{N\sum^N I_n I_n^* R_{sn}} \tag{7.26}$$

This can be written in terms of the isolated element gain g_{iso}, where

$$g_{\text{iso}}(\theta,\ \phi) = \frac{120F^2(\theta,\ \phi)}{R_{\text{iso}}} \tag{7.27}$$

Thus SEP becomes,

$$\text{SEP}(\theta,\ \phi) = \frac{R_{\text{iso}} g_{\text{iso}}(\theta,\ \phi)\sum^N I_n \sum^N I_n^*}{NR_s \sum^N I_n I_n^*} \tag{7.28}$$

This can be expressed as:

$$\text{SEP}(\theta,\ \phi) = \frac{120R_{\text{iso}}F^2(\theta,\ \phi)\sum^N I_n \sum^N I_n^*}{R_s(\theta,\ \phi)\sum^N I_n I_n^*} \times \left[1 - |\Gamma_*(\theta,\ \phi)|^2\right] \tag{7.29}$$

where R_s is the *scan resistance.*

The only approximation in Eq. (7.29) is that all element *scan resistances* are equal, which affects the power sum and the mismatch factor. For arrays that are not large, the mismatch factor can be calculated for each element, and then averaged. The effectiveness of this will be shown in Section 7.2.3.1.

7.2.3.1 Transmit versus Receive SEP. To determine exactly how the center element excited (transmit) SEP measurement process behaves with angle and frequency, a planar array code using thin dipoles on a square lattice was employed (Hansen, 2006). This code treats a finite square array, up to 100 on elements each axis, thereby replicating the measurement procedure. As the dipoles are thin, and the maximum length is half-wave, the current distribution is very closely simusoidal, thus Moment Methods solution is not necessary. All elements were terminated with a

resistance, and the center element was excited. Solution of the mutual impedance matrix equation gives the complex currents for all the array elements. Mutual impedances and matrix inversion were calculated in double precision. Far-field pattern was calculated by summing the element currents times the appropriate steering phases times the elements pattern. Power was calculated from Real (VI) for the driven element. Gain is simply $120E^2/P^*\ (1 - |\Gamma|^2)$, where Γ is the reflection coefficient. Two matching impedances were used: One for an infinite array of excited dipoles, and the other for only the center element excited. Calculations were performed on an HP 64 bit UNIX workstation.[2]

Results are given for a 50×50 element array (2500 unknowns) in Figure 7.3. The values of SEP (gain per element) are normalized to zero decibels at broadside as the absolute values are incorrect due to the passive mutual impedances. Absolute values are as discussed in later. The broadside array impedance of $63 + j23\ \Omega$ is used for matching. In comparison with the infinite array single element excited (called transmit), SEP values are roughly 2 dB higher in the range of θ from 40 to 80° for the *E*-plane. The *H*-plane results are slightly higher. The departure from absolute gain is much worse than would be indicated by the modest change in the center element impedance. The mutual coupling significantly reduces the current magnitudes, thus decreasing the gain per element. Significant oscillations in the *E*-plane SEP are due to edge effects, even for such a large array (Hansen, 1996). Note that in Figure 7.3 the *E*-plane SEP is higher than the *H*-plane, which is contrary to the infinite array Floquet results.

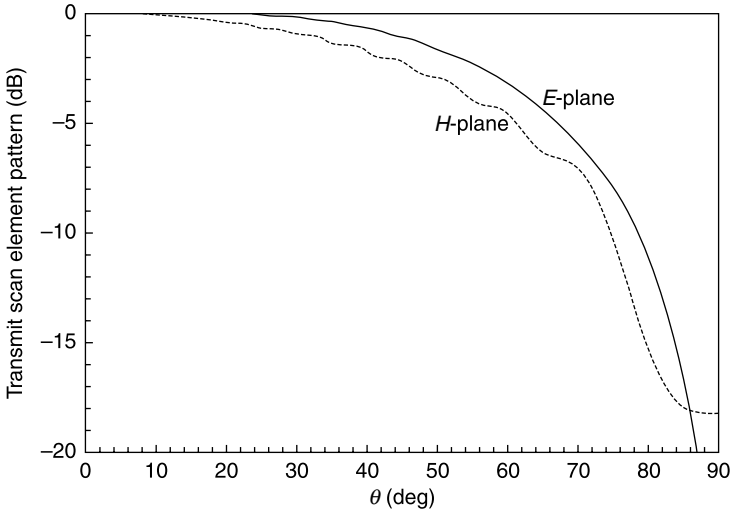


FIGURE 7.3 A 50×50 dipole array, $L = D_x = D_y = 0.05\lambda$.

[2]The CPU chips optimized for floating point operations tend to be several times as fast as PC chips optimized for integer handling, all for the same clock rate.

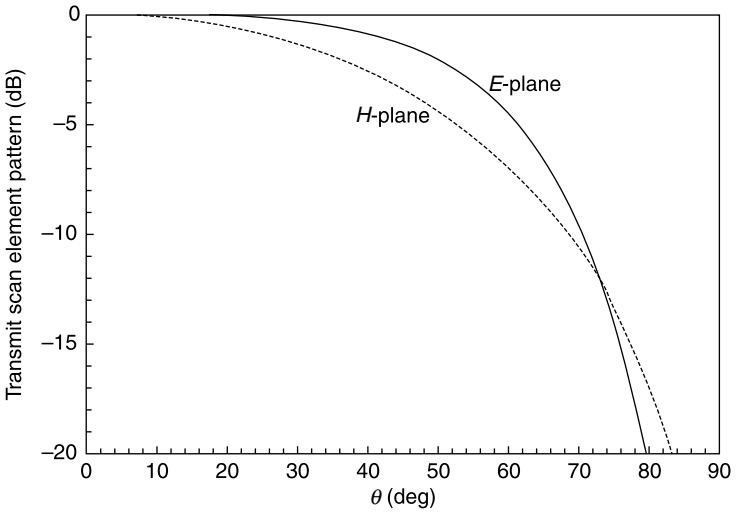


FIGURE 7.4 A 50×50 dipole/screen array, $L = D_x = D_y = 0.5\lambda$, $H = 0.25\lambda$.

With the ground screen added, at a spacing equal to one-half of the dipole spacing, the transmit SEP is as shown in Figure 7.4. The broadside array impedance of $70 + j58$ Ω is again used as match impedance. The *E*-plane result departs markedly from the infinite array result; the *H*-plane values are higher than the infinite array results to $\sim 60^\circ$, and lower for large angles. Note the large change in embedded impedance from the infinite array value of $153 + j32$. Not only is the absolute SEP at broadside incorrect, but the scan performance, normalized to 0 dB at broadside, is not a good predictor of array gain versus scan.

The same computer model was used to simulate the receive SEP measurement. All elements were excited by unit amplitude voltage and the proper scan phase. The SEP was calculated again from $120E^2/P \times (1 - |\Gamma|^2)$. Figure 7.5 shows SEP in an absolute value.

A matched array would have an SEP of $2\pi A/N\lambda^2$, which is $\pi/2 = 1.97$ dB for the half-wave case of Figure 7.5. The *E*-plane curve is a fair fit (but slightly lower) to infinite array results out to $\sim 70^\circ$, but there are edge effect oscillations. The *H*-plane curve is slightly higher for large angles. Calculations of 20×20 and 30×30 arrays (not shown) indicate that as the array size is larger, the match at large θ becomes better.

Figure 7.6 gives SEP with a ground screen, again spaced from the screen half of the dipole spacing. The *E*-plane SEP contains large oscillations, building up as θ increases. These oscillations occur as the *scan impedance* (SI) of the center element oscillates about the infinite array value. An average curve through the oscillations matches well the infinite array data. The *H*-plane data are roughly 1 dB high for angles $> 50^\circ$.

A better result comes from the average of the *scan reflection* coefficients of all elements. Figure 7.7 shows the SEP for the half-wave dipole array, using the average reflection coefficient. The oscillations in Figure 7.6 for the *E*-plane have been

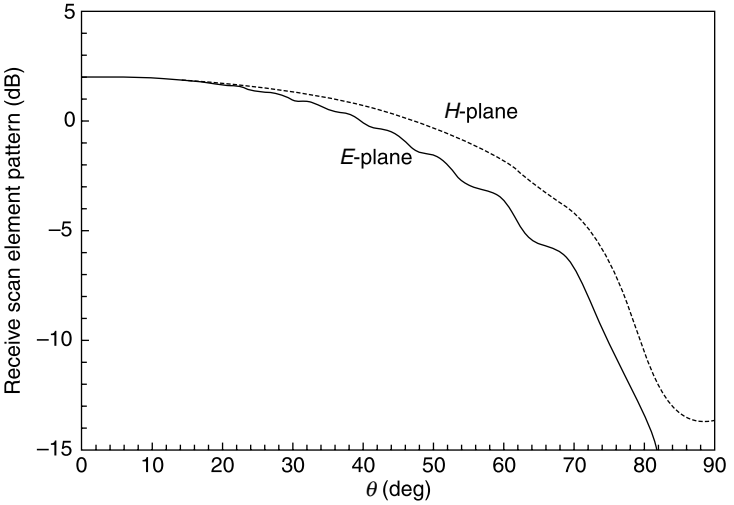


FIGURE 7.5 A 50×50 dipole array, $L = D_x = D_y = 0.5\lambda$.

smoothed out and the *H*-plane SEP is higher, as it should be. Figure 7.8 is for the dipole with a ground plane case with few oscillations. These two graphs compare well with the infinite array results of Section 7.4.1, but even for a 50 element wide array, there are some edge effects at larger scan angles. Thus Figures 7.7 and 7.8 are slightly different from the infinite array results.

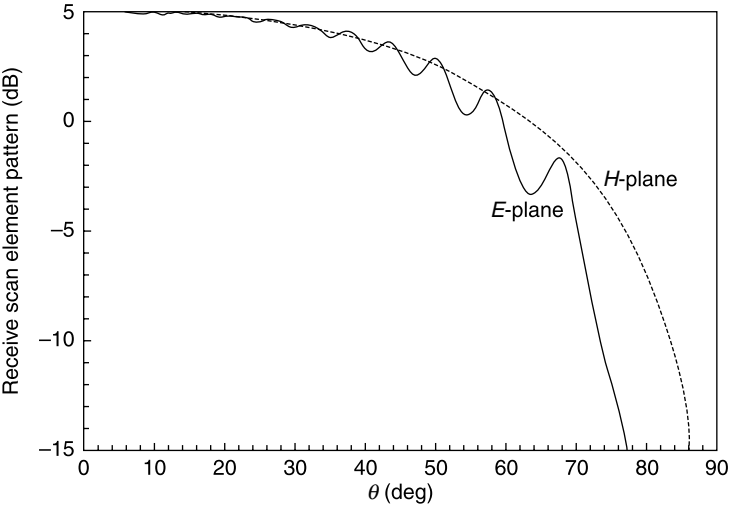


FIGURE 7.6 A 50×50 dipole/screen array, $L = D_x = D_y = 0.5\lambda$, $H = 0.25\lambda$.

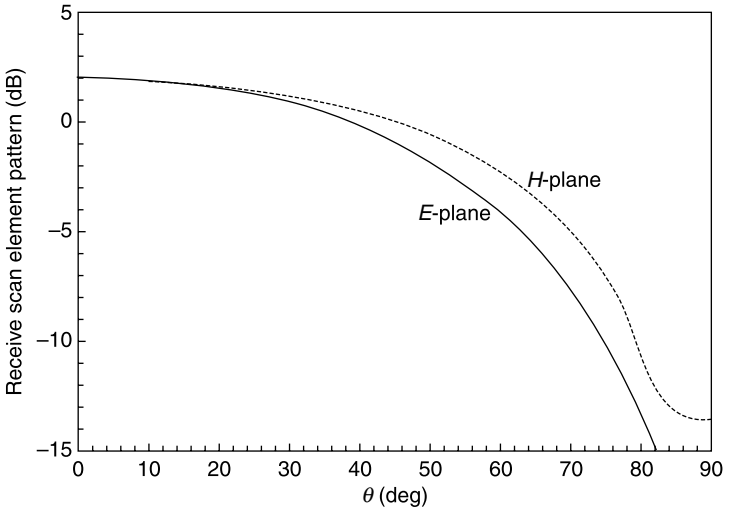


FIGURE 7.7 A 50×50 dipole array, $L = D_x = D_y = 0.5\lambda$, average CGAM.

The preceding transmit graphs were all normalized to 0 dB at broadside, to show scan behavior. Actual SEP, gain per element at broadside, is given in Table 7.1 for the 50×50 dipole array. For the (correct) receive case, with a center element match, the SEP is $\pi/2 = 1.96$ dB as expected. The error in the (incorrect) transmit case is 6.2 dB. The passive mutual coupling reduces all currents but that of the center element, resulting in grossly incorrect gain.

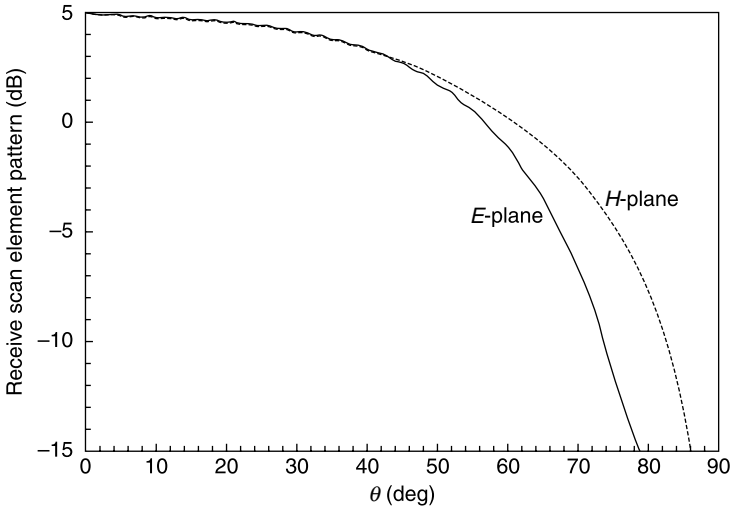


FIGURE 7.8 A 50×50 dipole/screen array, $L = D_x = D_y = 0.5\lambda$, $H = 0.25\lambda$, average CGAM.

TABLE 7.1 A 50 × 50 Dipole Array at Broadside

	SEP-T (dB)	SEP-R (dB)
Center element match	−4.26	+1.96
Infinite array match	−5.00	

TABLE 7.2 A 50 × 50 Dipole–Screen Array at Broadside

	SEP-T (dB)	SEP-R (dB)
Center element match	−6.42	+4.97
Infinite array match	−8.71	

The array over a ground plane gives the results in Table 7.2. Now the center element match case produces 4.97 dB as expected; the transmit case gives a gain of 11.4 dB in error. When the infinite array impedances are used as a match, the results are nearly the same.

In conclusion, measuring the SEP of an array with one element excited gives crude relative scan performance with significant errors, while the absolute values are grossly incorrect; due to only one element excitation leading to passive mutual coupling effects. The SEP should be measured with the arrays in the receiving mode in a standard gain test facility. The *E*-plane receiving SEP may show oscillations for small arrays, which can be smoothed out by measuring or calculating the SEP of several elements, and using averaging.

7.2.3.2 Measurement of Scan Impedance. It is important to determine SI so that an optimum match may be produced over the frequency and scan range. All elements must be excited, which means the array is in receive mode. However, a network analyzer cannot be used to measure SI as it produces a voltage. A solution was found for measuring the output impedance of power transistor amplifiers. It used "load pull", wherein an adjustable matching unit was connected to the transistor output; the match was adjusted for maximum power transfer; the output impedance was then the complex conjugate of the matching unit impedance.

This load pull technique is ideally suited to measurement of SI, as suggested by (Van Wagoner and Hansen, 2003). There are two companies that sell load pull instrumentation. The clever part of these instruments involves the algorithm for adjusting the matching unit in an efficient and accurate manner.

7.2.4 Minimum Scattering Antennas

In understanding the relationship between actual and ideal array element impedances, the concepts of minimum scattering antennas (MSA) and canonical minimum scattering antenna (CMS) are useful (Montgomery, 1947; Kahn and Kurss, 1965; Gately et al., 1968; Wasylkiwskyj and Kahn, 1970). Consider an antenna with N ports; for

example, these ports might be the dominant mode and higher modes (usually evanescent) in an open-ended waveguide element. When each port of an MSA is terminated in the proper reactance, the scattered power S_{ii} is zero. The CMS antenna is a special type of MSA in that an open circuit at each port produces $S_{ii} = 0$. The CMS antenna is lossless, which implies that the scattering matrix is unitary. The scattering matrix can be written as:

$$[S] = \begin{bmatrix} 0 & S_{ij}^{+} \\ S_{ij} & 1 - S_{ij}S_{ij}^{+} \end{bmatrix} \tag{7.30}$$

(Note that + denotes the conjugate transpose.) This N-port antenna possess N orthogonal radiation patterns, which are the S_{ij}. In the absence of nonreciprocal components, (e.g., ferrites, the antenna is reciprocal), which makes all patterns real and symmetric about the origin. Scattered and radiated patterns are equal. No impedance or admittance matrix exists, and the N eigenvalues of the scattering matrix are all -1. Unlike most scatterers (or antennas) the scattered field pattern of a CMS antenna is independent of the incident-field direction, although the amplitude of the pattern will depend on the incident wave. If, then, the pattern of an element is taken in the array environment with all other elements open circuited, this pattern is exactly the isolated array pattern only if the elements are CMS antennas. For example, when dipoles in an array are open-circuited, there is a small effect due to the half-length conducting rods that remain when each dipole is open-circuited. As expected, half-wave dipoles are not CMS antennas. Similarly, when slots in a ground plane are shorted across the feed terminals, the remaining half-length slots affect the current distributions on the ground plane.

Wasylkiwskyj and Kahn (1970) showed that the mutual impedance (admittance) between two identical MSAs can be written as an integral of the power pattern of an isolated element over certain real and complex angles (see also, Bamford et al., 1993):

$$Z_{12} = 2 \int_{\text{path}} \exp(-jkd \cos w) P(w)\, dw \tag{7.31}$$

where $w = u + jv$, and the power pattern is defined as $P(w) = f(w)f(w + \pi)$ and the integration path in the complex w plane is from $(-\pi/2, -\infty)$ to $(-\pi/2, 0)$ to $(3/2, 0)$ to $(3/2, \infty)$. The power pattern is normalized so that its integration over all real angles is unity. This integral can also be expressed in wavenumber coordinates. For example, with two antennas in the xy plane, the mutual impedance is given by

$$Z_{12} = \frac{2}{k} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{\exp[-j(k_x d_x + k_y d_y)]}{\sqrt{k^2 - k_x^2 - k_y^2}} P(k_x, k_y) dk_x\, dk_y \tag{7.32}$$

Mutual coupling, then, is specified by the element pattern and the lattice, and is completely independent of the means utilized to produce that pattern. This development for MSA is a generalization of that of Borgiotti, which is discussed in the Section 7.3 under grating lobe series.

Most multimode elements do not have orthogonal patterns, and hence are not MSAs. However, many single-mode antennas approximate to an MSA. Short dipoles, where the current distribution is essentially linear, are closely MSA. A resonant (near half-wave) dipole is approximately MSA if the radius is very thin; this thinness forces the current to be nearly sinusoidal. Mutual impedance between thin collinear half-wave dipoles is shown to agree with the Carter results (Wasylkiwskyj and Kahn, 1970). Andersen et al. (1974) show that crossed dipoles are approximately MSA in one pattern plane, and that the mutual impedance calculated from MSA theory is good if the center of one cross lies on a line bisecting the arms of the other cross. Small helices are also approximately MSA.

7.3 SPATIAL DOMAIN APPROACHES TO MUTUAL COUPLING

Except for arrays with only a few elements per side, array design is based on large or infinite array theory. This predicts the behavior of all except the edge elements; these are treated separately and subsequently (see Chapter 8). In the spatial domain, the array is simulated by an impedance (admittance) matrix that relates the voltages (currents) applied to the dipoles (slots) and the resulting currents (voltages). For most scan angle ranges a modest array size will provide *scan impedance* and *scan element pattern* that are nearly independent of that size. Before proceeding to this element-by-element approach, the mutual impedance between two elements will be examined; as seen in the preceding sections, mutual impedance is both the cause and explanation of all scanned array effects. These mutual impedances become the elements of the array impedance matrix.

7.3.1 Canonical Couplings

7.3.1.1 Dipole and Slot Mutual Impedance. There are several intuitive rules that concern mutual coupling between two antenna elements. First, the magnitude of mutual impedance decreases with distance between the elements. Second, if one antenna is in the pattern maximum of the other, coupling will be strong compared to that which the pattern null produces. If the antenna is in the radiating near-field of the other, then the near-field pattern is used. Third, if the electric fields are parallel, coupling will be stronger than if they are collinear. For wire antennas this can be restated in terms of shadowing: Large shadowing will correlate with large coupling. Fourth, larger antennas have smaller coupling. For example, large horns have lower mutual impedance than small horns. These rules, although useful, are no substitute for obtaining actual values of mutual impedance or admittance. The mutual impedance between dipoles is covered here in depth, as dipoles and slots are common array elements. These impedances are typical of resonant element mutual impedances,

such as microstrip patches,[3] so the dipole mutual impedance behavior is of general usefulness. Thin cylindrical dipoles are equivalent to flat strip dipoles, where the strip width is twice the wire diameter, and slot mutual admittance data can be derived from strip dipole results via Babinet's principle.

For mutual impedance the zero-order (sinusoidal current distribution) theory is usually adequate. For this case, the formulas of King (1957) based on the original work of Carter (1932) can be used. They give mutual impedance between parallel coplanar dipoles of unequal length in echelon. However, these formulas contain 24 sine and cosine integral terms, of various complicated and diverse arguments. Although the results of King could be rearranged and grouped into a computer algorithm, it is simpler to derive the formula directly, using the exact near-electric-field formulations of Schelkunoff and Friis (1952). The exact field from a sinusoidal current is written as three spherical waves, the field components in cylindrical coordinates. Mutual impedance is then the integral of this electric field (from each half-dipole) times the current distribution at the other dipole. With the geometry of Figure 7.9 the mutual impedance expression is

$$Z = \frac{-j30}{S_1 S_2} \int_0^{d_1} [\Psi_1 - 2\Psi_2 \cos(kd_2) + \Psi_3 + \Psi_4 - 2\Psi_5 \cos(kd_2) + \Psi_6] \times \sin[k(d_1 - x)]dx \tag{7.33}$$

where

$$S_1 = \sin(kd_1) \qquad S_2 = \sin(kd_2) \qquad \Psi_i = \exp(-jkR_i)/R_i$$

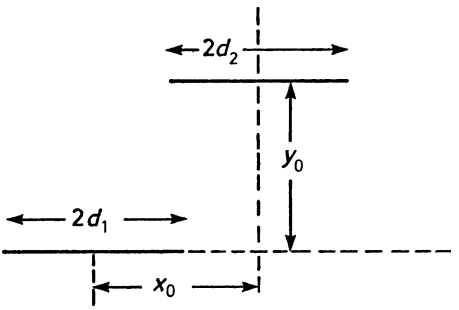


FIGURE 7.9 Echelon dipole geometry.

[3]For large substrate thickness, patch mutual impedance has a surface wave component so that the dipole results are no longer similar.

The R values are

$$\begin{aligned} R_1^2 &= y_0^2 + (x_0 + d_2 - x)^2 \\ R_2^2 &= y_0^2 + (x_0 - x)^2 \\ R_3^2 &= y_0^2 + (x_0 - d_2 - x)^2 \\ R_4^2 &= y_0^2 + (x_0 - d_2 + x)^2 \\ R_5^2 &= y_0^2 + (x_0 + x)^2 \\ R_6^2 &= y_0^2 + (x_0 + d_2 + x)^2 \end{aligned} \tag{7.34}$$

The integral could be evaluated numerically, and for large dipole separations this method is preferable. However, the adjacent and nearby dipoles are most important, and for these an exact evaluation is necessary.

The exact solution can be written as two sums (Hansen and Brunner, 1979), where n steps by increments of 2:

$$\begin{aligned} Z = \frac{-15}{S_1 S_2} \sum_{m=-1}^{1} \sum_{n=-1}^{1.2} C_m \{ &\exp(jkU)[E(kA + kU) - E(kB + kV)] \\ &+ \exp(-jkU)[E(kA - kU) - E(kB - kV)]\} \end{aligned} \tag{7.35}$$

The coefficients are

$$C_{-1} = C_1 = 1 \qquad C_0 = 2\cos kd_2$$

and $E(x)$ is the Exponential Integral (Abramowitz and Stegun, 1970) with arguments:

$$\begin{aligned} U &= d_1 + n(x_0 + md_2) \\ V &= n(x_0 + md_2) \\ A^2 &= y_0^2 + (x_0 + nd_2 + md_2)^2 \\ B^2 &= y_0^2 + (x_0 + md_2)^2 \end{aligned} \tag{7.36}$$

When the two dipoles are the same length the formula simplifies (Hansen, 1972) to

$$Z = \frac{15}{S^2} \sum_{m=-2}^{2} \sum_{n=-1}^{1,2} A_m \exp[-jkn(x_0 + md)]E(k\beta) \tag{7.37}$$

where

$$\beta = \sqrt{(x_0 + md)^2 + y_0^2} - n(x_0 + md)$$

and the coefficients are

$$\begin{aligned} A_{-2} = A_2 = 1 \qquad & A_{-1} = A_1 = -4\cos kd \\ A_0 = 2(1 + 2\cos^2 kd) \qquad & S = \sin kd \end{aligned} \tag{7.38}$$

$E(x)$ can be expressed as the Cosine and Sine Integrals, Ci and Si. When $y_0 = 0$, that is, the dipoles are collinear, y_0 is replaced by the dipole radius a. When the dipoles are half-wave, the $m = 2$ and 4 terms disappear and the exponential simplifies. This formulation is similar to that of Richmond and Geary (1970) for two equal-length thin dipoles with axes at an angle. Computer subroutines for Ci, Si are readily available in most computer libraries; with computation time comparable to that of trig functions. These subroutines use the economized series developed by Wimp and Luke (see Wimp, 1961).

Figure 7.10 shows Z_{ij} between two parallel half-wave dipoles on an impedance plot. The curve, with spacing–wavelength as a parameter, is similar to a Cornu spiral. It can be seen that the magnitude of impedance decreases as spacing increases. Data are normalized by the self-resistance, which, for a zero-order (zero-thickness) dipole, is 73.13 Ω. Figure 7.11 is a similar plot for collinear dipoles. The spacing here is

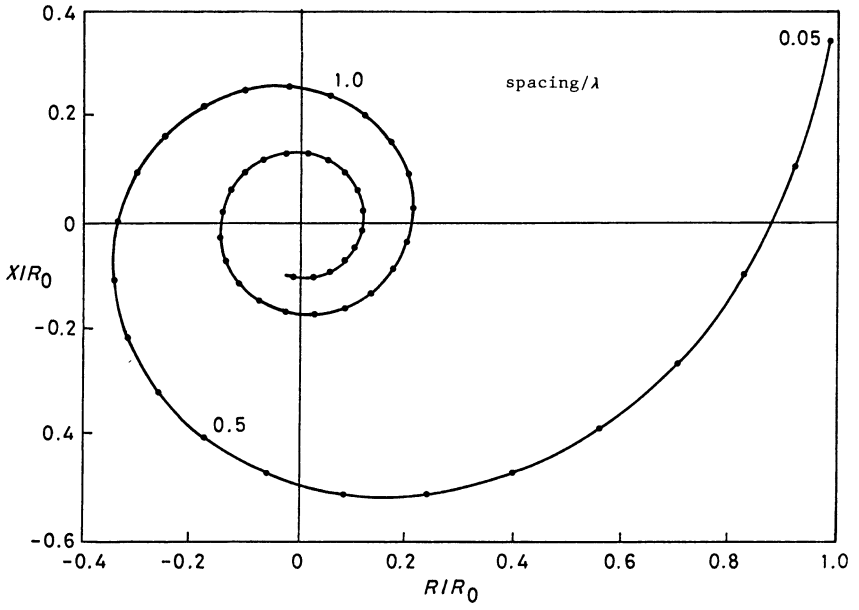


FIGURE 7.10 Parallel dipole mutual impedance.

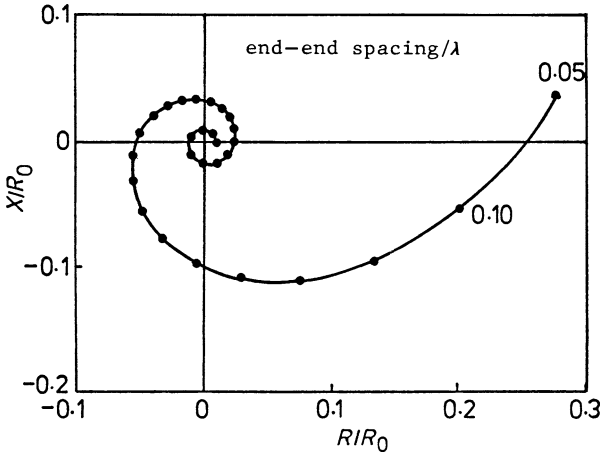


FIGURE 7.11 Collinear dipole mutual impedance.

between dipole tips; for center-to-center spacing add 0.5. As expected, the collinear coupling is less owing both to lower shadowing and a pattern null. Recall, however, that the dipole near-field has both axial and radial electric fields and that the latter does not have a pattern null along the axis.

When slots are located in a stripline surface or in contiguous waveguides the mutual admittance between two slots behaves as though the slots are located in an infinite ground plane. The slots can then be replaced by strip dipoles by Babinet's principle developed by Booker (1946) for both impedance and mutual impedance (Begovich, 1950). The slot mutual impedance Z_{12} and strip dipole mutual admittance, Y_{12} are related:

$$2Z_{12} = \eta^2 Y_{12} \tag{7.39}$$

where $\eta = 120\pi$. The factor of 2 has been reduced from 4 as the slots radiate one side only.

When the slots are narrow, which is usually the case, the equivalent strip dipoles are also narrow; they can be accurately replaced by cylindrical dipoles of radius equal to one-fourth the slot or strip dipole width (Lo, 1953). Thus the extensive development of dipole mutual impedance can be used for slots.

7.3.1.2 Microstrip Patch Mutual Impedance. Calculation of mutual impedance between two microstrip patches must include the effects of the grounded (dielectric) substrate, typically via a Green's function. This requires one, or often several, integrations of the Sommerfeld type, plagued by oscillating and unbounded behavior on the real axis, and branch cuts, surface wave, and other poles for the complex integration. The simplest procedure is based on the transmission line model; however, the patch edge gap formulation is critical. In general, this model has been

superseded by more accurate models. These are of two types: cavity model and moment method model. Both utilize the rigorous Green's function for the slab.

The moment method model establishes current expansion functions on the patch, such as rooftop functions; these and the test functions allow the integral equation to be discretized (Pozar, 1982; Newman et al., 1983). Several Green's function integrations are usually involved.

In the cavity model, the antennas are replaced by a grounded dielectric slab with magnetic current distributions at each patch cavity open wall. Calculations by Haneishi and Suzuki (1989) show the real and imaginary parts of mutual admittance oscillating and decaying with increasing separation, with the nulls of one occurring at the peaks of the other. The curves are close to those for dipole mutual impedance. An advantage of this model is that a single Green's function integration is needed, and this can be expedited via FFT (Mohammadian et al., 1989). This approach appears to be as accurate as the moment method approach, although comparative data for higher ϵ have not been available. Both sets of results compare well with *L*-band measurements of Jedlicka et al. (1981). Only a small change was seen as the thickness was doubled.

These measurements were repeated by Mohammadian et al. (1989) with good agreement; measurements were also made at 5 GHz on patches $0.282\lambda_0 \times 0.267\lambda_0$ with substrate of $\varepsilon = 2.55$ and $t = 0.0252\lambda_0$, and were compared with calculations. Figure 7.12 again shows good agreement. It may therefore be concluded that a relatively simple and accurate method exists for calculation of patch mutual admittance.

There have been many papers, over many years, on the subject of evaluation of Green's function integrals. Here a few references are given: Pearson (1983); Johnson and Dudley (1983); Marin et al. (1989); Chew (1989); Barkeshli et al. (1990); Barkeshli and Pathak (1990); Marin and Pathak (1992); Yla-Oijala and Taskinen (2003); Kourkoulos and Cangellaris (2006); Polimeridis et al. (2007).

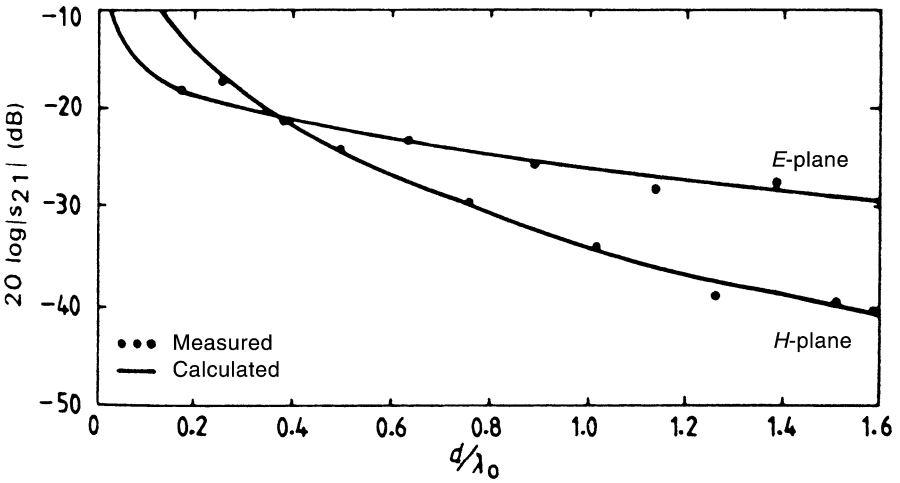


FIGURE 7.12 Patch coupling. (Courtesy Mohammadian, A. H., Martin, N. M., and Griffin, D. W., "A Theoretical and Experimental Study of Mutual Coupling in Microstrip Antenna Arrays", *Trans. IEEE,* Vol. AP-37, Oct. 1989, pp. 1217–1223.)

7.3.1.3 Horn Mutual Impedance. The limiting case of a waveguide horn is the open-end waveguide radiator. A single-mode excitation in one element tends to excite evanescent modes in addition to propagating modes in the second guide. Most analyses utilize aperture integration; see Luzwick and Harrington (1982) and Bird (1987) for rectangular guide elements; Bird (1990) for rectangular elements of different sizes; Bird and Bateman (1994) for rotated rectangular elements; Bailey and Bostian (1974), Bailey (1974), and Bird (1979) for circular guide elements.

Coupling between *E*-plane sectoral horns has been measured by Lyon et al. (1964). Horns of 8-dB gain with a separation of 0.73λ showed low coupling as seen in Figure 7.13. The *E*- and *H*-plane couplings alternate peaks and valleys as the orientation angle is changed. For large spacing, the parallel polarization decays as $1/r^2$, or 12 dB for each doubling of distance r, while the collinear coupling decays as $1/r$. Coupling data for pyramidal horns of a separation of 5.93λ are shown in Figure 7.14.

In a different approach, Hamid (1967) used Geometric Theory of Diffraction (GTD) to calculate mutual coupling between sectoral horns with 3λ apertures; the coupling showed a typical peak and null behavior with center-to-center separations, with the envelope varying roughly as coupling $\simeq(-49 - s/\lambda)$ dB.

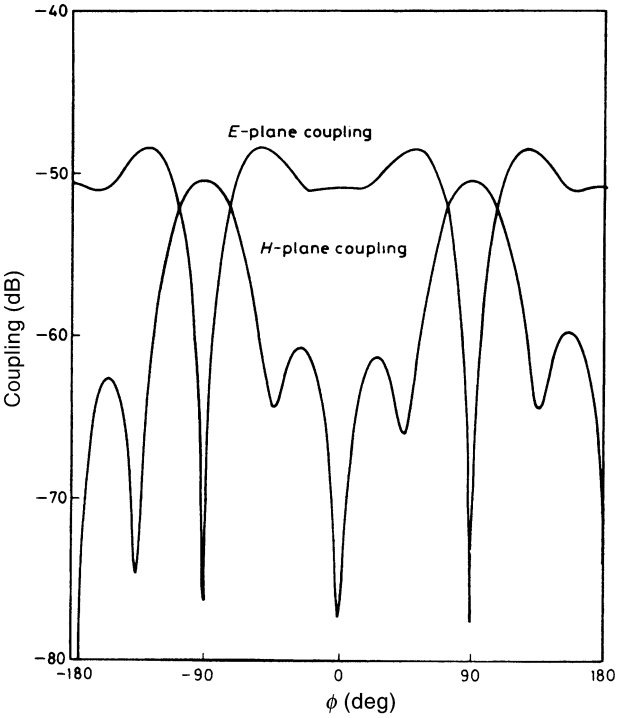


FIGURE 7.13 The *E*-plane sectoral horn coupling. (Courtesy Lyon, J. A. M. et al., "Interference Coupling Factors for Pairs of Antennas", *Proceedings 1964 USAF Antenna Symposium,* Allerton, IL, AD-609 104.)

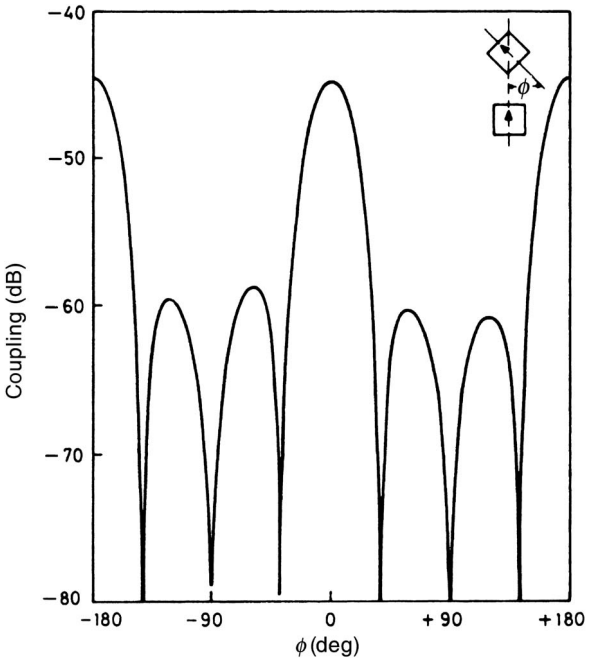


FIGURE 7.14 Pyramidal horn coupling. (Courtesy Lyon, J. A. M. et al., "Interference Coupling Factors for Pairs of Antennas", *Proceedings 1964 USAF Antenna Symposium,* Allerton, IL., AD-A609 104.)

7.3.2 Impedance Matrix Solution

Finite arrays of dipoles, slots, or patches can be analyzed simply by setting up a mutual impedance matrix. Carter-type mutual impedances (from the previous sections) are usually adequate. If an array uses dipoles over a ground plane, it is replaced by an image array, which makes the impedance matrix $2N \times 2N$ instead of $N \times N$. With a symmetric array and no scan the matrix can be folded to reduce its size to roughly one-half (exactly one-half if N is even). The scan angle appears in the progressive phase of the drive vector, of which the component from the nth row and mth column is

$$V_i = A_{nm} \exp[-j2\pi(nu + mv)] \tag{7.40}$$

As before, the steering vector components are

$$u = \frac{d_x}{\lambda} \sin\theta \cos\phi \qquad v = \frac{d_y}{\lambda} \sin\theta \sin\phi \tag{7.41}$$

The mutual impedance Z_{ij} is between the ith and jth element. When the complex simultaneous equations are solved[4] the result is the current vector

$$[V_i] = [I_j][Z_{ij}] \tag{7.42}$$

and from the current vector the *scan impedance* of each element in the array is determined. When the solution is desired for several scan angles, so that only the drive vector changes, but not the Z_{ij} matrix, a simultaneous equation solver can be used that stores a diagonalized matrix, allowing a rapid solution for new vectors after the first. From the current vector, of course, array patterns can be calculated. The impedance matrix solution has an intrinsic advantage: Array excitation tapers can be incorporated directly into the drive vector, unlike the infinite array methods of Section 7.4, where the excitation must be uniform.

For large arrays the *scan impedance* results are essentially the same as those for infinite arrays (see Section 7.4) except for scan angles near 90°. Diamond (1968) calculated an array of 65×149 half-wave dipoles with screen; no difference was noticed until the scan angle exceeded 70°, probably because there are more "edge elements" for larger scan angles.

Multimode elements in a finite array can, in principle, be handled by increasing the size of the matrix from $N \times N$ to $NM \times NM$, where each element has M modes, for example. Since the computer time for solution of simultaneous equations varies as the cube of the matrix size, large arrays may be difficult. An alternative solution has been offered by Goldberg (1972). The mode voltage at each element is written as a sum (over the elements) of drive voltages times coupling coefficients times scan phase factor. The coupling coefficients are approximately independent of array size (strictly true only for CMS elements) and may be determined from an infinite array. An infinite array with the proper spacing, scan range, and bandwidth is then designed as described elsewhere in this chapter; the results are mode voltages and *scan reflection coefficient* versus u. A Fourier inversion is used to obtain the coupling coefficients, which are then summed to give a set of mode voltages and reflection coefficient for each element in the array. Only the center elements in a large array experience essentially all the coupling; elements at or near the edge employ fewer strong coupling coefficients as they decay rapidly with element separation except near grating lobe incidence. Effects of amplitude taper can be determined simply by including the amplitude coefficients in each sum. From the reflection coefficient of each element, the *scan impedance* can be written, and the pattern of the array is found by adding the modal array patterns that use element modal patterns and mode voltages.

For large arrays, the *scan impedance* and *scan element pattern* are those of infinite arrays (see Section 7.4), with small oscillations; larger arrays have more oscillations and they are of smaller amplitude. Data for small arrays are given in Chapter 8.

[4]Note that solving simultaneous equations is roughly three times faster than matrix inversion (see Westlake, 1968).

7.3.3 The Grating Lobe Series

An intuitive understanding of the role of mutual coupling in array scanning is facilitated by the grating lobe series, which links the current sheet concepts of this section with the Floquet series approach of the next section. It could be discussed along with the periodic cell, Floquet material, but the kernel of the grating lobe series relates to an array element; thus it is located here.

It was observed by Wheeler (1966) that each term in the double series for dipole *scan impedance* is associated with a point in the u,v grating lobe lattice (e.g., Fig. 7.15). This is true for all types of elements in a regular lattice, but is most easily visualized by considering short dipoles. For short dipoles, the formulas for dipoles of general length simplify because the sinc and cosc factors become unity. If now Eqs. (7.41) and (7.42) (derived later in Section 7.4) are divided by the broadside *scan resistance*, the normalized impedance becomes

$$\begin{aligned} \frac{R(u,v)}{R(0,\,0)} &= H_{00} = \frac{k^2 - k_{x0}^2}{k\sqrt{k^2 - k_{x0}^2 - k_{y0}^2}} \\ \frac{X(u,\,v)}{R(0,\,0)} &= -\sum\sum H_{nm} = \sum\sum \frac{k^2 - k_{xn}^2}{k\sqrt{k_{xn}^2 + k_{ym}^2 - k^2}} \end{aligned} \tag{7.43}$$

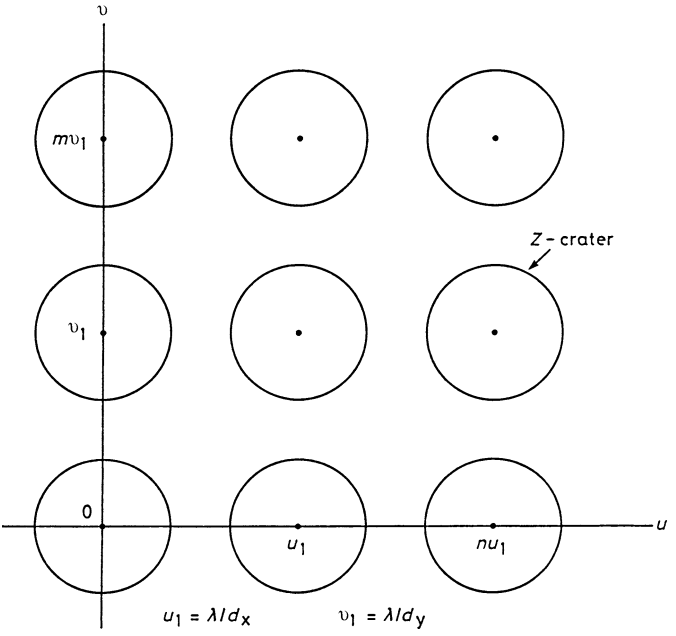


FIGURE 7.15 Grating lobe craters.

Here the expressions for H_{00} and H_{nm} are identical with Eq. (7.40). Wheeler correlated these terms with the u,v grating lobe points, and identified each term as an impedance crater, a concept developed by Rhodes (1974). The normalized impedance becomes

$$\frac{Z_s(u, v)}{R_s(0, 0)} = \sum\sum \frac{Z_c(u - n/d_x, v - m/d_y)}{R_s(0, 0)} \tag{7.44}$$

Here Z_s/R_s is the impedance crater (Frazita, 1967), which is a plot of the normalized impedance of an element in u,v space. Figure 7.16 shows the crater for a short dipole and Figure 7.17 is the crater for a half-wave slot. Wheeler discovered that mutual-coupling effects with scan, and the effects associated with grating lobe onset, could be visualized by placing an impedance crater at each (inverse) lattice point in the u,v plane. The term "crater" arises because the resistance presents the topological form of a crater. Thus the grating lobe series is a link between the current sheet concepts discussed earlier and the periodic structure approach.

The grating lobe series can also be expressed in terms of transforms of aperture fields (Borgiotti, 1968). *Scan admittance* of an infinite array on a rectangular lattice is written as an integral in wavenumber space of the Fourier transform of the aperture

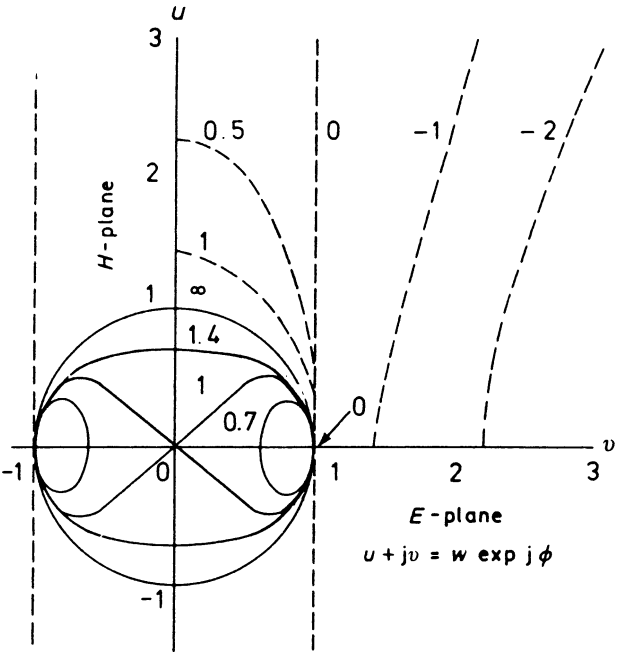


FIGURE 7.16 Impedance crater of short dipole. (Courtesy Wheeler, H. A., "The Grating-Lobe Series for the Impedance Variation in a Planar Phased-Array Antenna", *Trans. IEEE*, Vol. AP-14, Nov. 1966, pp. 707–714.)

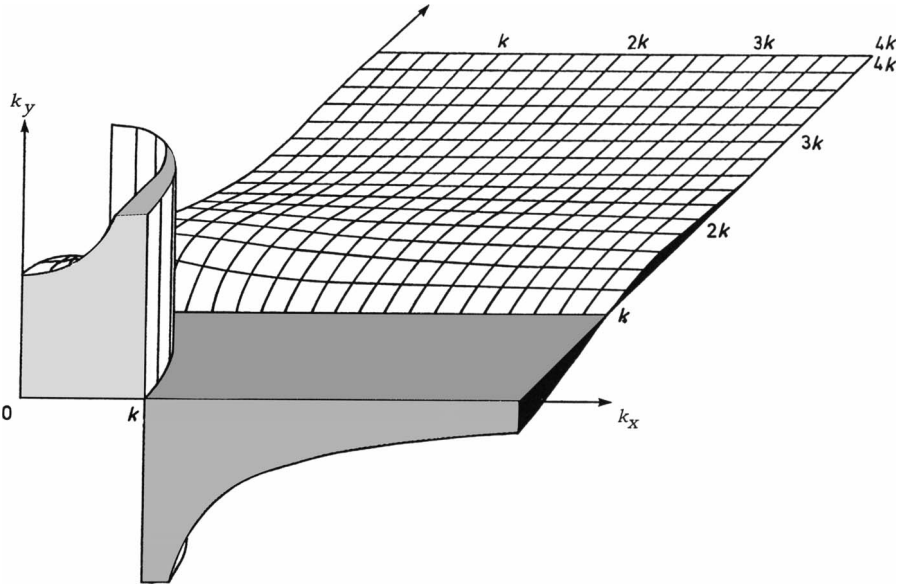


FIGURE 7.17 Admittance crater of half-wave slot. (Courtesy Rhodes, D. R., *Synthesis of Planar Antenna Sources,* Clarendon Press, 1974.)

field times the FT of the conjugate field for each element. The doubly infinite sum representing element phases can be expressed as a doubly periodic delta function, which reduces the integrals to

$$Y_s = \frac{4\pi\lambda^2}{k\eta d_x d_y} \sum_{-\infty}^{\infty} \sum_{-\infty}^{\infty} \left[\frac{k^2}{w} E_\rho E_\rho^* + w E_\psi E_\psi^* \right] \tag{7.45}$$

where $w = -j\sqrt{u^2 + v^2 - k^2}$ and the fields are evaluated at the u,v grating lobe points. Mutual impedance between two elements can similarly be written as an integral in wavenumber space of aperture of distribution FTs, which can be reduced to the aperture complex power equations developed by Borgiotti (1963) and Rhodes (1964).

7.4 SPECTRAL DOMAIN APPROACHES

7.4.1 Dipoles and Slots

Floquet's theorem for arrays states that an infinite regular periodic structure will have the same fields in each cell except for a progressive exponential multiplier (Amitay et al., 1972; Catedra et al., 1995). Further, the fields may be described as a set of orthogonal modes. In essence, the boundary conditions are matched in the Fourier transform domain, resulting in some cases in an integral equation reducing to an algebraic equation. Wheeler (1948) was probably the first to develop the unit cell or

periodic cell for fixed-beam arrays, where each element is contained in a cell, with all cells alike and contiguous. Edelberg and Oliner (1960) and Oliner and Malech (1966) extended the periodic cell approach to scanned arrays. It has proved to be the most powerful and perceptive technique for understanding and for designing sophisticated arrays. An equivalent development is by Diamond (1968). See also Stark (1966). Because of the Floquet symmetry, the single unit cell contains the complete admittance behavior of each element in the array. The unit cells are normal to the array face, centered about the elements, and contiguous (see Fig. 7.18). For broadside radiation with no grating lobes, the unit cell has two opposite electric walls (zero electric field) and two magnetic walls (zero magnetic field) with a TEM mode. However, when the beam is scanned, such a unit cell must be tilted along with the beam, and another unit cell must be added for each grating lobe. Because of this complexity, Oliner utilized a fixed normal unit cell that would accommodate any scan angle and any number of grating lobes. The unit cell walls are not in general electric or magnetic, and the modes inside are LSE and/or LSM, depending on the scan plane. Opposite walls support fields that differ by the Floquet phase shift. Each main lobe or grating lobe is represented by a propagating mode, with the evanescent modes contributing energy storage (susceptance) at the array face.

Although short slots or dipoles are simpler to analyze by the periodic-cell method, the resonant dipole is of most importance; so this case will be carried through in some detail. Scanning in a principal plane is also simpler, but arbitrary scan will be considered here. For this case, all four unit cell walls are scan dependent, with both E_z and H_z field components. A superposition of the dominant LSE and dominant LSM modes is necessary. The wavenumber in the z-direction is

$$k_z = \sqrt{k^2 - k_{x0}^2 - k_{y0}^2} = k \cos \theta_0 \tag{7.46}$$

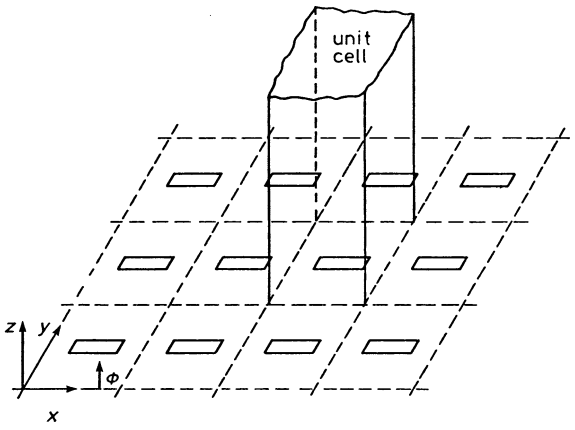


FIGURE 7.18 Slot array unit cell.

where

$$k_{x0} = k \sin\theta_0 \cos\phi_0 \qquad k_{y0} = k \sin\theta_0 \sin\phi_0$$

The unit cell dimensions are $D_x = d_x/\lambda$ and $D_y = d_y/\lambda$, where the element lattice spacings are d_x and d_y. The generalized wave numbers are

$$\begin{aligned} k_{xn} &= k\left(\sin\theta_0 \cos\phi_0 + \frac{n}{d_x}\right) \\ k_{ym} &= k\left(\sin\theta_0 \sin\phi_0 + \frac{m}{d_y}\right) \end{aligned} \tag{7.47}$$

and the generalized scan factor is

$$\begin{aligned} H_{nm} &= \frac{\left(\sin\theta_0 \cos\phi_0 + \dfrac{n}{d_x}\right)^2 - 1}{\sqrt{\left(\sin\theta_0 \ \cos\phi_0 + \dfrac{n}{d_x}\right)^2 + \left(\sin^2\theta_0 \ \sin^2\phi_0 + \dfrac{m}{d_y}\right)^2 - 1}} \\ H_{00} &= \frac{1 - \sin^2\theta_0 \ \cos^2\phi_0}{\cos\theta_0} \end{aligned} \tag{7.48}$$

Assuming dipole feed terminals at the center of the dipole, the reactance is in parallel with the unit-cell waveguide, but in series with the feed terminals. Consideration of the ground plane displays one of the advantages of the unit-cell approach. Whereas the dipole array in free space is represented by a unit cell one each side of the dipole, with both unit cells semiinfinite in length, the ground plane is simply represented by a short across one of the unit cells at height h from the dipole. Since the ground plane to dipole spacing is usually on the order of $\lambda/4$, the higher order unit-cell evanescent modes are sufficiently damped out in traveling from the dipole to the ground plane and back that they can be neglected. However, the ground plane affects both resistance and reactance through the dominant mode. Figure 7.19 shows the dimensions. The dipole current is assumed to flow only in the x-direction and to be constant in the y-direction. Along x, a cosine distribution is assumed. A match is then made between the electric field in the dipole plane produced by the dipole and that produced by a set of unit-cell waveguide modes. From this, the series for *scan reactance* is obtained. For the *scan resistance*, the ground plane is replaced by an image dipole at distance $2h$. The *scan resistance* is

$$R_s = \frac{480 a_s^2}{\pi \, d_x d_y \lambda^2} \operatorname{cosec}^2 \frac{k_{x0} a_s}{2} \operatorname{sinc}^2 \frac{k_{y0} b_s}{2} \sin^2 (k_{z0} h) H_{00} \tag{7.49}$$

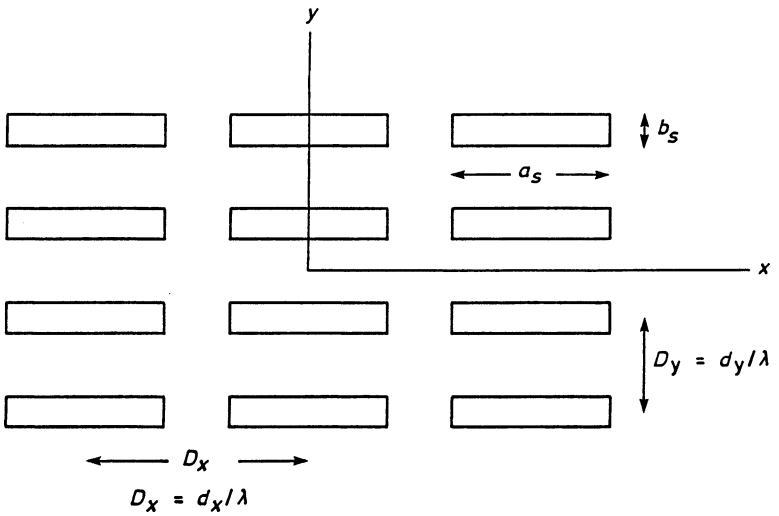


FIGURE 7.19 Strip dipole array geometry.

When a ground plane is not present, the $\sin^2 k_{z0}h$ factor is replaced by one-half. The expression for reactance is

$$X_s = \frac{240a_s^2}{\pi d_x d_y \lambda^2}\left[\operatorname{cosc}^2\frac{k_{x0}a_s}{2}\operatorname{sinc}^2\frac{k_{y0}b_s}{2}\sin^2(k_{z0}h)H_{00} - \sum\sum \operatorname{cosc}^2\frac{k_{xn}a_s}{2}\operatorname{sinc}^2\frac{k_{ym}b_s}{2}H_{nm}\right] \tag{7.50}$$

The sinc and cosc factors can be recognized as the Fourier transforms of the uniform and cosine current distributions.

Graphs of *scan impedance* and *scan element pattern* give insight into how arrays work as scan angle and lattice spacing are changed. Figures 7.20–7.22 show SEP and *scan impedance* for thin half-wave dipoles on a half-wave square lattice. As expected, the *H*-plane *scan resistance* and *scan reactance* increase with scan angle, while they decrease in the *E*-plane. Diagonal plane behavior is mixed. The SEP shows more decrease with scan angle in the *E*-plane, owing to the increasingly poor mismatch. A comparison of the SEP with powers of $\cos\theta$ shows that the best fit is $\cos^{1.5}\theta$; this power pattern lies roughly between the *H*- and *E*-plane curves. Note that slot parameters are obtained from those of the dipole by multiplying by a constant (Oliner and Malech, 1966). Adding a ground plane raises the broadside *scan resistance,* but most important it removes the *H*-plane trend to infinity see Figures 7.23–7.25. This result is offset by the screen pattern factor, with the result that the SEP for dipoles/screen (Fig. 7.25) is worse for all planes than that of the

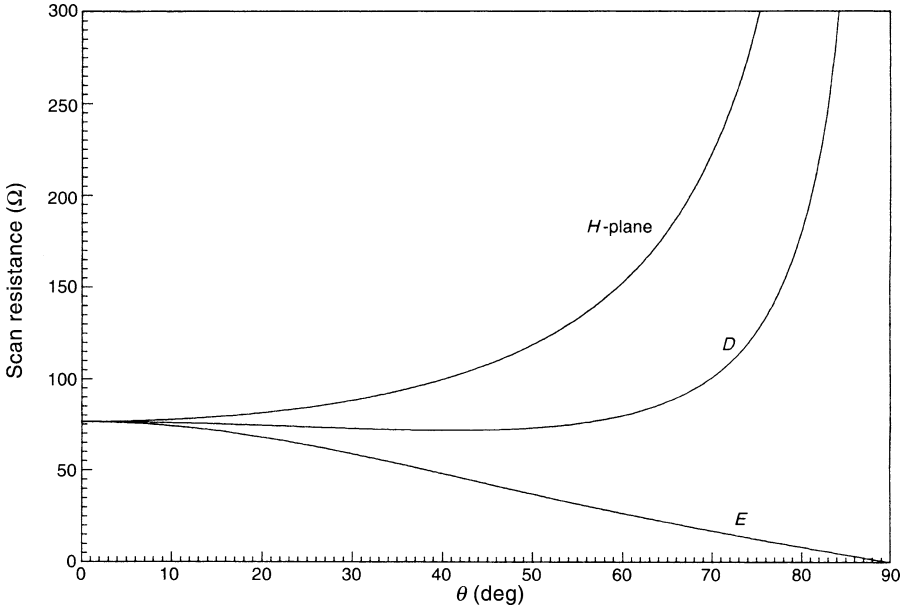


FIGURE 7.20 Dipole array, $D_x = D_y = 0.5$, $L = 0.5\lambda$.

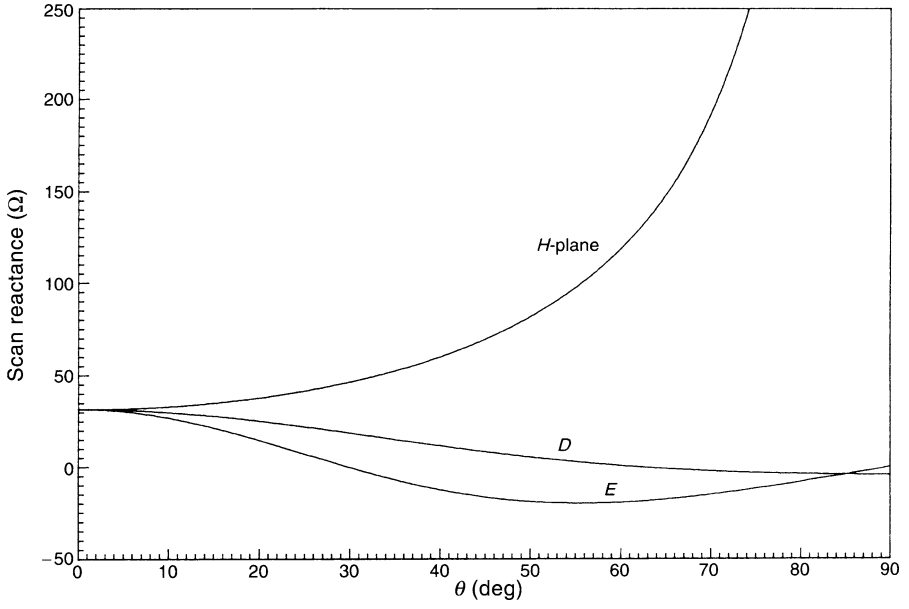


FIGURE 7.21 Dipole array, $D_x = D_y = 0.5$, $L = 0.5\lambda$.

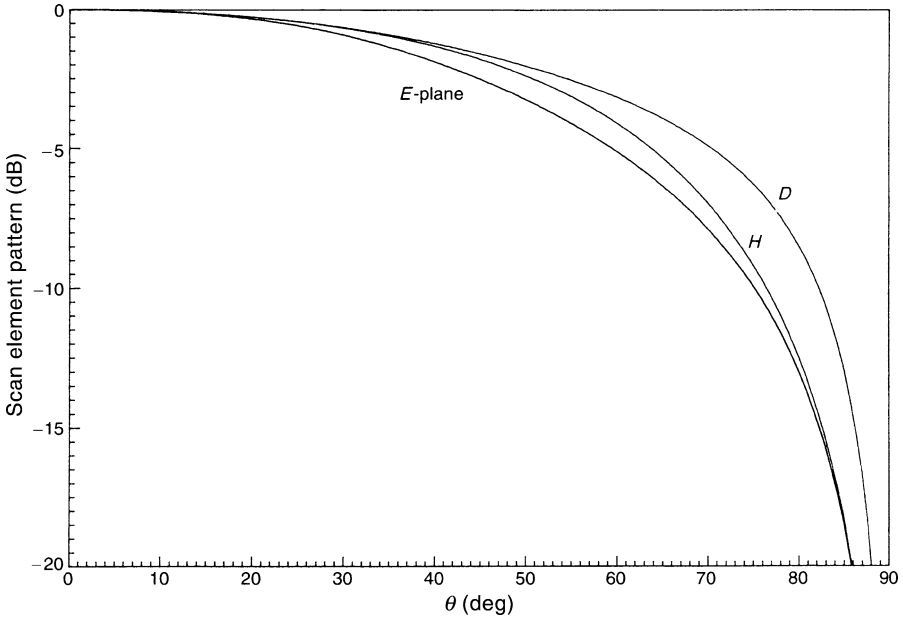


FIGURE 7.22 Dipole array, $D_x = D_y = 0.5$, $L = 0.5\lambda$.

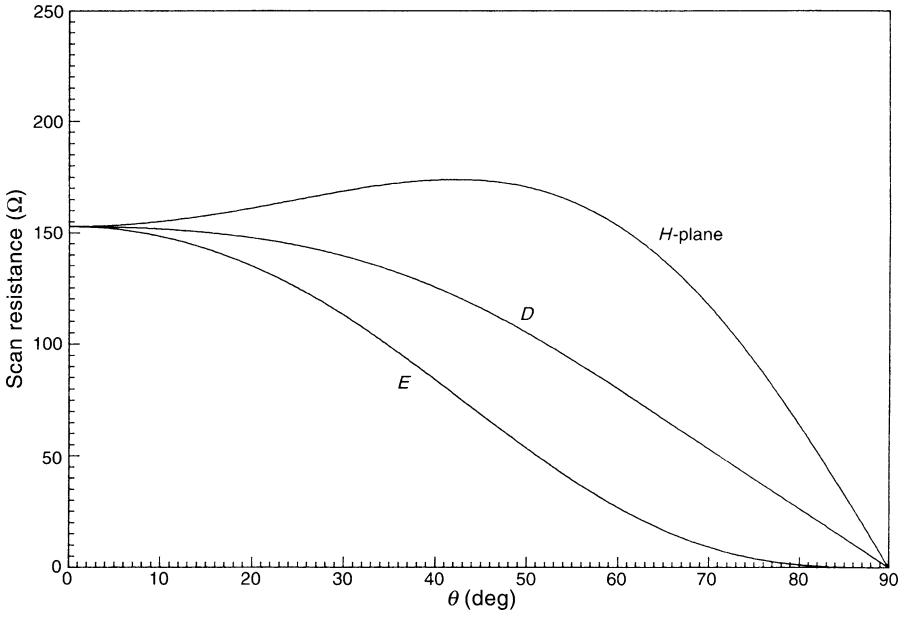


FIGURE 7.23 Dipole–screen array, $D_x = D_y = 0.5$, $L = 0.5\lambda$, $H = 0.25\lambda$.

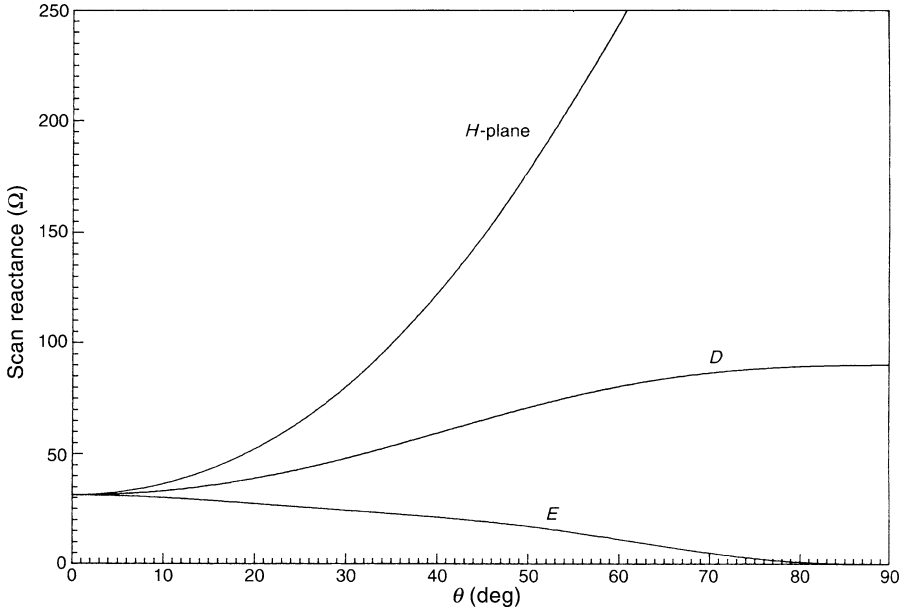


FIGURE 7.24 Dipole–screen array, $D_x = D_y = 0.5$, $L = 0.5\lambda$, $H = 0.25\lambda$.

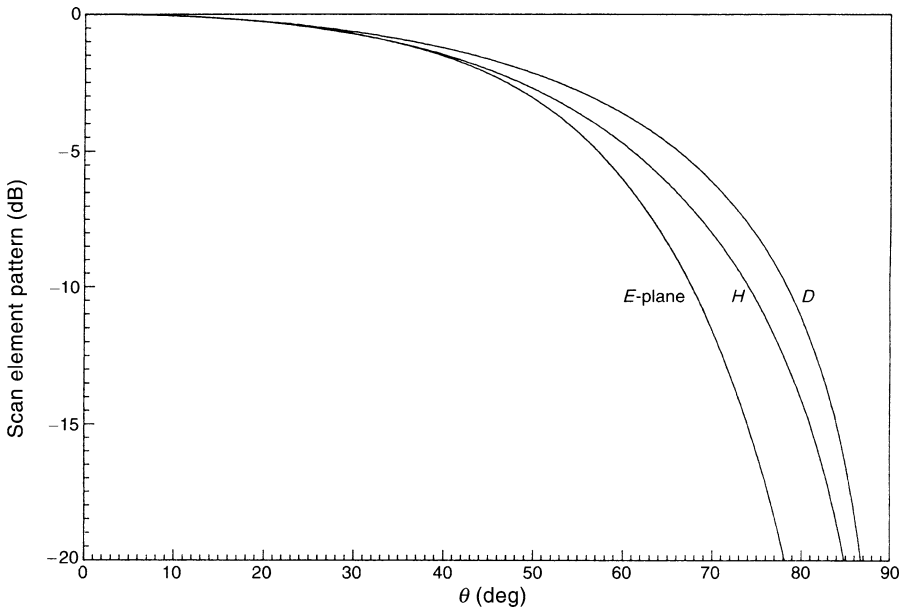


FIGURE 7.25 Dipole–screen array, $D_x = D_y = 0.5$, $L = 0.5$, $L = 0.5\lambda$, $H = 0.25\lambda$.

dipole array (Fig. 7.22). Again, the $\cos^{1.5}$ power pattern is a good fit out to $\sim 50^\circ$ scan; beyond this the SEP falls rapidly owing to the screen factor.

Impedance behavior at grating lobe incidence is illuminating; the cases just discussed are repeated for a square lattice of 0.7λ, which allows a grating lobe to occur at $\theta = 25.38^\circ$. Figures 7.26–7.28 show *scan impedance* and SEP for a dipole array; both parts of *scan impedance* have an infinite singularity at the grating lobe angle. The SEP shows a blind angle there for *H*-plane scan. The *E*-plane and diagonal plane scans are only slightly affected. Adding a screen replaces the infinite singularities by a steep but finite jump in *scan resistance,* and a cusp in *scan reactance* (see Figs. 7.29–7.31). The SEP has a roughly 3-dB dip at grating lobe incidence but the other planes are not much affected. Clearly for wideband operation, a screen is essential. Behavior for other lattice spacings is similar, about the grating lobe angle(s). The blind angle occurs because the reflection coefficient there is of unity magnitude. These data have been compared with spatial domain calculations on a large array using Carter mutual impedance data, with perfect agreement (Oliner and Malech, 1966). Typically 900 (30×90) terms have been used in the reactance series. All of the curves presented are for a dipole radius of 0.001λ; fatter dipoles reduce the broadside values of *scan impedance,* but the changes decrease with scan angle.

These spectral domain results can be derived directly from the Carter dipole mutual impedances. Consider an array of collinear coplanar thin dipoles along the y-axis and spaced by d_x and d_y. *Scan impedance* is obtained as the sum of mutual impedances between the array dipoles and another dipole spaced x_0 and y_0 from one of the array

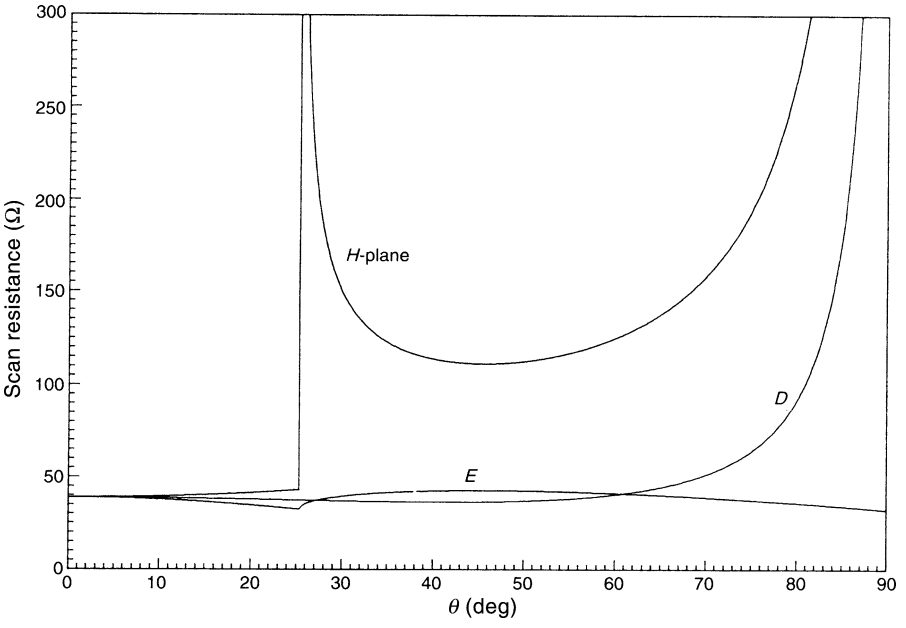


FIGURE 7.26 Dipole array, $D_x = D_y = 0.7$, $L = 0.5\lambda$.

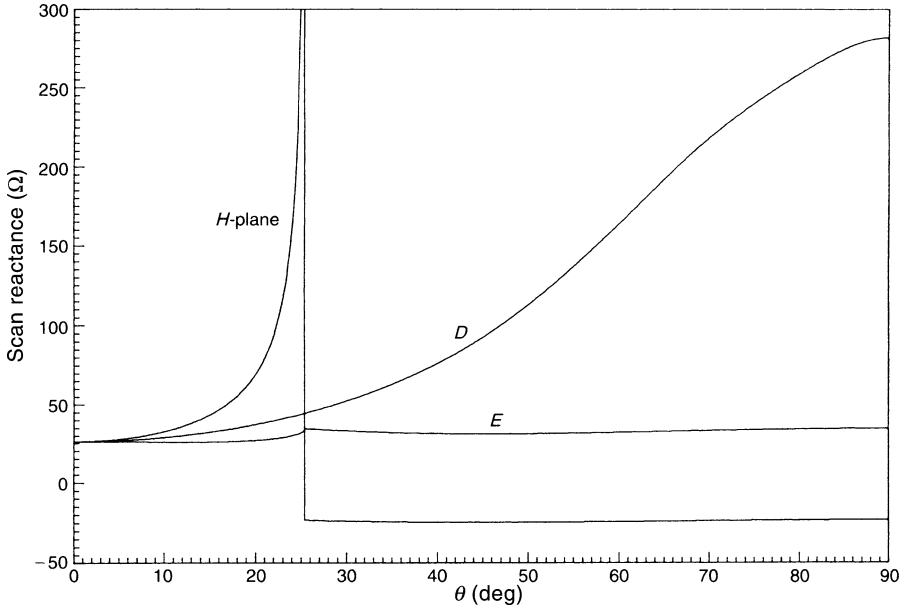


FIGURE 7.27 Dipole array, $D_x = D_y = 0.7$, $L = 0.5\lambda$.

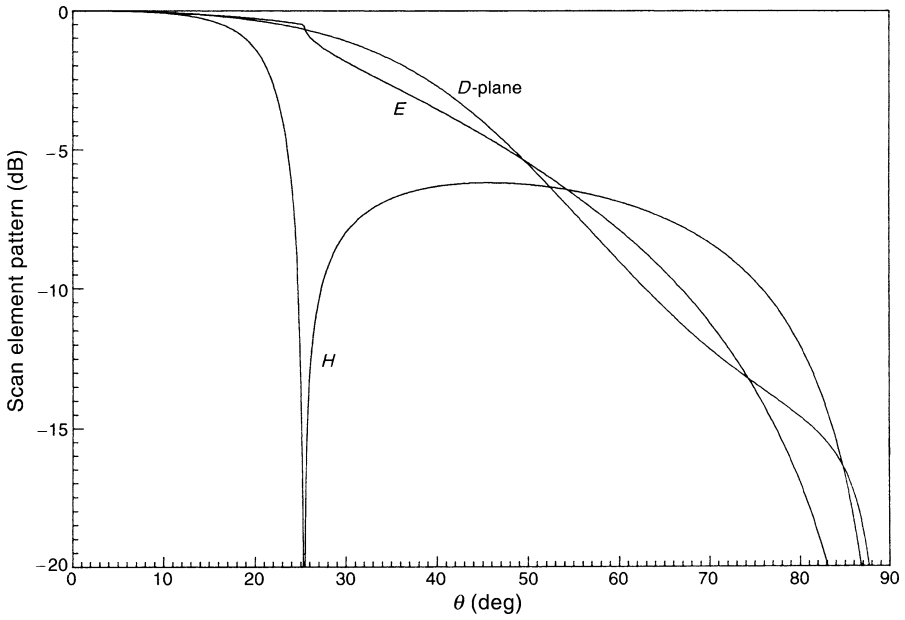


FIGURE 7.28 Dipole array, $D_x = D_y = 0.7$, $L = 0.5\lambda$.

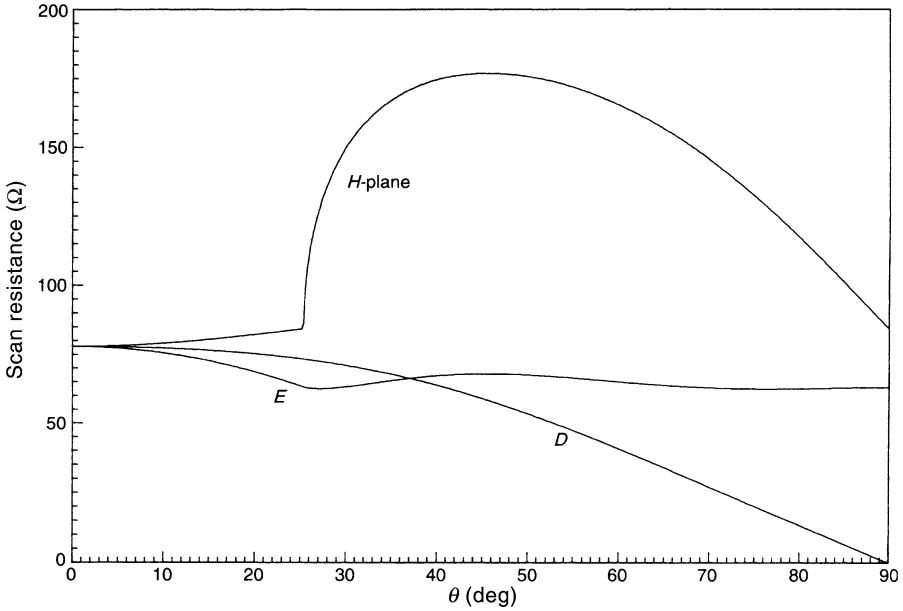


FIGURE 7.29 Dipole–screen array, $D_x = D_y = 0.7$, $L = 0.5\lambda$, $H = 0.25\lambda$.

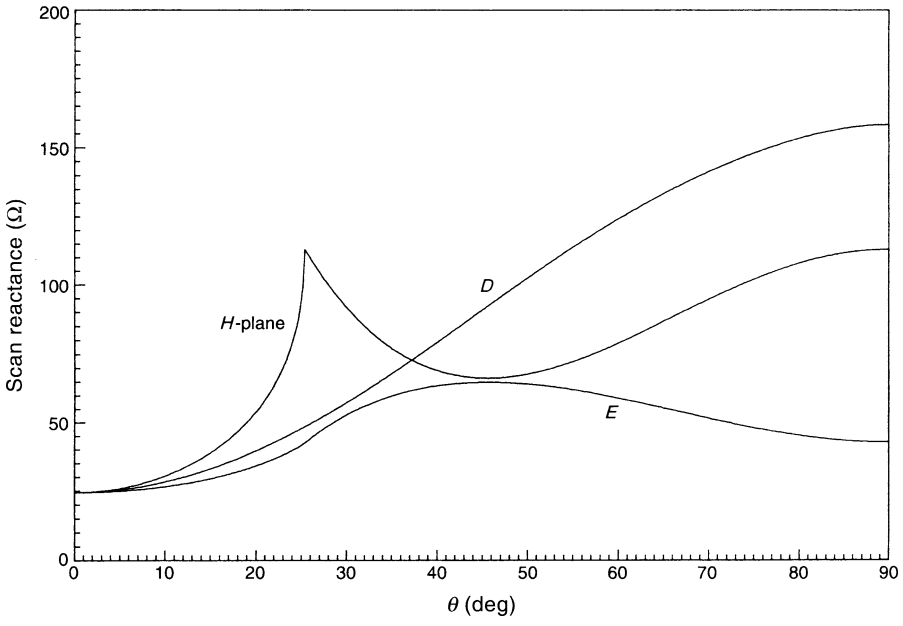


FIGURE 7.30 Dipole–screen array, $D_x = D_y = 0.7$, $L = 0.5\lambda$, $H = 0.25\lambda$.

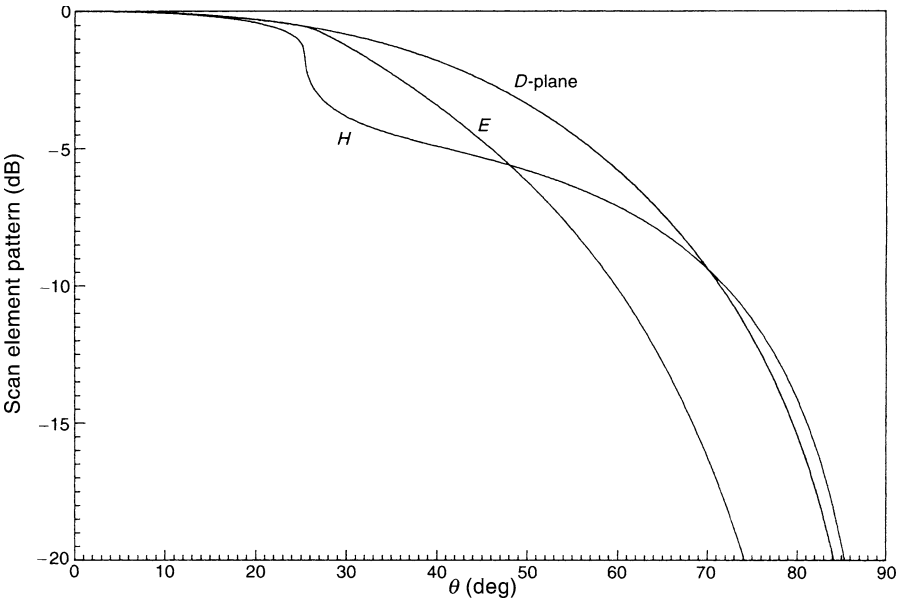


FIGURE 7.31 Dipole–screen array, $D_x = D_y = 0.7$, $L = 0.5\lambda$, $H = 0.25\lambda$.

dipoles. The spatial *scan impedance* is

$$Z = \sum\sum \exp[\,jk(nu + mv)]Z_{nm} \tag{7.51}$$

where Z_{nm} contains three spherical wave terms as derived by Carter, ~1932 (see Balanis, 1982). With dipole radius of a, the three distances, from dipole ends and center, are

$$\begin{aligned} r_1^2 &= (x_0 + a + nd_x)^2 + \left(y + y_0 + md_y - \frac{\ell_s}{2}\right)^2 \\ r_2^2 &= (x_0 + a + nd_x)^2 + (y + y_0 + md_y)^2 \\ r_3^2 &= (x_0 + a + nd_x)^2 + \left(y + y_0 + md_y + \frac{\ell_s}{2}\right)^2 \end{aligned} \tag{7.52}$$

where the single and array dipole lengths are ℓ_s and ℓ_t. The mutual dipole–dipole impedance becomes

$$\begin{aligned} Z_{nm} = \frac{j30}{S_s S_t} \int_{-l_t/2}^{l_t/2} &\left[\frac{\exp(-jkr_1)}{r_1} - 2C_s\frac{\exp(-jkr_0)}{r_0} + \frac{\exp(-jkr_2)}{r_2}\right] \\ &\times \sin k(\ell_t/2 - |y|)\,dy \end{aligned} \tag{7.53}$$

with $S_s = \sin k\ell_s/2$, $S_t = \sin k\ell_t/2$, $C_s = \cos k\ell_s/2$. One of the three mutual impedance terms is shown here:

$$Z_1 = \frac{j30}{S_s S_t} \sum_n \exp(jnk\, d_x u) \int \sin\left[k(\ell_t/2 - |y|)\right] \times \sum_n \exp(jmk\, d_y v) \frac{\exp(-jkr_1)}{r_1} \tag{7.54}$$

The Poisson sum formula combined with the frequency shift theorem provides

$$\sum_m \exp(jmk\, d_y v) \frac{\exp(-jkr_1)}{r_1} = \frac{j\pi}{d_y} \sum_m \exp[jk\beta(y + y_0 - \ell_s/2)] \times H_0^{(2)}\left[k\sqrt{1-\beta^2}(x_0 + a + nd_x)\right] \tag{7.55}$$

where $\beta = v + m\lambda/d_y$. This allows the integral in the Z_1 expression to be collected, and integrated.

$$Z_1 = \frac{-30\pi}{S_s S_t d_y} \sum_n \exp(jnk\, d_x u) \sum_m \exp[-jk\beta(y_0 - \ell_s/2)] \times H_0^{(2)}\left[k\sqrt{1-\beta^2}(x_0 + a + nd_x)\right] \int_{-\ell_t/2}^{\ell_t/2} \exp(jk\beta y) \sin\left[k(\ell_t/2 - |y|)\right] dy \tag{7.56}$$

The integral becomes a generalized pattern function:

$$-\frac{2[\cos(k\beta\ell_t/2) - \cos(k\ell_t/2)]}{k(\beta^2 - 1)} \tag{7.57}$$

The process for the other two Carter terms is the same. The coefficients and exponentials become

$$\exp(jk\beta\ell_s/2) - 2C_s + \exp(-jk\beta\ell_s/2) = 2[(\cos(k\beta\ell_s/2) - \cos(k\ell_s/2)] \tag{7.58}$$

Thus two generalized pattern functions are multiplied, giving

$$P_s P_t = \frac{[\cos(k\beta\ell_s/2) - \cos(k\ell_s/2)][\cos(k\beta\ell_t/2) - \cos(k\beta\ell_t/2)]}{1 - \beta^2} \tag{7.59}$$

The *scan impedance* becomes

$$Z = \frac{-120\pi}{S_s S_t k\, d_y} \sum_n \exp(jnk\, d_x u) \sum_m P_s P_t \exp(jk\beta y_0) \times H_0^{(2)}\left[k\sqrt{1-\beta^2}(x_0 + a + nd_x)\right] \tag{7.60}$$

The previous Poisson transformation can be used to change the Hankel sum into a spectral sum:

$$Z = \frac{2\eta_0}{S_s S_t k^2 d_x d_y} \sum_m P_s P_t \exp(jk\beta y_0) \times \sum_n \exp(jk\alpha x_0) \frac{\exp\left(-jka\sqrt{1-\alpha^2-\beta^2}\right)}{\sqrt{1-\alpha^2-\beta^2}} \tag{7.61}$$

where $\alpha = u + n\lambda/d_x$. This symmetric formula is the spectral domain result for *scan impedance*, developed by Munk and colleagues (see Luebbers and Munk, 1975, 1978; Munk and Burrell 1979, Munk et al., 1979). Similar techniques can be used for parallel dipoles (array of monopoles), or for dipoles at any 3D skew angle (Munk et al., 1979; Munk and Burrell, 1979). Slots are also readily handled (Luebbers and Munk, 1978). Moment method techniques can use this methodology as it applies directly to the highly efficient piecewise sinusoidal expansion and test functions (Larson and Munk, 1983). The resulting codes, such as the *periodic moment method*, have found wide use. This spectral domain method has also been applied to finite-by-infinite arrays (see Chapter 8).

7.4.2 Microstrip Patches

Large (and infinite) arrays of patch elements are also amenable to periodic cell, Floquet series-based analyses. This was pioneered by Liu et al. (1982) for an infinite array of patch strips with *E*-plane scan; a moment method approach was used in the spectral domain with sinusoidal current expansion functions. For a discussion of spectral moment methods for layered media, see Mosig (1989). It is not surprising that the *scan impedance* variation with angle follows closely that for dipole arrays (see Section 7.4.1). These computer simulations showed a leaky wave resonance that produces a peak in *scan resistance* versus angle; the peak becomes sharper and higher, and occurs at a smaller scan angle as the substrate thickness increases. This work was extended to printed dipole arrays and then to arrays of rectangular patches by Pozar and Schaubert (1984a). Entire domain trigonometric expansion functions were used in the spectral domain moment method. Blind angles due to both grating lobe incidence and surface wave excitation were found; several cases are of interest. A thin ($0.02\lambda_0$) substrate with $\varepsilon_r = 2.55$ and $\lambda_0/2$ spacings showed well-behaved *scan resistance* except for an *E*-plane blind angle at $\theta = 82.9^\circ$, which corresponds to the surface wave $\beta/k = 1.0076$, where $\sin\theta = k/\beta$. A thicker substrate ($0.06\lambda_0$) gives a surface

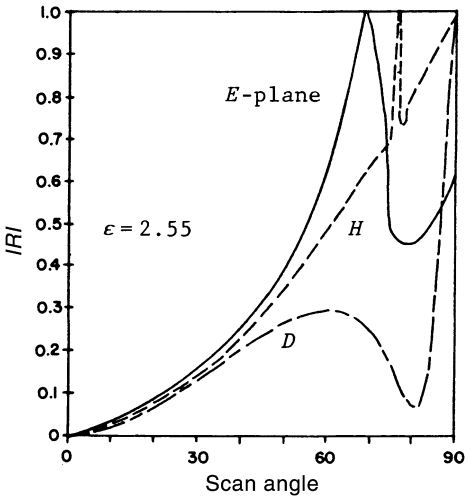


FIGURE 7.32 Patch array scan reflection coefficient. (Courtesy Pozar, D. M. and Schaubert, D. H., "Scan Blindness in Infinite Phased Arrays of Printed Dipoles", *Trans. IEEE*, Vol. AP-32, June 1984, pp. 602–610.)

wave $\beta/k = 1.028$, with blind angle of 68.8 in the E-plane, as seen in Figure 7.32. This calculation uses an E-plane element spacing of $0.51\lambda_0$, so a grating lobe appears at 73.9°. When a higher dielectric constant is used ($\varepsilon = 12.8$) for the thick substrate, with half-wave lattice, the blind angle is now 45.6°, as seen in Figure 7.33. This analysis by Pozar and Schaubert did not use probe current attachment modes, direct probe radiation, or edge condition expansion functions. The latter affects convergence, but the probe factors may here produce artifactual H-plane blind angles. Thus these have been eliminated in Figures 7.32 and 7.33.

Liu et al. (1988) used a moment method approach with probe attachment modes and with edge singularity matching expansion functions. Figure 7.34 shows, for square patches on a square lattice, that *scan resistance* becomes flatter and lower for thicker substrates. These resonant peaks are due to a phase match between the grating lobe and the leaky wave:

$$\sin \theta_{\ell w} = \left| \frac{\beta}{k} - \frac{\lambda}{d_E} \right| \tag{7.62}$$

where $\alpha + j\beta$ is the leaky wave propagation constant. The Q resonance, as seen in the figure, is given by

$$Q = \frac{\pi \sin \theta_{\ell w}}{\alpha \lambda} \tag{7.63}$$

Design guidelines from Liu et al. (1988) are as follows:

a. Too low an ε requires a large patch with light loading and rapid scan variation. Too high an ε gives a narrow bandwidth. A good range is $2 \leq \varepsilon \leq 5$.

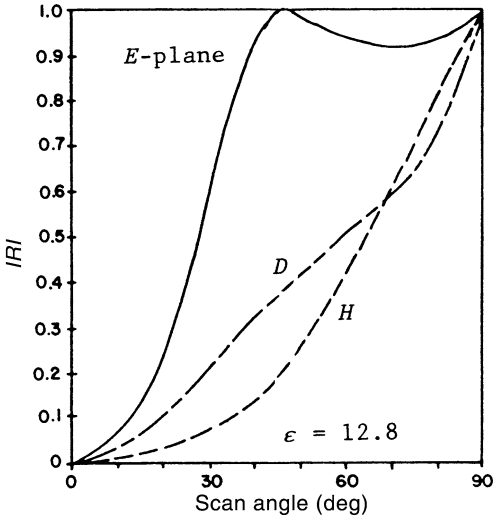


FIGURE 7.33 Patch array scan reflection coefficient. (Courtesy Pozar, D. M. and Schaubert, D. H., "Scan Blindness in Infinite Phased Arrays of Printed Dipoles", *Trans. IEEE*, Vol. AP-32, June 1984, pp. 602–610.)

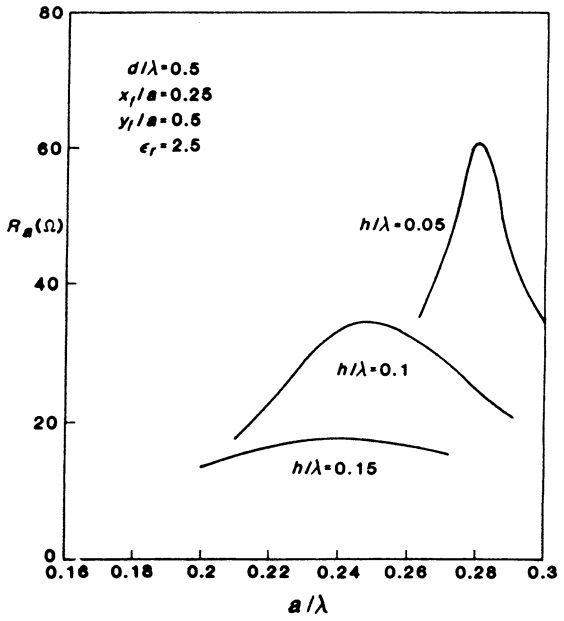


FIGURE 7.34 Patch array scan resistance. (Courtesy Liu, C. C., Hessel, A., and Shmoys, J., "Performance of Probe-Fed Microstrip-Patch Element Phased Arrays", *Trans. IEEE*, Vol. AP-36, Nov. 1988, pp. 1501–1509.)

b. A small thickness, such as $0.05\lambda_0$, reduces scan variation.
c. The lattice should preclude *E*-plane blind angles, and allow a reasonable slope there. The *H*-plane grating lobes should be precluded.
d. A transverse width roughly equal to one-half of the transverse spacing will avoid *H*-plane TE resonances.
e. Select patch length to resonate at optimum angle (may be broadside).
f. Examine *scan reflection coefficient* versus angle over frequency band.
g. Iterate design as needed.

Guidelines for improving the bandwidth performances of scanned patch arrays are given by Nehra et al. (1995).

In Chapter 5, it was shown how a parasitic (stacked) patch could roughly double the bandwidth. Arrays of patches with parasitic patches are analyzed by Lubin and Hessel (1991), again via the moment method in the spectral domain. The behavior of a stacked patch may be understood as follows. The driven and parasitic patches produce two resonances that should be separated (for good bandwidth). The driven patch with both dielectric layers present but no parasite approximately controls the lower resonant frequency. The parasitic patch on the upper dielectric layer, but parallel to the metallic sheet, approximately controls the upper resonant frequency. The patch size ratio affects mostly the resonant peak relative amplitudes. Element spacing should be selected to place resonance peaks beyond the upper frequency, at the maximum scan angle. Bandwidth can be adjusted by changing ε of upper and lower dielectric layers, with average held roughly constant. Adjusting the layer thicknesses allows the two resonance Q's to be equalized, to achieve a smoother performance. Spectral convergence has been studied by Aberle et al. (1994); they also utilize waveguide simulators (see Chapter 15).

Scan performance may be improved through use of a shorting pin on each patch; this obviates surface waves and thus eliminates blind angles due to them (Waterhouse, 1996). The cost of this is a larger patch, but still $< \lambda/2$, so these elements may be arrayed.

It may be said that the basic *scan impedance* behavior is controlled primarily by element lattice (grating lobe blindness) and by substrate ε and t (surface wave blindness); the type of element is less important. In all of these moment method array simulations, the convergence of the series is affected both by the number of spectral terms in x and in y and by the number of expansion functions in x and in y. As shown by Munk and others, these must be carefully chosen to assure accurate results. Acceleration techniques can often significantly reduce computation time. Two useful ones are by Shanks (1955) and Levin (1973)—the latter does not accumulate roundoff errors [see also Singh and Singh (1993)]. This author has had excellent success with Levin acceleration in spectral domain simulations.

7.4.3 Printed Dipoles

For convenience in manufacture, strip dipoles and feed lines are sometimes printed on dielectric sheets, with the sheets normal to the ground plane. Figure 7.35 sketches a

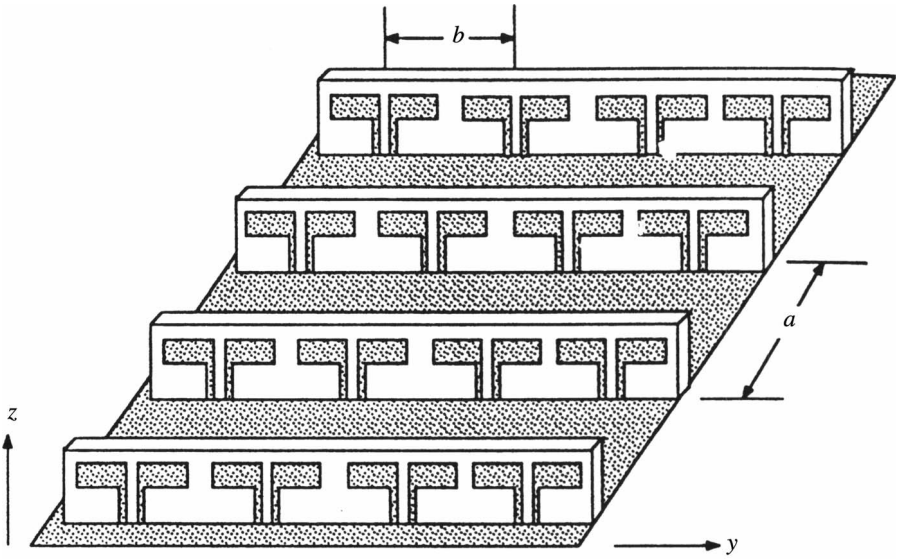


FIGURE 7.35 Array of printed dipoles.

typical configuration. The dielectric sheets, finite in height above the ground plane, must be taken into account. This has been done in the spectral domain using LSE and LSM hybrid waveguide modes by Chu and Lee (1987), where a current distribution on the dipoles was assumed. A combined spectral domain, moment method approach solves for the dipole currents, as well as for *scan impedance* (Bayard et al., 1991). Excellent agreement with the spectral results of Section 7.4.1 is shown in Figures 7.36 and 7.37, with $\varepsilon = 1$; these figures are for half-wave dipoles on a $0.6\lambda \times 0.6\lambda$ lattice. As expected, scan blind angles are observed for *E*-plane scan. A blind angle can be produced by a surface wave excited along each dielectric sheet, with the location controlled by the sheet thickness and dielectric constant. More serious is the blind angle caused by excitation of an asymmetric mode in the dipole and feed lines; this occurs even for free-standing dipoles. Figure 7.38 sketches the two current modes. A remedy was found by Reale (1974), wherein the dipole arms are bent into a vee, with the dipole ends closer to the ground plane. A number of large air defense radars have been built using fat (not printed) vee dipoles. A moment method analysis of strip vee dipole arrays has been given by Bayard and Cooley (1995). Dielectric radome covers over printed dipole arrays have been analyzed also, using the spectral domain technique (Bayard, 1994).

7.4.4 Printed TEM Horns

The TEM horns of Chapter 5 can be arrayed in both planes. Although earlier work was on printed circuit horns with a linear taper (Cooley et al., 1991), as one might expect from waveguide horn experience, tapers that are roughly exponential give better results. Figure 7.39 depicts a TEM horn array; each horn includes a slotline cavity

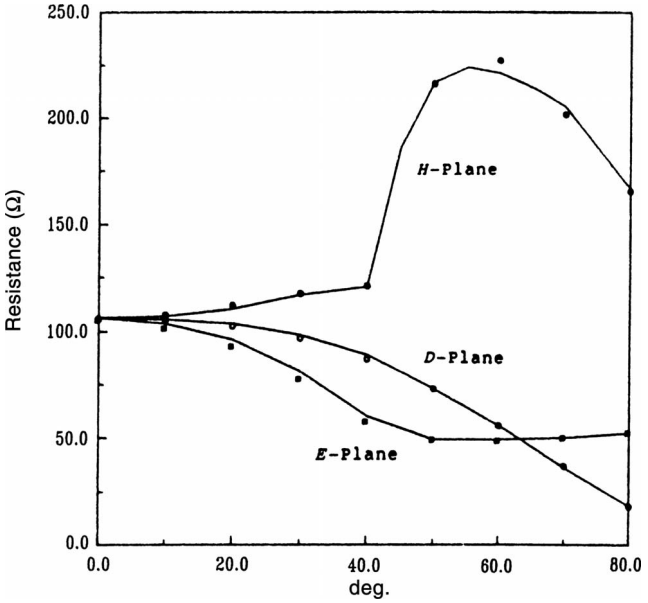


FIGURE 7.36 Printed dipole array scan resistance. (Courtesy Bayard, J.-P. R., Cooley, M. E., and Schaubert, D. H., "Analysis of Infinite Arrays of Printed Dipoles on Dielectric Sheets Perpendicular to a Ground Plane", *Trans. IEEE*, Vol. AP-39, Dec. 1991, pp. 1722–1732.)

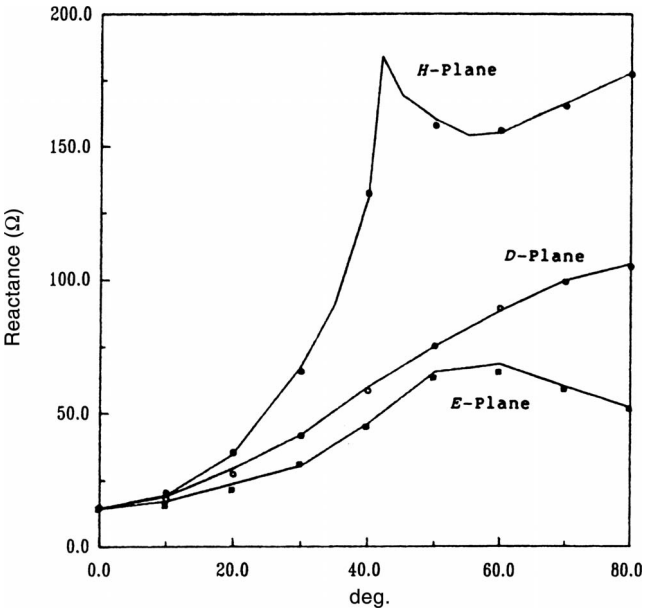


FIGURE 7.37 Printed dipole array scan reactance. (Courtesy Bayard, J.-P. R., Cooley, M. E., and Schaubert, D. H., "Analysis of Infinite Arrays of Printed Dipoles on Dielectric Sheets Perpendicular to a Ground Plane", *Trans. IEEE*, Vol. AP-39, Dec. 1991, pp. 1722–1732.)

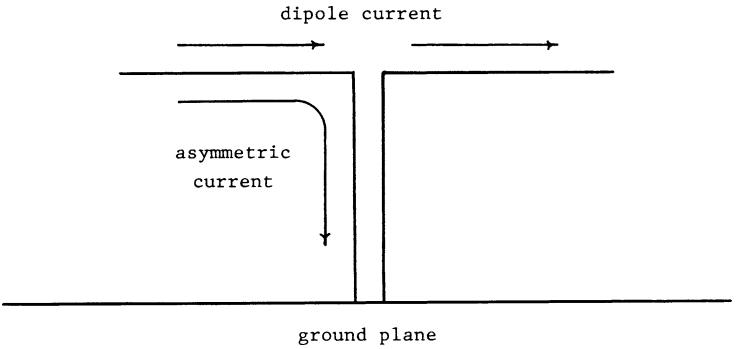


FIGURE 7.38 Dipole current modes due to scan.

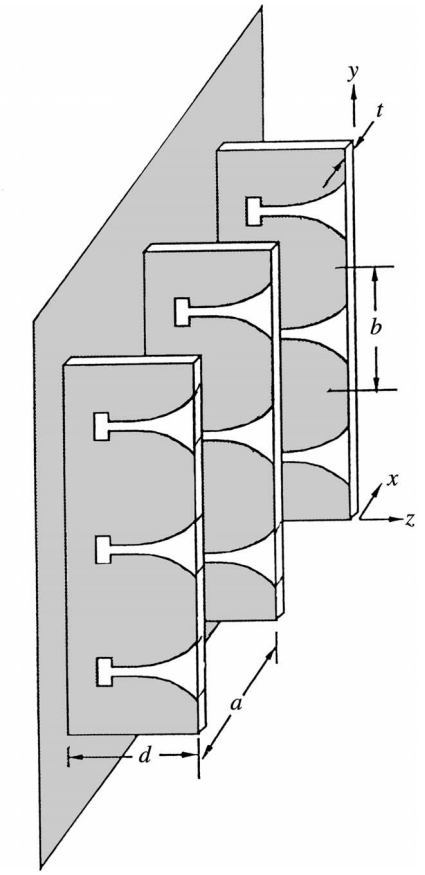


FIGURE 7.39 Array of flat TEM horns.

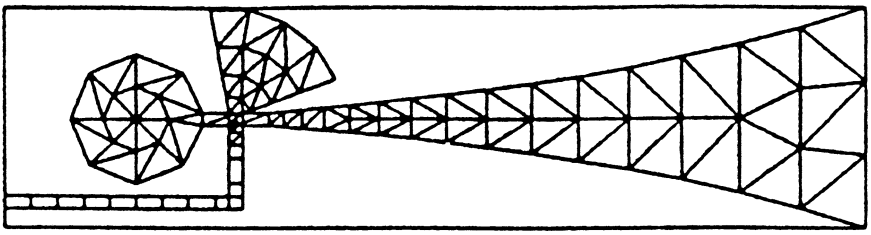


FIGURE 7.40 TEM horn modelling. (Courtesy Schaubert, D. H. and Shin, J., "Parameter Study of Tapered Slot Antenna Arrays", *IEEE AP 1995 Symposium Digest,* Newport Beach, CA, June 1995, pp. 1376–1379.)

at the narrow end of the horn, and a stripline feed line and stub. A patch moment method has been used to model the horn and feed, as sketched in Figure 7.40. Extensive parameter studies have been made by Schaubert and Shin (1995) and Shin and Schaubert (1995). Some typical results are very useful. For a broadside array, a VSWR ≤ 2 occurs over a two-octave bandwidth. At the lowest frequency, the *E*-plane lattice spacing is 0.139λ while the *H*-plane lattice spacing is 0.150λ. The horn length is roughly twice the horn mouth, while the latter is 0.095λ at the lowest frequency. Thus in an array, the horns can be smaller than a free-standing horn. A reduced array lattice allows scanning over a $\pm 45^\circ$ range, as shown in Figure 7.41. Here at the lowest frequency the lattice spacing and the mouth are

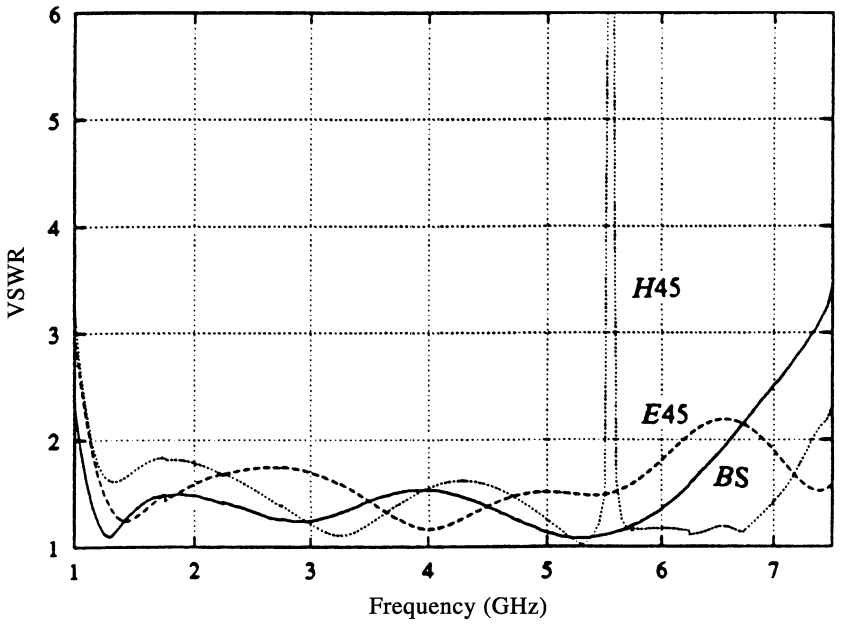


FIGURE 7.41 Flat TEM horn array. (Courtesy Schaubert, D. H. and Shin, J., "Parameter Study of Tapered Slot Antenna Arrays", *IEEE AP-1985 Symposium Digest*, Newport Beach, CA, June 1995, pp. 1376–1379.)

0.065λ; a remarkable result. A blind angle occurs in the *H*-plane, at roughly a lattice spacing of 0.312λ. This may be caused by a surface wave supported by the corrugated surface comprised of the metallic horn sheets; the blind angle is a strong function of *E*-plane spacing, and a less strong function of *H*-plane spacing (Shin and Schaubert, 1995). Another type of blindness occurs when the *H*-plane spacing exceeds half-wavelength. This *E*-plane scan blind angle is due to the corrugated nature along one horn sheet; the surface wave excitation depends on *E*-plane spacing and on the depth from horn mouth-to-ground screen (Schaubert, 1996).

The horn throat cavity and the feedline stub have significant effects upon the low and high ends of the useful frequency band, while the middle of the band is most affected by the horn taper (Schaubert et al., 1996). Care must be taken with any gaps in the printed conductors as they can induce resonances, with high VSWR (Schaubert, 1994). Gaps are sometimes introduced for ease of fabrication of 2D arrays, or to reduce mutual coupling.

7.4.5 Unit Cell Simulators

At certain scan angles and planes, the unit cell walls become perfectly conducting. This allows the unit cell, and thus the array performance, to be simulated in a waveguide with an iris containing an excerpt of the unit cells. These powerful measurement tools are discussed in Chapter 15.

7.5 SCAN COMPENSATION AND BLIND ANGLES

7.5.1 Blind Angles

Some arrays radiate no power at a certain angle; this is called a blind angle. This phenomenon can, like array performance in general, be viewed from the standpoint of either *scan reflection coefficient* or *scan element pattern*. The *scan reflection coefficient* may develop a magnitude near to or equal to unity, or the *scan element pattern* may be near to or equal to zero. First, taking the element *scan reflection coefficient* case, the blind angle occurs when a higher mode cancels the dominant mode in the element. Since the relative phase changes rapidly with scan angle, the two modes produce a resonance, with a resonant phase angle that can be defined on the decrease of $|\Gamma|$ with scan angle. The higher mode can be produced by external or internal structure. Examples of external structure causing a blind angle are a dielectric radome covering each element or row of elements (Lechtreck, 1968), a dielectric sheet placed on the array face (Bates, 1965; Byron and Frank, 1968; Wu and Galindo, 1968; Gregorwich et al., 1968), and a dielectric plug protruding from the mouth of each waveguide element (Hannan, 1967a). In all cases, the active wave impedance at the array face is modified by the external structure so as to produce a dominant and higher mode resonance at a particular scan angle; the higher mode exists primarily outside the face. Examples of internal structure that can allow a higher mode that can cause a blind angle are dielectric plugs flush with the array face (Amitay and Galindo,

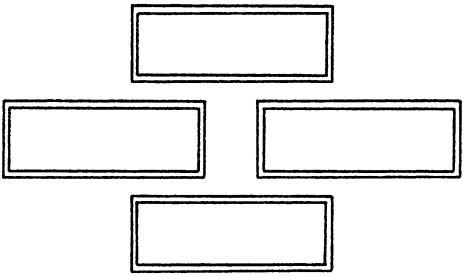


FIGURE 7.42 Brick waveguide array.

1968), dielectric-loaded wave guide elements (as opposed to a short plug) (Diamond, 1967), and a brick array of rectangular guide elements (Diamond, 1967). The latter is sketched in Figure 7.42. For these various structures, the waveguide element supports the next higher mode, which causes the resonance at the blind angle. For the brick array, the higher mode is just below cutoff owing to the large guide size. An equivalent circuit for the two modes has been used by Oliner (1969) and Knittel et al. (1968), to predict the reflection coefficient behavior. Complete reflection corresponds to an active conductance that is zero. Simulators (see Section 7.4) have been used both to verify the presence of blind angles and to preclude their occurrence through design.

From the *scan element pattern* viewpoint, the *scan impedance* at the array face, including the effect of external or internal structure, allows a leaky wave to be excited. This leaky wave is usually backward, and its amplitude depends on the *scan impedance* and on the length of the impedance surface (the array face). Thus small arrays usually have no blind angle or only a small dip instead of a null, while very large arrays are more susceptible as it is easier to excite the leaky wave. The leaky wave has a complex wavenumber, analogous to Wood anomalies on an optical reflection grating (Hessel and Oliner, 1965). When the angle and phase of the leaky waves are suitable, the direct radiation at the angle will be canceled, providing a *scan element pattern* null, or blind angle (see Fig. 7.43). The leaky-wave pole gives an approximate behavior of cancellation around the blind angle, where Γ is expanded into a Taylor series about the pole. Figure 7.44 shows the *scan reflection coefficient* for an array of narrow slots fed by parallel-plate waveguides and covered by a dielectric slab, for *E*-plane scan. This curve was calculated using 21 modes. Slot spacing was $\lambda/2$, waveguide width was 0.3λ, while the slab had $\varepsilon = 2.56$ and thickness 0.2λ. By using the leaky-wave pole, Knittel et al. (1968) calculated the dots shown in the figure, where the leaky-wave Q is given by

$$Q \simeq \frac{u^2 \ \mathrm{Re}(k_x/k)}{2(1-u^2)\mathrm{I_m}(k_x/k)} \tag{7.64}$$

where k_x is the wavenumber along the array. For this array the grating lobe occurs at 90°, and the *scan reflection coefficient* is also unity there. Another example used a brick array of rectangular guides $0.905 \times 0.4\lambda$, on a square brick lattice of 1.008λ

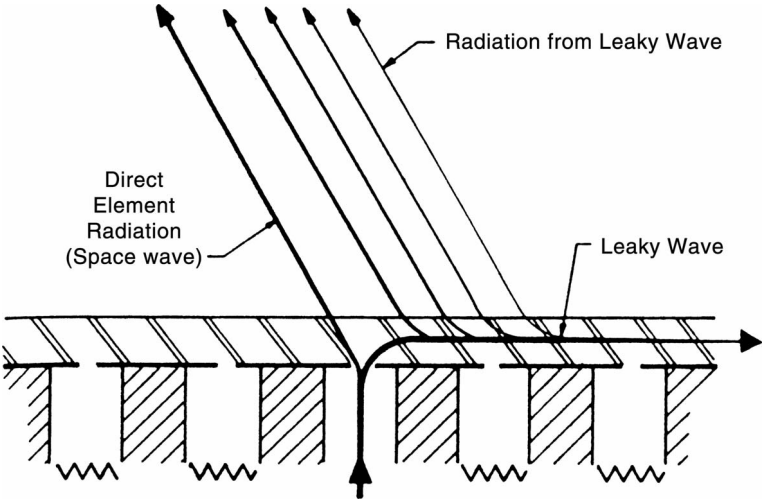


FIGURE 7.43 Blind angle due to wave cancellation. (Courtesy Knittel, G. H., Hessel, A., and Oliner, A. A., "Element Pattern Nulls in Phased Arrays and Their Relation to Guided Waves", *Proc. IEEE*, Vol. 56, Nov. 1968, pp. 1822–1836.)

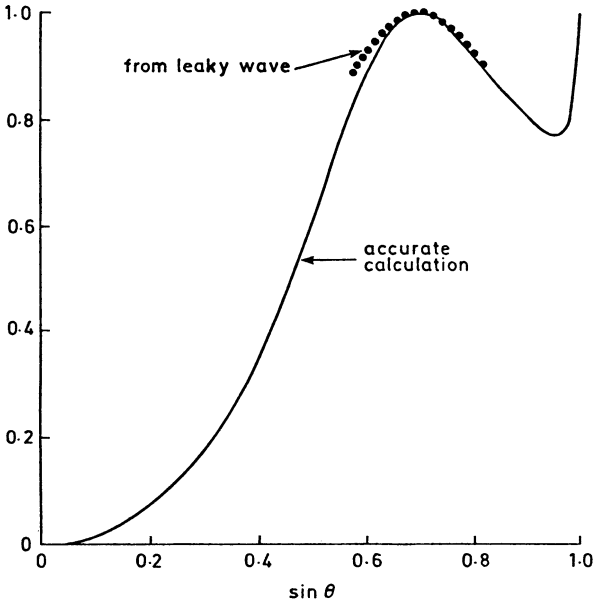


FIGURE 7.44 Blind angle of array of slits. (Courtesy Knittel, G. H., Hessel, A., and Oliner, A. A., "Element Pattern Nulls in Phased Arrays and Their Relation to Guided Waves", *Proc. IEEE*, Vol. 56, Nov. 1968, pp. 1822–1836.)

spacing. An array of only 7×7 elements exhibits a blind angle over 26 dB deep in the H-plane at 27° (Farrell and Kuhn, 1966). A grating lobe occurs on the 60° circle in u,v space; the blind angle is usually just inside the grating lobe, but in this example it is well inside (Oliner, 1972).

The understanding of how blind angles arise also gives indications of corrective measures to take (Knittel, 1972; Oliner, 1972). These include reducing dimensions to preclude–eliminate higher propagating modes, alternating external structure or external impedance, and changing various parameters to obtain a suitably low VSWR over a wide scan angle range.

7.5.2 Scan Compensation

Scan impedance changes can be compensated by any of four different methods. These are reduction of coupling, feed networks, multimode elements, and external wave transformers.

7.5.2.1 Coupling Reduction. The simplest way to reduce relative coupling is to pack the elements closely together in wavelengths. The results in Section 7.4 showed that the admittance variation with scan increases as grating lobe onset is approached. Reducing the element spacing below half-wave reduces the admittance variation owing to the grating lobe being farther away (in invisible space). This is, of course, difficult without going to narrowband elements. A more practical scheme uses H-plane baffles between slots or dipoles (Edelberg and Oliner, 1960). These baffles are simply metal strips perpendicular to the ground plane and running between the dipoles or slots rows in the H-plane, as shown in Figure 7.45. Figure 7.46 shows the improvement that can be realized. In this example, the dipoles are spaced 0.5λ apart and 0.25λ above a ground plane. Use of baffles $\sim 0.5\lambda$ high makes the E- and H-plane patterns closely alike. Over a 60° conical scan volume a VSWR ≤ 1.6 can be achieved. An analysis of the baffled array in terms of surface waves is given by Mailloux (1972). A moment method analysis of a dipole array with fences has been given by Lee (1990). The L-Shaped fences, with a flange parallel to the ground plane, are useful for a linear array (Forooraghi and Kildal, 1995).

Use of baffles is effective and inexpensive for arrays that scan only in one plane, provided the space, weight, and fabrication can be managed. However, some elements behave less well in the H-plane than the dipole–screen array of Figure 7.46. Rengarajan (1996), using a spectral domain moment method, shows the reduction in coupling provided by baffles.

Another scheme changes the impedance of the ground plane between elements by introducing corrugations (Dufort, 1968). There is a tradeoff here in that it may be necessary to increase element spacing to allow room for the corrugations.

7.5.2.2 Compensating Feed Networks. A complex matrix-type feed network that adapts its outputs according to the driving phase shifts could, in principle, compensate for most of the mutual impedance effects. Such a network connected between the antenna elements and the generators can be used either to make reflectionless



FIGURE 7.45 Edge slot array with baffles. (Courtesy Hughes Aircraft Co.)

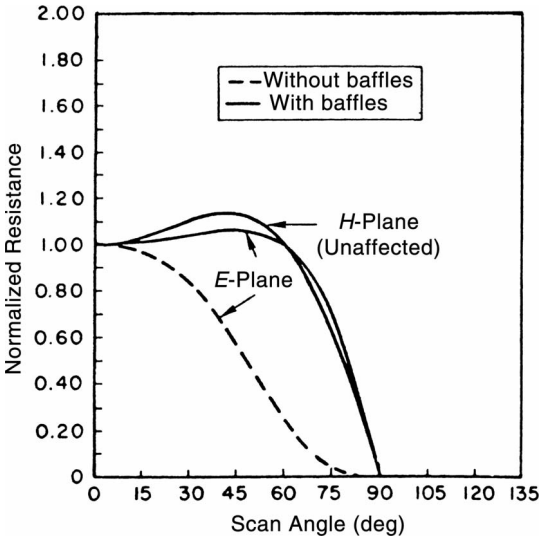


FIGURE 7.46 Scan resistance with baffles. (Courtesy Edelberg, S. and Oliner, A. A., "Mutual Coupling Effects in Large Antenna Arrays. Part I: Slot Arrays", and "Part II: Compensation Effects", *Trans. IRE*, Vol. AP-8, May and July 1960, pp. 286–297, 360–367.)

elements, or to make scatterless elements. The former is easier and will be discussed first. By use of successively more complex networks, the situation where $\Gamma = 0$ may be closely approached (Hannan, 1967b). However, an exact match at all angles can only be approximated (Varon and Zysman, 1968). Both methods improve the match by making the scan element power pattern closer to $\cos\theta$ (for a planar array with no grating lobes).

As a means of understanding the matching network process, consider the network to consist of three networks in cascade. The first network is a $2N$-port that transforms the array excitation vectors into eigenvectors of the *scan impedance* matrix. The second network consists of N separate 2-ports that equalize the eigenvalues. This process is analogous to the convergence speedup in adaptive arrays where the vectors are first orthogonalized, then their amplitudes are made equal. The third network is a $2N$-port that is the inverse of the first; it restores the array excitation vectors, which have now been matched. An excitation in an infinite array producing a beam in invisible space is represented by a singular (zero or infinite) eigen excitation. For finite arrays, such excitations produce very small or very large eigen excitations. Attempts to match such eigenvectors produce tolerance problems similar to those encountered in superdirectivity. Kahn (1977) has calculated *scan element patterns* for a linear array of uniform line sources. Figure 7.47 is for an element spacing of $\lambda/2$, for the center element of 25-element array. The solid line is the ideal $\cos\theta$ and the dashed line is the actual element pattern. The fit is excellent, and for larger arrays the oscillations will be more numerous and smaller. Care must be exercised for smaller spacings, where inclusion of small eigenvalues may cause instability.

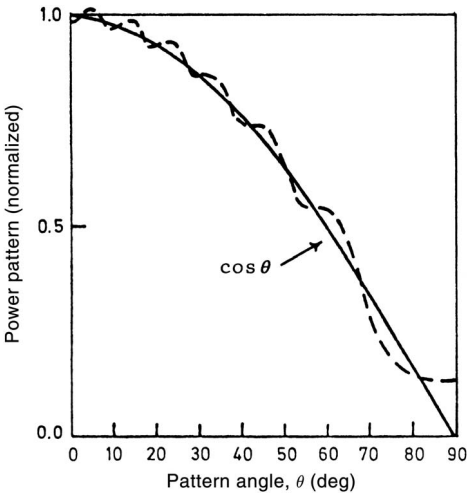


FIGURE 7.47 Scan element pattern of 25-element array. (Courtesy Kahn, W. K., "Impedance-Match and Element-Pattern Constraints for Finite Arrays", *Trans. IEEE*, Vol. AP-25, Nov. 1977, pp. 747–755.)

Descattering the array via a network requires that the element transmit and receive patterns be equal, that is, the elements are minimum scattering (see Section 7.2.4). In practice, descattering may be useful for single-mode elements, such as slots and dipoles. A further requirement is that all mutual impedances be reactive (Andersen and Rasmussen, 1976). A corollary is that the mutual resistance, since it is proportional to the beam orthogonality integral, must equal zero (Wasylkiwskyj and Kahn, 1970). Now the mutual impedance is related to the element power pattern; so the zero mutual resistance becomes a constraint on the element pattern. For example, rotationally symmetric power patterns of $\cos^{\mathrm{m}} \theta$ yield $R_{12} = 0$ for spacings that satisfy

$$j_{m/2}(kd) = 0 \tag{7.65}$$

where $j_{m/2}$ is the spherical Bessel function of order $m/2$. Pursuing the example further for isotropic elements ($m = 0$), the root gives $d = \lambda/2$, so that a half-wave spaced array may be completely decoupled and descattered. When a grating lobe just enters visible space, the *scan element pattern* has a null (at that angle), or in *scan impedance* terms the reflection coefficient magnitude is unity. Such a blind angle cannot be removed by a matching network (Wasylkiwskyj and Kahn, 1977). However, for those blind angles that occur at scan ranges inside the grating lobe scan contour (in u,v space), a suitable network can improve or even remove the null, provided that the open-circuited element pattern does not have a null at that angle.

A variation on the compensating network may be used with a receiving array with digital beam forming (Steyskal and Herd, 1990). Here the vector of array outputs is multiplied by a matrix designed to produce a scan element power pattern of $\cos \theta$. The matrix elements are based on a numerical simulation of the array *scan impedance* pattern.

Although the single-scan-plane compensation techniques can be effective, they are not often used. Instead, the array feeds can be matched to the *scan impedance* at an intermediate angle between broadside and maximum scan angle, so that the *scan element pattern* loss is equalized. This is frequently sufficiently good that no complicated techniques are warranted.

7.5.2.3 Multimode Elements. The ideal element pattern discussions presented earlier imply that both TE and TM modes are needed in the array unit cell to allow a $\cos \theta$ scan element power pattern. The additional modes can be produced by modifying the radiating element in any of several ways. Probably the most effective way of producing the optimum mix of modes uses open-end waveguide radiators, rectangular or circular, with the waveguide internal environment or the external environment suitably modified. Tang and Wong (1968) and Wong et al. (1972) use a dielectric plug at the rectangular guide mouth and a matching post. Tsandoulas (1972) uses rectangular waveguide width and height steps (in different planes). Tsandoulas and Knittel (1973) use both waveguide size steps and dielectric plugs at the mouth. Wu (1972) uses plugs in circular guide elements. The mix of

modes that is needed depends on the waveguide and lattice dimensions. A typical mode set might be TE_{10}, TE_{20}, TE_{30}, TE_{11}, TE_{01}, TM_{11}. With a dielectric plug in the waveguide the TE_{20} mode will likely be propagating; all other higher modes will be evanescent. The plug should be half-wave long (in loaded guide wavelengths) at center frequency to properly locate the *scan admittance* on the Smith chart. Then the guide dimensions are adjusted to give a high susceptance for the TE_{20} mode looking into the matching network. Steps in guide height can then be used for center angle impedance matching; additional dielectric plugs can also be used for matching. Figure 7.48 sketches a typical guide configuration, including two sheets over the array face. All of these are designed by a mode matching technique, where propagating and evanescent waveguide modes are matched at each boundary. This is conveniently performed in the spectral domain. Since the unit cell at scan angles other than broadside has impedance walls, the modes are LSE and LSM rather than TE and TM. It is necessary to assume values of all physical parameters to start the process: slab dielectric constants and thicknesses, plug dielectric constants and lengths, waveguide heights and widths and, of course, the element lattice dimensions. A mode set for the waveguide element is then assembled. Choosing a reasonably good set of all these parameters requires considerable experience, gained mostly by running various cases and then examining the results. With all parameters set, the unit-cell admittance is calculated for each waveguide mode, assuming unit amplitude of the latter. Many modes will be required in the unit cell to match the waveguide mode field at the aperture, and for each unit cell mode the admittance must be calculated. Starting with the value for large distance (from the slabs), this admittance is transformed through the dielectric slabs, and then combined with the other unit cell modes. Numbers of modes typically are

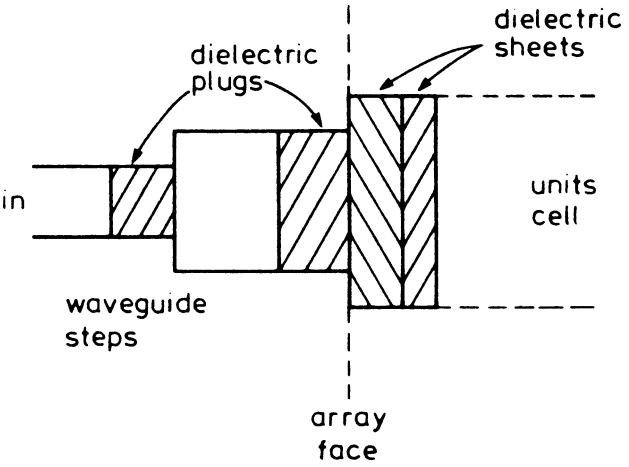


FIGURE 7.48 Typical open guide matching. (Courtesy Tsandoulas, G. N. and Knittel, G. H., "The Analysis and Design of Dual-Polarization Square-Waveguide Phased Arrays", *Trans. IEEE*, Vol. AP-21, Nov. 1973, pp. 796–808.)

100–300. Longitudinal-section modes must be converted to TE and TM modes so that mode admittances can be matched. Next, starting with an incident TE_{10} mode in the waveguide, the aperture admittance of each waveguide mode is calculated. Although the mode admittances must be equal across the aperture, the mode amplitudes depend on the total admittance. Thus, a set of simultaneous equations one for each waveguide mode, can be solved. Typical results are shown in Figure 7.49, which is for $0.403\lambda_0$ square waveguides in an equilateral triangular lattice (Tsandoulas and Knittel, 1973). Bandwidth for VSWR ≤ 3 is 25%, and the scan range is roughly a quarter hemisphere. The solid line is *E*-plane; the dashed line is *H*-plane. An interesting design principle resulting from many calculations by these authors is that polarization coupling (for a dual polarized array) is reduced as the element size is increased. Thus when multimode elements are used they should be as large as the lattice allows. For a single polarization, bandwidth of $>50\%$ with VSWR ≤ 4 and scan over nearly a quarter hemisphere has been obtained (Tsandoulas, 1972, 1980).

An iris can be used in each waveguide radiator, below the aperture, in addition to dielectric plugs, as an aid to matching. Excellent results have been obtained by Lee

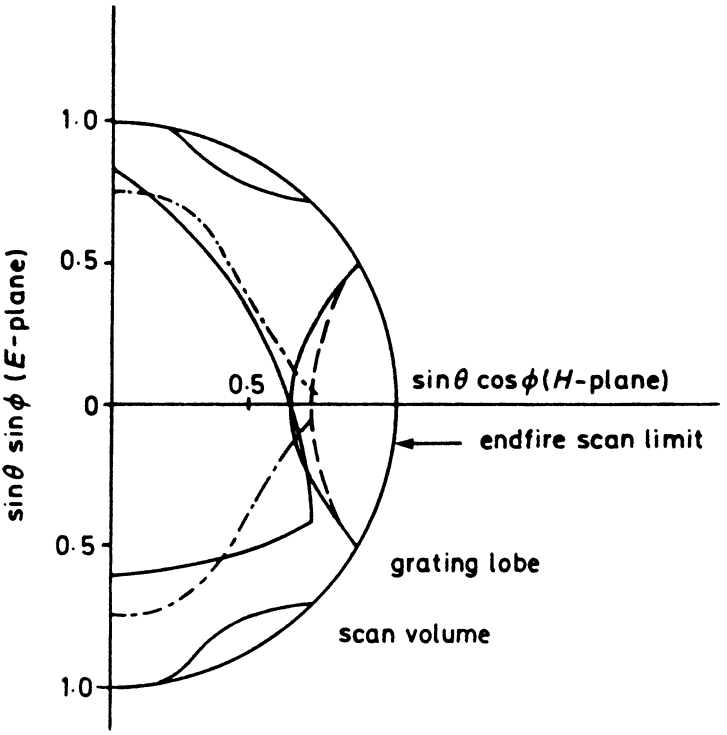


FIGURE 7.49 VSWR = 3 contour for waveguide array. (Courtesy Tsandoulas, G. N. and Knittel, G. H., "The Analysis and Design of Dual-Polarization Square-Waveguide Phased Arrays", *Trans. IEEE*, Vol. AP-21, Nov. 1973, pp. 796–808.)

and Jones (1971) and by Lee (1971). Excellent results have been demonstrated over wide scan angles. One or more dielectric sheets may be placed on the array face as part of the mode matching process (Wu, 1972).

Another scheme combines the TE and TM radiators. Figure 7.50 shows a longitudinal shunt slot array with parasitic monopoles with L-shaped extensions developed by Clavin et al. (1974). The monopoles reduce the *E*-plane beamwidth and add an electric-type mode to complement the slot magnetic-type mode. Patterns in *E*- and *H*-planes can be made equal (Hansen, 1985). Mutual coupling between elements is reduced 14 dB, and the array wide-angle sidelobes and backlobe are significantly reduced. These arrays have been analyzed by the periodic cell spectral domain technique, using a piecewise expansion on the monopole wires (Ng and Munk, 1988). The parasitic array can support surface waves and these can cause blind angles in some cases (Ng and Munk, 1989). With the spectral domain analysis, these problems can be obviated.

A different use of a dielectric sheet involves placing a high dielectric constant thin sheet near the array face. This technique, called wide-angle impedance matching (WAIM) by Wheeler (Magill and Wheeler, 1966), uses the susceptances of the sheet to equalize part of the aperture susceptance variation with scan. This is possible because the sheet susceptance varies differently in the *E*- and *H*-plane. Using the well-known formulas (Collin, 1991) for reflection from lossless dielectric sheet, the susceptance is obtained for a thin sheet, where this means $kt\sqrt{\varepsilon_r} \ll 1$. Thus higher dielectric

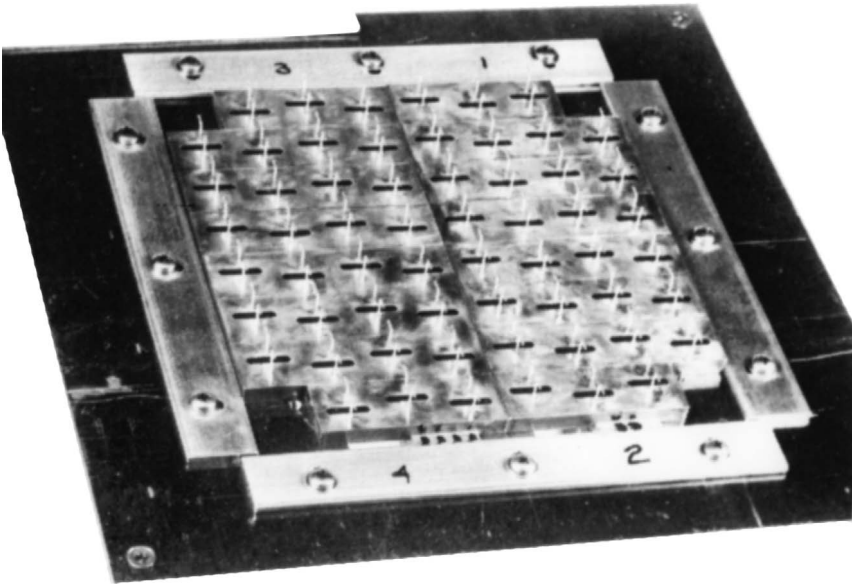


FIGURE 7.50 Slots with parasitic elements. (Courtesy Clavin, A., Huebner, D. A., and Kilburg, F. J., "An Improved Element for Use in Array Antennas", *Trans. IEEE*, Vol. AP-22, July 1974, pp. 521–526.)

constant sheets must be thinner. The normalized broadside conductance is simply $G_0 = kt(\varepsilon_r - 1)$. For H- and E-plane scans, the sheet susceptances are

$$\begin{aligned} H\text{-plane} \qquad & \frac{B}{G_0} = \frac{1}{\cos\theta} \\ E\text{-plane} \qquad & \frac{B}{G_0} = \cos\theta - \frac{\sin^2\theta}{\varepsilon_r \cos\theta} \end{aligned} \tag{7.66}$$

An optimum match is obtained by adjusting the sheet thickness t (for a given ε_r) and the sheet spacing from the array face to minimize *scan element pattern* variations over the scan volume. The Smith chart procedure described by Magill and Wheeler (1966; Kelly, 1966) may be suitable for some well-behaved elements; in general numerical optimization based on a periodic cell spectral model or on extensive measurements is necessary. A review of scan compensation techniques is given by Knittel (1972).

Another multimode technique employs dielectric coatings on slots that can support a partial surface wave; the effect is to make the E- and H-plane patterns more alike, and to extend them toward endfire (Villeneuve et al., 1982).

7.5.2.4 External Wave Filter. The array *scan element pattern* may be altered through use of a spatial filter or transformer placed in front of the array face, but sufficiently far away that the filter does not directly affect element impedances. The transfer function of the spatial filter is designed to compensate the SEP variations; polarization effects may also be improved. With dielectric sheets the transmission coefficient varies differently with angle in E- and H-planes, so that in principle one or more sheets could improve the SEP. With several sheets in cascade, matrices for each sheet may be cascaded. Note that impedance, admittance, and scattering matrices do not cascade. Two that do are the A–B–C–D matrix (Altman, 1964) and the transmission matrix (Collin, 1991; Section 3.4). Conversion of either with Z, Y, or S matrices is, of course, readily performed. Mailloux (1976) applied the T matrix to a stack of dielectric slabs to reduce far-out sidelobes. Munk, Kornbau, and Fulton (1979) cascaded in the spectral domain; this approach is more powerful as many arrays are best analyzed therein. As an example an array of short slots over a screen was matched by a single dielectric slab of $0.367\lambda_0$ thickness and $\varepsilon = 1.33$; the resulting VSWR was <1.5 for both planes of scan to 80° (Munk et al., 1979). Other useful references on stratified wave filters that can contain thin metallic inclusions, as well as dielectrics, are Ng and Munk (1988) and Munk and Kornbau (1988).

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Photograph courtesy of Dr. J. J. Lee.

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CHAPTER EIGHT

Finite Arrays

8.1 METHODS OF ANALYSIS

8.1.1 Overview

Although the infinite array techniques of Chapter 7 are excellent for system trades and preliminary design, final design requires a finite array simulation. Direct impedance (admittance) matrix methods were described in Section 7.3.2 These were developed by Oliner and Malech, 1966a; Galindo, 1972; Bailey, 1974; Bailey and Bostian, 1974; Cha and Hsiao, 1974; Steyskal, 1974; Bird, 1979; Luzwick and Harrington, 1982; Clarricoats et al., 1984; Pozar, 1985, 1986; Fukao et al., 1986; Deshpande and Bailey, 1989; Silvestro, 1989; Usoff and Munk, 1994; and others. When moment methods are not necessary (thin half-wave dipoles, e.g.), sizeable planar arrays may be solved. When the element current distribution is complicated, the number of moment method expansion functions needed for good convergence will restrict this method to small arrays.

Munk (2003) considered finite arrays as re-radiating arrays (scatterers).

An elegant solution to the semi-infinite array (i.e., an array extending to infinity on three sides) is given by Wasylkiwskyj (1973). The Weiner–Hopf factorization procedure is extended to finite Fourier transforms, resulting in an expression for *scan reflection coefficient* of the semi-infinite array in terms of *scan reflection coefficient* for the corresponding infinite array, and an integral of a phased sum of the infinite array *scan impedances*.

Still another approach embeds the finite array in a matrix of identical arrays with blank space between (Roederer, 1971; Ishimaru et al., 1985; Skriverik and Mosig, 1993; Roscoe and Perrott, 1994; Cátedra et al., 1995). For single-mode elements (thin dipoles, thin slots, thin patches) the procedure is simple. First, the *scan element pattern* (SEP) is computed for the corresponding infinite array. This SEP is then

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Fourier transformed [usually by discrete Fourier transform (DFT)] back to the aperture. Third, a periodic structure consisting of equally spaced finite arrays, with each array having the desired amplitude distribution and scan phase, is transformed. Fourth, these two transforms are convolved. Next the result is inverse transformed to get the SEPs over the periodic structure. Last, the SEP for one finite array is extracted, and multiplied by the isolated element pattern, yielding the final finite array *scan element pattern*.

When the type of element allows multiple modes or nonsimple currents, the changing mutual coupling over the finite array may change the mode mix or change the current distribution. For these cases, each element needs a moment method (or modal) expansion, with all of the equations representing the expansion functions and number of elements solved simultaneously. A term of the impedance matrix for one expansion function consists of a double sum (over the elements) containing an integral of the appropriate Green's function times the expansion and test functions, times the exponential scan factor. Through the Poisson sum formula this is transformed to the spectral domain. An example of this formula is given in Section 7.4.1 The term is now written as a convolution product of this spectral domain result and the transform of the array amplitude distribution. The simultaneous equations are solved in the spectral domain, separately for each array element, resulting in the *scan impedance* for each element. To reduce errors introduced by aliasing between nearby arrays, it is necessary for the blank spaces to be greater than the width of the finite array. But as the ratio of blank to array becomes larger, the convergence deteriorates. This method, like all the others, has both good and bad aspects.

A simple approximate analysis method is called the "large array method", wherein the element currents are assumed to be the Floquet infinite array currents, and the mutual impedances multiplied by the scan phase factors are summed over the finite array to get *scan impedance*. Amplitude tapers are readily included. Impedance of the center element is given by

$$Z = \sum \sum A_{nm} Z_{nm} \exp[\, jk(nd_x u + md_y v)] \tag{8.1}$$

The *scan element pattern* is obtained from this *scan impedance*, as shown in Section 7.2.3:

$$g(u, v) = \frac{R(0, 0) g_{\text{iso}}(u, v)}{R(u, v) g_{\text{iso}}(0, 0)} (1 - |\Gamma(u, v)|^2) \tag{8.2}$$

Here $R(u, v)$ and $\Gamma(u, v)$ are the *scan resistance* and *scan reflection coefficient*; $g_{\text{iso}}(u, v)$ is the isolated element (and backscreen) power pattern. Both resistance and pattern are normalized to broadside values.

Results from the large-array method show finite array oscillations superimposed onto infinite array results (Hansen, 1990). Figures 8.1 and 8.2 show *scan element pattern* for the 0.5λ lattice without screen, for array sizes 41×41 and 101×101. As expected, the number of oscillations in the pattern is about one-half of the number of

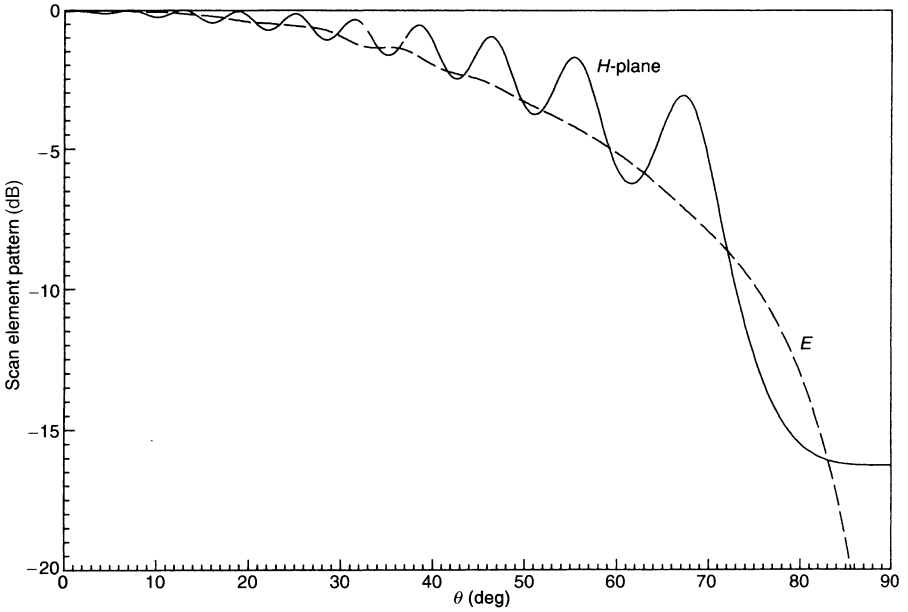


FIGURE 8.1 Dipole array, large array approximation, 41×41, $d_x = d_y = 0.5\lambda$, $L = 0.5\lambda$.

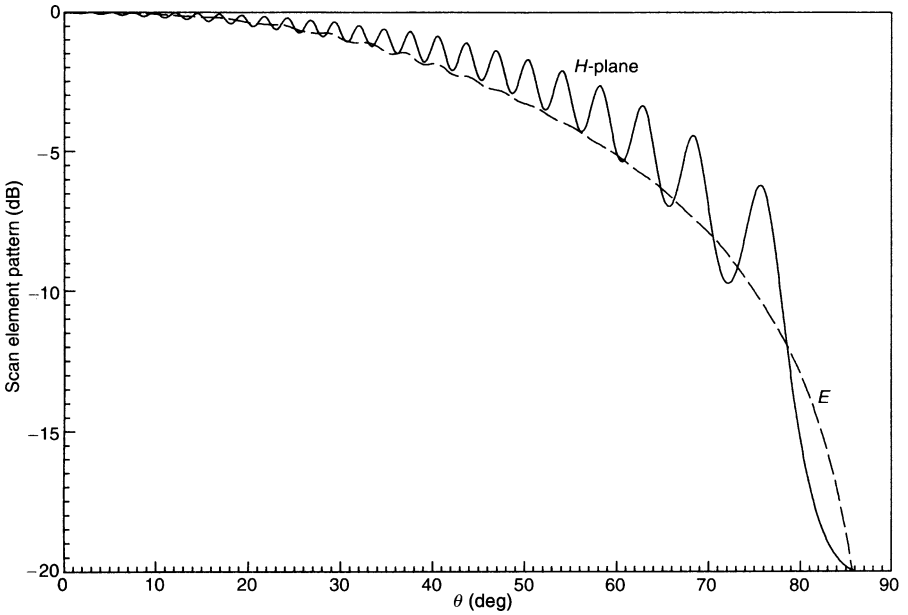


FIGURE 8.2 Dipole array, large array approximation, 101×101, $d_x = d_y = 0.5\lambda$, $L = 0.5\lambda$.

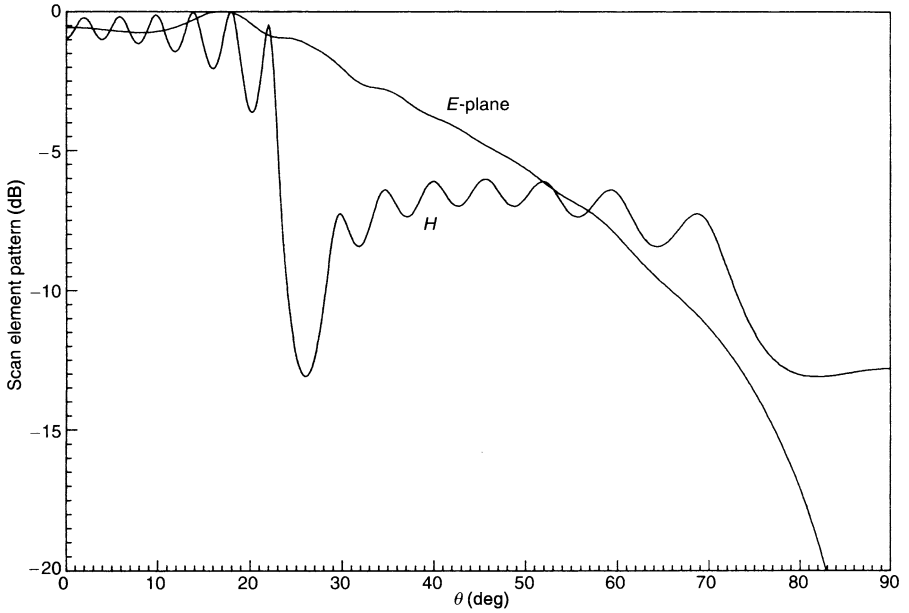


FIGURE 8.3 Dipole array, large array approximation, 41×41, $d_x = d_y = 0.7\lambda$, $L = 0.5\lambda$.

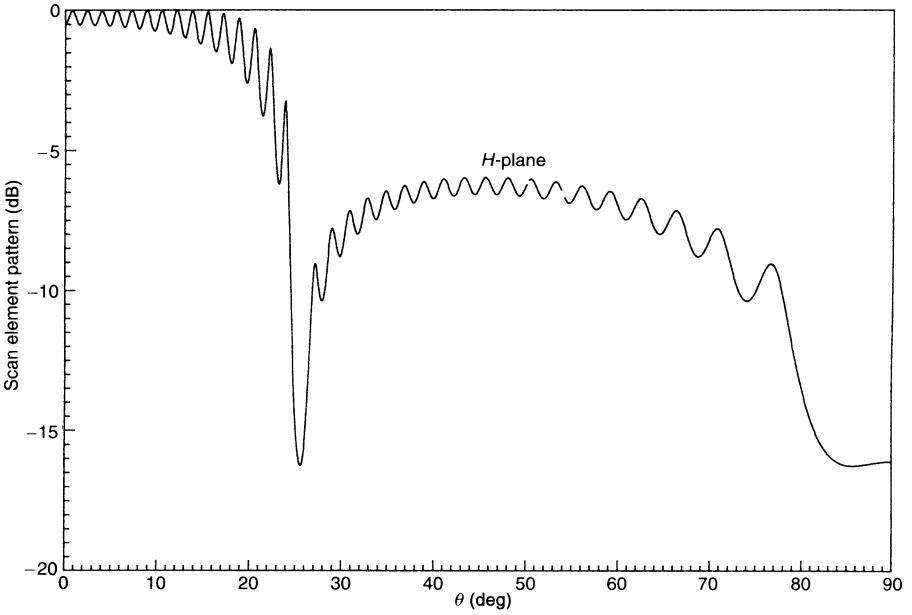


FIGURE 8.4 Dipole array, large array approximation, 101×101, $d_x = d_y = 0.7\lambda$, $L = 0.5\lambda$.

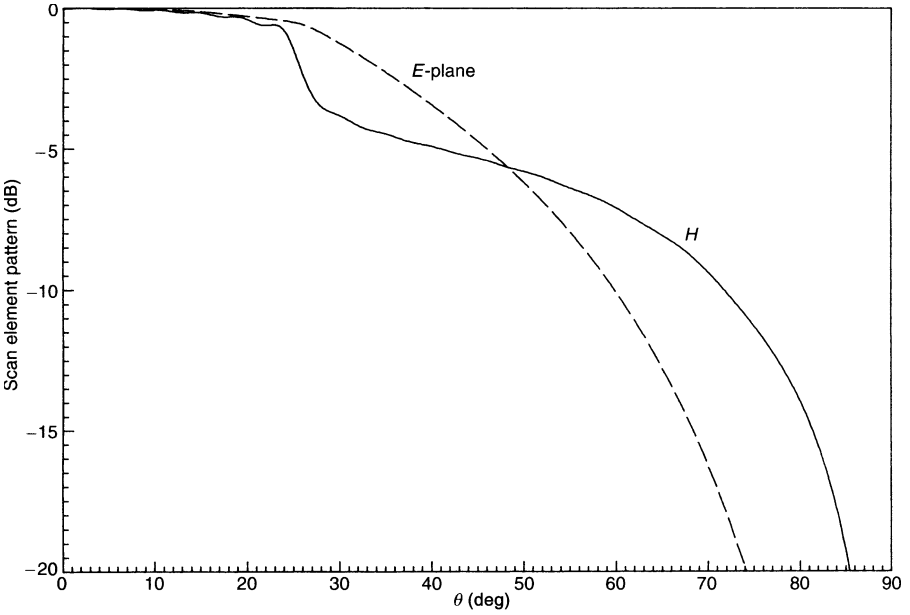


FIGURE 8.5 Dipole–screen array, large array approximation, 41×41, $d_x = d_y = 0.7\lambda$, $L = 0.5\lambda$, $H = 0.25\lambda$.

elements along x or y, but only one-half of the symmetric pattern is shown. Somewhat surprisingly, the wide-angle (largest) oscillations appear to have an amplitude independent of the array size. Because of the oscillatory effect of adding a row or column of elements to the array, it might be expected that adding one-half of the mutual impedances for the perimeter elements would reduce the oscillations. This proves to be the case, a 1-dB reduction is obtained, so all results shown utilize half-perimeter mutual impedances. Figures 8.3 and 8.4 show SEP for the 0.7λ lattice case. Again oscillations are superimposed on the infinite array results and, as expected, the number of oscillations is governed by the number of elements in the scan direction. The 41×41 array of Figure 8.3 provides a coarse approximation to the infinite array results; the 101×101 array of Figure 8.4 is better. The array size in the transverse direction is less critical; representation of deep grating lobe blind angles requires 41 or more elements.

Addition of a back screen appears to fill in the cyclic behavior of the dipole–dipole impedances, with the result that the large-array approximation gives very small oscillations. Figure 8.5 shows that very good correlation with infinite array results occurs for 41×41 array elements. For 21×21 array, the oscillations are <1 dB. The large-array method appears useful for arrays where the element pattern is of dipole–screen type.

8.1.2 Finite-by-Infinite Arrays

Computational cost is reduced when a finite-by-infinite array is considered, with scan across the finite width of the array (Denison and Scharstein, 1995). Only principal

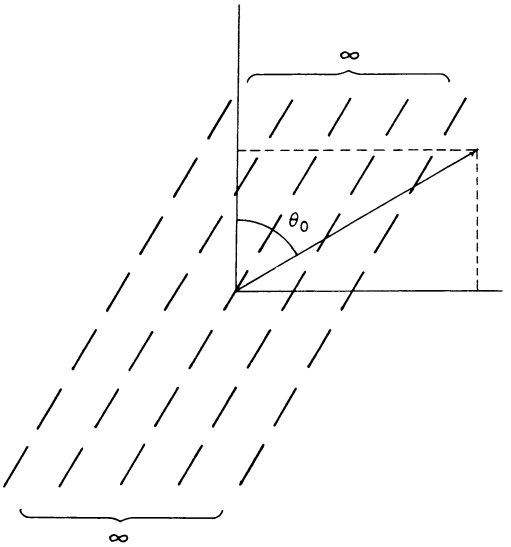


FIGURE 8.6 Finite-by-infinite dipole array with *H*-plane scan.

plane scans are allowed here. Even when moment methods are applied, relatively large arrays can be simulated. From this type of simulator, much can be gleaned about the behavior of edge elements in an array. As has been shown many times, half-wave thin dipole arrays exhibit all the pertinent features of mutual coupling. Thus complex array elements and moment method solutions are not needed at this stage. By considering collinear or parallel dipoles, both *H*- and *E*-plane scans can be accommodated. The

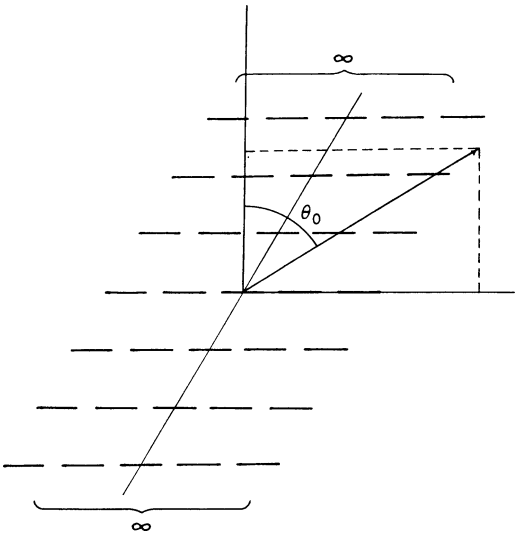


FIGURE 8.7 Finite-by-infinite dipole array with *E*-plane scan.

computational model then consists of N infinite linear arrays as sketched in Figures 8.6 and 8.7. Each infinite array (stick) is composed of half-wave thin dipoles. Thus mutual impedance between a dipole and the dipoles in one infinite "stick" is then computed. The set of these for all the sticks is used to form the usual impedance matrix, with the solution providing the *scan impedance* of a dipole in each stick.

The reference *scan impedance* for the infinite array is provided by the spectral domain formulation pioneered by Oliner and Malech (1996b). This result gives the *scan resistance* as a single term for each main lobe or grating lobe and a spectral summation of trigonometric-type functions for the *scan reactance*. For array spacings close to $\lambda/2$, the formulation converges extremely rapidly.

For both the H- and E-plane scan cases, the dipole stick impedances have been computed both in the spatial domain and in the spectral domain. In the spatial domain, a subroutine based on Carter's mutual impedance between two thin wire dipoles (Hansen, 1972) is used to obtain dipole–dipole impedance Z_a. This involves sine and cosine integrals that are calculated in double precision, to allow many terms to be summed. For the collinear dipole sticks (H-plane scan), the dipole to linear array impedance is just

$$Z = \sum Z_a\left(x_0 + nd, \sqrt{a^2 + y^2}\right) \tag{8.3}$$

As shown in Figure 8.8, the isolated dipole offset is a, the parallel shift is x_0, the perpendicular spacing is y_0, the array spacing is d, and dipole half-lengths are h. For parallel dipole sticks (E-plane scan), the dipole to linear array impedance is

$$Z = \sum Z_a\left(\sqrt{(x_0 + nd)^2 + a^2},\, y_0\right) \tag{8.4}$$

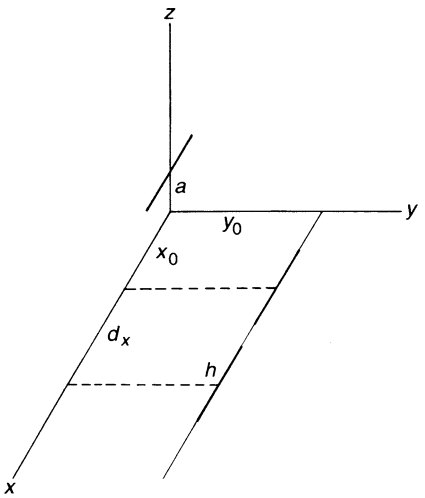


FIGURE 8.8 Collinear geometry.

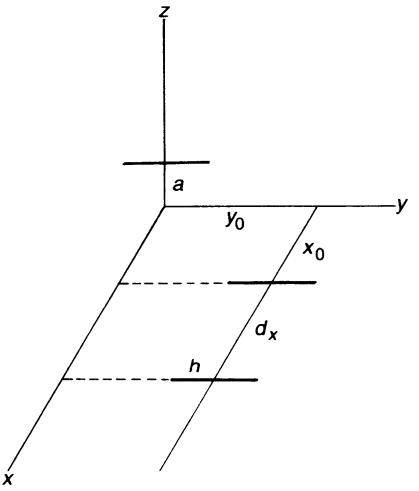


FIGURE 8.9 Parallel geometry.

Dimensions are shown in Figure 8.9. Difficulty occurs in that the sums oscillate, with this worsening as the dipole separation decreases. As the period of the sum oscillations changes with dimensions, no simple compensation method works. With these summations, no convergence was found after more than 1000 terms. Convergence acceleration techniques include those of Shanks (1955) and Levin (1973). Both have been used, but the Levin is superior as it does not concatenate roundoff errors. The Levin transformation is a rearrangement of partial sums (Singh et al., 1990; Singh and Singh, 1993):

$$S^n = \frac{\sum_{i=0}^{m} (-1)^i \binom{m}{i} \binom{n+i}{n+m}^{m-1} \dfrac{S_{n+1}}{S_{n+i+1} - S_{n+1}}}{\sum_{i=0}^{m} (-1)^i \binom{m}{i} \binom{n+1}{n+m}^{m-1} \dfrac{1}{S_{n+i+1} - S_{n+1}}} \tag{8.5}$$

This gives the transform of m order. $\binom{m}{i}$ is the binomial coefficient.

Typically, 21 Levin cycles gave excellent accuracy for all cases; because parallel dipole mutual impedance decays asymptotically as $1/r$, instead of the collinear value of $1/r^2$, more Levin cycles are needed to achieve convergence for that case. For both scan planes, larger element spacing gives faster convergence.

The spectral domain summation technique was also used; it is a simplification of the *periodic moment method*, developed by Ben Munk and colleagues at Ohio State University (Shubert and Munk, 1983). The mutual impedance is written as a spectral (Floquet) summation over the wavenumber, where the wavenumber includes the

element spacing. For the H-plane scan array, the summation contains Hankel functions and generalized pattern factors:

$$Z_i = -\frac{30}{d \sin^2 kh} \sum P^2 \exp(jkx_0 u)\, H_0^{(2)}\left(k\sqrt{(a^2 + y_0^2)(1 - u^2)}\right) \tag{8.6}$$

with

$$P = \frac{\cos khu - \cos kh}{k\sqrt{1 - u^2}} \qquad u = \sin\theta_0 \tag{8.7}$$

For this H-plane case, 401 terms gave a mutual impedance accuracy of 0.01 Ω. Larger element spacings require more terms.

For the parallel (E-plane) case, a significant difference exists from the collinear case: The Hankel function is now inside the generalized pattern function integral, as shown by Skinner and Munk (1992) and Skinner et al. (1995), and the integral cannot be integrated in closed form. The formulation, with details of the derivation omitted, is

$$Z_i = -\frac{30}{d_x \sin^2 kh} \sum \exp(jkx_0 u) \int_{-h}^{h} \sin[k(h - |y|)] \times H_0^{(2)}\left(k\sqrt{a^2 + (y + y_0 - h)^2}\sqrt{1 - u^2}\right) dy \tag{8.8}$$

If d_x does not allow a grating lobe, the argument of $H_0^{(2)}$ is imaginary except for $n = 0$; $H_0^{(2)}$ is replaced by $(2/\pi)K_0(\text{arg})$. Numerical integration was used for each term in the series above. For $y_0 > 1$, a complex 32-point Gaussian was used, with a 128-point Gaussian for the closer spacings (Stroud and Secrest, 1966). Excellent convergence was achieved with 501 terms, but the result was a slow program.

With proper choice of limits of both spectral and spatial summations, the plots of *scan impedance* from these two techniques lay on top of one another. In general, the spectral summation was slower and less satisfactory. Results are given in Section 8.3.1.

8.2 SCAN PERFORMANCE OF SMALL ARRAYS

Using the impedance matrix approach, Diamond (1972) computed *scan impedance* and *scan element pattern* of several small arrays of dipoles. Figure 8.10 shows *scan resistance* for the center element of an array of 7×9 half-wave dipoles on a $\lambda/2$ square lattice, with the infinite array results, shown dashed, for comparison. In Figure 8.11, the same array is over a ground screen at $\lambda/4$ spacing. The screen removes

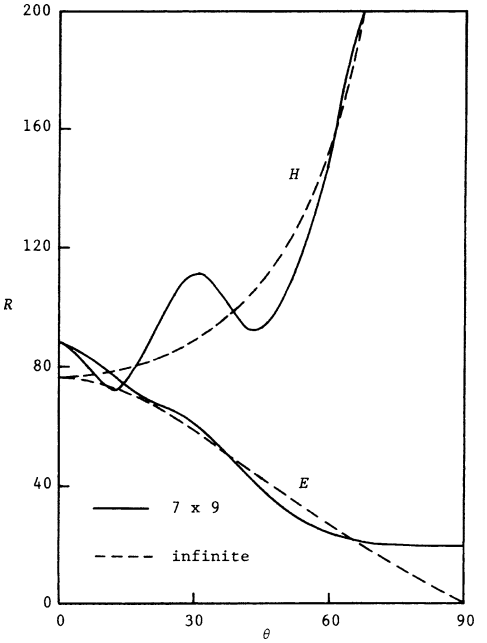


FIGURE 8.10 Scan resistance of center element of half-wave dipole array. (Courtesy Diamond, B. L., "Theoretical Investigation of Mutual Coupling Effects", Chapter 3 of TR381, Lincoln Laboratory, March 1965.)

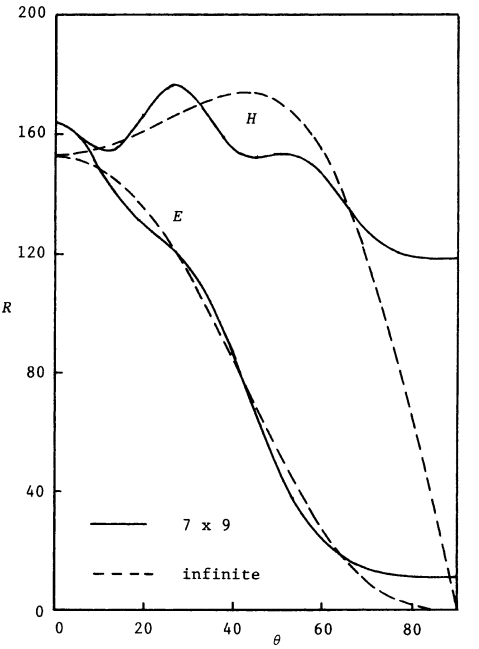


FIGURE 8.11 Scan resistance of center element of half-wave dipole array with screen. (Courtesy Diamond, B. L., "Theoretical Investigation of Mutual Coupling Effects", Chapter 3 of TR381, Lincoln Laboratory, March 1965.)

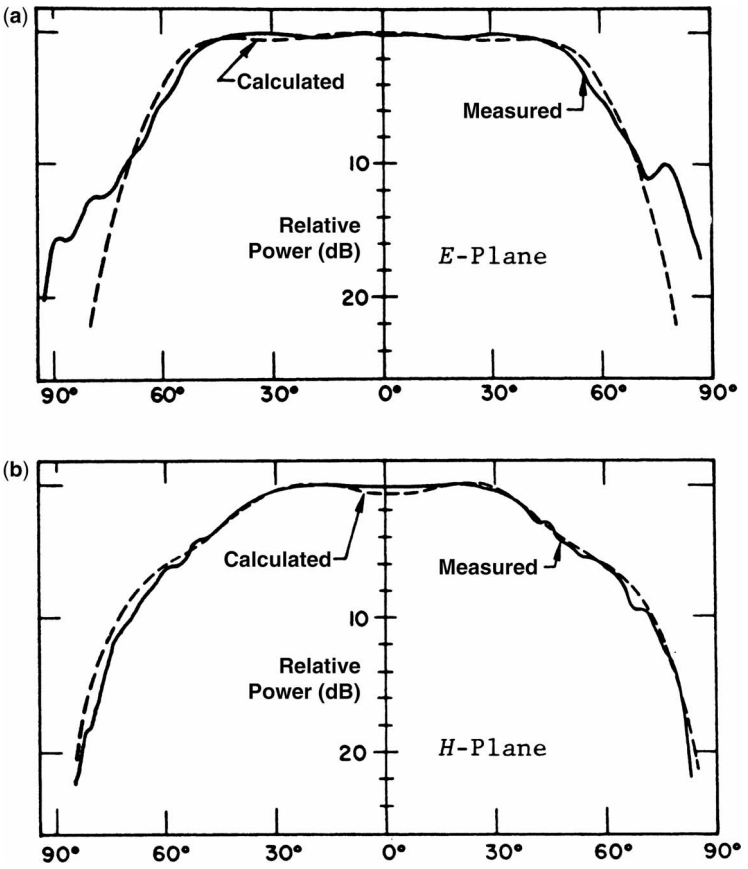


FIGURE 8.12 Scan element patterns of center element of a 5×5 dipole–screen array. (Courtesy W. E. Rupp.)

the high values of the *H*- and diagonal planes. Small arrays exhibit behavior that oscillates around that of a large array. Similar oscillations occur in the *scan reactance* of small arrays. In general, as might be expected, larger arrays have more oscillations, but of smaller amplitude.

Scan element patterns are of interest for small arrays although it is no longer valid to multiply the array factor by a single *scan element pattern* in order to get the overall array pattern. Figure 8.12 shows calculated and measured *scan element patterns* of the center dipole in a 5×5 array of half-wave dipoles in a half-wave lattice, and quarter-wave over a ground plane. The broadening and flattening effect on the *E*-plane pattern is quite noticeable. Figures 8.13 and 8.14 are *scan element patterns* for the center element of a 7×9 array of dipoles for various lattice spacings. From these curves it is apparent that the *scan element pattern* limits the scan angle range. As more elements are added to the array, the number of oscillations in the pattern increases and the curves

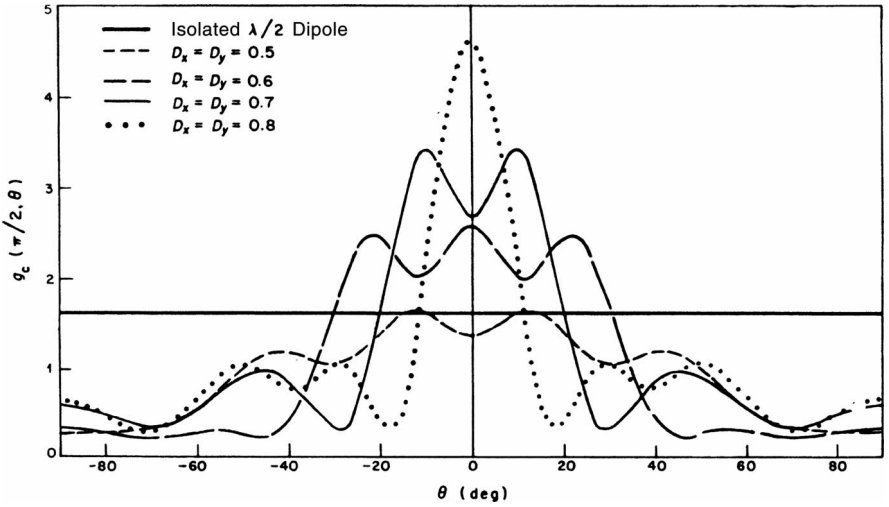


FIGURE 8.13 The *H*-plane scan element patterns of center element of 7 × 9 dipole array. (Courtesy Diamond, B. L., "Theoretical Investigation of Mutual Coupling Effects", Chapter 3 of TR381, Lincoln Laboratory, Mar. 1965.)

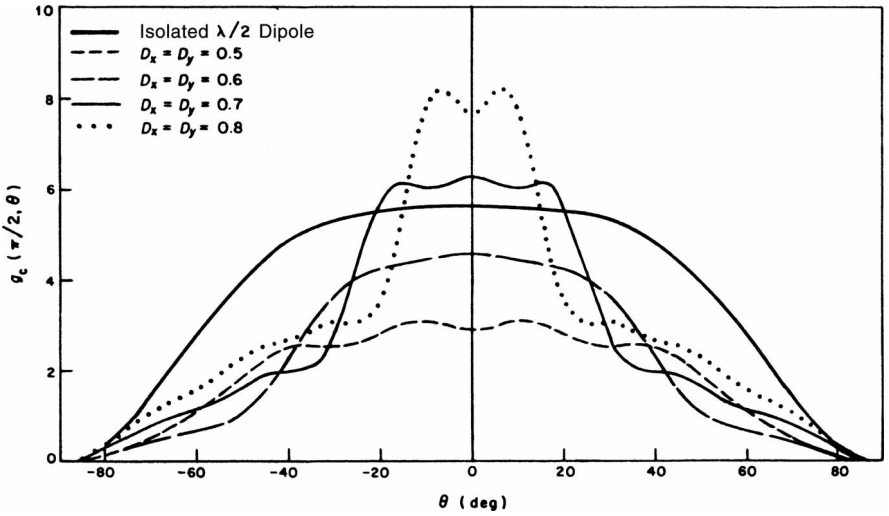


FIGURE 8.14 The *H*-plane scan element patterns of center element of 7 × 9 dipole/screen array. (Courtesy Diamond, B. L., "Theoretical Investigation of Mutual Coupling Effects", Chapter 3 of TR381, Lincoln Laboratory, Mar. 1965.)

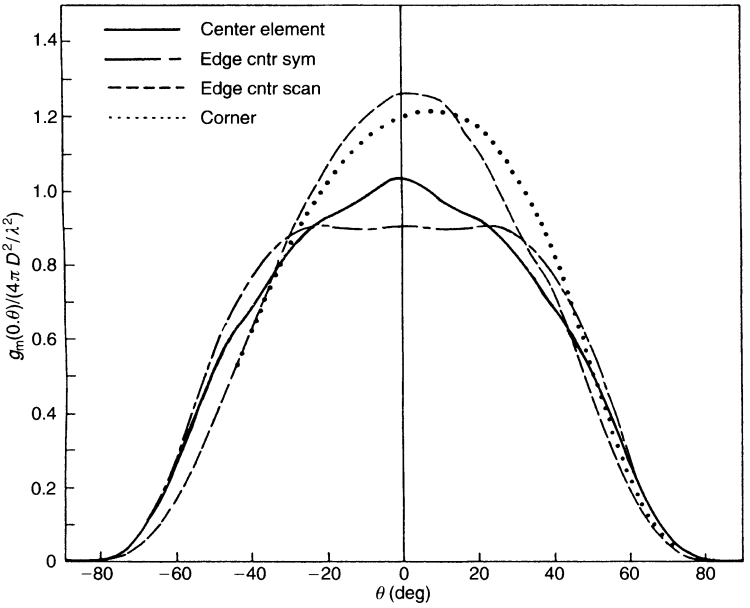


FIGURE 8.15 The *E*-plane scan element patterns for 9 × 11 dipole–screen array. (Courtesy Diamond, B. L., "Theoretical Investigation of Mutual Coupling Effects", Chapter 3 of TR381, Lincoln Laboratory, Mar. 1965.)

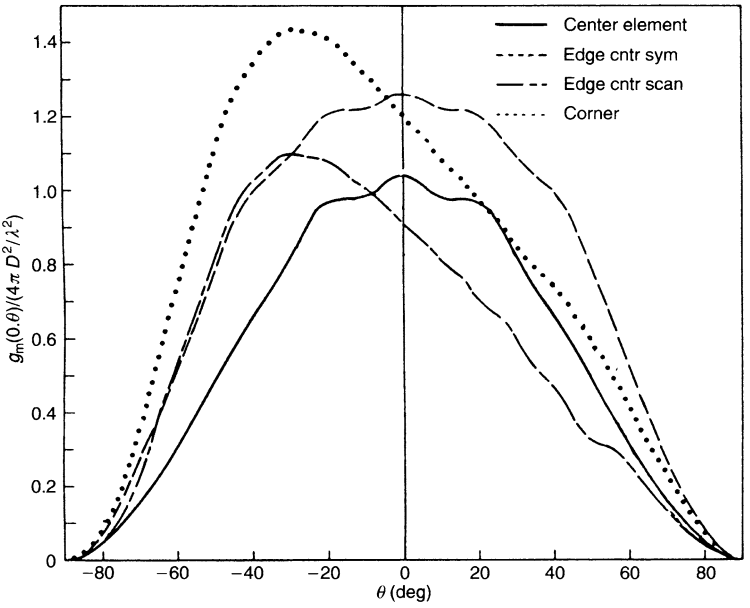


FIGURE 8.16 The *H*-plane scan element patterns for 9 × 11 dipole–screen array. (Courtesy Diamond, B. L., "Theoretical Investigation of Mutual Coupling Effects", Chapter 3 of TR381, Lincoln Laboratory, Mar. 1965.)

become smoother. Also, the falloff becomes steeper. Figures 8.15 and 8.16 compare center, edge center, and corner *scan element patterns* for a similar 9×11 array of dipoles. The corner elements do not show the symmetric mutual coupling effects that tend to flatten the *scan element pattern*. Dummy elements can be added around the edges of an array to be scanned if space is available.

An interesting example of edge effects in a linear array is given by Gallegro (1969), for half-wave dipoles with and without a ground plane. Spacing between elements is 0.586λ, which gives a just-visible grating lobe at $\theta = 90^\circ$ for 45° scan; dipole orientation is both parallel and collinear. The center element is matched at 0° scan angle, with all elements using the same matching network. All arrays have 51 elements, and the ground plane spacing is 0.25λ. The collinear arrays are well behaved even at grating lobe incidence; the parallel arrays are somewhat less well behaved up to grating lobe (GL) incidence, but at incidence there was a marked edge effect, especially for the array without ground plane. Figures 8.17 and 8.18 show element impedance by number with and without ground plane for scan angles of 0, 40, and 45° for parallel dipoles. It appears that the edge elements in the scan direction are

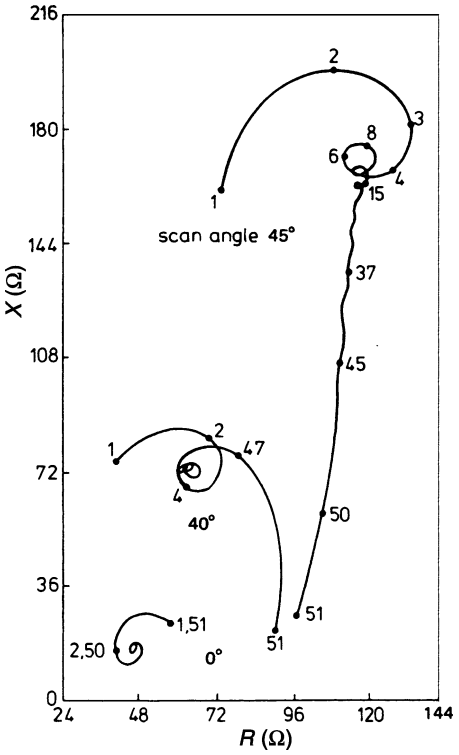


FIGURE 8.17 Scan impedance of linear array of parallel dipoles. (Courtesy Gallegro, A. D., Mutual Coupling and Edge Effects in Linear Phased Arrays, MS thesis, Polytechnic Institute of Brooklyn, 1969.)

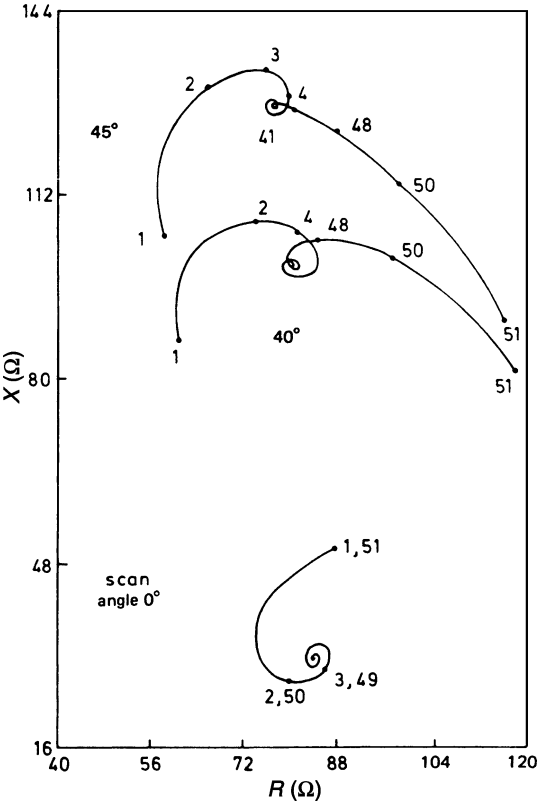


FIGURE 8.18 Scan impedance of a linear array of parallel dipoles–screen. (Courtesy Gallegro, A. D., Mutual Coupling and Edge Effects in Linear Phased Arrays, MS thesis, Polytechnic Institute of Brooklyn, 1969.)

more affected than the edge elements in the reverse direction, but the VSWR of the reverse edge elements is higher owing to the grating lobe pointing in that direction. A rough tabulation of the number of edge elements where the edge element impedance is outside the center cluster is shown in Table 8.1. It must not be supposed that these results apply to planar arrays, but the trends should be useful.

For TEM horn arrays, Holter and Steyskal (2002) have shown that the arrays size should be at least $5\lambda \times 5\lambda$ to minimize edge effects.

TABLE 8.1 Number of "Edge Elements": Grating Lobe for 45° Scan

Dipoles	Ground Plane	Scan Direction	Reverse Direction
Parallel	No	1–5–30	1–2–3
Parallel	Yes	1–3–6	1–2–3
Collinear	No	1–2–2	1–1–1
Collinear	Yes	1–1–2	1–1–2

8.3 FINITE-BY-INFINITE ARRAY GIBBSIAN MODEL

8.3.1 Salient Scan Impedance Characteristics

The finite-by-infinite array simulators described in Section 8.1.2 were used to compute *scan impedance* across the array, for several arrays from large to small. Results for *H*-plane for arrays of 201 resonant dipoles are shown as the solid lines in Figure 8.19–8.21. Scan angles are 0, 45, and 60°. The dotted line curves are the Gibbsian model that is discussed later. The dipole lengths were adjusted to give a resonant impedance in an infinite broadside array, and these dipole lengths were used. The array lattice was $\lambda/2 \times \lambda/2$, and dipole radii were 0.005λ (Hansen and Gammon, 1995). *Scan impedances* are normalized by the infinite array value at each scan angle. The dipoles are numbered 1–201 starting at the edge, with beam scan toward element 1. These figures show apparently continuous curves, but note that they have been plotted from 201 points. For Figure 8.19, only one-half of the array has been plotted; the broadside curves are symmetric.

Figures 8.22 and 8.23 show *scan impedance* across a smaller array of 51 sticks, for scan angles of 0 and 60°. Figure 8.24 showns *scan impedance* for a 21-stick array at broadside. Note that the smaller arrays fit the same pattern as the large array. The *scan impedance* oscillations are very small in the center of the array at broadside, and become larger as the scan angle increases. Edge values are larger for all scan angles, with 20% typical.

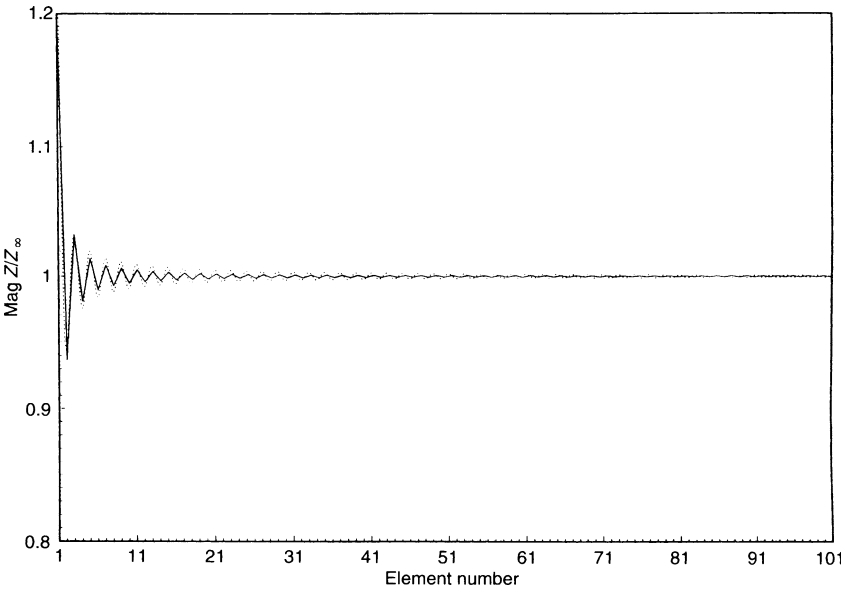


FIGURE 8.19 The 201 linear infinite arrays of resonant dipoles, *H*-plane scan at 0°.

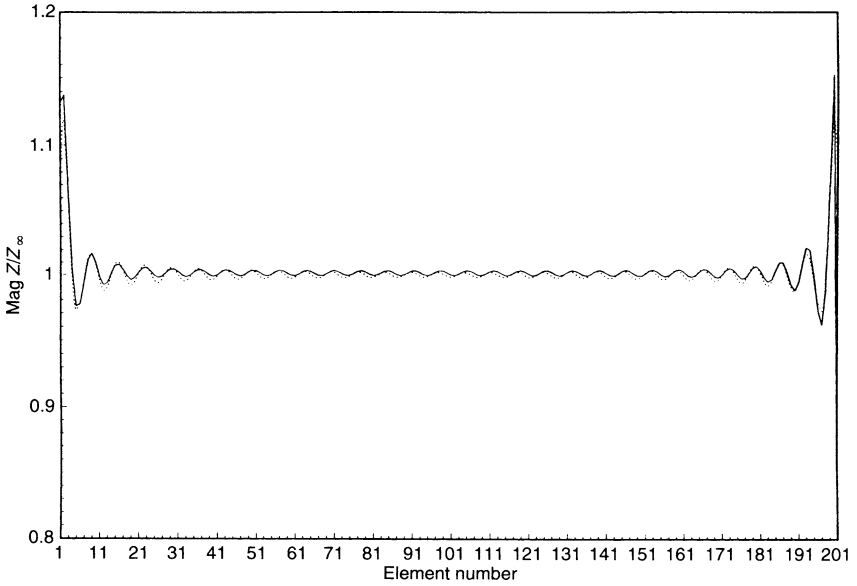


FIGURE 8.20 The 201 linear infinite arrays of resonant dipoles, *H*-plane scan at 45°.

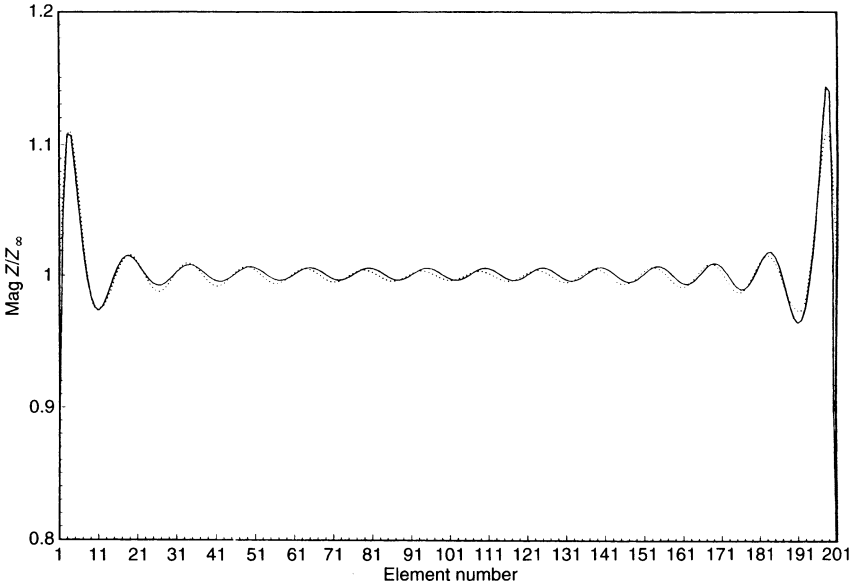


FIGURE 8.21 The 201 linear infinite arrays of resonant dipoles, *H*-plane scan at 60°.

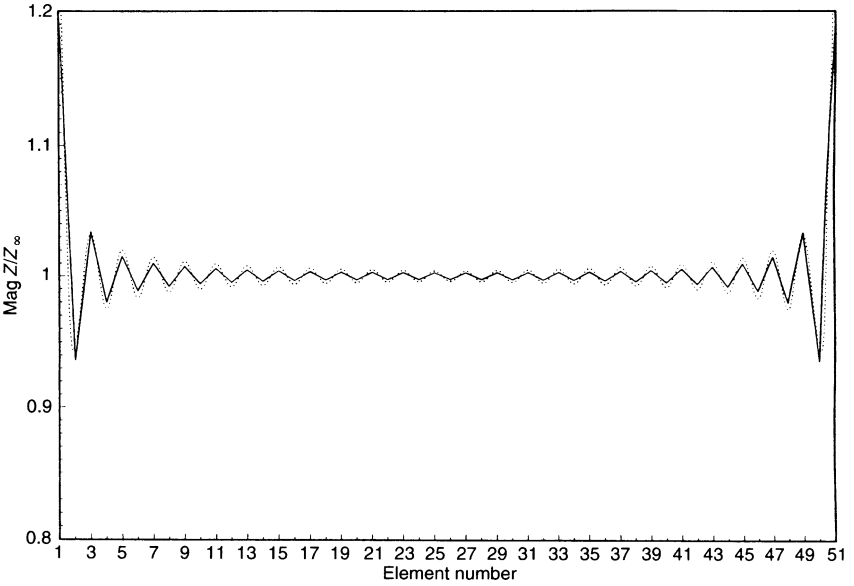


FIGURE 8.22 The 51 linear infinite arrays of resonant dipoles, *H*-plane scan at 0°.

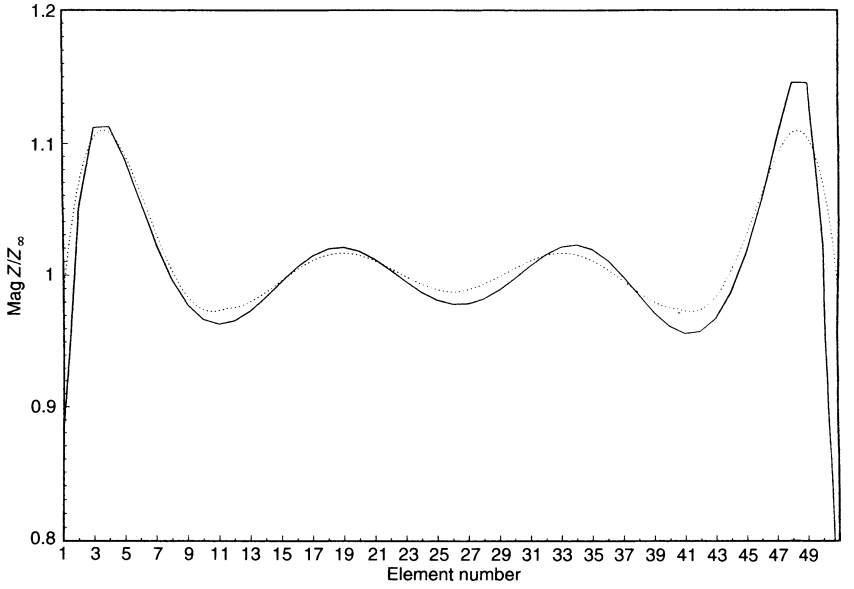


FIGURE 8.23 The 51 linear infinite arrays of resonant dipoles, *H*-plane scan at 0°.

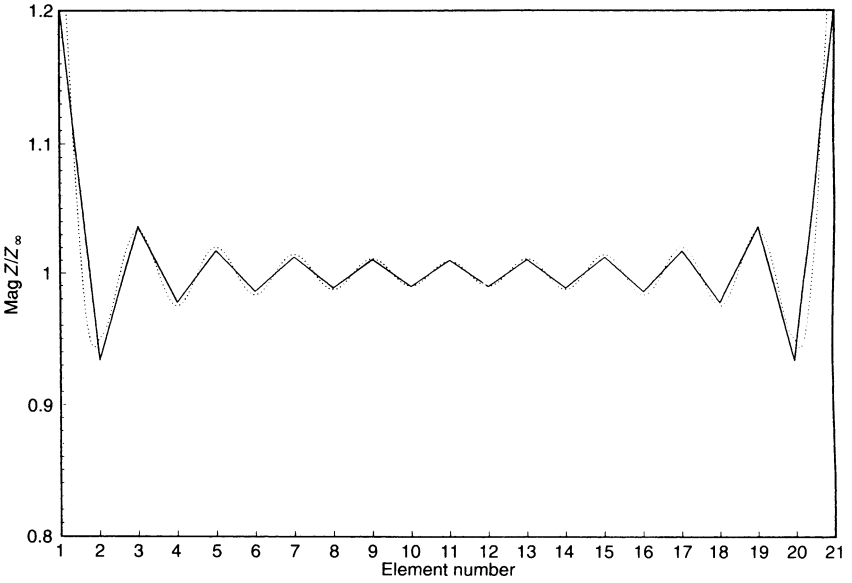


FIGURE 8.24 The 21 linear infinite arrays of resonant dipoles, *H*-plane scan at 0°.

These dipole array results are directly applicable to arrays of slots in a metallic sheet, through Babinet's principle. With slots in a waveguide wall, the internal susceptance must be added to the slot admittance.

Figures 8.25–8.27 are for the same 201-element array, but with a back screen at $\lambda/4$ spacing. Again dipole lengths were adjusted for resonance in an infinite broadside array. In these graphs, the oscillations decay more rapidly toward the array center, although the edge values are still large. Broadside results for smaller arrays of 51 and 21 elements are given in Figures 8.28 and 8.29. Again these follow the pattern for the 201-element array.

From these curves several things are evident. First a pseudo-standing wave in *scan impedance* exists over the face of the array, even for broadside radiation. Second, the period of the pseudo-standing wave is a function of the scan angle as one might expect. And third, the two pseudo-traveling waves that constitute the pseudo-standing wave are attenuated with distance from their respective edges.

The period across the array is almost exactly constant, and has been determined to be given closely by

$$\text{Period} = \frac{0.5}{1 - \sin\theta_0} \qquad \text{(in wavelengths)} \tag{8.9}$$

where θ_0 is the scan angle from broadside. This result may be interpreted as a grating lobe type formula from two pseudo-waves, one at the scan angle and a second along the array toward the edge.

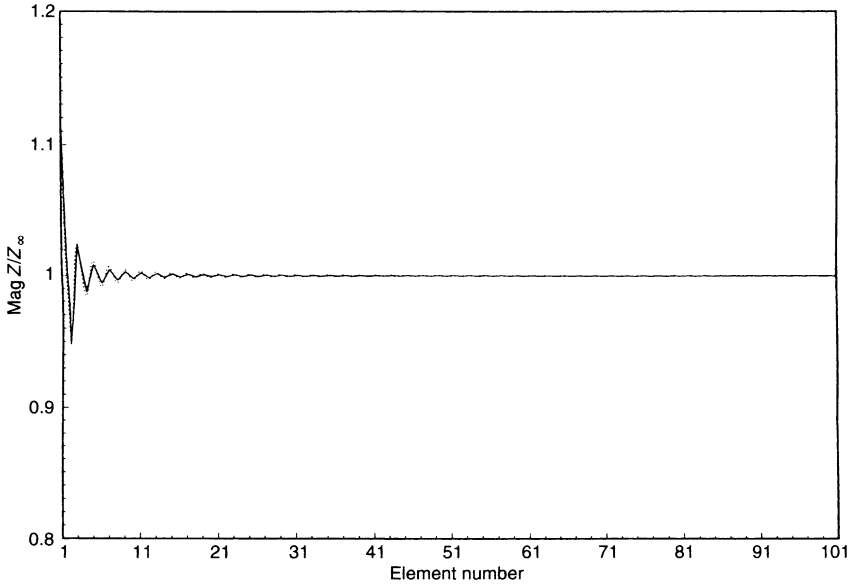


FIGURE 8.25 The 201 linear infinite arrays of resonant dipoles–screen, *H*-plane scan at 0°.

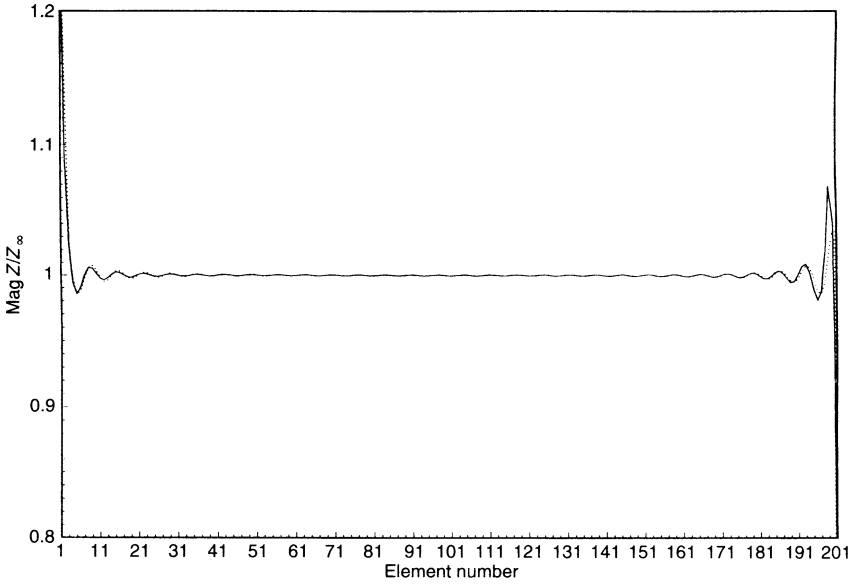


FIGURE 8.26 The 201 linear infinite arrays of resonant dipoles–screen, *H*-plane scan at 45°.

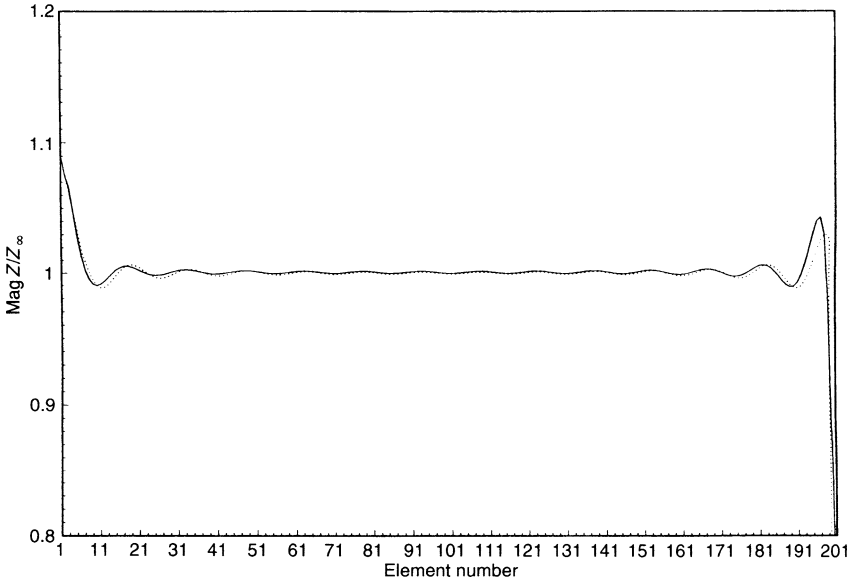


FIGURE 8.27 The 201 linear infinite arrays of resonant dipoles–screen, *H*-plane scan at 60°.

Turning now to *E*-plane scan, Figures 8.30 and 8.31 show arrays of 201 parallel dipole sticks, again a $\lambda/2 \times \lambda/2$ lattice, for scan angles of 0 and 60°. Dipole lengths were again adjusted for resonance in an infinite array. Now the decay of oscillations from edge to center is slower. In general, the excursions are larger than those for *H*-plane dipole arrays. At broadside, the *E*-plane scan edge value

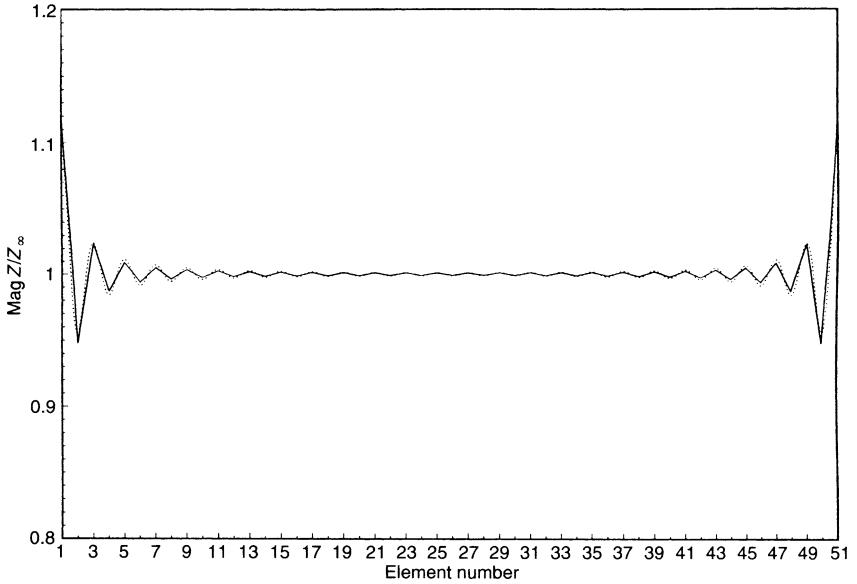


FIGURE 8.28 The 51 linear infinite arrays of resonant dipoles–screen, *H*-plane scan at 0°.

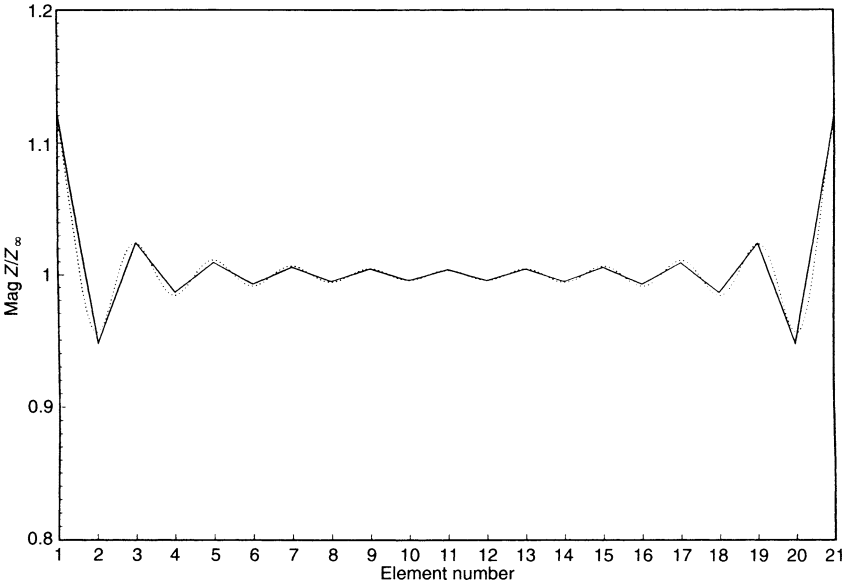


FIGURE 8.29 The 21 linear infinite arrays of resonant dipoles–screen, *H*-plane scan at 0°.

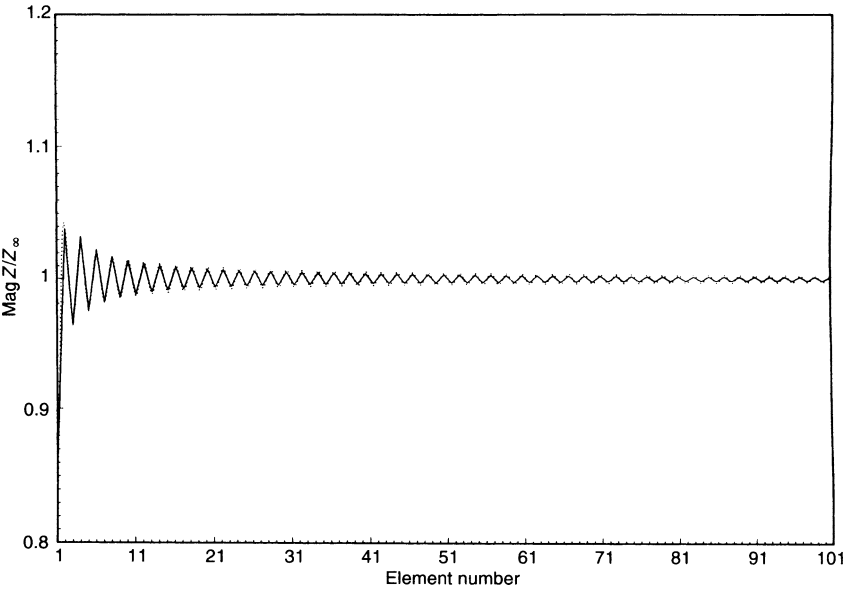


FIGURE 8.30 The 201 linear infinite arrays of resonant dipoles, *E*-plane scan at 0°.

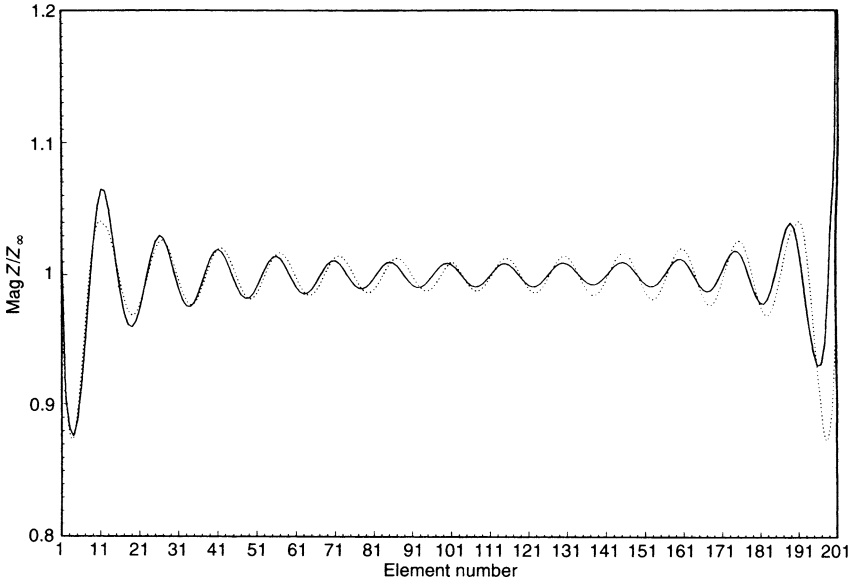


FIGURE 8.31 The 201 linear infinite arrays of resonant dipoles, *E*-plane scan at 60°.

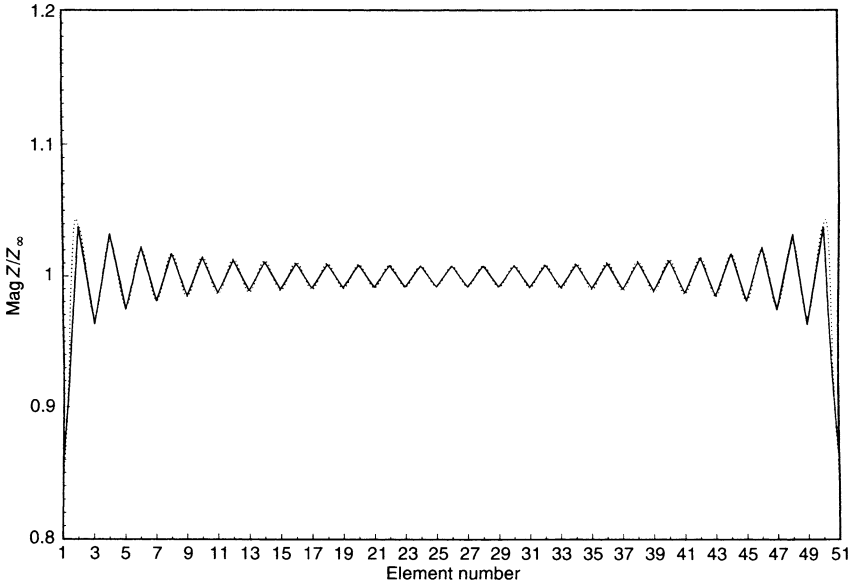


FIGURE 8.32 The 51 linear infinite arrays of resonant dipoles, *E*-plane scan at 0°.

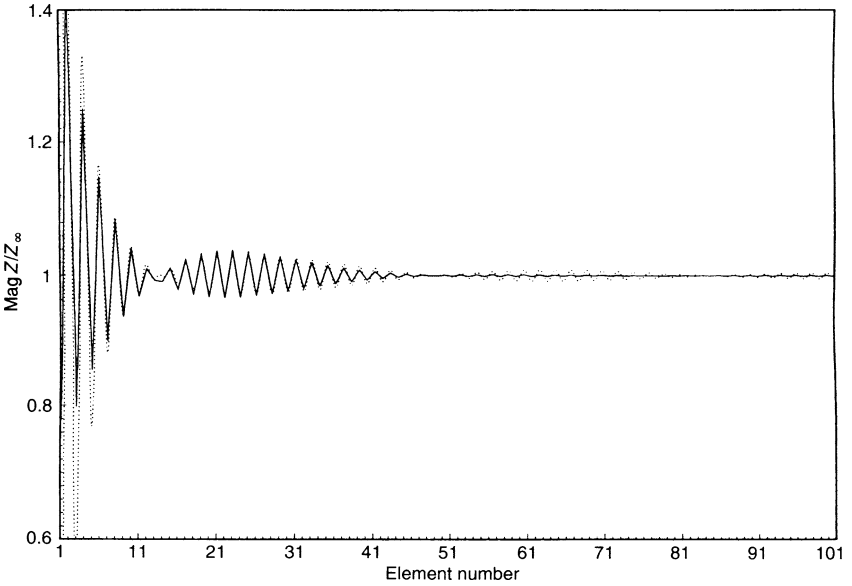


FIGURE 8.33 The 201 linear infinite arrays of resonant dipoles–screen, *E*-plane scan at 0°.

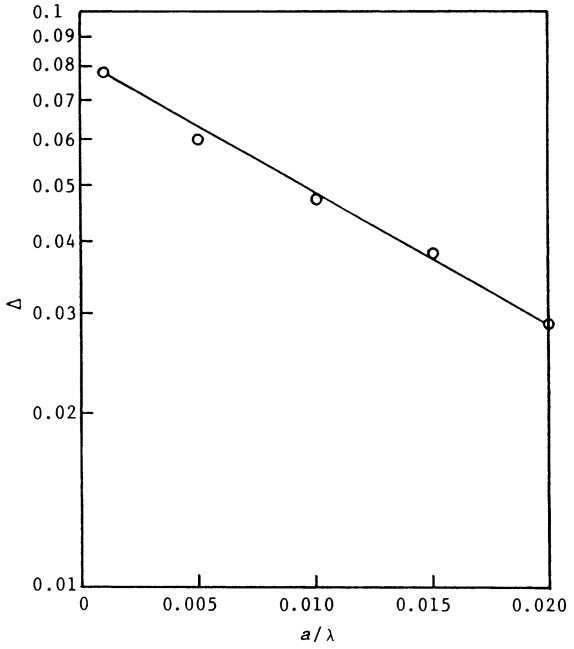


FIGURE 8.34 Modulation factor Δ versus radius.

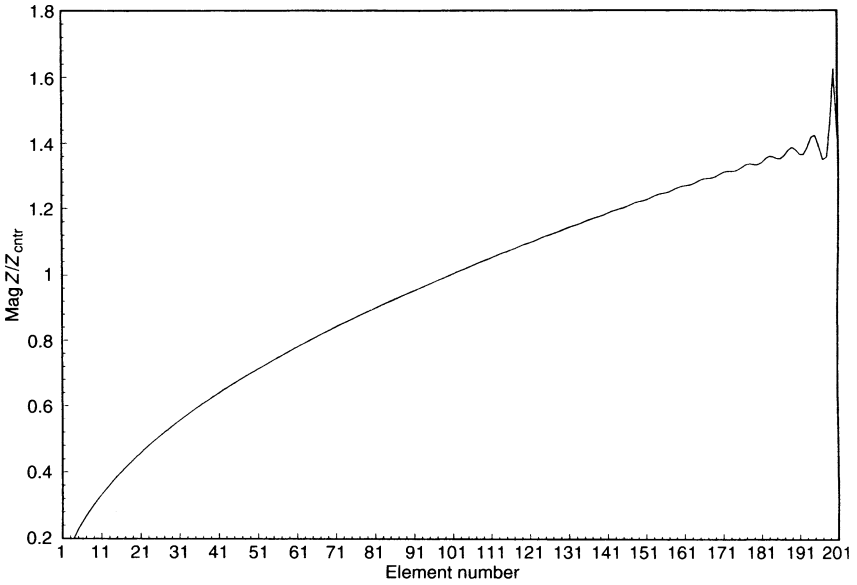


FIGURE 8.35 The 201 linear infinite arrays of resonant dipoles, *H*-plane scan at 45°.

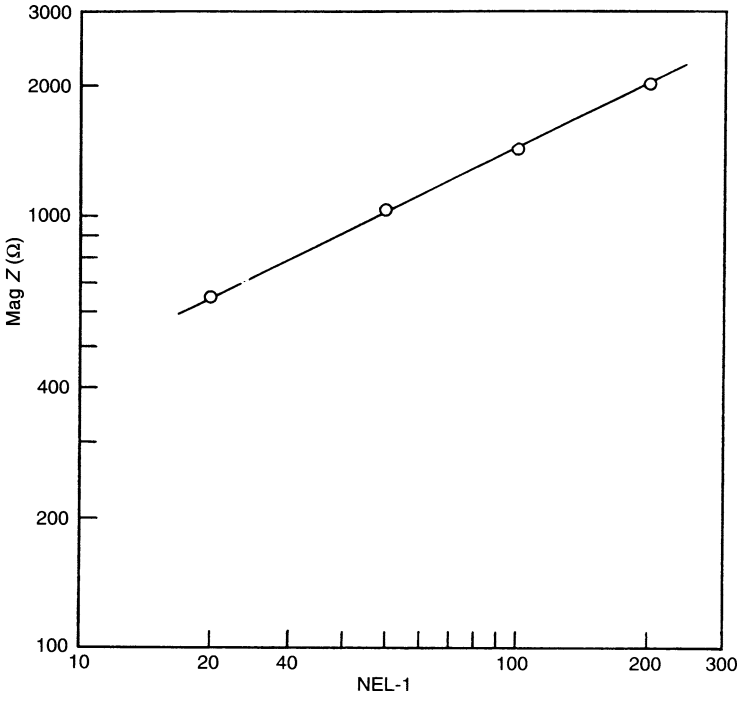


FIGURE 8.36 Array impedance center value at grating lobe angle.

is lower, while for *H*-plane it is higher than unity; the deviations are comparable (Hansen and Gammon, 1996a). Figure 8.32 gives data for a 51-element array at zero scan.

When a screen was added for *E*-plane scan cases, a new phenomenon appeared. At broadside, the expected half-wave-spaced oscillations appeared to be modulated by an oscillation of larger period, reminiscent of a heterodyne process. Figure 8.33 is for an array of 201 resonant dipole sticks, dipole length 0.4841λ, dipole radius 0.005λ, screen spacing 0.25λ and lattice $\lambda/2 \times \lambda/2$. The previous investigation of *H*-plane scan of dipoles and dipoles–screen, and of *E*-plane scan of dipoles, showed no significant change with dipole radius, as long as it was small. In the *E*-plane, dipole–screen case here, the radius affects the modulation period. Assuming a heterodyne process, with the "difference frequency" operating, the period P is simply $1/P = 1/P_1 - 1/P_2$, where P_1 is the basic *scan impedance* period of one wavelength (zero scan), and P_2 is P_1 modified by a factor: $P_2 = P_1(1 + \Delta)$. Figure 8.34 shows the parameter Δ for five radii from 0.001λ to 0.02λ (Hansen and Raudenbush, 1996). This log–linear relationship is due to the change in mutual impedances with change in radius, all reflected through the impedance matrix inverse. Why this appears only for *E*-plane dipoles with screen is not known.

The obvious question is how this phenomenon changes with scan angle. At 30° no modulation is apparent. At 15° scan there is modulation present, but the pattern of it is not clear. This new phenomenon represents a challenge in array modeling.

An infinite array of half-wave dipoles with element spacing sufficiently large to allow a grating lobe exhibits a blind angle at the grating lobe angle; that is, the *scan resistance* and *scan reactance* become infinite (see Chapter 7). As a result, the magnitude of the reflection coefficient is unity and no power is radiated. The obvious question is What happens to *scan impedance* in a finite array at GL incidence. The *H*-plane dipole array simulator was used with element spacing of 0.5858λ, which allowed a grating lobe to appear at 45°. Figure 8.35 shows the *scan impedance* magnitude for a 201-stick array at 45° scan; it increases smoothly from near zero to roughly 1.4, with a few oscillations at the rear. Smaller arrays show closely similar behavior, with of course fewer oscillations. The normalization factor, which is the value in the center of the array, shows a log–log behavior with number of elements, as seen in Figure 8.36. It can be seen that very large arrays approach a blind angle, with the infinite impedance. Phase of the *scan impedance* is roughly 45°; so the phase is approximately that of a conductor.

The same array at the other scan angles rapidly reverts to normal behavior; the smooth *scan impedance* behavior over the array occurs only for a narrow range of angles around the GL angle. At other scan angles, there is minor half-wave modulation.

8.3.2 A Gibbsian Model for Finite Arrays

It is well known that a bandlimited square wave exhibits oscillations that are termed Gibbs's phenomenon. That is, the Fourier transform with finite limits of the sinc

spectrum produces the square wave with these oscillations. An examination of various computed results reveals that the peaks and dips of the oscillations are regularly spaced, and that the amplitudes of the first, second, third, and as on, peaks are the same regardless of the upper band limit. There the first peak occurs at each edge of the square wave, with the second peak adjacent, and so on. As the upper band limit increases, the total number of peaks increases, being roughly the upper limit divided by π and, of course, the spacing changes, but the amplitudes remain the same. There is little mention in the literature regarding how to calculate these peaks for a square wave, but Hamming (1972) quotes results derived by Carslaw (1930). A clever technique is used by the latter to determine the extrema positions. The finite series representing the square wave is differentiated, and it is recognized that this sum has a closed-form result. When set to zero, the roots immediately give the extrema locations. These are $m\pi/2n$, where m is the extremum index and n is the upper limit on the summation. The amplitudes at these points are immediately expressed as an integral which, for large n, can be expressed as a sine integral; the result is $(2/\pi)\mathrm{Si}(m\pi)$. Thus for large band limit values, the peak amplitudes are readily determined to be 1.1790, 0.9028, 1.0062, 0.9499, and so on.

It might be expected that a single pulse with linear exponential phase across it would better represent the scanned finite array. The aperture distribution should be $\exp(-jtU\sin\theta_0)$ for $|t| \leq 1$, and zero for $|t| > 1$. The transform of this aperture distribution is the spectrum, which is (Sneddon, 1951)

$$f(t) = \mathrm{sinc}\left[(\omega - \alpha)\frac{\tau}{2}\right] \tag{8.10}$$

Here $\alpha = U\sin\theta_0$. This spectrum is then transformed with finite limits to produce the scanned pulse with Gibbs's-type oscillations. The transform is

$$f(t) = \int_{-U}^{U} \tau\,\mathrm{sinc}\left[(\omega - \alpha)\frac{\tau}{2}\right]\exp(j\omega\tau)\,dw \tag{8.11}$$

The transform integration is complex, and produces a real part involving sine integrals, and an imaginary part involving cosine integrals. The latter primarily contribute delta-like functions at the pulse edges, while the former constitute the oscillations over the pulse. After various changes of variables the real part becomes (Hansen and Gammon, 1996b)

$$\begin{aligned} f(t) = {} & \mathrm{Si}[(U+\alpha)(1+T)] + \mathrm{Si}[(U-\alpha)(1+T)] \\ & + \mathrm{Si}[(U+\alpha)(1-T)] + \mathrm{Si}[(U-\alpha)(1-T)] \end{aligned} \tag{8.12}$$

The extrema locations are found by setting the derivative of f w.r.t. T equal to zero using the previous value for U, and dropping the phase term:

$$f' = \frac{\sin[(U+\alpha)(1+T)]}{1+T} + \frac{\sin[(U-\alpha)(1+T)]}{1+T} - \frac{\sin[(U+\alpha)(1-T)]}{1-T} - \frac{\sin[(U-\alpha)(1-T)]}{1-T} = 0 \tag{8.13}$$

Because of the oscillations of each of these four terms, there are many clustered roots. A useful approximation assumes the first three Si to have large arguments, hence $\simeq \pi/2$. Then,

$$f' \simeq \frac{\sin[(U-\alpha)(1-T)]}{1-T} = 0 \tag{8.14}$$

which gives

$$T \simeq 1 - \frac{n\pi}{U-\alpha} = 1 - \frac{n\pi}{U(1-\sin\theta_0)} \tag{8.15}$$

Thus the period is increased by a pseudo-grating lobe factor $1/(1-\sin\theta_0)$. This reduction in number of oscillations with increasing scan angle closely fits the results from the computer simulations. Next, a model will be constructed for these *scan impedance* results (Hansen, 1996).

The method used here has been called "observable-based parametrization" (Carin and Felsen, 1993), where the observed data (from the simulators) is fitted with appropriate formulations, in this case the Gibbs's oscillations. These are nominally $(2/\pi)$ $[Si(\zeta\pi) - 1]$, where $\sim\pi L/\lambda$ extrema (at $\zeta = n$) occur from array edge to center. For H-plane scan of a dipole array, the $[Si(\zeta\pi) - 1]$ Gibbsian factor applies directly. When a screen is added, the decay from edge to center is much faster; it is fitted until $[Si(\zeta\pi) - 1]^{1.5}$. This is due to the different decay of mutual coupling with screen. The E-plane scan of dipoles shows a smaller decay; an exponent of 0.75 is used. These Gibbsian models fit the simulator data well except for the edge element. The fit is fair for the front element when the amplitude is doubled; a transition is used between the edge and the subsequent extrema. For the rear edge element, a polynomial fit was made to the simulated data.

Next, the Gibbsian model results will be compared with those from the simulators for 201 elements. Dipole array results, with H-plane scan, are shown in Figures 8.19–8.21, for scan angles of 0, 45, and 60°. Model data are dotted lines, while simulator data are solid lines. It can be seen that the agreement is excellent. Note that the match between the discrete array points and the dotted curve, as in Figure 8.20, is important, rather than the comparison between the dotted peak and the array solid line. Figures 8.22 and 8.23 give the comparison for 51 elements at 0 and 60°, while Fig. 8.24 is for 21 elements at broadside. Again excellent agreement.

Scan impedance of dipole with screen arrays of 201 elements for *H*-plane scan are displayed in Figures 8.25–8.27; again dotted lines are the Gibbsian model, and solid lines are simulator. The more rapid decay of the oscillations is apparent, although the edge values are comparable to those of the preceding set of data. Figures 8.28 and 8.29 show broadside arrays of 51 and 21 elements. Fits are excellent.

The *E*-plane scan produces a much slower decay of oscillations in *scan impedance*, as seen in Figures 8.30 and 8.31, which are for arrays of 201 dipoles. Figure 8.32 gives *scan impedance* for a 51-element broadside array. Again the match between model data and simulator data is excellent, except for edge elements. A dipole–screen array at broadside shows the modulated oscillations discussed in Section 8.3.1 in Figure 8.33. A large-period sine factor has been used to model these oscillations. The match is excellent over part of the array, and fair over other parts. For larger scan angles, where the modulation has disappeared, a Gibbsian factor exponent of 2.5 is used, with fair results. However for the broadside case above, the exponent was reduced to 1.5 because of the effect of the heterodyne factor.

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CHAPTER NINE

Superdirective Arrays

9.1 HISTORICAL NOTES

A useful operational definition of antenna array superdirectivity (formerly called supergain) is directivity (see Chapter 2) higher than that obtained with the same array length and elements uniformly excited (constant amplitude and linear phase). Excessive array superdirectivity inflicts major problems in low radiation resistance (hence low efficiency), sensitive excitation and position tolerances, and narrow bandwidth. In principle, superdirectivity applies to both arrays of isotropic elements, and to actual antenna arrays composed of nonisotropic elements, such as dipoles.

Probably the earliest work on the possibility of superdirectivity was by Oseen (1922): See Bloch et al. (1953) for a list of early references. A limited endfire superdirectivity using a monotonic phase function was accomplished by W. W. Hansen and Woodyard (1938). Another early contributor was Franz (1939). Schelkunoff (1943) in a classic paper on linear arrays discussed, among other topics, array spacings $<\lambda/2$, showing how equal spacing of the array polynomial zeros over that portion of the unit circle represented by the array gives superdirectivity. The field received wide publicity when La Paz and Miller (1943) purported to show that a given aperture would allow a maximum directivity, and when Bouwkamp and de Bruijn (1946) showed that they had made an error and that there was no limit on theoretical directivity. Thus the important theorem: A fixed aperture size can achieve (in theory) any desired directivity value. This theorem is now widely recognized, but the practical implications are less well known. Bloch et al. (1960) say that the theorem has been rediscovered several times; the practical limitations of superdirectivity occur as a surprise to systems engineers and others year after year! In 1946, a burst of wartime research reporting occurred. Reid (1946) generalized the Hansen–Woodyard endfire directivity as $d \to 0$ and Dolph (1946) invented the widely known Dolph–Chebyshev array distribution wherein the equal level oscillations of a Chebyshev

Phased Array Antennas, Second Edition. By R. C. Hansen
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polynomial are used to produce an array pattern with equal-level sidelobes. To follow this last development, Riblet (1947) developed Dolph–Chebyshev arrays for spacing below $\lambda/2$, that is, superdirective. DuHamel (1953) and Stegen (1953) developed complementary advances in the computation of Dolph–Chebyshev coefficients and directivity. Maximum directivity for an array with fixed spacings was derived, for acoustic arrays, by Pritchard (1953).

Superdirective aperture design thus requires a constraint; arrays with fixed number of elements and spacing, owing to the finite number of variables, do not, as clearly there is an excitation that provides maximum directivity. The Lagrangian process for determining this maximum will be discussed, followed by a discussion of Chebyshev array design and other constrained optimization schemes. Finally, matching problems will be examined.

9.2 MAXIMUM ARRAY DIRECTIVITY

9.2.1 Broadside Directivity for Fixed Spacing

An array with fixed length and number of elements represents a determinate problem. Clearly, a maximum directivity exists. The maximum value can be found by using the Lagrange multiplier method (Sokolnikoff and Sokolnikoff, 1941). The directivity G at broadside can be written as:

$$G = \frac{\left(\sum_{n=1}^{N} A_n\right)^2}{\sum_{n=1}^{N} \sum_{m=1}^{N} A_n A_m \ \mathrm{sinc}[(n-m)2\pi d/\lambda]} \tag{9.1}$$

Here it is assumed that N array elements are isotropic and equally spaced by d. In the formulation above, the array amplitudes are A_n and the sinc function $[(\sin x)/x]$ represents the mutual impedance between isotropic elements (Hansen, 1983): $120 \ \mathrm{sinc}\, kd$. Although the directivity expression could be maximized directly, it is convenient to constrain the sum of the coefficients to unity, and then to minimize the denominator. The Langrangian equations are

$$\begin{gathered} 2\sum_{n=1}^{N} A_n \ \mathrm{sinc}[(n-i)2\pi d/\lambda] - \beta = 0 \qquad i = 1, 2, 3, \ldots, N \\ \sum_{n=1}^{N} A_n = 1 \end{gathered} \tag{9.2}$$

β is the Lagrangian multiplier. Solving the first equation for $i = 1$ for β and substituting gives N equations in the unknown coefficients for an N-element array. These are easily solved using simultaneous equation computer subroutines with multiple precision as needed. These were solved by hand for $N = 3$, 5, 7 by Pritchard (1953).

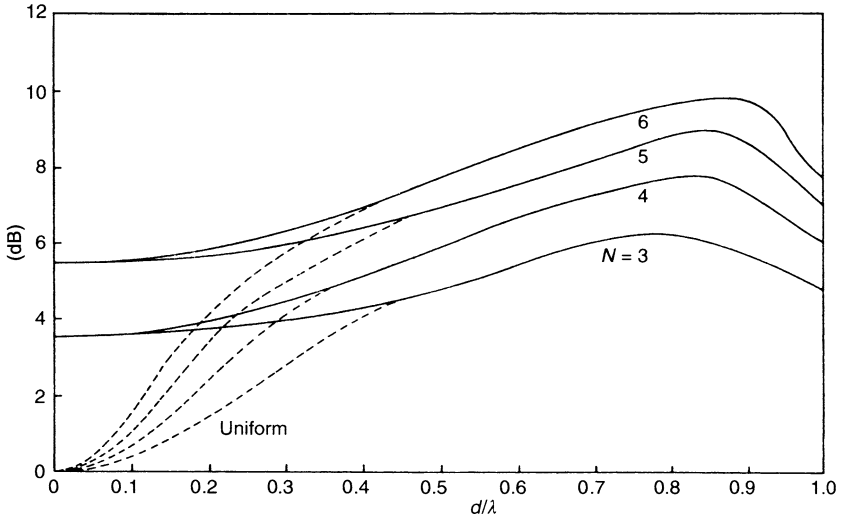


FIGURE 9.1 Maximum directivity for fixed spacing.

Hansen (1983) compares the maximum directivity of small arrays with the uniform amplitude directivity versus element spacing (see Fig. 9.1). Above $d/\lambda = 0.5$, the two are very close. Also, some minor oscillations in the directivity curves have been smoothed out, as they are not important here. The coalescing of pairs of curves at zero spacing occurs because arrays of $2N$ and $2N - 1$ elements have the same number of degrees of freedom.

To give an idea of the coefficients and pattern of a small array with modest superdirectivity, Table 9.1 shows the amplitude coefficients for an array of seven elements with quarter-wave spacing. Directivity is 5.21, and the pattern is the solid line in Figure 9.2. Also shown (dashed) is the pattern of the same length array with half-wave spacing. Directivity of a corresponding uniform amplitude array is 3.64. Note that some elements couple power into the feed network.

TABLE 9.1 Seven-Element Superdirective Array, $d = \lambda/4$

Element Number	Amplitude	Resistance (Ω)
1	1.443	0.13299
2	−3.933	−0.04879
3	7.122	0.02694
4	−8.264	−0.02322
5	7.122	0.02694
6	−3.933	−0.04879
7	1.443	0.13299

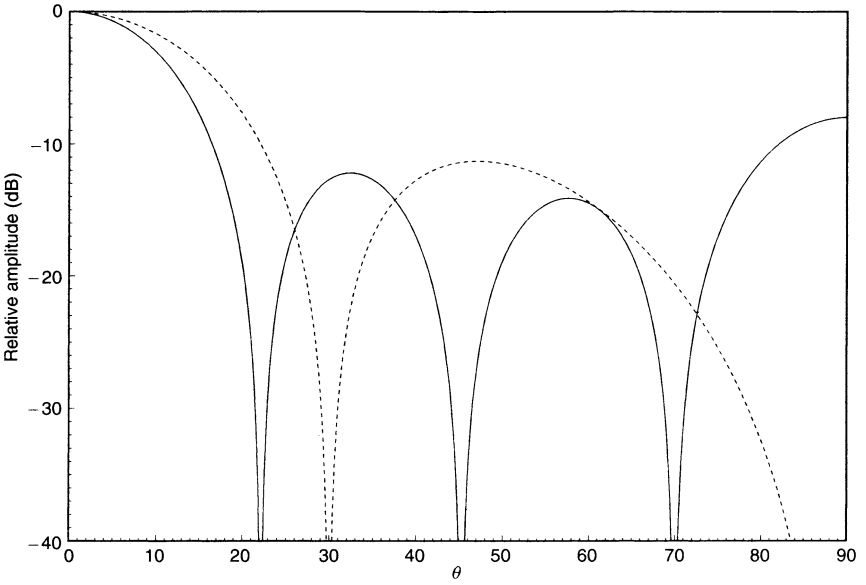


FIGURE 9.2 Maximum directivity array, $N = 7$, $d = 0.25\lambda$.

9.2.2 Directivity as Spacing Approaches Zero

Uzkov (1946) transformed the maximum directivity limit for an array as spacing goes to zero into a sum of Legendre polynomials. His result for an array of N elements is

$$G = \sum_{n=0}^{N-1} (2n+1)[P_n(\cos\theta)]^2 \tag{9.3}$$

where θ is the angle of the main beam from broadside. Tai (1964) independently developed this for broadside arrays, where $\cos\theta = 1$. Since the Legendre polynomial of argument 1 can be written as a product of factors, the result for maximum directivity is simply

$$G \longrightarrow \left(\frac{1 \cdot 3 \cdot 5 \cdots M}{2 \cdot 4 \cdot 6 \cdots (M-1)}\right) \qquad \text{where } M = 2\,AINT\left(\frac{N+1}{2}\right) - 1 \tag{9.4}$$

Here M is the number of odd elements or the number of even elements minus 1. Thus the interesting result that three- and four-element arrays have the same limiting value, five- and six-element arrays have the same value, and so on. After reflection this is not surprising since both three- and four-element arrays have two variables, both five- and six-elements have three variables, and so on. This maximum directivity is plotted in Figure 9.3. The circles show the corresponding limiting directivity for Chebyshev arrays with 10-dB SLR and the squares are for 20-dB SLR. The Chebyshev directivity is less, as the maximum directivity pattern does not

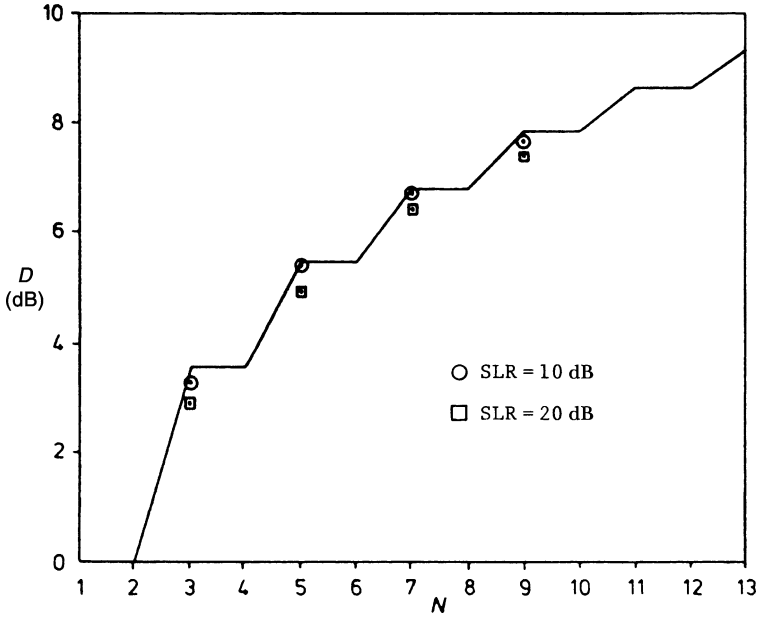


FIGURE 9.3 Maximum broadside directivity in zero-spacing limit.

have equal-level sidelobes, and even in the case where there is only one sidelobe (backlobe), the Chebyshev result would be equal to the maximum value only if the sidelobe ratio were properly chosen. See Section 9.2.4 for a discussion of Chebyshev arrays.

For endfire arrays, the maximum directivity in the limit of zero spacing is

$$G \longrightarrow \sum_{n=0}^{N-1} (2n+1)[P_n(0)]^2 = \sum_{n=0}^{N-1} (2n+1) = N^2 \tag{9.5}$$

9.2.3 Endfire Directivity

W. W. Hansen and Woodyard (1938) developed an endfire line source with modest superdirectivity. This finding is of interest because the distribution can be sampled to get array excitations and because the amplitude is constant, a feature that is attractive for arrays. They observed that, if the free-space phase progression along the aperture was increased, the space-factor power integral decreased faster than the peak value; thus the directivity increases up to a point. The endfire pattern, for a source of length L, is

$$f(\theta) = \operatorname{sinc}\left(\frac{L}{2}(k \sin\theta - \beta)\right) \tag{9.6}$$

where β is the wavenumber over the aperture. Inverse directivity is proportional to

$$\frac{1}{G} \propto \frac{1}{\operatorname{sinc}^2\left(\frac{L}{2}(k-\beta)\right)} \int \operatorname{sinc}^2\left(\frac{L}{2}(k\sin\theta-\beta)\right)\cos\theta\, d\theta$$

$$= \frac{1}{\operatorname{sinc}^2(\phi/2)}\left(\frac{\pi}{2} + \operatorname{Si}\phi + \frac{\cos\phi - 1}{\phi}\right) \tag{9.7}$$

Here Si is the sine integral, and $\phi = L(k-\beta)$, the additional phase along the aperture (in addition to the progressive endfire phase). Maximum directivity was determined to occur for $\phi = 2.922$ rad. In many books, it is carelessly stated that π extra radians of phase are needed, but there is no physical reason for this; a better approximation to 2.92 is 3. Directivity increase over normal endfire is 2.56 dB, and the sidelobe ratio is 9.92 dB. The distribution is suitable for long arrays; for short arrays a computer optimization of phase is recommended (see Hansen, 1992).

The Hansen–Woodyard distribution is endfire. In general, the maximum directivity does not occur there. The most general solution for uniform amplitude would allow any element phases needed to maximize directivity. Such a solution could be formally realized for a given number of elements and spacing, but the equations would require a numerical solution. A slightly simpler problem was worked by Bach (1970); he started with a uniform-amplitude array that was phased to produce a main beam at θ_0. The interelement phase factor is $\delta = kd\sin\theta_0$, and the directivity (see Chapter 2) is

$$G = \frac{N}{1 + \frac{2}{N}\sum_{n=1}^{N-1}(N-n)\operatorname{sinc}(nkd)\cos(n\delta)} \tag{9.8}$$

Calculations were made for 2-, 3-, 4-, and 10-element arrays for all beam angles, and for spacings up to λ. Figure 9.4 shows the results for 2- and 3-element arrays, while Figure 9.5 is for 4- and 10-element arrays. Figure 9.5b is striking in that high directivity occurs along a line roughly for $kd + \delta = 0$, or $\theta_0 = -\pi/2$, with peak directivity near endfire at λ spacing. Along the line roughly for $kd = \delta + 2\pi$ directivity is changing rapidly, perhaps owing to appearance of another lobe. Directivity values are shown at θ_0, but in some cases a "sidelobe" may have higher amplitude. Thus even for uniform amplitude an array is complex.

9.2.4 Bandwidth, Efficiency, and Tolerances

The first of three major difficulties with superdirective arrays is bandwidth, which rapidly becomes a problem as the element spacing decreases $<\lambda/2$. Thus the Q is of concern; for narrowband antennas, bandwidth $\simeq 1/Q$, and the impedance matched bandwidth $\simeq 2/Q$. The ratio of stored to dissipated energy, Q, can similarly

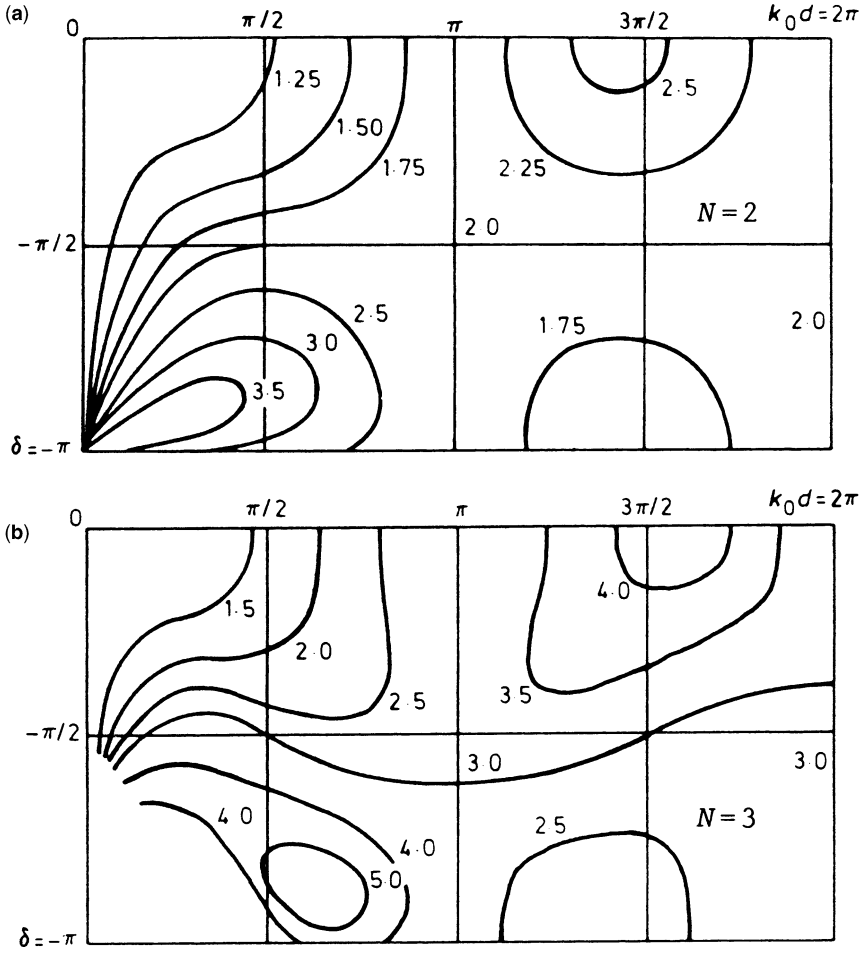


FIGURE 9.4 Uniform array directivity. (Courtesy Bach, H., "Directivity of Basic Linear Arrays", *Trans. IEEE*, Vol. AP-18, Jan. 1970, pp. 107–110.)

be written in terms of array coefficients and mutual coupling, which for isotropic elements is

$$Q = \frac{\sum_{n=1}^{N} A_n^2}{\sum_{n=1}^{N} \sum_{m=1}^{N} A_n A_m \operatorname{sinc}[(n-m)2\pi d/\lambda]} \tag{9.9}$$

Calculations have shown (Hansen, 1981b) that, for broadside arrays of fixed length, both directivity and Q increase with the number of elements as expected. An array with even number of elements has a slightly higher Q than the array with the next larger odd number of elements, possibly because the odd-element sampling is more

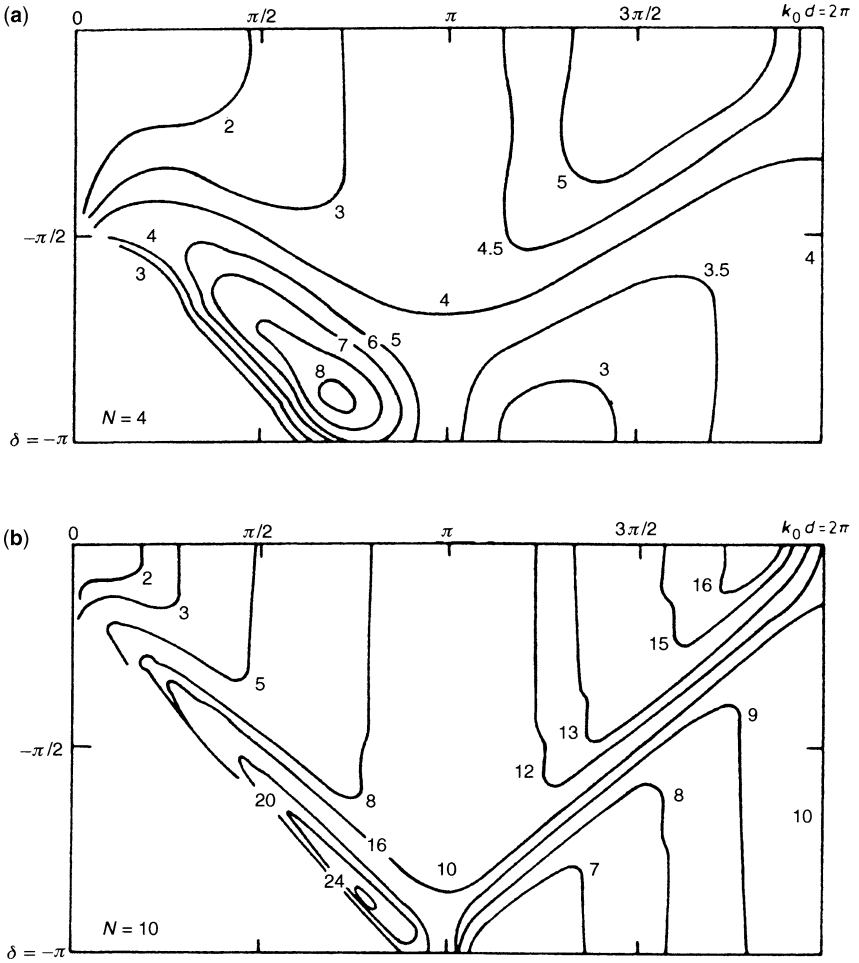


FIGURE 9.5 Uniform array directivity. (Courtesy Bach, H., "Directivity of Basic Linear Arrays", *Trans. IEEE*, Vol. AP-18, Jan. 1970, pp. 107–110.)

efficient. For all the cases computed, $\log Q$ varied approximately linearly with directivity. Figure 9.6 shows $\log Q$ versus directivity for odd arrays with lengths 1λ, 2λ, and 5λ. The circles represent points calculated in double precision; extended precision is required for arrays of more elements than those shown in the figure. The straight lines are drawn through the uniform excitation directivity point (equal to G_0) with a slope of π, where G_0 is the gain of the same length array with half-wave spacing: $G_0 = 1 + 2L/\lambda$. There is at this time no physical significance to using this value of slope, but it is suggested by calculations of Rhodes (1974) on superdirective line sources. In making a best straight-line fit to the set of points for each of the three arrays, the slopes were in fact clustered around the value of 3.14. However, it is difficult to perform this fit with precision because, as pointed out by Rhodes, the curve of $\log Q$ versus directivity has an oscillatory behavior for low values of Q. If the

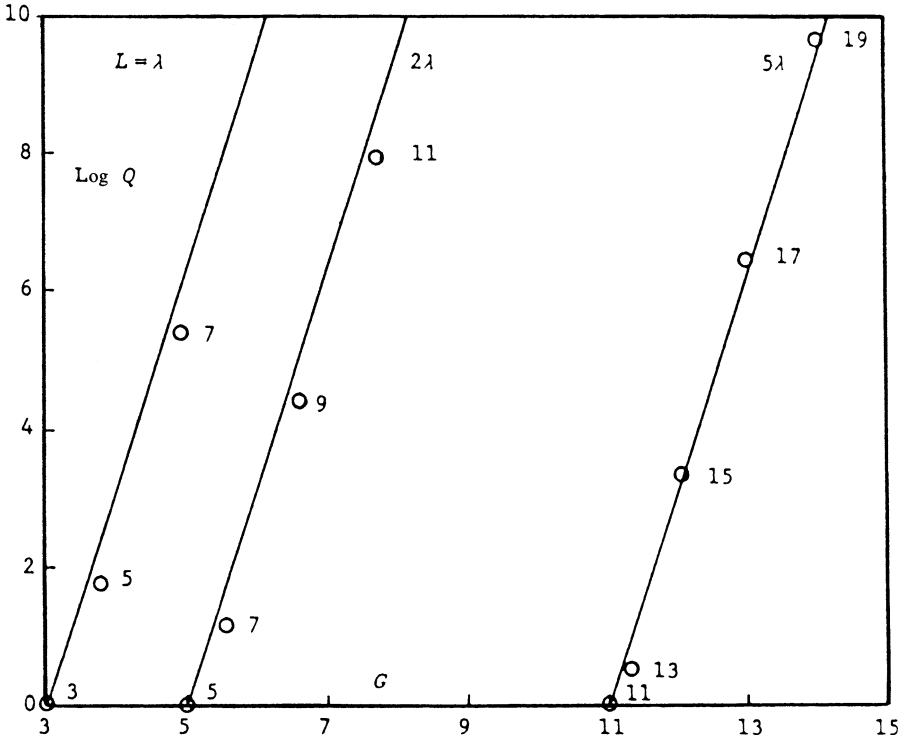


FIGURE 9.6 Plot of Q versus maximum directivity, odd arrays.

assumptions above are true, superdirective broadside array behavior is predicted by the equation

$$\log Q = \pi(G - G_0) \tag{9.10}$$

Thus the superdirective array clearly fits into the category of fundamental limitations in antennas (Hansen, 1981a). Whether the assumed slope of π can be physically justified remains an interesting problem at this time.

The Q to be expected from an array of isotropes is approximately a function of the number of elements divided by d/λ. Figure 9.7 shows this relationship for many arrays; each array is represented by a circle. Spacings $>0.3\lambda$ are not included, as the amount of superdirectivity achieved is small for these. Log Q is approximately linear with $N\lambda/d$; the straight-line fit in the figure is

$$\log Q \simeq 0.16043\left(\frac{N\lambda}{d}\right) - 1.53476 \tag{9.11}$$

These data allow the array designer to estimate the degree of superdirectivity (Q) for a given array geometry.

Calculations of performance have been made for superdirective linear arrays of parallel dipoles. Collinear dipoles are not considered as they would overlap neighbors.

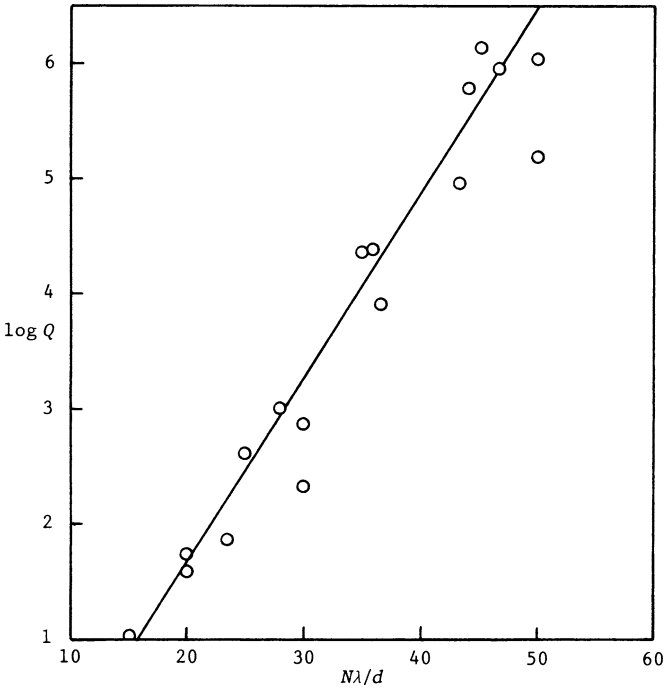


FIGURE 9.7 Plot of Q for broadside linear superdirective arrays.

Figure 9.8 shows log Q versus directivity for arrays 2 wavelengths long. Both 0.1λ and 0.5λ dipoles are used. Again log Q versus G is a straight line. As expected, the half-wave dipole array has higher Q than the corresponding isotropic array, owing to energy storage in the dipole near-field. With short (0.1λ) dipoles this energy storage is much larger, resulting in much higher Q.

For endfire linear parallel dipole arrays, the variation of log Q with directivity is again linear, but the slope changes as the length of the array changes. Figure 9.9 gives data for arrays of lengths 1λ, 2λ, and 5λ; the number of elements in each array is shown in the figure (Hansen, 1990).

A second undesirable feature of superdirective arrays is low efficiency, due both to matching network losses (see Section 9.4.1), and to losses in the antenna elements. Both are caused by the low-radiation resistances of these arrays. Typically, the elements at the array center show the lowest radiation resistance; calculations for many small arrays of isotropes shows that $R_{\text{rad}} \propto 1/\sqrt{Q}$. Figure 9.10 shows these data, where each circle represents one array. The straight-line log–log fit is

$$R_{\text{rad}} \simeq \frac{8.058}{\sqrt{Q}} \tag{9.12}$$

Similar data for endfire arrays are given in Figure 9.11. A nearly log–log dependence of radiation resistances and Q is observed.

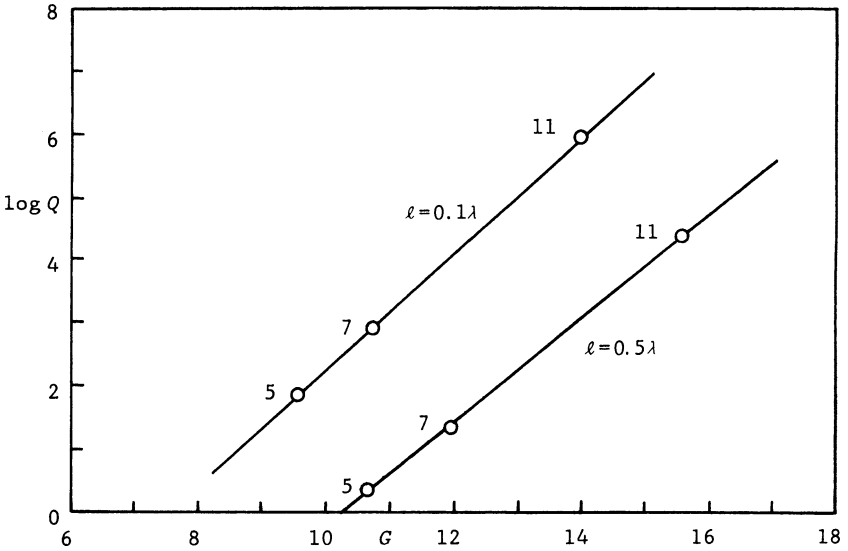


FIGURE 9.8 The Q plot of 2-wavelength broadside array of parallel dipoles.

Loss resistance of cylindrical or strip dipoles is easily computed: A dipole of length ℓ, radius a, made of material with surface resistance R_s, has a loss resistance R_{loss} of

$$R_{\text{loss}} = \frac{R_s \ell (1 - \operatorname{sinc} k\ell)}{4\pi a \sin^2 \frac{1}{2} k\ell} \tag{9.13}$$

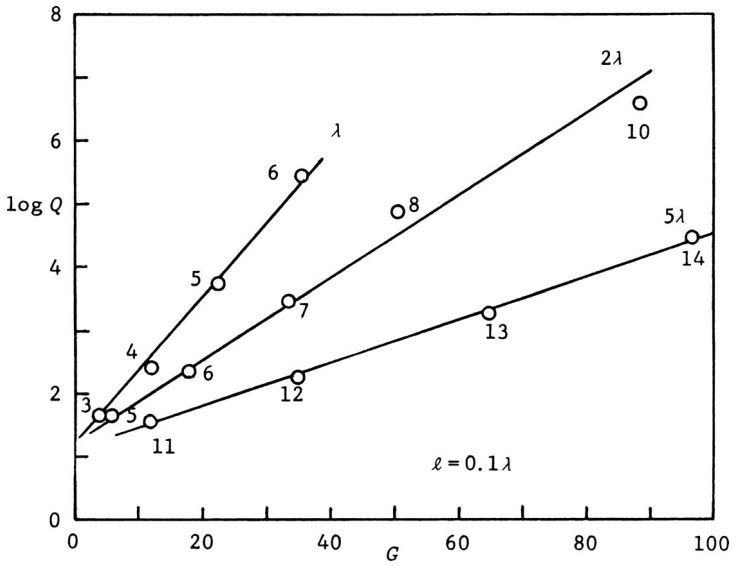


FIGURE 9.9 The Q plot of endfire arrays of parallel dipoles.

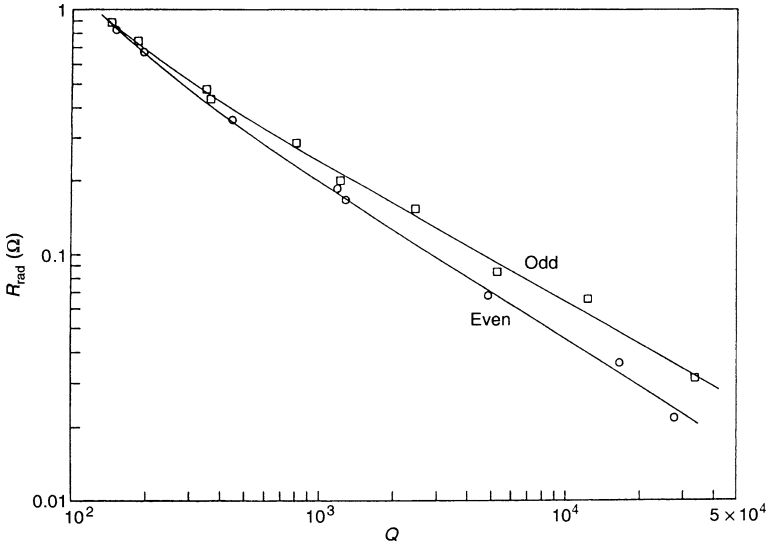


FIGURE 9.10 Plot of Q versus center element resistance for broadside arrays.

For half-wave or resonant dipoles, $R_{\text{loss}} = R_s \ell / 4\pi a$. Strip dipoles are equivalent to cylindrical dipoles, with strip width equal to $4a$. Copper wires have an ideal surface resistance of

$$R_s \simeq 0.000261 \sqrt{f_{\text{MHz}}} \tag{9.14}$$

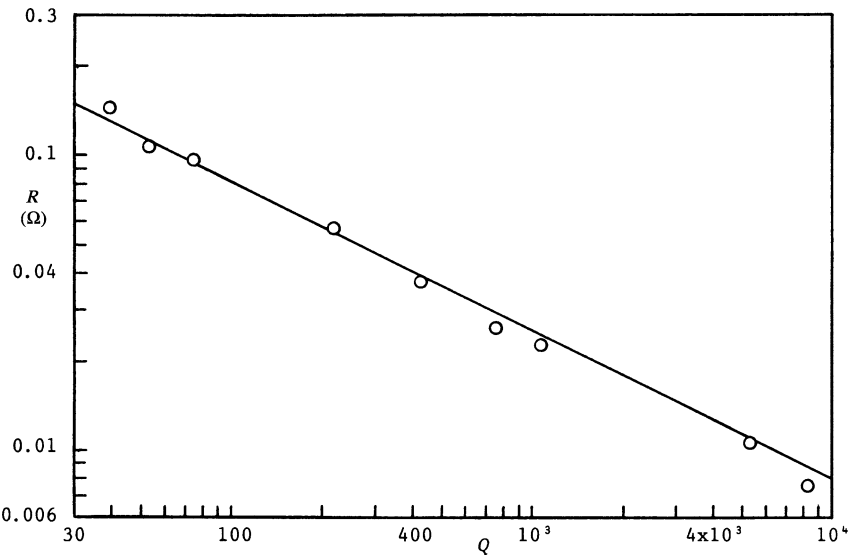


FIGURE 9.11 Plot of R versus Q for endfire superdirective arrays.

Over the range of 10–1000 MHz, R_s varies from 0.000825 to 0.00825 ohms/square. Using a value of $\ell/a = 25$, a moderately fat dipole, the loss resistance varies from 0.0066 Ω at 10 MHz–0.066 Ω at 1 GHz. When these typical loss resistance numbers are compared with the radiation resistance values of Figures 9.10 and 9.11, it is clear that superdirective array efficiency may be a severely limiting consideration. Use of high-temperature superconductor (HTS) materials in the array and feed network can produce high efficiencies, but now the Q is that from Figures 9.7 to 9.9.

The third significant problem with superdirective arrays is tight tolerances. Because superdirectivity involves a partial cancelation of the element contributions at the main beam peak, with more cancelation for more superdirectivity, the tolerance of each element coefficient (excitation) becomes smaller (tighter) with more superdirectivity (Uzsoky and Solymar, 1956). A simple calculation has been made of these effects for Dolph–Chebyshev arrays by perturbing the center element of an odd array, finding the tolerance to reduce the directivity by 0.5 dB. This is not expected to be sensitive to the sidelobe ratio. A value of 20 dB was used. Calculations were made for $N = 3$ and 5 as a function of spacing, with the results shown in Figure 9.12. Note that the percentage error versus d/λ curve is linear up to spacings of roughly 0.1λ. For $N = 3$, the slope is 2:1, and for $N = 5$ the slope is 4:1. Thus for $N = 3$, if the element spacing is halved, the tolerances must be four times tighter. The percentage tolerance for the center element, to maintain close (0.5 dB) to the expected directivity, is approximately given by

$$\frac{100}{\sqrt{Q}} \tag{9.15}$$

but the constant varies with the number of elements.

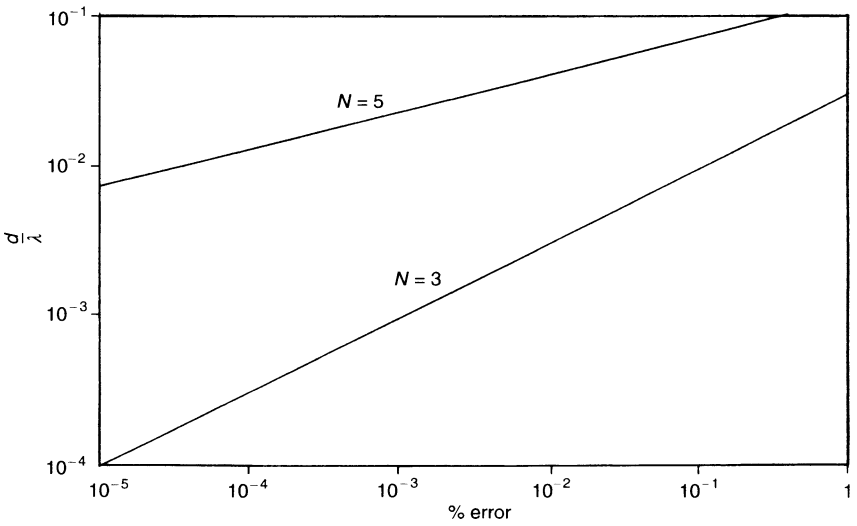


FIGURE 9.12 Center element tolerance for 0.5-dB directivity drop.

Thus with $Q = 1000$, for example, the tolerance is 3.2%. Bandwidth and radiation resistance are more serious limitations of superdirective arrays.

9.3 CONSTRAINED OPTIMIZATION

Directivity may be maximized subject to a constraint, which may be on pattern sidelobes, on bandwidth or Q, on allowable tolerances, the signal/noise ratio (S/N), and so on. The simplest is sidelobe constraint, after Dolph and Chebyshev.

9.3.1 Dolph–Chebyshev Superdirectivity

The principles and design equations for arrays with equal-level sidelobes, Dolph–Chebyshev arrays, were covered in detail in Chapter 3. However, Dolph's derivation and the formulas of Stegen are limited to $d \geq \lambda/2$. Riblet (1947) showed that this restriction could be removed, but only for N odd. For spacing below half-wave, the space factor is formed by starting at a point near the end of the Chebyshev ± 1 region,[1] tracing the oscillatory region to the other end, then retracing back to the start end and up the monotonic portion to form the main beam half. Since the Mth-order Chebyshev has $M - 1$ oscillations, which are traced twice, and since the trace from 0 to 1 and back forms the center sidelobe (in between the trace out and back), the space factor always has an odd number of sidelobes each side, or an even number of zeros. Hence, only an odd number of elements can be formed into a Chebyshev array for $d < \lambda/2$. The pattern is given by

$$\begin{aligned} &T_M(a \cos \Psi + b) \\ &a = \frac{z_0 + 1}{1 - \cos kd} \\ &b = \frac{z_0 \cos kd + 1}{\cos kd - 1} \end{aligned} \tag{9.16}$$

where as before $\Psi = kd \sin \theta$. The value of z_0 is

$$z_0 = \cosh \frac{\operatorname{arccosh} \text{ SLR}}{M} \tag{9.17}$$

The sidelobe ratio is

$$\text{SLR} = T_M(z_0) \tag{9.18}$$

Formulas have been developed by DuHamel (1953), Brown (1957, 1962), Salzer (1975), and Drane (1963, 1964). Those of Drane will be used here as they are suitable

[1]The exact starting point depends on N and kd.

for computer calculation of superdirective arrays. The array amplitudes are

$$A_n = \frac{\varepsilon_n}{4M} \sum_{m=0}^{M_1} \varepsilon_m \varepsilon_{M_2-m} T_n(x_n)[T_M(ax_n + b) + (-1)^n T_M(b - ax_n)] \tag{9.19}$$

where $\varepsilon_i = 1$ for $i = 0$ and is equal to 2 for $i > 0$; $x_n = \cos n\pi/M$. The integers M_1 and M_2 are, respectively, the integer parts of $M/2$ and $(M + 1)/2$. This result is valid for $d \leq \lambda/2$. Small spacings (highly superdirective arrays) may require multiple precision due to the subtraction of terms. Many arrays are half-wave spaced; for these the a and b reduce to

$$a = \frac{1}{2}(z_0 + 1) \qquad b = \frac{1}{2}(z_0 - 1) \tag{9.20}$$

For half-wave spacing, this approach and that of Chapter 2 give identical results! In fact, owing to the properties of the Chebyshev polynomial, the two space factors, in precursor form, are equal:

$$T_{N-1}\left(z_0^{N-1} \cos\frac{\Psi}{2}\right) \equiv T_M\left(\frac{z_0^M(\cos\Psi + 1) + \cos\Psi - 1}{2}\right) \qquad N - 1 = 2M \tag{9.21}$$

where the superscripts on z_0 indicate that each must be chosen for the proper form. Since many computers have no inverse hyperbolic functions, it is convenient to rewrite the z_0 as

$$z_0 = \frac{1}{2}\left[\text{SLR} + \sqrt{\text{SLR}^2 - 1}\right]^{1/M} + \frac{1}{2}\left[\text{SLR} - \sqrt{\text{SLR}^2 - 1}\right]^{1/M} \tag{9.22}$$

Directivity for arrays of 3, 5, 7, and 9 elements have been calculated, for sidelobe ratios of 10 and 20 dB. The superdirectivity can be seen in Figures 9.13 and 9.14 for spacing $<0.5\lambda$, as the ordinary directivity (using the Chebyshev coefficients that are independent of spacing) goes smoothly to 0 dB at zero spacing. The figures display these calculated directivities versus element spacing. Thus a three-element array offers roughly 3-dB directivity for small spacings, five elements offers roughly 5 dB. Next the disadvantages of superdirectivity will be discussed.

The parameter Q is given by Lo et al. (1966) as

$$Q = \frac{\sum_{n=1}^{N} A_n^2}{\sum_{n=1}^{N} \sum_{m=1}^{N} A_n A_m \operatorname{sinc}[(n - m)kd]} \tag{9.23}$$

Q, which is the inverse of fractional bandwidth, is plotted for Dolph–Chebyshev arrays versus d/λ in Figure 9.15. The $N = 3$ array has a log–log slope of Q versus spacing of 4:1, while the $N = 5$ array has a slope of 8:1. The $N = 7$ and 9 arrays have even higher slopes. For small values of Q, the curve is not too accurate.

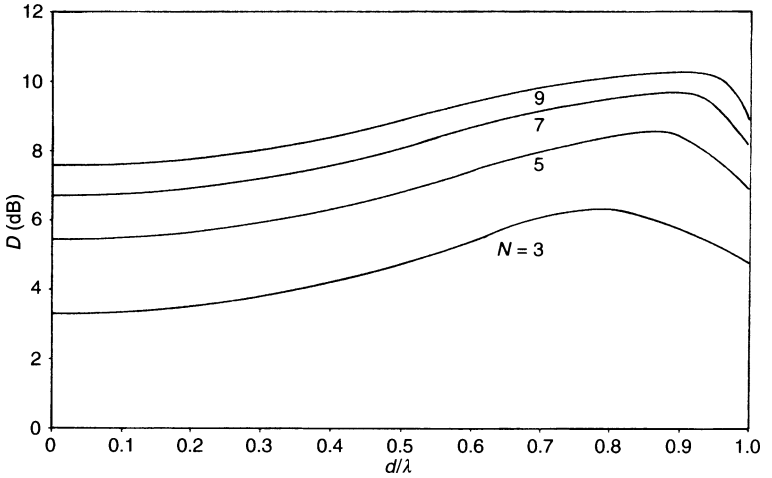


FIGURE 9.13 Chebyshev array directivity, SLR = 10 dB.

Bandwidth appears to be more restrictive than tolerances; the $N = 3$ array, to be practical, requires a spacing of the order of 0.1λ or larger. The $N > 3$ arrays are even less forgiving.

Directivity is given by

$$G = \frac{\left(\sum_{n=1}^{N} A_n\right)^2}{\sum_{n=1}^{N} \sum_{m=1}^{N} A_n A_m \ \text{sinc}[(n-m)kd]} \tag{9.24}$$

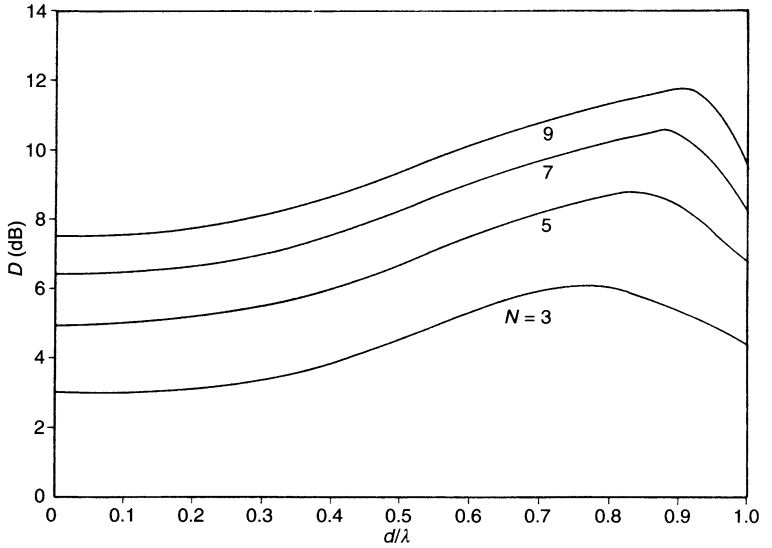


FIGURE 9.14 Chebyshev array directivity, SLR = 20 dB.

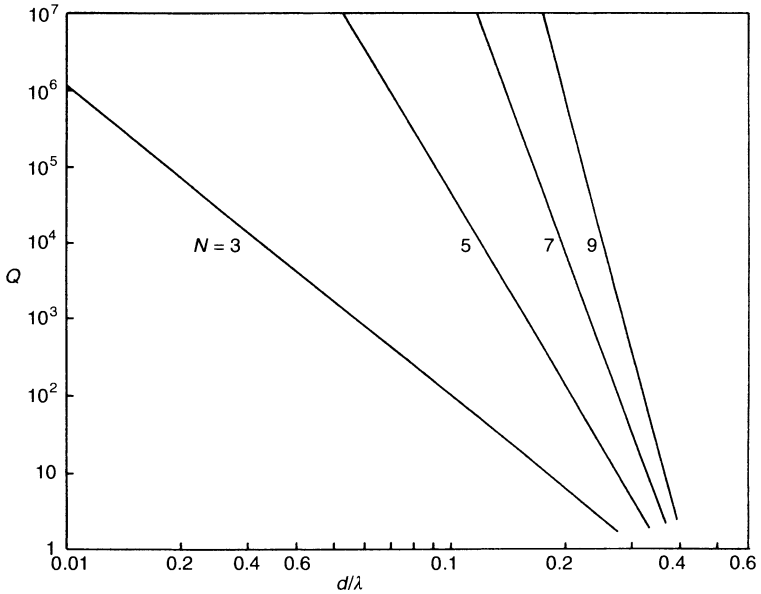


FIGURE 9.15 The Q plot of broadside arrays.

Note the similarity to the result for Q. Two cases of Chebyshev superdirective arrays have been calculated to illustrate the variation of Q with directivity increase. First is a two-wavelength array of isotropic elements, which with half-wave spacing has a minimum directivity of 5. This two-wavelength aperture is occupied by 7, 9, 11, and 13 elements. Table 9.2 shows the element spacing, directivity, and Q, for a design SLR of 20 dB. The Q values are shown in Figure 9.16, where the straight line has been fitted to the data, and is

$$\log Q \simeq 3.3175(G - 5) \tag{9.25}$$

The form of the equation, with the normal directivity ($G_0 = 5$) subtracted, was suggested by calculations of Rhodes. Rhodes (1971, 1974) maximized directivity of a line source subject to a constraint on Q using the eigenfunctions of the source: prolate spheroidal functions. These are particularly suited to allow a simple derivation as they are doubly orthogonal: over infinite limits, which fits the total energy integral

TABLE 9.2 Two-Wavelength Array, SLR = 20 dB

N	d/λ	G	Q
5	0.500	4.69	1.7
7	0.333	5.18	7.0
9	0.250	6.21	1.2×10^4
11	0.200	7.36	5.8×10^7
13	0.167	8.54	5.5×10^{11}

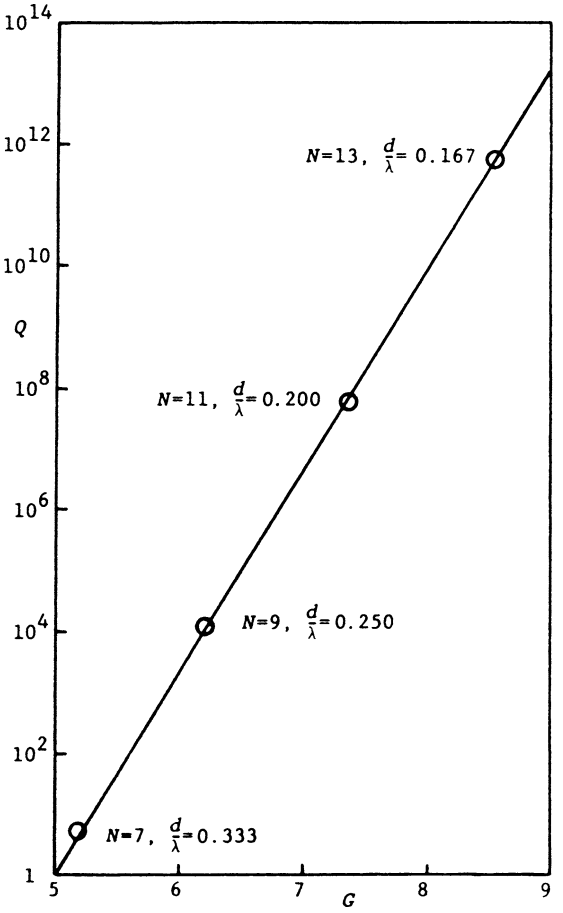


FIGURE 9.16 Plot of Q versus directivity, SLR = 20 dB.

over all angles, and over finite limits, which fits the radiated energy integral over real angles. He found a roughly linear relationship between $\log Q$ and directivity, with a coefficient of roughly 2. This substantiates the suspicion that the Chebyshev design may result in a higher Q than necessary for a given directivity, since the coefficients from Tables 9.2 and 9.3 are in the 3–4 range. To show the effects of

TABLE 9.3 Two-Wavelength Array, SLR = 10 dB

N	d/λ	G	Q
5	0.500	4.89	1.8
7	0.333	5.40	37
9	0.250	6.19	7.6×10^4
11	0.200	6.99	3.5×10^8
13	0.167	7.74	3.3×10^{12}

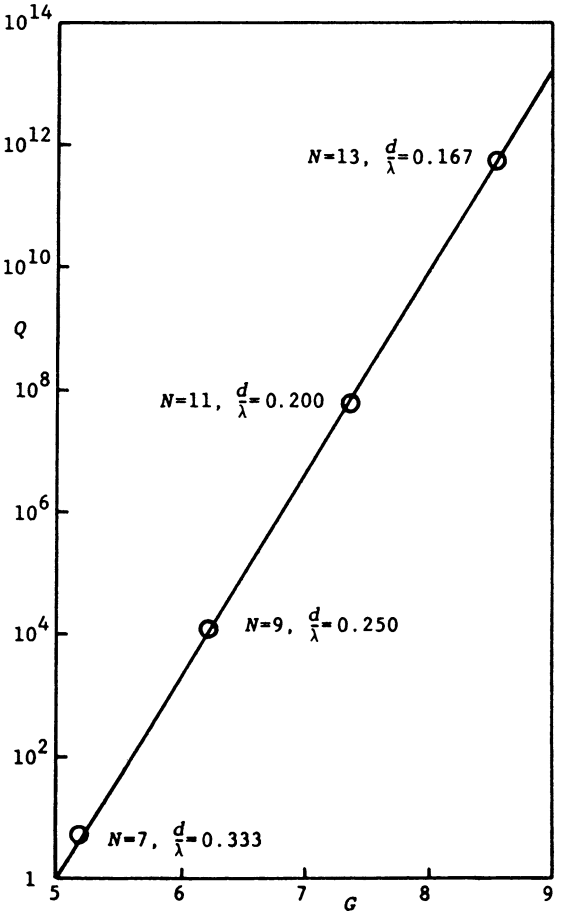


FIGURE 9.17 Plot of Q versus directivity, SLR = 10 dB.

SLR, Table 9.3 is a repeat for SLR = 10 dB and Figure 9.17 shows a straight-line fit, with coefficient 4.1475. Higher sidelobes increase directivity and Q. Calculated points for $N = 13$ are inaccurate, double precision on a 32-bit machine is inadequate. Thus the Q increases approximately exponentially with directivity above the normal value. For most increases above normal directivity, it may be inferred from Rhodes that the Q curve has minor oscillations. Note the very rapid increase of Q with directivity; even a 10% increase produces a bandwidth limited to a few percent. However, the Chebyshev design may not give the lowest Q; a constrained synthesis is apparently necessary to produce the lowest Q for a given directivity. Newman et al. (1978) did this for endfire arrays, but only for two cases.

The second case is a half-wavelength aperture, originally computed by Yaru (1951), which with two isotropic elements gives a maximum directivity of 2. Into this small aperture are placed three, five, seven, or nine elements. Table 9.4 gives

TABLE 9.4 Half-Wavelength Array, SLR = 20 dB

N	d/λ	G	Q
3	0.25	2.11	2.6
5	0.125	3.17	8.7×10^3
7	0.0833	4.40	5.7×10^8
9	0.0625	5.66	1.7×10^{14}

the pertinent parameters, again for a 20 dB SLR. A straight-line fit of the same type, $\log Q = a(G - 2)$, was tried but the fit is less good. It is not known whether this is due to computing errors or to the small size of the aperture in wavelengths.

DuHamel (1953) extended the Chebyshev design principle to endfire arrays, but only for $d < \lambda/2$. To avoid a back lobe, spacing is customarily made $\leq \lambda/4$. For any scan angle, Ψ is modified as usual to

$$\Psi = kd(\sin\theta - \sin\theta_0) \tag{9.26}$$

where θ_0 is the scan (main beam) angle, and the interelement phase shift is $kd \sin \theta_0$. Coefficients a and b become

$$\begin{aligned} a &= -\frac{3 + z_0 + 2\sqrt{2(z_0 + 1)}\cos kd}{2\sin^2 kd} \\ b &= \frac{\left(\sqrt{z_0 + 1} + \sqrt{2}\cos kd\right)}{2\sin^2 kd} \end{aligned} \tag{9.27}$$

9.3.2 Constraint on Q or Tolerances

Directivity may be maximized subject to constraints on bandwidth (Q) or tolerances. The basic framework of constrained optimization was developed by Gilbert and Morgan (1955) and Uzsoky and Solymar (1956), and was extended by Lo et al. (1966). Cheng (1971) wrote a review paper on this topic. Use of the braket notation introduced by Dirac (Friedman, 1956) allows the formulation to be simplified. Let $\langle A|$ be a row vector and $|A\rangle$ be a column vector. The scalar product then becomes

$$\langle A|A^*\rangle = \sum |A|_n^2 \tag{9.28}$$

where * indicates the complex conjugate. Call the row vector of complex array excitations J and the column vector of pathlength phases F:

$$F = \begin{vmatrix} \exp(-jkr_1 \sin\theta) \\ \vdots \\ \exp(-jkr_n \sin\theta) \end{vmatrix} \tag{9.29}$$

where r_n is the distance from the reference point to the nth element. For a uniformly spaced array, $r_n = (n-1)d$. Now define matrices A and B, where A is

$$|A| = |F^*\rangle\langle F| \tag{9.30}$$

and the elements of B are

$$B_{nm} = \frac{1}{4\pi}\int f_i(\theta, \phi)\exp[-jkd(n-m)\sin\theta]\, d\Omega \tag{9.31}$$

The pattern of the ith element is f_i. Isotropic elements and a uniformly spaced array allow simplifications of A and B. The elements then become

$$\begin{aligned} A_{nm} &= \exp[-jkd(n-m)\sin\theta] \\ B_{nm} &= \operatorname{sinc}[(n-m)kd] \end{aligned} \tag{9.32}$$

Now the directivity can be written in abbreviated form:

$$G = \frac{\langle J|A|J^*\rangle}{\langle J|B|J^*\rangle} \tag{9.33}$$

The tolerance sensitivity S is defined as the ratio of variance of peak field strength produced by errors of variance σ_T^2 (Uzsoky and Solymar, 1956):

$$S = \frac{\sigma_E^2}{\sigma_f^2} = \frac{\overline{(\Delta E)^2}}{\overline{E^2}}\frac{\sigma_f^2\langle J|J\rangle}{\sigma_f^2} = \frac{\sigma_f^2\langle J|J\rangle}{\langle J|A|J^*\rangle}\frac{1}{\sigma_f^2} = \frac{\langle J|J^*\rangle}{\langle J|A|J^*\rangle} \tag{9.34}$$

Thus sensitivity is also written in abbreviated form. The parameter Q is conveniently found from $Q = SG$, or

$$Q = \frac{\langle J|J^*\rangle}{\langle J|B|J^*\rangle} \tag{9.35}$$

The directivity is a ratio of two Hermitian quadratic forms, with B positive definite and A at least-positive semidefinite. Thus all the eigenvalues of the associated equation

$$|A|J\rangle = G_{\max}|B|J\rangle \tag{9.36}$$

are zero or positive real. Since A is a single-term dyad there is one nonzero eigenvalue. The eigenvector is found to be

$$|J_{\max}\rangle = |B|^{-1}F\rangle \tag{9.37}$$

The corresponding maximum directivity is given by

$$G_{\max} = \langle F|B|^{-1}F^* \rangle \tag{9.38}$$

In many applications it is important to maximize G/T, directivity system noise temperature, which, is equivalent to maximizing S/N. To do this the element pattern in the integral for B_{nm} is multiplied by the antenna noise temperature $T(\theta, \phi)$. Then the excitation vector that maximizes S/N or G/T is that of Eq. (9.37).

Another ratio that can be directly minimized is beam efficiency: The fraction of power contained within the main beam, null to null. This is

$$\eta_b = \frac{\langle J|A|J^* \rangle}{\langle J|B|J^* \rangle} \tag{9.39}$$

Optimization by adjustment of only phases is also possible (Voges and Butler, 1972). This might be attractive for arrays with active modules, where uniform amplitude is desirable. Unfortunately, one-half of the degrees of freedom are lost, and amplitude plays a much stronger role in pattern shaping.

In practice, because of the limited ensemble of superdirective array design available before unusably high Q occurs, it is simpler to calculate Q, R_{rad}, and G for a few arrays directly. This allows satisfactory design trades to be made, without the complexity of constrained optimization.

9.4 MATCHING OF SUPERDIRECTIVE ARRAYS

Superdirective arrays, like electrically small antennas, tend to have radiation resistances $\ll 50\,\Omega$, and reactance $\gg 50\,\Omega$. In both cases the matching network presents a serious problem: a low intrinsic loss is greatly increased, as will be shown.

9.4.1 Network Loss Magnification

A matching network for a superdirective array must cancel the large reactance, and must transform the small radiation plus loss resistance to the nominal value, usually $50\,\Omega$. Generally, each element will require a different matching network, although symmetric (broadside) arrays need fewer. If the matching network is composed of discrete L and C components, the low R, high X requirements will produce very large circulating currents in the overall circuit, consisting of the array element, the matching network, and the generator. These circulating currents will magnify the intrinsic loss to a much larger realized loss. Similarly, a distributed matching network under low R, high X conditions will have very large standing waves along the stubs and transformer sections. Again the intrinsic loss is magnified. In both cases, the larger loss is due to power being proportional to voltage (or current) squared; the circulating currents or standing waves are large.

A transmission line transformer is typical. Let the matched loss be L, and the antenna voltage standing wave ratio be VSWR. Then the actual loss L_a is (Moreno, 1948)

$$L_a = \frac{(\text{VSWR}+1)^2 L^2 - (\text{VSWR}-1)^2}{4 \cdot L \cdot \text{VSWR}} \tag{9.40}$$

For VSWR $\gg 1$, the actual loss is

$$L_a \simeq \frac{\text{VSWR} \cdot (L^2 - 1)}{4L} \tag{9.41}$$

Figure 9.18 shows the actual loss versus VSWR for intrinsic (matched) losses of 0.1, 0.2, 0.3, 0.5, 1, 2, 3, and 5 dB. For $R \ll R_0$, the VSWR is VSWR $\simeq (R_0^2 + X^2)/RR_0$, while for $X \gg R_0$,VSWR $\simeq X^2/RR_0$. The latter holds also for $R \ll R_0$ and $X \gg R_0$. Some typical values are given in Table 9.5. For example, a case of $R/R_0 = 0.1$ and $X/R_0 = 10$ produces a VSWR = 1010. A matched loss of 0.01 dB becomes an actual loss of 3.35 dB; a matched loss of 0.1 dB becomes an actual loss of 11.01 dB. A more likely case, since the reactance ratio is usually higher, is $R/R_0 = 0.1$ and $X/R_0 = 30$, giving VSWR = 9010. A matched loss of 0.01 dB now yields an actual loss of 10.56 dB; a matched loss of 0.1 dB gives an actual loss of 20.2 dB. These actual losses of the order of 10 dB are in agreement with measurements on HTS and copper matching networks (Khamas et al., 1988, 1990; Lancaster et al., 1992).

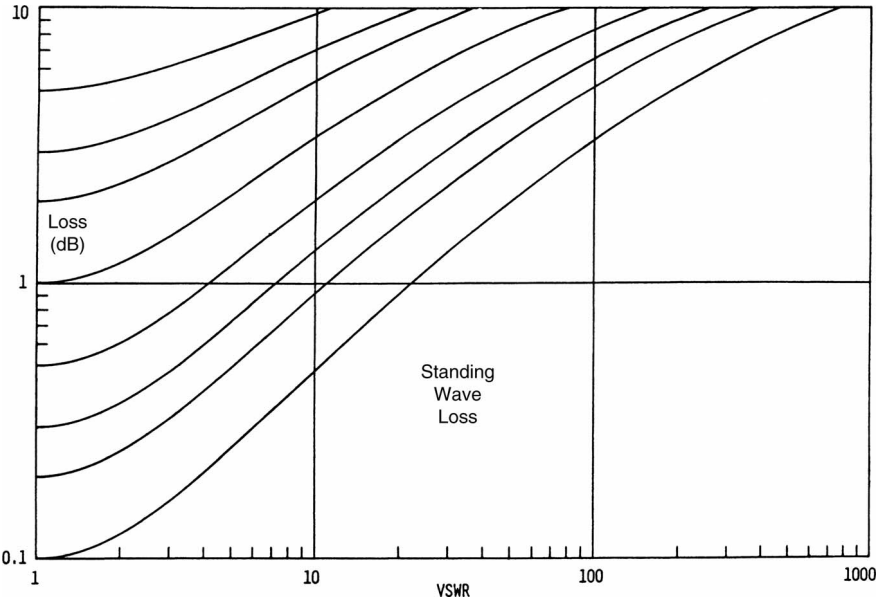


FIGURE 9.18 Line loss with mismatch.

TABLE 9.5 VSWR from Impedance Mismatches

	X/R_0				
R/R_0	1	3	10	30	100
1	1.000	10.91	102.0	902.0	1.000×10^4
0.3	6.820	33.60	337.0	3004.0	3.334×10^4
0.1	20.05	100.1	1010.0	9010.0	1.000×10^5
0.03	66.68	333.4	3367.0	3.003×10^4	3.334×10^5
0.01	200.0	1000.0	1.010×10^4	9.010×10^4	1.000×10^6

9.4.2 HTS Arrays

Because the radiation resistance of a superdirective array decreases roughly inversely with $\sqrt{Q}$, it is easy to design an array where R_{rad} is comparable to or $< R_{\text{loss}}$, which makes the efficiency 50% or less. In addition, there is matching network loss, as just described (Hansen, 1991). In practice, the usability of these arrays is limited by the Q or bandwidth; an antenna with Q of several hundreds may be usable, but one with Q of several thousands is generally not. Even if the signal bandwidth is $\ll 1\%$, the environmental instability of very high-Q antennas makes their use difficult. Adaptive tuning is possible but carries its own disadvantages. For these few cases, where the antenna Q is manageable but the efficiency suffers, an HTS implementation of the array may be feasible (Dinger et al., 1991).

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CHAPTER TEN

Multiple-Beam Antennas

10.1 INTRODUCTION

Multiple simultaneous beams exist when an array of N elements is connected to a beamformer with M beam ports, where N and M may be different. Multiple-beam systems have many uses: in electronic countermeasures, in satellite communications, in multiple-target radars, and in adaptive nulling, for example. The last application uses adaption in beam space as it avoids several serious difficulties that arise with adaption in array space (Mayhan, 1978).

Not included in this chapter are array feeds for parabolic reflectors, where amplitude and phase control of the elements allows coma correction, beam switching, adaptive nulling, and to a limited degree correction of reflector shape errors (Rudge and Davies, 1970; Blank and Imbriale, 1988; Smith and Stutzman, 1992) (see Chapter 14).

Also not included are horn arrays feeding reflector antennas on communication satellites to provide tailored earth coverage patterns (Clarricoats and Brown, 1984; Richie and Kritikos, 1988; Bird, 1990).

10.2 BEAMFORMERS

This section deals with the methods for forming multiple beams. Multiple beamformers are either networks or quasioptical lenses. It is customary to call either type a beamforming network (BFN). In general, with a network BFN, beam crossover levels are independent of frequency, while beamwidths and beam angles change with frequency. With a lens BFN, beam angles are fixed, while beamwidths, hence crossover levels, change with frequency. Each class of BFN is discussed.

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10.2.1 Networks

10.2.1.1 Power Divider BFN. Probably the simplest BFN uses power dividers to split the received signal from each array element into N outputs; the No. 1 outputs from the dividers are connected through fixed phasers to a combiner to provide beam No. 1; and so on. Figure 10.1 sketches such a configuration. Since there is a signal decrease to $1/N$, preamplifiers are almost always used at each element to maintain S/N. This type of beamformer is attractive for a few beams, but is in use for as many as 16 beams. The Globalstar satellite S-band transmit array has 91 printed circuit elements, with each connected to an amplifier–isolator–filter chain. Each amplifier is driven by a 16-way power combiner; the No. 1 inputs to the combiners are connected to a 16-way power divider through fixed phasers; each power divider input represents a transmit beam (Metzen, 1996). The 91-element array is shown in Figure 10.2.

Another satellite communications system is Iridium; Figure 10.3 shows an array of 106 patch elements. Each patch is connected to an electronic TR module; the BFN forms 16 shaped beams for earth coverage.

Beamforming networks were originated in the late 1950s by Jesse Butler; thus it is appropriate to discuss the Butler matrix next.

10.2.1.2 Butler Matrix. A Butler matrix BFN connects 2^n array elements to an equal number of beam ports (Butler and Lowe, 1961). It consists of alternate rows of hybrid junctions and fixed phasers (phase shifters), with a typical diagram shown in Figure 10.4 for an 8-element Butler. This figure exhibits both good and bad features of this BFN: It is a simple network using components easily implemented in stripline or microstrip, but conductor crossovers are required. It has been observed many times that Butler matrix diagrams, such as Figure 10.4, are exactly the same as the calculation flow chart of the fast Fourier transform (FFT). In fact the Butler matrix is the analog circuit equivalent of FFT, so we may expect that this BFN has the minimum number of components and minimum path length of all uniform excitation BFNs. Useful references are Butler (1966), Shelton and Kelleher (1961), and Moody (1964).

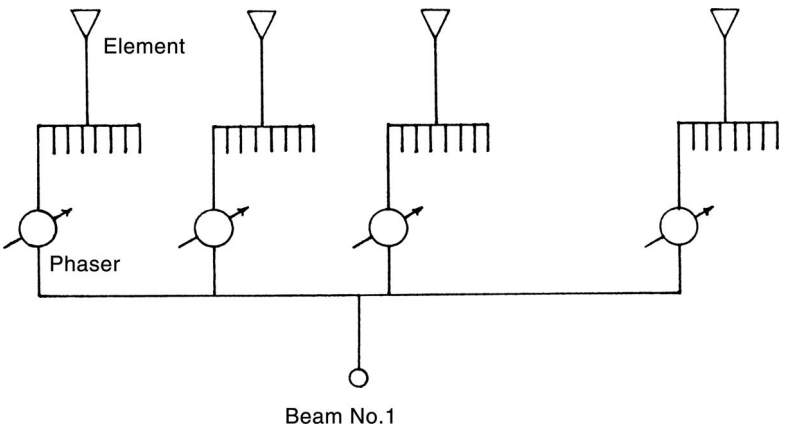


FIGURE 10.1 Beamforming network using power dividers.

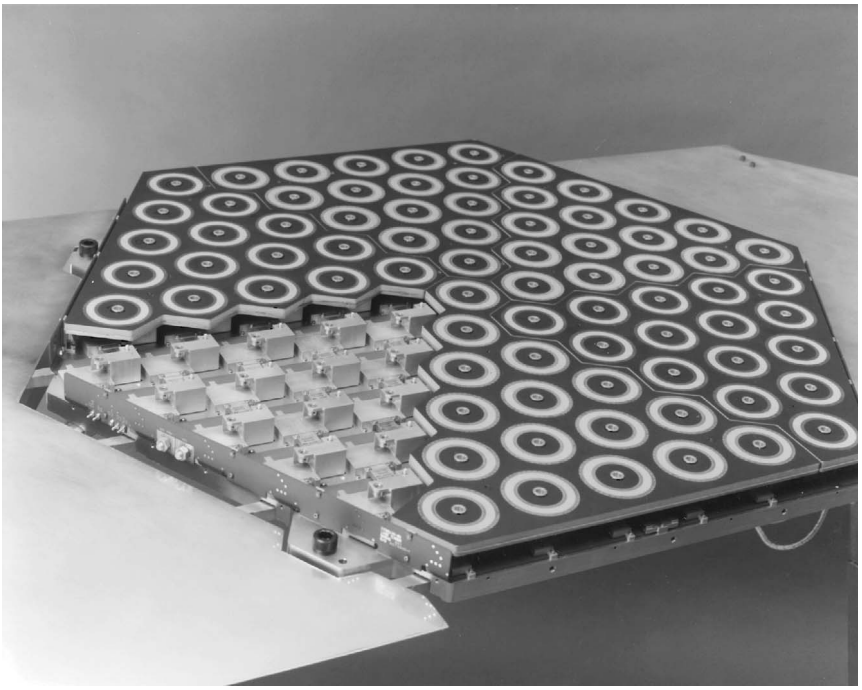


FIGURE 10.2 Globalstar multibeam array. (Courtesy Space Systems/Loral.)

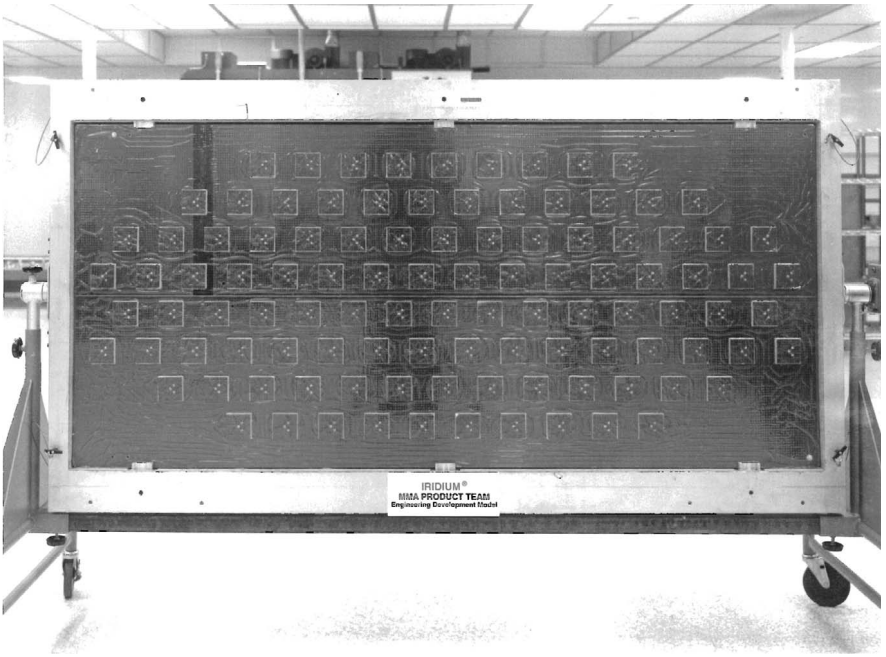


FIGURE 10.3 Iridium multibeam array. (Courtesy Motorola.)

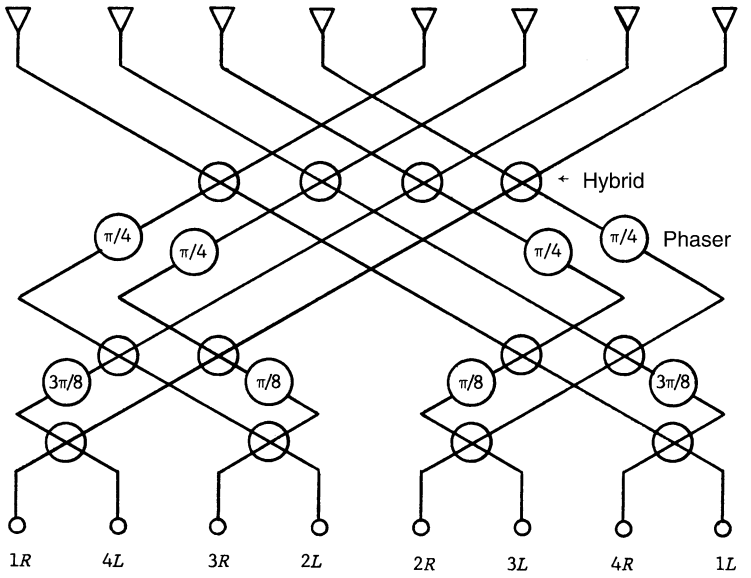


FIGURE 10.4 Butler matrix BFN.

Using the universal variable $u = (d/\lambda)(\sin\theta - \sin\theta_i)$, where d/λ is the element spacing in wavelengths and θ_i is the axis of the ith beam measured from broadside, the Butler beam patterns are

$$F(u) = \frac{\sin(N\pi u)}{N\sin(\pi u)} \tag{10.1}$$

All beams have the same shape in u space. For half-wave spacing and large N (number of array elements), the beams are orthogonal. Beam position for any spacing is

$$\sin\theta_i = \pm\frac{i\lambda}{2Nd} \qquad i = 1, 3, 5, \ldots, (N-1) \tag{10.2}$$

and the beams are spaced apart $1/N$ in u. For half-wave element spacing the beams fill up visible space as sketched in the rosette of beams in Figure 10.5. The array is uniformly excited and the sidelobe level is -13.2 dB. When the element spacing is increased, the beamwidths become narrower and the beams move closer together. The crossover level, discussed below, does not change, as these effects cancel out. Beam coverage, from the center of the left-most beam to the center of the right-most beam, is

$$\theta_{\text{coverage}} = 2\arcsin\frac{(N-1)\lambda}{2Nd} \tag{10.3}$$

This shows the filling of visible space for $d = \lambda/2$, and the smaller coverage for $d > \lambda/2$. As expected, the larger element spacing produces grating lobes, with a grating lobe (GL) corresponding to the right-most beam located just left of the

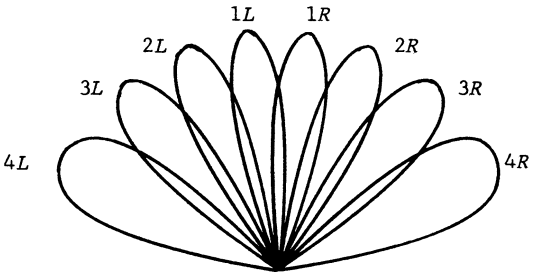


FIGURE 10.5 Butler BFN beam rosette.

left-most beam, and so on. In other words, a Butler BFN fills visible space with beams, and if $d > \lambda/2$ there will be aliasing and a gain loss due to GLs. When larger element spacing is used, it often is because the elements are larger—sub-arrays, horns, and so on. For this case the GL are replaced by quantization lobes whose levels are reduced by the element pattern (see Chapter 2).

Adjacent beam crossovers occur for $u_x = \frac{1}{2}(u_i + u_{i+1}) = (m+1)\pi/2N$, which gives $u = u_x - u_i = 1/2N$. The crossover level is now

$$F_{\text{xov}} = \frac{1}{N \sin(\pi/2N)} \tag{10.4}$$

Table 10.1 shows this for several powers of 2, along with the beam sidelobe ratios. In both instances, the uniform continuous aperture values are obtained for large N; the crossover level approaches $2/\pi$ and the space factor approaches sinc $N\pi u$ for main beam and close-in sidelobes. The nonorthogonal effects for small N are discussed later.

Frequency changes are equivalent to changing d/λ. Increasing frequency narrows the beams and moves them closer together, while grating lobes may appear if the frequency is raised even a modest amount. As discussed above, these grating lobes are aliased with the primary lobes; the grating lobe beam appearing at -90° is associated with the beam port closest to $+90^\circ$, and so on.

Butler matrix improvements include bandwidth improvement (24%) using under band phase using open stubs on a delay line, and wider band sequated disk hybrids, Zheng et al. (2007). Denidni and Nedil (2008) use slotline technology to avoid cross over. Sidelobes have been reduced by using 8 elements with unequal split directional couples with 8 elements (Wincza et al., 2006).

TABLE 10.1 Butler Array Parameters

N	SLR (dB)	Crossover Level (dB)
4	11.30	−3.70
8	12.80	−3.87
16	13.15	−3.91
32	13.23	−3.92
∞	13.26	−3.92

10.2.1.3 Blass and Nolen Matrices. The Blass matrix uses a set of array element transmission lines which intersect a set of beam port lines, with a directional coupler at each intersection (Blass, 1960). See the sketch in Figure 10.6. The top feed line couples out a broadside beam, and is relatively unaffected by the other feed lines. The second feed line tends to couple out a beam at an angle off broadside but the coupling is affected by the top line. A successful analysis must consider each beam port line as a traveling wave (TW) array (of couplers), with interaction between the several TW arrays. See Butler (1966) for an analysis of the Blass matrix. The design can be implemented on a computer, to allow an efficacious trade between network efficiency and beam coupling. However, these arrays are difficult to construct as the interaction between couplers makes coupler adjustment and the correlation with measured data difficult. Broadband versions of the Blass matrix incorporating time delay have been discussed by Butler (1966).

The Nolen matrix (Nolen, 1965) is a generalization of both Blass and Butler matrices, and is a canonical BFN configuration. It derives from the fact stated later that orthogonal beams require a unitary coupling matrix, where the matrix elements are the coupling from the nth antenna element to the mth beam port. Unitary matrices can be factored into a product of elementary unitary matrices, of which the type of interest here is

$$\begin{bmatrix} j\cos\alpha\exp j\beta & -j\sin\alpha & & \\ & & 0 & \\ j\sin\alpha\exp j\beta & j\cos\alpha & & \\ & & 1 & \\ 0 & & & 1 \\ & & & & 1 \end{bmatrix} \tag{10.5}$$

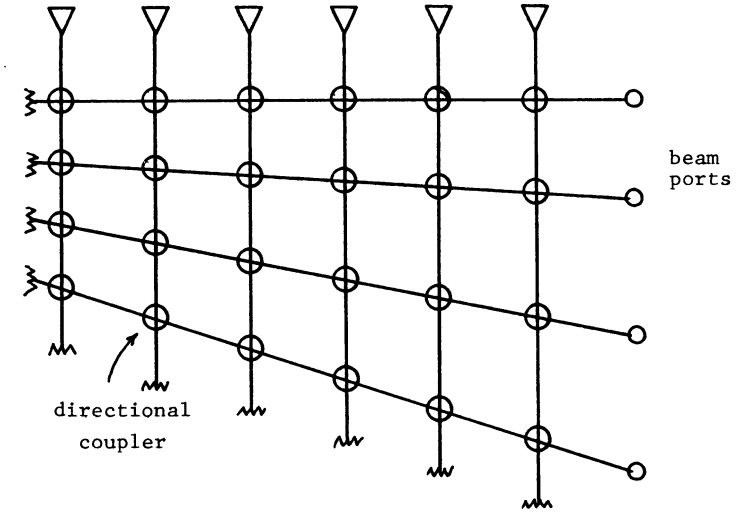


FIGURE 10.6 Blass matrix BFN.

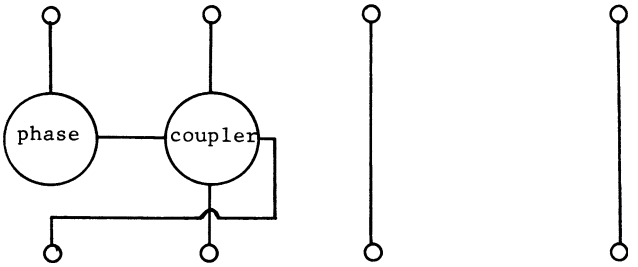


FIGURE 10.7 Unitary coupling matrix circuit.

This equation is useful because it represents a simple circuit containing a phaser and a directional coupler, as seen in Figure 10.7. The concatenation of these prototype network sections produces the Nolen matrix shown in Figure 10.8. Cummings (1978) has shown that the Nolen matrix can be reduced to the Butler. As in the case of the Blass matrix, the Nolen matrix can have a number of beam ports not equal to the number of antenna elements, although the conceptual development just presented requires them to be equal. The Nolen array is an implementation of the general DFT algorithm and works for any number of elements, including prime numbers. As the number of factors in the number of array elements increases, the Nolen matrix can be reduced to simpler forms, culminating in the Butler FFT analog when $N = 2^n$. Owing to the high parts count and the difficulties connected with network adjustments, the Nolen BFN is seldom used.

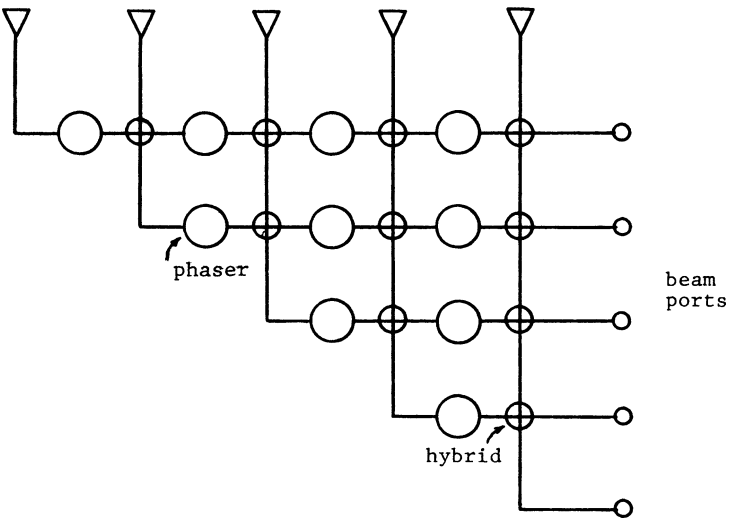


FIGURE 10.8 Nolen matrix BFN.

10.2.1.4 The 2D BFN. Multiple beams in both planes can be achieved by cross-connecting two stacks of Butler matrices or of Rotman lenses (see Section 10.2.2.1). Figure 10.9 sketches this arrangement (Sole and Smith, 1987). A stack of Rotman lenses, without the connecting cables, is shown in Figure 10.10 (Archer, 1975). The hexagonal lattice McFarland matrix of next section is a more efficient (fewer components) matrix BFN. Another 2D system uses two stacks of 8-port Butler matrices, as sketched in Figure 10.11 (Detrick and Rosenberg, 1990). The beam cluster from this array is shown in Figure 10.12. A sophisticated antenna using crossed Butler matrices is for the Iridium satellites. Sixteen-element matrices are connected to the array elements, with 8-element matrices cross-connected (Schuss et al., 1996). Because the coverage area of each array is roughly sector shaped, only one-half of the normal 256 matrices are needed, and some beam ports are not used. In the next section the more efficient (fewer components) 2D matrix is described.

10.2.1.5 McFarland 2D Matrix. The idea of a 2D matrix beamformer utilizing a hexagonal lattice of radiating elements apparently originated at Radiation Systems Inc. in the early 1960s, with work continued at TRW (Williams and Schroeder, 1969). McFarland (1979) discovered how to synthesize a network of power dividers and phase shift units for a 2D array of beams. This beamformer is based on the odd hexagonal lattice where the center is not a single element but rather a triad of elements as seen in Figure 10.13. This results in every other side of the hexagon of elements having one more element than the adjacent sides. If M is one-half of the number of elements in the largest diagonal, the total number of elements N is equal to $3M^2$. It turns out that this odd lattice can be divided into three symmetric sets of elements as shown in the figure, and each set can be connected to a beamformer. The three interconnected beamformers then provide the hexagonal lattice of beams, and they are connected to a hexagonal lattice of elements. These beamformers are more complex, of course, than a 1D matrix. They contain multiple-arm power dividers, such as three-way dividers, and phase shift sections (Chadwick et al., 1981). Figures 10.14 and 10.15 show two

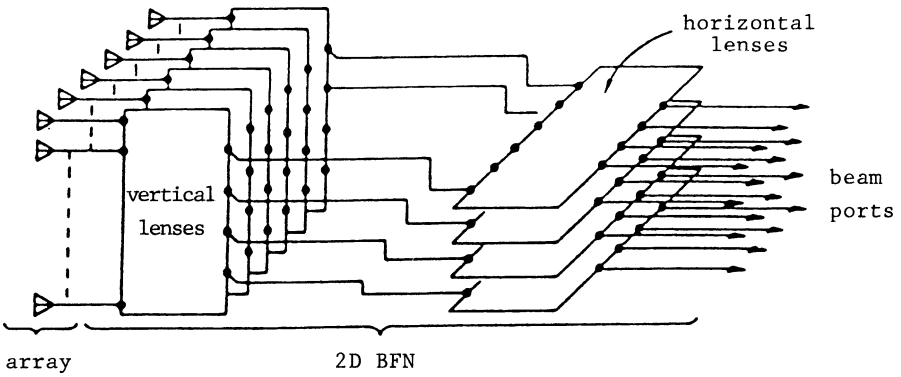


FIGURE 10.9 The 2D Rotman lenses. (Courtesy Sole, G. C. and Smith, M. S., "Multiple Beam Forming for Planar Antenna Arrays Using a Three-Dimensional Rotman Lens", *Proc. IEE*, Vol. 134, Pt. H, Aug. 1987, pp. 375–385.)

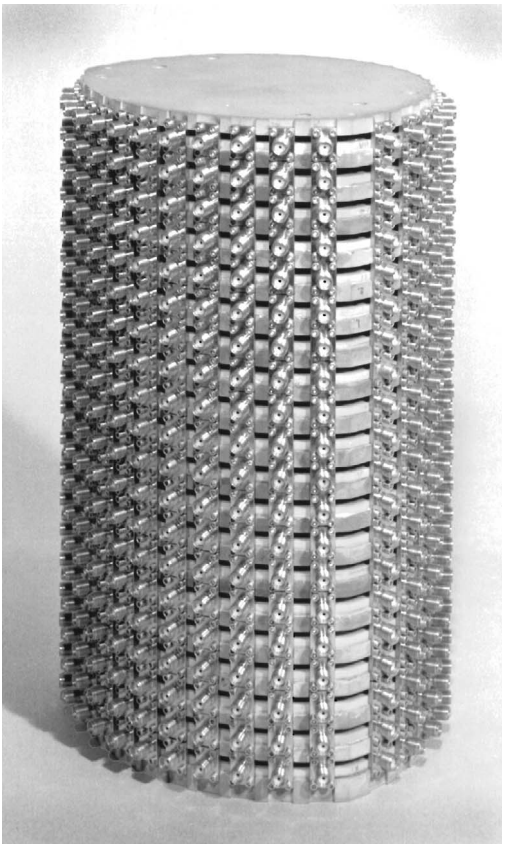


FIGURE 10.10 Stack of Rotman Lenses. (Courtesy Raytheon Co.)

schemes for connecting the elements to the beam ports (McFarland, unpublished notes, ~1978). In Figure 10.14, a triad of array elements is connected to a set of divider–phaser boxes, with the outputs providing a triad of beam ports. Figure 10.15 reverses the element triads and the beam port triads. Relatively little work has been done on this type of array, and general synthesis methods are not yet known. In fact, since the untimely death of McFarland, the entire subject appears to have languished, in spite of its promise. The 2D beamformer has several advantages. First, it has fewer parts than a cascade of 1D Butlers. Second, a synergistic integration of the beamformer with a switch matrix appears possible. Finally, the geometry fits both the 2D roughly circular array shape and spatial coverage more efficiently.

10.2.2 Lenses

10.2.2.1 Rotman Lens BFN. A bootlace lens is a constrained lens wherein the path lengths from the lens output ports to the radiating element ports are prescribed. An early constrained lens was the R-2R (Brown, 1953, 1969) where the inner- and

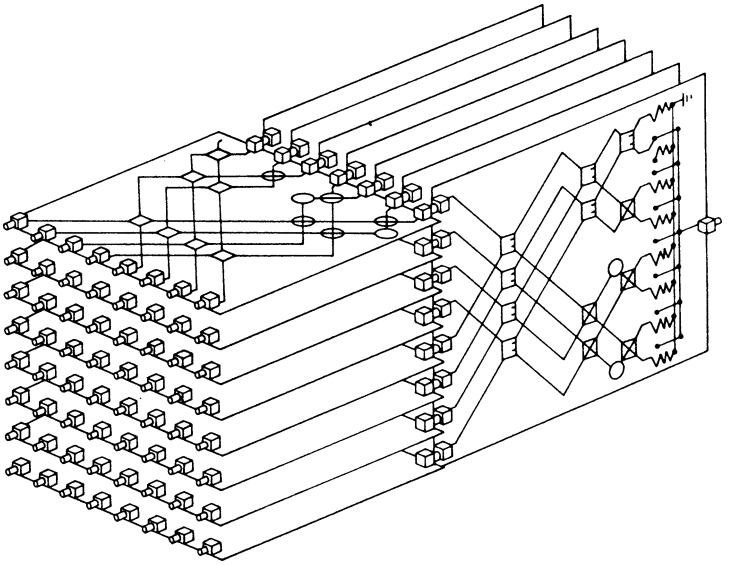


FIGURE 10.11 The 2D Butler matrices. (Courtesy Detrick, D. L. and Rosenberg, T. J., "A Phased-Array Radiowave Imager for Studies of Cosmic Noise Absorption", *Radio Sci.*, Vol. 25, No. 4, July–Aug. 1990, pp. 325–338.)

outer-lens surfaces are circular arcs, with the outer radius twice the inner. This is the only path length lens with perfect collimation for a feed point anywhere on the focal arc (Kales and Brown, 1965). However, the amplitude asymmetry for beams off-axis raises the sidelobes. This, with the circular array shape, has limited the usefulness of

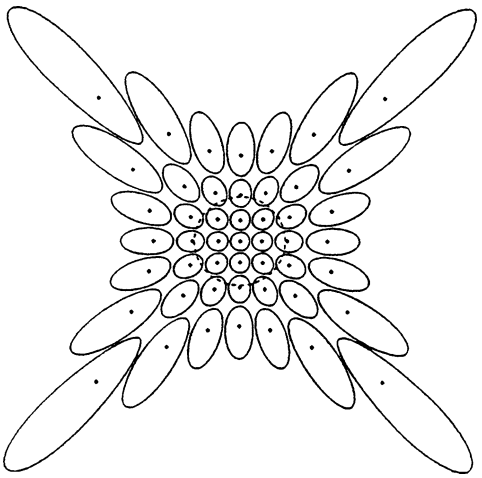


FIGURE 10.12 2D Butler beam rosette. (Courtesy Detrick, D. L. and Rosenberg, T. J., "A Phased-Array Radiowave Imager for Studies of Cosmic Noise Absorption", *Radio Sci.*, Vol. 25, No. 4, July–Aug. 1990, pp. 325–338.)

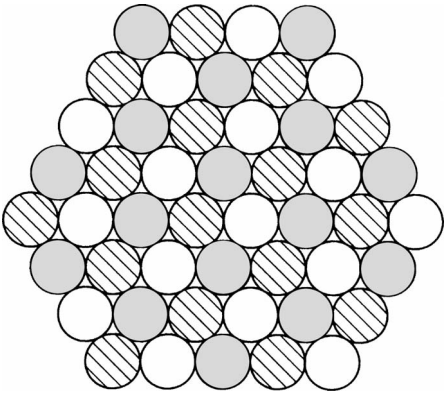


FIGURE 10.13 The $3M^2$ horn cluster triads for $M = 4$, $N = 48$.

the R-2R. It was originally implemented in geodesic form (deformed parallel plates; Tin Hat). A general theory of bootlace lenses was given by Gent (1957). Another one-dimensional lens (called 1D because it produces beams in a single plane) designed for a moving feed and a linear radiating array was developed by Rotman and Turner (1963). This soon led to a Rotman lens with an array of fixed feeds for beam ports. Figure 10.16 sketches a Rotman lens; only one beam port is shown.

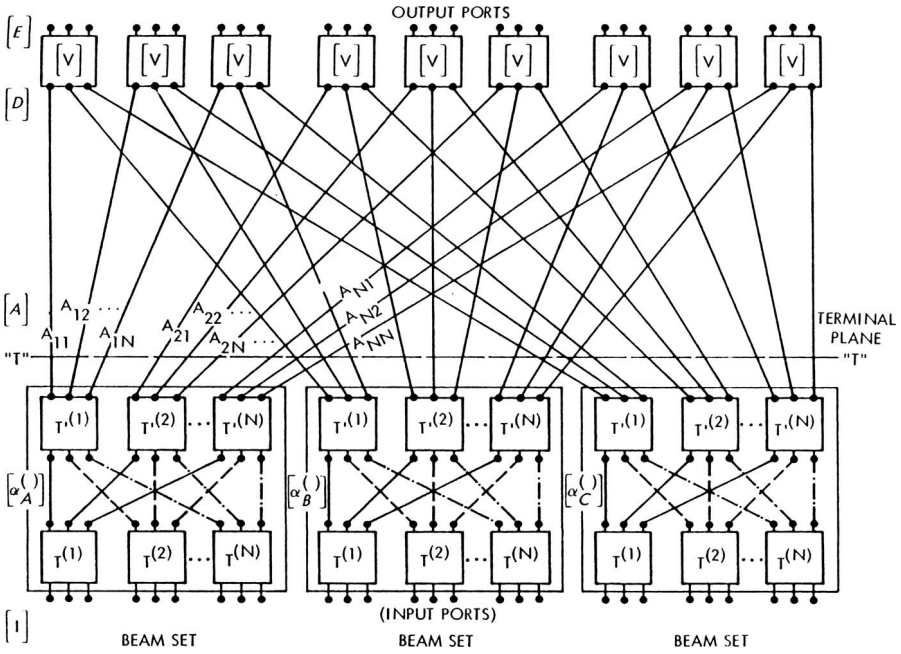


FIGURE 10.14 Hexagonal lattice BFN triad. (Courtesy J. L. McFarland.)

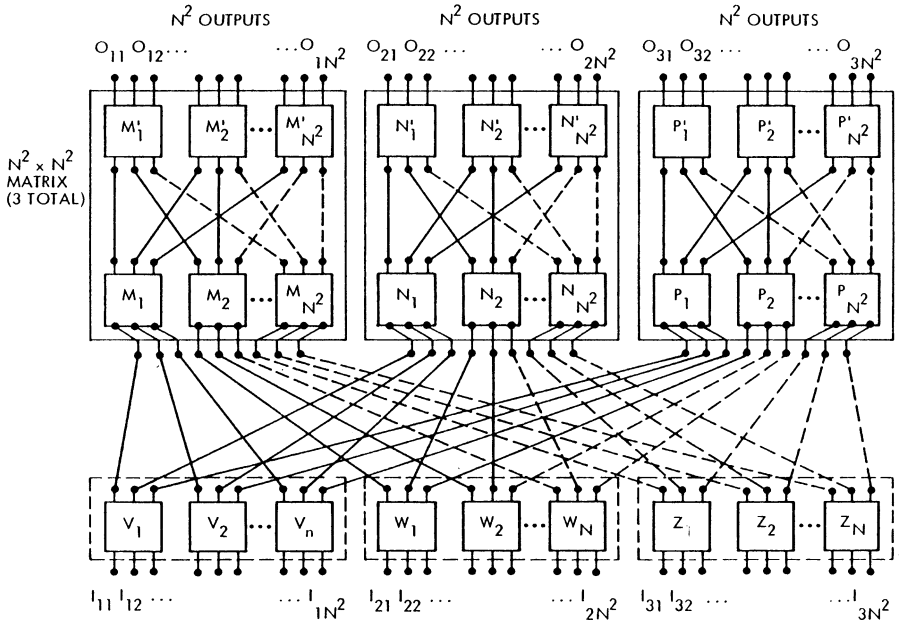


FIGURE 10.15 Hexagonal lattice BFN triad. (Courtesy J. L. McFarland.)

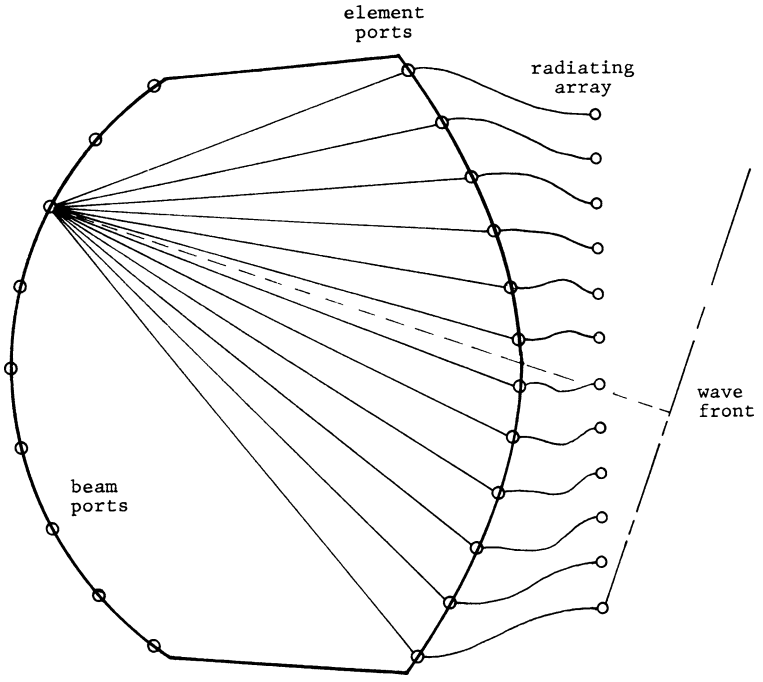


FIGURE 10.16 Rotman lens beamformer.

The original lens used parallel plate waveguide for the feed portion, and coax cables for the bootlace portion. Implementation now is typically microstrip or stripline for the feed portion, and either coax cables or printed circuit lines for the bootlace portion (Archer, 1975). Strictly speaking, the lens is the bootlace portion, but it is common practice to refer to the entire assembly as the lens. These lenses have three (perfect) focal points, usually at the center of the feed array, and at two symmetric points between the center and the edge of the feed arc. They are thus a major improvement on the designs of Ruze (1950).

Design of these lenses must involve both geometric trades and mutual coupling effects between the lens ports. The latter is relatively difficult to control, but the former is crucial to the realization of an efficient and compact lens. Thus a careful geometric optics design should be accomplished first; then adjustments must be made to reduce mutual coupling effects. The geometric design trades are described here. The Rotman lens has six basic design parameters: focal angle α, focal ratio β, beam angle to ray angle ratio γ, maximum beam angle Ψ_m, focal length f_1, and array element spacing d. The last two are in wavelengths, and γ is a ratio of sines. A seventh design parameter allows the beam port arc to be elliptical instead of circular. Since the design equations are implicit and transcendental, with only one sequence of solution, the interplay of design parameters is difficult to discern. A series of lens plots is used to show the effects of each parameter. Geometric phase and amplitude errors over the element port arc vary primarily with α and β, and with an implicit parameter that is the normalized element port arc height. Representative plots show how these errors depend on the parameters.

For lenses where the beam port arc and feed port arc are identical, resulting in a completely symmetric lens, the design equations are greatly simplified (Shelton, 1978). However, these lenses are seldom used because their design options are much more constrained.

The lens equations equate path lengths from the foci to the array elements; see Rotman and Turner (1963) for a derivation of these. By using the nomenclature of Figure 10.17, it is convenient to normalize all dimensions by the principal focal length f_1. This is also the lens width at the center. The focal angle α is subtended by the upper and lower foci at the center of the element port curve. It is assumed here that the foci are symmetrically disposed about the axis, and that the lens is also symmetric. Then the parameter β is the ratio[1] of upper (and lower) focal length f_2 to f_1:

$$\beta = \frac{f_2}{f_1} \tag{10.6}$$

Clearly, the lens width in wavelengths, f_1/λ, is another parameter. Now the angle of the beam radiated by the array is Ψ, and if one of the off-axis foci is excited, then the

[1]Note that β is the inverse of the ratio g used by Rotman and Turner (1963) and by McGrath (1988); β is more convenient due to the f_1 normalization.

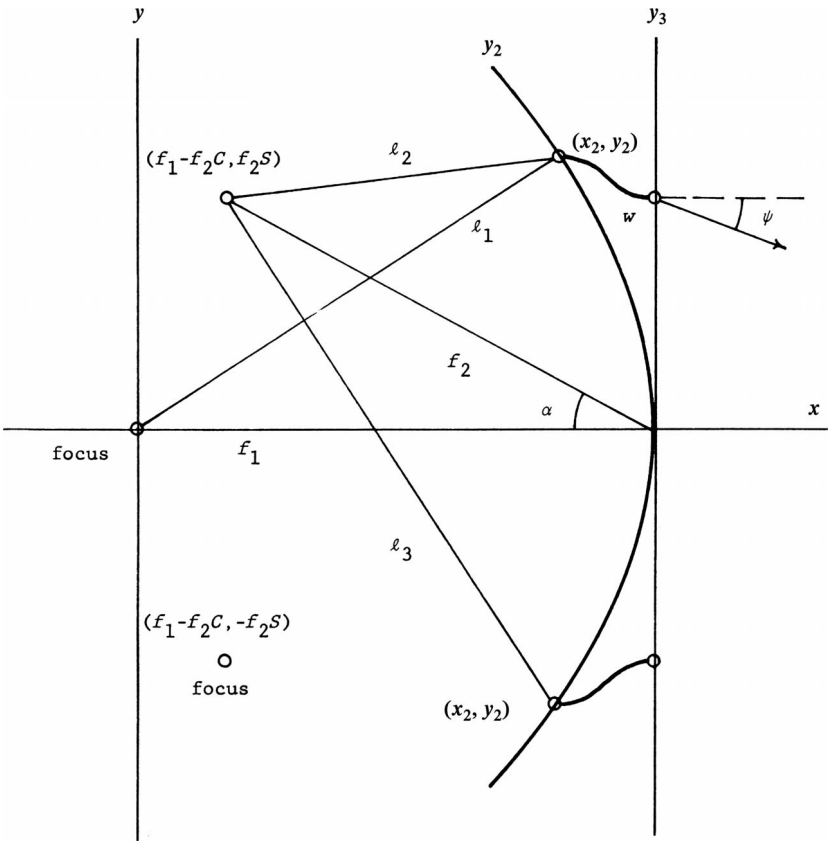


FIGURE 10.17 Ray geometry.

ratio of lens ray angle α to array beam angle Ψ is γ:

$$\gamma = \frac{\sin \Psi}{\sin \alpha} \tag{10.7}$$

An indirect parameter of utility is ζ, which relates the distance y_3 of any point on the array from the axis to f_1. This parameter controls the portion of the phase and amplitude error curves that the lens experiences. It is expressed as

$$\zeta = \frac{y_3 \gamma}{f_1} \tag{10.8}$$

Note that the line lengths w of Figure 10.16 are an integral and essential part of the lens. The maximum beam angle, Ψ_m, is an important parameter, as is the array element

spacing in wavelengths d/λ. The $\zeta_{\max}$ that corresponds is given by

$$\zeta_{\max} = \frac{(\mathrm{NE} - 1)\gamma d}{2f_1} \tag{10.9}$$

where NE is the number of elements in the linear array, since $y_{\max} = (\mathrm{NE} - 1)/2$. An upper limit on ζ occurs when the tangent to the element port curve is vertical; this also gives $w = 0$. This value of ζ is given by

$$\zeta_{w=0} = \frac{2\sqrt{1 - \beta C}}{S}\sqrt{1 - \frac{1 - \beta C}{S}} \tag{10.10}$$

Figure 10.18 gives this limiting value versus β, for several values of α. Since the useful range of ζ is roughly from 0.5 to 0.8, a range of β appropriate for a given α may be inferred.

The geometric lens equation is a quadratic in the line length w that connects an element port to the corresponding array element:

$$a\left(\frac{w}{f_1}\right)^2 + \frac{bw}{f_1} + c = 0 \tag{10.11}$$

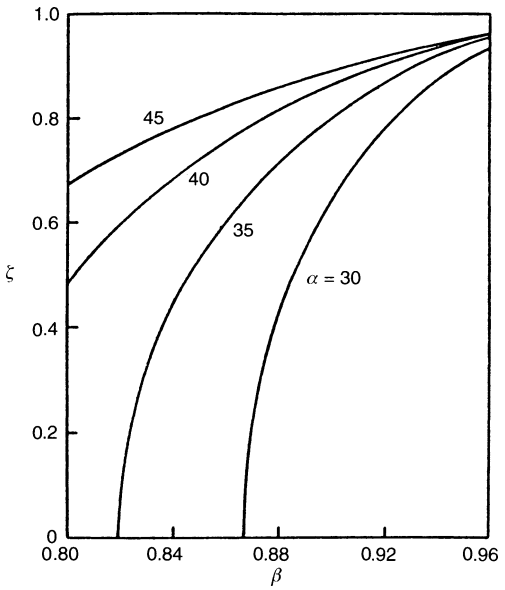


FIGURE 10.18 Upper limit on parameter ζ.

where the coefficients involve the parameters α, β, and γ:

$$
\begin{aligned}
a &= 1 - \frac{(1-\beta)^2}{(1-\beta C)^2} - \frac{\zeta^2}{\beta^2} \\
b &= -2 + \frac{2\zeta^2}{\beta} + \frac{2(1-\beta)}{1-\beta C} - \frac{\zeta^2 S^2 (1-\beta)}{(1-\beta C)^2} \\
c &= -\zeta^2 + \frac{\zeta^2 S^2}{1-\beta C} - \frac{\zeta^4 S^4}{4(1-\beta C)^2}
\end{aligned}
\tag{10.12}
$$

and $C = \cos\alpha$, $S = \sin\alpha$.

Usually the number of beams and elements, the maximum beam angle, and element spacing are specified from the system requirements. Thus, the task is to select the optimum α, β, γ, and f_1/λ.

The lens shape is important, both in conserving space and in reducing loss. For example, a flatter lens tends to have path lengths that are more nearly equal, and allows the beam port curve and the element port curve to have different heights, and even different curvatures. As in Figure 10.17, the width is along the lens axis. Wide lenses have large spillover loss, and higher transmission line loss. A compact lens tends to minimize spillover losses; roughly equal port curve heights now become important, to avoid severe asymmetric amplitude tapers and large phase errors. Curvatures of the two-port curves may be different; use of array element spacing greater than half-wavelength allows more beam ports than element ports to be used. For this case, the beam port curve may be more curved, and the element port curve flatter.

The effects of the seven parameters will be shown through a series of charts (Hansen, 1991). Six charts are shown in Figures 10.19–10.24. Beam ports, which are ticked, are on the left. Element ports, also ticked, are on the right. Foci are indicated by asterisks. The focal length is normalized to unity, so that each tick mark on the axis (and on the ordinate) is 0.05. From the ordinate scale the element positions may be inferred. Each lens curve is extended past the outermost port by half the width of that port. These examples have 9 beam ports and 11 element ports, and of course 11 elements in an equally spaced linear array.

With all other variables fixed, increasing α opens the beam port curve, and closes the element port curve. Port positions are roughly unchanged. But the outer foci locations change markedly as expected. The three lens plots of Figure 10.19 illustrate these effects. Note that a value of α can be selected that roughly equalizes the heights of the two curves. Of course, α must be selected in conjunction with the other variables to minimize phase errors over the aperture. The outer foci should be comfortably inside the beam port curve.

Increasing β has an effect similar to increasing α; the beam port curve opens, and the element port curve closes. Figure 10.20 contains three lens plots to show this. Again, port positions are roughly unchanged. Also the focal locations change relatively little. Again, a value of β can be selected that roughly equalizes the curve heights.

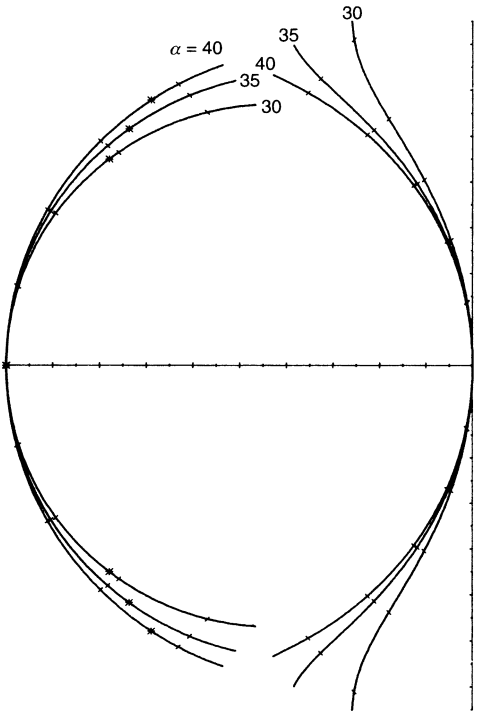


FIGURE 10.19 Effect of focal angle: $\beta = 0.9$, $\gamma = 1.1$, $\Psi_m = 50$, $f_1 = 4\lambda$, $d = 0.5\lambda$.

There are pairs of α and β that produce closely the same lens shape, and port positions. However, the foci vary with α, and the connecting lines (from element ports to elements) are different. Table 10.2 shows three $\alpha-\beta$ lens pairs that have common lens curves and ports, all for $\gamma = 1.1$, $\Psi_m = 50$, $f_1/\lambda = 4$, and $d = \lambda/2$. One may thus infer that the phase error over the aperture for each beam will be different, depending on α. This will be shown in the next section. For any set of the other four parameters, there are probably some $\alpha-\beta$ pairs that behave similarly.

Increasing γ leaves both lens curves unchanged, but the beam ports are moved closer together, while the element ports are spread apart. A three-lens set in Figure 10.21 shows this trend. Although the foci remain fixed, the ends of the curves change, so that the relative position of the foci changes. γ also can affect the relative heights of the two curves.

Values of γ here are 1 or greater, as the cases used are all for large beam angles. When the beam cluster subtends a more modest angle (e.g., 30°) values of $\gamma < 1$ are appropriate as they allow a "fat" or curved lens.

When Ψ_m is changed, only the beam port spacings change. Increasing Ψ_m spreads the beam ports and extends the port curve, so that this parameter helps produce a lens with roughly equal heights of beam and element port curves. The three lens plots of Figure 10.22 depict this behavior.

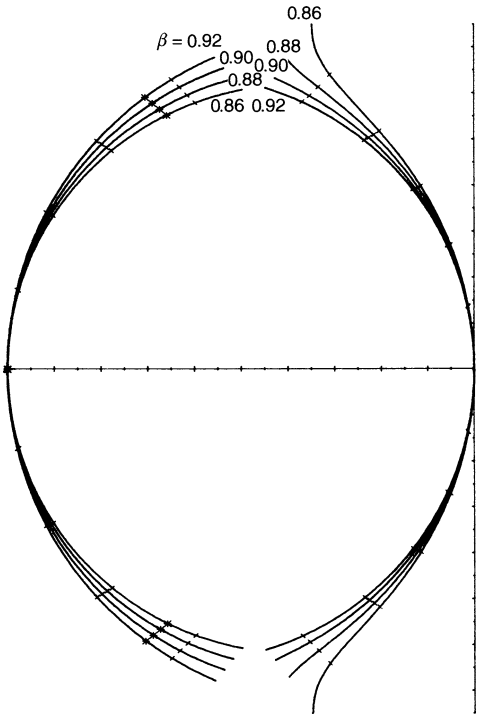


FIGURE 10.20 Effect of focal ratio: $\alpha = 40$, $\gamma = 1.1$, $\Psi_m = 50$, $f_1 = 4\lambda$, $d = 0.5\lambda$.

Element spacing is critical as it controls the appearance of grating lobes (see Chapter 2). For a maximum beam angle of Ψ_m, the spacing that just admits a grating lobe is

$$\frac{d}{\lambda} = \frac{1}{1 + \sin \Psi_m} \tag{10.13}$$

In general, spacings are kept below this value.

When d is changed, only the element port spacings and the extent of the port curve change, analogous to Ψ_m changing beam port spacing. Figure 10.23 uses two lens plots to show this.

Increasing the lens focal length (width) in general increases the separation between the end ports as well. But changing f_1/λ also changes all spacings, as the lens equations are normalized by f_1. Thus, as shown in the two lens plots of Figure 10.24, changing f_1/λ also changes the element port arc and element port spacings. The minimum value of f_1 is smaller for Rotman lenses than for other types of lenses (Smith, 1982).

Next, phase and amplitude errors will be examined.

Aperture errors depend on α and β, and upon eccentricity, but only indirectly on the other parameters. Thus the most insight results from plotting phase and amplitude

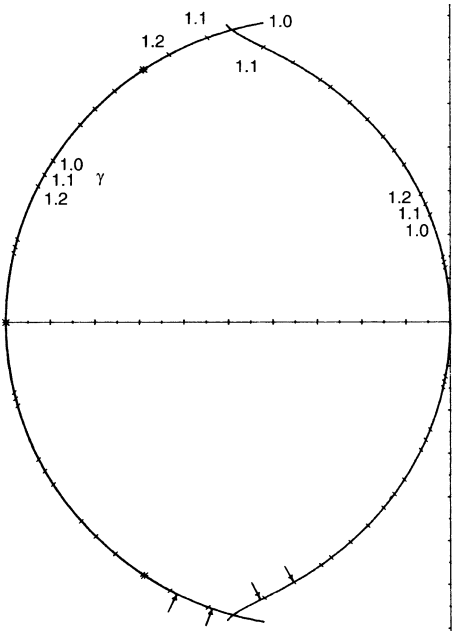


FIGURE 10.21 Effect of beam angle ratio: $\alpha = 40$, $\beta = 0.9$, $\Psi_m = 50$, $f_1 = 4\lambda$, $d = 0.5\lambda$.

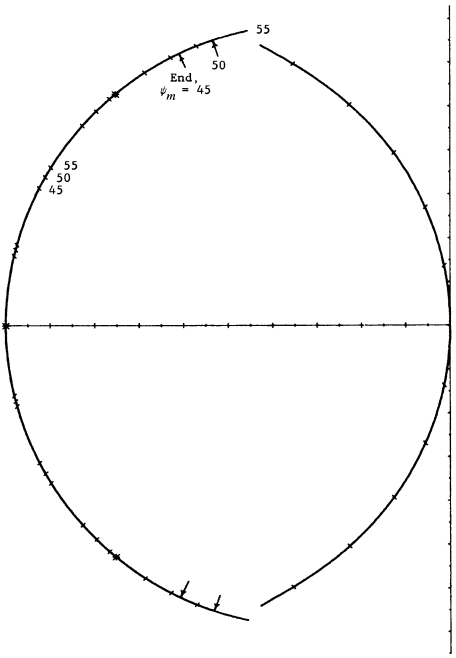


FIGURE 10.22 Effect of maximum beam angle: $\alpha = 35$, $\beta = 0.92$, $\gamma = 1.1$, $f_1 = 4\lambda$, $d = 0.5\lambda$.

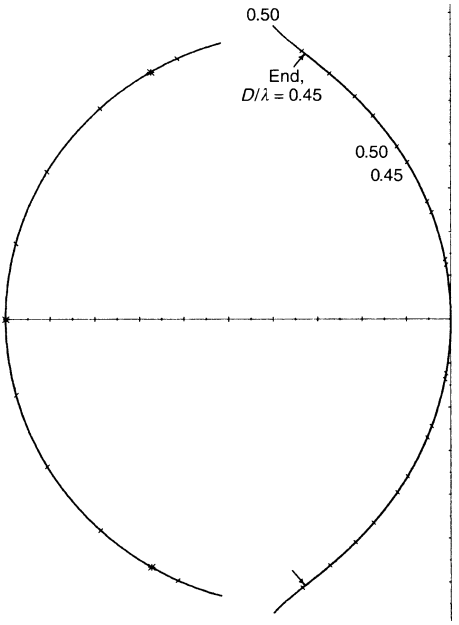


FIGURE 10.23 Effect of array element spacing: $\alpha = 40$, $\beta = 0.88$, $\gamma = 1.1$, $\Psi_m = 50$, $f_1 = 4\lambda$.

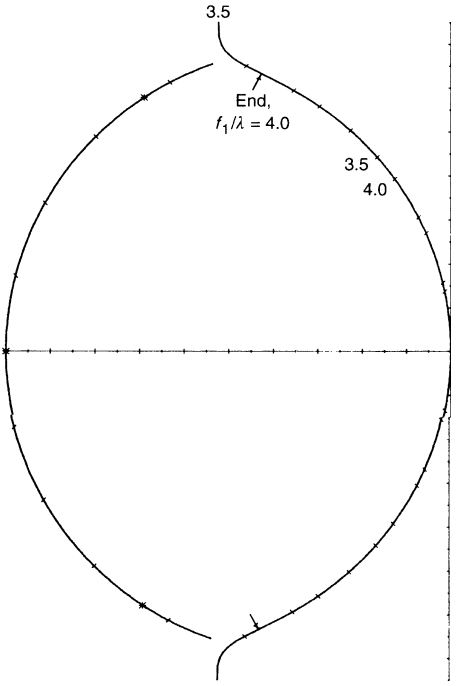


FIGURE 10.24 Effect of focal length: $\alpha = 40$, $\beta = 0.9$, $\gamma = 1.1$, $\Psi_m = 50$, $d = 0.5\lambda$.

TABLE 10.2 The $\alpha-\beta$ Pairs

Number	α	β
1	30	0.94
2	35	0.92
3	40	0.90

errors versus the normalized parameter ζ [see Eq. (10.8)]. Since phase errors are zero at angles corresponding to the three foci, a satisfactory approach uses one beam position midway between the central and edge foci, and a second beam position beyond the edge focus. Amplitude errors occur at all beam ports, so more cases are needed to display amplitude error behavior.

Figures 10.25–10.28 show phase error versus ζ for lenses with α of 35, 40, and 45°, for various values of β. Note that to get phase error, the values from the figures are to be multiplied by f_1/λ. For the midfoci beams, the phase error is small, except for very large lenses. Phase errors for the wider-angle beams are still modest, and will not be important except for large lenses, or designs with $\zeta > 0.75$. In general, the phase errors increase as α is increased, for all beam positions.

Although Rotman and Turner indicated an optimum value of β, which was $2/(2+\alpha)^2$, examination of Figures 10.25–10.28 show that an optimum β exists only for one range of ζ and one ray angle. Best values of β are different for between-foci rays and rays outside the foci. Since the latter usually have larger phase error, the designer could optimize, but the value would vary with both ray angle and $\zeta_{\max}$.

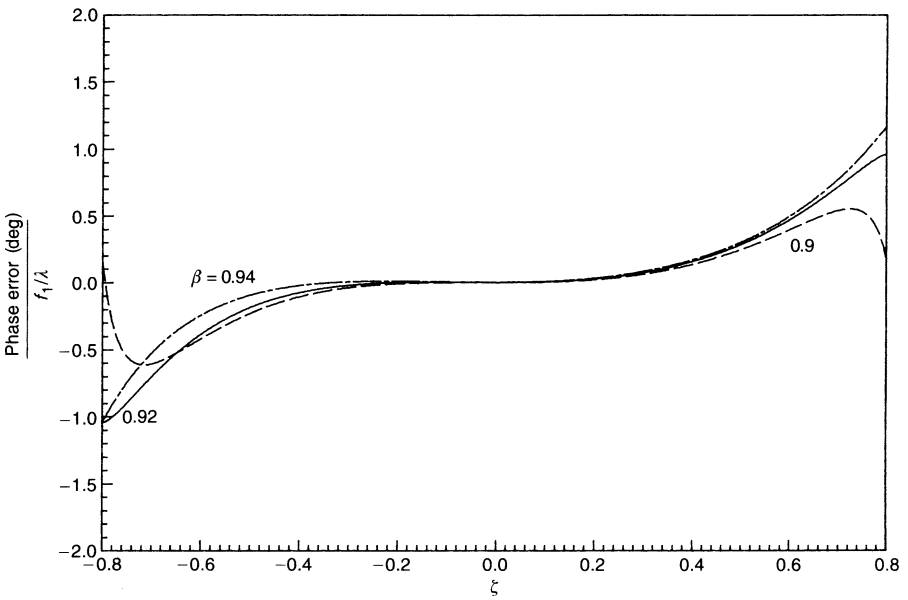


FIGURE 10.25 Rotman lens, $\alpha = 35$, ray angle = 17.5°.

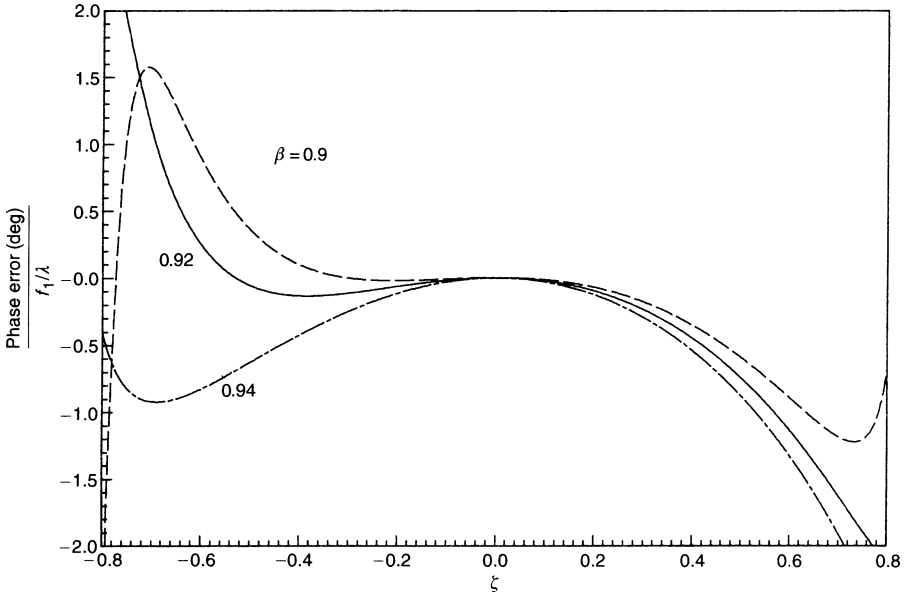


FIGURE 10.26 Rotman lens, $\alpha = 35$, ray angle $= 45^\circ$.

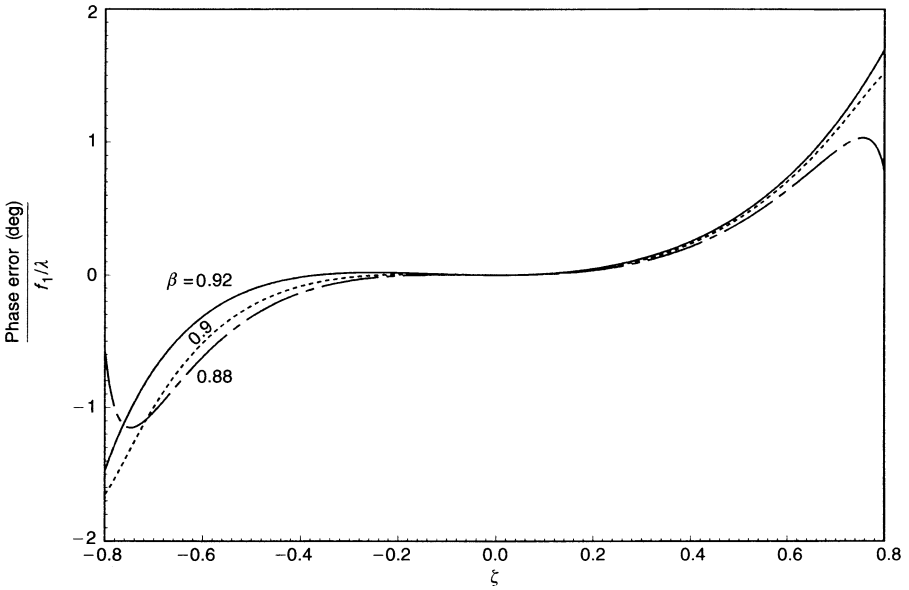


FIGURE 10.27 Rotman lens, $\alpha = 40$, ray angle $= 20^\circ$.

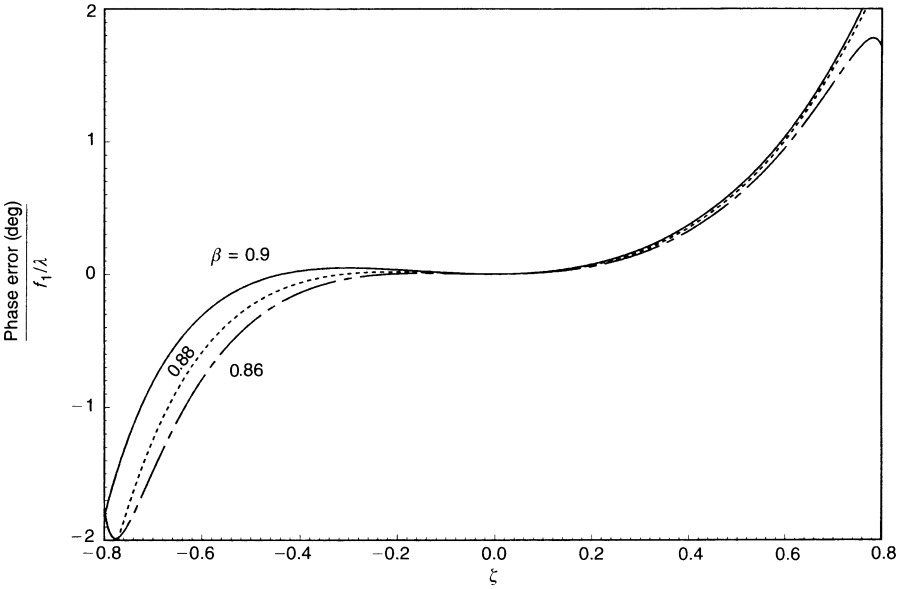


FIGURE 10.28 Rotman lens, $\alpha = 45$, ray angle $= 22.5^\circ$.

This old value also gives poor lens shapes when the numbers of beam and element ports are roughly equal.

The results of using an elliptical beam port curve are shown in Figure 10.29, where the phase errors are equalized at $\zeta = \pm 0.7$ for a ray angle of 45°. Note that the elliptical beam port curve is a simple way of realizing the optimum curve of Katagi et al. (1984). This gives a 13% reduction in phase error; the phase errors for the midfoci rays at 17.5° become slightly more asymmetric, but are still well below the 45° ray errors. Note that the ellipticity of -0.3 only changes the principal radius by 5%, so amplitudes are essentially unchanged. The beam port ellipse major axis is along the lens axis for this ellipticity.

Amplitude errors are calculated using beam port and element port horn patterns of sinc πu, where horn widths are all set to a nominal $\lambda/2$. Each port horn has its axis normal to the port curve. Figure 10.30 shows amplitude error, normalized to 0 dB for the axial ray, for a lens with $\alpha = 35$, $\beta = 0.92$. Curves for ray angles of 0, 17.5, and 45°, are given. Similarly, Figure 10.31 is for a lens with $\alpha = 40$, $\beta = 0.9$, for ray angles of 0, 20, and 50°. These examples are two of the α–β pairs of Table 10.2, and thus the amplitude errors are similar. As expected, for wide ray angles the near and far ends of the element port curve experience modest amplitude changes. Compared to the amplitude taper needed to produce 25-dB sidelobes, these amplitude errors are small. Actual lenses may have port widths $>\lambda/2$, so the amplitude tapers can be expected to increase, especially for edge beams.

The asymmetry of amplitude for the off-axis beams can be reduced by pointing each port horn at the opposite apex instead of normal to the port curve (Musa and

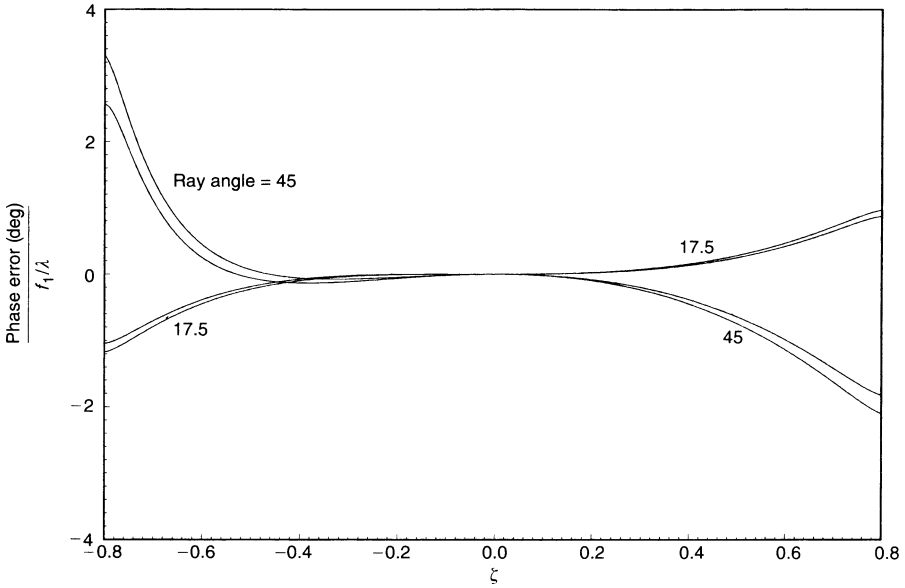


FIGURE 10.29 Rotman lens, $\alpha = 35$, $\beta = 0.92$.

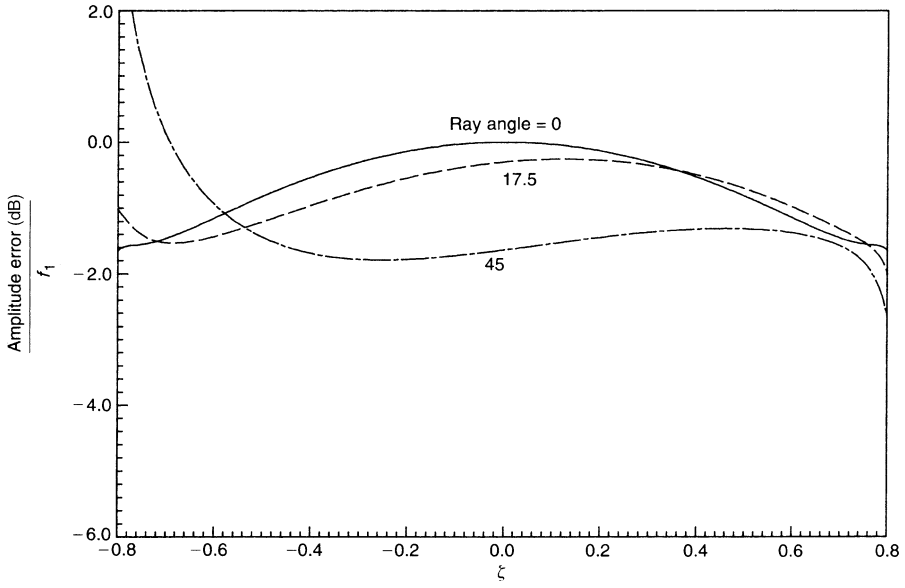


FIGURE 10.30 Rotman lens, $\alpha = 35$, $\beta = 0.92$.

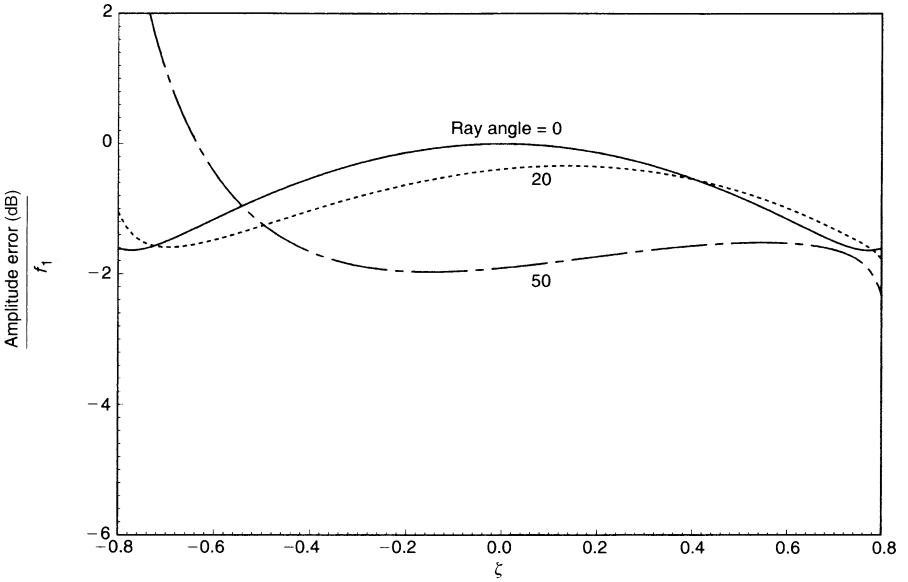


FIGURE 10.31 Rotman lens, $\alpha = 40$, $\beta = 0.9$.

Smith, 1989). For example, a 9-beam, 11-element lens with $\alpha = 40$, $\beta = 0.9$, $\gamma = 1.1$, $\Psi_m = 50$, $d = 0.5\lambda$, and $f_1 = 4\lambda$ has amplitude taper for the outside beam, as shown in Table 10.3. Also shown is the taper for apex pointed horns. Use of apex pointing produces appreciable improvement. Gain is slightly improved.

Element and beam port spillover, phase and amplitude errors, port impedance mismatches, and transmission line loss all contribute to reducing lens gain. Note that, as in the case of a horn feeding a reflector antenna, there is no feed horn spreading loss, owing to the equal-path property through the foci. The small inequality of other

TABLE 10.3 Beam One Amplitude Taper; $f_1 = 4\lambda$

Element Number	Axes Normal to Arc (dB)	Axes Through Apex (dB)
1	−8.49	−2.26
2	−7.03	−1.84
3	−5.50	−1.46
4	−4.29	−1.31
5	−3.33	−1.37
6	−2.58	−1.63
7	−2.04	−2.08
8	−1.70	−2.73
9	−1.57	−3.60
10	−1.68	−4.71
11	−2.08	−5.95

paths is subsumed in the path phase and amplitude errors. Gain will be calculated here based on port spillover and on aperture errors. Since the amplitude error calculation includes both beam port and element port horn patterns, spillover is included (Smith and Fong, 1983). The phase and amplitude errors at the element ports are transferred to an array of isotropic elements. Then the problem reduces to that of calculating gain of a symmetric linear array of isotropic elements with complex coefficients, using the formula from Chapter 2. Actual gain is then that of the isotropic array factor multiplied by the element gain, times the impedance mismatch factors.

Variation of gain with parameters is very small. For example, for a typical small lens only 0.2 dB change occurs from the center beam to the edge beam, and the gain values are roughly independent of α, β, γ, and so on. With a larger array the gain increases just as expected. Using the same 9-beam, 11-element example of Section 10.2.1.5, gain ranges from 10.2 to 10.4 dB; the latter value, for the center beam, is within 0.1 dB of the gain for a similar uniformly excited array. For this lens, the ζ range is ± 0.756.

Effects of feed horn spillover and internal lens reflections can be reduced either by employing dummy (terminated) feed horns adjacent to the edge horns, or through the use of absorber between the ends of the beam port arc and the element port arc (Musa and Smith, 1989).

The design process starts by the selection of a center frequency, at which all dimensions are computed. Then, a starting value of f_1/λ is selected, to keep $\zeta_{\max}$ well below 0.8. The focal length will be somewhat less than the array length. Next, using the guidelines, α, β, and γ are selected, to (1) locate the outer-beam port a modest amount past the outer focus; (2) produce beam port and element port arcs of comparable heights; and (3) yield acceptable phase and amplitude errors at each port. Achieving this may require adjustment of f_1/λ or d/λ, and of α, β, and γ. Use of an elliptical beam port arc is usually not warranted except for large lenses.

When a satisfactory design is realized at the center frequency, phase and amplitude errors at each port are calculated at representative frequencies, to assess performance over the frequency range. Of course, the actual beam and element port horn widths are used. At this stage, calculation of a beam rosette (a set of beam patterns) at each frequency is appropriate. Some compromise and iterative adjustment of parameters may be necessary to obtain good wideband results, and to best accommodate mutual coupling effects.

Although the lens width f_1 is less than the array length, the lens height is always greater than the array length. Lens dimensions are reduced by the square root of effective dielectric constant for either stripline or microstrip implementation. See Archer (1984) for examples. Low-sidelobe designs are discussed in Section 10.3. Figure 10.32 shows a typical microstrip Rotman lens.

10.2.2.2 Bootlace Lenses. Although the Rotman lens is a bootlace lens, this section is devoted to 2D bootlace lenses; the multiple beams occur in two dimensions. Unfortunately, rotating a Rotman lens to form a symmetric lens does not give good results; a circular focus is not possible. A 2D bootlace lens may have up to four discrete perfect focal points, but they cannot be located with quadrantal symmetry, as the

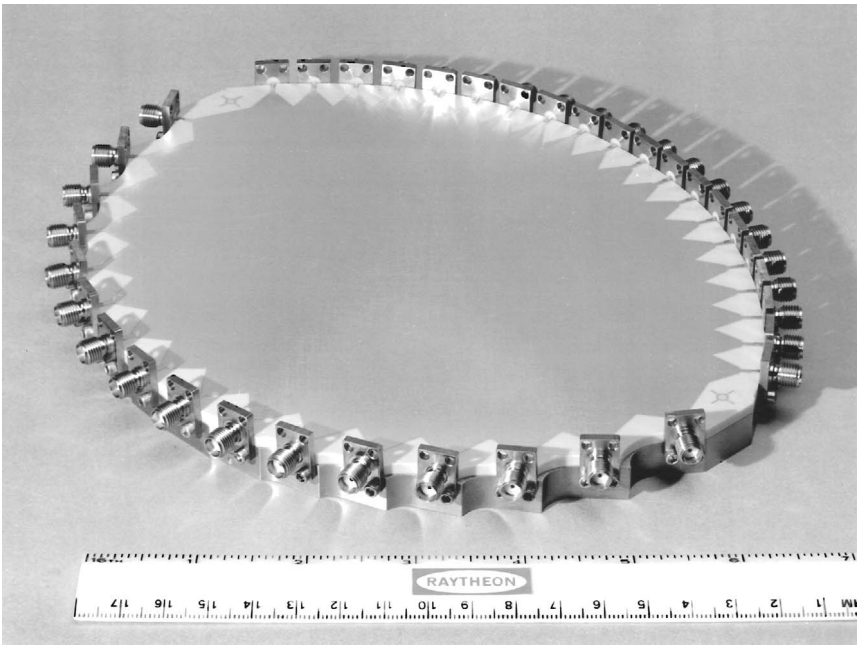


FIGURE 10.32 Microstrip Rotman lens. (Courtesy Raytheon Co.)

discussion below will show. In general, the shape of the lens and array surfaces is affected by the choice of focal points (Cornbleet, 1994).

In practice, bootlace lenses and other lenses are not designed with discrete foci; the optimum design is a MINIMAX solution for circles of least confusion. That is, the aberrations are minimized over the aperture in some sense, such as least-mean-squares. Consideration of foci, however, allow lenses to be demarcated into classes, and gives an understanding of how errors depend on lens geometry.

The simplest bootlace lens has a flat array face and a flat lens face, and has two foci that are symmetrically disposed about the axis in a plane containing the axis. The foci can be coalesced on axis with the result that the lens faces are circular and symmetric (Rao, 1982a). However this arrangement provides the largest errors when scanned.

When the lens surface becomes ellipsoidal, but still with a flat array face, three foci are allowed. These consist of an on-axis focus and two conjugate foci. The three foci are in a plane containing the axis. The lens (and array) outlines are elliptical to allow the cross-plane path lengths to be nearly the same as in-plane path lengths from any focus.

The quadrifocal lens, as expected, has four foci that are disposed in two conjugate pairs about the axis, with all foci on the plane containing the axis. This lens has an ellipsoidal lens surface, as does the trifocal lens, but the array surface is now much more complex. In-plane the cross sections are ellipsoidal, while in cross planes they are circular arcs (Rao, 1982b). This fourth-order surface is easily specified, but difficult to construct.

The following is a simple explanation of how a bootlace lens focuses. Consider a bootlace lens with flat array and lens face, with the feed surface a spherical cap. For a focal point off axis, the lace lengths are chosen so that the path length from the focus to the center array element is the same as that for other elements in the plane containing the axis and the focus. The radiated beam angle exactly compensates for the shorter path to the closest element, and the longer path to the farthest element.

However, for elements in the cross plane, if the lens face is symmetric, the path lengths from the focus to elements in the cross plane are always longer than the in-plane path lengths. This is corrected by bringing the lens face (and also array face) elements in the cross plane closer. The result is that the shape of the lens face is now elliptical, with the major axis in-plane and the minor axis cross plane. Correspondingly, the array face is also elliptical. Thus for a bifocal flat face lens, the flat faces must be elliptical in shape.

The bifocal and trifocal lenses always have the focal points in a plane containing the axis of the lens. It is not possible to design a lens with two or three focal points where one focal point is on the x-axis and a second focal point is on the y-axis, and at the same time have small path length errors for all angles of scan.

Asymmetric phase errors over an array or aperture tend to shift the main beam position. Symmetric errors tend to raise sidelobes, reduce gain, and reduce difference pattern slope at the center. These symmetric error effects are similar to those that are observed in the near-field, where a quadratic error typically occurs. It has been shown (in Chapter 15) that as the phase (or path length) error increases, the first sidelobe rises and with further increases is absorbed into the side of the main beam. Meanwhile, the next null is filling in and the next sidelobe is rising. As sidelobes are shouldered into the main beam, the gain decreases correspondingly. For difference patterns, a similar phenomenon occurs. Again the first sidelobe null fills and the sidelobe rises as error increases. The slope changes, but only with significant errors.

Bootlace lenses tend to have an error that is maximum at the lens center, and maximum at the lens edge but with opposite sign. This is a result of minimizing the errors over the aperture. This type of error produces similar effects upon the sidelobe and main beam structure. However, the change of difference pattern slope appears to be more significant, probably out of phase with the aperture edge.

Table 10.4 shows typical gain decrease caused by a path length error. Note that these errors are given in wavelengths.

TABLE 10.4 Path Length Error Gain Decrease

Δ/λ	Gain Decrease (dB)
0.05	0.14
0.1	0.50
0.15	1.1
0.2	1.9
0.25	2.8

Lens path length errors of less than roughly 0.1λ have small effect. The small gain decrease is due to the shouldering of the first sidelobes. When the error is about a quarter wavelength, several sidelobes have been shouldered each side, and the main beam is significantly broadened. The difference pattern sidelobe structure is not much altered, as the sidelobes were originally high. There is not yet significant slope change. Path length errors on the order of a half-wavelength produce a very broad main beam incorporating many sidelobes. The difference pattern now exhibits significant slope change. These give an indication of how path length errors affect lens performance.

The calculation of bootlace lens path length errors is complicated, even when the lenses are designed with discrete foci. However, there are some general rules of thumb. The path length error is directly proportional to aperture diameter in wavelengths. Error is proportional to the scan angle in the form of $\sin^2 \Psi$. Error is directly proportional to the beam angle ratio γ; this is the ratio of beam-to-feed angle. It is related to the ratio of array height to lens height and to the foci locations and maximum scan angle. In general, γ is around unity and the array lens dimensions are comparable. The path error is roughly inversely proportional to f/D.

In general, having more foci reduces errors when the plane of scan is in the plane of foci. For scan in all planes, the best design appears to be one with all foci on the axis, thereby providing a symmetric lens and array.

A simpler bootlace lens is made with a flat face and a spherical face, with corresponding elements on the faces connected by phasers. Figure 10.33 sketches such a lens without the feed horns. With a fixed four-horn feed, monopulse tracking is achieved. The phasers provide beam scan. An advantage of this lens is that the aperture

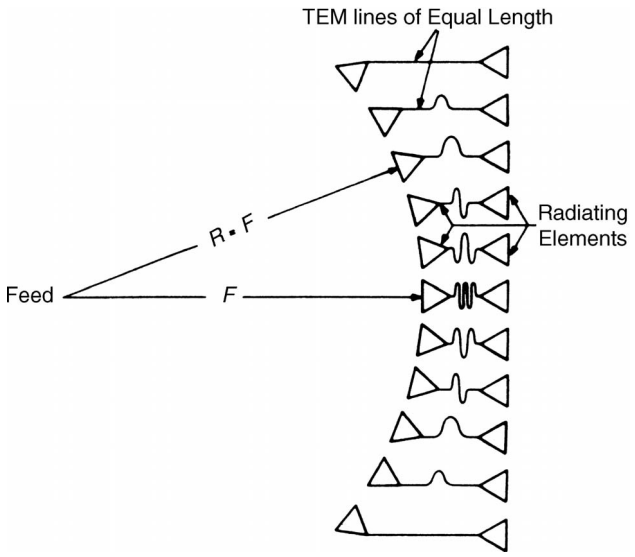


FIGURE 10.33 Bootlace lens.

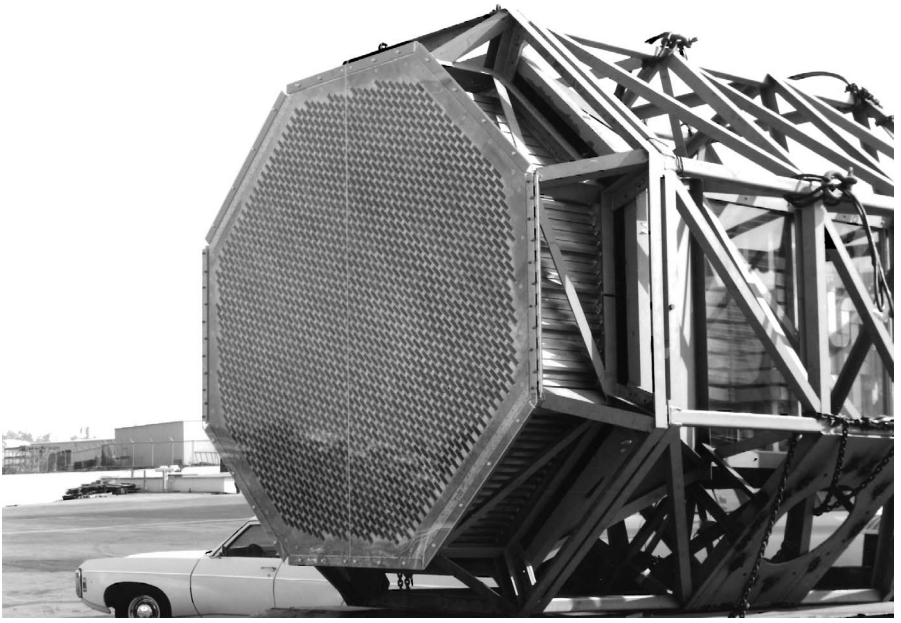


FIGURE 10.34 The S-Band bootlace lens. (Courtesy Hughes Aircraft Co.)

amplitude distribution is independent of scan. Figure 10.34 shows an experimental S-band lens with diameter of roughly 8 ft (Wong et al., 1974).

A double lens system can be used to provide wide-angle scanning where the radiating array amplitude distribution does not change with scan angle. Time-delay paths are also provided. In one such scheme (Ajioka and McFarland, 1988), a cluster of feeds excites a hemispherical bootlace lens with a hemispherical output face. The outputs in turn excite a transfer array that contains the radiating array (see Fig. 10.35). Figure 10.36 shows part of this wideband, wide angle array system (Lee, 1988).

10.2.2.3 Dome Lenses. A hemispherical, or near-hemispherical, bootlace lens has been used to refract the scanned beam produced by a planar array across the diameter of the lens. That is, if the array radiates at angle θ_0, the lens will radiate the beam at $K\theta_0$, where $K > 1$ (see Fig. 10.37). Thus coverage near the horizon is obtained without the problems common to arrays with scan ranges approaching $80-90^\circ$. The dome bootlace lens uses fixed phase or delay "laces", designed to produce a specific K. A $K = 1.5$ dome will provide zenith to horizon coverage with a feed array scanning $\pm 60^\circ$. For shipborne applications, $K = 2$ gives coverage 30° below the horizon (to accommodate roll) with the same array scan range (Schwartzman and Stangel, 1975; Schwartzman and Liebman, 1979). The phase function in the dome is given by

$$\sin \theta_{\text{inc}} + \sin \theta_{\text{refr}} = \frac{\partial \phi}{\partial s} \tag{10.14}$$

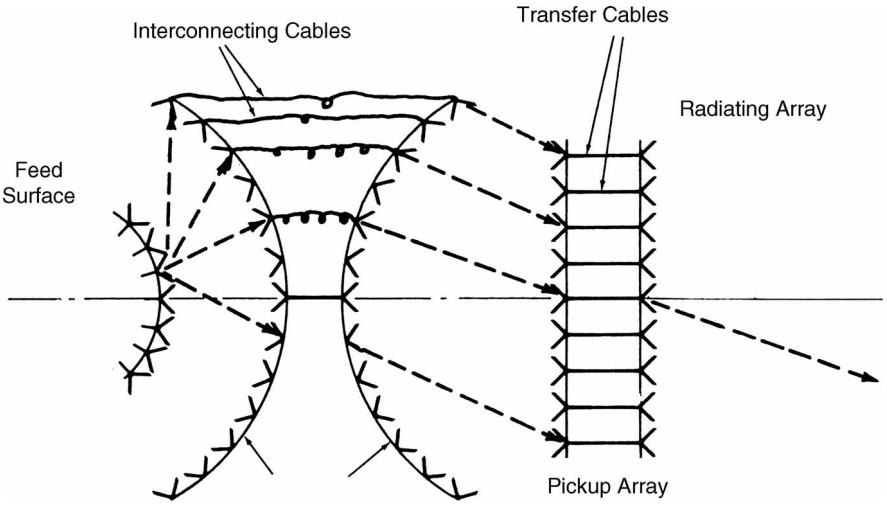


FIGURE 10.35 Dual hemispherical lens array.

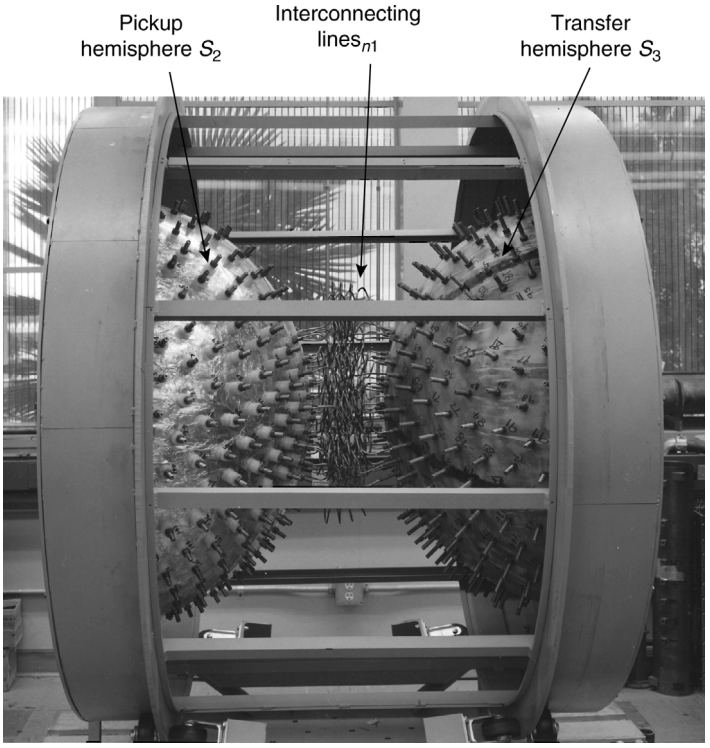


FIGURE 10.36 Dual hemispherical lens. (Courtesy Hughes Aircraft Co.)

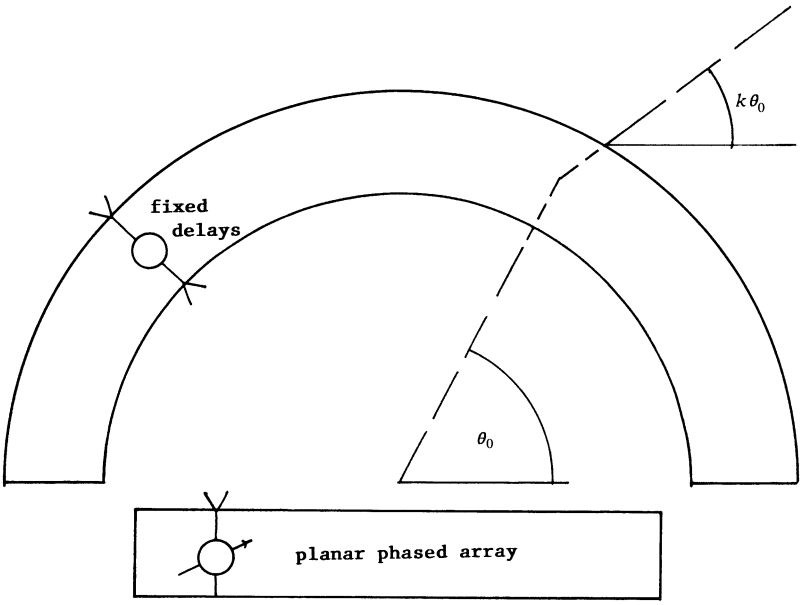


FIGURE 10.37 Dome lens concept.

where the incident and refracted angles are with respect to the dome normal, and the derivative is that of the insertion phase w.r.t. the arc length (Susman and Mieros, 1979). Because of the circular geometry, a nonlinear phase is required, so that the planar array phase is adjusted with scan angle to provide this correction. An important performance criterion is the area under the gain versus scan angle curve; this is controlled primarily by the planar array and K. However, the maximum value of the curve is controlled by the dome parameters (Steyskal et al., 1979). Figure 10.38 shows a dome bootlace lens antenna. Dielectric dome lenses have also been used; the wall thickness and/or dielectric constant varies with angle (Valentino et al., 1980).

10.2.2.4 Other Lenses. Two schemes that provide scanning in 1D via a variable medium will be described. The first uses a stack of ferroelectric dielectric plates with metal strip in between close to the fixed beam array face. As shown in Figure 10.39, the stack is the same size as the array, and is roughly one free-space wavelength thick; a voltage gradient is applied across the metal strips to scan the beam in the E-plane. Experimental loss at X-band is $\sim$1 dB (Rao et al., 1996).

In the second scheme, a similar stack of metal strips is close to the array face, but the dielectric plates have been replaced by an array of diodes. Because the diode medium produces less phase shift than the ferroelectric medium, the stack is thicker, perhaps $3-4\lambda_0$, with several diode rows normal to the array face. This type of lens is called the radant lens. The diodes are turned on–off to control the phase shift through the medium (Rao et al., 1996). Loss data for this lens were not available.

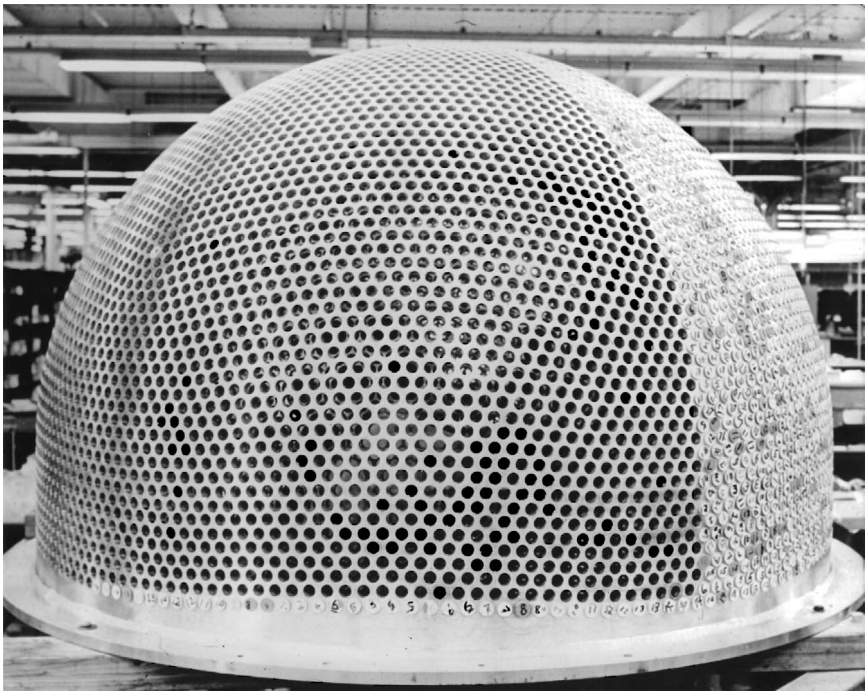


FIGURE 10.38 Dome lens under construction. (Courtesy Sperry Corp.)

A completely different type of lens is the constrained lens, where the rays are restricted to specific paths. Waveguide lenses are a good example, where the ray paths are through waveguides whose axes are parallel to the lens axis. Because of the dispersive nature of waveguides, phase shift is a function of length and frequency. Thus a waveguide lens would have longest guides at the periphery, and shortest guides at the center. Of course, such lenses are usually zoned, to reduce thickness and weight, at a cost of reduced bandwidth. A cluster feed provides a multiple-beam cluster. Such a lens is shown in Figure 10.40 (Dion and Ricardi, 1971). Detailed design data are given by Ruze (1950), and Lee (1988).

Another multiple beam arrangement uses completely overlapped sub-arrays. Implementation can be a quasioptical or beamforming network (BFN). Radiating elements are connected to a BFN, such as a Rotman lens or a Butler matrix. Beam ports of the BFN are connected to beam ports of the second BFN. The outputs of this BFN represent sub-arrays without quantization lobes. These sub-array terminals can be combined as desired to provide beams. These "dual transform" systems are elegant, but are complex, lossy, and costly. Their domain has almost completely been antenna research labs. For furthur information see: Tang (1972); Borgiotti (1977); Dufort (1978); Fante (1980); Southall and McGrath (1986); Ajioka and McFarland (1988).

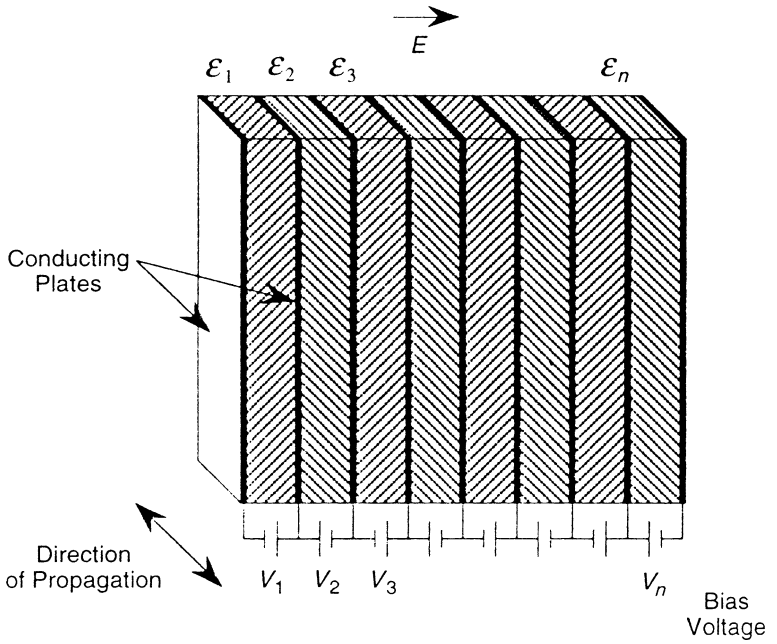


FIGURE 10.39 Ferroelectric lens. (Courtesy Rao, J. B. L., Trunk, G. V., and Patel, D. P., "Two Low-Cost Phased Arrays", *Proceedings IEEE Symposium Phased Array Systems and Technology*, Boston, MA, Oct. 1996, pp. 119–124.)

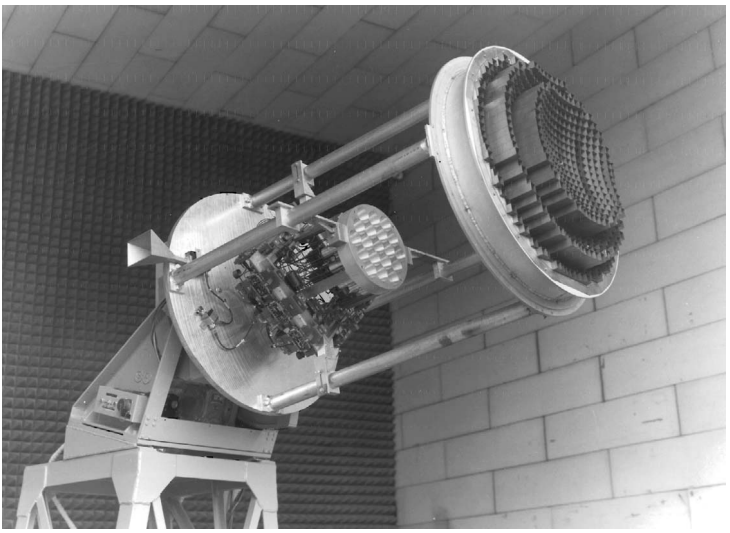


FIGURE 10.40 Multiple-beam waveguide lens. (Courtesy Lincoln Labs.)

A limited scan array, but not utilizing overlapped sub-arrays, uses a spherical bootlace lens fed by a corrective lens (Dufort, 1986). In operation, the bootlace lens is excited with phases (and amplitude adjustment if desired) to produce a series of sinc beams that excite the radiating lens or matrix. Each sinc beam excitation produce a flat-top sub-array excitation at the radiating array. Together these produce a radiated sinc beam at the desired angle. The configuration is complex, but provides 100% aperture efficiency, with no spillover. It is closely related to the double lens arrangement of Section 10.2.2.2.

10.2.3 Digital Beamforming

Digital beamforming for receive is conceptually the simplest and at the same time the most capable configuration. Each element is connected to a preamp and then to an A/D converter, as shown in Figure 10.41. The outputs of the A/D's are connected to a digital data bus, which would probably be a coded serial bus. The computer can then form any number of multiple beams, perform rapid beam scan, produce low-sidelobe beams, perform adaptive nulling, implement algorithms for target classification and for multipath discrimination, and do monopulse-type tracking with algorithms that simulate monopulse. Because the element combining algorithms can be formulated in time delay form, there are no bandwidth limitations in the digital beamformer. The serious limitation is, of course, the bit-bandwidth product of the A/D. Table 10.5 gives dynamic range versus number of bits in the A/D converter. At the present time, hardware appears suitable only for low microwave frequencies and below. Advances in superconducting A/D converters are promising, but practical realization of this hardware is years away.

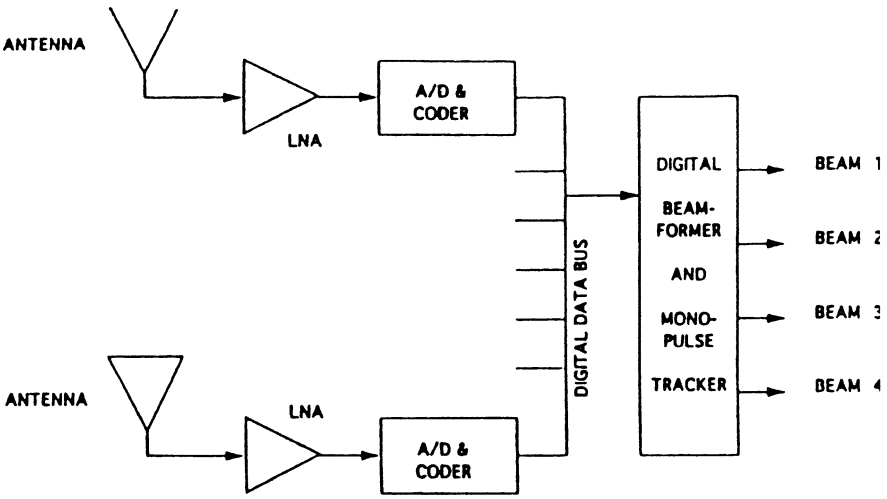


FIGURE 10.41 Digital topology.

TABLE 10.5 The A/D Converter Dynamic Range

Number of Bits	Dynamic Range (dB)
10	60.21
12	72.25
14	84.29
16	96.33

The computer (processor) for a digital array has two speed regimes: The handling of the digitized RF requires coherent beam formation at the A/D converter upper frequency, while the calculation–change of amplitude tapers for sidelobe control, formation of monopulse tracking data, and so on, can proceed at a much slower rate. That is, the RF data handling requires a very fast processor; for the array control functions a much slower processor will suffice. Some array functions, such as adaptive nulling, may require a specialized and/or separate processor for such tasks as matrix inversion, singular value decomposition, and so on. All of the digital hardware and software is available; only the A/D capability is awaited.

10.3 LOW SIDELOBES AND BEAM INTERPOLATION

10.3.1 Low-Sidelobe Techniques

This section discusses various ways of providing decoupled beams so that low sidelobes and high crossover level may both be enjoyed. All of these schemes involve loss. The first uses two separate arrays with beam interlace that could be considered to involve a 3-dB loss. A single array is more practical and the loss can be directly applied as an array taper, it can be applied through a hybrid network, or it can be applied through beam superposition.

10.3.1.1 Interlaced Beams. A direct way of achieving orthogonal beams with low sidelobes and high crossover level employs two sets of interlaced beams. Take a beam set using cosine excitation, where the space factor is

$$F(u) = \frac{\cos \pi u}{1 - 4u^2} \tag{10.15}$$

The beams are spaced at $u = 0, \pm 2, \pm 4, \ldots$ and the crossover level is $F(1) = 1/3 = -9.54$ dB. Now form a second similar beam set spaced at $u = \pm 1, \pm 3, \pm 5, \ldots$. A beam from set one crosses the adjacent beams from set two at $F(0.5) = \pi/4 = -2.10$ dB, thus achieving high crossover and modestly low (−23 dB) sidelobes. Similarly, interlaced cosine square beam sets yield a −3.27-dB crossover with −31.5-dB sidelobes. Unfortunately, each beam set requires its own array and BFN. Aside from the extra hardware requirements, it may be difficult to get the two array phase centers close together (see sketch in Fig. 10.42).

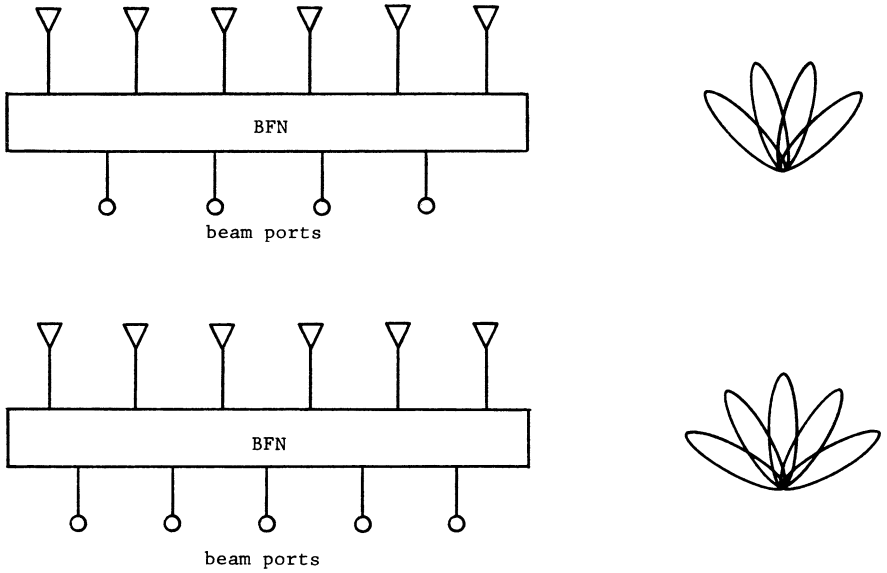


FIGURE 10.42 Interlaced beams from two multiple-beam antennas.

10.3.1.2 Resistive Tapering. A tapered amplitude, low-sidelobe distribution can be achieved simply by attenuating the array element outputs. For example, a cosine distribution has an aperture efficiency of 0.811. However, to obtain orthogonal beams, the crossover level would be -9.5 dB as mentioned. Moving the beams closer to raise the crossover will couple the beams. This type of "loss" is of no help in reducing the beam coupling. Another way of forming a cosine amplitude taper does decouple the beams, but with higher loss as next described.

10.3.1.3 Lower Sidelobes via Lossy Networks. The relationship between sidelobe ratio and crossover level for a lossless BFN does not apply to lossy networks. An important question is how much loss must be accepted to raise the crossover level. There is not at this time a definitive answer, but some key results can be given. White (1962) showed that cosine orthogonal beams can be formed from two sets of displaced uniform beams (see Section 10.4). This can be accomplished with an additional network of hybrids, as shown in Figure 10.43. The row of hybrids combines the uniform beams to form the cosine beams, and as shown in the network a BFN output of N beams produces $N-1$ cosine beams. Another way of looking at the operation of the network is that the hybrids isolate the beam ports. Tracing a signal from one beam port shows that one-half of the power goes into hybrid loads, so the efficiency is -3.01 dB and the sidelobe ratio is 23.0 dB. Lower sidelobes can be achieved by combining three sets of uniform beams, as shown in Section 10.4 to get cosine squared beams. Figure 10.44 shows this configuration, which yields $N-2$ beams from N BFN outputs. The SLR $= 31.5$ dB and the efficiency is $3/8$, or -4.26 dB. Since the only orthogonal aperture distributions appear to be $\cos^M$, performance of

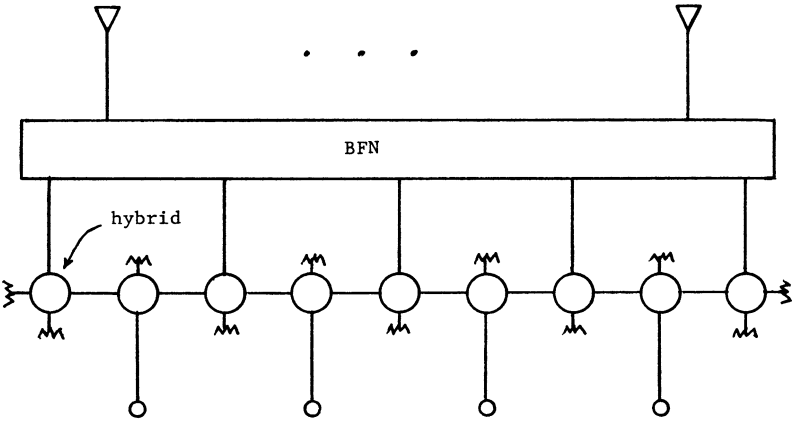


FIGURE 10.43 Formation of orthogonal cosine beams.

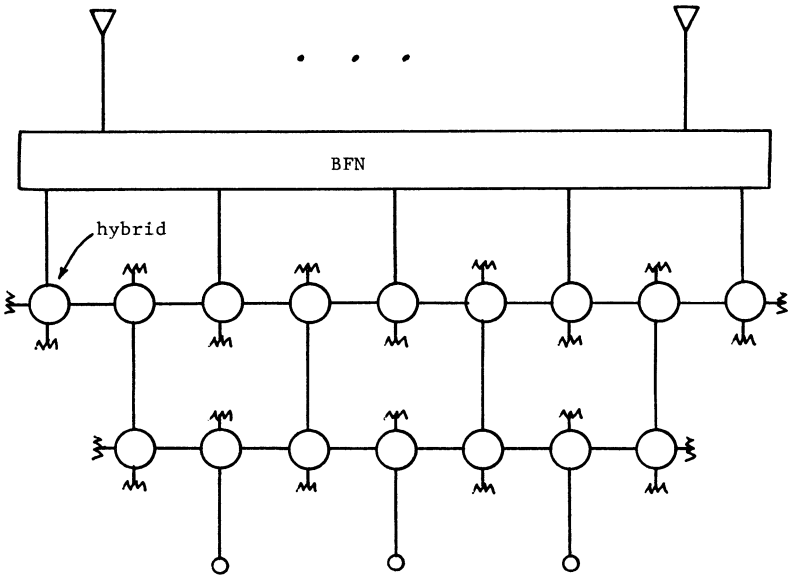


FIGURE 10.44 Formation of orthogonal cosine square beams.

TABLE 10.6 Lossy Network Formation of Orthogonal Beams with High Crossovers

Distribution	SLR (dB)	Crossover Level (dB)	η (dB)
Uniform	13.26	−3.92	0
Cosine	23.00	−2.10	−3.01
Cosine square	31.47	−3.27	−4.26
Cosine cubed	39.30	−4.44	−5.05

several of these is summarized in Table 10.6. Figure 10.45 shows efficiency versus sidelobe level. The network efficiency is

$$\eta = \int_0^1 \cos^{2M}(\pi p/2)dp = \frac{1 \cdot 3 \cdot 5 \cdots (2M-1)}{2^M M!} \tag{10.16}$$

Although the network in the figures will produce the desired orthogonal low-sidelobe, high-crossover beams, it is not clear whether more efficient networks are feasible. Early work by Stein (1962), Allen (1961), and Shelton (1965) have given only general results. Thomas (1978) uses a Rotman lens BFN, but the "network" losses are inseparable from the lens losses see also Winter (1979, 1980). Thus at this time there appears to be no data that indicates that lower losses can be achieved, but also no proof that lower losses are not possible.

10.3.1.4 Beam Superposition. Low sidelobes can be produced by appropriate beam superposition using the Taylor $\bar{n}$ distribution of Chapter 3. This distribution

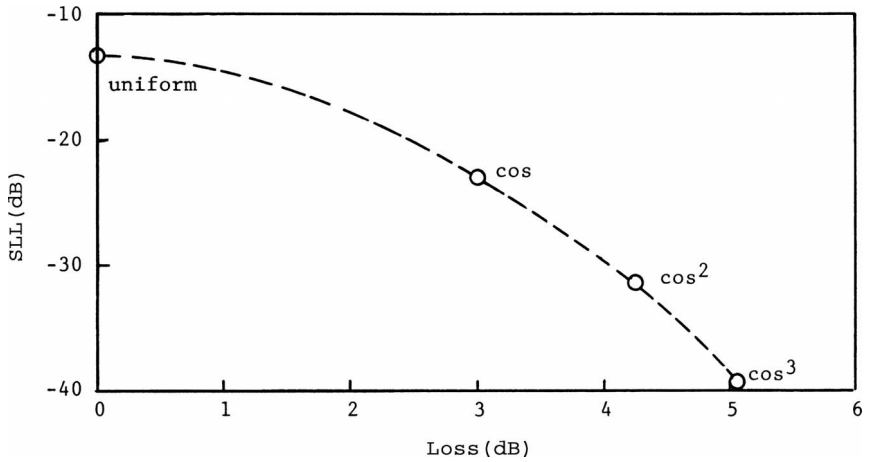


FIGURE 10.45 Orthogonal distribution sidelobe levels.

provides near-optimum directivity and beamwidth for a given SLR by making the first $\bar{n}$ sidelobes of equal height, with remaining sidelobes decaying as $1/u$, where

$$u = \frac{L}{\lambda}(\sin\theta - \sin\theta_m) \tag{10.17}$$

with θ_m the direction of the mth beam. A transition region exists around $\bar{n}$, so the $\bar{n}$ sidelobes are not precisely at the design SLR, but taper smoothly into the $1/u$ envelope. The $1/u$ envelope taper is necessary to assure a robust (low-Q) aperture. The Taylor $\bar{n}$ distribution is given by

$$g(p) = 1 + 2\sum_{n=1}^{\bar{n}-1} F(n, A, \bar{n})\cos n\pi p \tag{10.18}$$

where the coefficients and terms are defined in Chapter 3. The Taylor $\bar{n}$ distribution is interesting because of the possibility of approximating $g(p)$ with a few terms.

Except for very small $\bar{n}$, the only significant term in the $g(p)$ series is the first: $n = 1$. For example, consider the 30-dB space factor with the largest $\bar{n}$ resulting in a monotonic distribution. The coefficients are given in Table 10.7, where it can be seen that only two coefficients are necessary. This distribution is a "cosine-on-a-pedestal", but not the usual one since here the cosine has value -1 at the aperture ends. $\bar{n}$ can be increased to the value giving maximum aperture efficiency ($\bar{n} = 23$ for this case) without degrading this two-coefficient approximation. Since the change of efficiency from max monotonic $\bar{n}$ to max efficiency $\bar{n}$ is $<2\%$ for SLR ≤ 40 dB, the monotonic $\bar{n}$ is a good choice.

Taylor showed that the space factor can be written as (see Chapter 3)

$$F(u) = \sum_{n=-\bar{n}-1}^{\bar{n}+1} \varepsilon_n F(n, A, \bar{n})\,\mathrm{sinc}[\pi(u+n)] \tag{10.19}$$

Thus the space factor consists of a central beam with $\bar{n} + 1$ beams on either side, all spaced at integer values of u. These constituent beams are each the result of uniform excitation. With only $n = \pm 1$ needed to approximate the Taylor $\bar{n}$ distribution, only

TABLE 10.7 Distribution Coefficients for SLR = 30 dB, $\bar{n} = 7$

$F_0 = 1$
$F_1 = 0.28266$
$F_2 = -0.01313$
$F_3 = -0.00164$
$F_4 = 0.00468$
$F_5 = -0.00414$
$F_6 = 0.00225$

three beams are needed to approximate the Taylor $\bar{n}$ space factor. The BFN then must add three contiguous beams with the proper weighting to produce a single low-sidelobe beam. To have N low-sidelobe beams the BFN must have $N + 2$ uniform excitation beams of course. A simple implementation would utilize a three-way power divider at each uniform beamport, with two arms at $2F(1, A, \bar{n})$ and one arm at 1. Two arms of each outside beamport are terminated, and one arm of each of the adjacent beam-ports. The efficiency is then a function of the number of beamports:

$$\eta = \frac{N-2}{N} = \frac{M}{M+2} \tag{10.20}$$

where there are N uniform beamports, and M low-sidelobe beams. For 8 beams, the loss is 0.97 dB and for 16 beams it is 0.51 dB, for example. This dissipation loss is separate from the aperture efficiency that pertains to the low-sidelobe beams of course. Sidelobes below -32 dB have been achieved by Thomas (1978) using a Rotman lens BFN, with -28-dB sidelobes over a 10% bandwidth. When the BFN is a Rotman or bootlace lens, beam superposition is sometimes called "block feeding".

10.3.2 Beam Interpolation Circuits

With beams having low crossover levels, or for accurate angle of arrival determination, it is desirable to be able to move the beam cluster in space, or to form interpolated beams that can be moved. One simple method of performing the latter will be described. Consider two adjacent beams from a uniformly excited half-wave spaced multiple-beam antenna (MBA). Beam peak angles occur at

$$\sin \theta_i = \frac{m_i}{N} \tag{10.21}$$

where m_i is an odd integer $\leq$ half the number of array elements N. Each beam has phase of $(\omega t - kd \sin \theta_i)$, which gives $m_i \pi / N$ with ωt suppressed. Now two adjacent beam ports are connected through phasers to a hybrid junction, as shown in Figure 10.46. In this figure, the beam phases are $\phi_i = m_i \pi / N$, while the phasers are

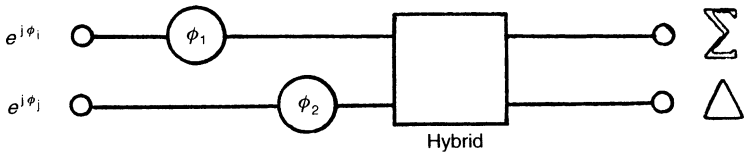


FIGURE 10.46 Beam interpolation circuit.

ϕ_1 and ϕ_2. The sum and difference outputs from the junction are

$$\begin{aligned}\Sigma &= 2\cos\left[\frac{1}{2}(\phi_i - \phi_j + \phi_1 - \phi_2)\right]\exp\left[\frac{j}{2}(\phi_i + \phi_j + \phi_1 + \phi_2)\right] \\ \Delta &= 2\sin\left[\frac{1}{2}(\phi_i - \phi_j + \phi_1 - \phi_2)\right]\exp\left[\frac{j}{2}(\phi_i + \phi_j + \phi_1 + \phi_2)\right]\end{aligned} \tag{10.22}$$

First, take the sum output. If only one phaser is used, the beam phaser versus phaser value is shown below, along with the amplitude loss. The lost power goes into the difference arm.

Beam Phase	Phaser ϕ_1	Amplitude
ϕ_i	$\phi_i - \phi_j$	$\cos(\phi_i - \phi_j)$
$\frac{1}{2}(\phi_i + \phi_j)$	0	$\cos\left[\frac{1}{2}(\phi_i - \phi_j)\right]$
ϕ_j	$\phi_j - \phi_i$	0

(10.23)

Since $\phi_i - \phi_j = 2/N$ there is an appreciable loss in small arrays when the beam is moved close to position ϕ_i. This loss can be reduced by using both phasers; the cosine factor may be kept at unity. Then,

Beam Phase	Phaser ϕ_1	Phaser ϕ_2
ϕ_i	0	$\phi_i - \phi_j$
$\frac{1}{2}(\phi_i + \phi_j)$	$\frac{1}{2}(\phi_i - \phi_j)$	$\frac{1}{2}(\phi_i - \phi_j)$
ϕ_j	$\phi_i - \phi_j$	0

(10.24)

Although only three beam positions are shown above, two phasers can be adjusted to position the sum beam anywhere from beam position ϕ_i to beam position ϕ_{i+1} without loss. The actual beam position is found from the beam phase, from

$$\pi \sin\theta_i = \phi_i \tag{10.25}$$

The difference pattern that can be used to steer a monopulse type null for accurate angle tracking behaves similarly:

Beam Phase	Phaser ϕ_1	Phaser ϕ_2
ϕ_i	$\pi/2$	$\frac{\pi}{2} + \phi_i - \phi_j$
$\frac{1}{2}(\phi_i + \phi_j)$	$\frac{1}{2}(\pi - \phi_i + \phi_j)$	$\frac{1}{2}(\pi + \phi_i - \phi_j)$
ϕ_j	$\frac{\pi}{2} - \phi_i + \phi_j$	$\pi/2$

(10.26)

Again the amplitude factor is unity but the phaser values are $\pi/2$ larger than those needed for shifting the sum beam peak. Using adequate phasers, a tracking loop can be constructed that will operate as long as the signal is between beam angles ϕ_i and ϕ_{i+1} (Mailloux et al., 1970).

Beam and null shifting can also be accompanied at IF using one receiver channel per beam port (Mailloux, 1994).

10.4 BEAM ORTHOGONALITY

10.4.1 Orthogonal Beams

10.4.1.1 Meaning of Orthogonality. When beams in a multiple-beam antenna (MBA) are orthogonal, a signal injected into a given beam port appears only in that beam pattern, with no signal in other beam patterns or other beam ports. By using a scattering matrix representation of the array (of N isotropic elements) and beamforming network, the independence of M ports requires that the submatrix representing the beam ports have zero elements. Reciprocal BFNs imply a symmetric matrix and a lossless network requires a unity matrix:

$$\sum_{n=1}^{N+M} S_{ni} S_{nj}^{*} = \delta_j^i \tag{10.27}$$

By using this relationship, Allen (1961) showed that uncoupled beams (and ports) lead to

$$\int_{-1/2}^{1/2} F_i(\pi u) F_j^{*}(\pi u)\, du = \delta_j^i \tag{10.28}$$

where $u = (d/\lambda) \sin \theta$ with θ measured from broadside, and the space factor is

$$F_i(\pi u) = \sum S_{ni} \exp(j2n\pi u) \tag{10.29}$$

Thus the space factors must be orthogonal over an angular range of

$$-\arcsin\frac{\lambda}{2d} \leq \theta \leq \arcsin\frac{\lambda}{2d} \tag{10.30}$$

in order to yield uncoupled beams and uncoupled ports (Kahn and Kurss, 1962). When $d/\lambda = 1/2$ (or $d/\lambda = 1$ since the integration limits could have been $= 1$), the beams are orthogonal over visible space. Thus for spacings that are multiples of $\lambda/2$ the beams and ports are both uncoupled for proper beam spacing.

In an extension of the network approach to multibeam analysis, Andersen and Rasmussen (1976) showed that decoupling and descattering (among ports) requires that the array mutual resistances be zero. For isotropic elements, the mutual resistance is proportional to sinc mkd, which is zero for half-wave spacing. This is necessary, but not sufficient, as it implies decoupling (orthogonality) for all beam spacings. Beam

orthogonality requires either mutual impedance that is zero for the array spacing, and this is never true even for isotropic or minimum scattering antennas (see Wasylkiwskyj and Kahn, 1970), or that the drive vectors of the array be eigenvectors of the mutual impedance matrix inverse. The eigenvalue relationship only appear to hold for amplitude distributions consisting of $\cos^M x$, with the proper beam spacings. Thus patterns and crossover levels are not independent.

10.4.1.2 Orthogonality of Distributions. While an MBA is either an array or uses an array as primary feed, much can be learned from a study of continuous (line) sources. A uniform line source has the space factor

$$F(u) = \operatorname{sinc}[\pi(u - u_1)] \tag{10.31}$$

where $u = (L/\lambda)\sin\theta$, $u_1 = (L/\lambda)\sin\theta_1$. Here θ_1 is the beam peak. The aperture distribution, from $p = -1$ to $+1$, is

$$g(p) = \exp(-jp\pi u) \tag{10.32}$$

Orthogonality of the space factors can be simply written (Lewin, 1975)

$$\int_{-\infty}^{\infty} \operatorname{sinc}[\pi(u - u_1)]\operatorname{sinc}[\pi(u - u_2)]du = \operatorname{sinc}[\pi(u_1 - u_2)] = \delta_2^1 \tag{10.33}$$

where the line source length has been assumed long so that the limits can be approximated by ∞. Orthogonal beams then can be spaced apart in units of $u_\Delta = \pm 1, \pm 2, \ldots$, but only $u_\Delta = 1$ allows adjacent main beams to cross over. This occurs at $u = u_1 + \frac{1}{2}$ and gives a crossover level of $2/\pi = -3.92$ dB. Alternatively, the orthogonality integral could be written over the aperture, since the Fourier transforms of orthogonal functions are also orthogonal, with one conjugated (Harmuth, 1970). For the uniform aperture this gives

$$\int_{-1}^{1} \exp[jp\pi(u - u_1)]\exp[jp\pi(u - u_2)]du = \operatorname{sinc}[\pi(u_1 - u_2)] = \operatorname{sinc}(\pi u_\Delta) \tag{10.34}$$

The general class of orthogonal amplitude tapered distributions is $\cos^M(p\pi/2)$. Since the space factors and orthogonality integrals can be evaluated in closed form, it is easy to compare these distributions. The cosine distribution will be used as an example. The space factor can be written either of two ways, with $F(0) = 1$ in both cases:

$$F(u) = \frac{\pi}{4}\left\{\operatorname{sinc}\left[\pi\left(\frac{1}{2} - u\right)\right] + \operatorname{sinc}\left[\pi\left(\frac{1}{2} + u\right)\right]\right\} = \frac{\cos\pi u}{1 - 4u^2} \tag{10.35}$$

Space factor zeros occur for $u = 3/2, 5/2, 7/2, \ldots$. The first sidelobe peak is at $u = 1.8894$ and the sidelobe ratio is 23 dB. Half-beamwidth, in u, is $u_3 = 0.5945$, while aperture efficiency (directivity with respect to uniform excitation) is 0.8106. The orthogonality integral reduces to

$$2\,\mathrm{sinc}(\pi u_\Delta) + \mathrm{sinc}[\pi(1 - u_\Delta)] + \mathrm{sinc}[\pi(1 + u_\Delta)] = 0 \tag{10.36}$$

which can be simplified to

$$\frac{\sin \pi u_\Delta}{u_\Delta(1 - u_\Delta)^2} = 0 \tag{10.37}$$

Values of beam separation that give beam orthogonality then are the solutions of this equation

$$u_\Delta = 2,\ 3,\ 4, \ldots \tag{10.38}$$

In all cases, minus u can be substituted for plus u, but this has not been indicated. Table 10.8 gives pertinent parameters for four aperture distributions.

Beamwidth and aperture excitation efficiency are u_3 and η_t, while the sidelobe peak angle is u_{SL}. The crossover angle and level are u_x and F_x.

It is sometimes stated that linear combinations of orthogonal space factors can also be orthogonal, for example, a cosine-on-a-pedestal distribution. This finding is not valid, as examination will show. The orthogonality integral over the aperture, where $u_\Delta = u_1 - u_2$, is

$$\begin{aligned}\frac{1}{2}\int_{-1}^{1}\left(a + b\cos\frac{1}{2}\pi p\right)^2 \exp(jp\pi u_\Delta)\,dp = &\left(a^2 + \frac{1}{2}b^2\right)\mathrm{sinc}(\pi u_\Delta) \\ &+ \frac{1}{2}ab\left\{\mathrm{sinc}\left[\pi\left(\frac{1}{2} - u_\Delta\right)\right] + \mathrm{sinc}\left[\pi\left(\frac{1}{2} + u_\Delta\right)\right]\right\} \\ &+ \frac{1}{4}b^2\{\mathrm{sinc}[\pi(1 - u_\Delta)] + \mathrm{sinc}[\pi(1 + u_\Delta)]\}\end{aligned} \tag{10.39}$$

TABLE 10.8 Orthogonal Space Factors

Distribution	Zeros	U_{SL}	SLR (dB)	$\frac{1}{2}U_3$	η	U_x	F_x (dB)
Uniform	1, 2, 3	1.4303	13.26	0.4429	1	0.5	−3.94
Cosine	1.5, 2.5, 3	1.8894	23.00	0.5945	0.8106	1	−9.54
Cosine square	2, 3, 4	2.3619	31.47	0.7203	0.6667	1.5	−15.40
Cosine cubed	2.5, 3.5, 4.5	2.8420	39.30	0.8292	0.5764	2	−21.34

This reduces to

$$\left(a^2+\frac{1}{2}b^2\right)\operatorname{sinc}\pi u_\Delta+\frac{2ab\cos\pi u_\Delta}{\pi(1-4u_\Delta)^2}+\frac{b^2u_\Delta\sin\pi u_\Delta}{2\pi(1-u_\Delta^2)}=0 \tag{10.40}$$

For the special cases of uniform excitation ($b=0$) and cosine excitation ($a=0$), the correct values of u_Δ appear. The general case has values of u_Δ that make the beams orthogonal, but the beam spacings are not uniform. For example, taking a distribution with 10-dB pedestal (edge excitation -10 dB) that has $a=0.46248$, $b=1$, the values of u_Δ are[2]

$$u_\Delta=1.648,2.271,3.143,4.097,5.074,6.060,\ldots \tag{10.41}$$

The implication of this nonuniform sequence is that all adjacent beams are orthogonal ($u_\Delta=1.648$), but none of the beams are orthogonal to any beams past their neighbor, since $2\times1.648=3.143$, $3\times1.648=5.074$, and so on. In contrast, the cosine alone distribution has $u_\Delta=2,3,4,\ldots$, so that if adjacent beams are 2 apart, all beams are multiples of 2 apart, hence orthogonal. That the cosine-on-a-pedestal is not orthogonal is not surprising, since the orthogonality integral function of u is related to the space factor: If the space factor zeros are evenly spaced, and the beam peaks are located at the zeros of the other beams, the beams can be orthogonal. Only the $\cos^M$ distributions have this characteristic. Unfortunately, these distributions are inefficient compared with Taylor space factors. Such modern space factors are designed by locating the close-in zeros to yield the proper sidelobe level, then allowing the zeros to transition to the integer spacing for large u, as required for low aperture Q. The resulting nonuniform spacing makes the space factors not orthogonal.

Orthogonality is then difficult to achieve for several reasons: apertures must be large in wavelengths and the $\cos^M$ distributions are inefficient; arrays must have half-wave spacing. It is possible to modify these constraints by introducing loss into the BFN. This, of course, will also reduce efficiency. Before this is considered, it is necessary to determine the effects of nonorthogonal beams.

10.4.1.3 Orthogonality of Arrays. The preceding work addressed continuous aperture distributions since they are more tractable. It is well known that arrays of many elements can accurately approximate a continuous distribution simply by sampling it at the array element locations. To realize a given space factor with an array of few elements it is usually necessary to adjust the array polynomial zeros (Elliott, 1977). Unfortunately, this adjustment of zeros destroys the beam orthogonality. To see whether orthogonal beams can be formed from small arrays, the

[2]These are calculated using a Wegstein rooter.

orthogonality integral could be examined directly. For example,

$$\int \frac{\sin(N\pi u_i)}{N\sin(\pi u_i)} \cdot \frac{\sin(N\pi u_j)}{N\sin(\pi u_j)} du \tag{10.42}$$

represents two uniform excitation beams produced by an N-element array. Integrals of this type appear to be intractable. In addition, there is no theorem for discrete functions that indicates orthogonality of the DFT of orthogonal functions. We are then left with the practical solution of evaluating nominally orthogonal arrays as a function of number of elements. Many elements should give the continuous distribution sidelobes, while fewer elements may show a sidelobe rise owing to the less accurate representation. This is calculated below.

10.4.2 Effects of Nonorthogonality

10.4.2.1 Efficiency Loss. When aperture distributions that are orthogonal are used, but with a nonorthogonal beam spacing, BFN efficiency is lost owing to beam coupling. Stein (1962) derived the limits on beam coupling and the resultant efficiency. Let the BFN beam overlap parameter β be

$$\beta_{ij} = \int E_i(\theta, \phi) E_j(\theta, \phi) d\Omega \tag{10.43}$$

with $B_{ii} = 1$. The maximum efficiency now becomes

$$\eta_{\max} = \frac{1}{1+\beta^2} \tag{10.44}$$

The integral allows the BFN efficiency to be determined easily; DuFort (1985) showed that the BFN efficiency is also equal to the ratio of the average to peak aperture (array) power distribution. Thus for uniform, cosine, and cosine square distributions, the maximum BFN efficiency is 0 dB, -3.01 dB, and -4.26 dB, respectively. Discrete BFN efficiency is given by

$$\eta_{\text{BFN}} = \frac{1 + 2\sum B_n^2}{(1 + 2\sum B_n)^2} \tag{10.45}$$

where the B_n, for example, are Taylor $\bar{n}$ coefficients. A Taylor distribution with 25-dB sidelobe ratio and $\bar{n} = 5$ has a BFN efficiency of -1.75 dB, much better than that of the cosine distribution, which has a comparable sidelobe level. For large BFN, the

beam overlap parameter is given by

$$\beta = \frac{1}{\pi} \int_{-\infty}^{\infty} \operatorname{sinc} x \operatorname{sinc}(x + \pi u_\Delta) dx = \operatorname{sinc} \pi u_\Delta \tag{10.46}$$

This parameter and the corresponding beam overlap efficiency are plotted in Figure 10.47. A similar equation for a cosine squared distribution is

$$\begin{aligned} \beta &= \frac{1}{\pi} \int_{-\infty}^{\infty} \operatorname{sinc}^2 x \operatorname{sinc}^2(x + \pi u_0) dx \\ &= \frac{4}{\pi} \{2 \operatorname{sinc} \pi u_\Delta + \operatorname{sinc}[\pi(1 + u_\Delta)] + \operatorname{sinc}[\pi(1 - u_\Delta)]\} \end{aligned} \tag{10.47}$$

This parameter and its efficiency are also plotted in Figure 10.47, where clearly the cosine square efficiency is considerably poorer. Note, however, that the Taylor $\bar{n}$ distribution can be expected to be much better than even the cosine distribution.

10.4.2.2 Sidelobe Changes. Beam coupling in an actual antenna is produced by mutual coupling changing the array excitation amplitude and phase. The earlier discussions of, and calculations about, orthogonality all assumed isotropic array elements. For these hypothetical elements, certain amplitude tapers and corresponding

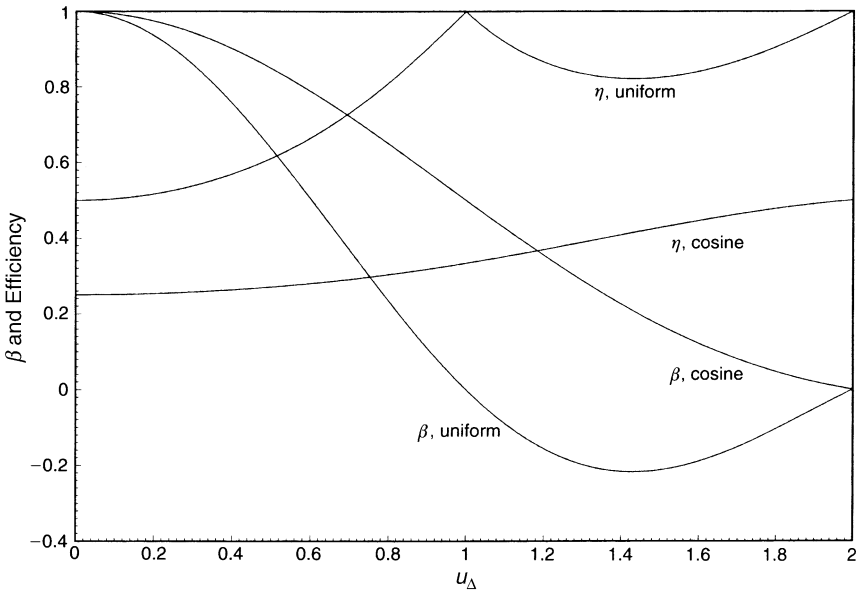


FIGURE 10.47 Beam overlap coefficient and BFN efficiency.

proper beam locations produced orthogonal beams. But actual arrays use slots, dipoles, loops, patches, and so on, when low-gain elements are desired, or horns, flat spirals, conical spirals, log-periodics, and so on, when moderate-gain elements are needed. None of these approximate the self- and mutual impedances of isotropic elements, where $Z = 120 \text{ sinc } kd$, and in consequence there are no exactly orthogonal beams from arrays utilizing practical elements. Now the important question becomes: What is the sidelobe degradation (beam coupling) due to beam nonorthogonality? The part of this that is due to mutual coupling affecting the array distribution is answered below.

A dipole array will be used as an example, but results for slot arrays are similar: A dipole array without ground screen is equivalent, in terms of reflection coefficient variation, to a slot array. Call the applied drive voltages V_n and the resulting currents I_n. These are related through the impedance matrix:

$$[V] = [I][Z] \tag{10.48}$$

where an element Z_{ij} of the latter is the mutual impedance between the ith and jth elements. The array pattern of the mth beam is given by

$$F_m(\theta) = \sum_{n=1}^{N} I_n \exp\{j[nkd(\sin\theta - \sin\theta_m)]\} \tag{10.49}$$

where d is the element spacing. Thus sidelobe degradation is computed by placing a beam at the desired angle and computing the pattern. To do this, the complex simultaneous equations are first solved, with the voltage drive vector containing both the amplitude taper and the phasing to position the mth beam. The resulting solution is the current vector that is used in the sum above to calculate the embedded pattern of the mth beam. Mutual impedances are computed from Carter's zero-order theory, using a version arranged for efficient computation (Hansen and Brunner, 1979). As shown in Section 3.1, the beams, for uniform excitation, are centered at angles given by

$$\sin\theta_t = \frac{2i-1}{2Nd/\lambda} \qquad i = 1, 2, 3, \ldots, N/2 \tag{10.50}$$

For the uniform case, all $I_n = 1$. Low sidelobes are produced by use of a Taylor $\bar{n}$ distribution (see Chapter 3). This highly efficient distribution is particularly suitable for multiple-beam antennas. All patterns are normalized to unity at the nominal beam angle.

Calculations were made for half-wave dipoles, in both parallel and collinear configurations. The highest sidelobe is adjacent to the main beam, but can be on either side; the patterns are asymmetric. To determine any effect of placing the beams at the prescribed locations (for a uniform excitation array), beams were placed at broadside, next to broadside, at the outside position, and next to the outside. For the wider element spacings grating lobes will exist, but in all cases the outside beam is part of the

TABLE 10.9 Sidelobe Level of 8-Element Array of Parallel Dipoles

	d/λ, Sidelobe Level (dB)					
Beam	0.5	0.6	0.7	0.8	0.9	1.0
Broadside	−13.3	−13.3	−13.5	−15.0	−12.8	−11.5
Next broadside	−12.9	−12.9	−13.0	−13.2	−12.3	−10.9
Next outside	−10.4	−10.0	−9.8	−12.6	−12.7	−12.8
Outside	−10.0	−9.4	−11.1	−12.7	−12.8	−12.8

original set of beams. No grating lobe beams were evaluated. For the four beams the highest sidelobe level was found, as shown in Table 10.9 for parallel dipole elements. Sidelobe level varies cyclically with element spacing, increasing slowly then rapidly falling, in a manner similar to that of array directivity with spacing. This is not surprising since both effects are caused by mutual coupling. As the beam position moves out, the sidelobe level break point moves to a smaller d/λ, and as expected the performance of the far-out beams is poorer. There is no apparent difference in performance as the beam moves through the prescribed positions; apparently half-wave dipoles are sufficiently different from minimum scattering antennas to overcome any beam position effects. Table 10.10 gives results for collinear dipoles but only for the center beam and outside beam. Except for the strong coupling effects at half-wave spacing for the outside beam, the collinear arrays show less sidelobe level variation with spacing and from beam to beam. For the collinear arrays, all sidelobe levels are with respect to the peak of that beam. These calculations have utilized 8 elements as this is a frequently used size of multiple-beam array. Table 10.11 gives sidelobe results for arrays of 16 parallel dipoles. Only the outside beam position was calculated as it represents the worst case. Note that there is somewhat less variation for these larger arrays: As expected for 32-element arrays, the realized sidelobes are very close to the design values.

It was expected that the high sidelobes of uniformly excited arrays would be affected to a minor degree while low-sidelobe arrays are more seriously affected. In

TABLE 10.10 Sidelobe Level of 8-Element Array of Collinear Dipoles

	d/λ, Sidelobe Level (dB)					
Beam	0.5	0.6	0.7	0.8	0.9	1.0
Central	−12.8	−12.9	−12.8	−12.9	−13.0	−13.1
Outside	−17.5	−13.7	−12.4	−12.4	−12.5	−12.6

TABLE 10.11 Sidelobe Level of 16-Element Array of Parallel Dipoles

	d/λ, Sidelobe Level (dB)					
Beam	0.5	0.6	0.7	0.8	0.9	1.0
Outside	−10.8	−11.6	−13.1	−13.1	−13.1	−13.2

TABLE 10.12 Sidelobe Level of Array of Parallel Dipoles, SLR = 25 dB

	Sidelobe Levels (dB)	
Beam	$d/\lambda = 0.5$	$d/\lambda = 1.0$
	$N = 8$	$N = 8$
Broadside	−24.0	−20.2
Outside	−20.3	−22.7
	$N = 16$	
Broadside	−25.1	
Outside	−21.1	

calculating low-sidelobe arrays, a Taylor $\bar{n}$ distribution was used. The value of $\bar{n}$ for each value of sidelobe level was chosen as the largest that gave a monotonic distribution. Table 10.12 shows sidelobe level for Taylor arrays of 25-dB SLR with $\bar{n} = 5$. The poor broadside results for the 8-element array are due to the inadequate sampling of the Taylor distribution that an 8-element array allows[3]. It can be observed that the 16-element array sidelobe level is satisfactory at broadside. As expected, there is sidelobe level degradation at large beam angles. This degradation can be reduced by impedance matching the elements at an intermediate angle so that the broadside and outside beams are equally mismatched. From these calculations, two conclusions can be drawn. First, small arrays are inefficient producers of low-sidelobe patterns; 32 elements closely approximate the efficient continuous distribution, while 16 elements are a fair approximation. Second, sidelobe levels will degrade at extreme beam positions, so the array must be overdesigned; that is the design SLR must be larger than the expected value.

Degradation of sidelobe level for outside beams can be improved by compensating the *scan impedance* versus scan angle (see Chapter 7).

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[3]For N so small, a Villeneuve $\bar{n}$ distribution should be used instead of the Taylor $\bar{n}$ (see Chapter 3).

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CHAPTER ELEVEN

Conformal Arrays

11.1 SCOPE

The essential constituent of a conformal array is curvature. Authorities disagree on whether the array must be part of a curved metallic structure; in this chapter curvature alone is sufficient. Arrays of one or more concentric rings of elements, here called "ring arrays", are treated first. The term "circular array" is not used, as it often means a planar array of circular perimeter. The following sections deal with arrays on curved metallic bodies. Most simple is the cylinder; Section 11.3 treats cylindrical arrays with elements around the full circumference. Partial cylindrical arrays, where only part of the perimeter has elements, are the subject of Section 11.4. Finally, arrays on surfaces with 2D curvature, such as spheres and cones, are examined.

Many applications exist where the array occupies only a small part of a curved body, and where the conformal effects are primarily on patterns and excitation rather than on impedance. For these applications, the array can often be designed as a planar array (but with proper conformal phasing), with pattern perturbations predicted by techniques such as geometric theory of diffraction (McNamara et al., 1990). "Smart skin" flush-mounted antennas for aircraft and missiles often are of this type.

One commom application of conformal array is an aircraft or helicopter satellite link, where the signal is circularly polarized, and zenith to horizon coverage is desired. Essentially all elements that radiate circular polarization at broadside radiate linear polarization at $\theta = 90^\circ$, or 3-dB directivity drop. Array directivity also changes. A rectangular planar array $L \times W$ has broadside directivity of $4\pi LW\eta/\lambda^2$, where η is the area taper efficiency. The same array at endfire, in the L direction, has directivity of $3\pi W/\lambda\sqrt{2L/\lambda}$. Thus at the horizon the data link is worse by roughly $1/\sqrt{L/\lambda}$; for example, for a uniform amplitude array with a zenith directivity of 23 dB ($L = W = 4\lambda$), the horizon value is 20 dB, and 3 dB lower.

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11.2 RING ARRAYS

A ring array, where the elements are located in a circle, offers advantages of many fewer elements than a filled planar array, and a geometry that facilitates 360° scanning. Obvious applications of the latter are direction finding in the HF–VHF–UHF bands, navigation systems, communications, and military electronic support systems. Of course, the directivity is well below that of a filled array, and many grating lobes may appear.

Analytical efforts on ring arrays existed before 1940, but significant progress was made during World War II by Page (1948a,b), who considered a single azimuthal Fourier mode, and by Knudsen (1951, 1953, 1956), who analyzed a ring array as a series of Bessel function terms and showed that the azimuthal mode number and ring diameter must be compatible to avoid superdirectivity. The properties of individual azimuthal modes were pursued by Tillman and colleagues (Hickman et al., 1961; Hilburn and Hickman, 1968; Tillman, 1968; Hilburn, 1969). The last work is a book containing extensive modal data, both pattern and impedance. Beam cophasal excitation (see Section 11.2.1), in which all elements contribute to the main beam in phase, is almost always used. The harmonic (Bessel) series analysis has been used for sidelobe control through excitation tapering, and for pattern synthesis (Fenby, 1965; James, 1965; Royer, 1966; Longstaff et al., 1967; Redlich, 1970; Lim and Davies, 1975). A harmonic analysis without explicit Bessel functions has also been used (Blass, 1974) as has a polynomial approach (Gerlin, 1974). DuHamel developed a procedure for producing Chebyshev patterns for ring arrays (DuHamel, 1951). Effects of ground for HF ring arrays have been reported by Tillman et al. (1955) and by Ma and Walters (1970). An examination of current distribution on dipoles in a ring array via moment method has been made by Sinnott and Harrington (1973).

A ring array can produce a near-omnidirectional pattern, if the element spacing is not too large. Carter (1943) computed the amplitude of pattern oscillations versus element spacing.

Circular arc arrays, which result when tapered excitation is used, are reported by Lo and co-workers (Lee and Lo, 1965). Gobert and Yang (1974) consider a ring array of nonparallel dipoles.

11.2.1 Continuous Ring Antenna

Assume a local "element" pattern G where ϕ is the azimuth angle, and β is the angular location of the local element. A continuous current density I then produces the space factor (pattern)

$$E(\phi,\ \theta) = \frac{M}{2\pi} \int_0^{2\pi} I(\beta) G(\phi - \beta, \theta) d\beta \tag{11.1}$$

where M is the number of elements that the continuous distribution replaces. If $I(\beta)$ is symmetric, it may be written

$$I(\beta) = \sum_0^{\infty} I_n \cos n\beta \tag{11.2}$$

The I_n are the complex current mode amplitudes. The element factor $G(\phi - \beta, \theta)$ may also be expanded as:

$$G(\phi - \beta, \theta) = f(\theta) \sum_0^\infty F_m(\theta) \cos[m(\phi - \beta)] \tag{11.3}$$

Substitution of Eqs. (11.2) and (11.3) into Eq. (11.1) and integrating yields

$$E(\phi, \theta) = Mf(\theta) \sum \frac{1}{\varepsilon_n} I_n F_n(\theta) \cos n\phi \tag{11.4}$$

where $\varepsilon_n = 1$ if $n = 0$, $\varepsilon_n = 2$ if $n \neq 0$. The I_n are determined from the desired azimuth pattern expanded in a Fourier series. Call the desired pattern $T_n(\phi)$ to indicate an Nth-order Chebyshev pattern, though any pattern may be used if it can be put into the form

$$T_N(\phi) = \sum_0^\infty C_n^N \cos n\phi \tag{11.5}$$

The computation of coefficients C_n^N was developed by DuHamel (1951), but his algorithm has poor accuracy for large N. Better formulas are given by Hansen (1981).

Equating (11.4) and (11.5) at $\theta = \theta_0$ yields

$$I_n^{\theta_0} = \frac{\varepsilon_n C_n^N}{Nf(\theta) F_n(\theta)} \tag{11.6}$$

and

$$I^{\theta_0}(\beta) = \sum_0^\infty I_n^{\theta_0} \cos n\beta \tag{11.7}$$

The superscript (θ_0) has been introduced to emphasize that I_n and $I(\beta)$ give the optimum azimuth pattern $T_n(\phi)$ only in the cone $\theta = \theta_0$. At a general elevation angle θ, the pattern is

$$E^{\theta_0}(\phi, \theta) = Mf(\theta) \sum_0^\infty I_n^{\theta_0} F_n(\theta) \cos n\phi \tag{11.8}$$

The choice of $N_0 = k\rho \sin \theta_0$ avoids superdirectivity, and allows efficient excitation of azimuthal modes. Thus $N^{\theta_0} = N_0 \cos \theta_0$, where $N_0 \simeq k\rho$ is N at $\theta_0 = 0$. This selection of N^{θ_0} gives approximately the same amplitude distribution for all θ_0, and a cophasal beam. For a given sidelobe level, the beamwidth is approximately proportional to $1/N$. Thus

$$\text{Azimuth beamwidth} \simeq \frac{\text{Beamwidth at broadside}}{\cos \theta_0}$$

This broadening of the beamwidth is only apparent, however. It is due to the fact that the azimuth pattern is measured on a cone $\theta = \theta_0$, and the ratio of the perimeter of the cone base at θ_0 to that at $\theta = 0$ is just $(\cos \theta_0)^{-1}$. The actual spatial extent of the beam

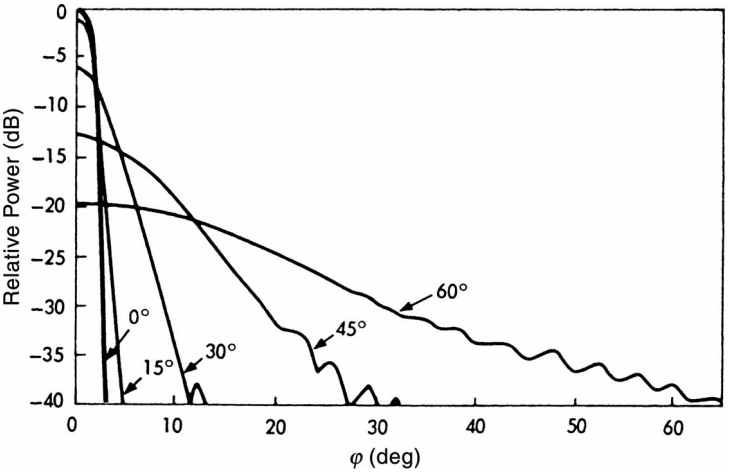


FIGURE 11.1 Ring-array pattern for $\theta_0 = 0^\circ$.

in the plane perpendicular to the $\phi = 0$ plane remains approximately constant to near zenith.

Figures 11.1 and 11.2 show the behavior of the azimuth main beam for $\theta_0 = 0$ and $\theta_0 = 45^\circ$; there are 128 axial slots in a cylinder of diameter 52.6λ (Munger and Gladman, 1970). The distribution is for a -50-dB Chebyshev pattern, so only the main beam shape appears in the plot (all sidelobes are at -50 dB for $\theta = \theta_0$). The extensive beam broadening [beyond the $(\cos \theta_0)^{-1}$ factor] and gain loss for $\theta \neq \theta_0$ are due to the deviation of the phase from the optimum (which is essentially

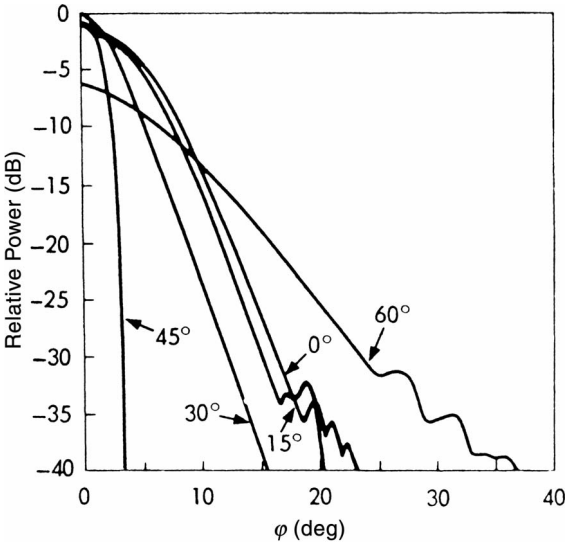


FIGURE 11.2 Ring-array pattern for $\theta_0 = 45^\circ$.

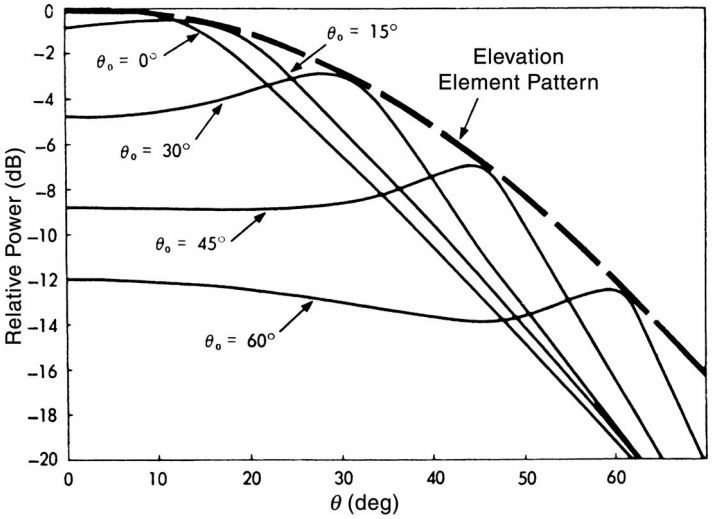


FIGURE 11.3 Ring-array elevation patterns for azimuth pattern optimized at various θ_0.

$-jk \cos \theta_0 \cos \beta p$). Figure 11.3 shows elevation patterns through the beam $\phi = 0$ direction with the azimuth pattern optimized at various θ_0. They are compared with the element elevation pattern, since this represents the maximum possible for each curve. For a linear array the curves would all coincide with the elevation element pattern. For the ring array they coincide with the elevation element pattern only at θ_0, where all elements add in phase to form the beam.

Bandwidth of these arrays of isotropic elements is limited by changes in the Bessel coefficients; a change of $\pi/8$ gives a bandwidth of $D/32\lambda$ (Davies, 1983), where D is the ring diameter. However, use of directional elements greatly increases bandwidth (Rahim and Davies, 1982).

11.2.2 Discrete Ring Array

The continuous distribution can be replaced by M elements located at

$$\beta_p = \frac{2\pi(p+f)}{M} \qquad p = 0, 1, 2, \ldots, M-1 \tag{11.9}$$

where f is a fraction that indicates the position of the beam with respect to the first element. It can be shown (Hansen, 1981) that

$$E(\phi, \theta) = M \sum_{n=0}^{N} \frac{1}{\varepsilon_n} I_n^{\theta_0} \left\{ F_n(\theta) \cos n\phi + \sum_{r=1}^{\infty} F_{rM-n}(\theta) \cos[(rM-n)\phi - 2\pi rf] \right\} \tag{11.10}$$

Equation (11.10) is the desired pattern plus an error term. For spacing $s = 2\pi p/m$ of less than a half-wavelength, only the $r = 0$ term is significant. For spacing <1 wavelength, only $r = 0$ and $r = 1$ contribute, and so forth. The error term's primary contribution to the pattern is in the form of a grating lobe.

A convenient way of evaluating the grating lobe is to represent the terms in Eq. (11.9) by integrals and evaluate by the method of stationary phase (Biswell, 1968). An early approximation was given by Walsh (1951). The pattern for an arc array of $2P + 1$ active elements is

$$E(\phi,\ \theta) = \sum_{p=-P}^{P} I_p G(\phi - \beta, \theta) \tag{11.11}$$

This may be written as a series of integrals according to the Poisson sum formula:

$$E(\phi,\ \theta) = \sum_{r=-\infty}^{\infty} \int_{-P}^{P} I(\beta_p) G(\phi - \beta, \theta) \exp(j2\pi rp) dp \tag{11.12}$$

By change of variable $\beta = 2\pi p/M$,

$$E(\phi,\ \theta) = \frac{M}{2\pi} \sum_{r} \int I(\beta) G(\phi - \beta, \theta) \exp(jMr\beta) d\beta \tag{11.13}$$

$$E(\phi,\ \theta) = \frac{M}{2\pi} \sum_{r} \varepsilon_r \int I(\beta) G(\phi - \beta, \theta) \cos(Mr\beta) d\beta \tag{11.14}$$

where $2\beta(p)$ is the active segment of the ring, M is the number of elements on the full ring, and $I(\beta)$ is a continuous representation of the distribution $I(\beta_p)$.

Now, substitute Eqs. (11.10) and (11.6) into Eq. (11.12), and let $\beta(p) = \pi$, it may be shown that the rth term of Eq. (11.12) is identical to the rth term of Eq. (11.10). The $r = 0$ term is just the pattern due to the continuous distribution $I(\beta)$, and the higher terms are the error introduced by letting $\beta = 2\pi(p + f)/M$.

The method of stationary phase can be applied to evaluate approximately the integrals in Eq. (11.12) — in particular to find the contribution to the grating lobe. From Eqs. (11.32) and (11.36), the rth integral of Eq. (11.12) is

$$I_r(\phi,\ \theta) = \int_{-\beta(p)}^{\beta(p)} |I(\beta)| \cdot |G(\phi - \beta, \theta)| \exp(jk\rho\beta u) d\beta \tag{11.15}$$

where

$$u(\beta) = \cos\theta \cos(\phi - \beta) - \cos\theta_0 \cos\beta + \frac{r\beta}{s} \tag{11.16}$$

and $s = k\rho/M$ is the interelement spacing.

The integral is to be evaluated under the assumption that the largest contribution is from the neighborhood of the point(s) β_0, where the phase term $u(\beta)$ is constant. This

assumption is best for $k\rho$ large. The condition for β_0 is now $u'(\beta) = 0$, or

$$\cos\theta \sin(\phi - \beta) + \cos\theta_0 \sin\beta + \frac{r}{s} = 0 \tag{11.17}$$

In the region around β_0 since $u'(\beta_0) = 0$,

$$u(\beta) \simeq u(\beta_0) + \frac{1}{2}(\beta - \beta_0)^2 u''(\beta_0) \tag{11.18}$$

Furthermore, if β_0 is well inside the range $-\beta(P)$ to $\beta(P)$, the limits may be extended to $-\infty$ and ∞. Finally, assume that $I(\beta)$ and $G(\phi - \beta, \theta)$ are reasonably constant over the region near β_0 so that they may be taken out of the integral. The integral is now in a standard form and becomes

$$I_r(\phi, \theta) = |I(\beta)| \cdot |G(\phi - \beta, \theta)| \sqrt{\frac{2\pi}{k\rho\,|u''(\beta_0)|}} \tag{11.19}$$

with

$$u''(\beta_0) = -\cos\theta \cos(\phi - \beta) + \cos\theta_0 \cos\beta \tag{11.20}$$

For simplicity, consider $\theta = \theta_0$. Now $u''(\beta_0) = 0$ for $\phi_0 = 2\beta_0$, which gives the maximum of Eq. (11.16) corresponding to a grating lobe. The position of the grating lobe is now given by Eq. (11.14), which becomes

$$\sin\frac{\phi_0}{2} = -\frac{r}{2s}\cos\theta_0 \tag{11.21}$$

$r = 0$ gives the main beam at $\phi_0 = 0$, while positive and negative r give grating lobes at positive and negative ϕ_0 angles under the condition

$$\left|\frac{r}{2s}\cos\theta\right| \leq 1 \tag{11.22}$$

However, Eq. (11.16) is not a valid representation of the integral at $u'(\beta_0) = u''(\beta_0) = 0$, for Eq. (11.15) must be replaced by

$$u(\beta) = u(\beta_0) + \frac{1}{6}(\beta - \beta_0)^3 u'''(\beta_0) \tag{11.23}$$

Now evaluating the integral thus obtained at $\phi_0 = 2\beta_0$,

$$\begin{aligned} I_n(\phi, \theta) &= |I(\beta_0)| \cdot G(\phi_0 - \beta_0, \theta_0) \cdot \left(\frac{6}{k\rho|u'''(\beta_0)|}\right)^{1/3} \frac{\Gamma(1/3)\exp(jk\rho\beta_0 u)}{\sqrt{3}} \\ |I_r(\phi_0, \theta_0)| &= \left| I\left(\frac{\phi_0}{2}\right) G\left(\frac{\phi_0}{2}, \theta_0\right) \frac{\Gamma(1/3)}{\sqrt{3}} \left(\frac{12}{k}\right)^{1/3} \left(\frac{s\cos\theta_0}{r\rho}\right)^{1/3} \right| \end{aligned} \tag{11.24}$$

Equation (11.20) is subject to the restrictions that $I(\beta)$ and $G(\phi - \beta, \theta)$ are slowly varying, $k\rho$ is large, and β_0 is not near 0, but is inside the range $-\beta(P)$ to $+\beta(P)$. To compute the grating lobe height compared with the main beam height, compare the grating lobe computed from Eq. (11.20) with the main beam computed from

$$I_0 = \int_{-\beta(p)}^{\beta(p)} |I(\beta)| \cdot |G(\beta, \theta)| d\beta \tag{11.25}$$

Figure 11.4 shows the grating lobe height relative to the beam as a function of spacing ($s \cos \theta_0$) for various $\cos^n$ amplitude tapers, with $\rho = 26\lambda$ and $G(\phi - \beta, \theta_0) = \cos(\phi - \beta)$ (Munger, 1969). The equation $\beta(P) = \pi/2$ was used to ensure that $0 < \beta_0 < \beta(P)$. For comparison, actual patterns were computed from Eq. (11.10) and the grating lobe heights are shown. The maximum difference between the actual grating lobe and that computed from Eq. (11.20) is 2 dB. Since the approximation from Eq. (11.20) represents the worst case, it appears to be a useful guide for controlling the grating lobe as a function of spacing, current distribution, element pattern, radius, and arc length. The position of the grating lobe (for $r = -1$ and $\theta_0 = 0$)

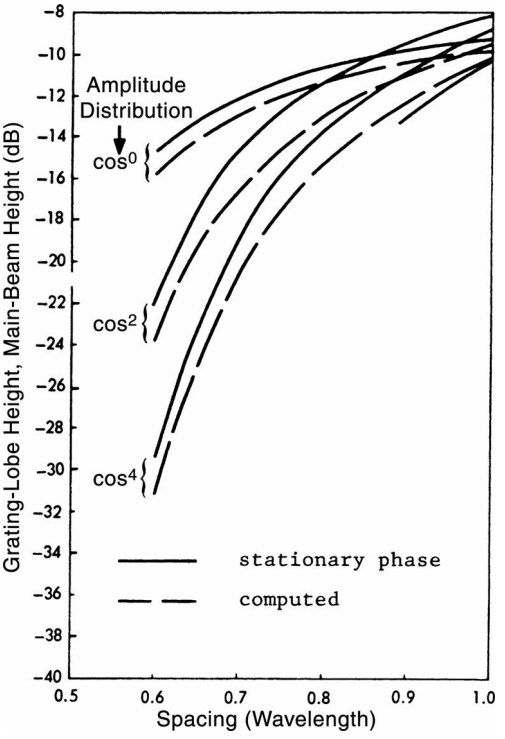


FIGURE 11.4 Grating lobe height versus spacing.

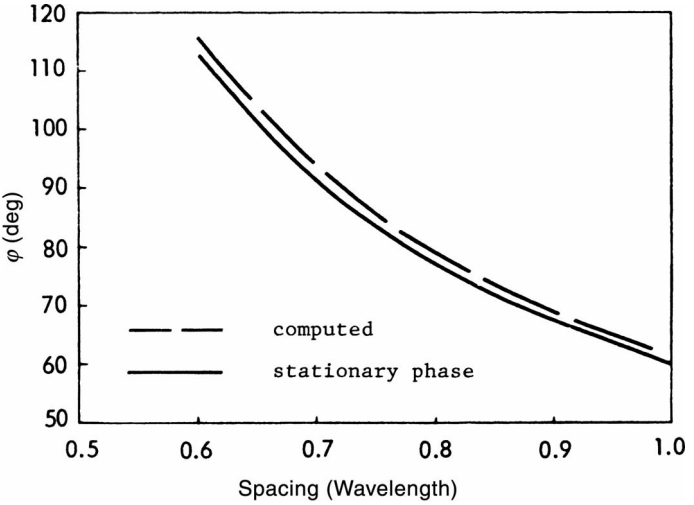


FIGURE 11.5 Grating lobe position versus spacing.

predicted by Eq. (11.18) is shown in Figure 11.5 in comparison with the actual position, with good agreement.

11.2.3 Beam Cophasal Excitation

The direct way of achieving a narrow beam from a ring array is to phase all elements so that the element contributions are in phase at the far-field angle ϕ_0. This is called beam cophasal excitation, and requires phases around a ring of radius r_0 of

$$\exp[-jkr_0 \cos(\phi - \phi_0)] \tag{11.26}$$

The discrete array samples this continuous phase distribution. If the element spacing is half-wave or less, and the number of elements is large, the pattern is approximately that of a continuous ring source, Eq. (11.1).

The application of azimuthal modal theory to ring arrays was pioneered by Tillman et al. (1961), and extended by Davies (1965). In concept, the phase and amplitude distribution around a ring array can be expressed in a series of azimuthal modes, where the DFT of each mode gives the mode pattern. One can also obtain the self- and mutual impedances for each mode (Tillman, 1968). These modes allow synthesis of a desired pattern, and allow the impedances, directivity, and so on, to be determined. They are analogous to the symmetrical components used in analysis of multiphase power systems.

Let the excitation of the mth mode on the ring array of diameter D be

$$g_m(\beta) = \sum_{n=1}^{N} A_n \exp(jkdm \cos \beta) \tag{11.27}$$

where $d = 2\pi D/N$ and β is the angle around the ring array. The pattern of this mode is given by the DFT:

$$f_m(\phi) = J_0\left(2kr_0 \sin\frac{\phi}{2}\right) J_0[kr_0(1 - \cos\theta)] \tag{11.28}$$

where the first Bessel function gives the azimuth pattern, and the second provides the elevation pattern. The J_0 pattern has a poor sidelobe ratio of 7.9 dB, and a slowly decreasing sidelobe envelope, as shown in Figure 11.6. The abscissa is a universal variable, in this case $u = 2kr_0 \sin(\phi/2)$. Because of the higher sidelobes one might expect the directivity to be higher than that of a linear array across the ring array diameter, and this proves to be true.

It is obvious that sidelobe reduction requires turning off the rear elements, and deweighting the side elements. An amplitude taper is thus introduced such that the elements are excited over an angle $<180^\circ$, typically 120°. To scan the beam, this amplitude taper (and the corresponding cophasal phase) must be rotated. Wullenweber arrays for HF–DF have been used for many decades; typically element cables are brought to a central control point where a rotating goniometer provides the desired phase and amplitude (Gething, 1991). Large Wullenwebers have been constructed: the University of Illinois antenna covers 4–16 MHz and uses 120 monopole elements (40 at a time) on a ring of diameter 955 ft (Hayden, 1961). Since ring arrays tend to be used at long wavelengths, with corresponding relatively narrow bandwidths, the DFT processing can sometimes be performed digitally for receive arrays, using A/D

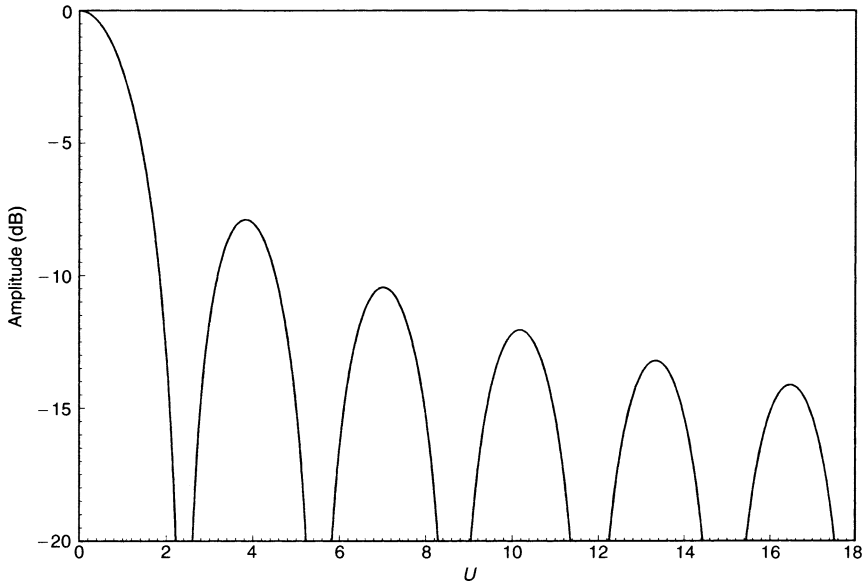


FIGURE 11.6 The J_0 Bessel function versus argument.

converters. Electronic commutators have been described by Fenby and Davies (1968), Longstaff and Davies (1968), Bogner (1974), and Irzinski (1981). Ring arrays for TACAN also use a rotating distribution (Christopher, 1974; Shestag, 1974). Electronic despin of a spacecraft antenna is described by Gregorwich (1974). The azimuthal amplitude taper will increase the elevation plane beamwidth somewhat. Although the goniometer has been replaced by electronic phase and amplitude control, modern ring arrays are more likely to use modal beamformers. Use of directional elements instead of isotropes (e.g., vertical HF monopoles) can improve performance if the excited azimuthal sector is not too large. For a 90° sector some improvement may be expected, but less for a 120° sector. The prinicipal improvement from directional elements is a potential reduction in mutual coupling effects, but this too is nicely used advantageously with modal beamforming.

Recall from Chapter 10 that the Butler matrix generates the DFT of the input, so it is logical that the Butler matrix can be used with a ring array to produce a narrow beam, and to rotate it. Here an $N \times N$ Butler matrix is connected with equal-length lines to an N-element ring array; Figure 11.7 shows this for $N = 16$. Each Butler port will excite one azimuthal mode, so the proper selection of amplitudes and phases on the ports will

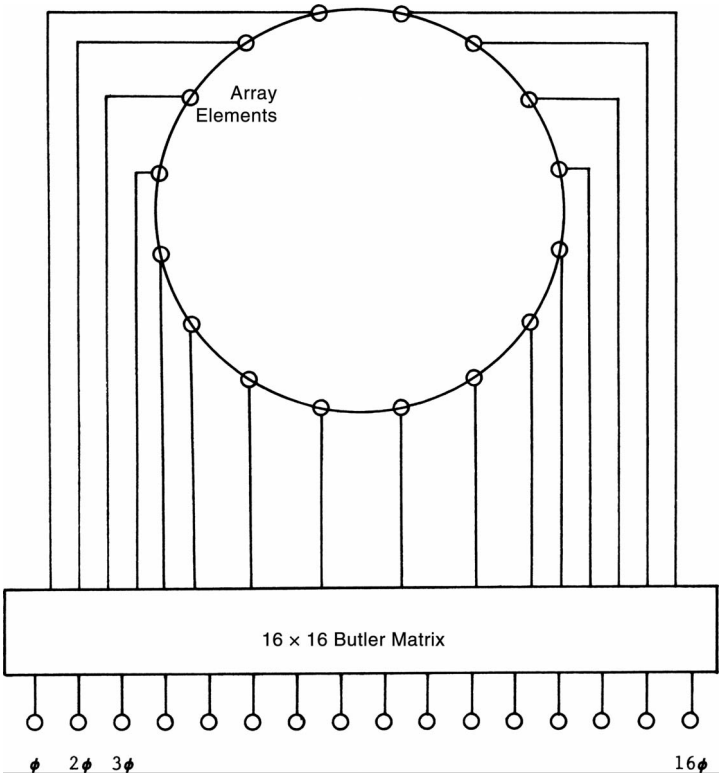


FIGURE 11.7 Azimuthal beam scan.

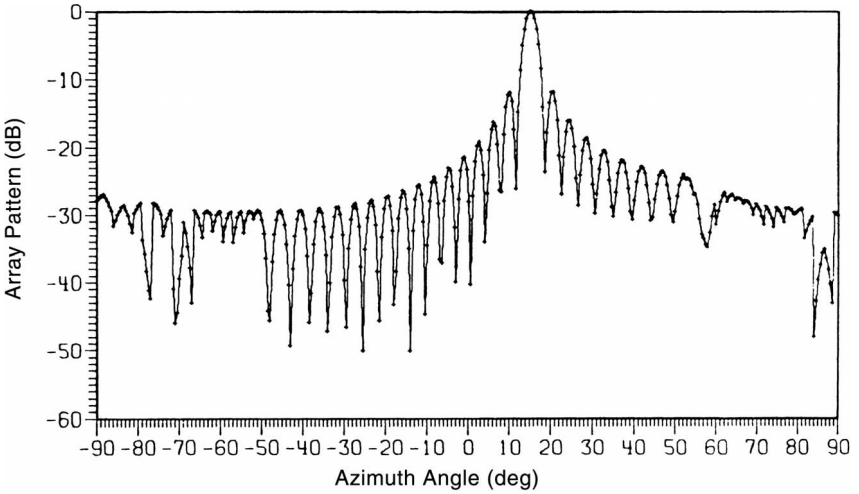


FIGURE 11.8 Pattern of arc array.

synthesize the desired pattern using the modes. A further advantage is that adding a progressive phase of $n\phi_0$ to the ports, where $n = 0, 1, \ldots, N-1$, will rotate the beam to azimuth angle ϕ_0 (Davies, 1965, 1983; Sheleg, 1968, 1973; Skahill and White, 1975). A further advantage of the Butler matrix as a ring array beamformer is that while the modal mutual impedances are different, the effects of each mode appear only at the corresponding Butler port, and hence may be compensated there.

Arc arrays, where the radiating elements are fixed, also suffer from a higher sidelobe envelope at wide angles. Figure 11.8 shows the complete pattern of an arc array

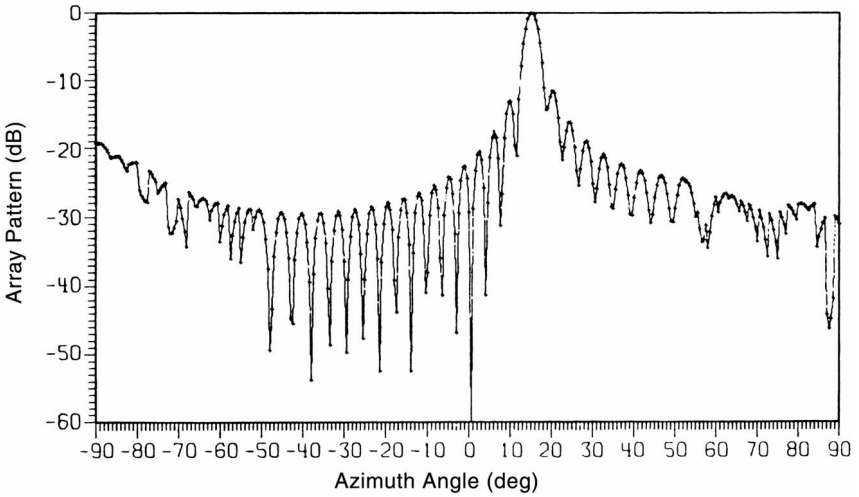


FIGURE 11.9 Pattern of arc array with equal projected spacing.

covering a subtended angle of 63°, with a radius of 12.7λ, and scanned to 15°. Note that the wide-angle sidelobe level is roughly −30 dB. When the element spacing is adjusted to make the projected virtual array equally spaced, the sidelobes at −90° are now only −20 dB, (as shown in Fig. 11.9) (Hansen, 1981). These are due to phase asymmetries that appear on beam scan; they can be reduced by combining a small amount of difference pattern with the sum pattern (Antonucci and Franchi, 1985).

11.3 ARRAYS ON CYLINDERS

In this section, cylinders with arrays covering the circumference, and extending a distance L axially are treated. The modal analysis of Section 11.3.3.1 is applicable, and the use of Butler matrix beamformers is also. However, there are several crucial differences. The axial extent of the array allows elevation pattern control, and higher directivity. Because of the metallic cylinder the element patterns are, of necessity, more direct. A major difficulty occurs in the calculation of mutual impedances. In a ring array simple methods, such as the Carter approach of Chapter 7, are used. However, with curved metal surfaces, the mutual impedance expressions produce integrals that are difficult to evaluate, as will be seen below. First the influence of the curved surface on element pattern is examined.

11.3.1 Slot Patterns

The boundary value problem of radiation from a slot (with a sinusoidal distribution) on a metallic circular cylinder was solved exactly several decades ago (Carter, 1943; Harrington and Lepage, 1952; Knudsen, 1959). The solution is a harmonic series of functions appropriate for the cylinder: Bessel and Hankel functions. A standard cylindrical coordinate system is used, where z, θ, ϕ are the axial, polar, and circumferential coordinates. Finite width of the slot has a small effect upon the patterns, but the appropriate factor can be included if desired (Compton and Collin, 1969). With slot voltage V and cylinder radius a, the field for an axial half-wave slot (Wait, 1959) are $E_\theta = 0$ and

$$E_\theta = \frac{V \cos\left(\frac{1}{2}\pi \cos\theta\right)}{2\pi^2 ka \sin^2\theta} \sum_{m=0} \frac{\varepsilon_m j^m \cos m\phi}{H'_m(ka \sin\theta)} \tag{11.29}$$

This pattern is that of a half-wave dipole modified by the summation. For a circumferential half-wave slot, again thin, the fields are

$$E_\theta = \frac{2kaV}{\pi^2 \sin\theta \cos\theta} \sum_{m=0} \frac{\varepsilon_m j^{m+1} \cos m\phi \cos(m\pi/2ka)}{(k^2a^2 - m^2)H_m(ka \sin\theta)} \tag{11.30}$$

$$E_\phi = \frac{2V \cos\theta}{\pi^2 \sin^2\theta} \sum_{m=0}^{\infty} \frac{m j^{m+1} \sin m\phi \cos(m\pi/2ka)}{(k^2a^2 - m^2)H'_m(ka \sin\theta)} \tag{11.31}$$

Convergence of these series is controlled by ka. Between ka and $2ka$ terms are usually needed. Large cylinders thus require many terms, but with current computer capability many tens of terms can be included. However, for such large cylinders the Watson transformation has been used to convert the harmonic series above, which results from an evaluation of an integral at real poles, to a residue series of complex-order Hankel functions, evaluated at the complex poles. For large cylinders the residue series, which is analogous to the Regge pole series in quantum mechanics, converges with only a few terms (Compton and Collin, 1969; Lee and Eichmann, 1980). Returning to the harmonic series, Wait (1959) has computed the sum for the azimuth patterns of primary interest: E_ϕ for an axial slot, and E_θ for a circumferential slot. His results are given in Figures 11.10 and 11.11. It is interesting to note the axial slot electric field produces diffraction interference behind the cylinder, whereas the circumferential slot pattern exhibits a smooth decay. These patterns are used with the array factor and *scan reflection coefficient* to produce the complete pattern.

11.3.2 Array Pattern

The cylindrical array can be considered to consist of a stack of identical ring arrays. Denote the complex excitation of the pth element in the qth ring by $I_{pq} = I(\beta_p, z_q)$, where β_p is the angular location of the pth and z_q is the z-axis location of the qth ring. The coordinate system is shown in Figure 11.12. The beam is assumed to be pointed in the $\phi = 0$ reference point of the element location. The beam is stepped around the cylinder by redefining the $\beta = 0$ reference to the desired position.

All elements are assumed identical, symmetrical, equally spaced, and pointed along the radius vector. Thus, the azimuth element pattern can be expressed as a function of $|\phi - \beta|$. In general, the azimuth pattern depends on the elevation angle θ. The complex element pattern is denoted by $G(\phi - \beta, \theta)$, with the phase referenced to the center of the ring in which it lies. Thus, if it is assumed that the phase center is at the element,

$$G(\phi - \beta, \theta) = |G(\phi - \beta, \theta)| \exp[jk\rho \cos\theta \cos(\phi - \beta)] \tag{11.32}$$

The far field is

$$E(\phi, \theta) = \sum_p \sum_q I_{pq} G(\phi - \beta_p, \theta) \exp(jqu) \tag{11.33}$$

where $u = kd \sin\theta$ and $d =$ spacing between elements in the axial direction.

A beam can be formed in the direction $\phi = 0$, $\theta = \theta_0$ by exciting all elements to add in phase in that direction (beam cophasal excitation). Thus, in view of Eqs. (11.32) and (11.33):

$$I_{pq} = |I_{pq}| \exp[jk\rho \cos\theta_0 \cos\alpha_p - jqu_0] \tag{11.34}$$

where $u_0 = kd \sin\theta_0$.

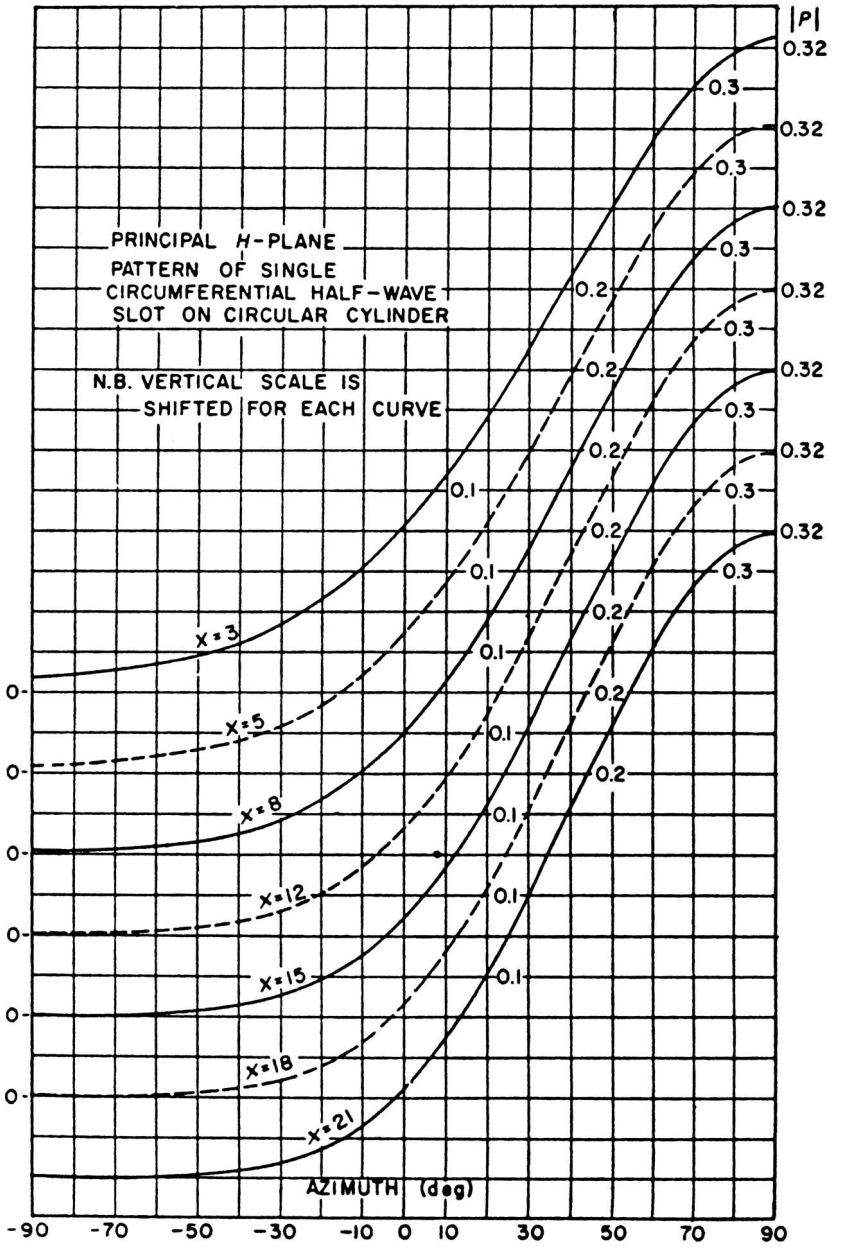


FIGURE 11.10 Circumferential slot patterns; $x = ka$. (Courtesy Wait, J. R., *Electromagnetic Radiation from Cylindrical Structures*, Pergamon Press, 1959, p. 45.)

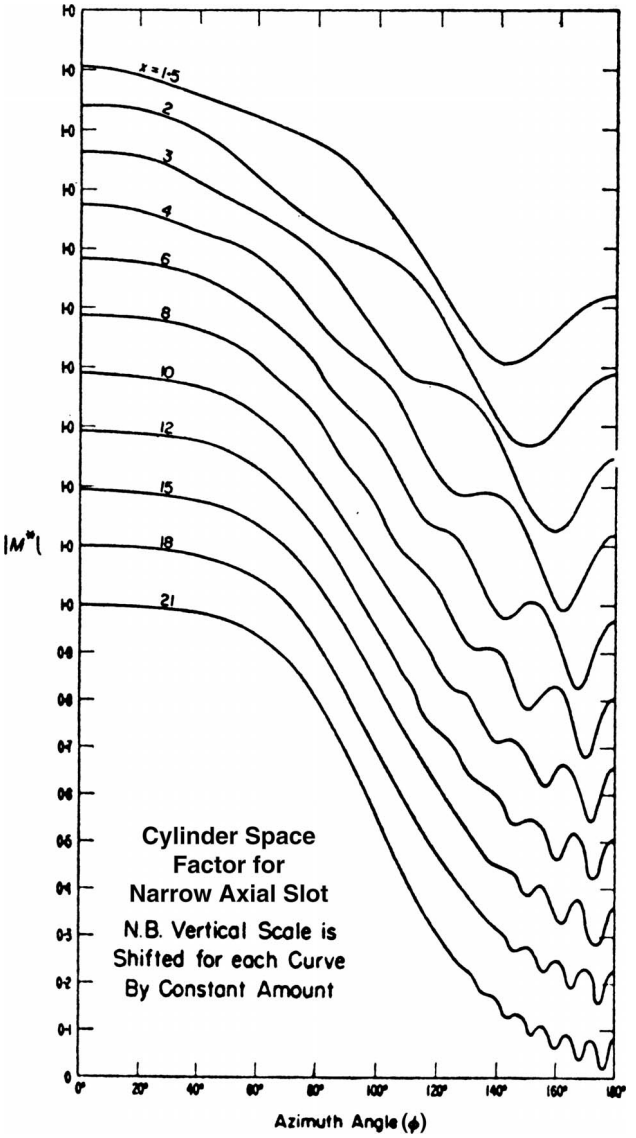


FIGURE 11.11 Axial slot patterns; $x = ka$. (Courtesy Wait, J. R., *Electromagnetic Radiation from Cylindrical Structures*, Pergamon Press, 1959, p. 30.)

In Eq. (11.34), the phase terms are separated in β_p and z_q, where $z_q = qd$. This allows the assumption of a current distribution of the form

$$I(\beta_p, z_q) = I^a(\beta_p)I^e(z_q) = I_p^a I_q^e \tag{11.35}$$

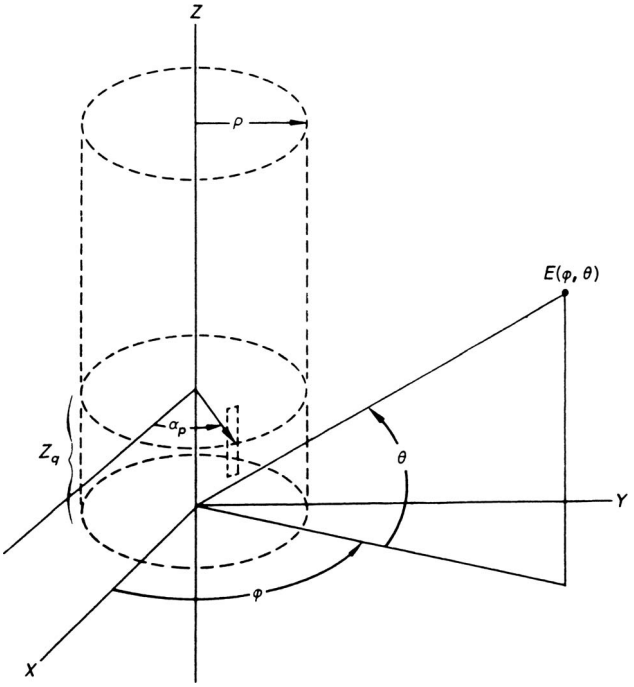


FIGURE 11.12 Coordinate system.

with

$$I_p^a = I_p^a \exp(-jk\rho \cos\theta_0 \cos\beta_p) \tag{11.36}$$

and

$$I_q^e = |I_q^e| \exp(-jqu_0) \tag{11.37}$$

The superscripts a and e indicate azimuth and elevation distributions, respectively. Note that the azimuth distribution depends on the beam-pointing angle in both azimuth and elevation, whereas I_q^e depends only on θ_0. Writing the current distribution in the form of Eq. (11.35) allows the pattern (11.33) to be written in the form

$$E(\phi, \theta) = E^a(\phi, \theta)E^e(\theta) \tag{11.38}$$

where

$$E^a(\phi, \theta) = \sum_p I_p^a G(\phi - \beta_p, \theta) \tag{11.39}$$

and

$$E^e(\theta) = \sum_q I_q^e \exp(jqu) \tag{11.40}$$

where $E^a(\phi, \theta)$ is just the pattern of a single ring excited by I_p^a; $E^e(\theta)$ is the space factor of a single vertical column of elements excited by I_q^e.

Thus, the analysis can be simplified by assuming the separable aperture distribution and considering the cylindrical-array pattern to be the product of a ring array pattern and a linear array pattern. A pencil beam can be formed by selecting I_p^a to form a beam at $\phi = 0$ in the azimuth cone $\theta = \theta_0$, and by selecting I_q^e to form a beam at $\phi = 0$ in the azimuth cone $\theta = \theta_0$, and by selecting I_q^e to form a beam at θ_0 in the elevation plane $\phi = 0$. Since the pencil beam is the product of two fan beams, the principal side-lobes will lie on the cone and plane in which the fan beams were shaped.

The patterns do not include the effects of mutual coupling. *Scan element patterns*, which do, and the corresponding element efficiencies have been calculated by Schwartzman and Kahn (1964) and by Kahn (1971).

11.3.2.1 Grating Lobes. A cylinder can be covered with a regular lattice, but the projection in any direction produces unequal spacing in azimuth. In elevation, however, the projected element spacings are uniform, so that conventional grating lobe theory can be used for elevation. There are no sharply defined, high-amplitude grating lobes due to azimuth spacing, but the side-lobes of a low sidelobe design may be raised if the element spacing is too large. Since the cylinder must have elements "all around" to allow 360° azimuth scanning, there are two variables involved in grating lobe type calculations; element spacing and cylinder radius. In addition, each element pattern usually has its axis in a radial direction so that the element pattern cannot be factored out. Amplitude tapering is typically used to produce moderately low sidelobes, so that any meaningful pattern calculation must include element spacing, cylinder radius, element pattern, and amplitude taper. For this reason, there are no simple grating lobe-type results for the cylindrical array. Cylindrical arrays that are phased to produce narrow beams tend to be more susceptible to grating lobe problems than do comparable planar arrays because of two factors: (1) On the sides of the active portion of the array the element patterns do not point in the direction of the beam; and (2) in this same region it is necessary to introduce a large interelement phase shift into the excitation to compensate for the curvature of the cylinder. The latter factor is equivalent to scanning the side portion of the array to some angle off the normal to the cylinder at that point; hence, the interelement spacing must be kept correspondingly small to prevent the formation of grating lobes.

Elevation scan to θ_0 is achieved by adding the phase $-qkd \sin \theta_0$ to the qth ring. The performance of the array pattern in the plane $\phi = 0$ is now the same as for a linear array with ring array "elements" with patterns, such as those in Figure 11.3. In particular, a grating lobe appears at θ when

$$\frac{d}{\lambda}(\sin \theta - \sin \theta_0) = \pm m, \quad m = 1, 2, \ldots \tag{11.41}$$

The lobe is at $\phi = 0$ in azimuth because $E^a(\phi, \theta)$ has its maximum at $\phi = 0$ (or nearly so) for all θ. The grating lobe, as it arises from $E^e(\theta)$ has unit magnitude (equal to main beam), but is reduced by $E^a(\phi = 0, \theta)$.

Staggering alternate columns of elements on the cylinder is an effective means of extending the elevation scanning angle for a given ring-to-ring spacing d and maintaining a small grating lobe. Consider the staggered array as a superposition of two regular arrays, each with the normal number of rings Q but only half the number of elements in each ring, $M/2$. The sub-arrays are identical except one is rotated by half a spacing in azimuth and is raised by half a spacing in the vertical direction, with the phase compensating for the dislocation. The ring-array patterns for the sub-arrays can be written as follows:

Array I. The beam in the direction of the first element; that is, $f = 0$:

$$\begin{aligned} E_{\mathrm{I}} = f(\theta) \sum_n I_n(\theta_0) F_n(\theta) \cos n\phi + f(\theta) \sum_n I_n(\theta_0) F_{M/2-n}(\theta) \\ \times \cos[(M/2 - n)\phi] + f(\theta) \sum_n I_n(\theta_0) F_{M-n}(\theta) \cos[(M - n)\phi] \end{aligned} \tag{11.42}$$

The $r = 2$ term, which ordinarily would not contribute (for $s < \lambda$) is included because the spacing is now double the normal spacing.

Array II. Beam is in a direction halfway between two elements; that is, $f = 0.5$:

$$\begin{aligned} E_{\mathrm{II}} = f(\theta) \sum_n I_n(\theta_0) F_n(\theta) \cos(n\phi) \\ - f(\theta) \sum_n I_n(\theta_0) F_{M/2-n}(\theta) \cos[(M/2 - n)\phi] \\ + f(\theta) \sum_n I_n(\theta_0) F_{M-n}(\theta) \cos[(M - n)\phi] \end{aligned} \tag{11.43}$$

If the array were not staggered, each ring would have M elements and a pattern

$$\begin{aligned} E^a(\phi, \theta) = f(\theta) \sum_n I_n(\theta_0) F_n(\theta) \cos(n\phi) \\ + f(\theta) \sum_n I_n(\theta_0) F_{M-n}(\theta) \cos[(M - n)\phi] \end{aligned} \tag{11.44}$$

The grating lobe term that arises because of the doubling of the spacing is

$$E_{gl}(\phi, \theta) = \sum_n I_n(\theta_0) F_{M/2-n}(\theta) \cos[(M/2 - n)\phi] \tag{11.45}$$

Assume array II is raised with respect to array I by $d/2$ and multiplied by $\exp[\, jk(d/2)(\sin\theta - \sin\theta_0)]$ to account for the phase. The patterns of cylindrical arrays I and II are then

$$E_{\mathrm{I}}(\phi, \theta) = \sum_{q=1}^{Q} |I_q| [E^a(\phi, \theta) + E_{gl}(\phi, \theta)] \exp[\, jqkd(\sin\theta - \sin\theta_0)] \tag{11.46}$$

and

$$E_{\mathrm{II}}(\phi,\ \theta) = \exp\left(\frac{jkd}{2}(\sin\theta - \sin\theta_0)\right) \times \sum_{q=1}^{Q} |I_q|[E^a(\phi,\ \theta) - E_{gl}(\phi,\ \theta)] \exp[jqkd(\sin\theta - \sin\theta_0)] \tag{11.47}$$

The sum of E_{I} and E_{II} is the pattern of the staggered array. It may be put in the form:

$$E(\phi,\ \theta) = E^a(\phi,\ \theta) \sum_{q=1}^{2Q} |I_q| \exp\left(\frac{jqkd}{2}(\sin\theta - \sin\theta_0)\right) + E_{gl}(\phi,\ \theta)\left[1 - \exp\left(\frac{jkd}{2}(\sin\theta - \sin\theta_0)\right)\right] \times \sum_{q-1}^{Q} |I_q| \exp[jqkd(\sin\theta - \sin\theta_0)] \tag{11.48}$$

The first term of Eq. (11.48) is the pattern of an array of $2Q$ rings spaced at half the normal spacing with M elements on each ring; that is, the staggered array with the "holes" filled in. This term should give no grating lobe because of the half-spacing in elevation and the normal spacing in azimuth.

The second term accounts for the grating lobe. The factor

$$\sum_{q=1}^{Q} |I_q| \exp[jqkd(\sin\theta - \sin\theta_0)] \tag{11.49}$$

is the linear array pattern for normal elevation spacing, and gives a grating lobe when

$$\frac{d}{\lambda}(\sin\theta - \sin\theta_0) = \pm 1 \tag{11.50}$$

This gives

$$1 - \exp\left(\frac{jkd}{2}(\sin\theta - \sin\theta_0)\right) = 2 \tag{11.51}$$

Thus, the grating lobe of the cylindrical staggered array is equal to the grating lobe of the linear array, with spacing d, times the grating lobe of a ring array, with spacing $2s$. The elevation and azimuth positions of the product lobe are the positions of the linear and ring array lobes, respectively. The staggered-array lobe appears at the same elevation angle as the lobe of the regular array, but is removed from $\phi = 0$ to $\phi = \phi_{gl}$ as determined from a ring array with every other element removed. The advantage gained is the amount the grating lobe of the ring sub-arrays is down from the main beam.

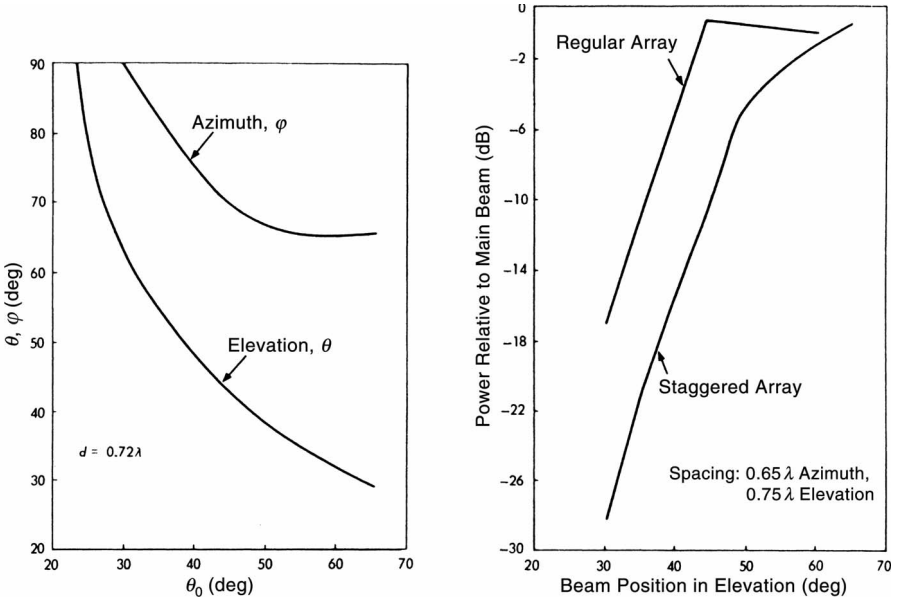


FIGURE 11.13 Grating lobe position and height versus scan angle.

Figure 11.13 shows the grating lobe height as a function of scan angle for regular and staggered configuration, and the position in elevation and azimuth for the staggered array lobe. The parameters are $N_0 = 128$, $\theta_0 = 0^\circ$ and a -50-dB Chebyshev distribution (Hansen, 1981).

In obtaining Eq. (11.48), Eq. (11.9) was the start. However, the start could equally well have been Eq. (11.13), using $\beta = 2\pi(p + f)/M$. Also, it was assumed that $f = 0$ for array I and $f = 0.5$ for array II. Equation (11.48) depends only on the fact that $E_{gl}(\phi, \theta)$ for array I is equal to $-E_{gl}(\phi, \theta)$ for array II, which is true for any orientation of the first element with respect to the main beam. Thus Eq. (11.48) is valid for cylindrical arc arrays in general. Furthermore, $E_{gl}(\phi, \theta)$ may easily be identified from patterns computed directly from Eq. (11.10) with alternate elements excited.

Figure 11.14 shows $E^a(\phi, \theta) + E_{gl}(\phi, \theta)$ for various θ_0. These patterns are computed from Eq. (11.11) and the contribution from $E_{gl}(\phi, \theta)$ is easily identified. The effect of staggering is easily seen. For example, if the rings are spaced at $d = 0.72\lambda$, a beam at $\theta_0 = 30^\circ$ gives a grating lobe at about -60°. A regular array gives the grating lobe height at -11 dB, which is the difference between the beams at $\theta = 30^\circ$ and $\theta = 60^\circ$. For a staggered array, the grating lobe height is the difference between the beam at $\theta = 30^\circ$ and the grating lobe at $\theta = 60^\circ$, or about -28 dB.

11.3.2.2 Principal Sidelobes. For the regular array the principal sidelobes will lie on the plane $\phi = 0$ and the cone $\theta = \theta_0$, because the regular-array pattern, for the separable distribution, can be thought of as the product of two fan beams.

For the staggered array, we can consider Eq. (11.48). The first term is the product of a linear array fan beam and a ring array fan beam, giving principal sidelobes as a

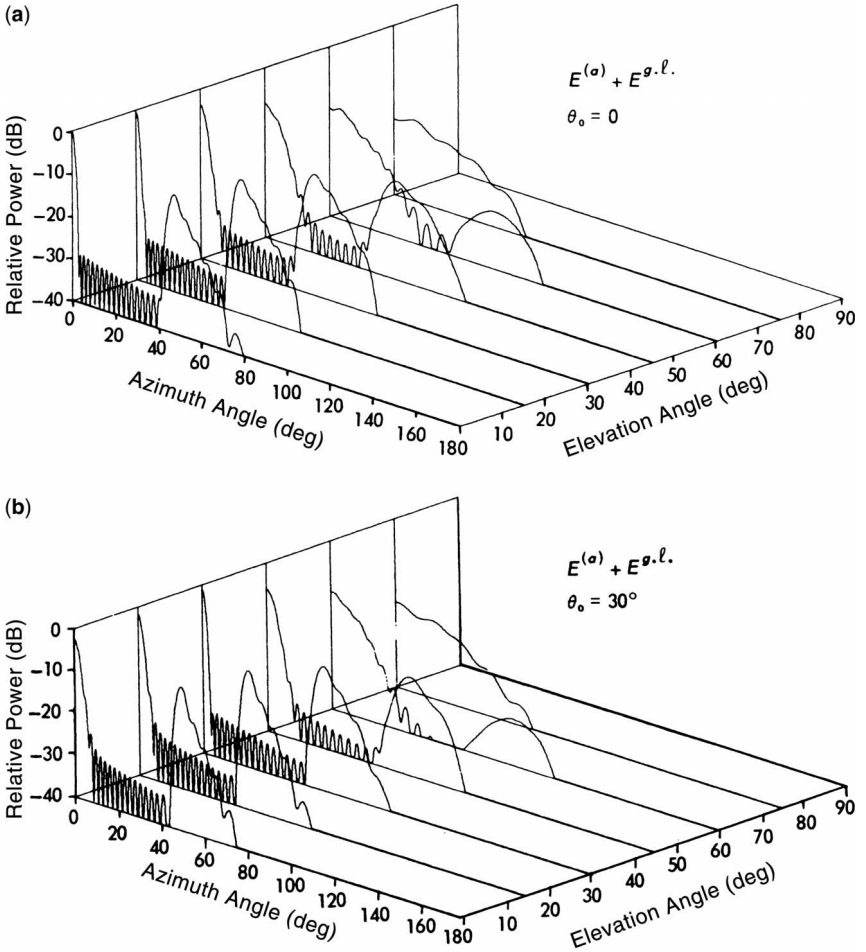


FIGURE 11.14 (a) The 30-dB Chebyshev patterns for $\theta = 0$. (b) The 30-dB Chebyshev patterns for $\theta = 30°$. (c) The 30-dB Chebyshev patterns for $\theta = 60°$.

regular array on the plane $\phi = 0$ and the cone $\theta = \theta_0$. The second term gives another set of axes, however. The first two factors of the second term give the linear array grating lobe (without the main beam); the third factor gives the ring array grating lobe (with double spacing). Thus, another set of principal sidelobes lies on the cone $\theta = \theta_{gl}$ and the warped plane $\phi = \theta_{gl}(\theta)$ where θ_{gl} is given by Eq. (11.41) and $\theta_{gl}(\theta)$ gives a maximum to $E_{gl}(\phi, \theta)$.

For the regular array, then, the full cylindrical array pattern is well represented by two contours through ϕ, θ space: (ϕ, θ_0) and $(0, \theta)$. For the staggered array, the full pattern is well represented by the four contours (ϕ, θ_0), $(0, \theta)$, (ϕ, θ_{gl}), and $(\phi_{gl}(\theta), \theta)$. Figure 11.15 presents the patterns for regular and staggered arrays at various θ_0. The same single ring parameters are assumed as for Figure 11.14; in addition, 32

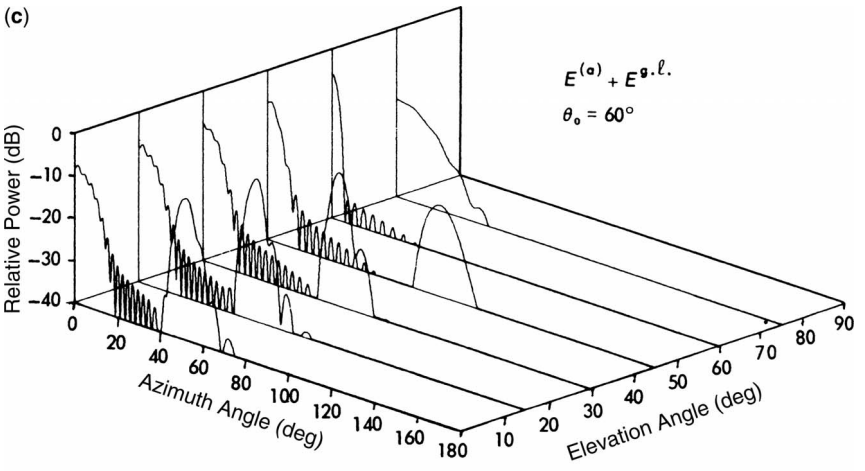


FIGURE 11.14 (*Continued*).

rings spaced at 0.72λ are used with a 30-dB Chebyshev distribution for $I_q(e)$. The contour $[\phi_{gl}(\theta), \theta]$ was determined from the patterns of Figure 11.14 in interpolation between the maximum point on the grating lobe.

The grating lobe can be reduced and elevation scan extended by reducing the azimuth and/or elevation spacing of the staggered array. For example, reducing the azimuth spacing from 0.65λ to 0.5λ (with $d = 0.72\lambda$) increases the scan angle limit from 30 to $\sim 40^\circ$ to maintain a grating lobe of 30 dB, and further reduction to 0.4λ allows scanning to $>75^\circ$ with the grating lobe <40 dB. Reduction of elevation spacing (with $s = 0.65\lambda$) from 0.72 to 0.6λ allows scanning to above 50° for a grating lobe <40 dB. In the array being implemented, however, the elevation spacing is restricted to a minimum of 0.72λ because of the element size.

11.3.2.3 Cylindrical Depolarization. The cylindrical surface, like any curved surface, depolarizes an incident wave. For example, if a linearly polarized wave is incident in the plane of incidence (the plane containing the cylinder axis and the direction of incidence), polarizations that are parallel or normal to this plane behave differently. With electric field parallel to the plane of incidence, the axial component of field on the cylindrical surface is in the same direction, but the circumferential components oppose. Of course, at normal incidence the latter are zero. For normal polarization, the axial components are in the same direction; and the circumferential components are also in the circumferential direction. Thus incident circular polarization produces fields that have the polarization axes oriented right on one side, and left on the other side, with all polarization elliptical except at points of incidence. Hence, a cylindrical array should use elements that radiate both axial and circumferential components. Further, the ratio of these components in general for any given element will change as the angle of incidence changes. Arrays on large (in wavelengths) cylinders with

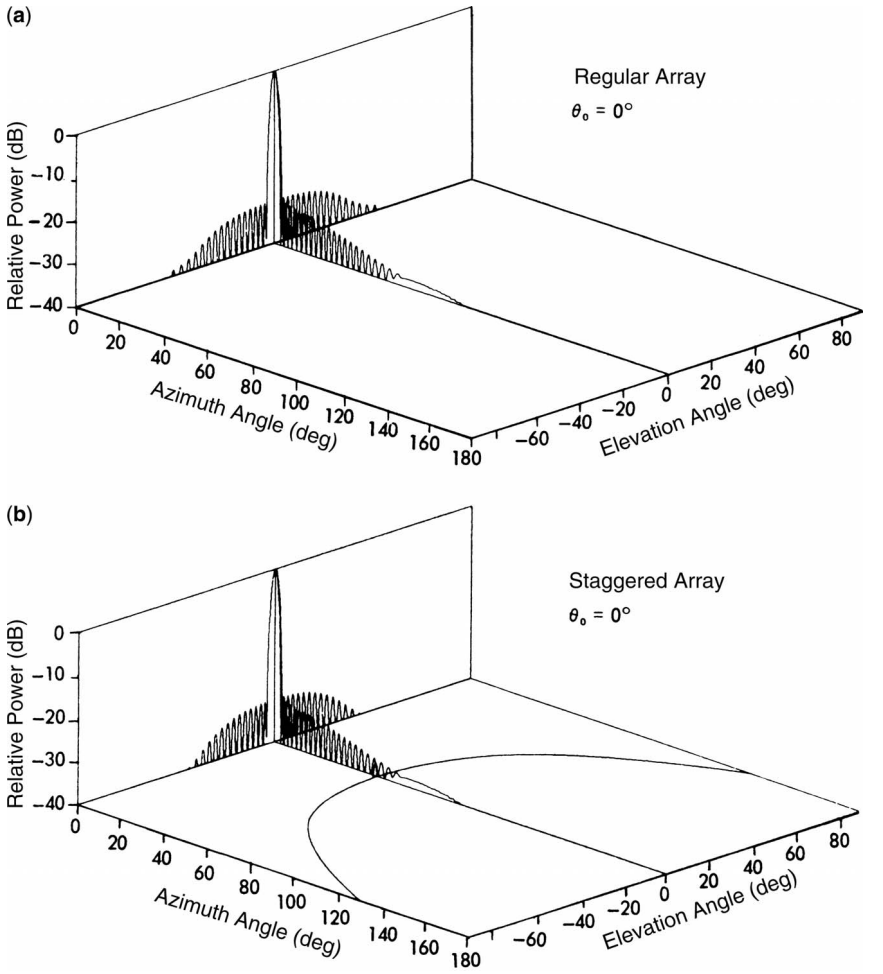


FIGURE 11.15 The 30-dB Chebyshev patterns.

tapered circumferential distributions will experience a lower and perhaps negligible level of depolarization as the amplitude at elements significantly away from a projected aperture plane may be small. Conical arrays do not share this advantage unless only the large-diameter portion of the surface is utilized. The general conical case will be discussed later.

11.3.3 Slot Mutual Admittance

In the design of a conformal slot array on the surface of a conducting cylinder, the calculation of the mutual admittance Y_{12} is a crucial step, which has been studied extensively in recent years. Referring to Figure 11.16, two identical slots, circumferential or

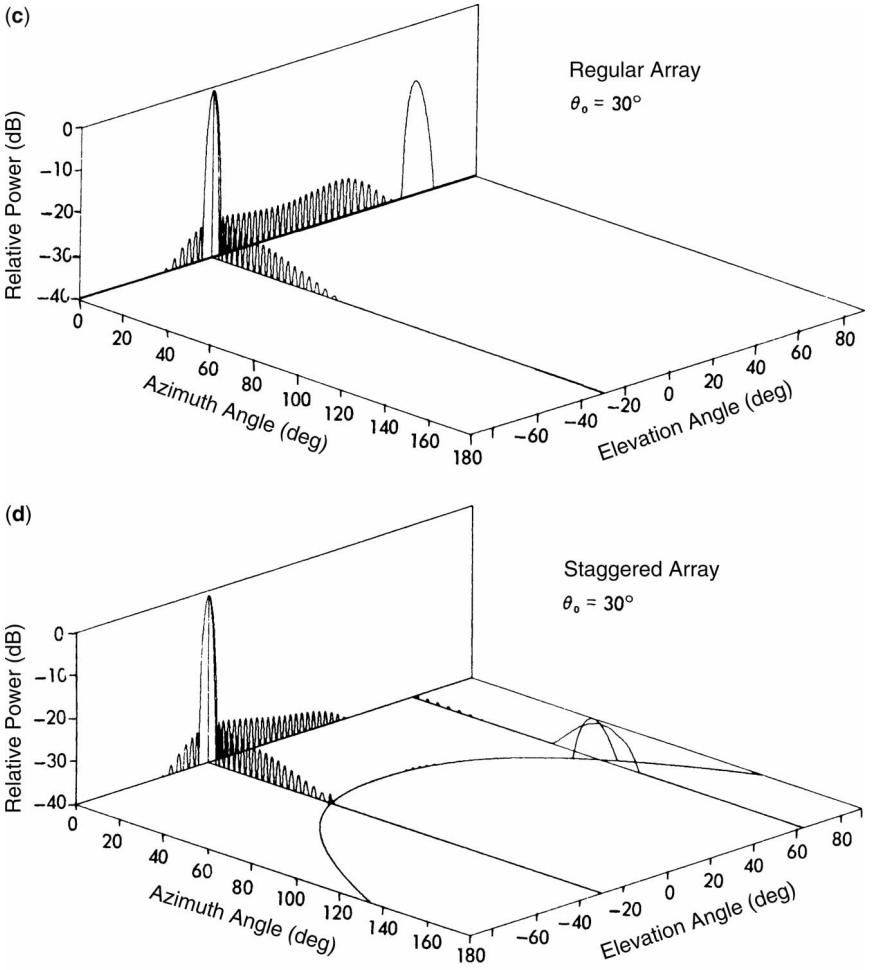


FIGURE 11.15 (*Continued*).

axial, are located on the surface of an infinitely long cylinder. The geometrical parameters are

R = radius of the cylinder

(a, b) = dimensions of the slot along (ϕ, z) directions (a is the arc length along the cylinder)

$(z_0, R\phi_0)$ = center-to-center distances between slots

$s_0 = \sqrt{z_0^2 + (R\phi_0)^2}$

$\theta_0 = \arctan(z_0/R\phi_0)$

The problem is to determine the mutual admittance between these two slots when kR is large. Throught this work it is assumed that (1) the slots are thin, and (2) their

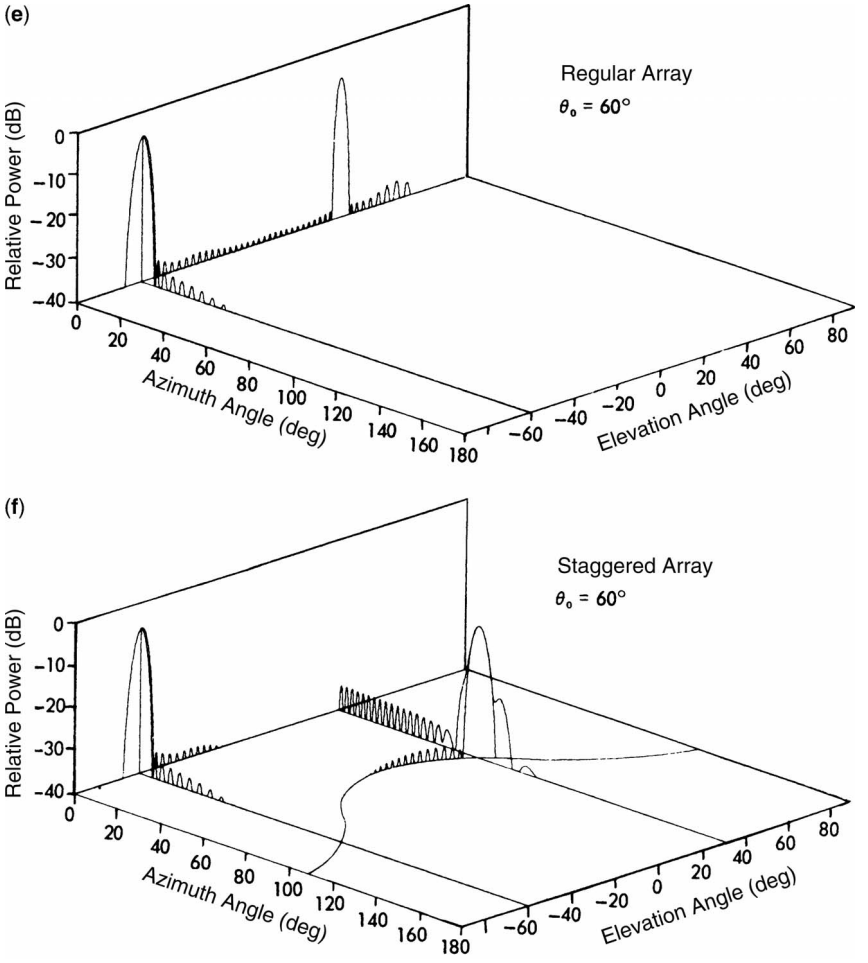


FIGURE 11.15 (*Continued*).

length is roughly a half-wavelength. Then, the aperture field in each slot can be adequately approximated by a simple cosine distribution, which is the so-called "one-mode" approximation. For example, if slot 1 is circumferential, its aperture field under the "one-mode" approximation is given by

$$\bar{E} = V_1 \bar{e}_1 \qquad \bar{H} = I_1 \bar{h}_1 \tag{11.52}$$

where

$$\begin{aligned} \bar{e}_1 &= \left(\hat{z} \frac{2}{ab} \cos \frac{\pi y}{a} \right) \qquad y = R\phi \\ \bar{h}_1 &= \hat{x} \cdot \bar{e}_1 \end{aligned} \tag{11.53}$$

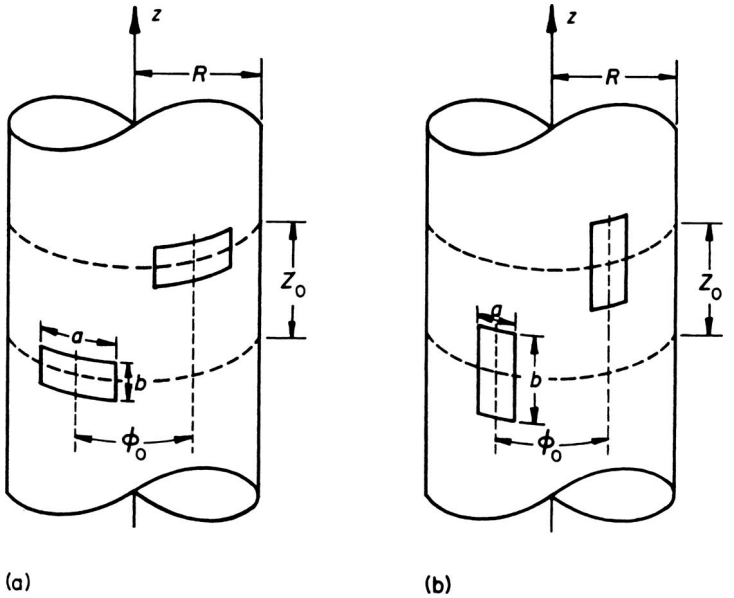


FIGURE 11.16 Slot geometry: (a) circumferential slots; (b) axial slots.

(V_1, I_1) are, respectively, the modal (voltage, current) of slot 1. The mutual admittance Y_{12} is defined by

$$Y_{12} = Y_{21} = \frac{I_{21}}{V_1} \tag{11.54}$$

where I_{21} is the induced current in slot 2, when slot 1 is excited by a voltage V_1 and slot 2 is short-circuited.

There is an alternative definition of mutual admittance. Instead of Eq. (11.52), a modal voltage $\underline{V}_1$ (with a bar) may be defined through the expression for the aperture field of slot 1 as follows:

$$\bar{E} = \hat{z}\frac{\underline{V}_1}{b}\cos\frac{\pi y}{a} \tag{11.55}$$

or equivalently

$$\underline{V}_1 = \int_{-b/2}^{b/2} (\hat{z}\cdot\bar{E})_{y=0}\, dz \tag{11.56}$$

Then, a different mutual admittance $\underline{Y}_{12}$ is defined after replacing (V_1, V_2) by $(\underline{V}_1, \underline{V}_2)$. It can easily be shown that

$$\underline{Y}_{12} = \frac{a}{2b} Y_{12} \tag{11.57}$$

Three remarks are in order: (1) In the limiting case that $b \to 0$, Y_{12} goes to zero as b, whereas $\underline{Y}_{12}$ approaches a constant independent of b. (2) For the special case $a = \lambda/2$ and $R \to \infty$, it is $\underline{Y}_{12}$ that is identical to the mutual impedance Z_{12} between two corresponding dipoles calculated by the classical Carter's method (Hansen, 1972). (3) When the slots are excited by waveguides (transmission lines), one often uses Y_{12}. From here on, attention will be given to Y_{12} instead of $\underline{Y}_{12}$.

The mutual admittance defined in Eq. (11.54) includes the self-admittance Y_{11} as a special case that occurs when two slots coincide. (All the formulas of Y_{12} given in this section, except for the one developed by Lee and Mittra (see Hansen, 1981) in Section 11.3.3.1, can be used for calculating Y_{11} by setting $\phi_0 \to 0$ and $z_0 \to 0$.)

11.3.3.1 Modal Series. A modal series was developed by Golden, Stewart, and Pridmore-Brown (1974), utilizing a series for the azimuth poles in the Green's function, and an integral for the continuous spectrum in z. See also Golden and Stewart (1971) and Stewart and Golden (1971). This series has been used extensively in obtaining Y_{12} for slots on a cylinder (Bargeliotes et al., 1976, 1977). The infinite integral must be approximated, and this is facilitated by assumption of a small loss in the medium via a small negative imaginary part of k. This becomes less satisfactory as z becomes larger, but a modified solution for that regime will be discussed later. See also the modal approach of Balzano and Dowling (1974).

The mutual admittance formulas are as follows:

Circumferential Slots

$$Y_{12} = \int_{-\infty}^{\infty} dk_z \sum_{m=-\infty}^{\infty} \psi(m, k_z) G(m, k_z) \exp[-j(m\phi_0 + k_z z_0)] \tag{11.58}$$

where

$$\begin{aligned}
\psi(m, k_z) &= \frac{ab}{8\pi^2 R} \operatorname{sinc}^2 \frac{k_z b}{2} \left[\operatorname{sinc}\left(m\phi_a + \frac{\pi}{2}\right) + \operatorname{sinc}\left(m\phi_a - \frac{\pi}{2}\right) \right]^2 \\
\phi_a &= \frac{a}{2R} \\
G(m, k_z) &= Y_0 \frac{jkH_m^{(2)\prime}(k_t R)}{k_t H_m^{(2)}(k_t R)} + \left(\frac{mk_z}{k_t^2 R}\right)^2 \frac{k_t H_m^{(2)}(k_t R)}{jkH_m^{(2)\prime}(k_t R)} \\
k_t &= \begin{cases} \sqrt{k^2 - k_z^2} & k \geq k_z \\ -j\sqrt{k_z^2 - k^2} & k \leq k_z \end{cases}
\end{aligned} \tag{11.59}$$

Axial Slots

$$Y_{12} = \int_{-\infty}^{\infty} dk_z \sum_{m=-\infty}^{\infty} \phi(m, k_z) F(m, k_z) \exp[-j(m\phi_0 + k_z z_0)] \tag{11.60}$$

where

$$\phi(m, k_z) = \frac{ab}{8R}\left[\text{sinc } m\phi_a \frac{\cos(k_z b/2)}{\left(\frac{k_z b}{2}\right)^2 - \left(\frac{\pi}{2}\right)^2}\right]^2$$
$$F(m, k_z) = \frac{Y_0 k_t H_m^{(2)}(k_t R)}{jk H_m^{(2)\prime}(k_1 R)} \tag{11.61}$$

Lee and Mittra (see Hansen, 1981) developed an alternate modal series, suitable for large z. First, consider circumferential slots. Rewrite Y_{12} in terms of its real and imaginary parts: $Y_{12} = G + jB$. It can be shown that G is given by

$$G = \int_0^{k_0} dk_z \sum_{m=0}^{\infty} \phi \frac{1}{\varepsilon_m} \cos(m\phi_0) \cos(k_z z_0) \psi(m, k_z) R(m, k_z) \tag{11.62}$$

where

$$R(m, k_z) = \frac{2k_0}{\pi k_t^2 R}\left[\frac{1}{M_m^2(k_t R)} + \left(\frac{mk_z}{k_0 k_t R}\right)^2 \frac{1}{N_m^2(k_t R)}\right]$$
$$M_m^2(x) = J_m^2(x) + Y_m^2(x) \tag{11.63}$$
$$N_m^2(x) = J_m'^2(x) + Y_m'^2(x)$$

Note that G contains a *finite* integral and can be evaluated in a straightforward manner by standard numerical integration techniques. The imaginary part of Y_{12} is given by

$$B = \int_{C_1} dk_z \sum_{m=0}^{\infty} \frac{1}{\varepsilon_m} \cos(m\phi_0) \cos(k_z z_0) \psi(m, k_z) W(m, k_z) \tag{11.64}$$

where the integration contour C_1 is shown in Figure 11.17 and

$$W_m(m, k_z) = \begin{cases} \dfrac{k_0}{k_t}(J_m J_m' + Y_m Y_m')\left[\dfrac{1}{M_m^2(k_t R)} - \left(\dfrac{mk_z}{k_0 k_t R}\right)^2 \dfrac{1}{N_m^2(k_t R)}\right] & k_0 > k_z, \\ \dfrac{-k_0}{|k_t|}\left[\dfrac{K_m'(|k_t|R)}{K_m(|k_t|R)} - \left(\dfrac{mk_z}{k_0 |k_t| R}\right)^2 \dfrac{K_m(|k_t|R)}{K_m'(|k_t|R)}\right] & k_0 < k_z \end{cases} \tag{11.65}$$

The computation of B given in Eq. (11.64) can be quite laborious because (1) the integration with respect to k_z is of infinite range, and the factor $\cos k_z z_0$ is highly

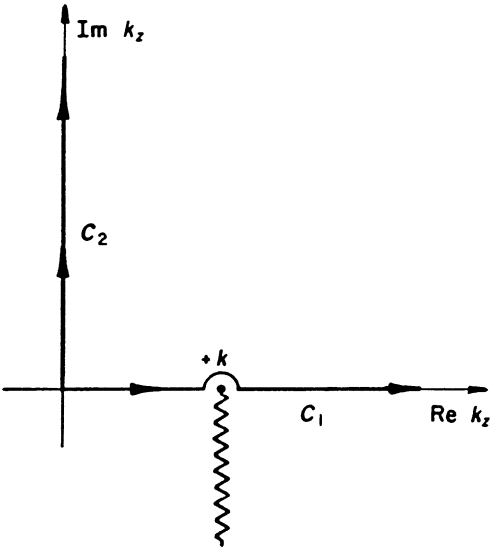


FIGURE 11.17 Contours in the complex k_z plane.

oscillatory for large $k_0 z_0$; (2) $W(m, k_z)$ has nonintegrable singularities of opposite sign on both sides of $k_z = k_0$; (3) $W(m, k_z)$ decays slowly with respect to m and k_z.

To circumvent the above difficulties in evaluating B, a method introduced by Duncan (1962) in the study of cylindrical antenna problems is adopted. Rewrite Eq. (11.64) as

$$B = \text{Im}\left[\sum_{m=0}^{\infty} \frac{1}{\varepsilon_m} \cos m\phi_0 \left(-j \int F(m, k_z) \sin(k_z z_0) dk_z + \int_{C_1} F(m, k_z) \exp(jk_z z_0) dk_z \right)\right] \tag{11.66}$$

where

$$F(m, k_z) = [R(m, k_z) + jW(m, k_z)]\psi(m, k_z) \tag{11.67}$$

The imaginary part of the first term inside the bracket of Eq. (11.66) is

$$\text{Im}\left[-j \int_{C_1} F(m, k_z) \sin(k_z z_0) dk_z\right] = -\int_0^{k_0} R(m, k_z)\psi(m, k_z) \sin(k_z z_0) dk_z \tag{11.68}$$

In order to compute the imaginary part of the second term of Eq. (11.66), the integration contour C_1 is deformed into C_2 (Fig. 11.17) according to the theory of complex

variables. This manipulation leads to

$$\mathrm{Im}\int_{C_1} F(m, k_z)\exp(jk_z z_0)dk_z = \mathrm{Im}\int_{C_2} F(m, k_z)\exp(jk_z z_0)dk_z \tag{11.69}$$

Make the change of variable $k_z = j\eta$ in Eq. (11.69). Substitution of the resultant equation and Eq. (11.68) into Eq. (11.66) gives

$$B = \sum_{m=0}^{\infty}\frac{1}{\varepsilon_m}\cos(m\phi_0)\left(-\int_0^{k_0} R(m, k_z)\psi(m, k_z)\sin(k_z z_0)dk_z + \int_0^{\infty} R(m, j\eta)\psi(m, j\eta)\exp(-\eta z_0)d\eta\right) \tag{11.70}$$

The final expression for Y_{12} is given by the real part G in Eq. (11.62) and the imaginary part B in Eq. (11.70). Several remarks are in order: (1) Not only G, but also B is determined by $R(m, k_z)$, which is much simpler than $W(m, k_z)$ defined in Eq. (11.65). (2) The parameter G contains only a finite integral. (3) The infinite integral in B, that is, the second integral in Eq. (11.58), contains an exponentially decaying factor $\exp[-(z_0 - a)\eta]$ in its integrand. Thus the larger z_0, the faster convergence in the evaluation of B. This is in contrast to the original expression for Y_{12}. (4) There is no nonintegrable singularity in Eq. (11.62) or (11.70).

The same method applies to the derivation of an alternative expression for Y_{12} for two axial slots ($a < b$, as shown in Fig. 11.16b). Only the final result is given

$$Y_{12} = -\frac{abY_0}{\pi k_0 R^2}\sum_{m=0}^{\infty}\frac{\cos m\phi_0}{\varepsilon_m}\left[\int_0^{k_0}\Phi(m, k_z)\frac{\exp(-jk_z z_0)}{N_m^2(k_t R)}dk_z + j\int_0^{\infty}\Phi(m, j\eta)\frac{\exp(-\eta z_0)}{N_m^2\left(R\sqrt{k_z^2 + k_0^2}\right)}dk_z\right] \tag{11.71}$$

where

$$\Phi(m, k_z) = \left[\mathrm{sinc}\frac{1}{2}m\phi_a\frac{\cos\dfrac{k_z b}{2}}{\left(\dfrac{k_z b}{2}\right)^2 - \left(\dfrac{\pi}{2}\right)^2}\right]^2 \tag{11.72}$$

$$\phi_a = 2\arcsin\frac{a}{2R}$$

Asymptotic approaches, useful for large cylinder diameters in wavelengths, are included in Section 11.3.4.

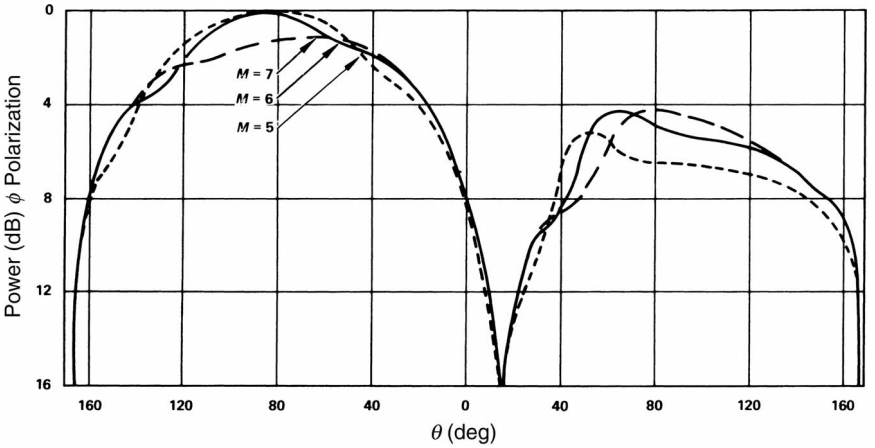


FIGURE 11.18 Pattern of radial slot on cone. (Courtesy Hughes Aircraft Co.)

Calculated patterns of a radial slot in a cylinder of 20° total angle, at a location where $ka = 39$, are shown in Figure 11.18 for 5–7 terms in the modal series. These can be compared with the patterns in Figure 11.36 based on many terms, and on measurements.

For cylindrical arrays, where the finite length of the cylinder affects element impedance, conventional analytical techniques are inadequate. A hybrid Moment Method, spectral approach has been used by Rengarajan (1996).

11.3.3.2 Admittance Data. Using the formulas described, slot mutual impedance data have been calculated for both axial and circumferential slots (Hansen, 1981). All data are at 9 GHz, for slot dimensions 0.9 × 0.4 in. Figure 11.19 shows mutual admittance between two circumferential slots as a function of rotation angle between the slots. Slot offset is 1.524λ, while cylinder radius is 1.517λ. Both the asymptotic result and the exact modal series result are shown. Two circumferential slots along the same generator, with axial separation a variable, provide the results of Figure 11.20. Cylinder radius and slot dimensions are unchanged for this figure. The mutual admittance magnitude is smaller for slots on a cylinder than for slots in an infinite plane. Figure 11.21 is again for in-line slots, with an axial separation of 6.09λ, as a function of cylinder radius. Y_{12} is normalized by the plane value of $5.37 \times 10^{-5} \exp(j53.55)$mho.

Axial slot behavior is simpler, and is typified by Figure 11.22, which gives magnitude of mutual impedance versus axial spacing. Note that the cylindrical curvature has little effect, even though the cylinder diameter is only 3 wavelengths.

11.3.4 Scan Element Pattern

A canonical geometry that yields insight into the behavior of slots upon a cylinder is an array of infinitely long axial thin slots (slits) deployed all around the cylinder. Sureau

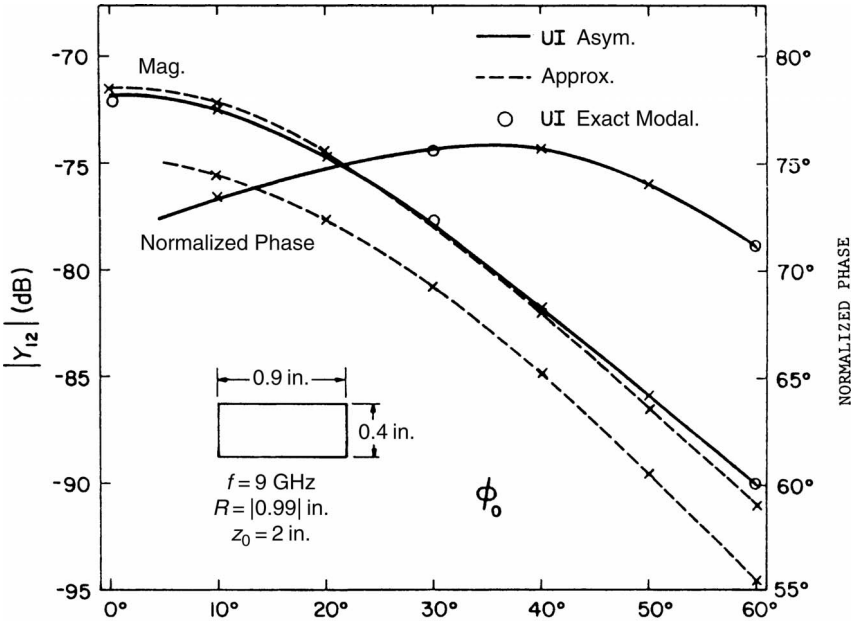


FIGURE 11.19 Circumferential slot mutual admittance versus rotation.

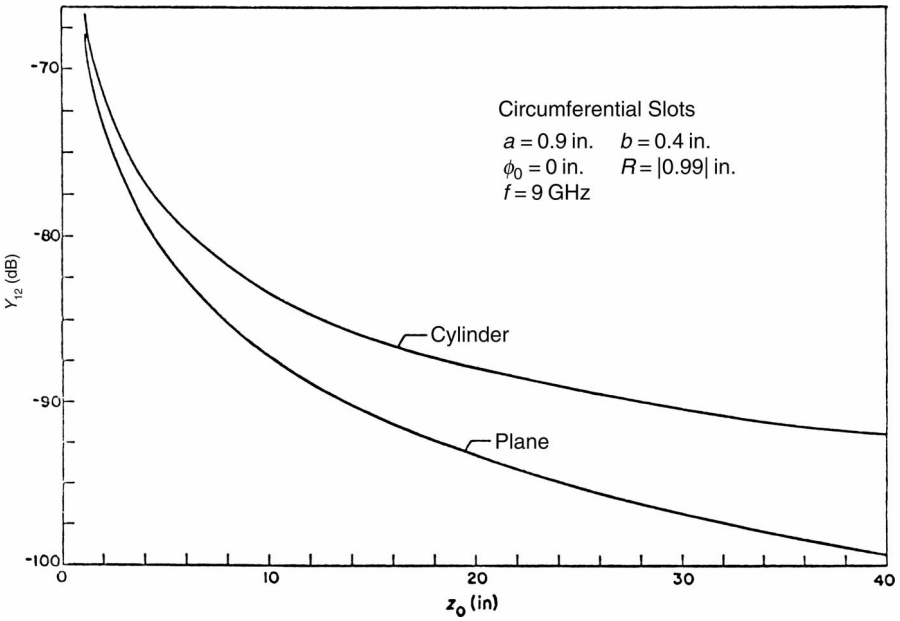


FIGURE 11.20 Circumferential slot mutual admittance versus axial separation.

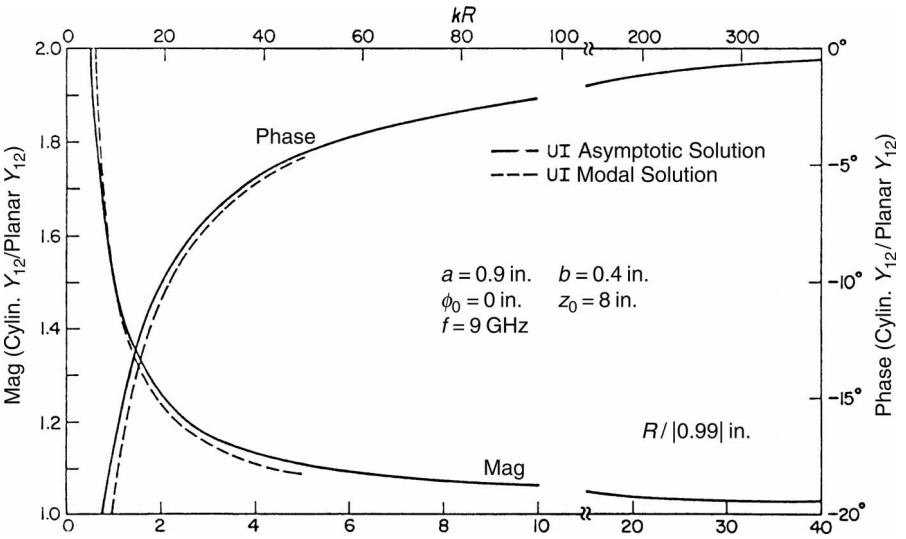


FIGURE 11.21 Circumferential slot mutual admittance versus radius.

and Hessel (1969, 1971, 1972), Hessel and Sureau (1971, 1973), and Hessel (1972) derived *scan element patterns* using the unit cell approach of Oliner and Malech (1966), identifying a direct component associated with the integrated saddle point and a set of leaky waves representing complex poles (natural resonances) of the periodic cylinder. Pattern nulls due to external dielectric sheet covers were also exhibited. The unit cell approach was extended by Munger et al. (1971), who considered an

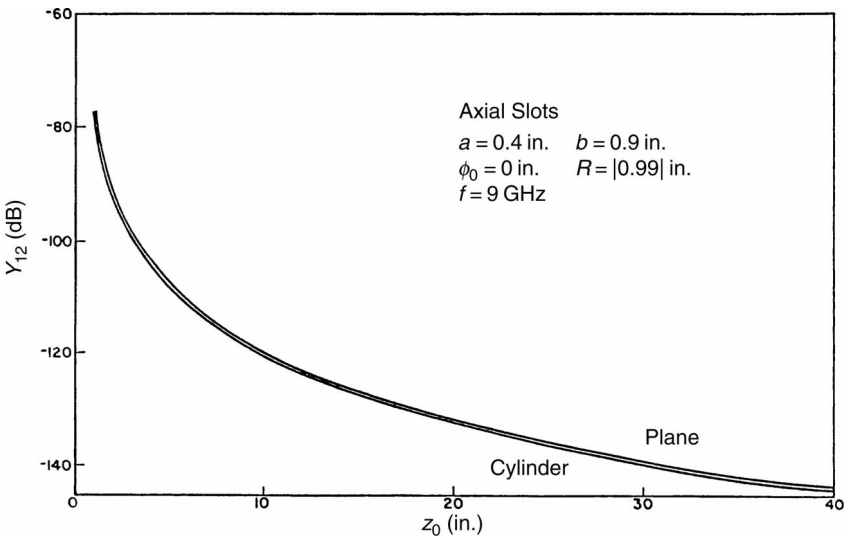


FIGURE 11.22 Axial slot mutual admittance versus separation.

infinitely long cylinder covered with a regular array of axial slots. A different approach gives insight into coupling between azimuthal modes. Borgiotti and Balzano (1970, 1972a,b) and Balzano (1974) decomposed a ring of slots into a sum of eigen excitations (modes) and then computed coupling between modes on adjacent rings. If this were generalized to obtain mutual admittance between two slots, each on a different ring, the result would be the modal series admittance, using the cylindrical Green's function (Bowman et al., 1969). This approach, developed by Pridmore-Brown and Stewart, and extended by Lee is the most useful as it gives mutual admittance between slots directly. It was described in detail in Section 11.3.3.1.

Scan element patterns of elements disposed around a cylinder can be obtained by solving an equation for each azimuthal mode separately (Herper et al., 1985); Figure 11.23 shows SEP for circumferential dipoles around a cylinder of diameter $(120/\pi)\lambda$, with axial dipole spacing of 0.72λ, and circumferential spacings of 0.5, 0.6, and 0.72λ. As expected from the analogous *H*-plane planar array SEP, the wider spacings show oscillations at broadside, leading to a drop at a pseudo-grating lobe angle. The drops occur at angles smaller than those for the planar case, but are less steep [see also Borgiotti (1983)].

An asymptotic approach has been used by Sureau and Hessel (1972) with the SEP calculated from a combination of a space wave and creeping waves [see also Lee (1978), Lee and Safavi-Nairi (1978), and Bird (1985)]. Asymptotic methods for general cylindrical structures have been developed by Shapira et al. (1974a,b). Figure 11.24 shows SEP for an array of rectangular waveguide radiators around a cylinder of diameter $(185/\pi)\lambda$, with axial spacing of 0.8λ and circumferential spacing

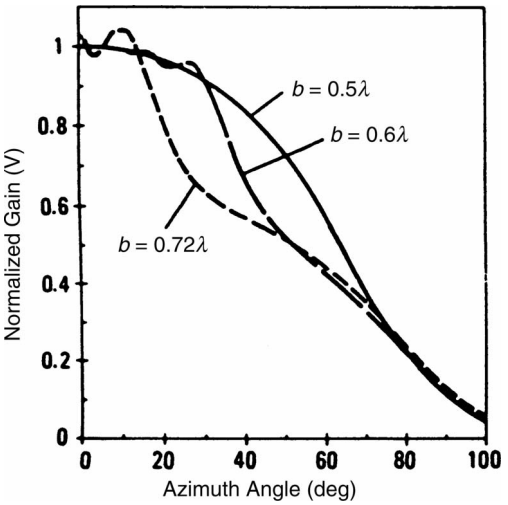


FIGURE 11.23 Scan element patterns for several spacings. (Courtesy Herper, J. C., Hessel, A., and Tomasic, B., "Element Pattern of an Axial Dipole in a Cylindrical Phased Array—Part I: Theory", "Part II: Element Design and Experiments", *Trans. IEEE*, Vol. AP-33, Mar. 1985, pp. 259–272, 273–278.)

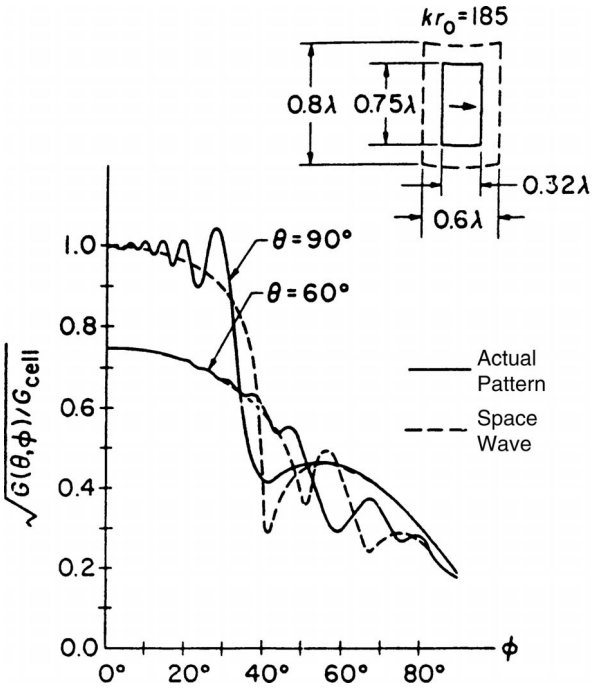


FIGURE 11.24 Scan element pattern. (Courtesy Sureau, J. C., and Hessel, A., "Realized Gain Function for a Cylindrical Array of Open-Ended Waveguides", in *Phased Array Antennas*, A. A. Oliner and G. H. Knittel, Eds., Proceedings of 1970 Symposium at PIB, Artech House, 1972, pp. 315–322.)

of 0.6λ. The *E*-field was circumferential, with guide dimensions of 0.32 by 0.75λ. This larger cylinder shows a steeper drop, more like the planar array results of Chapter 7.

Mutual coupling for a concave cylindrical array has been investigated using Floquet theory by Tomasic and Hessel (1989), and for asymptotic theory by Tomasic et al. (1993). This geometry is useful for Rotman and bootlace lenses.

11.4 SECTOR ARRAYS ON CYLINDERS

11.4.1 Patterns and Directivity

In many applications, such as missiles and aircraft, full azimuth scanning is not needed. Sector arrays, where the elements occupy a sector of subtended angle ϕ_a, are appropriate. When the sector angle is small, perhaps $<60^\circ$, the array can be designed as a planar array with minor adjustments. However, large sectors require an examination of all the curvature effects. Figure 11.25 shows relative directivity versus sector included angle for several element pattern variations. The directivity is projected area times the element directivity. Many phased arrays have a $\cos^{1.5}$ behavior in the strong coupling plane,

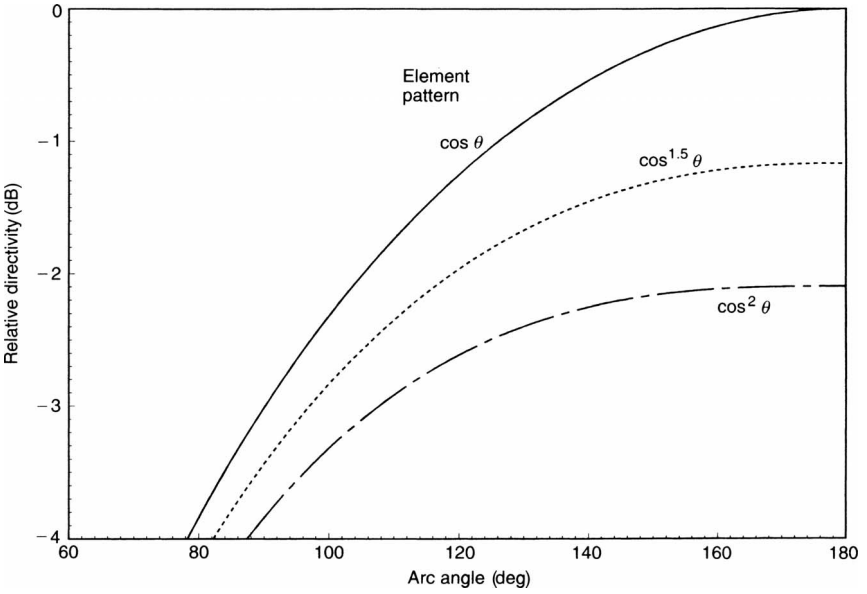


FIGURE 11.25 Arc array directivity relative to flat array across diameter.

and a cos behavior in the weak coupling plane, owing to *scan impedance* variation. An example of an array with 0.65λ spacing on a 27.4λ diameter cylinder, for sector angles of 60, 90, and 120°, is shown in Figure 11.26 (Hessel, 1972). Larger projected element spacings at large angles allow the grating lobe to increase with larger sector angle. This lobe could be suppressed with closer element spacings of course. However, the directivity penalty cannot. From a 60° sector to one of 120° the projected area has doubled, but the directivity has increased only 1.0 dB.

When the cylinder radius is large, microwave optics approaches are suitable. The far field is divided into three regions: illuminated, shadow, and a transition region between (see Fig. 11.27) (Pathak and Kouyoumjian, 1974). In the illuminated region, GTD is used to obtain the pattern; in the shadow region, creeping waves exist; and the transition region is difficult. The latter region covers an angular region each side of the tangent line proportional to $(kr_0)^{-1/3}$. An axial slot (circumferential electric) field is roughly constant, but drops to ~ 0.7 at $\phi = 90°$, independently of cylinder diameter as long as it is large. A circumferential slot (axial electric) field in contrast drops to roughly $0.4(2/kr_0)^{1/3}$. A slow dependence on cylinder diameter exists. A cylinder of 20λ diameter gives field of -17.9 dB, while one of 40λ diameter gives -19.9 dB. The GTD analysis used FOCK functions: [see Fock (1965) and Bowman et al. (1969)]. The complexity of the creeping-wave Watson transformation formulation puts it outside the scope of this book; useful references are Watson (1918), Bremmer (1949), and Felsen and Marcuvitz (1973).

The nonseparability of the elevation and azimuth factors in the pattern expression means that, for a fixed amplitude taper, both the elevation and azimuth patterns change with elevation scan angle θ_0. The practical implications of this are serious: to maintain

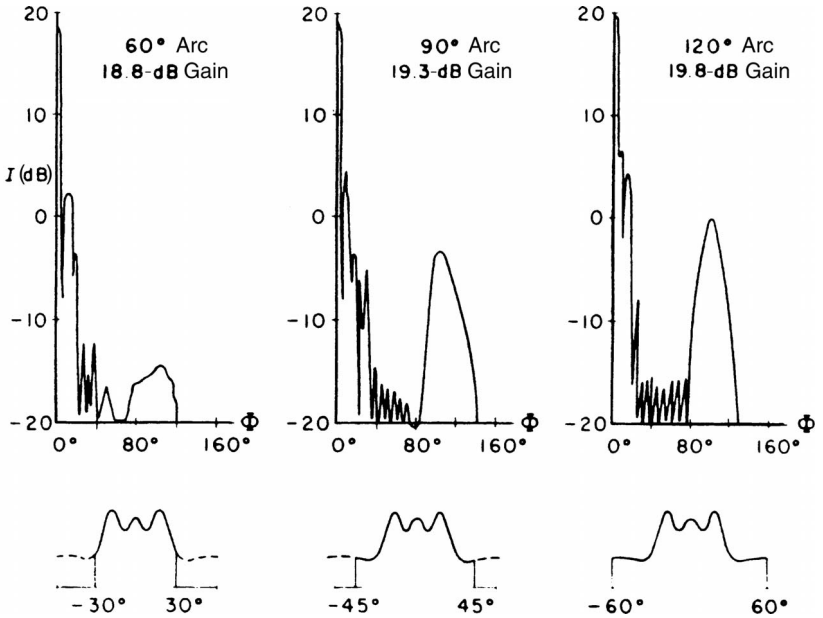


FIGURE 11.26 Scan element patterns of arc arrays. (Courtesy Hessel, A., "Mutual Coupling Effects in Circular Arrays on Cylindrical Surfaces—Aperture Design Implications and Analysis", in *Phased Array Antennas*, A. A. Oliner and G. H. Knittel, Eds., Proceedings of 1970 Symposium at PIB, Artech House, 1972, pp. 273–291.)

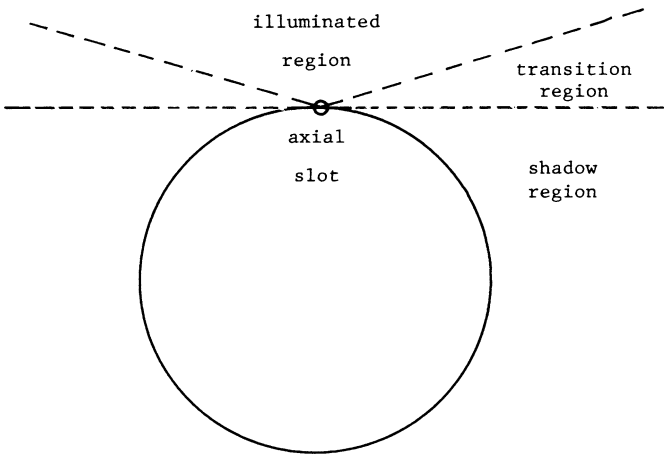


FIGURE 11.27 Field regions.

low sidelobes in azimuth, as the elevation scan angle changes, the array excitation must change.

In addition to the switching and phasing configurations for rotating the beam in azimuth previously mentioned, lenses, such as the R-2R, have been used for cylindrical array feeds (Boyns et al., 1968; Holley et al., 1974).

11.4.2 Comparison of Planar and Sector Arrays

The question often arises whether the cylindrical array makes efficient use of aperture and hardware—in particular, when compared with the standard planar-array approach. For 360° azimuth coverage, four planar arrays, each scanning $\pm 45^\circ$, are generally used, so the cylinder is compared with the four-sided planar configuration. Identical elements are assumed. For elevation scanning and elevation pattern, the two configurations give nearly identical results, assuming that both use a separable cophasal distribution and identical distributions in elevation. The planar array elevation pattern is the array factor multiplied by the elevation element pattern, while the cylindrical array elevation pattern is the array factor multiplied by the "ring array element" elevation pattern. Figure 11.3, for example, shows the "ring array element" elevation patterns compared with the element elevation pattern. If anything, it is an advantage of the cylinder that the ring array elevation patterns tend to suppress sidelobes more than does the element pattern alone.

The first-order comparison, then, can be reduced to that of the ring array azimuth pattern to the linear array azimuth pattern.

Assume that no more than 180° of the arc will be excited on the ring. Then the amplitude excitation of a linear array can be projected[1] onto an arc whose chord is equal to the length of the linear array, with the phase of the elements on the arc corrected to give a linear phase front. For small angles off broadside—that is, the main beam and first few sidelobes, the arc can be expected to give about the same results as the linear array, because for small angles the curvature has a negligible effect on the phased contribution from each element. Also, effects of element spacing become apparent only at larger angles. By this reasoning, the projection of, say, a Chebyshev distribution is a convenient means for forming the desired beamwidth and constraining the inner sidelobes of an arc array. Computations bear this out. Furthermore, the farther-out sidelobes tend naturally to be lower, with the exception of the grating lobe. Computations indicate that if the grating lobe is controlled, all sidelobes will be below the inner sidelobes.

The relative performance of the linear and arc arrays can be evaluated in terms of the number of elements and the overall antenna size required. Consider Figure 11.28. The active apertures are shown in dark lines, and are of the same projected length. The element spacing on the linear array can be fixed at $d = 0.586\lambda$, which is the spacing required to scan $\pm 45^\circ$ with grating lobe just coming in at $\pm 90^\circ$. For the arc, the grating lobe can be controlled by placing the stationary point outside the active arc,

[1]The increased element density as a function of α on the arc means the amplitude should be reduced by $1/\cos\alpha$; however, this is exactly compensated by the assumed cosine element pattern.

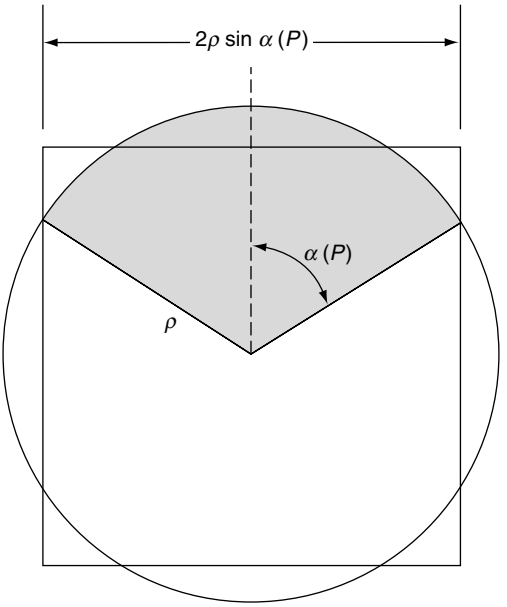


FIGURE 11.28 Four-sided array versus circular array.

say at $\alpha_0 = \alpha(P) + \delta$. Then from Eq. (11.21) with $\theta_0 = 0$ and $r = -1$, since δ is small,

$$s = \frac{\lambda}{2\sin(\alpha(P) + \delta)} \simeq \frac{\lambda}{2[\sin\alpha(P) + \delta\cos\alpha(P)]} \tag{11.73}$$

The total number of elements on the four-sided array is about

$$\frac{8\rho\sin\alpha(P)}{0.586\lambda} \tag{11.74}$$

The total number on the ring is

$$\frac{2\pi\rho}{s} = \frac{4\pi\rho}{\lambda}[\sin\alpha(P) + \delta\cos\alpha(P)] \tag{11.75}$$

The ratio is

$$\frac{\text{Number of ring elements}}{\text{Number of linear elements}} = 0.920[1 + \delta\cot\alpha(P)]$$

From this, one may conclude that $\sim$92–100% of the elements required for a four-sided linear array are required to obtain the equivalent ring array.

Results (Hansen, 1981) indicate that the circular array of size and number of elements equivalent to a four-sided linear array can be made approximately equivalent

in broadside performance. Since the linear array beam broadens for scan off broadside, however, the overall performance of the ring array is superior.

In addition, there are some disadvantages of the planar array that the cylindrical array inherently avoids. The ring array beam is identical for all beam positions, while the planar array beam is broader in scanning off broadside. As the ring array beam is scanned by, in effect, commutating the distribution, it is always formed by a distribution that is symmetrical in phase and amplitude. This results in superior beam pointing accuracy independent of frequency change. Finally, the cylindrical array gives 360° coverage in azimuth with none of the handover problems associated with the use of several planar arrays. In some applications, these advantages can be very important.

The cylindrical array, however, has some disadvantages. For scanning, the amplitude as well as the phase must be switched in azimuth, and the feeding system that results will be more complex than that of a planar array system. (However, computer control is not a problem in view of the separable aperture.) The greatest disadvantage would appear to be that the cylindrical array cannot be physically separated as can the four planar arrays. This means that the cylinder must be in a position to look 360°, while each planar array need see only a 90° sector. More important, it means that the cylinder cannot be tilted back to increase the elevation coverage, as is common

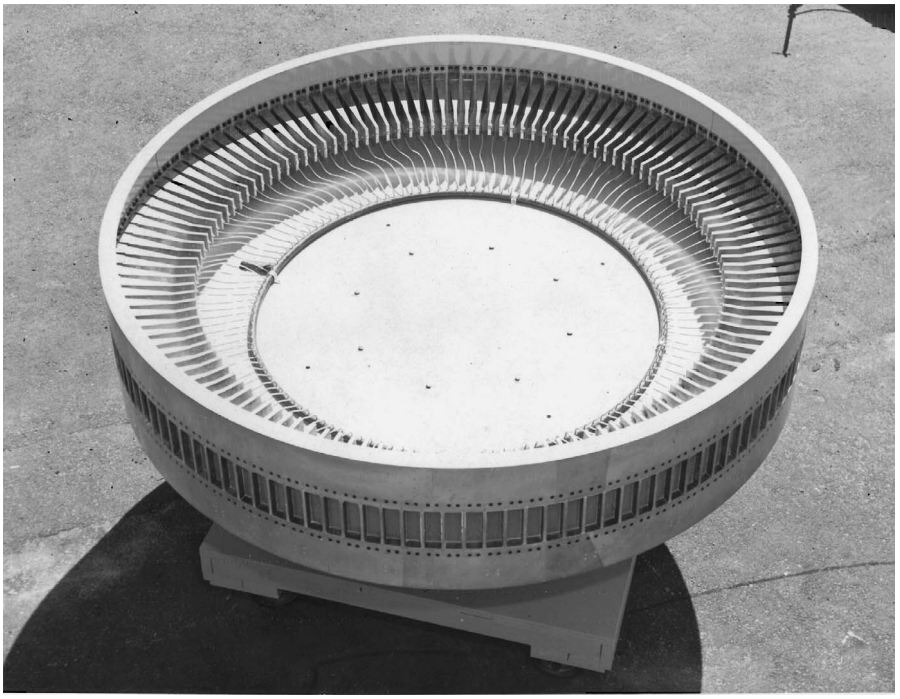


FIGURE 11.29 Ring array. (Courtesy Naval Electronics Lab. Center.)



FIGURE 11.30 Doppler scan edge slot array. (Courtesy Hazeltine Corp.)

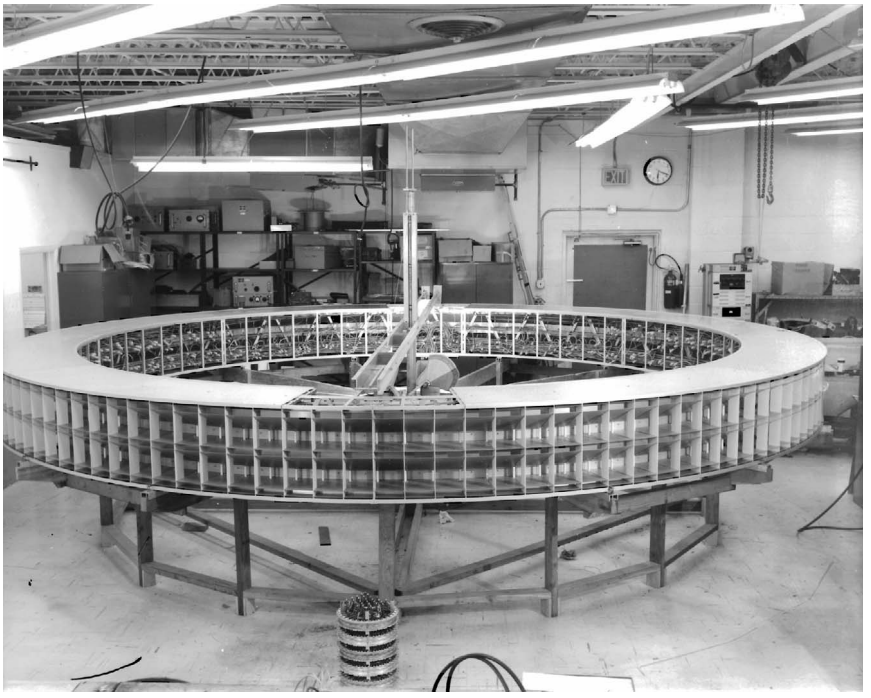


FIGURE 11.31 Electronic scanned dual-beam Wullenweber. (Courtesy Scanwell Labs.)

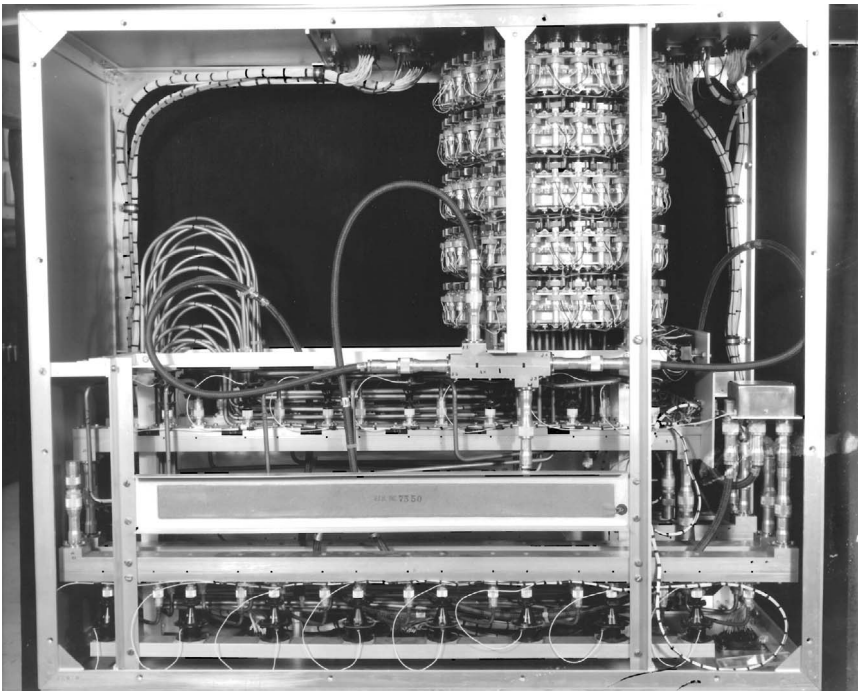


FIGURE 11.32 Electronic scanned dual beam ring switch. (Courtesy Scanwell Labs.)

practice with planar arrays. For this reason, a truncated cone might be considered to extend the elevation coverage and still retain the advantages of circular symmetry. Comparison with tilted face planar arrays is slightly more complex [see Knittel (1965) and Corey (1985)].

11.4.3 Ring and Cylindrical Array Hardware

A few examples of experimental arrays are given here. Figure 11.29 shows a ring array of 128 sectoral horns, designed to be fed by a parallel plate circular lens. Ring diameter is 26.4λ; horns are $1/2 \times 3/2 \times 3\lambda$ (Boyns et al., 1970a,b).

Figure 11.30 shows an edge slot Doppler scan Microwave Landing System array; each column is excited by a phaser (Hannan, 1976). Another L-band ring array is pictured in Figure 11.31. It uses two rings of 124 elements each, with 30 excited to form a beam. Elements are probe fed flared horns with chokes to reduce ring-to-ring coupling. This 20-ft diameter array uses an elaborate switching and phasing unit shown in Figure 11.32. (Gabriel and Cummings, 1970). Provencher (1972) reviews ring array practice. A cylindrical array with 256 elements in 32 columns is shown in Figure 11.33. Each column of eight dipoles incorporates a serpentine feed for frequency scanning in elevation. The switch matrix utilizes eight active columns, and provides an amplitude taper for 25-dB sidelobe level (Sheleg, 1975).

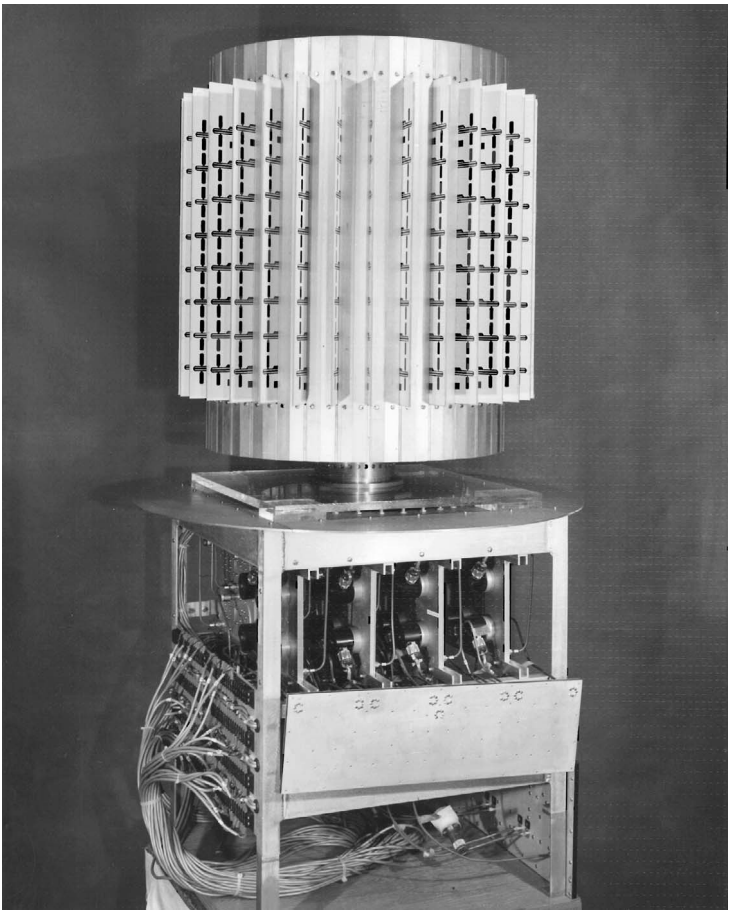


FIGURE 11.33 Cylindrical frequency scanning array. (Courtesy Naval Research Lab.)

11.5 ARRAYS ON CONES AND SPHERES

Cones and spheres are sometimes useful conformal array structures. A sphere, or more usually a hemisphere, can in principle scan in all directions, with only modest degradation at angles near 90° where the other hemisphere would have contributed. Conical arrays fit missile front ends, aircraft fuselage noses, smart munitions, and so on. The difficulties, however, are enormous. A sphere cannot be populated by a regular lattice smaller than a dodecahedron (Fullerine), nor can a cone, owing to the 2D curvature. When polarization is taken into account, the picture is far worse. An array on a cone, designed for linear circumferential polarization on-axis, requires slot polarizations that rotate as the observer rotates around the cone. This severe distortion is due to the projection of the polarization.

With modern computer control of phase and amplitude, the use of a somewhat irregular lattice on a sphere is acceptable. But the fabrication, assembly, and feed

network are more costly due to these irregularities. A special case is the spherical dome array. This is a hemispherical bootlace lens, with a conventional scanned planar array in the diametrical plane. Limited scan of the planar array is magnified to wide-angle scan by the lens (see Chapter 10 for details).

Considerable analytical and experimental work has been done on conical arrays (Hansen, 1981), as described in Section 11.5.1.

11.5.1 Conical Arrays

In the design of electronically scanned conical arrays, problems arise that are different than those for arrays on planar and cylindrical surfaces. The gain of conical arrays changes as a function of the scan angle, and also depends on the part of the radiating structure that is visible from the far field at the beam pointing position. The polarization in the far field changes as a function of scan angle for radiating elements whose polarization is fixed with respect to the surface of the cone. Sum and difference patterns become sensitive to the incident polarization. The manner in which these quantities vary with scan angle will be discussed. The problem of pattern synthesis and analysis is also examined, and several techniques are discussed.

The general configuration for conical arrays consists of a set of radiating elements placed on a conical surface. The far-field patterns to be radiated from this array are pencil beams with suitably controlled sidelobes. The pencil beam is summed from a direction perpendicular to the generatrices through the axis of the cone. The radiating elements comprising the conical array are assumed to have symmetry in the plane perpendicular to the axis of the cone, the ϕ plane, and the array placed on the conical surface is assumed to have symmetry in the ϕ plane. Whatever the shape of the active part of the array, the projected aperture of the active part will be a function of the position angle defining the beam pointing direction in the plane defined by the axis of the cone and a generatrix (the θ plane). The active part is defined as that part that is turned on to receive (or transmit) energy. The projected aperture will be constant for any ϕ scan at constant θ because of the symmetry mentioned above.

The beamwidth of a planar array changes as a function of the beam pointing in direct proportion to the projected aperture perpendicular to the beam pointing direction. This relation is an approximation, but is quite accurate for angles between broadside and $50-60^\circ$. In a conformal array the beamwidth is not so easily related to the geometry of the array because both the shape of the active part of the array and its projected area change as a function of the beam pointing direction. An estimate of the beam and sidelobe shapes will be given below.

For the beam pointed perpendicular to the generatrix of the cone, the radiation characteristics may be approximated using an equivalent planar array whose area is the projected area of the active part of the conformal array. Thus one would expect beamwidths and sidelobe levels commensurate with the projected aperture dimensions. At the other extreme, for a beam pointed along the axis of the cone, a different condition exists. For small angles near the cone axis, the array when projected forward would look approximately like a concentric ring array. The center rings of the array are missing, since the extreme tip of the cone is probably not usable in practice. For a

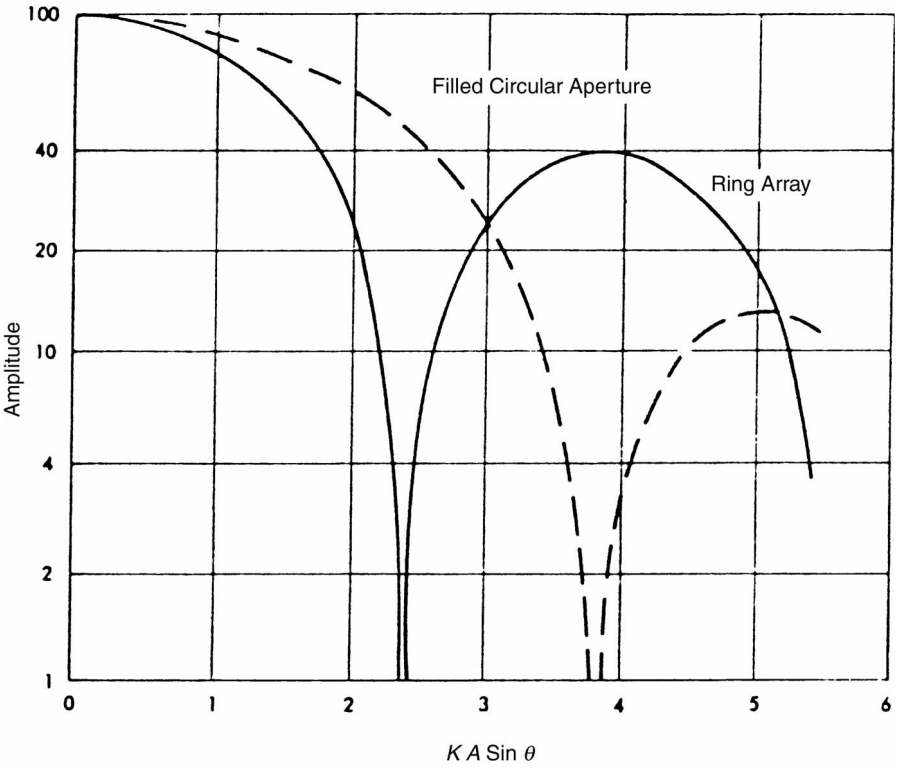


FIGURE 11.34 Circular aperture patterns.

6-ring by 24-element–ring array, the pattern would be somewhat as shown in Figure 11.34. For comparison, the figure also shows the pattern of an array of the same size on the same cone but with the center rings filled. The latter is the usual pattern for a uniform circular array. Note that the ring array has a narrower beamwidth and higher sidelobes than the filled array. This narrower beamwidth can be explained with reference to an interferometer that has a beamwidth one-half of that of a completely filled linear array of the same length. The "sidelobes" are as high as the main beam. This particular case is an intermediate one. Thus it can be seen that there is a tradeoff between array filling, beamwidth, and sidelobe level.

Patterns of individual slots are of interest. Figure 11.35 shows measured and computed elevation patterns of a circumferential $\lambda/2$ slot on a cone of 20° total angle; the slot is 6.22λ from the tip. Flattening of the pattern by the cone in the lit region is marked. Azimuthal patterns of a radial slot at the same position are given in Figure 11.36. The back lobe is -14 dB.

11.5.1.1 Lattices on a Cone. In a conical (and also a cylindrical) array, advantage can be taken of the circular symmetry of the surface to reduce the steering problem essentially to that of scanning in 1D only with switching in azimuth. Figure 11.37

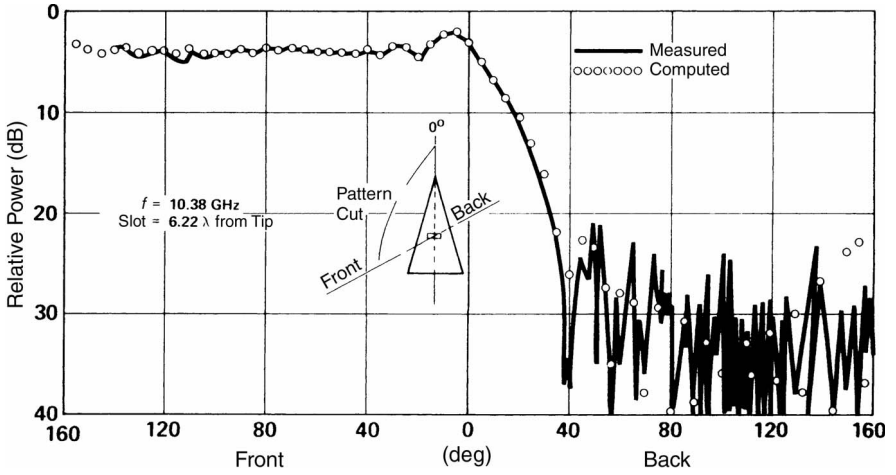


FIGURE 11.35 Pattern of circumferential slot on cone. (Courtesy Hughes Aircraft Co.)

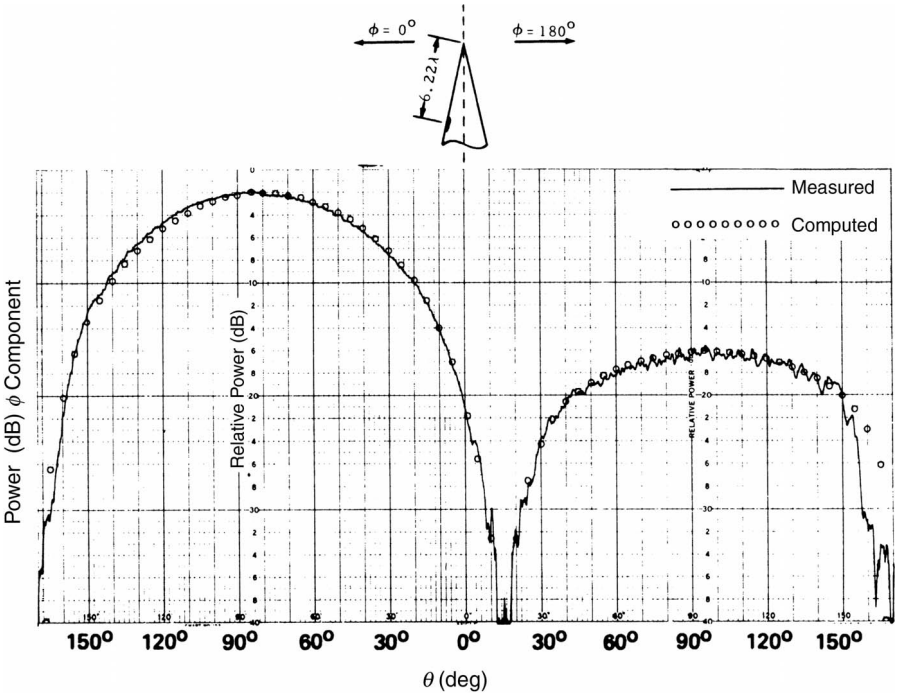


FIGURE 11.36 Pattern of radial slot on cone. (Courtesy Hughes Aircraft Co.)

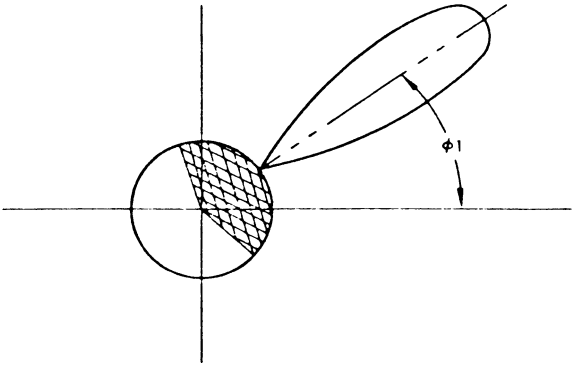


FIGURE 11.37 Excited part of cone.

shows an end-on view of the cone with the shaded portion representing the excited area of the surface. If the beam lies in the plane perpendicular to the cone axis, its position will be as shown in the figure—and symmetrical with respect to the active area of the array. If the beam is scanned toward endfire while ϕ_1 is held fixed, it can be seen that the projection of the beam onto the plane of the paper will still fall at ϕ_1. Hence, the active area is still symmetrically located with respect to the beam. The beam can be thought of as being broadside to the cone in the ϕ-plane and capable of being electronically scanned in the plane that passes through the cone axis (the θ-plane).

The beam is now steered around in the ϕ-plane to ϕ_2, by switching the active portion of the array to maintain a symmetrical relationship with the new beam pointing direction. Here also, the beam can be electronically steered toward endfire without changing the angle ϕ_2 or disturbing the symmetry conditions. Hence, it may be concluded that for any angle of ϕ, in the ideal case of continuously illuminated aperture, the beam can be considered to be broadside to the cone in the ϕ-plane and need be electronically scanned only in the θ-plane, with switching in the ϕ-plane.

With a discrete number of elements on the cone, the ideal situation outlined above can only be approximated. However, if a reasonably large number of elements is used, the approximation will be quite close. For example, if 16 elements are used in the larger rings around the cone, the largest portion of the area of excitation can be switched in 22.5° steps. Thus, it is necessary to electronically scan the beam only $\pm 11.25°$ off the perpendicular to the excited area to achieve full coverage around the axis of the cone by a sequence of switching and phasing operations. Since any one active area of the cone scans only $\pm 11.25°$ in the ϕ plane, the interelement spacing in that plane need be only slightly less than the spacing required for an array that does not scan at all in that plane. Hence, the interelement spacing problem is reduced approximately to that associated with scanning in the θ-plane only. It may also be attractive to provide amplitude and phase control at each element instead of switching.

The conical array can be thought of as a set of cylindrical arrays of different radii as a first-order approximation. Hence, it is to be expected that it, too, will have a tendency to have grating lobes in the ϕ plane. Therefore, it is anticipated that the interelement spacing in that plane will have to be kept smaller than would normally be necessary

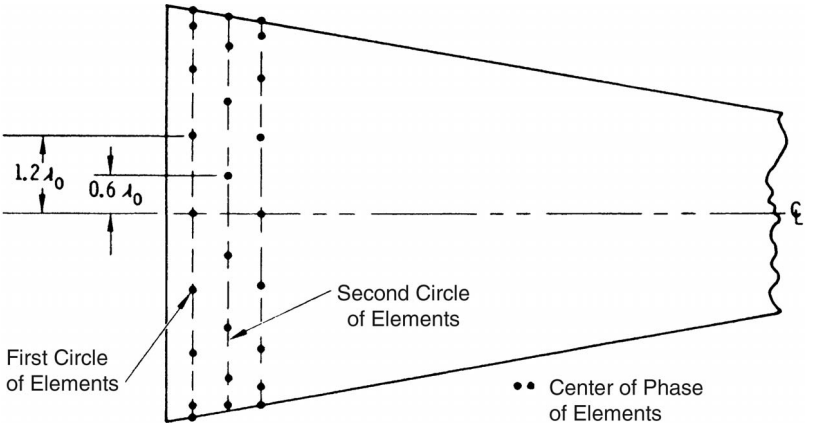


FIGURE 11.38 Staggered elements on cone.

for an equivalent planar array. However, the fact that each circle of elements on the cone nearer the tip is smaller than the previous one may tend to alleviate this problem by introducing a modest amount of quasirandomness into the element placement.

A more nearly continuous illumination in both principal axes of the cone can be achieved by staggering the elements in alternate circles, as shown in Figure 11.38. Thus, although the actual spacing between any two elements on a circle with a 6λ diameter is $1.2\lambda_0$ (assuming 16 elements per circle), the effective spacing is only one-half that value. An interelement spacing of $0.6\lambda_0$ is rather large for a circular array; however, each circle nearer the tip of the cone will bring the elements closer together until the point of physical interference is reached. The average interelement spacing in the ϕ-plane for the section of the cone that has 16 elements per ring should thus be $<0.5\lambda_0$.

In the smaller regions of the cone, fewer elements will be needed per circle. The type of element to be used will influence the decision on just how many should be used in each circle. The crossed waveguide elements (Kummer et al., 1971, 1973; Kummer, 1972) can be fitted together quite closely on a flat surface provided that they are rotated at an angle of $\sim 20-25^\circ$, depending on wall thickness, to the principal axes of the lattice (see Fig. 11.39). They can be brought most closely together on a curved surface if they still maintain that angle. When the elements in alternate circles are staggered in a symmetrical fashion, the lattice is rotated 45°, as shown in Figure 11.40.

11.5.1.2 Conical Depolarization and Coordinate Systems. A cone is less hospitable to an array than a cylinder in that its surface cannot be covered with a regular lattice. The question of orienting the radiators on the cone so that optimum performance in all directions is obtained involves the geometry of the radiator. The case of a beam pointing directly ahead is considered first. Since the array is conformal, some sort of flush-mounted radiator will be required. The slot and the open-ended waveguide are the two most commonly used flush-mounted radiators. The polarizations of these radiators are nearly identical; hence the polarization of the slot radiator is considered.

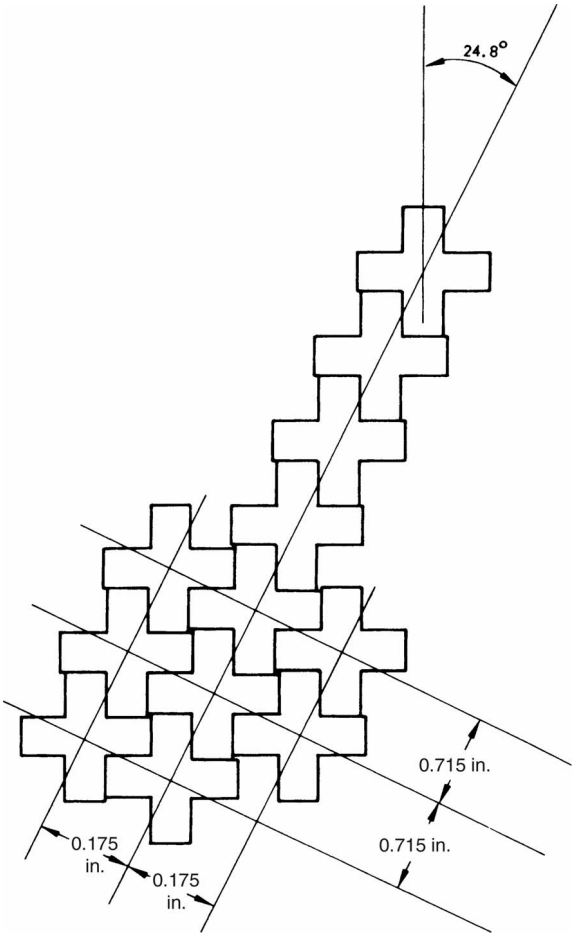


FIGURE 11.39 Titled lattice for crossed-guide elements.

The *E*-field in one quadrant of the far field of a slot is shown in Figure 11.41. In this figure, the ground plane is assumed to be in the *x*, *y*-plane. Note that the polarization of the field is everywhere perpendicular to the ground plane in the immediate vicinity of that plane. If the slots are imagined to be placed on a 10° half-angle cone, then in the forward direction each slot will be viewed from an angle of 10° above its nominal ground plane. The polarization at that angle can be visualized by constructing a surface on Figure 11.41 that is 10° above the *x*,*y*-plane. This surface is shown in the figure by dashed lines.

It is instructive to start near the *x*-axis and note the change in polarization angle as the 10° surface is followed around to the *y*-axis. (The scale of angles in *x*,*y*-plane is used for convenience.) At 0°, the polarization (shown by the dark arrows) is found to be parallel to the ground plane and, hence, is termed horizontal. As the progression continues around the quadrant toward *y*-axis, the polarization vector at first turns quite

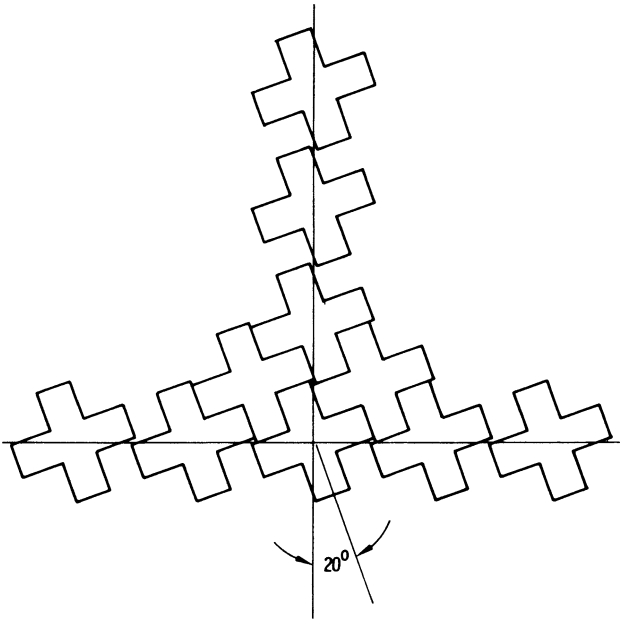


FIGURE 11.40 Staggered and tilted elements.

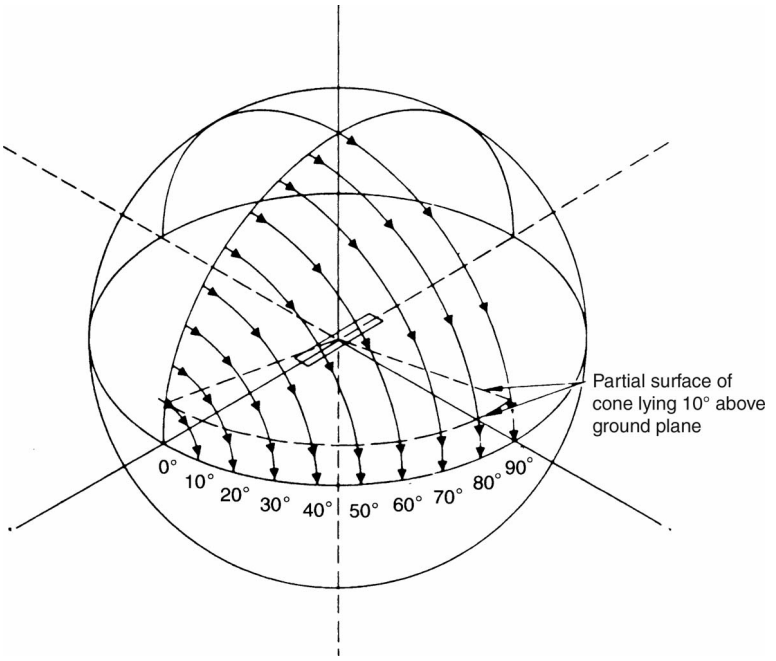


FIGURE 11.41 Field polarization 10° above ground plane.

rapidly downward, then turns increasingly more slowly in the same direction, until at the y-axis it is vertical. At 45°, the polarization is still very nearly vertical instead of nearly 45° as might be expected. The result of this unequal rate of rotation of the polarization vector is that, when the slots are placed on a cone, they must rotate at an uneven rate with their position around the cone in order for them to all have the same polarization directly ahead. It can be shown that, in the quadrants, the slots must be oriented so that their projections on the endfire view are also parallel with the projections of the principal axis slots. This requirement is due to the fact that the far-field polarization of the E-field of a slot is always perpendicular to the projection of the long dimension of the slot onto a plane perpendicular to the line-of-sight.

To obtain a picture of how the slots oriented to favor endfire operation would look from other angles in space, a paper cone was made with a number of slots drawn on it (Fig. 11.42). When the cone is viewed from endfire, the projections of all these slots are parallel as desired (see Fig. 11.42a). When the slots are viewed from a broadside position nearest the axial slots, they are predominantly lined up in an approximately axial fashion (Fig. 11.42b); hence, cross-polarization would not be too much of a problem. However, when the slots are viewed from the broadside region nearest the transverse slots, they are not properly lined up at all (Fig. 11.42c). With this slot arrangement, large amounts of the available power would go into cross-polarized lobes, and the effective aperture would be much smaller than the projected area of the cone in this direction.

There are a number of ways in which radiators providing variable polarization might be mounted on a cone, but complete symmetry cannot be maintained with a minimum number of elements or with simple control functions for beam formation and steering. An arrangement that simplifies the control task has the slots located

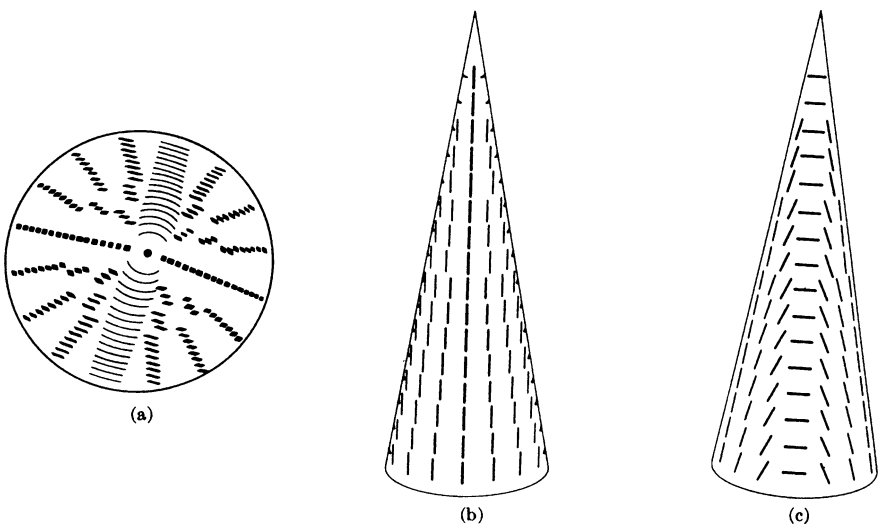


FIGURE 11.42 Conical array designed for endfire. (Courtesy Hughes Aircraft Co.)

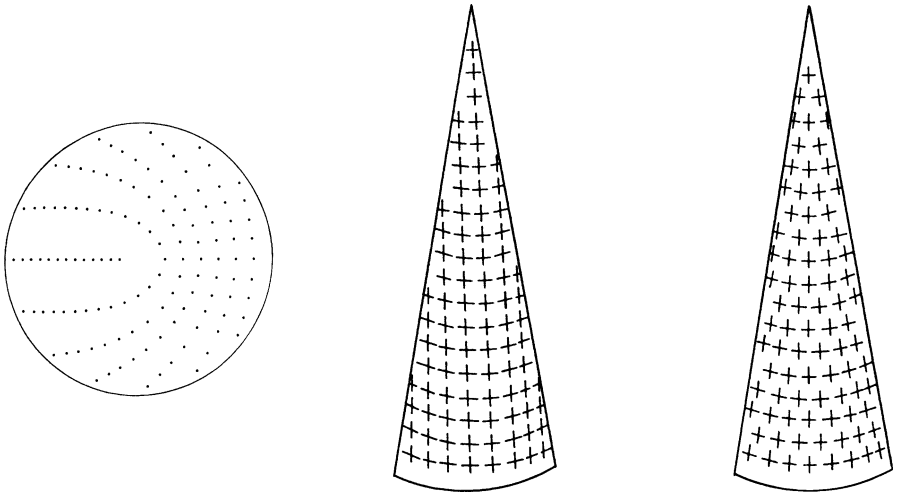


FIGURE 11.43 Cone with elements on rings.

on rings at fixed interelement spacings about each ring, and oriented along both generatrices and circumferential circles (see Fig. 11.43). Only the radial and circumferential components of the excitation must be computed, and the phase computation is simple (Kummer, 1972).

The unusual geometry of the cone requires special consideration of the coordinate system used to represent the patterns. It has been found that in some coordinate systems it is impossible to properly define the two principal planes of the beam when they are scanned to the nose-fire position (Howard et al., 1969). Hence, a special coordinate system was devised that follows the peak of the beam as it is scanned (Howard et al., 1969). The major requirement for the new coordinate system is that two fixed planes of the new system intersect the main beam of the antenna pattern at right angles. This requirement is satisfied by a variable spherical coordinate system, with angular coordinates ϕ' and θ'. The system is positioned so that $\phi' = 0$, $\theta' = 90^\circ$ corresponds to the main beam pointing direction $\phi = \phi_1$, $\theta = \theta_1$. This variable coordinate system can be related to the fixed conventional coordinate system through the transformations presented in Kummer et al. (1971). In this reference,

$$\cos\theta = \cos\theta' \sin\theta_1 + \sin\theta' \cos\phi' \cos\theta_1 \tag{11.76}$$

$$\sin\theta = \sqrt{\sin^2\theta' \sin^2\phi' + (\sin\theta' \cos\phi' \sin\theta_1 + \cos\theta' \cos\theta_1)^2} \tag{11.77}$$

$$\sin\phi = \frac{\sin\theta' \sin\phi'}{\sin\theta} \tag{11.78}$$

$$\cos\phi = \frac{\sin\theta' \cos\phi' \sin\theta_1 - \cos\theta' \cos\phi_1}{\sin\theta} \tag{11.79}$$

The transformations have little effect on the pattern representations for broadside beam pointing directions (θ_1 near 90°), but have considerable effect for endfire beam pointing directions (θ_1 near 0°).

Another transformation is necessary to preserve the polarization of the test point used to perform the pattern plots. The test point polarization must be fixed in the Cartesian frame of the physical antenna cone. Again using Kummer et al. (1971), the new polarizations are related to the old polarizations through

$$\underline{l}_{\theta'} = \frac{\underline{l}_{\theta}(\cos\theta\cos\phi\cos\theta_1 + \sin\theta\sin\theta_1) - \underline{l}_{\phi}\sin\phi\cos\theta_1}{\sin\theta'} \tag{11.80}$$

$$\underline{l}_{\phi'} = \frac{\underline{l}_{\theta}\cos\theta_1\sin\phi + \underline{l}_{\phi}(\cos\theta\cos\phi\cos\theta_1 + \sin\theta\sin\theta_1)}{\sin\theta'} \tag{11.81}$$

For convenience, the new coordinates were renormalized as $\phi = \phi'$ and $\theta = \theta' - 90°$ so that the main beam peak is always centered at $\phi = 0°$, $\theta = 0°$.

Using these modifications to the coordinate system, a series of patterns was calculated for a crossed-slot configuration that partially filled a cone. The slot arrangement consisted of six rings of crossed slots with 24 slots per ring. The two arms of each crossed slot were fed in such amplitude and phase that linear polarization of the proper orientation resulted at the peak of the beam. The large end only of the cone was filled with elements, and the ring nearest the tip was 10 wavelengths from it. Total length of the cone was $12.2\lambda_0$ and the base had a diameter of $4.4\lambda_0$. The elements thus extended only ~20% of the way from the base of the cone to its tip, and the computer patterns only present a rough indication of those of a completely covered cone (Bargeliotes and Villeneuve, 1975; Bargeliotes et al., 1976, 1977).

In the calculations, each element was weighted by the gain that it has in the direction of the peak of the beam. It was determined that this weighting yields the highest S/N ratio when the array is operating in the receiving mode. Figure 11.44 presents a series of patterns computed for a beam steered 20° off the noise-fire position ($\theta_1 = 20°$. These patterns represent only the desired component of polarization. There is a sizable cross-polarized component at some angles. The asymmetry of the conical geometry impedes a straightforward presentation of the antenna patterns, as evidenced by the complexity of the set of pattern plots presented in Figure 11.44. Pattern computations were next made for elements arranged in a staggered configuration similar to that shown in Figure 11.43. Sixteen elements per ring were assumed for the initial computation. With that number of elements in each ring, it was estimated that 10 rings would fit on the cone. Thus, a total of 160 elements "filled" the large end of the cone. The small end of the cone was then left "empty". The diameter of the base of the cone was 4.724λ and its total length was 13.335λ. The spacing between rings was $0.45\lambda_0$ to prevent grating lobes. The 10 rings of elements thus covered the lower four wavelengths of the cone. In the nose-fire direction, the projected radius of the empty area was over twice as large as the projected thickness of the annulus of the filled area.

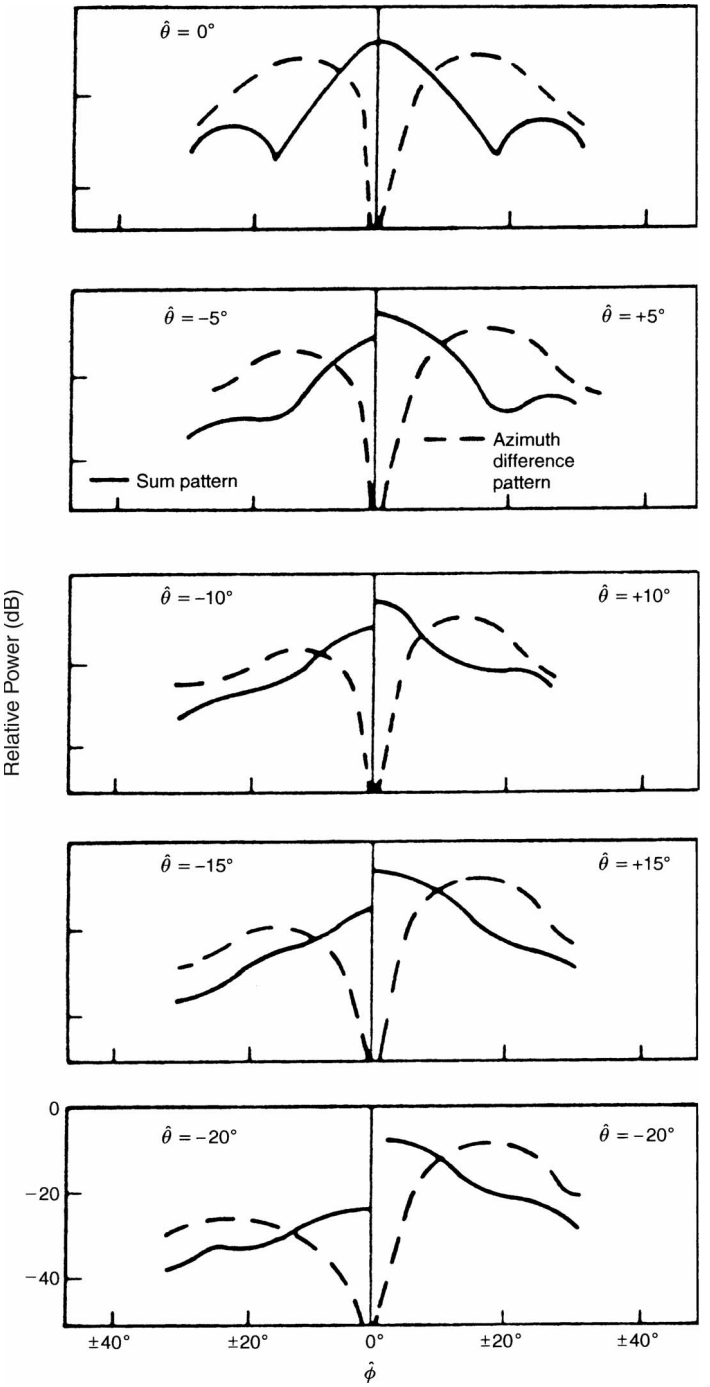


FIGURE 11.44 Antenna patterns optimized for $\theta = 20^{\circ}$.

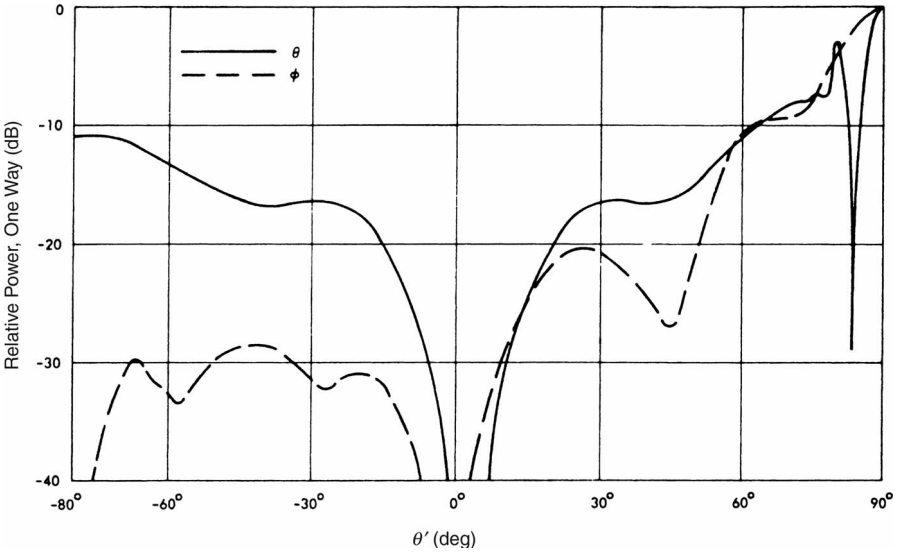


FIGURE 11.45 Patterns for staggered element array.

The broadside patterns of an array of this type will be quite normal because the visible surface of the array at any one point is approximately rectangular. Further, the majority of the slots are "seen" from favorable angles in regard to polarization.

In the nose-fire direction, however, a marked difference in beamwidths for the two principal planes appears, as can be seen in Figure 11.44. Three reasons were determined to explain this difference:

1. The weighting applied to the elements tends to "turn off" the elements that are being viewed from an unfavorable angle. Slots in two areas of the annulus are excited at a much reduced level. This different excitation tends to create an interferometer effect in one principal plane while the effective width of the useful aperture in the other plane is reduced to considerably less than the full diameter of the base.
2. The geometry of the cone is such that when the slots are oriented in a position to optimize the polarization of the signal in the nose-fire position, they are no longer optimum at any other angle. In the process of computing a pattern it is necessary to fix the polarizations of the slots so that there is no cross-polarized component at the peak of the beam. At other angles, more or less energy will go into this component depending on the steepness of the cone and the direction of the cut for the computed pattern. These two factors are related to the rapid change in polarization angle in the radiated pattern of a slot when viewed from a point close to either null in that pattern. This rapid change in polarization further aggravates the interferometer effect.
3. The large empty area in the center of the "working" annulus is in essence aperture blockage that contributes also to the interferometer effect, making it somewhat worse than in either of the other situations alone.

11.5.1.3 Projective Synthesis. Because of the effects of the conical surface upon element patterns, no direct synthesis technique is feasible. The study of depolarization in Section 11.5.1.2 utilized projection; this method proved successful in conical array synthesis. A virtual planar array, with the desired size and amplitude tapers, is placed normal to the desired conical array beam. Then each planar array linearly polarized element is projected onto the conical surface including the rotation of polarization, amplitude, and phase (Kummer et al., 1973; Bargeliotes et al., 1977).

The projective synthesis result may, if necessary, be improved by an iterative method that includes a rigorous calculation of element patterns on a cone. These calculations are discussed briefly in Section 11.5.1.4.

To increase array efficiency, the element patterns can be broadened by dielectric loading (Villeneuve et al., 1975, 1982).

Quantization effects of digital phasers and of digital polarization rotators have been studied (Bargeliotes et al., 1977). Results are similar to those obtained for planar arrays with phasers. Four-bit polarization rotators are adequate unless cross-polarization levels or difference pattern null depths must be very low.

11.5.1.4 Patterns and Mutual Coupling. As was the case of arrays of elements on a cylinder, arrays of elements on a cone can be attacked either by harmonic series or by asymptotic methods, the latter for portions of the cone where the radius is large in wavelengths. Early work on calculation of near-fields of slots on cones was done by Bailin and Silver (1956); the series include associated Legendre functions and spherical Bessel functions. Considerable effort has been expanded on these calculations, as they are useful for patterns and for mutual admittance (Pridmore-Brown, 1972;

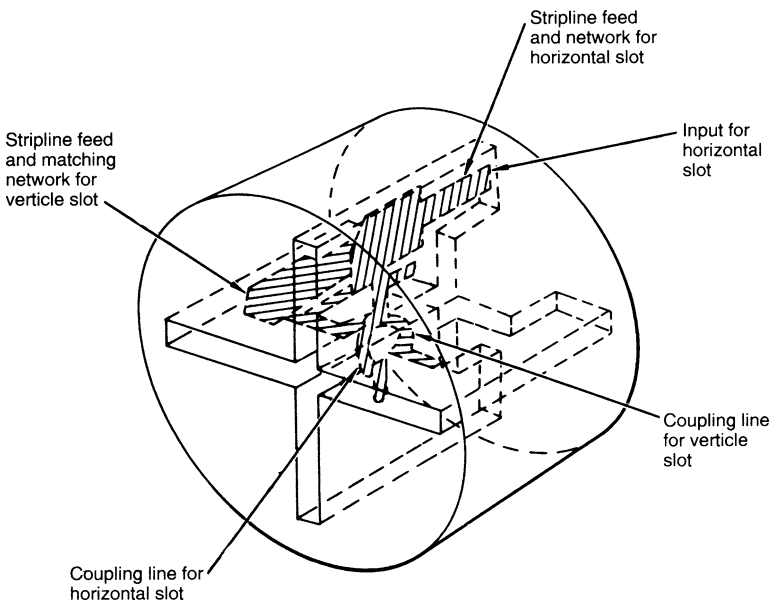


FIGURE 11.46 Crossed-slot element.

Pridmore-Brown and Stewart, 1972). The reader may also wish to refer to work on asymptotic analyses of elements on cones (Pridmore-Brown, 1973; Chan et al., 1977). Damiano uses a computer algebra package to compute zeros of the harmonic functions (Damiano et al., 1996).

11.5.1.5 Conical Array Experiments. Experiments with arrays on cones include a large array on a cone (Munger et al., 1974) and a small array (Thiele and Donn, 1974). The most extensive work was performed at Hughes Aircraft, Culver City. Initial experiments utilized a linear array along a generatrix (Behnke et al., 1972). Following this, an experimental array using crossed slot elements on a cone was constructed and extensively tested (Villeneuve, 1979). The crossed-slot element is sketched in Figure 11.46, and a close-up view of the array is shown in Figure 11.47. Details of this research program are given in Hansen (1981), Bargeliotes et al. (1974, 1977), Bargeliotes and Villeneuve (1975), Behnke et al. (1972), Howard et al. (1969), Kummer et al. (1971, 1973), Villeneuve et al. (1974, 1975, 1982), and Villeneuve (1979).

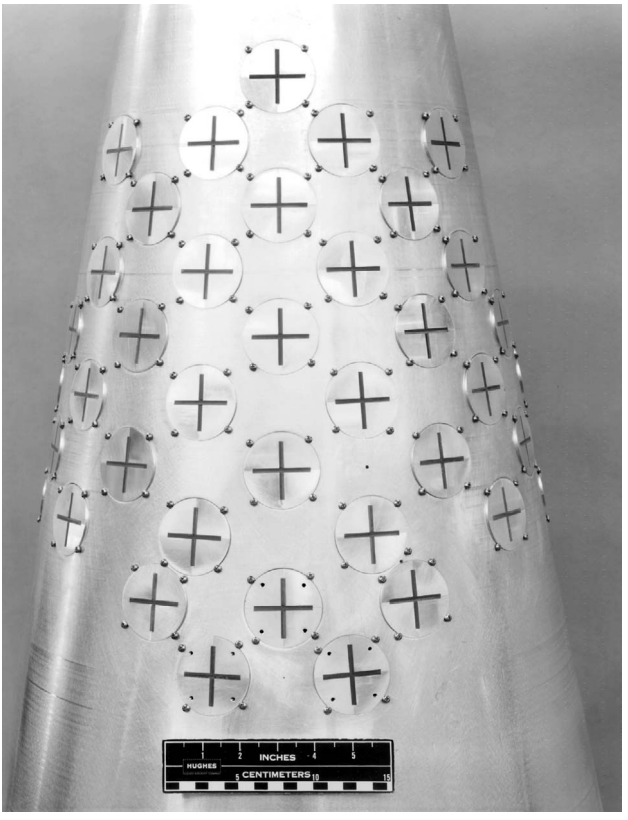


FIGURE 11.47 Crossed elements on a cone. (Courtesy Hughes Aircraft Co.)

11.5.2 Spherical Arrays

A spherical array behaves much like a cylindrical array scanned only in azimuth, but the lattice and depolarization problems are unique to the sphere. Geometry, pattern coverage, and grating lobes in spherical arrays have been investigated by Hoffman (1963), MacPhie (1968), Chan et al. (1968), Sengupta et al. (1968a,b), and Schrank (1972). Thinned arrays on a sphere have been probabilistically evaluated by Panicali and Lo (1969). A spherical patch array is shown in Figure 11.48. (Hockensmith and Stockton, 1977; Stockton and Hockensmith, 1979).

An early work applying the harmonic (modal) series to a slot on a sphere and a cone was given by Bailin and Silver (1956). Much of the mutual coupling work for cones is, with minor changes applicable to coupling of elements on a sphere. Both modal series and asymptotic methods are useful (Hessel et al., 1979).

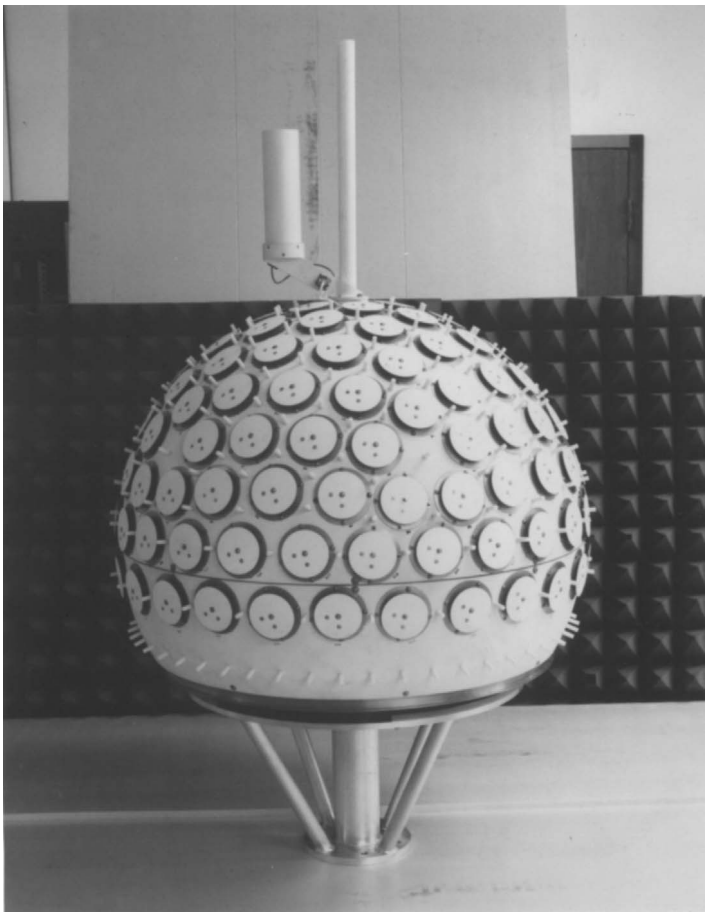


FIGURE 11.48 Switched patch spherical array. (Courtesy Ball Aerospace and Communications.)

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CHAPTER TWELVE

Connected Arrays

Arrays of dipoles, whose collinear ends are connected together by conductors or by impedances, are the subject of this chapter.

12.1 HISTORY OF CONNECTED ARRAYS

The precursor of the connected array was invented by Baum (1970) as a timed array radiating a short pulse along the array in endfire direction. Figure 12.1 (from Baum,

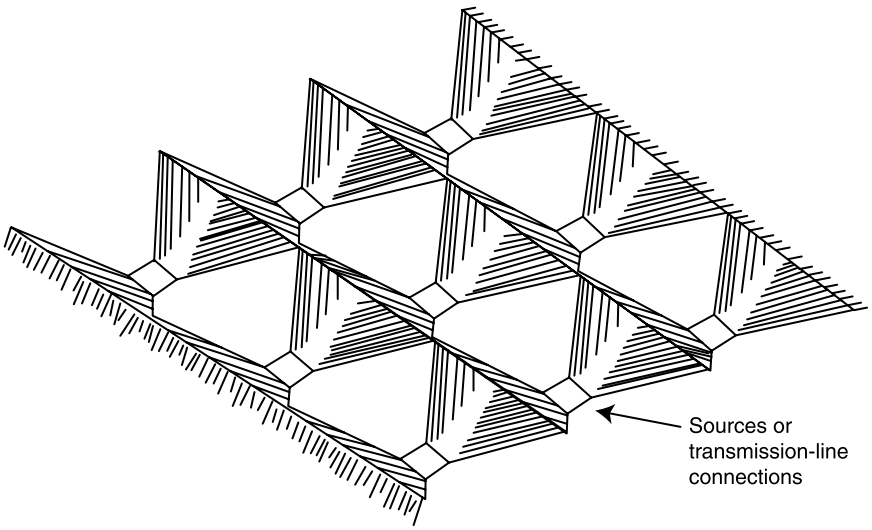


FIGURE 12.1 Connected TEM horn array.

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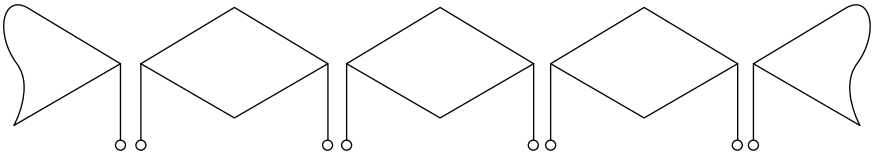


FIGURE 12.2 Linear connected array.

1977) shows connected 3D TEM horns, with timed sources exciting each horn at the feed. In 1979, Inagaki invented the Ichimatsu Moyou (self-complementary) antenna (Inagaki et al., 1979). In its simplest form, this is a checkerboard of triangular elements that are back-to-back, with feeds across the apices. Figure 12.2 sketches a bowtie connected array, shown in linear array form. Inagaki shows that the nominal input impedance at resonance is $60\pi\ \Omega$; one-half of the free space value. McGrath and Baum (1999) reaffirmed the nearly constant input resistance value of $60\pi\,\Omega$.

The Ingaki self-complementary antenna was mentioned by Mushiake (1996), which may have led to connected array work at GTRI (Friederich et al., 2001). A separate path was followed by Munk from FSS theory; his current sheet antenna (Munk, 2000) uses capacitive coupling between dipole ends. Later it will be shown that this is the wrong physics.

12.2 CONNECTED ARRAY PRINCIPLES

The impedance bandwidth of a conventional planar array of dipoles is limited because below dipole resonance ($\ell \sim \lambda/2$) the scan impedance becomes highly reactive and the real part becomes small, even at broadside (no scan). By connecting the collinear dipole (or bowtie) ends, each line of dipoles in the array acts as a single dipole even though there are many feed points. The impedance change of a conventional dipole array is largely avoided; the low-frequency limit occurs when the entire collinear line of dipoles becomes on the order of $\lambda/4$. For a 2D connected array, this results in input resistance of $60\pi\ \Omega$ over a wide frequency range. At the upper end of the frequency band, where each dipole length approaches $\lambda/2$, the performance is somewhat degraded, since at the upper end of the band the dipoles would like to be unconnected. This will be discussed later. For a conventional planar array of dipoles the center frequency is where $d = \lambda/2$. With a connected dipole array the collinear dipole "stick" length at the lowest frequency is roughly $\lambda/4$. Thus the useable frequency range is extended by $2N$, for N collinear elements. This performance is extraordinary.

The limiting factor on connected array bandwidth is of course the metallic ground plane. Because the practical phase bandwidth is $\pm 45^\circ$ (Hansen, 2002), and because of the fundamental limitation of Brewitt-Taylor (2007) on artificial magnetic conductors, the usable bandwidth is constrained. However a low loss magnetic layer between dipoles and ground plane can increase the bandwidth by μ (Metaferrite, from Metamaterials LLC).

12.3 CONNECTED DIPOLE CURRENTS

When elements are connected, three significant questions arise. First, since in scanning there is a phase-step progression from each unit cell to the next unit cell, is it possible to maintain these phase progressions, that is, will arrays of connected elements scan properly, and with reasonable gain and sidelobes? Second, how does current amplitude vary over the array length? Third, how does impedance behave over a wideband?

Since linear arrays of coupled dipoles exhibit many of the characteristics of planar arrays of coupled dipoles, a linear array has been modeled (Hansen, 2004a). The array consists of a wire many wavelengths long, interrupted by feeds at regular nominally half-wave intervals. Alternatively, the array model is a collinear array of half-wave dipoles connected at the contiguous ends. Although for computational convenience the dipoles are thin, the electromagnetic characteristics are not significantly different for thick dipoles, and the model should well represent such fat elements as bowties.

The amplitude and phase of the current distribution along the wire is of crucial importance, so it has been modeled by the Moment Method. A Galerkin Moment Method code using piecewise sinusoidal expansion and test functions was used; mutual impedances were calculated using Carter's method (Hansen, 1972). Six expansion functions were used per dipole; this gave excellent results. Each feed point connected to a load (generator). Results of the code were validated against codes for conventional dipole arrays by inserting large values of resistance half-way between the feeds, thus simulating a conventional dipole array. The mutual impedances and the equation solver used double precision complex, and the program was run on an HP 64 bit UNIX Workstation.

The simulation code allowed results to be calculated for a number of angles with a specified feed spacing, or for a number of feed spacings in wavelengths at a specific scan angle. Parameters calculated were the magnitude and phase of the current distribution along the wire, the impedance at the center feed, and the gain per element (Scan Element Pattern). The code allowed up to 100 feed points representing 600 complex double precision simultaneous equations.

A most important finding was that the load impedance Z_0 must be included in the simulation, in the impedance matrix. In conventional dipole arrays, whether Z_0 is part of the impedance matrix, has little effect upon current distributions and impedance. Of course, in all cases the Z_0 affects power transfer. Without Z_0 in the matrix, the impedance versus frequency is unstable, and the edge wave currents are less predictable. In all results that follow, the simulation includes Z_0 in the impedance matrix.

12.3.1 Simulation Results: Current Phases

The current distribution and the impedance at the central feed point were direct results from the Moment Method procedure. Figure 12.3 shows phase along a 10λ wire with resistances of $10^5\ \Omega$ separating half-wave dipoles, for a 30° scan. In Figure 12.3, the dipoles go from expansion segments 1–6, 6–12, . . . , 108–114, and 114–119. Feeds are located at expansion segments 3, 9, . . . , 117. Phase slope closely matches the 30°

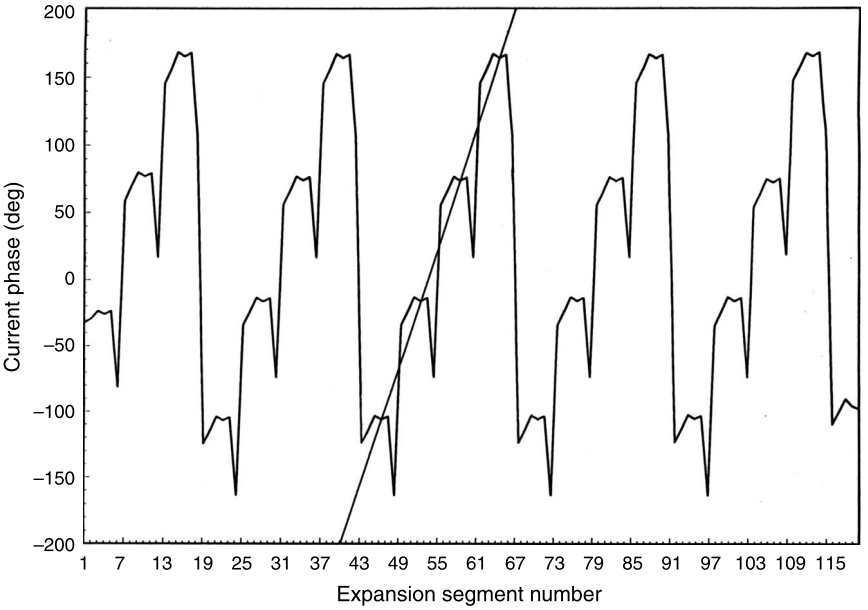


FIGURE 12.3 Twenty collinear dipoles, $L = 0.5$ wv, 30° scan.

scan linear behavior, modulo π of course, as shown by the inclined line. This result is as expected for an array of dipoles. Figure 12.4 shows phase along the connected wire. A close examination shows that the sharp peaks in Figure 12.4 correspond to a scan angle of 30°, however, the phase slope at other expansion segments on the connected dipoles appears to be irregular.

Additional insight is given by examining array behavior where the feed spacing is halved (0.25λ) and the number of feeds is doubled (40), resulting in a 10λ wire again. Figure 12.5 shows phase along the dipole array (dipoles separated by 10^5 Ω). Again the phase slope matched the 30° scan phase perfectly, as it should. Connected wires are shown in Figure 12.6, again for 30° scan. Again the peaks roughly follow the scan phase slope; now the phase varies linearly over all expansion segments, unlike Figure 12.4, where the dipole lengths are evident.

12.3.2 Simulation Results: Current Amplitudes

Figure 12.7 showns amplitude over the half-wave dipole array (connect resistance 10^5 Ω). Expansion segment numbering is as in Figure 12.3. Feeds are located at expansion segments 6, 12, . . . , 114. For each of the 20 dipoles, it is apparent that the current is near zero at the ends with peaks in the middle. This distribution shows minor edge effects, even at broadside (Hansen, 1996a, 1999; Hansen and Gammon, 1995, 1996). These edge effects are not well defined; in dipole arrays they are manifested as oscillations in current amplitude and impedance that are largest

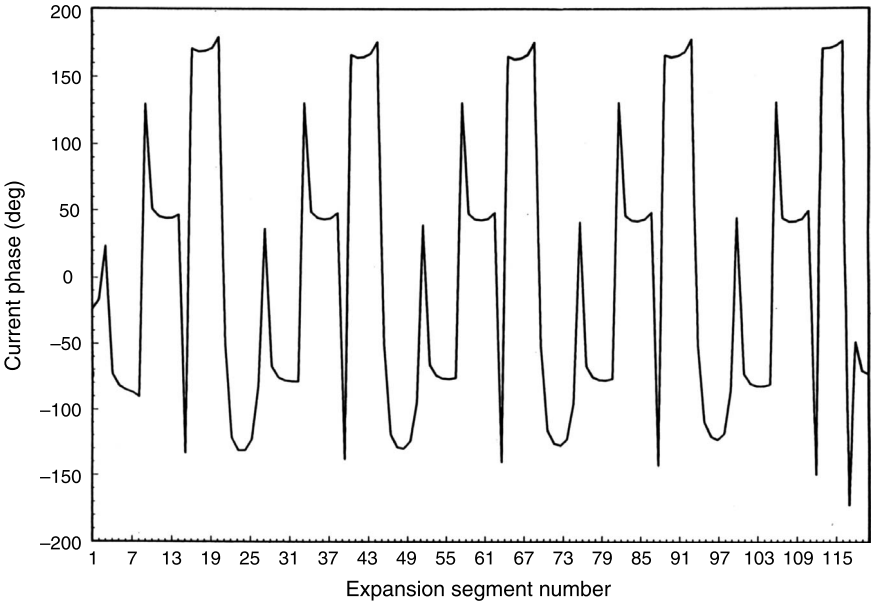


FIGURE 12.4 A plot of a 10 wv wire, feeds at 0.5 wv, 30° scan.

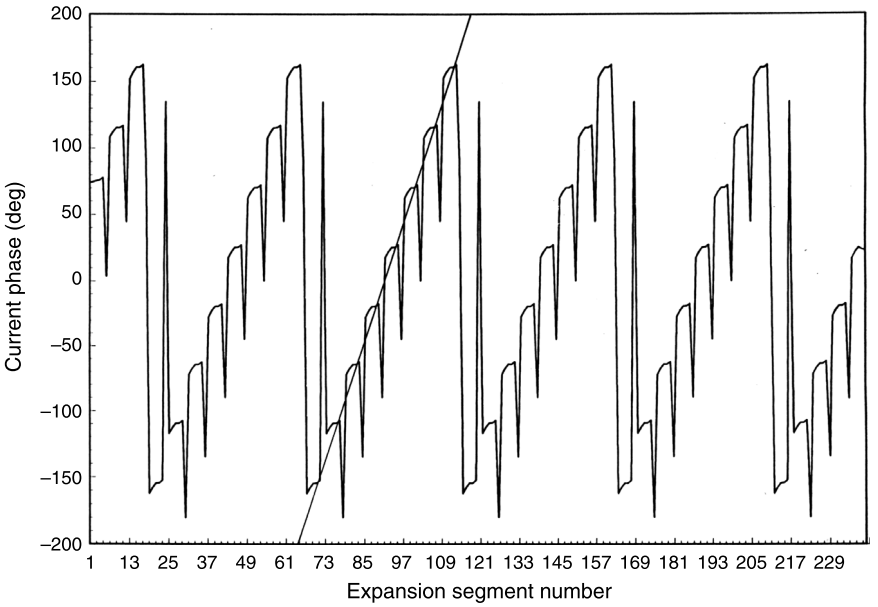


FIGURE 12.5 Plot of 40 collinear dipoles, $L = 0.25$ wv, 30° scan.

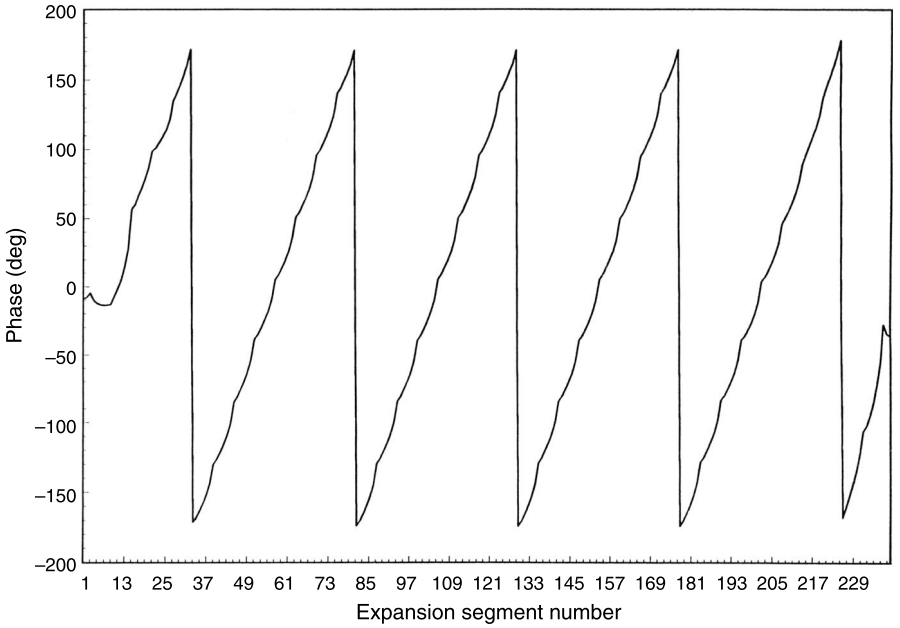


FIGURE 12.6 Plot of a 10 wv wire, feeds at 0.25 wv, 30° scan.

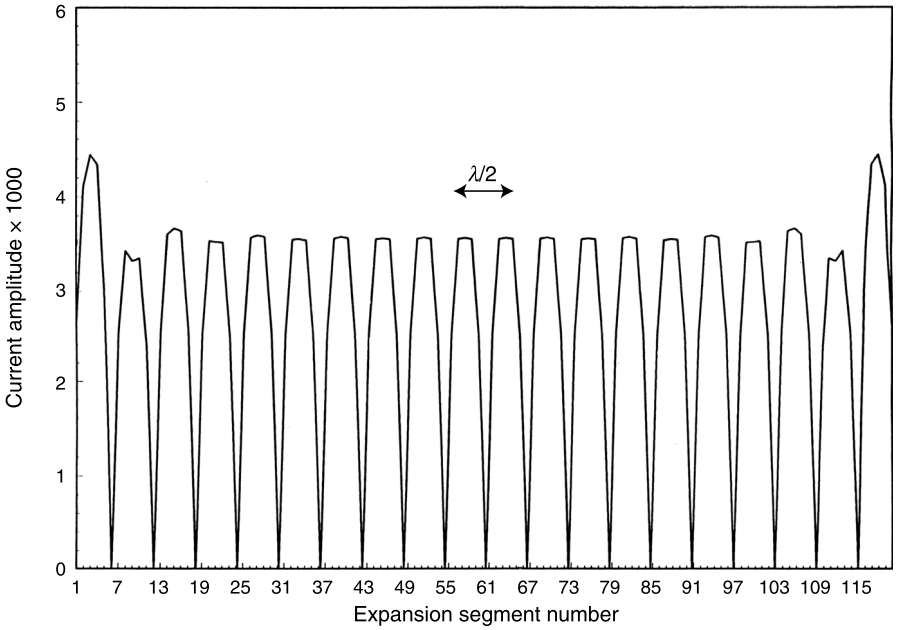


FIGURE 12.7 A plot of 20 collinear dipoles, $L = 0.5$ wv, 0° scan.

at array edges, and decay toward the array center. The current amplitudes result from 1 V applied to each dipole. However, since the impedance changes with frequency, the magnitude of current is of less interest than the shape.

When the dipoles are connected, the 10λ wire with $\lambda/2$ spacing at broadside exhibits a surprising current distribution. Over the entire wire every other "dipole" has very small current (see Fig. 12.8). Current is peaked, and peaks are one wavelength apart. At broadside, with all feeds in phase, a nearly constant current might have been expected, simulating the current sheet "gedanken" of Wheeler. It appears that the connected array current exhibits cuspidation, without attenuation or edge effects. With a 30° scan every fourth $\lambda/2$ section of wire has a small current. Peaks of current are now 2λ apart (see Fig. 12.9). This increase in the period of the currents with scan angle agrees with observations of Hansen and Gammon (1995) and lends some support to the unattenuated edge current hypothesis (Hansen, 2003a). Changing the wire length to 9.75λ and 9.5λ, with 20 feeds, still shows a basic picture of peaks 2λ apart. The intermediate peaks are irregular and partially raised. Thus the edge current "resonance" is not a function of element spacings in free space wavelengths. Because the edge current velocity appears to be that of light, these currents are not related to surface waves. Surface waves are slow waves and are not supported by collinear dipoles.

When twice the number of feeds is used with $\lambda/4$ spacing, the 30° scan current distribution is markedly changed, as seen in Figure 12.10. Now the dipoles go from expansion segments 1–6, 6–12, . . . , 234–239. Now, each dipole has a roughly sinusoidal distribution and the dips are modest. The latter is accentuated with a dipole

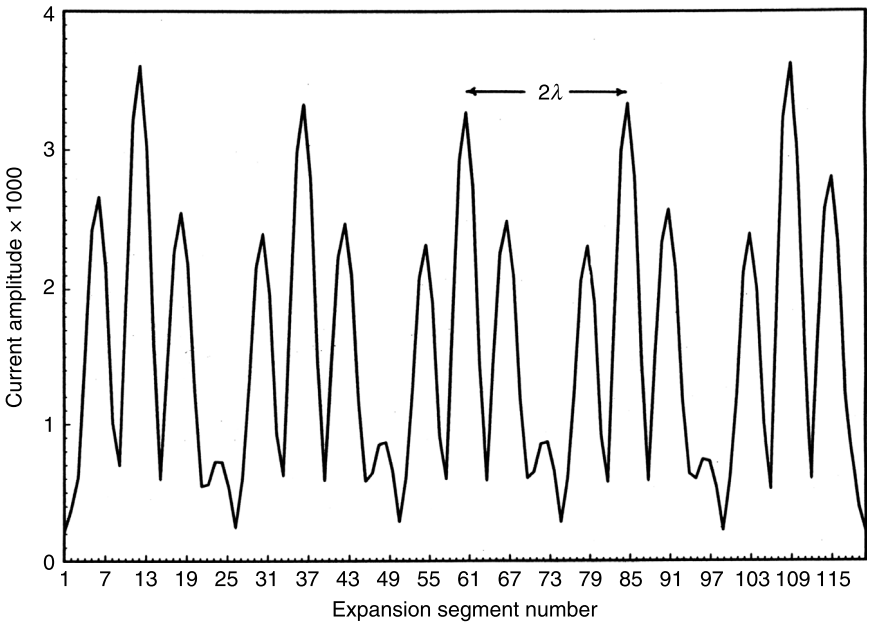


FIGURE 12.8 A plot of 10 wv wire, feeds at 0.5 wv, 30° scan.

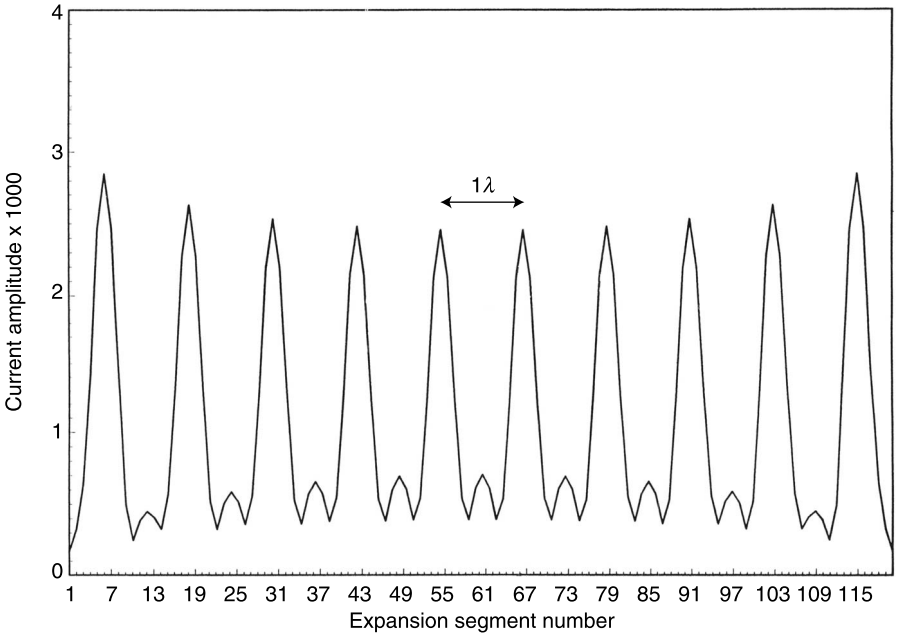


FIGURE 12.9 A plot of 10 wv wire, feeds at 0.5 wv, 0° scan.

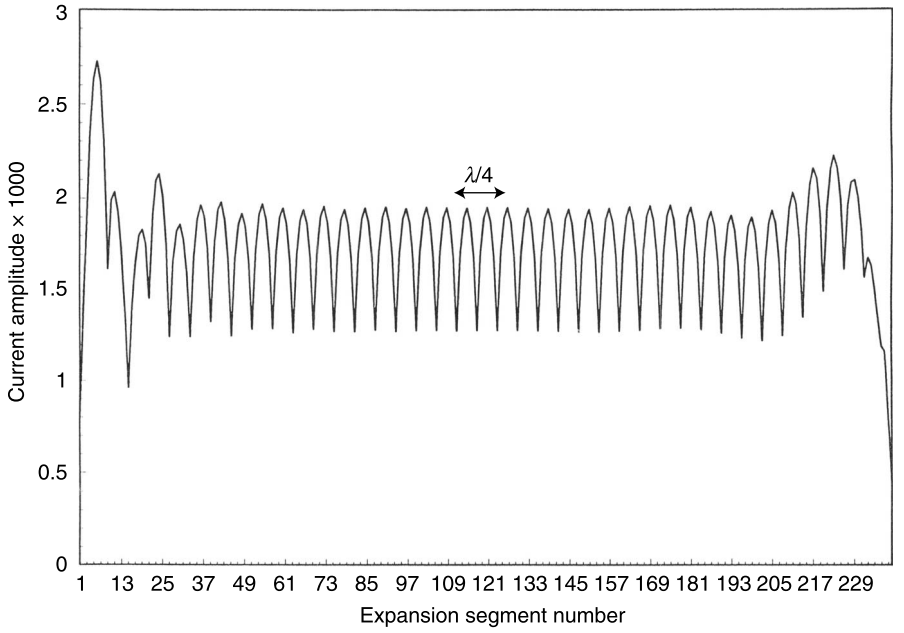


FIGURE 12.10 A plot of 10 wv wire, feeds at 0.25 wv, 30° scan.

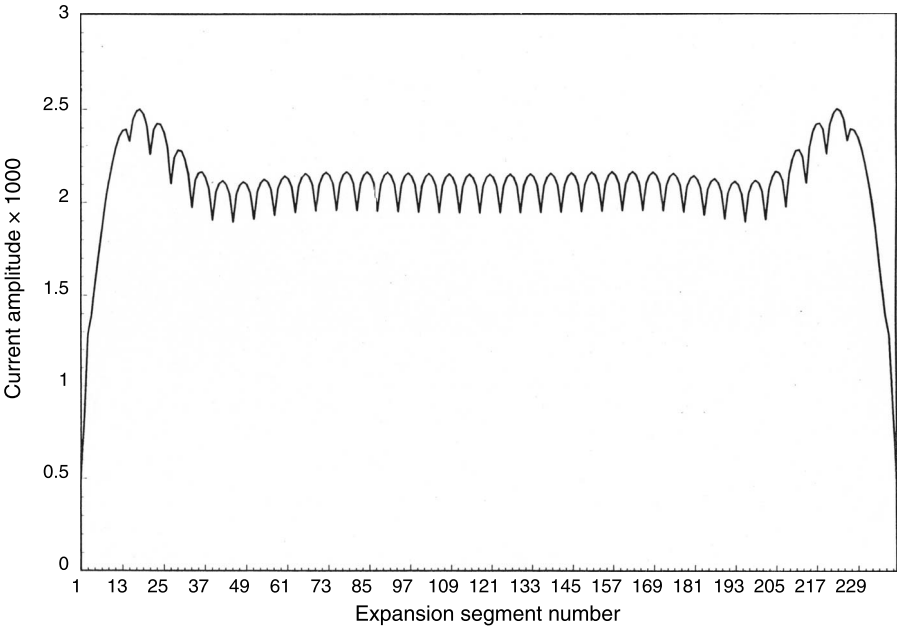


FIGURE 12.11 A plot of 40 connected dipoles, broadside, $L = 0.1$ wv.

spacing of 0.1λ; the current is roughly constant over the 4λ wire (see Fig. 12.11). Thus at low frequencies the continuity of current approximates the current sheet of Wheeler; this improves the low-frequency impedance as will appear later. The edge currents, which are of modest amplitude in dipole arrays (Hansen, 1996a,b), are much stronger in coupled wires; there is no collinear mutual impedance to attenuate the currents.

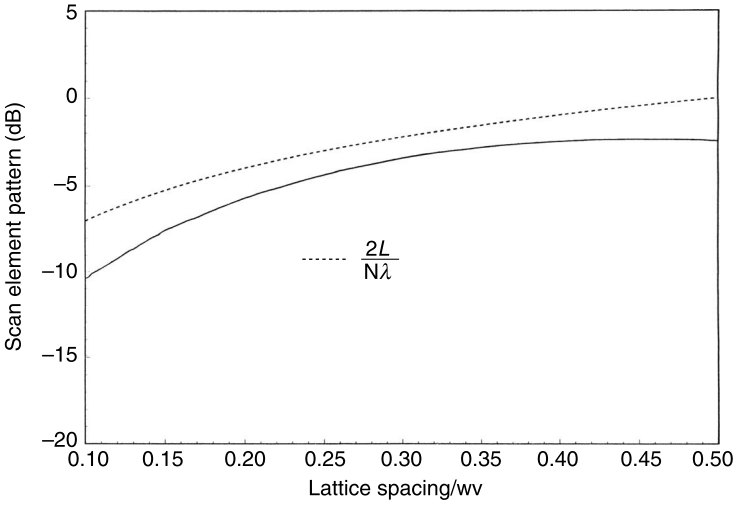


FIGURE 12.12 A plot of 40 connected dipoles, broadside, $Z_0 = 400\,\Omega$.

12.3.3 Simulation Results: SEP

The SEP (gain per element) is shown in Figure 12.12 for the 5 : 1 frequency range. The array is 20λ long at the upper frequency, where the "dipoles" are $\lambda/2$. Also shown is the normalized ideal line source gain, $2L/N\lambda$, where L is the array length. The drop in gain ranging from 3.5 dB at $L = 0.1\lambda$ to 2.5 dB at $L = 0.5\lambda$ is mostly due to impedance mismatch. Note that the term gain used here is that used in industry: Impedance mismatch loss is included. The IEEE definition, which does not include impedance mismatch, is at odds with antenna vendors and users, and with systems engineers.

12.4 CONNECTION BY REACTANCE

Because connecting the dipoles produces a forced resonance in current, it is useful to consider whether a reactance connecting the dipoles would perform better. At half-wave feed spacing, the ideal reactance would be infinite, as isolated dipoles are robust. For small feed spacings, the ideal reactance would be small to allow continuity of current. Clearly, use of capacitive coupling provides the wrong physics; at low frequencies the capacitors tend to uncouple the dipoles, while near half-wave the capacitors tend to connect the dipoles. Both are wrong [see Hansen (2004b)].

The ideal coupling would use Non-Foster circuits. These are active circuits that provide negative resistance, negative inductance, or negative capacitance. All employ positive feedback, so that stability must always be addressed. These are termed Non-Foster as they violate Foster's reactance theorem. These circuits are not new; in the early 1930s Bell Laboratories produced long line amplifiers that introduced a modest negative resistance to partially offset long line loss. These made transcontinental telephony possible, and they were widely used until the microwave towers appeared after WW II. Non-Foster capability was greatly increased with the development of transistors; now the limits imposed by tube capacitances, and so on, were largely overcome.

Non-Foster was first applied to antennas by Poggio and Mayes (1971) using several Non-Foster circuits as loads along the arms of a dipole. Following this was a sporadic collection of paper studies and patents. Practical Non-Foster antenna impedance matching was initiated by Skahill et al. (1998). Use of Non-Foster circuits to connect dipoles in a connected array was suggested and studied by Hansen (2003b, 2004b). Such couplings preserve the middle and lower frequency performance of a connected array, but also restore the gain drop near half-wave that occurs with direct connections. Figure 12.13 shows SEP for a 20×20 array at broadside versus dipole length/λ. A Z_0 of 400 Ω gave the best match. Also shown is the area gain per element. The connected array has a 5-dB drop at the upper band edge. For comparison, the same array with Non-Foster (negative inductor) connections SEP is shown in Figure 12.14. Note the excellent match to the ideal, from 0.05λ dipole length to 0.30λ dipole length. Near $\lambda/2$ the discrepancy is 1.7 dB, because near resonance the connected

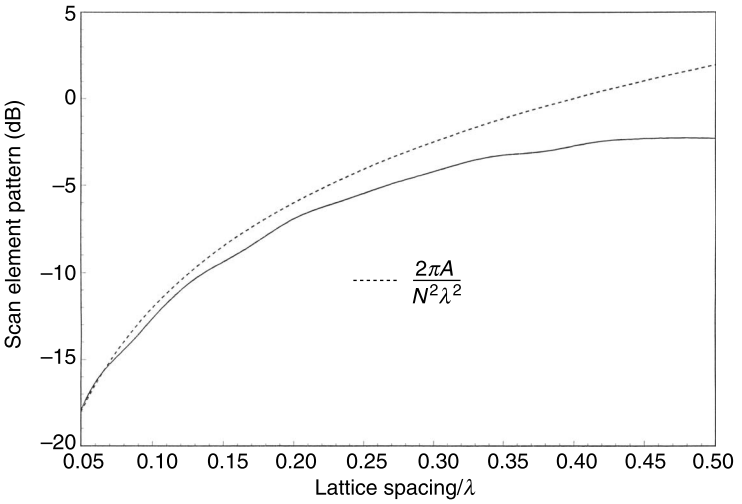


FIGURE 12.13 A plot of 20×20 connected array at broadside, $Z_0 = 400\,\Omega$.

array no longer acts as an inductance. These data show the potential for Non-Foster connected arrays.

In summary, connected dipole arrays offer excellent multioctave bandwidth; they are a good competitor to TEM horn arrays. Non-Foster coupling improves high band performance, but the component cost is a negative aspect.

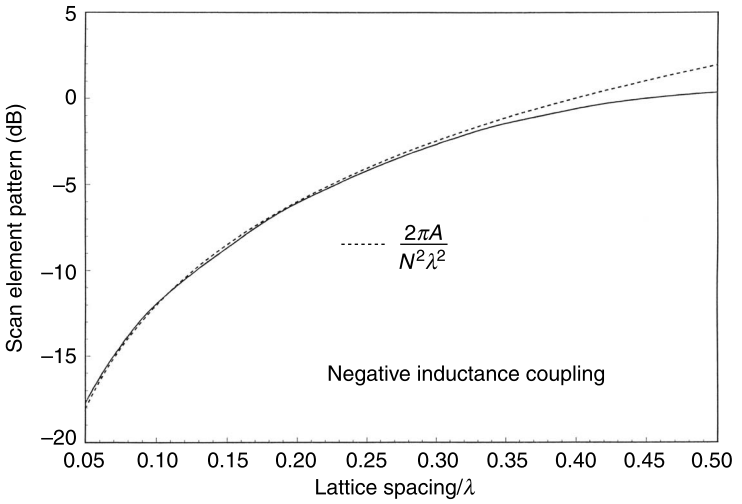


FIGURE 12.14 A plot of 20×20 array, Non-Foster coupling, $L_{\text{match}} = 0.25\lambda$.

12.5 CONNECTED ARRAY EXTENSIONS

Neto and Lee (2005) transformed the planar array of connected dipoles to a planar array of parallel slots. Each slot is excited by an array of short dipoles. Thus the long-slot array is the Babinet equivalent of the connected dipole array. The title of the referenced paper implies infinite bandwidth, but this only occurs for infinitely long slots; of course, the same bandwidth limits apply as for the dipole connected array. Neto and Lee (2006) have given a Green's function analysis of the long-slot arrays. A UHF version was built using four slots with eight balanced feed loops, for a 4×8 array (Lee et al., 2006). With ground plane the bandwidth for VSWR ≤ 2 was 4:1. Use of low loss magnetic materials, such as Metaferrite, next to the ground plane will reduce the array thickness. Since the slots are in a parallel plate region, leaky waves can be excited, and these must be controlled. The magnetic material will help, as reducing height of the parallel plate region tends to reduce leaky waves.

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CHAPTER THIRTEEN

Reflectarrays and Retrodirective Arrays

Reflectarrays are sometimes a replacement for phased arrays and small reflector antennas; they offer a more compact envelope, and simpler circuitry. This chapter provides a brief overview of reflectarrays; more detailed data are in Huang and Encinar (2008) and Bhattacharyya (2006). Retrodirective arrays, both passive and active, have recently become of interest, and are also covered.

13.1 REFLECTARRAYS

13.1.1 History of Reflectarrays

The reflectarray was invented by Malech et al. (1962, 1963), although a patent on array lenses makes a brief mention of a mirror array (Jones et al., 1961). The first reflectarray consisted of a closely packed planar array of open-ended waveguides fed by a horn in front of the waveguide array. Each guide was shorted and the guide lengths were chosen to compensate for the feed-to-guide path; the reflected wave was then a plane wave. Figure 13.1 shows one of their experimental reflectarrays. The next step occurred in the early 1970s with the invention of the spiraphase reflectarray by Phelan. This work is described in a four part series of papers in *Microwave Journal* (Phelan, 1974, 1977). Here the array of open-end waveguides is replaced by an array of four-arm spiral antennas. By controlling the spiral feed points by diodes, the phase of the reflected field can be rotated. This system is inherently wideband, but diode performance restricts spiraphase capabilities. The jump from waveguides to patches was made by Malagisi (1977) in 1976; patch reflection phases were controlled by diodes on the patch edges; the diodes allowed electronic scanning.

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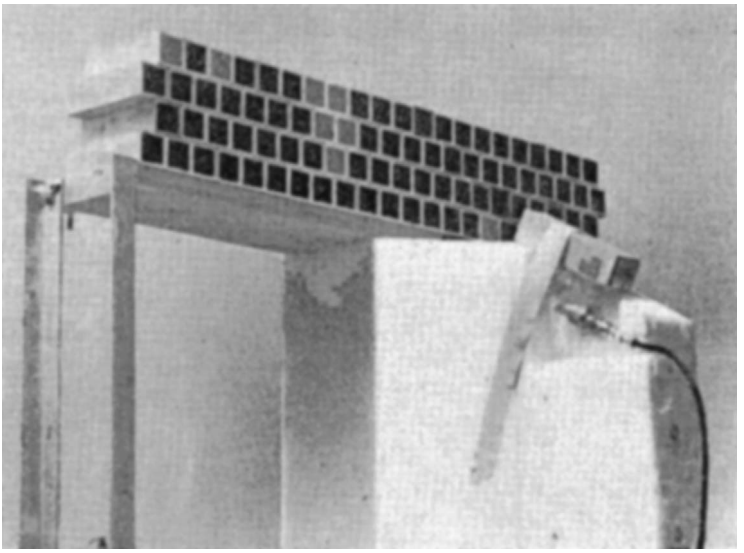


FIGURE 13.1 Waveguide reflectarray.

This work was continued by Montgomery (1978). The next step occurred ~1987 when Munson (1987) devised a fixed-beam patch reflectarray, where each element had a shorted feed line, with the feed line lengths adjusted to collimate the reradiated beam. Work continued on patch reflectarrays, but little was published until the mid-1990s. The most useful of these are discussed in the following sections.

13.1.2 Geometric Design

For flat reflectarray with a feed subtended angle of $2\theta_0$, the f/D is

$$f/D = \frac{0.5}{\tan \theta_0} \tag{13.1}$$

Feeds are usually horns and the electric field pattern is commonly approximated by $\cos^n \theta$. Spillover efficiency is given by

$$\eta_s = 1 - \cos^{2n+1} \theta_0 \tag{13.2}$$

Aperture taper efficiency is (Pozar et al., 1997):

$$\eta_t = \frac{4n[1 - \cos^{n-1} \theta_0]^2}{(n-1)^2 \tan^2 \theta_0 [1 - \cos^{2n} \theta_0]} \tag{13.3}$$

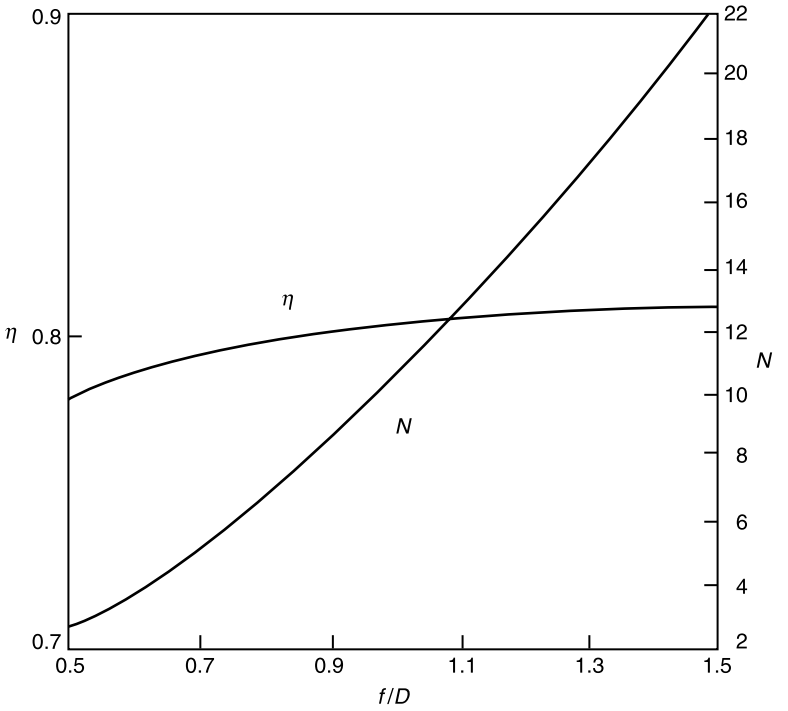


FIGURE 13.2 Efficiency and feed pattern exponent.

Finally, the overall optical efficiency is

$$\eta = \eta_s \cdot \eta_t \tag{13.4}$$

Figure 13.2 shows the peak overall efficiency η, and the feed pattern exponent N, all versus f/D. It is apparent that for a wide range of f/D, from 0.5 to 1.5, the efficiency varies only slightly, from 0.78 to 0.81. This allows the reflectarray designer to choose f/D on the basis of feed choice ($\cos^n \theta_0$) and mechanical convenience. Probably small f/D and low η are the best choices.

13.1.3 Elements

The original reflectarray elements were, of course, open-ended waveguides. Printed circuit elements have largely superceded them. Concentric circular ring elements were developed for FSS (Huang et al., 1994). It required many years for ring elements to transition to reflectarrays. Chaharmir et al. (2006a,b) use double square ring elements, and double cross loops. Printed strip dipoles are also used (Chaharmir et al., 2006c). Patches and strip dipoles have been combined for dual band operation (Shaker et al., 2000).

13.1.4 Phasing of Elements

The direct way of phasing a patch reflectarray starts with calculating the excess path length of an edge element compared with an element at radial distance, r. An edge patch would have *feed line length* $= 0$; a center patch would have feed line length of

$$1/2\left[\sqrt{D^2/4+f^2}-f\right] \tag{13.5}$$

while a patch at radial distance r from the reflectarray center needs a feed line length of

$$1/2\left[\sqrt{r^2+f^2}-f\right] \tag{13.6}$$

The factor of $1/2$ occurs as the feed line and is shorted, while the wave is reflected. In the simplest implementation, all feed line lengths are decreased by multiples of λ until line length is $<\lambda$. This produces a phasing bandwidth, as shown in Figure 13.3. Figure 13.4 sketches a patch reflectarray phased by shorted feed lines.

A second technique phases the reflectarray by rotating the patches, analogous to the spiraphase reflectarray (Huang and Pogorzelski, 1998; Chaharmir et al., 2009). Figure 13.5 sketches this example.

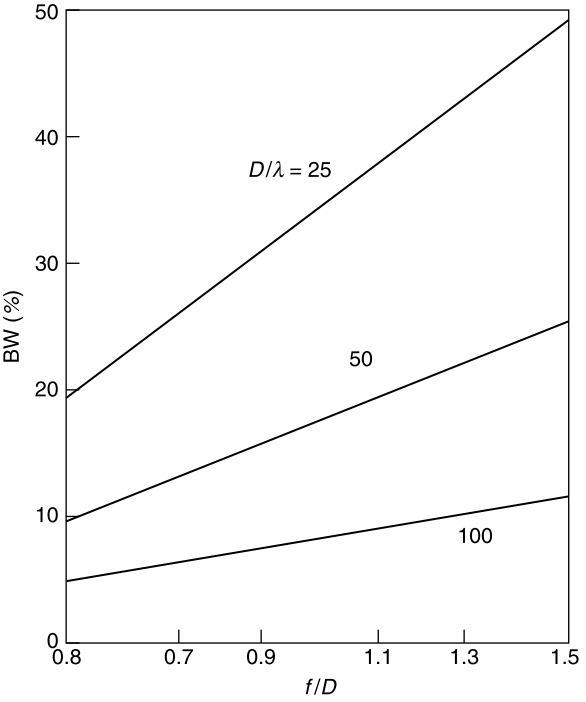


FIGURE 13.3 Bandwidth using phase shift.

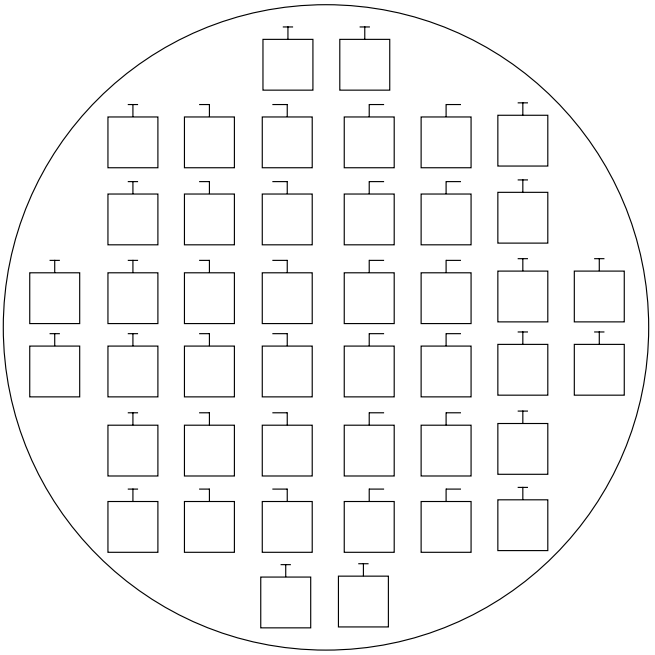


FIGURE 13.4 Phasing by line length.

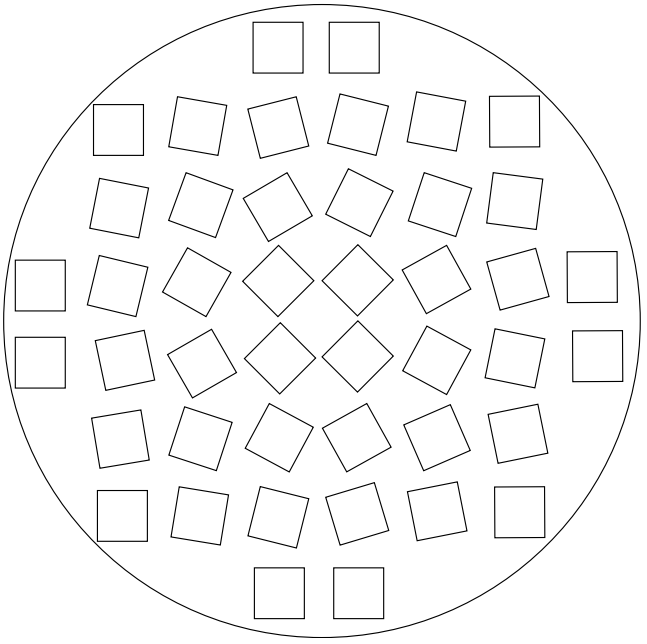


FIGURE 13.5 Phasing by patch rotation.

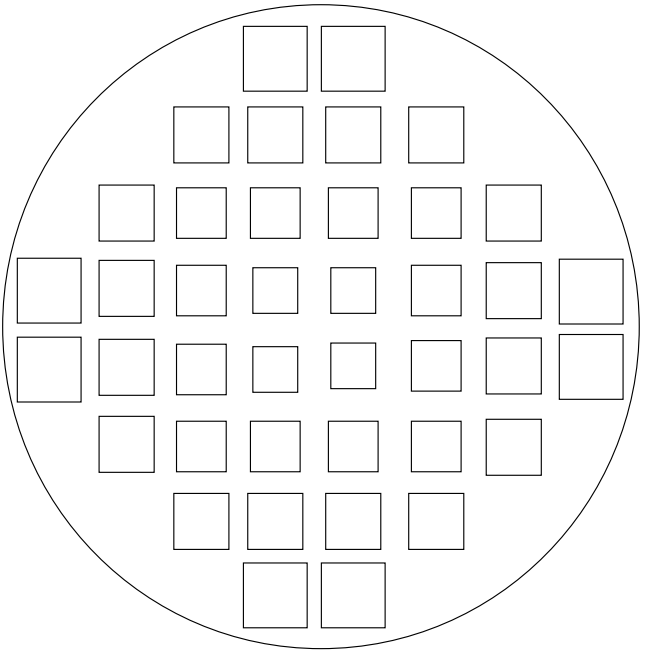


FIGURE 13.6 Phasing by patch size.

A third technique uses patch size to control phase (Encinar, 2001). Here the patches do not have feed lines; the phase of the reflected field is related to the patch size. A variable patch size reflectarray is sketched in Figure 13.6. Phase versus patch size is given by Pozar and Metzler (1993). A Floquet unit cell analysis has been given by Tsai and Bialkowski (2003). Phase characteristics were investigated by Bialkowski and Sayidmarie (2008).

A novel technique uses patches of equal size, with each patch fed by a slot in the ground plane (Chaharmir et al., 2003). Varying the slot lengths varies the phase of the coupling.

All of these involve phase; if the bandwidth of Figure 13.3 is not comfortably larger than the patch bandwidth, time delay can be used. Here each patch has a shorted feed line of length equal to one-half of Eq. (13.6). This equation provides beam collimation independent of frequency.

Cross-polarization can be significantly reduced by placing the feed lines and shorts symmetrically (Chang and Huang, 1995).

Polarization may be changed by rotating the feed of the array (Wu et al., 2006).

13.1.5 Bandwidth

Two factors control reflectarray bandwidth (BW): element and path length BW. With a flat reflectarray, the path length from the feed phase center to each element varies (for

broadside radiation). The phase error is $(k - k_0)(R - f)$, where the wave number is k. Allowing the phase error to be π at both upper and lower frequencies gives (Pozar, 2003):

$$\pi = (k - k_0)(r - f) = (k_0 - k_1)(r - f) \tag{13.7}$$

Here, $r = \sqrt{D^2/4 + f^2}$. The bandwidth is then:

$$\text{BW} = \frac{1}{\dfrac{D}{\lambda}\left[\sqrt{f^2/D^2 + 1/4} - f/D\right]} \tag{13.8}$$

Figure 13.3 shows *bandwidth versus* f/D for $D/\lambda = 25$, 50, *and* 100. For example, with $f/D = 0.5$ for $D/\lambda = 25$ the bandwidth is 19%. Clearly, this exceeds most patch bandwidths. Large reflectarrays (large D/λ) may not allow sufficient bandwidth using phase; the simple solution is to use lengths of transmission line to terminate the patches. This, just as in a phased array, eliminates path length bandwidth restrictions.

Patch reflectarray bandwidth is usually limited by the element (patch) bandwidth. For a resonant patch over a dielectric slab of thickness t and dielectric constant ε_r, the bandwidth is approximately:

$$\text{BW} \simeq \frac{4t}{\lambda_0\sqrt{2\varepsilon_r}} \tag{13.9}$$

Several methods exist for BW improvement. The simplest uses a low-loss magnetic substrate instead of the dielectric slab. Hansen and Burke (2000) showed that a substrate with modest μ and low ε would provide a significant BW increase over the patch with dielectric substrate. For example, a patch with $\varepsilon = 1$, $\mu = 20$ provides more than three times the BW of a patch with $\varepsilon = 1$, $\mu = 1$. A low-loss magnetic material is Metaferrite, which consists of a stack of very thin laminated sheets, each with a pattern of magnetic materials (proprietary product of Metamaterials LLC, Austin, TX).

Another technique for increasing BW uses patches with one or two parasitic patches above the primary patches (Bhattacharyya, 1999). This reference appears to be the first two-layer patch reflectarray work. Of course, two-layer patches for direct radiating arrays have been known for years. Books that discuss parasitic patch design include James and Hall (1989), Zurcher and Gardiol (1995), Sainati (1996), Lee and Chen (1997), Wong (2003), and Kumar and Ray (2003). Related papers are Encinar (2001), Tsai and Bialkowski (2003), and Encinar and Zornoza (2003).

13.1.6 Reflectarray Extensions

Reflectarrays can be configured for beam scanning and amplification. Just as a phased array is scanned, the reflectarray is scanned by insertion of phasers (phase shifters) or delayers (time delay) in the element feed lines. Typically, diode switches (Martynyuk

et al., 2003) or MEMS (Legay et al., 2003) are used to select one of several lengths of line to provide the requisite phase shift. This complexity and loss removes much of the advantage of the reflectarray. Another configuration uses varactor diodes to connect a split patch thereby controlling the phase (Hum et al., 2005). An exotic scheme uses a high-resistivity silicon slab under the patch elements; optical excitation of the slab controls a photoinduced plasma and thus controls the phase (Chaharmir et al., 2006d). Still another scheme uses a hybrid and diode mixers to steer the beam (Cabria et al., 2006).

Another configuration uses patch elements, each contains a slot that couples to delay lines underneath (Carrasco et al., 2008).

When dual polarized elements with separate feeds are used in the reflectarray, low-power amplifiers may be applied at each element, receiving a signal at one port (polarization), amplifying it, and transmitting it out the second port (cross-polarization) (see Robinson et al., 2000). Amplifier gain must be comfortably less than the element cross-polarization level in order to prevent instability. Alternatively, the incurring modulation can be removed, and applied to a new carrier frequency (Cutler et al., 1963).

Cassegrain reflectarrays have also been used (Pozar et al., 1997). A Gregorian system can also be used, where the subreflector is hyperboloidal instead of paraboloidal. Due to aperture blockage, feeds are sometimes offset using Cassegrain or Gregorian subreflectors or even a direct offset feed. For all these geometries, the usual design constraints exist. The direct offset feed exacerbates the phasing bandwidth problem, but for modest D/λ the bandwidth may be limited by the element bandwidth (see Chapter 5).

Multiple feeds can provide multiple beams, or a shaped-beam for particular earth coverage (Pozar et al., 1999; Zornoza and Bialkowski, 2003; Encinar and Zornoza, 2004; Carrasco et al., 2008).

Shaped-beam coverage can be provided by adjusting the element phases (Encinar and Zornoza, 2004), and then in a second step adjusting element sizes to improve the BW while maintaining the desired phase at each patch.

13.2 RETRODIRECTIVE ARRAYS

13.2.1 History of Retrodirective Arrays

The retrodirective array (Fig. 13.7) was invented by Van Atta (1959). Each array element is connected to another array element by transmission lines of equal lengths. Almost immediately a difficulty arose, when Sharp and Diab (1960) showed that the backscatter RCS is comparable to the reradiated wave. This difficulty is readily overcome by changing polarization, frequency, or modulation, or by adding amplification. Several of these were explored for communications satellites by Hansen (1961). Steady interest in these arrays resulted in a March 1964 special issue of the IEEE AP Transactions. Van Atta arrays are sometimes called "self-focusing"; a better term would be "self-collimating". Sichelstiel and Waters (1964) discuss an array

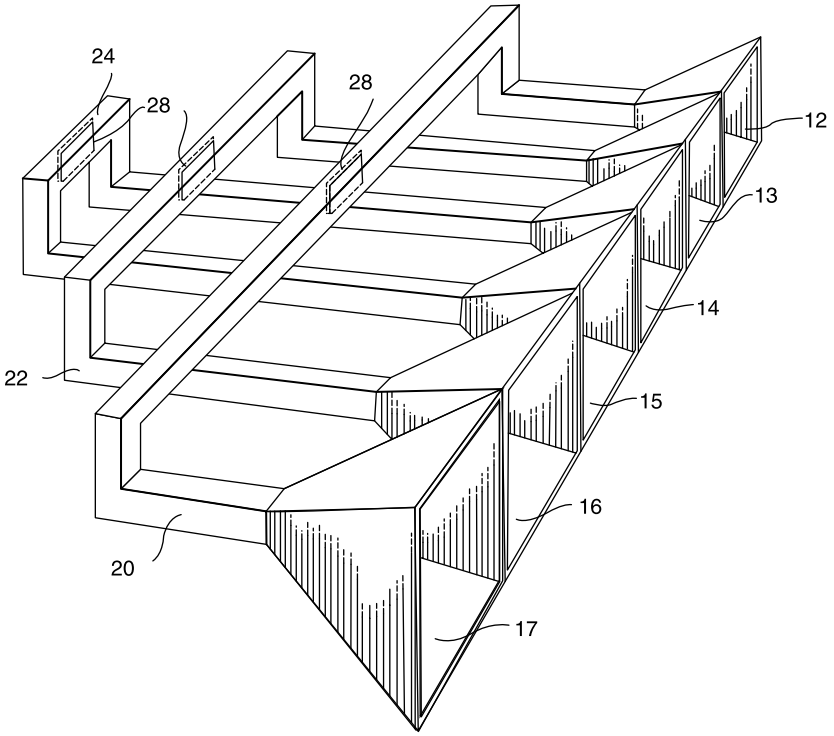


FIGURE 13.7 Van Atta retrodirective array.

with frequency shift. Of course, with any change in frequency the BW must be considered. Andre and Leonard (1964) consider a retrodirective array for satellite use; both frequency offset and polarization change are considered. This requires elements that handle both polarizations. A more common technique, which trades element complexity for circuit complexity, is phase inversion at each antenna element. This inversion allows the retrodirective array to have any shape, although grating lobes must always be considered. Gruenberg and Johnson (1964) use a balanced mixer, where the LO frequency is close to the signal frequency. The second harmonic of the LO and the signal frequency are mixed; the lower sideband frequency has the inverted phase [see also, Pon (1964)]. A tunnel-diode has also been used instead of the mixer (Rutz-Phillipp, 1964). Use of dual mixers (see Section 13.2.2) was pioneered by Skolnik and King (1964).

13.2.2 Recent Progress

As sometimes happens in science, a large time gap occurred after the 1964 papers. Tausworthe and Chernoff (1979) introduced a chain-phase conjugation circuit where one antenna element is used to generate a phase reference, which is then passed onto the next element; each element has a phase conjugation circuit using

mixers [see also Luxey and Laheurte (1999)]. However, the complexity outweighs any advantages. Another variation utilizes an incoming signal plus an incoming pilot carrier, using mixers (Brennan, 1989). The mixer-phase inverter previously described (see Fig. 13.8) is susceptible to reflection at the antenna terminals due to impedance mismatch, and to leakage from the input port of the mixer to the output port. The effect of leakage has been assessed by Toh et al. (2002). Major leakage can be avoided thru use of two mixers (Chang et al., 1998). Good RF–IF isolation is provided by a dual channel-phase conjugator (see Fig. 13.9). The delay line causes RF cancellation at the RF–IF port, and the two channels have a 180° delay at the RF–IF port, thereby providing good IF isolation (Miyamoto et al., 2000, 2001, 2003).

Subharmonic mixing avoids the problem of an LO frequency that is too high. In this arrangement, dual mixers are used with the LO frequency one-half of the signal frequency (Lim et al., 2005; Lim and Itoh, 2008). As sketched in Figure 13.10, the first mixer output is connected to the second mixer input through a low pass filter that removes the LO leakage and the incoming RF signal.

Another scheme to avoid using an LO twice the signal frequency uses a double-balanced harmonic mixer (Brabetz et al., 2001) (see Fig. 13.11). The LO signal is canceled in the rat-race hybrid, providing good LO–IF isolation. The LO is at the signal frequency, easing the LO requirement. The RF leakage thru the two mixers is in phase; this can be remedied thru use of two harmonic mixers in a circuit that cancels both RF and LO signals (see Fig. 13.12).

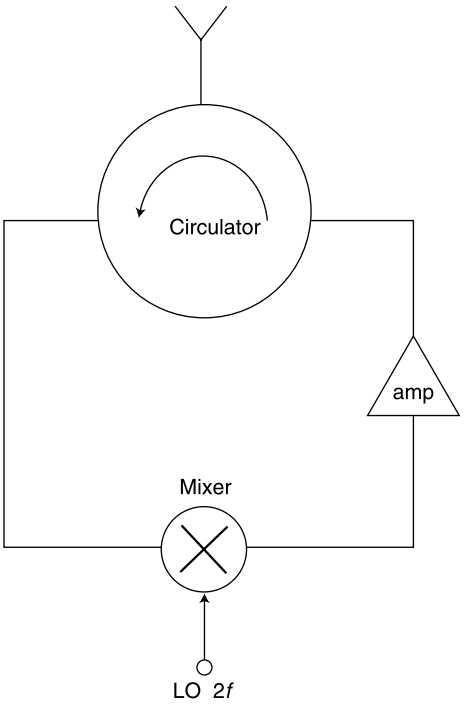


FIGURE 13.8 Rudimentary mixer-phase inverter.

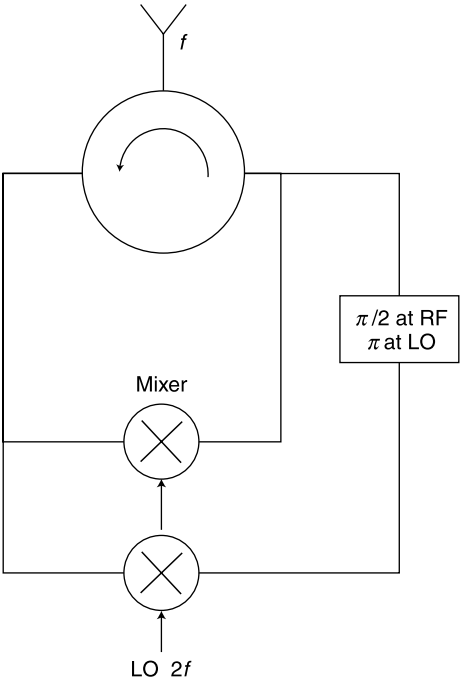


FIGURE 13.9 Mixer-phase inverter with delay line.

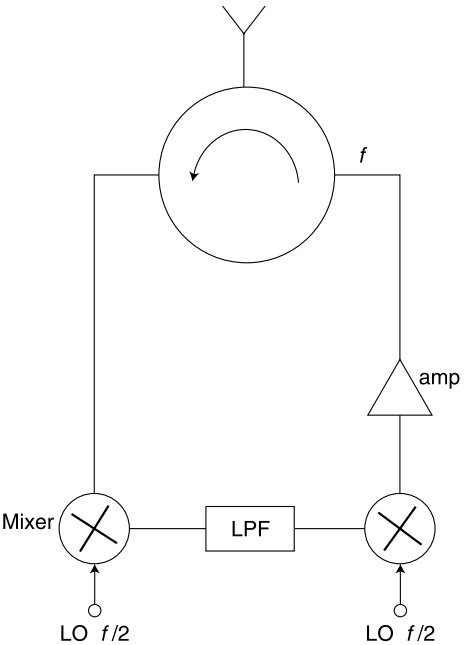


FIGURE 13.10 Subharmonic mixing-phase inversion.

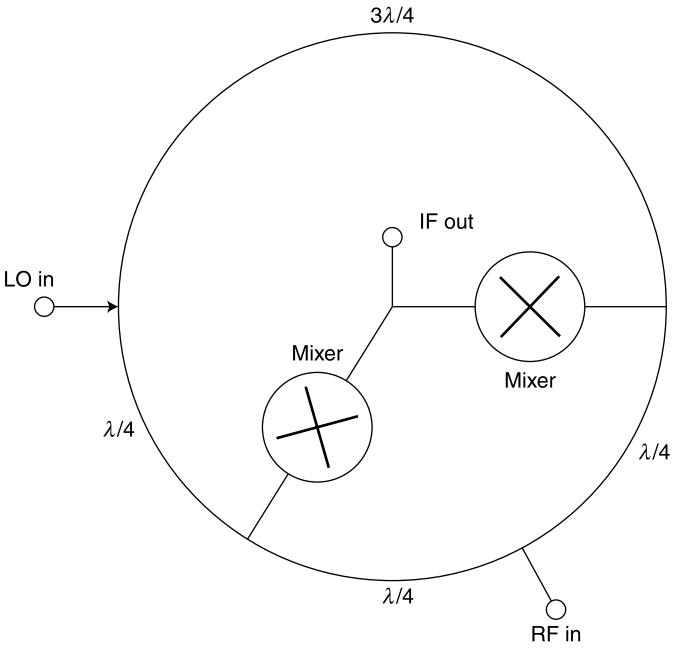


FIGURE 13.11 Subharmonic phase-inverter using ratrace hybrid.

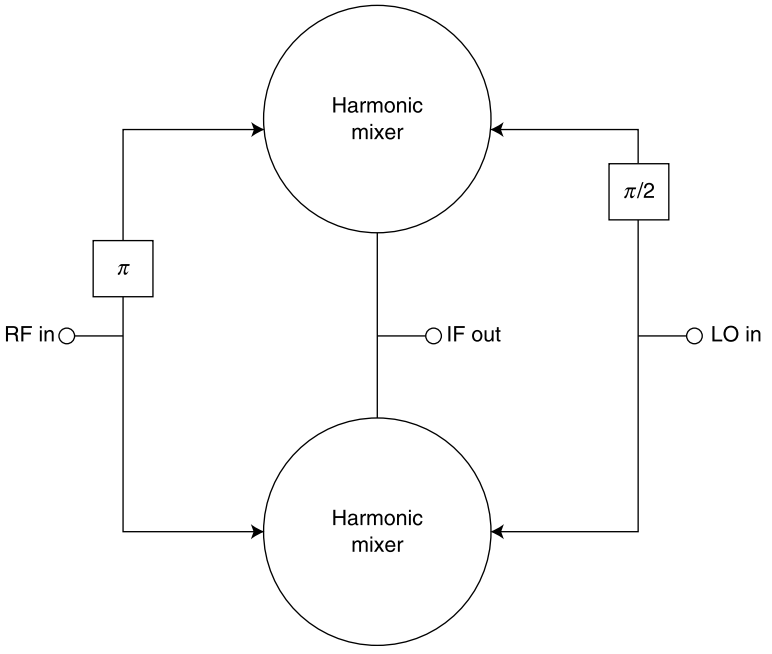


FIGURE 13.12 Phase inversion with RF and LO cancellation.

Still another scheme for obtaining phase inversion utilizes quadrature hybrids terminated with loads similar to those used in reflection phasers. With this arrangement the input and output lines are reversed so crossover circuits are used (Hsieh and Chu, 2008). This technique has not yet been fully evaluated. When the hybrid terminations are reflection amplifiers, a bidirectional amplifier results (Chung et al., 2003).

Returning to the array as a scatterer, the performance of a retrodirective array has been analyzed in terms of the field scattered from the patches, the field scattered from the ground plane, and the reradiation field (Chung and Chang, 1998; Tseng et al., 2000). The scattered field introduces oscillations in the reradiated field, and slightly increases the peak value of the total field.

13.2.3 Advanced Applications

Use of a retrodirective array in a full duplex communications system requires special design to avoid retransmitting the received signal. A PSK signal can be received and the modulation stripped off. Then the carrier can be phase inverted and remodulated with a new PSK signal (DiDomenico and Rebeiz, 2001). Another approach uses orthogonal modulations for receive and transmit (Leong and Itoh, 2005). Frequency offset between receive and transmit is another option (Cutler et al., 1963; Karode and Fusco, 1999).

When a retrodirective array is on a moving platform, the frequency transmitted is related to the phase-inversion circuit open-loop gain. The high gains required for good retrodirection may be incompatible with certain platform motions (DiDomenico and Rebeiz, 2000).

An unusual application of a retrodirective array with phase inversion uses noise correlation to accelerate the detection of targets. In this system, the LO frequency is twice the carrier frequency. Correlator circuits augment the noise from the target direction (Gupta and Brown, 2007).

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CHAPTER FOURTEEN

Reflectors with Arrays

The combination of arrays and reflectors has become of considerable interest, primarily for limited scan applications. Both focal plane arrays, and near-field Cassegrain and Gregorian systems are treated here.

14.1 FOCAL PLANE ARRAYS

This section is concerned with arrays for coma correction, when the beam is scanned to sufficiently large angles. Clusters of feeds used for area coverage or for other pattern shaping, and multiple-beam antennas, are not treated here.

14.1.1 Focal Plane Fields and Coma

Most paraboloidal dish antennas operate with a feed at the focus of the dish. Figure 14.1 shows focal plane field intensities for a square aperture. As expected, the "sidelobes" are small (Stamnes, 1986). When the feed is moved off-axis the beam scans in the opposite direction. The ratio of beam angle to feed angle is called beam deviation factor, and is (Lo, 1960; Ruze, 1965):

$$\mathrm{BDF} = \frac{\displaystyle\int_0^1 \frac{\rho^3 g(\rho)\, d\rho}{1 + (D\rho/4f)^2}}{\displaystyle\int_0^1 \rho^3 g(\rho)\, d\rho} \tag{14.1}$$

when $g(\rho)$ is the aperture amplitude distribution. Both Lo and Ruze used obsolete and inefficient aperture distributions. A good fit to most dish antennas is the

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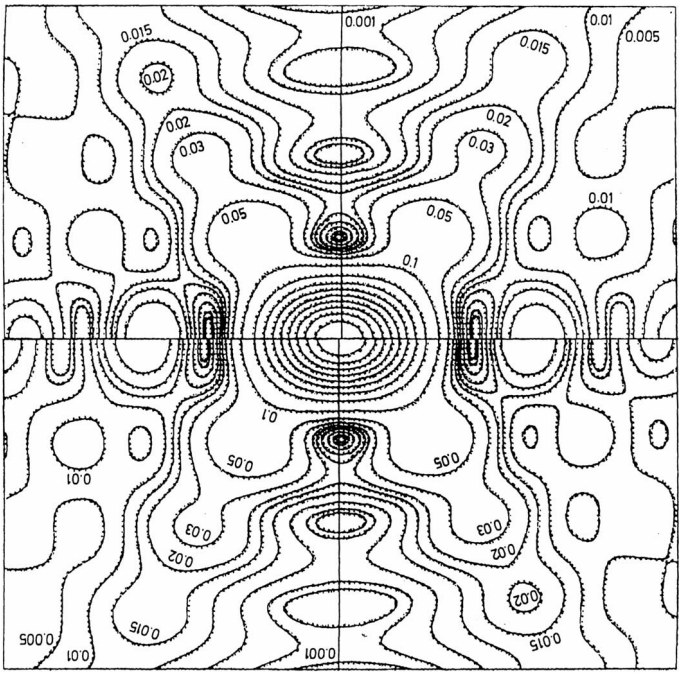


FIGURE 14.1 Focal plane field, on axis.

Hansen one-parameter circular distribution (Hansen, 1976) with 25-dB sidelobe level:

$$g(\rho) = I_0\left(\pi H\sqrt{1-\rho^2}\right) \tag{14.2}$$

where $H = 0.8899$ for 25-dB sidelobes. Figure 14.2 shows BDF for edge tapers of -40, -25 and -17.57 dB (uniform).

As the feed moves, the optimum location is not on the focal plane, but on a curved surface. Ruze (1965) showed, using geometrical optics, that the sharpest nulls occur on the Petzval surface, a paraboloid of focal length one-half that of the main dish; other aberrations are absent. The feed surface for maximum gain is close to, but not the same as, the Petzval surface (Rusch and Ludwig, 1973). Also, the off-axis feed should be parallel to the dish axis, not pointed at the dish vertex. As the beam is scanned, the pattern is degraded by astigmatism and coma, primarily the latter. As the beams move into the coma region, the main beam broadens and drops, and high sidelobes appear on the axis side of the main beam. Figure 14.3 is a sketch of a main beam on-axis (normalized to 0 dB) and a coma beam (Imbriale et al., 1974). Coma is caused by rays from the outer part of the dish focusing over a circle instead of at the shifted focal point. Thus the focal plane field has a comet-like shape, as shown in Figure 14.4. The coma is a third-order optical aberration, due to the asymmetry of the field in the focal plane. As the feed angle increases, the energy located

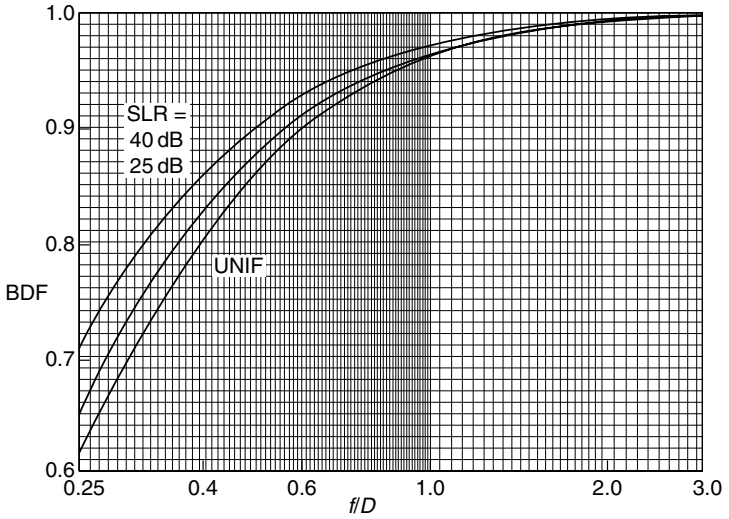


FIGURE 14.2 Beam deviation factor of circular one-parameter distribution.

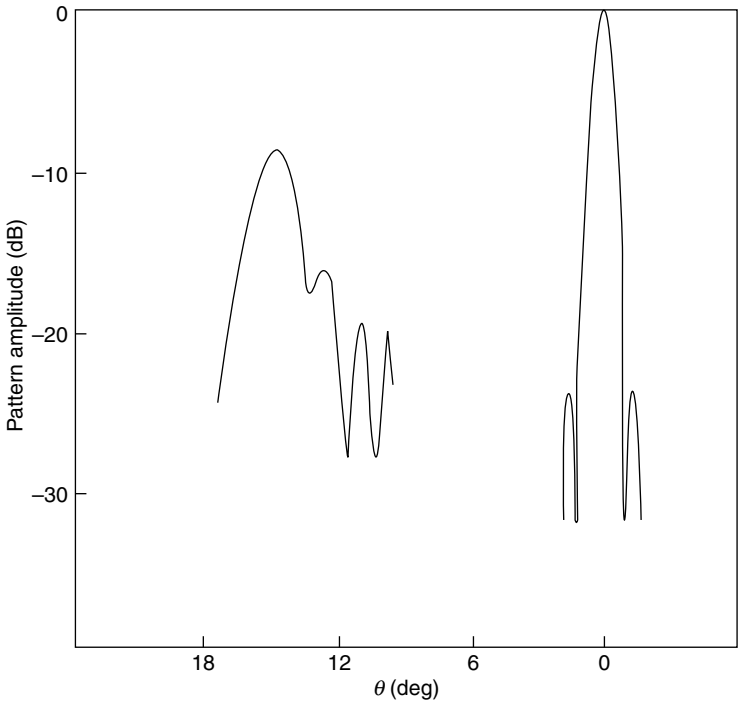


FIGURE 14.3 Patterns: On-axis and coma.

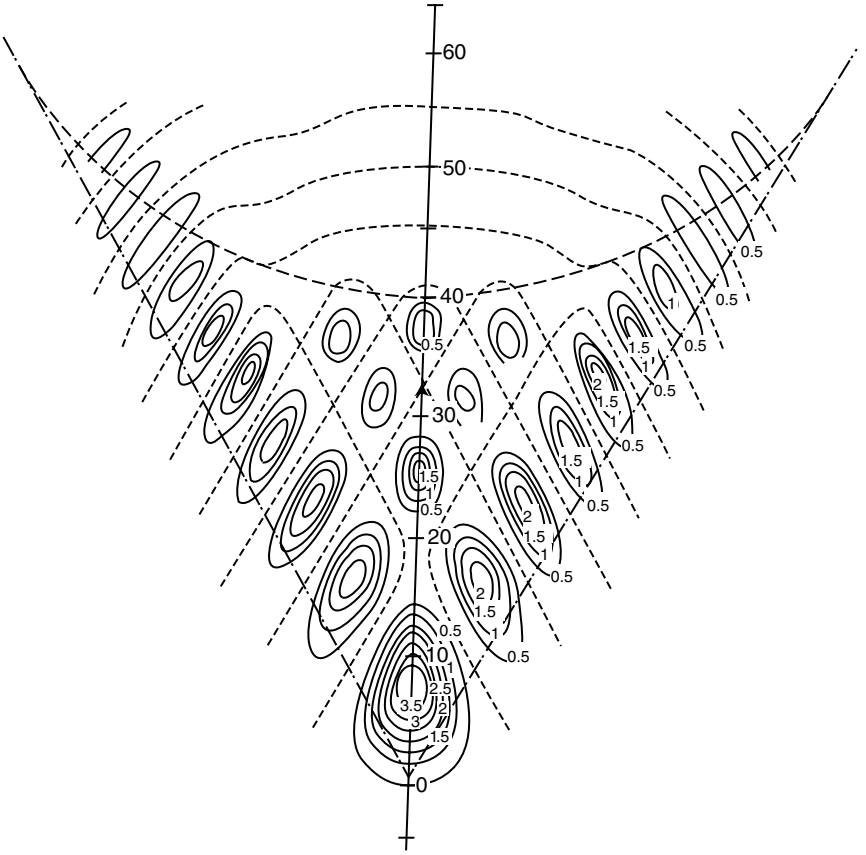


FIGURE 14.4 Typical coma focal plane field.

along the $\pm 30^\circ$ line increases, and the antenna gain decreases. The coma "wings" are always located at $\pm 30^\circ$ from the axis, but how this might be derived from the field integrals is unknown.

Scan loss was computed by Ruze (1965), but he used a now obsolete aperture distribution. Scan loss is given approximately by

$$G = G_0 \int_0^1 g(\rho) J_0(\arg) \rho \, d\rho \tag{14.3}$$

where $g(\rho)$ is the aperture distribution, and the Bessel argument is

$$\arg = \pi \theta_{\text{scan}} \rho \left[1 - \frac{1}{\text{BDF}\left[1 + (\rho D/4f)^2\right]} \right] \tag{14.4}$$

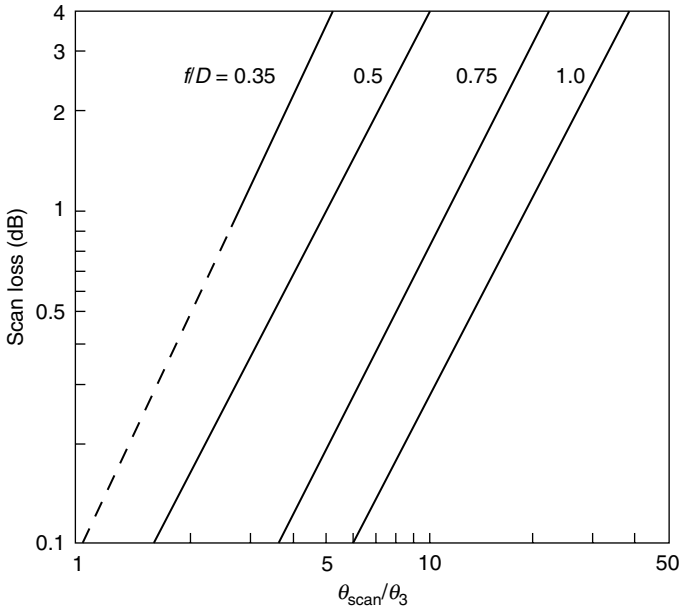


FIGURE 14.5 Approximate parabolic reflector scan loss.

The scan loss has been calculated using the more representative Hansen one-parameter circular distribution, with Romberg integration (Stark, 1970). Figure 14.5 shows scan loss versus the scan angle in terms of half-power beamwidths. It is apparent that small f/D suffer higher scan loss. Note that the Ruze formulation, Eq. (14.3), was based on the decrease in the beam peak as the beam scans; this is only approximate. A more accurate calculation would integrate the entire 3D coma pattern, but this has apparently not been done.

For a parabolic cylinder reflector with line source feed that moves off axis, Eq. (14.3) is modified.

$$G = G_0 \int_0^1 g(\rho) \exp(j \arg) \rho \, d\rho \tag{14.5}$$

where the argument is again Eq. (14.4). Romberg integration was used again, with the results shown in Figure 14.6. Note that, as expected, the parabolic cylinder scan loss is much less than that of a paraboloid.

An analysis technique for large parabolic cylinder antennas with line source feeds first calculates the physical optics field over the reflector from a feed. Next the reflector surface is divided into narrow strips (along the cylinder axis); the fields from the strips are decomposed into spatial and spectral integrals, which are readily evaluated (Tap and Pathak, 2006).

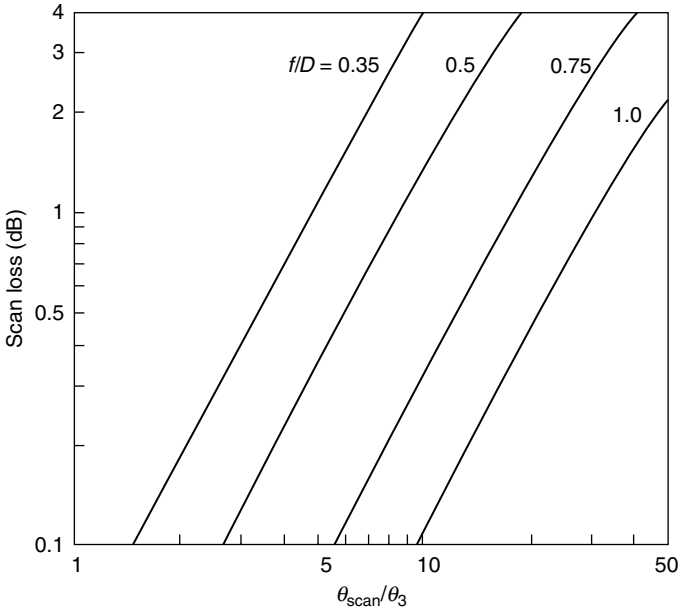


FIGURE 14.6 Approximate cylindrical parabola reflector scan loss.

14.1.2 Recovering Coma Scan Loss

Most of the gain lost due to coma can be recovered through use of a focal plane array. Such an array would have elements located at each of the elliptical "hot spots" of Figure 14.4. Amplitudes at each hot spot are not uniform, and the phasers are not linearly related as they are in a conventional phased array. To realize the most gain recovery, maximal ratio combining (MRC) is used. This was developed for multiantenna communications systems (Jakes, 1974), and weighs each signal proportional to its power S/N before combining. The MRC was originally proposed by Kahn (1954). When applied to coma correction, each element signal is weighted by the complex conjugate of the fields at that element when the antenna system is receiving. There is no simple relationship among either the element amplitudes or phases.

For modest gain drops, cluster feeds can be used. Here feed elements are arranged in a ring with or without a center element, with the ring center at the focal point, or displaced (Galindo-Israel et al., 1978). As the beam is moved off-axis, the element coupling must be switched or reduced to maintain good efficiency. For wideband systems, such as focal plane arrays for radio astronomy, use of fixed-feed element positions and weights degrades S/N (Veidt and Dewdney, 2005).

14.1.3 Coma Correction Limitations

There are several serious difficulties in focal plane arrays for coma correction. First, as the ϕ angle changes, the entire coma pattern (Fig. 14.4) rotates with it. So the array

must also rotate. Of course, a larger array covering $\phi = 2\pi$ would be stationary, but then the appropriate coma elements must be switched on or off, and blockage would be large. Second, as the θ angle changes the focal plane energy pattern changes, from a single circle (at or near broadside), to a beginning coma distribution with a few hot spots, to a mature coma distribution, such as shown in Figure 14.4. With coma focal plane distributions, as the θ angle changes all of the hot spots move and new ones are created. A further complication is that the hot-spot spacing along the $\pm 30^\circ$ lines is not uniform and is not linear with frequency. Time delay, which is used in wideband phased arrays, is not applicable here. The MRC complex coefficients vary with angle θ, with hot-spot position, and with frequency. These serious difficulties have limited the use of focal plane arrays for coma correction.

14.2 NEAR-FIELD ELECTROMAGNETIC OPTICS

14.2.1 Near-Field Cassegrain

This section is concerned with near-field Cassegrain, Gregorian, and focused antennas. These are sometimes called "limited scan" antennas. The traditional Cassegrain antenna has a paraboloidal main dish (reflector) of diameter D and a hyperboloidal subdish of diameter d. These are typically fed by a horn antenna. In near-field reflector systems, a plane wave is used to illuminate the subdish; for the near-field Cassegrain the subdish is also paraboloidal. Figure 14.7 is a sketch of a near-field Cassegrain

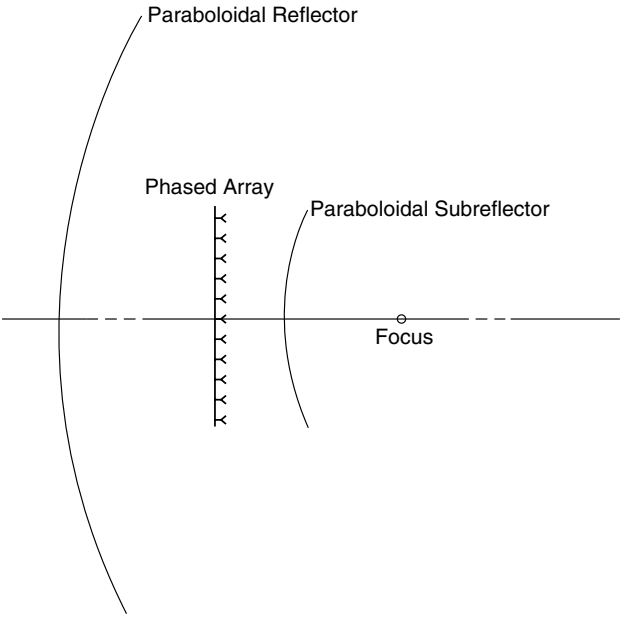


FIGURE 14.7 Near-field cassegrain antenna.

antenna. The origin of this concept is not clear, but it is at least as early as 1963 (Profera and Sciambi). They used a horn-parabola to feed a paraboloidal subdish, at a distance of $W^2/6\lambda$; the feed aperture was W. Hogg and Semplak (1964) tested a 16-ft dish with a horn parabola feed at 6 GHz, with particular emphasis on antenna noise temperature. A landmark study was made by Fitzgerald (1971a, 1972). He used a phased array to illuminate the subdish; the array allows limited electronic scan of the near-field Cassegrain system. Other references are Winter (1968) and Dragone and Hogg (1974), Mrstik (1979, 1981, 1982), and Smith and Mrstik (1982). Note that the subdish is in the near-field of the phased array; the near-field is a plane wave where the phase front contains the beam direction. Here the array diameter is W. If the array is sufficiently large (in W/λ), for structural reasons the array can be located at the apex of the main dish, with the array structure behind the main dish and the array face located in an aperture in the apex. Note that the dual reflector system demagnifies the beam direction; the array scan angle is given by

$$\tan \theta_{\text{arr}} = \frac{D}{d} \tan \theta_{\text{beam}} \tag{14.6}$$

The amplitude of the field in the focal plane is symmetric and the phase is linear. It was shown by Linfoot (1955) that in this case the third-order aberration is very small, and that Coma lobes are negligible. There are some limitations and restrictions on phased array feeds for reflectors. If the reflector diameter and focal length are D and f, and the

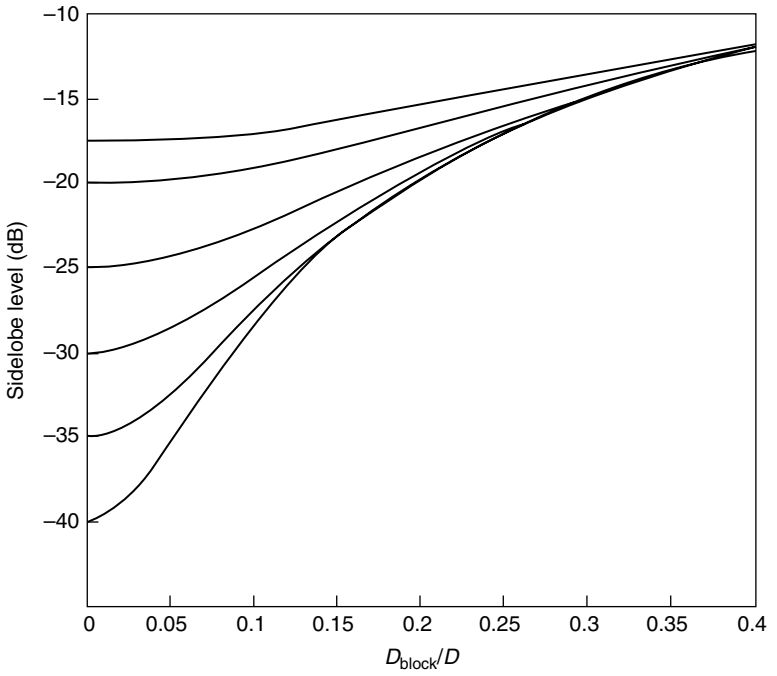


FIGURE 14.8 Sidelobe level versus blockage diameter ratio for one-parameter circular distribution.

array diameter and distance from the array to the focus are W and f_{arr} to map the array onto the dish, which gives maximum efficiency, requires $f/D = f_{\text{arr}}/W$.

The central blockage provided by the subdish and array raises the sidelobes and reduces the gain. Calculations have been made for a system using the one-parameter circular aperture distribution (Hansen, 1976). This distribution for 25-dB sidelobe ratios fits most reflector antennas very well. Figure 14.8 shows sidelobe level versus d/D for several unblocked sidelobe levels. For 25 dB, a subdish–main dish diameter ratio of 0.25 gives a sidelobe level of -16 dB. Data on gain changes are not available. Placing an amplitude taper on the array will produce an amplitude taper over the subdish, but this will not reduce the sidelobes, except through the loss of gain, 0.5 dB for a 25-dB circular one-parameter taper. The system sidelobe level is controlled by the blockage diameter ratio d/D. Calculations and measurements by Fitzgerald show sidelobe levels of roughly -12 dB for both uniform and tapered excitations.

Array element spacings $>\lambda/2$ may produce grating lobes, but as observed by Fitzgerald they are outside the cone of the subdish and thus they are not increased by the reflector gain. Because of this element spacings of λ or more are feasible. However, the Shannon sampling theorem states that two samples per wavelength are needed for faithful replication of the near-field. The array elements, typically dipoles or patches, are not point samplers; how these larger element spacings affect sidelobe level and gain should be determined. It is probable that spacings up to λ will be possible.

In any reflector system, as the beam scans the feed coverage pattern moves off the reflector (spillover). This vignetting reduces the gain. As the coverage dish produced by the feed moves off the reflector dish, the effective reflector area is two back-to-back circular segments (see Fig. 14.9). Using the sector area formula from Burington (1973), the efficiency versus scan is approximately:

$$\text{eff} = \frac{2}{\pi}\left[\text{arc cos}\left(1 - \frac{h}{a}\right) - \left(1 - \frac{h}{a}\right)\sqrt{\frac{h}{a}\left(2 - \frac{h}{a}\right)}\right] \tag{14.7}$$

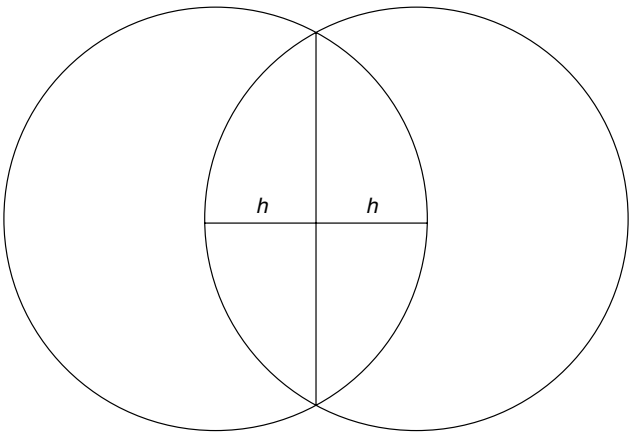


FIGURE 14.9 Vignetting due to scan.

where h is the segment central thickness, and $a = D/2$. With a main beam swing about the focus f to angle θ, $(a - h)/f = \tan \theta/2$. Converting to f/D, a universal variable is

$$\frac{1}{2}\left(1 - \frac{h}{a}\right) = \frac{f}{D}\tan\frac{\theta}{2} \tag{14.8}$$

Figure 14.10 gives vignetting efficiency (gain drop) versus $f/D \cdot \tan \theta/2$. For example, a 3-dB loss is produced by

$$\frac{f}{D}\tan\frac{\theta}{2} = 0.20147 \tag{14.9}$$

An $f/D = 0.40$ allows scan of $\pm 53.47^\circ$, while $f/D = 0.5$ allows $\pm 43.89^\circ$ scan.

The question arises: How close must the array be to the subdish to be in the near-field? There have been many papers on near-fields of aperture and array antennas, see, for example, Hansen and Libelo (1993). A good rule of thumb (Hansen, 1991) is to have the array subdish distance no more than $0.05 \times 2W^2/\lambda$, where W is the array diameter. For example, if $W/\lambda = 100$, the separation should be no more than $10W$.

It is sometimes stated that a near-field Cassegrain system has a fixed field of view. However, when phase errors are included, the number of beamwidths scanned is limited. This finding is not surprising; most systems are so limited. Fitzgerald (1971a) offers a good rule of thumb for scan capabilities: The number of half-power

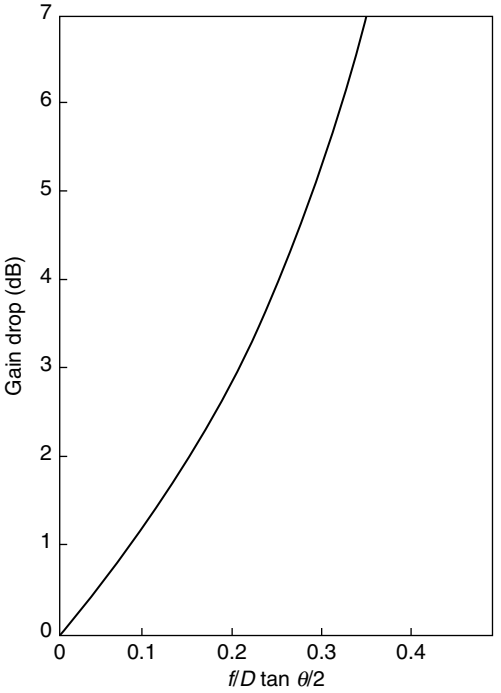


FIGURE 14.10 Gain drop versus scan.

beamwidths of scan from boresight is given by $0.5d/\lambda$. This approximate formula is for large D/λ, and uniform excitation of the array. As mentioned, a tapered excitation is not advantageous. This scan range formula varies slightly for f/D from 0.3 to 0.5. As an example, main dish of $D/\lambda = 600$ would have $\theta_3 \sim 0.1^\circ$. With $d/D = 0.2$, the subdish is $d/\lambda = 120$ and the scan range to 3-dB drop would be 60 beams each side of broadside, or 6° scan each side. With 1λ element spacing the feed array would have roughly 11,310 elements, for the case of array at dish apex.

This is all for a narrowband system. For multiple octaves of bandwidth, the element spacing would need to be the design value (here 1λ) at the highest frequency. The array feed manifold would need to include phasers at each element, and appropriate time delay. Note that a corporate feed with doubled delay at each successive feed level may only provide correct delay at some frequencies (Hansen, 2002).

With a uniformly excited array, boresight efficiency is typically 50–55%; a 25-dB amplitude taper reduces these numbers to 29–32% (Fitzgerald, 1972).

As the subdish is in the (close) near-field of the array, there will be scattering from the subdish back to the array, and this will change as the scan angle changes. We can expect the scan impedance to be affected, and the near-center elements may be affected differently from the edge elements. Some very careful calculations will be needed to evaluate the impedance mismatch data. Note that it is not practical to measure scan impedance directly (Van Wagoner and Hansen, 2003). Path length error can be minimized by collimating the beam in two directions instead of one (Rao, 1984). This is analogous to a lens, such as a Rotman lens, with two focal points. Bicollimation can significantly reduce path length error. A more sophisticated approach to optimization includes focal length adjustment, and reshaping the surfaces (Martinez-Lorenzo et al., 2008).

14.2.1.1 System Trades and Restrictions

1. The array scan angle is demagnified to produce the beam scan angle: $D \tan \theta_{\text{beam}} = W \tan \theta_{\text{arr}}$.
2. Grating Lobes do not appear in the reflector pattern. However, they affect array Scan Impedance.
3. Aperture blockage by array (or lens) raises sidelobes. $W/D = 0.3$ gives *sidelobe level* ~ -15 dB.
4. As beam scans, aperture is reduced (vignetting or spillover). For -3-dB loss f/D $\tan \theta_{\text{beam}}/2 = 0.2015$.
5. The focus must be in the array (lens) near-field. A safe value is $f/W \leq 0.1\ W/\lambda$.

14.2.2 Near-Field Gregorian

When the phased array and subreflector are placed on the opposite side of the focal point form the main reflector, the system is called a near-field Gregorian antenna (see sketch in Fig. 14.11), the subreflector focuses energy upon the focal point, and then onto the main reflector. Again aperture blockage is a problem. Radio

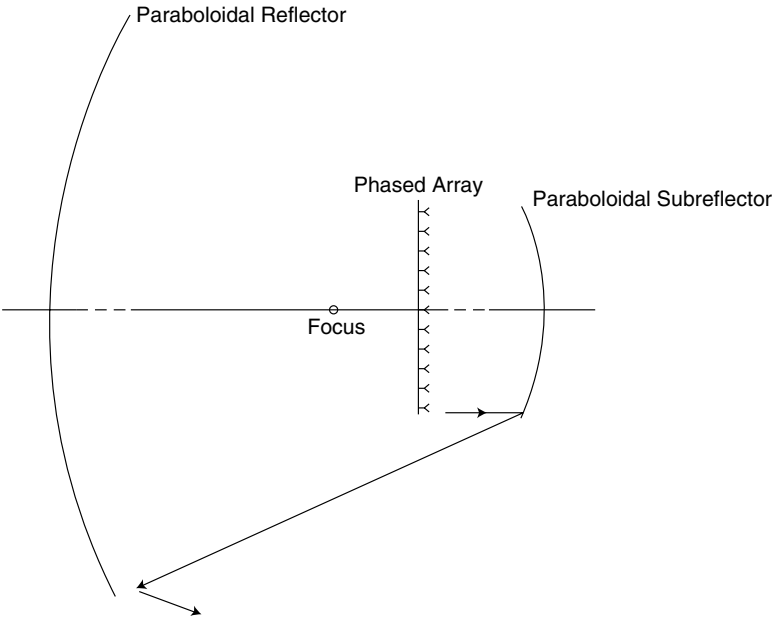


FIGURE 14.11 Near-field Gregorian antenna.

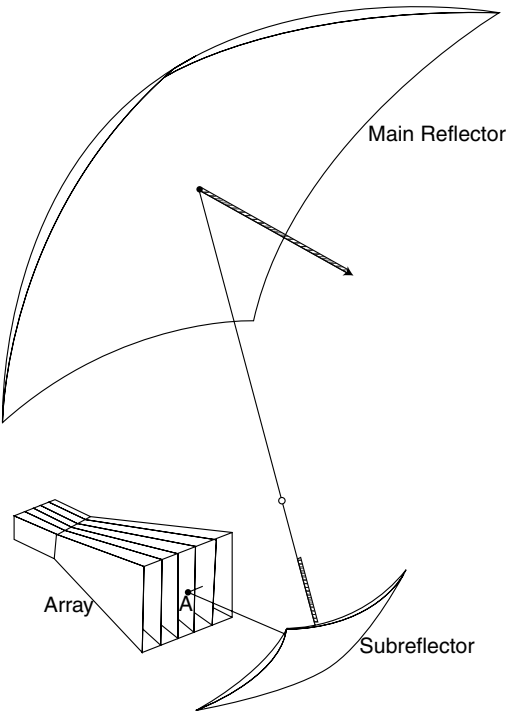


FIGURE 14.12 Offset Gregorian.

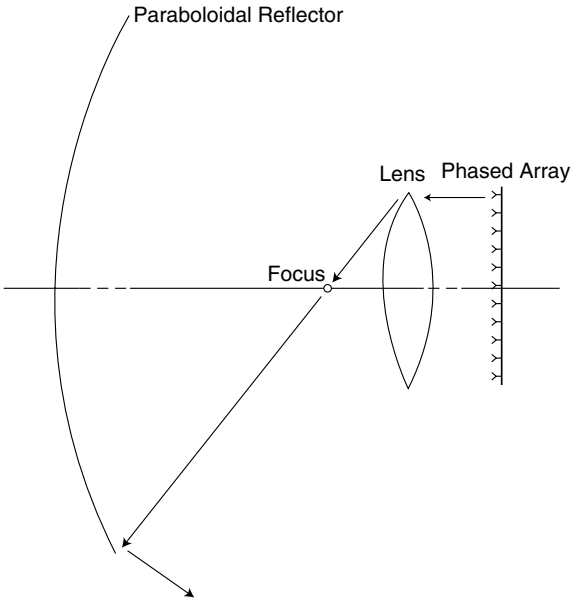


FIGURE 14.13 Near-field reflector-lens antenna.

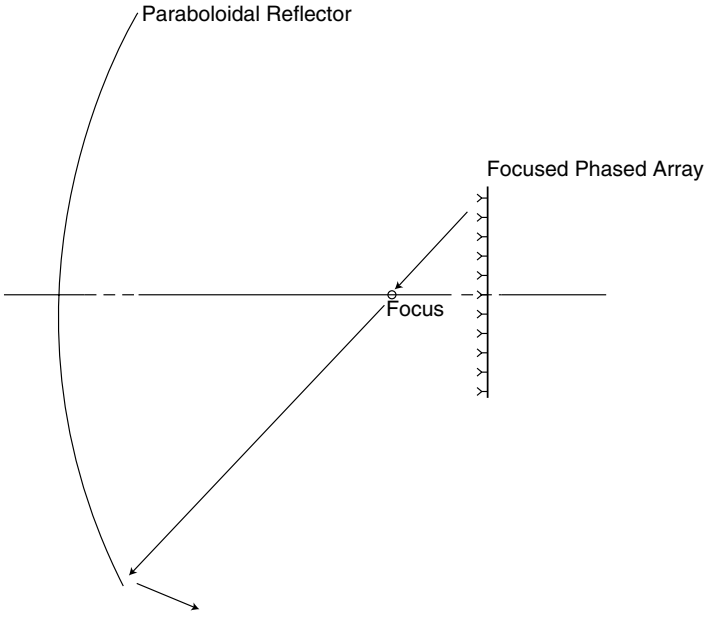


FIGURE 14.14 Near-field reflector focused array antenna.

astronomers have used offset near-field Gregorian antennas for certain applications; this obviates aperture blockage. Figure 14.12 sketches an offset Gregorian system. The array should be within the near-field distance (see Section 14.2.1), but not too close to cause subdish blockage. The offset geometry creates elevation beam squint when scanning in the azimuth plane. It also creates cross-polarized fields. Grating lobes can be reduced by inserting a filter in the focal plane (Dragone and Gans, 1980). A unique offset Gregorian uses pins in a parallel plate region for main reflector and subreflector, with a 2D array of radiating slots on the upper plate (Ettore et al., 2008).

Returning to symmetric Gregorian systems, the beam angle and array angle are related.

$$\frac{\theta_{\text{beam}}}{\theta_{\text{array}}} = \frac{f_{\text{main}}}{f_{\text{subdish}}} \tag{14.10}$$

Just as in the near-field Cassegrain system, in the near-field Gregorian system, the array produces a plane wave incident on the subdish, where the phase front contains the beam angle (Fitzgerald, 1971b).

A modification of the Gregorian antenna uses a lens to focus the array radiation on the focal point, as sketched in Figure 14.13. Again the phase front contains the implicit beam angle. Here there is a trade between weight, lens loss, mounting, and so on for the subreflector versus the lens.

A further advance omits the lens and uses a focused array. The array is focused on the system focus (see Fig. 14.14). Again the beam direction is implicit in the phase front. Characteristics of focused arrays have been given by Hansen (1985). Because the focus is fixed, each array element needs only a fixed time delay. Use of delay lines insures wideband focusing. Note that time delay is neither true nor false. The differing attenuations in the various delay lines are easily compensated by adding resistors to the shorter lines. Delay lines are usually located after preamps so that delay line loss should have only minor effect on noise figure.

The focused array system is advantageous in that no lens, with its weight and loss, is needed. The array can be closer to the focal point, thus reducing the overall antenna volume. The minor disadvantage is that each array element must have a fixed delay line. This reflector near-field focused phased array antenna system looks like an excellent choice.

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CHAPTER FIFTEEN

Measurements and Tolerances

There are excellent books on measurements, which are listed below. Thus only special topics pertinent to arrays will be discussed herein. These are measurement of low sidelobe antennas; array diagnostics; and *scan impedance* simulators. Because array performance is affected by random errors, tolerances are also discussed in this chapter.

A. E. Bailey, *Microwave Measurement*, IEE/Peregrinus, 1985.

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15.1 MEASUREMENT OF LOW-SIDELOBE PATTERNS

As mentioned in Chapter 3, the conventional far-field distance of $2D^2/\lambda$ is usually inadequate for accurate measurement of patterns with low sidelobes; antenna diameter is D. Consider patterns, such as the Taylor $\bar{n}$. As the observation distance R moves in from infinity, the first sidelobe rises and the null starts filling. Then the sidelobe becomes a shoulder on the now wider main beam, and the second null rises (see Fig. 15.1). This process continues as the distance decreases. To first order, the results are dependent only on design sidelobe level $U = (L/\lambda)\sin\theta$ Figure 15.2 shows the distance at which the first sidelobe is raised a given amount, for values of SLR from 13 (uniform) to 60 dB (Hansen, 1984). This graph also shows distances at which the first sidelobe is shouldered.

Phased Array Antennas, Second Edition. By R. C. Hansen

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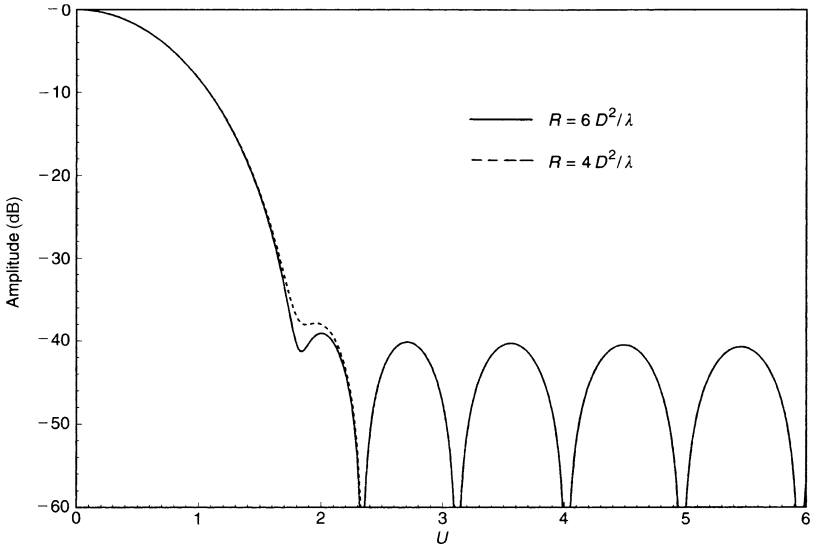


FIGURE 15.1 Taylor pattern for SLR = 40 dB, $\bar{n} = 11$.

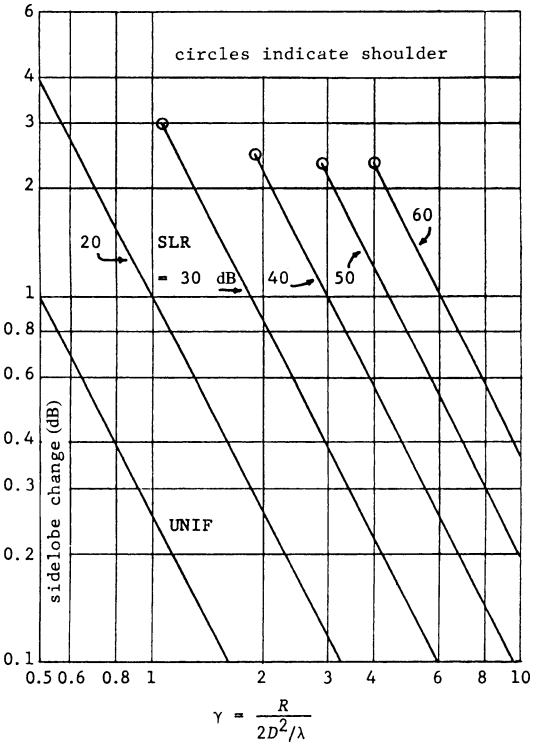


FIGURE 15.2 Sidelobe change versus normalized measurement distance for Taylor $\bar{n}$ line source.

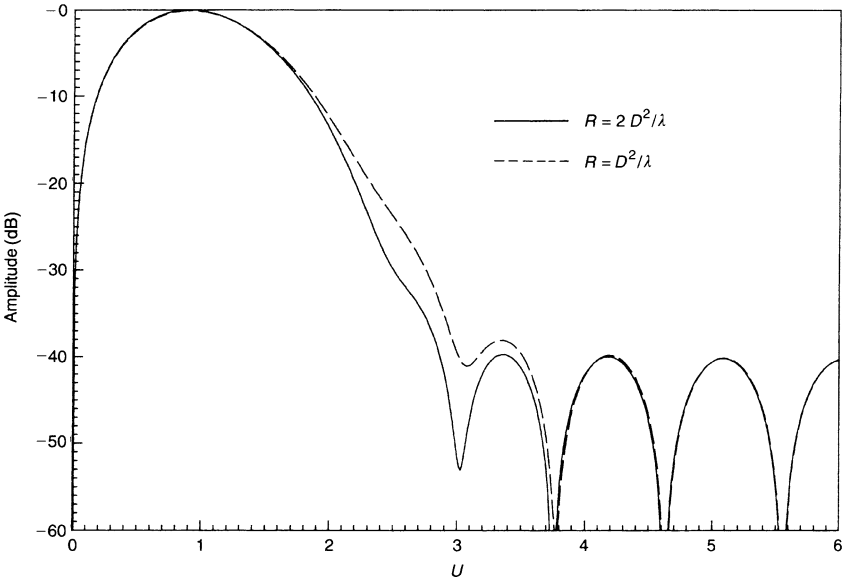


FIGURE 15.3 Bayliss pattern for SLR = 40 dB, $\bar{n} = 11$.

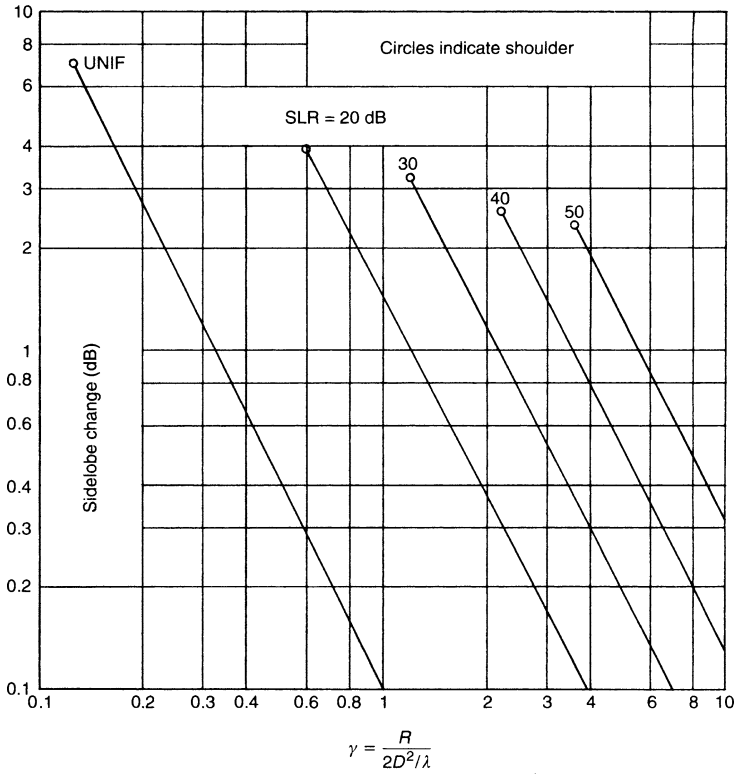


FIGURE 15.4 Sidelobe change versus normalized measurement distance for Bayliss line source.

Thus the accuracy of measurement may be traded against the measurement distance.

Bayliss and other difference patterns possess the same type of extended far-field distances as the low-sidelobe designs of Chapter 3. Again, the first sidelobe height and null depth deteriorate as the distance decreases from $2D^2/\lambda$. The progressive raising and shouldering of sidelobes continues as R decreases. Figure 15.3 shows a case where the first sidelobe has already been absorbed by the main beam, and the second sidelobe and null are raised. A universal chart (Hansen, 1992), Figure 15.4, gives first sidelobe increase, and shoulder value, versus normalized distance, for SLR from 13 (uniform) to 50 dB.

Notice that the difference patterns require a slightly larger measurement distance than the corresponding sum patterns.

15.2 ARRAY DIAGNOSTICS

Element errors in arrays are seldom due to mechanical tolerances, but instead are due to less tangible factors, such as mutual coupling affecting both phase and amplitude of

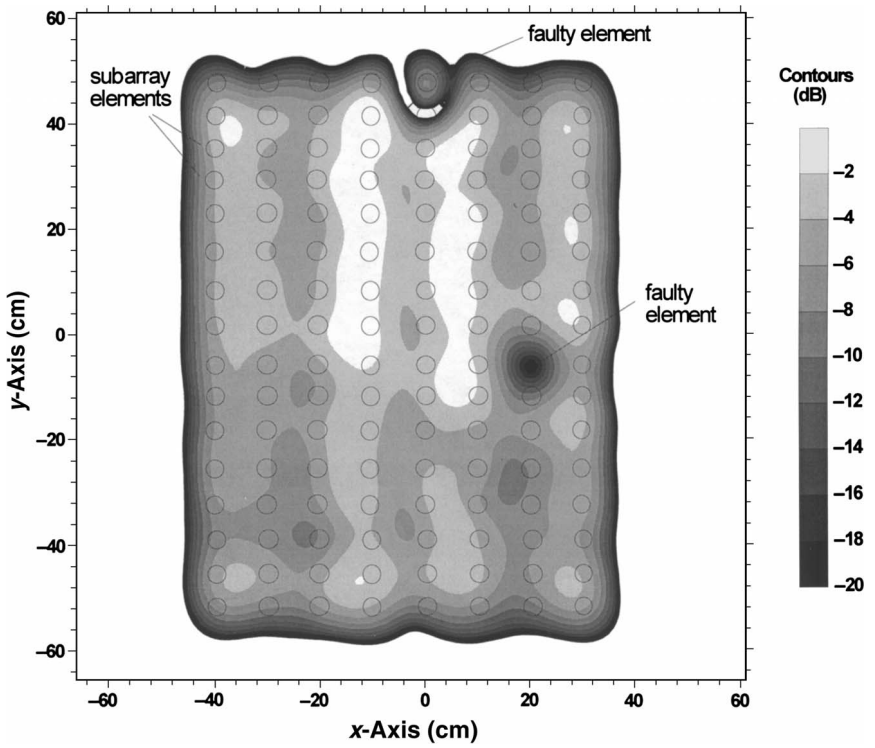


FIGURE 15.5 Array face amplitude. (Courtesy Repjar, A. G. et al., "Determining Faults on a Flat Phased Array Antenna Using Planar Near-Field Techniques", *Proceedings AMTA Symposium*, Boulder CO, Oct. 1991, pp. 8-11–8-19.)

excitation, uncertain loss in phasers and feed networks, and phase errors in phasers. The most direct way of determining these is through probing, where a movable probe is coupled closely to one element at a time. An obvious problem is the effect of the element–probe coupling. Modulated small scatterers have been used instead of a probe to ease this difficulty. A review of near-field measurements with an extensive bibliography has been given by Dyson (1973). See also Aumann and Willwerth (1987) and Gillespie (1988).

In all of these procedures, the measured near-field is transformed to a complex far-field pattern. This pattern is then back-transformed to the array aperture. When the near-field measurements are over a planar surface, the FFT can be used for both transformations. It is important to include a correction for the probe 2D pattern. To this end, the probe field is expressed as a plane wave expansion. The probe effects can then be handled (Paris et al., 1978; Joy et al., 1978). Useful references on near-field measurements as a diagnostic tool include Lee et al. (1988), Garneski (1990), Repjar et al. (1991), MacReynolds et al. (1992), Friedel et al. (1993), and Guerrieri et al. (1995).

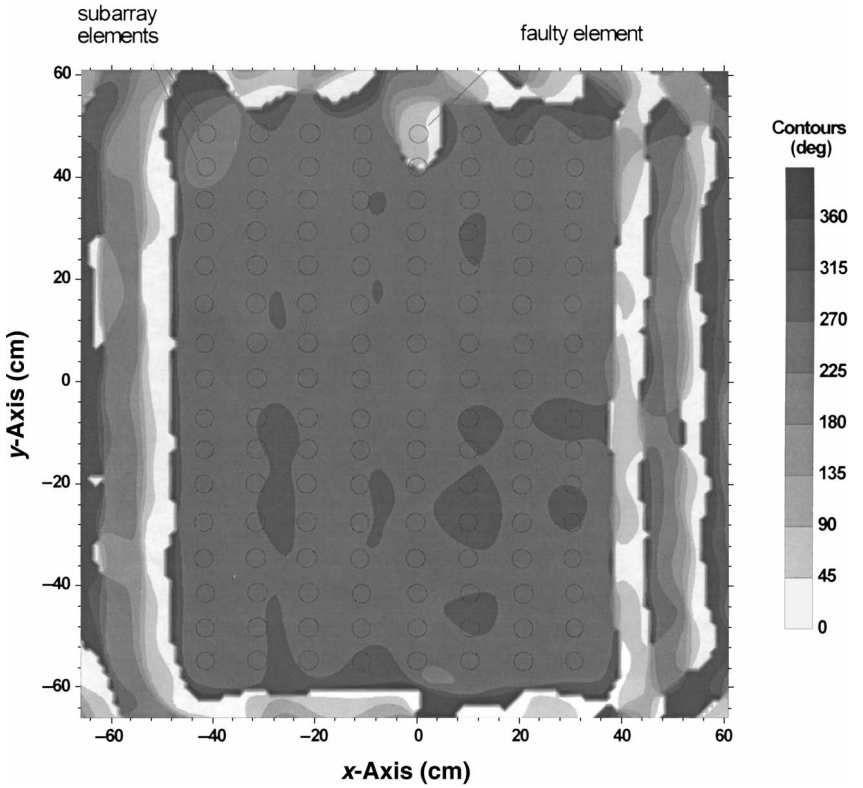


FIGURE 15.6 Array face phase. (Courtesy Repjar, A. G. et al., "Determining Faults on a Flat Phased Array Antenna Using Planar Near-Field Techniques", *Proceedings AMTA Symposium*, Boulder, CO, Oct. 1991, pp. 8-11–8-19.)

Similar transformations can be applied to cylindrical near-field measurements (Langsford et al., 1989) and to bipolar near-field measurements (Williams et al., 1994; Yaccarino et al., 1994). Probe correction and transformations are more complicated with spherical near-field measurements, but are readily manageable with modern computers (J. E. Hansen, 1988; Rahmat-Samii and Lemanczyk, 1988). Of course, far-field measurements can substitute for the near-field measurements (Ekelman, 1987; Johansson and Svensson, 1990; Guler et al., 1992); however, it is necessary to measure over a plane extending well beyond the array, or to measure over close to a hemisphere. In addition, both phase and amplitude are needed; many far-field ranges or chambers have marginal coverage for this.

Examples of planar near-field diagnostics are shown in Figures 15.5 and 15.6; these are calculated array face errors for an 8×16 element microstrip patch sub-array. In Figure 15.5 element excitation amplitude is shown, and two defective elements are noted. Excitation phase is given in Figure 15.6, where one element indicates improper amplitude and phase (Repjar et al., 1991). Near-field diagnostics has become a powerful tool for verifying the realization of the desired aperture distribution. Another way of validating, and in this case monitoring, the aperture distribution utilizes a second array located near to, but weakly coupled to, the primary array (Acoraci, 1990). Measured or calculated mutual couplings are then used in an algorithm to determine the primary array distribution. Finally, with a digitally beamformed receiving array, element failure can be partially corrected for one or two signals by calculating the fields at adjacent elements, and then interpolating to supply the missing element input (Mailloux, 1996).

15.3 WAVEGUIDE SIMULATORS

A simple microwave tool that allows the measurement of *scan impedance* in one scan direction and for one frequency is the waveguide simulator, developed by Brown and Carberry (1963) and Hannan et al. (1963). Although the unit cell virtual walls are, in general, impedance walls, there are some symmetry planes and scan angles that allow perfectly conducting walls. These walls can then be a metallic waveguide, terminated in a section of the array containing a small number of elements. Another way of visualizing the process utilizes the decomposition of a TE_{01} waveguide mode into two plane waves that reflect off the side walls, as sketched in Figure 15.7. The angle of the plane wave with the guide axis corresponds to the θ_0 scan angle. When neither mode index is zero, an additional pair of plane waves reflect off the broad walls

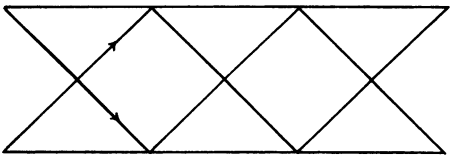


FIGURE 15.7 Plane waves comprising TE_{10} mode.

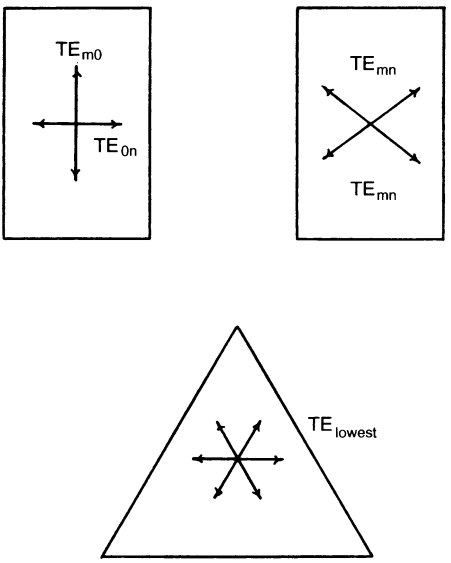


FIGURE 15.8 Simulator scan planes.

(Collin, 1966). For the less often used triangular waveguide simulator, there are three sets of plane waves involved. Allowable scan planes are sketched in Figure 15.8. In a rectangular waveguide simulator, the dominant modes allow principal plane scans, while diagonal plane scans require higher modes. For the triangular waveguide simulator the lowest TE mode provides scan planes along the symmetry planes. Again, higher modes allow scan planes at other angles. Figure 15.9 shows a square lattice array, with two of the many possible simulators using rectangular guide. Note the use of half and quarter elements. A symmetric hexagonal lattice array is depicted in Figure 15.10, along with possible simulators. Note that rectangular waveguide simulators can be used for triangular lattice arrays. Also note that simulators may include only partial elements.

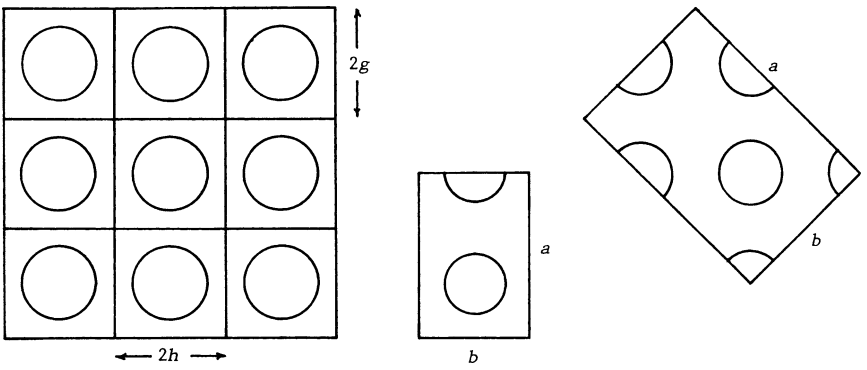


FIGURE 15.9 Square-lattice array with simulators.

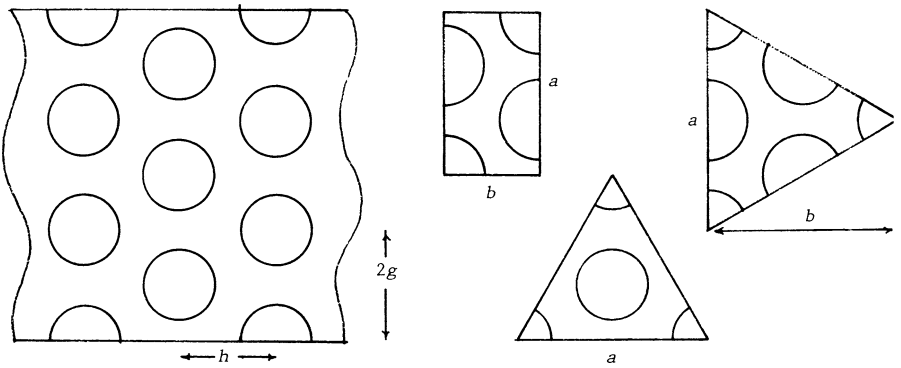


FIGURE 15.10 Hexagonal lattice array with simulators.

A typical simulator consists of a transition section from standard to the simulator guide, the simulator waveguide, and the element port. Across the end of the simulator waveguide is placed a thin metallic sheet with the slot elements appearing as holes. Behind this sheet is a waveguide that includes all the complete elements, and that is terminated in a matched load; this is called the element port. For other elements than slots, or for complex element structures, the element plate is modified accordingly. The simulator waveguide must be sufficiently long to allow decay of evanescent modes. All diagonal plane simulators contain some fractional elements, as the simulator waveguide is always larger than the array unit cell. Triangular waveguide simulators also contain fractional elements for any scan plane. Since regular arrays have rectangular or triangular lattices, the simulator waveguide is limited to polygonal cross sections with three or four sides, and internal angles of 30, 45, 60, or 90°. Of course, most simulators use rectangular or square waveguide because the modes are simple and the connection to standard guide is easy. Because the waveguide mode constituent plane waves reflect in the *H*-plane, simulation of *H*-plane scan, either in a principal plane or in a diagonal plane, is direct. Conversely, since there is no situation where there are only two plane wave constituents that reflect in the *E*-plane, there are no linearly polarized *E*-plane simulators. Approximate simulation in the *E*-plane will be discussed below.

From the previous discussion, it is clear that each simulator provides limited *scan impedance* data, although frequency variation may provide limited additional data. Thus the choice of the fewest and most economical simulators for a given application is like a chess game, as observed by Wheeler (1972). Consider a rectangular simulator waveguide of dimensions a and b, and call the spacing between symmetry planes in the array g and h; see Figure 15.9 where the array lattice is square, and the unit cell is $2g \times 2h$. The cutoff wavelength in the simulator waveguide is

$$\lambda_c = \frac{2}{\sqrt{(m/a)^2 + (n/b)^2}} \tag{15.1}$$

and the scan angles are

$$\sin\theta = \frac{\lambda}{\lambda_c} \qquad \tan\phi = \frac{na}{mb} \tag{15.2}$$

Then,

$$\frac{m}{a} = \frac{2}{\lambda}\sin\theta\cos\phi \qquad \frac{n}{b} = \frac{2}{\lambda}\sin\theta\sin\phi \tag{15.3}$$

Simulator waveguide dimensions are now chosen as integral multiples of symmetry plane spacings:

$$a = pg \qquad b = qh \tag{15.4}$$

where p and q are integers. Finally,

$$\frac{m}{p} = \frac{2g}{h}\sin\theta\cos\theta \qquad \frac{n}{q} = \frac{2h}{\lambda}\sin\theta\sin\phi \tag{15.5}$$

Figure 15.11 shows simulators for a square lattice array (Hannan et al., 1963). The H-plane tools use the TE_{10} mode, and are principal plane (C) or diagonal plane (IC). The TM_{11} mode is used in the E-plane tools. All are for an array with element spacing 0.575λ. Calculation of the beam angle is simple: For the near-broadside simulator, $p = 8$ and $1/8 = 0.575 \sin 12.56$. For the principal E-plane simulator, $p = 3$ and $1/3 = 0.575 \sin 35.43$. Finally for the diagonal E-plane simulator $p = 4$

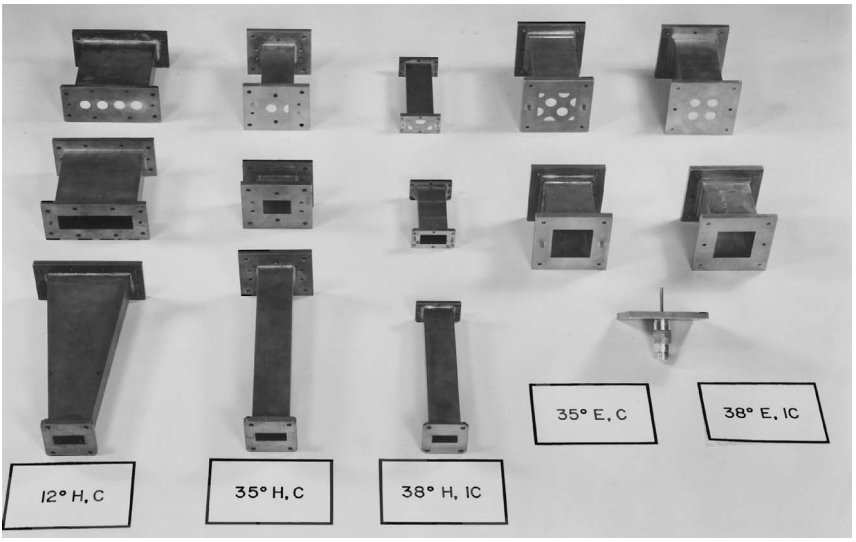


FIGURE 15.11 Square-lattice waveguide simulators. (Courtesy Hazeltine Corp.)

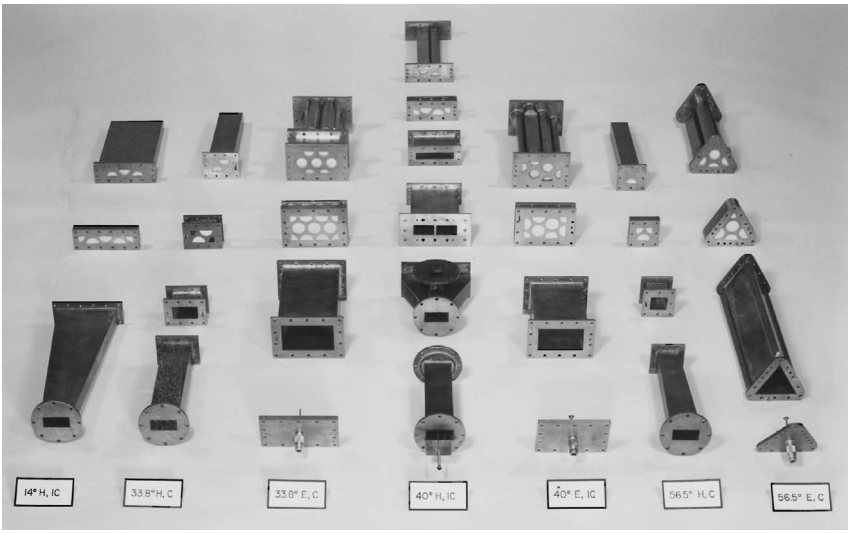


FIGURE 15.12 Triangular lattice waveguide simulators. (Courtesy Hazeltine Corp.)

and $1/4 = (0.575/\sqrt{2})\sin 37.94$. Equilateral triangular lattice array simulators are shown in Figure 15.12 (Balfour, 1965). The TE_{10} mode is used in the 14, 34, and 57° *H*-plane simulators, while the TE_{20} mode is used in the 40° *H*-plane model. The rectangular *E*-plane simulators use the TM_{11} mode; the triangular *E*-plane simulators use the lowest TM mode (see Schelkunoff, 1943). Scan angle calculation is again simple. Element spacing is 0.6λ, and for the near-broadside tool, $p = 4$ and $1/4 = 0.6\sqrt{3}\sin 13.92$. The adjacent tool uses $p = 3$ and $1/3 = 0.6\sqrt{3}\sin 18.71$. For the TE_{20} mode simulator, $p = 3$, and $2/3 = 0.6\sqrt{3}\sin 39.90$. Note in the photographs that some of the partial elements are not loaded.

To use the simulator, the desired mode(s) is excited in the simulator waveguide, with the element port(s) match loaded. Partial elements need not be connected to an element port, or loaded, if they do not couple to the waveguide mode. However, all elements must include all structures that affect stored energy (dielectrics, iris, etc.). The reflection coefficient is now measured in the simulator waveguide; this gives the impedance seen by the incident plane wave. For multielement simulators, a transformation is necessary to get *scan impedance* from the wave impedance. This is based on the junction of two waveguides and involves line lengths and a transformer (Hannan and Balfour, 1965). Later, one-port simulators that give *scan impedance* directly will be discussed. Surface waves have been observed in simulators of dielectric loaded elements (Hannan, 1967).

Simulation of a broadside beam is not possible; small angles require large simulators. Also, *E* polarization produced by a linear element cannot be simulated exactly. The *E*-plane simulators previously shown use circularly polarized or cross-linear-polarized elements. The *E*-plane simulation in a diagonal plane can utilize a pair of

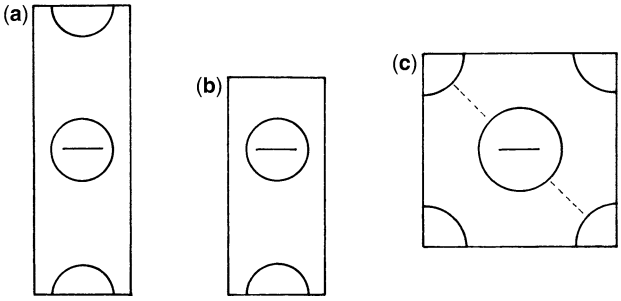


FIGURE 15.13 Square-lattice single port simulators.

modes, TE and TM, with identical nonzero indices. A detailed review of waveguide simulator design and practice has been given by Wheeler (1972). See also Hannan and Balfour (1965) and Gregorwich et al. (1971). Byron and Frank (1968) show excellent agreement with direct measurements.

"One-port" simulators have only one whole element port, along with partial ports. Thus only the single port needs to be terminated. These simple simulators are useful for *H*-plane, wide-angle, principal plane work, where no grating lobes appear (Balfour, 1967). Figure 15.13 shows three simulators for an array with a square lattice. All use the TE_{01} mode. Figure 15.13a uses $p = 4$, $q = 2$, with a principal plane beam at 30°. Figure 15.13b uses $p = 3$, $q = 2$ with a principal plane beam at 41.8°. Figure 15.13c uses $p = q = 2$ at 45°. Element spacing is $\lambda/2$. Equilateral triangular (symmetric hexagonal) array simulators are shown in Figure 15.14. Element spacing to the nearest neighbor is $1/\sqrt{3}$. Figure 15.14b is for an *H*-plane principal plane beam at 60° $p = q = 2$. Figure 15.14a shows a diagonal plane simulator, again with $p = q = 2$, and with beam at 30°. A triangular simulator is shown in Figure 15.14c, where the index is 3, and a diagonal plane beam is at 41.81°.

Microstrip patch arrays have also been simulated. Solbach (1986) used two *H*-plane waveguide simulators, corresponding to Figure 15.13a and b. Here it was important to carefully reproduce the patch feeds and load resistances. These simulators gave

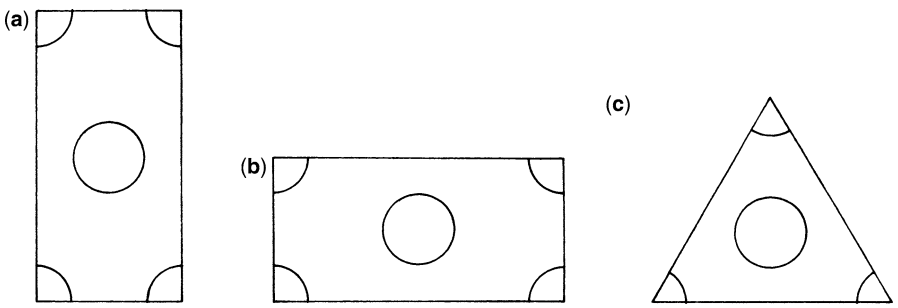


FIGURE 15.14 Hexagonal lattice single-port simulators.

angles of 11.92 and 43.53°. This work was continued by Aberle and Pozar (1990) and Aberle et al. (1994). Arrays of probe fed patches, and patches with parasitic patches were simulated using a single centered patch with four corner quarter patches, with TE_{10} (see Fig. 15.14). The array lattice was symmetrical hexagonal, and the element spacing gave a beam angle of 22.48°.

A more general-purpose simulator was developed by Gustincic (1972) and Derneryd and Gustincic (1979), which allows *scan admittance* at various scan angles to be determined. Grating lobes can also be encompassed. The simulator uses many elements, with all elements and partial elements, but two terminated. Transmission measurements are then made between these two elements. In turn, the two active elements occupy all positions in the simulator, which allows *scan admittance* data to be calculated for N scan angles if there are N simulator elements.

A waveguide simulator can also be modeled in a computer, where the waveguide walls and elements are represented by moment method patches, or by grids as used by finite element methods and FDTD procedures.

15.4 ARRAY TOLERANCES

Errors in arrays include those in element position and orientation, and element excitation phase and amplitude. All errors are of two types: systematic and random. Many errors, of course, are combinations of the two. Systematic errors (correlated and bias errors) usually require an array calculation including the specific errors; there are no general rules for these effects because there are many types and combinations of these errors. Random errors, however, are subject to statistical manipulation, and this allows general, and often simple, results to be obtained.

In well-designed and -manufactured arrays, element position and orientation errors are small, and can be replaced by equivalent phase (and, if needed, amplitude) errors. Thus this section is concerned with the effects of phase and amplitude errors only. Quantization of phase, and the associated phenomena, are treated in detail in Chapter 2. Array optimization subject to constraints on tolerances is discussed in Chapter 4. Array errors are observed in three areas: directivity (gain) reduction; raising of average sidelobe level and of some individual sidelobes; and main beam pointing and perhaps shape change. Each of these is analyzed separately. All random errors will be assumed to be normally distributed (Gaussian) with zero mean and variance of σ^2. Arrays will be assumed sufficiently large that the central limit theorem applies. This means for a linear array $N \geq 10$, and desirably $N \geq 20$. Linear arrays are analyzed; extensions to planar arrays are shown later.

15.4.1 Directivity Reduction and Average Sidelobe Level

Consider a linear array of isotropic elements, with space factor

$$f_0(u) = \sum A_n \exp jn\pi u \tag{15.6}$$

With errors, the coefficient becomes

$$A_n(1+\delta_a)\exp j\delta_\phi \tag{15.7}$$

where the amplitude error is δ_a and the phase error is δ_ϕ in radians. The array power pattern becomes

$$f^2 = \sum\sum A_n A_m^*(1+\delta_{an})(1+\delta_{am})\exp\{j[\delta_{\phi n} - \delta_{\phi m} + (n-m)\pi u]\} \tag{15.8}$$

Under the assumptions above, the mean of the sum of independent variables equals the sum of the means, giving the average pattern

$$f^2 = f_0^2 + \frac{\sigma^2}{G_0} \tag{15.9}$$

where G_0 is the error-free directivity $\sigma^2 = \sigma_a^2 + \sigma_\phi^2$, where σ_a is the amplitude standard deviation, and σ_ϕ is the phase standard deviation, in radians. Directivity, normalized to G_0 is

$$\frac{G}{G_0} \simeq \exp(-\sigma^2) \simeq \frac{1}{1+\sigma^2} \tag{15.10}$$

For example, 1σ errors of 10° phase and 1-dB amplitude reduce directivity by 0.193 dB. Ruze (1952, 1966) developed this theory, which also shows that the pattern is raised slightly, although the increase is apparent only in the sidelobe region. This increase is less for higher directivity arrays. Clearly, the sidelobe increase affects low-sidelobe designs more than uniform amplitude -13-dB sidelobes. From Eq. 15.10, the mean sidelobe level (SLL) is related to the design sidelobe level (SLL_0) by

$$\text{SLL} = \text{SLL}_0\sqrt{1+\sigma^2/(G_0\cdot \text{SLL}_0^2)} \tag{15.11}$$

This is shown in Figure 15.15 versus the universal parameter $\sigma/\sqrt{G_0}$. Note that the sidelobe levels are in voltage in this expression. Clearly, larger (higher directivity) arrays are less sensitive. An approximate result for average sidelobe level with respect to isotropic is based on a filled aperture of area A, where directivity is $4\pi A/\lambda^2$ (Mailloux, 1994):

$$\text{SLL} \simeq \pi\left(\sigma_a^2 + \sigma_\phi^2\right) \tag{15.12}$$

Here SLL is in power. Because the variances of phase and amplitude add, the plot of SLL versus σ_a and σ_ϕ is a circle, as seen in Figure 15.16. From this plot, it is easy to find combinations of errors that produce a specified average sidelobe level.

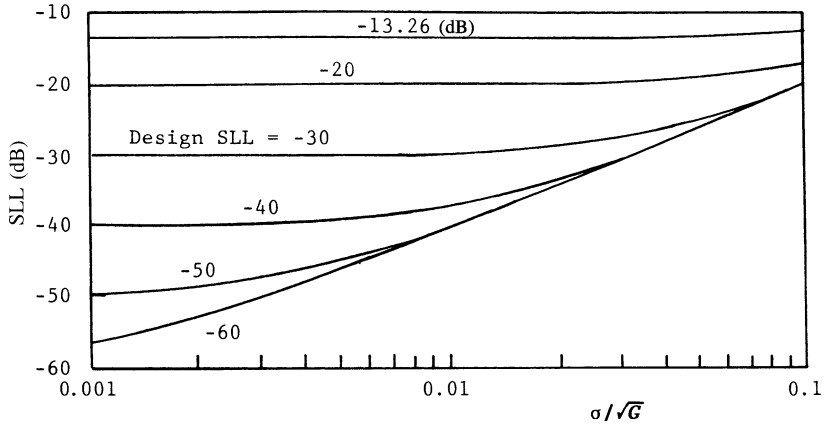


FIGURE 15.15 Mean sidelobe level versus universal factor.

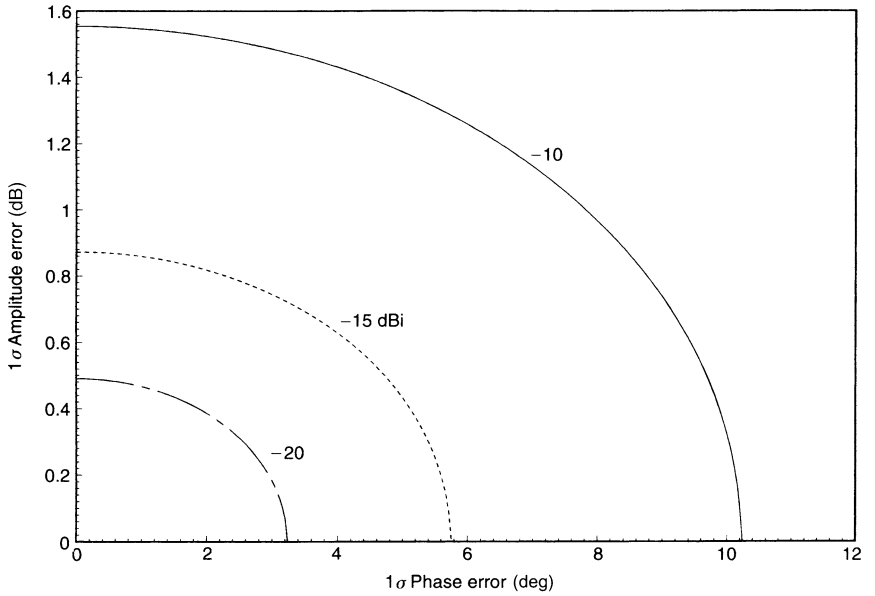


FIGURE 15.16 Average sidelobe level due to random errors.

15.4.2 Beam Pointing Error

Assuming that element position and orientation errors are subsumed into phase and amplitude errors, the beam pointing variance for linear array becomes (Rondinelli, 1959; Carver et al., 1973; Steinberg, 1976):

$$\sigma_\theta^2 = \frac{\sigma^2 \sum_{-N/2}^{N/2} A_n^2 k^2 d^2 n^2}{\left[\sum_{-N/2}^{N/2} A_n k^2 d^2 n^2\right]^2} \tag{15.13}$$

This is for a symmetric equally spaced array at broadside. An array with uniform excitation has $\sigma_\theta^2 = 3\sigma^2/(\pi^2 N^3 d^2/\lambda^2)$; larger arrays have smaller beam pointing error, but the beamwidth is correspondingly smaller. Since the uniform linear array beamwidth is $0.886\lambda/Nd$ with d the element spacing, the standard deviation of the beam pointing error normalized to the 3-dB beamwidth is

$$\frac{\sigma_\theta}{\theta_3} = \frac{\sqrt{3}\sigma}{0.886\pi\sqrt{N}} = \frac{0.622\sigma}{\sqrt{N}} \tag{15.14}$$

where σ^2 is the variance of phase and amplitude errors. Typically, the beam pointing errors are small, and as a fraction of beamwidth they decrease slowly with number of elements, for a uniform linear array. For a uniform square planar array of N elements per side, the standard deviation of the beam pointing error, in terms of the 3-dB beamwidth is

$$\frac{\sigma_\theta}{\theta_3} = \frac{\sqrt{3}\sigma}{0.886\pi N} = \frac{0.662\sigma}{N} \tag{15.15}$$

Note that the error now decreases as $1/N$ owing to the averaging of a larger number of errors.

15.4.3 Peak Sidelobes

The probability that peak sidelobes will exceed a specified SLL is found from the probability density function (PDF); this is a Ricean distribution, as the field at any angle is the vector sum of the error-free field E_0 and the error components (Bennett, 1956; Skolnik, 1969). This PDF is (Rondinelli, 1959; Hsaio, 1984)

$$\text{PDF} = \frac{\text{SLL}\cdot\text{SLL}_0}{\Delta^2}\exp\left(-\frac{\text{SLL}^2+\text{SLL}_0^2}{2\Delta^2}\right) I_0\left(\frac{\text{SLL}\cdot\text{SLL}_0}{\sigma^2}\right) \tag{15.16}$$

where the variance $\sigma^2 = \Delta^2/2G_0$ and I_0 is a modified Bessel function. Now the probability of individual sidelobes equal to or less than a specified sidelobe level is given by the integral of the PDF from zero to the specified SLL:

$$P(\text{SLL} \le \text{SLL}_0) = \int_0^{\text{SLL}} \text{PDF}\ d\text{SLL} \tag{15.17}$$

Through a change of variable this can be written as a Marcum Q function (Marcum, 1960):

$$P = 1 - Q(A\cdot B) \tag{15.18}$$

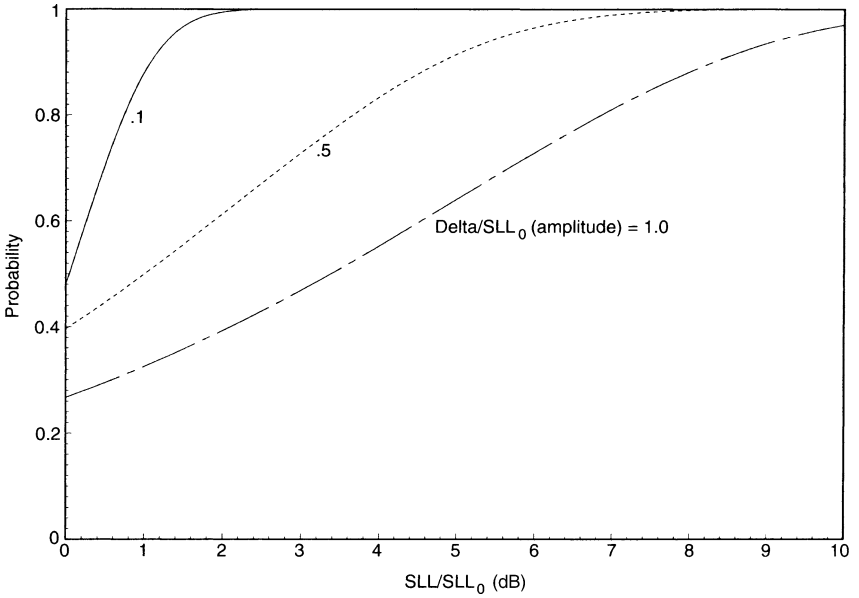


FIGURE 15.17 Probability of increase in sidelobe level.

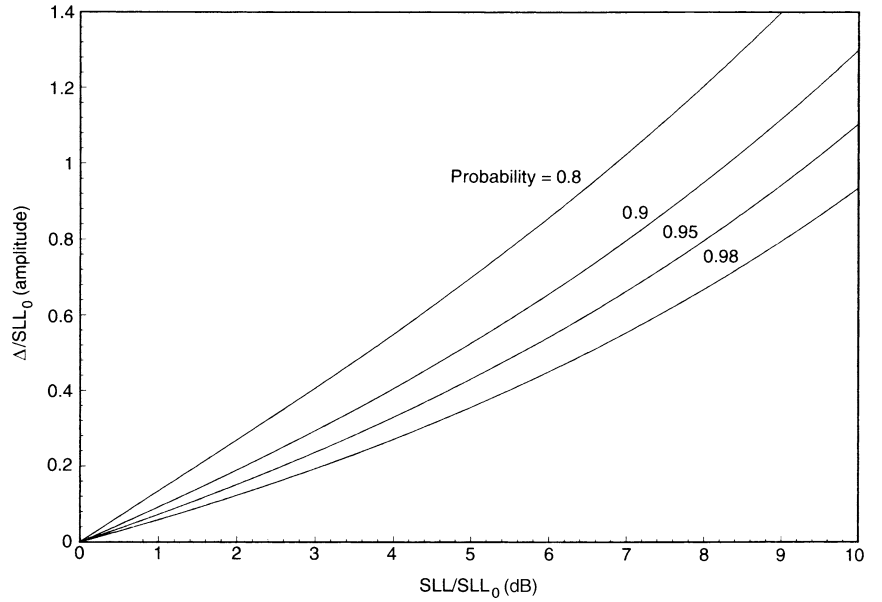


FIGURE 15.18 Tolerance versus sidelobe level increase.

Here the parameters are $A = \mathrm{SLL}_0/\Delta$ and $B = \mathrm{SLL}/\Delta$. The Q function definition is

$$Q(A,B) = \int_B^\infty t \exp\left[\frac{1}{2}\left(A^2 + t^2\right)\right] I_0(At)dt \tag{15.19}$$

When the relative errors are small, the Ricean distribution reduces to a Rayleigh distribution; this allows the cumulative probability to be simply integrated. However, when the sidelobe region of interest is low, or when the errors are not small, the more accurate results above should be used.

An iterative algorithm for Q function calculation of Brennan and Reed (1965) has been improved by McGee (1970). With double-precision arithmetic this routine is suitable for all but small Q values and large A, B values. Asymptotic formulas (Helstrom, 1968) have proved disappointing.

Figure 15.17 shows the probability that sidelobes will be below a given increase over the design sidelobe level SLL_0, as a function of the increase in decibels, for values of Δ/SLL_0 of 0.1, 0.5, and 1. Since probability is more likely to be specified, in Figure 15.18, curves of constant probability are plotted against the sidelobe increase, versus Δ/SLL_0. To give an example, let the probability be 0.9 that all sidelobes will be no larger than 3 dB above SLL_0. Then the Q function solutions are $A = 3.40731$ and $B = 4.81295$, giving $B/A = 3.0$ dB. From this, $\Delta/\mathrm{SLL}_0 = 1/A = 0.2935$, a value that can be obtained directly from Figure 15.18. Further assume an array gain of 20 dB and a design sidelobe level of -30 dB. Then $\Delta = 0.009281$, and since $\sigma^2 = \sigma_a^2 + \sigma_\phi^2 = \Delta^2/2G_0$, the total error variance $\sigma^2 = 0.01723$. If now the amplitude and phase variances are equal, $\sigma_a = 0.0928$ and $\sigma_\phi = 5.32^\circ$. Thus error standard deviations of 9% in amplitude and 5° in phase should have a 90% probability of raising the sidelobe level 3 dB, for this example. In general, as mentioned above, longer arrays are more tolerant, as are higher design sidelobes. Figure 15.18 allows a rapid determination of acceptable tolerances.

ACKNOWLEDGMENT

Photographs courtesy of Peter Hannan.

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